

AOS-CX 10.07 Command-Line Interface Guide

6100 Switch Series



a Hewlett Packard
Enterprise company

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Contents	3
About this document	23
Applicable products	23
Latest version available online	23
Command syntax notation conventions	23
About the examples	24
Identifying switch ports and interfaces	24
Introduction to the AOS-CX CLI	26
CLI access	26
Getting CLI help	26
Authority levels	27
Command contexts	28
Operator context (>)	28
Navigating to the operator context (>)	28
Auditor context	28
Manager context (#)	29
Navigating to the manager context (#)	29
Global configuration context (config)	29
Navigating to the config context	30
Other configuration command contexts	30
Support for range contexts	31
Rules for range contexts	33
Command history	33
Command completion	34
Pipe () support in show commands	34
Command syntax notation conventions	35
Service OS CLI commands	36
boot	36
cat	36
cd path	37
config-clear	37
cp	38
du	39
erase zeroize	40
exit	41
format	42
identify	42
ls	43
md5sum	44
mkdir	45
mount	45
mv	46
password	47
pwd	47
reboot	48

rm	48
rmdir	49
secure-mode	49
sh	51
umount	51
update	52
version	53
ACL commands	54
ACL application	54
access-list copy	54
access-list ip	57
access-list ipv6	65
access-list log-timer	72
access-list mac	74
access-list resequence	79
access-list reset	81
apply access-list control-plane	84
apply access-list (to interface or LAG)	85
apply access-list (to VLAN)	86
clear access-list hitcounts	87
clear access-list hitcounts control-plane	88
show access-list	89
show access-list control-plane	92
show access-list hitcounts	94
show access-list hitcounts control-plane	95
show capacities	96
show capacities-status	99
ACL and Policy hardware resource commands	101
show resources	101
ARP commands	103
arp inspection	103
arp inspection trust	103
arp ipv4 mac	104
clear arp	105
ip local-proxy-arp	105
ipv6 neighbor mac	106
ip proxy-arp	107
show arp	107
show arp inspection interface	108
show arp inspection statistics	109
show arp state	110
show arp summary	111
show arp timeout	112
show arp vrf	113
show ipv6 neighbors	114
show ipv6 neighbors state	114
Aruba Central commands	116
aruba-central	116
aruba-central support-mode	116
configuration-lockout central managed	117
disable	118
enable	118

location-override	119
show aruba-central	120
show running-config current-context	120
Banner commands	122
banner	122
show banner	123
Boot commands	125
boot set-default	125
boot system	125
show boot-history	127
Cable diagnostic commands	129
diag cable-diagnostic	129
CDP commands	131
cdp	131
clear cdp counters	132
clear cdp neighbor-info	132
show cdp	132
show cdp neighbor-info	133
show cdp traffic	134
Checkpoint commands	135
checkpoint auto	135
checkpoint auto confirm	136
checkpoint diff	136
checkpoint post-configuration	137
checkpoint post-configuration timeout	138
checkpoint rename	139
checkpoint rollback	139
copy checkpoint <CHECKPOINT-NAME> <REMOTE-URL>	140
copy checkpoint <CHECKPOINT-NAME> {running-config startup-config}	141
copy checkpoint <CHECKPOINT-NAME> <STORAGE-URL>	142
copy <REMOTE-URL> checkpoint <CHECKPOINT-NAME>	142
copy <REMOTE-URL> {running-config startup-config}	143
copy running-config {startup-config checkpoint <CHECKPOINT-NAME>}	144
copy {running-config startup-config} <REMOTE-URL>	145
copy {running-config startup-config} <STORAGE-URL>	146
copy startup-config running-config	147
copy <STORAGE-URL> running-config	148
erase {checkpoint <CHECKPOINT-NAME> startup-config all}	149
show checkpoint <CHECKPOINT-NAME>	149
show checkpoint post-configuration	152
show checkpoint list	152
write memory	153
Classifier policy commands	155
Classifier policy application	155
apply policy (Context: config)	155
apply policy (Contexts: config-if, config-vlan)	156
class copy	157
class ip	158
class ipv6	164

class resequence	169
class reset	170
clear policy hitcounts	170
policy	171
policy copy	175
policy resequence	176
policy reset	177
show class	177
show policy	178
CLI session commands	182
alias	182
auto-confirm	183
configure terminal	184
disable	184
do	185
enable	185
end	186
exit	186
list	187
page	187
Pipe () command	188
repeat	189
session-timeout	190
show session-timeout	190
show alias	191
show history	191
CLI user session management commands	193
cli-session	193
Clock commands	196
clock date	196
clock datetime	196
clock time	197
clock timezone	197
show clock	198
CoPP commands	199
Classes of traffic	199
apply copp-policy	199
class	200
clear copp-policy statistics	202
copp-policy	202
default-class	203
reset copp-policy	204
show copp-policy	205
show copp-policy factory-default	206
show copp-policy statistics	207
show tech copp	208
Debug logging commands	210
clear debug buffer	210
debug {all <MODULE-NAME>}	210
debug db	212
debug destination	214

show debug	215
show debug buffer	215
show debug destination	216
Device profile commands	218
aaa authentication port-access allow-cdp-bpdu	218
aaa authentication port-access allow-lldp-bpdu	219
associate cdp-group	220
associate lldp-group	221
associate mac-group	222
associate role	222
disable	223
enable	224
ignore (for CDP groups)	224
ignore (for LLDP groups)	225
ignore (for MAC groups)	226
mac-group	231
match (for CDP groups)	232
match (for LLDP groups)	233
match (for MAC groups)	235
port-access cdp-group	238
port-access device-profile	239
port-access device-profile mode block-until-profile-applied	240
port-access lldp-group	241
show port-access device-profile	242
DHCP client commands	244
ip dhcp	244
http proxy	244
DNS client commands	246
ip dns domain-list	246
ip dns domain-name	246
ip dns host	247
ip dns server address	248
show ip dns	249
Fault monitor commands	250
(enabling, disabling faults)	250
action	251
apply fault-monitor profile	253
fault-monitor profile	254
show fault-monitor profile	255
show interface fault-monitor profile	256
show interface fault-monitor status	257
show running-config	258
threshold	260
vsx-sync (fault monitor)	262
Firmware management commands	263
copy {primary secondary} <REMOTE-URL>	263
copy {primary secondary} <FIRMWARE-FILENAME>	263
copy primary secondary	264
copy <REMOTE-URL>	264
copy secondary primary	265
copy <STORAGE-URL>	266

URL formatting for copy commands	267
TFTP URL	267
SFTP URL	267
USB URL	268
HTTPS server commands	269
https-server rest access-mode	269
https-server session close all	270
https-server vrf	270
show https-server	271
ICMP commands	273
ip icmp redirect	273
ip icmp throttle	273
ip icmp unreachable	274
IGMP commands	275
ip igmp	275
ip igmp apply access-list	275
ip igmp last-member-query-interval	276
ip igmp querier	277
ip igmp querier interval	277
ip igmp querier query-max-response-time	278
ip igmp robustness	279
ip igmp static-group	279
ip igmp version	280
ip igmp version strict	281
no ip igmp	281
show ip igmp	282
show ip igmp counters	283
show ip igmp group	284
show ip igmp groups	286
show ip igmp interface	288
show ip igmp interface counters	289
show ip igmp interface group	289
show ip igmp interface groups	291
show ip igmp interface statistics	292
show ip igmp static-groups	293
show ip igmp statistics	293
In-System Programming commands	295
clear update-log	295
show needed-updates	295
Interface commands	296
allow-unsupported-transceiver	296
default interface	297
description	298
energy-efficient-ethernet	298
flow-control	299
interface	299
interface vlan	300
ip address	300
ip mtu	301
ip source-interface	302
ipv6 address	303

ipv6 source-interface	304
mtu	305
show allow-unsupported-transceiver	306
show interface	306
show interface dom	309
show interface energy-efficient ethernet	309
show interface transceiver	310
show ip interface	312
show ip source-interface	313
show ipv6 interface	314
show ipv6 source-interface	315
shutdown	316

IP Client Tracker commands 317

client track ip	317
client track ip { enable disable auto }	317
client track ip client-limit	318
client track ip update-interval	319
client track ip update-method probe	319
show capacities	320
show client ip { count port vlan }	321

IPv6 RA commands 322

ipv6 address <global-unicast-address>	322
ipv6 address autoconfig	322
ipv6 address link-local	323
ipv6 nd cache-limit	324
ipv6 nd dad attempts	325
ipv6 nd hop-limit	325
ipv6 nd mtu	326
ipv6 nd ns-interval	326
ipv6 nd prefix	327
ipv6 nd ra dns search-list	328
ipv6 nd ra dns server	329
ipv6 nd ra lifetime	330
ipv6 nd ra managed-config-flag	330
ipv6 nd ra max-interval	331
ipv6 nd ra min-interval	332
ipv6 nd ra other-config-flag	333
ipv6 nd ra reachable-time	333
ipv6 nd ra retrans-timer	334
ipv6 nd router-preference	334
ipv6 nd suppress-ra	335
show ipv6 nd global traffic	336
show ipv6 nd interface	336
show ipv6 nd interface prefix	339
show ipv6 nd ra dns search-list	340
show ipv6 nd ra dns server	340

IRDP commands 342

diag-dump irdp basic	342
ip irdp	342
ip irdp holdtime	343
ip irdp maxadvertinterval	344
ip irdp minadvertinterval	344
ip irdp preference	345

show ip irdp	346
Key chain commands	347
accept-lifetime	347
key	348
keychain	349
key-string	349
send-lifetime	350
show capacities keychain	351
show keychain	352
show running-config keychain	353
L1-100Mbps downshift commands	355
downshift enable	355
show interface	356
show interface downshift-enable	358
show running-config interface	359
LACP and LAG commands	361
description	361
hash	361
interface lag	362
ip address	363
ipv6 address	364
lacp hash	365
lacp mode	365
lacp port-id	366
lacp port-priority	367
lacp rate	367
lacp system-priority	368
lag	369
show interface	370
show lacp aggregates	371
show lacp configuration	372
show lacp interfaces	372
shutdown	375
vlan trunk native	376
LLDP commands	378
clear lldp neighbors	378
clear lldp statistics	378
lldp	379
lldp dot3	379
lldp holdtime	380
lldp management-ipv4-address	381
lldp management-ipv6-address	381
lldp med	382
lldp med-location	383
lldp receive	384
lldp reinit	384
lldp select-tlv	385
lldp timer	386
lldp transmit	387
lldp txdelay	388
lldp trap enable	389
show lldp configuration	391

show lldp local-device	392
show lldp neighbor-info	393
show lldp neighbor-info detail	395
show lldp statistics	397
show lldp tlv	399
Local AAA commands	400
aaa accounting all-mgmt	400
aaa authentication console-login-attempts	401
aaa authentication limit-login-attempts	402
aaa authentication login	403
aaa authentication minimum-password-length	404
aaa authorization commands	405
show aaa accounting	407
show aaa authentication	407
show aaa authorization	408
show ssh authentication-method	409
show user	409
ssh password-authentication	410
ssh public-key-authentication	411
user authorized-key	411
Log rotation commands	414
logrotate maxsize	414
logrotate period	414
logrotate target	415
show logrotate	416
Loop protect commands	418
loop-protect	418
loop-protect action	418
loop-protect re-enable-timer	419
loop-protect transmit-interval	420
loop-protect trap loop-detected	420
loop-protect vlan	421
show loop-protect	421
Loopback commands	423
interface loopback	423
ip address	423
ipv6 address	424
show interface loopback	424
MAC address table commands	426
clear mac-address	426
mac-address-table age-time	426
mac-lockout	427
show mac-address-table	428
show mac-address-table address	428
show mac-address-table count	429
show mac-address-table dynamic	430
show mac-address-table lockout	431
show mac-address-table port	431
show mac-address-table static	432
show mac-address-table vlan	433
static-mac	433

Mirroring commands	435
clear mirror	435
comment	435
destination interface	436
disable	437
enable	438
mirror session	438
show mirror	439
source interface	441
source vlan	442
 MLD snooping global configuration commands	 445
ipv6 mld snooping	445
 MLD snooping VLAN configuration commands	 446
ipv6 mld snooping	446
ipv6 mld snooping fastlearn	446
ipv6 mld snooping fastleave vlan	447
ipv6 mld snooping forced fastleave	448
ipv6 mld snooping [version <ver>]	449
ipv6 mld snooping apply access-list	449
ipv6 mld snooping auto vlan	450
ipv6 mld snooping blocked vlan	451
ipv6 mld snooping forward vlan	451
ipv6 mld snooping [static-group <X:X::X:X>]	452
 MLD snooping show commands	 454
show ipv6 mld snooping	454
show ipv6 mld snooping [counters]	454
show ipv6 mld snooping [groups]	455
show ipv6 mld snooping [statistics]	456
show ipv6 mld snooping [vlan <vlan-id> [counters]]	457
show ipv6 mld snooping [vlan <vlan-id> [statistics]]	458
show ipv6 mld snooping [vlan <vlan-id> [group [<group-ip>] [source <source-ip>]]]	458
show ipv6 mld snooping [vlan <vlan-id> [group [port <port_id>]]]	460
show ipv6 mld snooping [static-groups]	461
 MLD configuration commands for interface VLAN	 463
ipv6 mld {enable disable}	463
ipv6 mld apply access-list	463
no ipv6 mld	464
ipv6 mld querier	464
ipv6 mld querier [interval <interval-value>]	465
ipv6 mld last-member-query-interval <interval-value>	465
ipv6 mld querier query-max-response-time <response-time>	466
ipv6 mld robustness	467
ipv6 mld static-group <MULTICAST-GROUP-IP>	467
ipv6 mld version <VERSION>	468
ipv6 mld version <VERSION> [strict]	468
 MLD show commands for interface VLAN	 470
show ipv6 mld	470
show ipv6 mld [interface <intf_id> vlan <vlan-id>]	470
show ipv6 mld [all-vrfs]	471
show ipv6 mld [interface <intf-id> vlan <vlan-id>] [counters]	472

show ipv6 mld [interface <intf-id> vlan <vlan-id>] [groups]]	473
show ipv6 mld [interface (<intf-id> vlan <vlan-id>) [group <group_ip>] [source <source_ip>]]]	474
show ipv6 mld [groups]	475
show ipv6 mld groups [all-vrfs]	476
show ipv6 mld [interface <intf-id> [counters]]	477
show ipv6 mld [interface <intf-id> [statistics]]	478
show ipv6 mld [interface <intf-id> [groups]]	478
show ipv6 mld [interface (<intf-id> vlan <vlan-id>) [group <group_ip>] [source <source_ip>]]]	479
show ipv6 mld [group <group_ip> [all-vrfs]]	480
show ipv6 mld [group <group_ip> [source <source_ip> [all-vrfs]]]	481
show ipv6 mld [interface vlan <vlan-id> [statistics]]	482
show ipv6 mld [static-groups [all-vrfs]]	483
show ipv6 mld counters	484

MLD configuration commands for interface 486

ipv6 mld {enable disable}	486
ipv6 mld apply access-list	486
no ipv6 mld	487
ipv6 mld querier	487
ipv6 mld querier [interval <interval-value>]	488
ipv6 mld last-member-query-interval	488
ipv6 mld querier query-max-response-time	489
ipv6 mld robustness	490
ipv6 mld static-group	490
ipv6 mld version	491
ipv6 mld version [strict]	491

MSTP commands 493

show spanning-tree	493
show spanning-tree detail	494
show spanning-tree inconsistent-ports	495
show spanning-tree mst	496
show spanning-tree mst-config	498
show spanning-tree mst detail	498
show spanning-tree mst <INSTANCE-ID>	501
show spanning-tree mst <INSTANCE-ID> detail	502
show spanning-tree mst interface	504
show spanning-tree summary port	504
show spanning-tree summary root	505
spanning-tree	506
spanning-tree bdp-filter	506
spanning-tree bpdu-guard	507
spanning-tree bpdu-guard timeout	508
spanning-tree config-name	509
spanning-tree config-revision	509
spanning-tree cost	510
spanning-tree forward-delay	511
spanning-tree hello-time	512
spanning-tree instance cost	512
spanning-tree instance port-priority	513
spanning-tree instance priority	514
spanning-tree instance vlan	515
spanning-tree link-type	515
spanning-tree loop-guard	516

spanning-tree max-age	517
spanning-tree max-hops	518
spanning-tree mode	519
spanning-tree port-priority	519
spanning-tree port-type	520
spanning-tree priority	521
spanning-tree root-guard	522
spanning-tree rpvt-filter	522
spanning-tree rpvt-guard	523
spanning-tree rpvt-mstp interconnect vlan	524
spanning-tree tcn-guard	524
spanning-tree transmit-hold-count	525
spanning-tree trap	526
MVRP commands	528
clear mvrp statistics	528
mvrp	528
mvrp registration	529
mvrp timer	530
show mvrp config	530
show mvrp state	531
show mvrp statistics	532
NTP commands	534
ntp authentication	534
ntp authentication-key	534
ntp disable	535
ntp enable	536
ntp server	536
ntp trusted-key	538
ntp vrf	538
show ntp associations	539
show ntp authentication-keys	540
show ntp servers	540
show ntp statistics	541
show ntp status	542
Ping commands	544
ping	544
ping6	548
PKI commands	552
crypto pki application	552
crypto pki certificate	553
crypto pki ta-profile	554
enroll self-signed	554
enroll terminal	555
import (CA-signed leaf certificate)	556
import (self-signed leaf certificate)	558
key-type	559
ocsp disable-nonce	560
ocsp enforcement-level	561
ocsp url	562
ocsp vrf	563
revocation-check ocsp	563
show crypto pki application	564

show crypto pki certificate	564
show crypto pki ta-profile	566
ta-certificate	568
subject	569
PKI EST commands	571
arbitrary-label	571
arbitrary-label-enrollment	572
arbitrary-label-reenrollment	573
crypto pki est-profile	574
enroll est-profile	575
reenrollment-lead-time	575
retry-count	576
retry-interval	577
show crypto pki est-profile	578
url	579
username	580
vrf	581
PoE commands	583
lldp dot3 poe	583
lldp med poe	583
power-over-ethernet	584
power-over-ethernet allocate-by	585
power-over-ethernet assigned-class	586
power-over-ethernet pre-std-detect	586
power-over-ethernet priority	587
power-over-ethernet threshold	588
power-over-ethernet priority	588
show lldp local	589
show lldp neighbor	590
show power-over-ethernet	590
Port access general commands	595
aaa authentication port-access auth-mode	595
aaa authentication port-access auth-precedence	595
aaa authentication port-access auth-priority	596
aaa authentication port-access client-limit	597
aaa authentication port-access (role)	598
port-access allow-flood-traffic	599
port-access client-move	599
port-access fallback-role	600
port-access log-off client	600
port-access onboarding-method precedence	601
port-access onboarding-method concurrent	602
port-access reauthenticate interface	603
port-access security violation action	604
port-access security violation action shutdown auto-recovery	605
port-access security violation action shutdown recovery-timer	606
show aaa authentication port-access interface client-status	606
show port-access clients	607
show port-access clients detail	609
show port-access clients onboarding-method	611
show port-access security violation client-limit-exceeded interface	612
Port access 802.1X authentication commands	614

aaa authentication port-access dot1x authenticator	614
aaa authentication port-access dot1x authenticator auth-method	615
aaa authentication port-access dot1x authenticator cached-reauth	615
aaa authentication port-access dot1x authenticator cached-reauth-period	616
aaa authentication port-access dot1x authenticator discovery-period	617
aaa authentication port-access dot1x authenticator eapol-timeout	617
aaa authentication port-access dot1x authenticator max-eapol-requests	618
port-access dot1x authenticator max-retries	619
aaa authentication port-access dot1x authenticator quiet-period	620
aaa authentication port-access dot1x authenticator radius server-group	620
aaa authentication port-access dot1x authenticator reauth	621
aaa authentication port-access dot1x authenticator reauth-period	621
clear dot1x authenticator statistics interface	622
show aaa authentication port-access dot1x authenticator interface client-status	623
show aaa authentication port-access dot1x authenticator interface port-statistics	624

Port access MAC authentication commands 627

aaa authentication port-access mac-auth	627
aaa authentication port-access mac-auth addr-format	627
aaa authentication port-access mac-auth auth-method	628
aaa authentication port-access mac-auth cached-reauth	629
aaa authentication port-access mac-auth cached-reauth-period	629
aaa authentication port-access mac-auth password	630
aaa authentication port-access mac-auth quiet-period	631
aaa authentication port-access mac-auth radius server-group	631
aaa authentication port-access mac-auth reauth	632
aaa authentication port-access mac-auth reauth-period	633
clear mac-auth statistics	633
show aaa authentication port-access mac-auth interface client-status	634
show aaa authentication port-access mac-auth interface port-statistics	636

Port access policy commands 638

port-access policy	638
port-access policy copy	642
port-access policy resequence	643
port-access policy reset	643
clear port-access policy hitcounts	645
Description	645
show port-access policy	647
show port-access policy hitcounts	648

Port access role commands 650

associate policy	650
auth-mode	650
cached-reauth-period	651
client-inactivity timeout	652
description	652
mtu	653
poe-priority	653
port-access role	654
reauth-period	655
session timeout	655
show aaa authentication port-access interface client-status	656
show port-access role	657
stp-admin-edge-port	658
trust-mode	659

vlan	660
Port access VLAN group commands	662
associate-vlan	662
port-access vlan-group	662
show running-config port-access vlan-group	663
Port filtering commands	665
portfilter	665
show portfilter	666
Port security commands	668
port-access port-security	668
port-access port-security client-limit	668
port-access port-security mac-address	669
show port-access port-security interface client-status	670
show port-access port-security interface port-statistics	671
sticky-learn enable	672
show port-access security violation sticky-mac-client-move interface	672
QoS commands	674
apply qos	674
map queue	676
min-bandwidth	677
name queue	678
qos cos	679
qos dscp	679
qos dscp-map	681
qos queue-profile	682
qos schedule-profile	683
qos trust	684
rate-limit	685
show interface queues	687
show interface qos	688
show qos dscp-map	689
show qos queue-profile	690
show qos schedule-profile	691
show qos trust	692
strict queue	693
Configurable RADIUS attribute commands	695
aaa radius-attribute group	695
nas-id request-type	695
nas-id value	696
tunnel-private-group-id request-type	697
tunnel-private-group-id value	698
RADIUS dynamic authorization commands	700
radius dyn-authorization enable	700
radius dyn-authorization client	700
radius dyn-authorization port	701
show radius dyn-authorization	702
show radius dyn-authorization client	704
Remote AAA (TACACS+, RADIUS) commands	706

aaa accounting all-mgmt	706
aaa authentication allow-fail-through	708
aaa authentication login	708
aaa authorization commands	710
aaa group server	712
radius-server auth-type	713
radius-server host	714
radius-server host secure ipsec	716
radius-server key	720
radius-server retries	721
radius-server timeout	721
radius-server tracking	722
server	724
show aaa accounting	725
show aaa authentication	727
show aaa authorization	729
show aaa server-groups	730
show accounting log	732
show radius-server	734
show radius-server secure ipsec	737
show tacacs-server	738
show tacacs-server statistics	740
show tech aaa	741
tacacs-server auth-type	744
tacacs-server host	745
tacacs-server key	747
tacacs-server timeout	748
tacacs-server tracking	748

Remote syslog commands 751

logging	751
logging filter	753
logging facility	756
logging persistent-storage	756

RPVST+ commands 758

show spanning-tree	758
show spanning-tree inconsistent-ports	760
show spanning-tree summary port	761
show spanning-tree summary root	762
show spanning-tree vlan	762
spanning-tree bpduguard timeout	764
spanning-tree extend-system-id	765
spanning-tree ignore-pvid-inconsistency	766
spanning-tree link-type	767
spanning-tree mode	767
spanning-tree pathcost-type	768
spanning-tree rpvst-mstp interconnect vlan	769
spanning-tree tcnguard	770
spanning-tree vlan	770
spanning-tree vlan cost	771
spanning-tree vlan port-priority	772
spanning-tree trap	773

Runtime diagnostic commands 775

diagnostic monitor	775
--------------------------	-----

diag on-demand	775
show diagnostic	776
show diagnostic events	778
Selftest commands	780
fastboot	780
show selftest	781
sFlow agent commands	783
clear sflow statistics	783
sflow	783
sflow agent-ip	785
sflow collector	785
sflow disable	786
sflow header-size	787
sflow max-datagram-size	787
sflow polling	788
sflow sampling	788
show sflow	789
SNMP commands	791
event-trap-enable	791
lldp trap enable	791
rmon alarm	793
rmon alarm {enable disable} {index all}	794
show rmon alarm	795
show snmp agent-port	797
show snmp community	797
show snmp system	798
show snmp trap	798
show snmp vrf	799
show snmpv3 context	799
show snmpv3 engine-id	800
show snmpv3 security-level	800
show snmpv3 users	801
snmp-server agent-port	801
SNMP server community commands	802
snmp-server community	802
access-level	803
access-list	804
snmp-server historical-counters-monitor	805
snmp-server host	805
snmp-server system-contact	807
snmp-server system-description	808
snmp-server system-location	808
snmp-server vrf	809
snmp-server trap-source interface vrf	810
snmp-server trap	811
snmp-server trap vsx	812
snmpv3 context	813
snmpv3 engine-id	814
snmpv3 security-level	814
snmpv3 user	815
Source-interface selection commands	818
ip source-interface	818

ip source-interface interface	819
ipv6 source-interface	821
ipv6 source-interface interface	822
show ip source-interface	823
show ipv6 source-interface	825
show running-config	826
SSH client commands	828
ssh (client login)	828
SSH server commands	830
show ssh host-key	830
show ssh server	831
show ssh server sessions	832
ssh ciphers	833
ssh host-key	834
ssh host-key-algorithms	835
ssh key-exchange-algorithms	836
ssh known-host remove	837
ssh macs	838
ssh maximum-auth-attempts	839
ssh public-key-algorithms	840
ssh server vrf	841
Static routing commands	842
ip route	842
ip route distance	842
ip route tag	843
ipv6 route	844
ipv6 route distance	845
ipv6 route tag	846
show ipv4 rib	847
show ipv6 rib	849
Supportability copy commands	852
copy command-output	852
copy diag-dump feature <FEATURE>	853
copy diag-dump local-file	853
copy show-tech feature	854
copy show-tech local-file	855
copy support-files	856
copy support-files local-file	857
copy support-log	858
Switch system and hardware commands	860
clear events	860
domain-name	860
hostname	861
led locator	862
show capacities	862
show capacities-status	865
show core-dump	866
show domain-name	868
show environment fan	868
show environment led	870
show environment power-supply	870

show environment temperature	871
show events	872
show hostname	875
show images	875
show module	876
show running-config	877
show running-config current-context	881
show startup-config	882
show system	883
show system resource-utilization	884
show tech	885
show usb	886
show usb file-system	887
show version	888
system resource-utilization poll-interval	889
top cpu	889
top memory	890
usb	890
usb mount unmount	891
Terminal monitor commands	893
terminal-monitor {notify severity filter}	893
show terminal-monitor	894
Traceroute commands	895
traceroute	895
traceroute6	897
UDLD commands	899
clear udld statistics	899
show udld	899
udld	901
udld interval	902
udld mode	903
udld retries	905
UDP commands	906
ip forward-protocol udp	906
ip udp-bcast-forward	907
show ip forward-protocol udp	907
User and group commands	909
user	909
user-group	911
user password	914
service export-password	916
show user-group	917
show user information	918
show user-list	919
VLAN commands	921
description	921
name	921
show capacities svi-count	922
show vlan	922

show vlan port	923
show vlan summary	924
shutdown	924
vlan	925
vlan access	926
vlan trunk allowed	927
vlan trunk native	928
vlan trunk native tag	929
voice	930
Zeroization commands	931
erase all zeroize	931
ZTP commands	933
show ztp information	933
ztp force provision	936
Support and Other Resources	938
Accessing Aruba Support	938
Accessing Updates	938
Aruba Support Portal	938
My Networking	939
Warranty Information	939
Regulatory Information	939
Documentation Feedback	939

This document describes features of the AOS-CX network operating system. It is intended for administrators responsible for installing, configuring, and managing Aruba switches on a network.

Applicable products

This document applies to the following products:

- Aruba 6100 Switch Series (JL675A, JL676A, JL677A, JL678A, JL679A)

Latest version available online

Updates to this document can occur after initial publication. For the latest versions of product documentation, see the links provided in [Support and Other Resources](#).

Command syntax notation conventions

Convention	Usage
example-text	Identifies commands and their options and operands, code examples, filenames, pathnames, and output displayed in a command window. Items that appear like the example text in the previous column are to be entered exactly as shown and are required unless enclosed in brackets ([]).
example-text	In code and screen examples, indicates text entered by a user.
Any of the following: ■ <i><example-text></i> ■ <code><example-text></code> ■ <i>example-text</i> ■ example-text	Identifies a placeholder—such as a parameter or a variable—that you must substitute with an actual value in a command or in code: ■ For output formats where italic text cannot be displayed, variables are enclosed in angle brackets (< >). Substitute the text—including the enclosing angle brackets—with an actual value. ■ For output formats where italic text can be displayed, variables might or might not be enclosed in angle brackets. Substitute the text including the enclosing angle brackets, if any, with an actual value.
	Vertical bar. A logical OR that separates multiple items from which you can choose only one. Any spaces that are on either side of the vertical bar are included for readability and are not a required part of the command syntax.
{ }	Braces. Indicates that at least one of the enclosed items is required.
[]	Brackets. Indicates that the enclosed item or items are optional.

Convention	Usage
... or ...	Ellipsis: ■ In code and screen examples, a vertical or horizontal ellipsis indicates an omission of information. ■ In syntax using brackets and braces, an ellipsis indicates items that can be repeated. When an item followed by ellipses is enclosed in brackets, zero or more items can be specified.

About the examples

Examples in this document are representative and might not match your particular switch or environment. The slot and port numbers in this document are for illustration only and might be unavailable on your switch.

Understanding the CLI prompts

When illustrating the prompts in the command line interface (CLI), this document uses the generic term `switch`, instead of the host name of the switch. For example:

```
switch>
```

The CLI prompt indicates the current command context. For example:

```
switch>
```

Indicates the operator command context.

```
switch#
```

Indicates the manager command context.

```
switch (CONTEXT-NAME)#
```

Indicates the configuration context for a feature. For example:

```
switch(config-if) #
```

Identifies the `interface` context.

Variable information in CLI prompts

In certain configuration contexts, the prompt may include variable information. For example, when in the VLAN configuration context, a VLAN number appears in the prompt:

```
switch(config-vlan-100) #
```

When referring to this context, this document uses the syntax:

```
switch(config-vlan-<VLAN-ID>) #
```

Where `<VLAN-ID>` is a variable representing the VLAN number.

Identifying switch ports and interfaces

Physical ports on the switch and their corresponding logical software interfaces are identified using the format:

```
member/slot/port
```

On the 6100 Switch Series

- *member*: Always 1. VSF is not supported on this switch.
- *slot*: Always 1. This is not a modular switch, so there are no slots.
- *port*: Physical number of a port on the switch.

For example, the logical interface 1/1/4 in software is associated with physical port 4 on the switch.

CLI access

Access the CLI through the following interfaces:

Console port

Connect the management port on the switch to your computer using a serial cable and then use terminal emulation software to reach the switch from the computer. Typically, the console port is used when first installing the switch and performing initial configuration tasks.

Management port (out-of-band connection)

Connect the management port on the switch to your network, and then use SSH client software to reach the switch from a computer connected to the same network. This requires that a DHCP server is installed on the network.

In the switch factory default state, the management port and SSH on the management VRF (`mgmt`) are enabled.

Data port (in-band connection)

Connect a data port on the switch to your network, and then use SSH client software to reach the switch from a computer connected to the same network. Management traffic ingresses and egresses switch data ports with rest of the traffic on the network, therefore it can be affected by traffic congestion and other issues impacting the network.

Getting CLI help

To show the available commands that you can execute in the current command context, enter the `?` symbol. For example:

```
switch# ?
  boot           Reboot all or part of the system
  checkpoint     Checkpoint information
switch#
```

The `?` symbol does not display on the screen when you enter it.

The commands that are available to you depend on your authority and the command context. In a given command context, you can only list and execute the commands available in that context.

To show the available parameters for a command, enter the command followed by a space and then enter the `?` symbol.

For example:

```
switch(config)# access-list ?
  all          All access-lists
  ip           Internet Protocol v4 (IPv4)
  ipv6         Internet Protocol v6 (IPv6)
  log-timer    Set ACL log timer length (frequency)
  mac          Ethernet MAC Protocol
switch(config)# access-list
```

After the CLI displays the information, it automatically displays the text you entered before you entered the **?** symbol.

If there is no `<cr>` symbol at the end the command help output, the command is not complete as displayed. You must specify one of the listed parameters.

The `<cr>` symbol alone in the command help output indicates that there are no additional parameters and that you must press the enter key to complete the command. For example:

```
switch# list ?
  <cr>
switch# list
```

The `<cr>` symbol at the end of the command help output indicates that the parameters preceding the `<cr>` are optional and you can enter the command as is displayed. For example:

```
switch# configure ?
  terminal     Configuration terminal (default)
  <cr>
switch# configure
```

To show information about a parameter for a command, enter the command and parameter followed by a space, then enter the **?** symbol.

For example:

```
switch(config)# access-list log-timer ?
  <30-300>     Specify value (in seconds)
  default      Default value (300 seconds)
switch(config)# access-list log-timer
```

Authority levels

In command descriptions, the authority level indicates the user role that is required to execute a command:

Administrators

Users with the role: `administrators`

Users with administrator rights can execute any command.

Operators

Users with the role: `operators`.

Users with operator rights can execute commands in the operator context (`>`) only.

Auditors

Users with the role: `auditors`.

Users with auditor rights can execute commands in the auditor context (`auditor>`) only.

Local user group members with execution rights for a command

You can create up to 29 user-defined local user groups on the switch. Each group can be defined to allow execution of up to 1024 specific CLI commands.

Command contexts

The command context determines the following:

- Which parts of the switch can be managed
- Which commands are available to users with the appropriate authority

Command contexts have a parent-child tree structure in which contexts might themselves contain nested contexts.

Operator context (>)

The operator context enables you to execute commands to view—but not change—the configuration.

The operator context requires the least user privilege to execute commands.

In command descriptions, this context is listed as: Operator (>)

Switch prompt example

```
switch>
```

Authority

Operators or Administrators

Showing the available commands in this context

At the command prompt, enter the `?` symbol.

Navigating to the operator context (>)

To navigate to the operator command context (>), do one of the following:

- Log in to the switch CLI with a user ID that has the `operator-group` role.
- From the manager context (`#`), enter the `disable` command.

Auditor context

When you log in to the switch as user with auditor rights, you have access to the auditor command context only.

Users with auditor rights have access to a limited set of commands. for more information about auditors, see the *Security Guide* for your switch and software version.

Switch prompt example

```
auditor>
```

Showing the available commands in this context

At the command prompt, enter the `?` symbol.

Manager context (#)

From the manager context (#), you can execute commands that do not require saving changes to the configuration.

In command descriptions, this context is listed as: Manager (#)

Switch prompt example

```
switch#
```

Authority

Administrators or local user group members with execution rights for this command.

Showing the available commands in this context

At the command prompt, enter the ? symbol.

Access to manager context commands from descendant contexts

The `do` command enables you to access commands from the manager context while you are in a child or descendent context, such as `config` or `config-if`.

For example, to execute the `clear` command from the `config` context, enter the following: `do clear`.

The `show` command can be executed from configuration contexts as well as the manager context, so using the `do` command with the `show` command is deprecated. Support for `do show` might be discontinued in a future software release.

Navigating to the manager context (#)

To navigate to the manager command context (#), do one of the following:

- Log in to the switch CLI with a user ID that has the `administrators` role.
- From the operator context (>), enter the `enable` command.

You must have administrator authority to enter the `enable` command.

```
switch> enable
switch#
```

- From the configuration context (`config`), enter either the `exit` or the `end` command.

For example:

```
switch(config)# exit
switch#
```

- From any child or descendent context, enter the `end` command.

For example:

```
switch(config-vlan-100)# end
switch#
```

Global configuration context (config)

From the global configuration context (`config`), you can execute commands that change the configuration of the switch.

In command descriptions, this context is listed as: `config`

Switch prompt example

```
switch(config)#
```

Authority

Administrators or local user group members with execution rights for this command.

Showing the available commands in this context

At the command prompt, enter the `?` symbol.



You can use the `do` command to execute some manager context commands—such as the `clear` command—from the global configuration context.

Navigating to the config context

To navigate to the `config` command context, do one of the following:

- From the manager context (`#`), enter the `configure terminal` command:

```
switch# configure terminal
switch(config)#
```

- From a child configuration context, enter the `exit` command.

For example:

```
switch(config-vlan-100)# exit
switch(config)#
```

Other configuration command contexts

All other configuration command contexts are descendants of the global configuration command context (`config`).

From these command contexts, you can execute commands that apply to that specific context, such as an interface or a VLAN.

Switch prompt examples

- `switch(config-if)#`
- `switch(config-router)#`
- `switch(config-vlan-100)#`

Authority

Administrators or local user group members with execution rights for this command.

Showing the available commands in this context

At the command prompt, enter the `?` symbol.

Support for range contexts

Some switch features enable you to use a single command to apply configuration settings to multiple items. You specify the multiple items by creating a type of command context called a `range` context. Then you can execute commands that are applied to every item in the range. For example:

```
switch(config)# interface 1/1/1-1/1/5  
switch(config-if-<1/1/1-1/1/5>)# no shutdown
```

You can use a range context to specify multiple items for the following:

Physical interfaces

- Command example: `interface 1/1/1-1/1/8,1/1/10,1/1/12`
- Switch prompt example: `switch(config-if-<1/1/1-1/1/8,1/1/10,1/1/12>)#`

LAG interfaces

- Command example: `interface lag 1-10`
- Switch prompt example: `switch(config-if-lag-<1-10>)#`

Loopback interfaces

- Command example: `interface loopback 1-10`
- Switch prompt example: `switch(config-if-loopback-<1-10>)#`

VLAN interfaces

- Command example: `interface vlan 1,2,3-6`
- Switch prompt example: `switch(config-vlan-if-<1,2,3-6>)#`

VLANs

- Command example: `vlan 1-10,15,20-25`
- Switch prompt example: `switch(config-vlan-<1-10,15,20-25>)#`

Commands entered in a range context are applied to each item in the range individually:

- Each item in the range has its own entry in the output of `show running-config` commands.

For example, you can configure a range of interfaces as follows:

```
switch(config)# interface 1/1/1-1/1/5  
switch(config-if-<1/1/1-1/1/5>)# no shutdown
```

In the output for the `show running-config` command, the interfaces are displayed individually:

```
switch(config-if-<1/1/1-1/1/5>)# show running-config  
Current configuration:  
...  
interface 1/1/1  
    no shutdown  
interface 1/1/2
```

```

        no shutdown
interface 1/1/3
        no shutdown
interface 1/1/4
        no shutdown
interface 1/1/5
        no shutdown
...
switch(config-if-<1/1/1-1/1/5>)#

```

- If you specify a range context for interfaces, you cannot execute commands that create a context within the range context. For example, you cannot execute the `vrrp` command from an interface range context, even though you can execute the command from the `config-if` context for a single interface.
- If error is encountered during the execution of a command for an item in the range, the error message returned includes a prefix that identifies the item to which the error applies. However command execution does not stop until the command is attempted on all the items in the range.

For example, attempting to set an IP address in a range context of loopback interfaces results in the IP address being applied to the first loopback interface in the range, but results in errors for the subsequent interfaces:

```

switch(config)# interface loopback 1-4
switch(config-loopback-if-<1-4>)# ip address 10.1.11.11/24
[loopback2] Overlapping networks observed for "10.1.11.11/24". Please configure
non overlapping networks.
[loopback3] Overlapping networks observed for "10.1.11.11/24". Please configure
non overlapping networks.
[loopback4] Overlapping networks observed for "10.1.11.11/24". Please configure
non overlapping networks.

switch(config-loopback-if-<1-4>)# show running-config | begin 4 "loopback 1"
interface loopback 1
    ip address 10.1.11.11/24
interface loopback 2
interface loopback 3
interface loopback 4

```

- The range context is created only if every item in the range is successfully created or already exists in configuration. If an error occurs during the creation of an item in a range, the items that are created successfully are added to the configuration, but the range context is not created. The switch prompt.

For example, in the following sequence:

1. VLANs 1 through 100 are created successfully, so the switch prompt reflects the range of VLANs:
`switch(config-vlan-<1-100>)#`
2. The command `interface vlan 95-105` fails for VLANs 101 through 105, so the range context is not created and the switch prompt remains in the global configuration context: `switch(config)#`
3. The configuration includes all the VLANs and VLAN interfaces that are created successfully.

```

switch(config)# vlan 1-100
switch(config-vlan-<1-100>)# exit
switch(config)# interface vlan 95-105
VLAN 101 should be created before creating interface VLAN101.
VLAN 102 should be created before creating interface VLAN102.

```



```

VLAN 103 should be created before creating interface VLAN103.
VLAN 104 should be created before creating interface VLAN104.
VLAN 105 should be created before creating interface VLAN105.
switch(config)# show running-config
Current configuration:
...
vlan 1-100
interface vlan95
interface vlan96
interface vlan97
interface vlan98
interface vlan99
interface vlan100
...
switch(config)#

```

- If the `no` form of the command can be used to remove an item from the configuration, you can use a range context with the `no` form of the command to remove multiple items from the configuration.

For example, you can remove VLANs 95 through 100 from the configuration by entering: `no vlan 95-100`

Rules for range contexts

For interfaces that use the *member/slot/port* notation, items in the range must be specified in ascending order.

Contiguous items in the range are represented by the smallest and largest values separated by a hyphen.

For example:

Command: `interface 1/1/1-1/1/8`

Switch prompt: `switch(config-if-<1/1/1-1/1/8>)#`

Command: `vlan 1-10`

Switch prompt: `switch(config-vlan-<1-10>)#`

Noncontiguous items in the range must be separated by commas.

For example:

Command: `interface 1/1/1-1/1/8,1/1/10,1/1/12`

Switch prompt: `switch(config-if-<1/1/1-1/1/8,1/1/10,1/1/12>)#`

Command: `vlan 1-10,15,20-25`

Switch prompt: `switch(config-vlan-<1-10,15,20-25>)#`

The switch prompt is truncated to 50 characters.

Command history

You can use the **up arrow** key or **Ctrl+P** to display the previous command in the session history, if any.

You can use the **down arrow** key or **Ctrl+N** to display the next command in the session history, if any.

You can use the `show history` command to show a numbered list of the commands executed during this session. You use the command numbers to specify commands to repeat using the `repeat` command. The `show history` and `repeat` commands are not saved in the history buffer.

The commands saved in the history command buffer are in the same format in which you entered the commands. If you enter an incomplete command, the command saved in the history command buffer is also an incomplete one.

If you execute the same command repeatedly, the switch saves only the earliest record. However, if you execute the same command in different formats, the switch saves them as different commands.

For example, if you execute the `show startup-config` command repeatedly, the system saves only one command in the history command buffer. If you execute the command in the format of `show start` and `show startup` respectively, the system saves them as two commands.

Command completion

The CLI supports both command abbreviation and command completion:

- If you enter enough letters to match a valid command, the CLI accepts the command.

For example, you can enter `con` instead of `configure` to navigate from the manager context to the global configuration context.

```
switch# con
switch(config)#
```

- If you enter part of a command word and then press the **Tab** key, one of the following occurs:
 - If you have entered enough letters to match a valid command, the CLI displays the remainder of the word.
 - If you have not entered enough letters to match a valid command, the CLI does not complete the command.

If you press the **Tab** key a second time, the CLI displays commands that match the letters you entered.

For example:

```
switch(config)# cl
class clear clock
switch(config)# cl
```

- If you press the **Tab** key twice after a completed word, the CLI displays the command options.

For example, if you enter the word `clock` followed by a space and then press the **Tab** key twice, the CLI displays the commands available in that command context that start with that word, and then displays the prompt—including the characters you entered—enabling you to complete the command without retyping.

```
switch(config)# clock
date          datetime    time    timezone
switch(config)# clock
```

Pipe (|) support in show commands

The pipe (|) command is a CLI session command that filters the output of `show` commands according to the criteria specified by the parameter `include`, `exclude`, `count`, `begin`, or `redirect`.

- The pipe (|) command is supported for use with the `show` command only.
- You can use multiple pipe commands with a single `show` command.

For example: `show running-config | include "vlan" | exclude "vlan2" | count`

- You can use the pipe command with the `page` command.
- Command completion by pressing the **Tab** key is not supported for pipe commands.

Command syntax notation conventions

Convention	Usage
<code>example-text</code>	Identifies commands and their options and operands, code examples, filenames, pathnames, and output displayed in a command window. Items that appear like the example text in the previous column are to be entered exactly as shown and are required unless enclosed in brackets (<code>[]</code>).
example-text	In code and screen examples, indicates text entered by a user.
Any of the following: ■ <code><example-text></code> ■ <i><example-text></i> ■ <code>example-text</code> ■ <i>example-text</i>	Identifies a placeholder—such as a parameter or a variable—that you must substitute with an actual value in a command or in code: ■ For output formats where italic text cannot be displayed, variables are enclosed in angle brackets (<code>< ></code>). Substitute the text—including the enclosing angle brackets—with an actual value. ■ For output formats where italic text can be displayed, variables might or might not be enclosed in angle brackets. Substitute the text including the enclosing angle brackets, if any, with an actual value.
	Vertical bar. A logical OR that separates multiple items from which you can choose only one. Any spaces that are on either side of the vertical bar are included for readability and are not a required part of the command syntax.
{ }	Braces. Indicates that at least one of the enclosed items is required.
[]	Brackets. Indicates that the enclosed item or items are optional.
... or ...	Ellipsis: ■ In code and screen examples, a vertical or horizontal ellipsis indicates an omission of information. ■ In syntax using brackets and braces, an ellipsis indicates items that can be repeated. When an item followed by ellipses is enclosed in brackets, zero or more items can be specified.

boot

Syntax

boot

Description

Presents you with the boot menu prompt. You can then specify which boot profile: primary, secondary, or Service OS console.

Command context

ServiceOS (svos>)

Authority

Administrators or local user group members with execution rights for this command.

Example

Presenting the boot menu prompt:

```
SVOS> boot

ServiceOS Information:
  Version:      PL.01.07.0004-internal
  Build Date:   2020-11-23 18:07:42 PST
  Build ID:     ServiceOS:PL.01.07.0004-internal:133137f635df:202011231807
  SHA:         133137f635dff5778bf3e109eb75825b68d64789

Boot Profiles:

0. Service OS Console
1. Primary Software Image [PL.10.06.0001]
2. Secondary Software Image [PL.10.08.0000-168-g3089099c34e6]

Select profile(primary):
```

cat

Syntax

cat <FILENAME/DIRECTORY-NAME>

Description

Prints the contents of a file to the console. The Service OS does not allow command output redirection, so this command is only useful for reading short text files.

Command context

ServiceOS (svos>)

Parameters

<FILENAME/DIRECTORY-NAME>

Shows the contents of the specified file or directory.

Authority

Administrators or local user group members with execution rights for this command.

Example

Showing the contents of /nos/hosts:

```
SVOS> cat /nos/hosts
127.0.0.1      localhost.localdomain      localhost

SVOS>
```

cd path

Syntax

cd path

Description

Changes the current working directory.

Command context

ServiceOS (svos>)

Authority

Administrators or local user group members with execution rights for this command.

Example

Changing the current working directory:

```
cd /
```

config-clear

Syntax

config-clear

Description

Configures the switch to set all configuration settings to factory default when the switch is restarted. The next time the switch starts, the current `startup-config` is renamed to `startup-config-fixme`, and a new `startup-config` is created with factory default settings.



Using this command is not the same as performing zeroization, which securely erases the entire primary storage and other devices, and not just the configuration.

Command context

ServiceOS (svos>)

Authority

Administrators or local user group members with execution rights for this command.

Example

Configuring the system to clear the switch configuration:

```
SVOS> config-clear

The switch configuration will be cleared.

Continue (y/n)? y
The system has been configured to clear the startup-config on the next
boot. Please execute the 'boot' command to complete this action.
SVOS>
```

cp

Syntax

cp [options] <SOURCE-FILENAME/SOURCE-DIRECTORY> <DESTINATION-FILENAME/DESTINATION-DIRECTORY>

Description

Copies files or directories.

Command context

ServiceOS (svos>)

Parameters

[options]

Selects the options for the command.

-d, -P

Specifies the preservation of symlinks (default if -R).

-a

Same as -dpR.

R, -r

Specifies recursiveness, all files, and subdirectories are copied.

-L

Specifies the following of all symlinks.

-H

Specifies the following of symlinks on command line.

-p

Specifies the preservation of file attributes if possible.

-f

Specifies the overwriting of a file or directory.

-i

Specifies the prompting before an overwrite.

-l, -s

Specifies the creation of (sym) links.

<SOURCE-FILENAME/SOURCE-DIRECTORY>

Specifies the name of the source file or directory.

<DESTINATION-FILENAME/DESTINATION-DIRECTORY>

Specifies the name of the destination file or directory.

Authority

Administrators or local user group members with execution rights for this command.

Example

Copying /home/customers directory to the /home/clients directory:

```
SVOS> cp /home/customers /home/clients
```

du

Syntax

du [options] <FILENAME/DIRECTORY-NAME>...

Description

Shows estimated disk space used for each file or directory or both.

Command context

ServiceOS (svos>)

Parameters

[options]

Selects the options for the command.

-a

Show file sizes.

-L

Shows all symlinks.

-H

Shows symlinks on a command line.

-d, N

Shows limited output to directories (and files with -a) of depth less than N.

-c

Shows the total disk space usage of all files or directories or both.

-l

Shows the count sizes if hard linked.

-s

Shows only a total for each argument.

-x

Does not show directories on different file systems.

-h

Show sizes in human readable format (1K, 243M, and 2G).

-m

Show sizes in megabytes.

-k

Show sizes in kilobytes (default).

<FILENAME/DIRECTORY-NAME>

Specifies the file or directory or both for displaying a size estimate.

Authority

Administrators or local user group members with execution rights for this command.

Example

Estimating disk space for the /nos directory:

```
SVOS> du -ah /nos
196.4M /nos/primary.swi
196.4M /nos
SVOS>
```

erase zeroize

Syntax

erase zeroize

Description

Securely erases any user data contained on the eMMC or other storage devices on the management module.



Back up all data before running this command or all user/config data will be lost.

Command context

ServiceOS (svos>)

Authority

Administrators or local user group members with execution rights for this command.

Example

Erasing user data:

```
SVOS> SVOS> erase --help
Usage: erase zeroize

Securely erases storage devices on the management module.
SVOS>
^^^

^^^

SVOS> erase zeroize
#####WARNING#####
```



```

This will securely erase all customer data and reset the switch
to factory defaults. This will initiate a reboot and render the
switch unavailable until the zeroization is complete.
This should take several minutes to one hour to complete.
#####WARNING#####

Continue (y/n)? y
reboot: Restarting system

ServiceOS Information:
  Version: PL.01.07.0004-internal
  Build Date: 2020-11-23 18:07:42 PST
  Build ID: ServiceOS:PL.01.07.0004-internal:133137f635df:202011231807
  SHA: 133137f635dff5778bf3e109eb75825b68d64789

##### Preparing for zeroization #####

##### Storage zeroization #####
##### WARNING: DO NOT POWER OFF UNTIL #####
##### ZEROIZATION IS COMPLETE #####
##### This should take several minutes #####
##### to one hour to complete #####

##### Restoring files #####

```

exit

Syntax

exit

Description

Logs the user out from the `svos>` prompt.

Command context

ServiceOS (`svos>`)

Authority

Administrators or local user group members with execution rights for this command.

Example

Logging the user out from the `svos>` prompt:

```

SVOS> exit

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Confidential computer software. Valid license from Hewlett Packard Enterprise
Development LP required for possession, use or copying. Consistent with FAR
12.211 and 12.212, Commercial Computer Software, Computer Software
Documentation, and Technical Data for Commercial Items are licensed to the
U.S. Government under vendor's standard commercial license.

To reboot without logging in, enter 'reboot' as the login user name.

```

```
ServiceOS login:
```

format

Syntax

```
format
```

Description

Configures the primary storage device with the correct partition and file system formatting. This command removes all pre-existing data on the primary storage device.

Command context

ServiceOS (svos>)

Authority

Administrators or local user group members with execution rights for this command.

Example

Configuring the primary storage device with the correct partition and file system formatting:

```
SVOS> format
#####WARNING#####
The following action will cause all data on
the primary storage device to be lost. After
formatting has completed, a reboot will be
initiated to complete storage initialization.
#####WARNING#####

Continue? (y/n): y

Working...This may take a few minutes...
```

identify

Syntax

```
identify
```

Description

Command context

ServiceOS (svos>)

Authority

Administrators or local user group members with execution rights for this command.

Example

ls

Syntax

```
ls [<OPTIONS>] [<FILE-NAME>]
```

Description

This command lists directory contents.

Command context

ServiceOS (svos>)

Parameters

<OPTIONS>

Specifies options for the command.

-l

Shows one-column output.

-a

Shows entries which start with a period (.).

-A

Shows output similar to -a, but excludes a period (.) and a double period (..).

-C

Shows output list by columns.

-x

Shows output list by lines.

-d

Shows listing of directory entries instead of contents

-L

Follows symlinks.

-H

Follows symlinks on the command line.

-R

Recurse.

-p

Appends a slash (/) to directory entries.

-F

Appends an indicator to entries. An indicator can be as an asterisk (*) or slash (/) or equal sign (=) or at sign (@) or pipe (|).

-l

Shows the output in a long listing format.

-i

Shows the list inode numbers.

-n

Shows a list of numeric UIDs and GIDs instead of names.

-s

Shows a list of allocated blocks.

-e

Shows in one column a list with the full date and time.

-h

Shows list sizes in human readable format (1K, 243M, 2G) with a one-column output.

-r

Shows in one column a sort in reverse order.

-S
Shows in one column a sort by size.

-X
Shows in the output sort by extension.

-v
Shows in one column a sort by version.

-c
With -l, it shows a sort in one column by `ctime`.

-t
With -l, it shows a sort by `mtime`.

-u
With -l, sort by `atime`.

-C
With -l, it shows a sort in one column by `ctime`

-w <N>
Assumes that the terminal has the number of columns wide as specified by <N>.

--color[={always | never | auto}]
Controls color in the output.

<FILE-NAME>
Specifies the name of the file to list.

Authority

Administrators or local user group members with execution rights for this command.

Example

Listing directory contents:

```
SVOS> ls -la /nos
drwxr-xr-x   3 0      0          4096 Nov 21 03:19 .
drwxr-xr-x  11 0      0          220 Nov 21 03:21 ..
drwx-----   2 0      0        16384 Nov 21 03:20 lost+found
-rwxr-xr-x   1 0      0    205957424 Nov 21 03:19 primary.swi
SVOS>
```

md5sum

Syntax

```
md5sum [-c | -s | -w] [<FILE-NAME>]
```

Description

This command computes and checks the MD5 message digest.

Command context

ServiceOS (svos>)

Parameters

```
[-c | -s | -w]
```

Selects the options for the command.

-c

Specifies to check the sums against the list in files.

-s

Specifies not output anything, status code shows success.

-w

Specifies to warn about improperly formatted checksum lines.

<FILE-NAME>

Specifies the file name to run the checksum against.

Authority

Administrators or local user group members with execution rights for this command.

Example

Computing and checking the MD5 message digest for /nos/primary.swi:

```
SVOS> md5sum /nos/primary.swi
93ffc89e7ec357854704d8e450c4b7ab /nos/primary.swi
SVOS>
```

mkdir

Syntax

```
mkdir [-m | -p] [<DIRECTORY-NAME>]
```

Description

This command makes directories.

Command context

ServiceOS (svos>)

Parameters

[-m | -p]

Specifies the options for the command.

-m

Specifies the mode.

-p

Specifies to make parent directories as needed with no errors for pre-existing directories.

<DIRECTORY-NAME>

Specifies the directory to create.

Authority

Administrators or local user group members with execution rights for this command.

Example

Making the dir directory:

```
SVOS> mkdir dir
```

mount

Syntax

`mount <DEVICE>`

Description

This command mounts the eMMC partitions to the following locations: `/coredump`, `/logs`, `/nos`, `/selftest`, and mounts the USB device to `/mnt/usb`.

Users can mount USB flash drives formatted as either FAT16 or FAT32 with a single partition.

Command context

ServiceOS (svos>)

Parameters

`<DEVICE>`

Specifies the device to be mounted. Supported device options include `all` and `usb`.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Mounting all of the eMMC partitions:

```
SVOS> mount all
SVOS> mount usb
```

mv

Syntax

`mv [-f | -i | -n] <TARGET-DIRECTORY>`

Description

This command moves (renames) files.

Command context

ServiceOS (svos>)

Parameters

- `-f`
Specifies not to prompt before overwriting.
- `-i`
Specifies to prompt before overwriting.
- `-n`
Specifies to not overwrite an existing file.

Authority

Administrators or local user group members with execution rights for this command.

Example

Moving the file named myfile:

```
SVOS> mv myfile
```

password

Syntax

password

Description

Sets the admin user account password for both Service OS and AOS-CX once the user boots into AOS-CX and saves the configuration. This will overwrite the previous password if one exists. User input is masked with asterisks.

This command is not available if enhanced secure mode is set.

Command context

ServiceOS (svos>)

Authority

Administrators or local user group members with execution rights for this command.

Example

Setting the admin account password:

```
SVOS> password
Enter password:*****
Confirm password:*****
SVOS>
```

pwd

Syntax

pwd

Description

Displays the current working directory.

Command context

ServiceOS (svos>)

Authority

Administrators or local user group members with execution rights for this command.

Example

Displaying the current working directory:

```
SVOS> pwd
/home
SVOS>
```

reboot

Syntax

reboot

Description

Reboots the Management Module.

Command context

ServiceOS (svos>)

Authority

Administrators or local user group members with execution rights for this command.

Example

Rebooting the management module:

```
SVOS> reboot
reboot: Restarting system
```

rm

Syntax

rm [-f | -i | -R | -r] <FILE-NAME>

Description

Removes files or directories.

Command context

ServiceOS (svos>)

Parameters

[-f | -i | -R | -r]

Selects the options for removing files or directories.

-f

Never prompt before removing files or directories.

-i

Always prompt before removing files or directories.

-R | -r

Recursive.

Authority

Administrators or local user group members with execution rights for this command.

Example

Removing the file named `foo`:

```
SVOS> rm foo
```

rmmdir

Syntax

```
rmmdir [-p] <DIRECTORY-NAME>
```

Description

Removes empty directories.

Command context

ServiceOS (svos>)

Parameters

`-p`

Specifies to remove parent directories.

Authority

Administrators or local user group members with execution rights for this command.

Example

Removing the empty `foo` directory:

```
SVOS> rmmdir foo  
SVOS>
```

secure-mode

Syntax

```
secure-mode <enhanced | standard | status>
```

Description

Sets the secure mode to enhanced or standard secure mode. Also can display the current secure mode. A zeroization is required before switching between enhanced and standard secure modes.

The command also displays a message notifying the user that they are already in the targeted secure mode.

Command context

ServiceOS (svos>)

Authority

Administrators or local user group members with execution rights for this command.

Example

Setting the secure mode to enhanced or standard:

```
SVOS> secure-mode --help
Usage: secure-mode <enhanced | standard | status>

Set or retrieve the secure mode setting. Requires a zeroization to change modes.
SVOS>
...

...

SVOS> secure-mode enhanced
#####WARNING#####
This will set the switch into enhanced secure mode. Before
enhanced secure mode is enabled, the switch must securely erase
all customer data and reset the switch to factory defaults.
This will initiate a reboot and render the switch unavailable
until the zeroization is complete.
This should take several minutes to one hour to complete.
#####WARNING#####

Continue (y/n)? y
reboot: Restarting system

...

...

SVOS> secure-mode standard
#####WARNING#####
This will set the switch into standard secure mode. Before
standard secure mode is enabled, the switch must securely erase
all customer data and reset the switch to factory defaults.
This will initiate a reboot and render the switch unavailable
until the zeroization is complete.
This should take several minutes to one hour to complete.
#####WARNING#####

Continue (y/n)? y
reboot: Restarting system

...

...

SVOS> secure-mode standard
#####WARNING#####
Secure mode is already set to standard. Setting it again will
repeat the zeroization process. The switch must securely erase
all customer data and reset the switch to factory defaults.
This will initiate a reboot and render the switch unavailable
until the zeroization is complete.
This should take several minutes to one hour to complete.
#####WARNING#####

Continue (y/n)? y
reboot: Restarting system

...

...

SVOS> secure-mode status
enhanced secure mode is set.
SVOS>
```

sh

Syntax
sh

Description

Launches a bash shell for support purposes. To quit bash, enter `exit`.
This command is not available if enhanced secure mode is set.

Command context

ServiceOS (svos>)

Authority

Administrators or local user group members with execution rights for this command.

Example

Launching a bash shell:

```
SVOS> sh
switch:/cli/fs/home#
```

umount

Syntax

umount <DEVICE>

Description

Unmounts the eMMC partitions mounted to the following locations: `/coredump`, `/logs`, `/nos`, `/selftest`, and unmounts the USB device mounted to `/mnt/usb`.

Command context

ServiceOS (svos>)

Parameters

<DEVICE>

Specifies the device to be unmounted. Supported device options include `all` and `usb`.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Unmounting all devices:

```
SVOS> umount all
SVOS> umount usb
```

Unmounting a USB device:

```
SVOS> umount all
SVOS> umount usb
```

update

Syntax

```
update {primary | secondary} <IMAGE>
```

Description

Verifies and installs a product image. The user can select the primary or secondary boot profile to update and the location of the file.

Command context

ServiceOS (svos>)

Parameters

{primary | secondary}

Selects either the primary or secondary image.

<IMAGE>

Specifies the image name.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Updating the software image using TFTP:



The OOBM port is disabled on first boot and must be enabled using the `ip` command.

```
SVOS> ip dhcp
SVOS> ip show
Interface   : Link Up
IP Address  : 192.0.2.22
Subnet Mask : 255.255.200.20
Gateway     : 10.0.24.1
SVOS> tftp -g -r XL.10.00.0001.swi -l image.swi 192.4.8.10
XL.10.00.0001.swi 100% |*****| 178M 0:00:00 ETA
SVOS> ls
image.swi
SVOS> update primary image.swi
Updating primary software image...
Verifying image...
Done
```

Update the software image using USB:



This example assumes that the user has preloaded a USB flash drive with the image to be updated. The image name on the flash drive is not important.

```
SVOS> mount usb
SVOS> ls /mnt/usb
image.swi
SVOS> update primary /mnt/usb/image.swi
Updating primary software image...
Verifying image...
Done
```

version

Syntax

version

Description

Displays the following build strings:

- Version.
- Build date.
- Build time.
- Build ID.
- SHA.

Command context

ServiceOS (svos>)

Authority

Administrators or local user group members with execution rights for this command.

Example

Displaying version build strings:

```
SVOS> version
ServiceOS Information:
  Version:      GT.01.01.0001
  Build Date:   2017-07-19 14:52:31 PDT
  Build ID:     ServiceOS:GT.01.01.0001:461519208911:201707191452
  SHA:          46151920891195cdb2267ea6889a3c6cbc3d4193
SVOS>
```



On the 6100 Switch Series, only the vrf named `default` is available. Replace any references to the `mgmt` or other VRFs with `default`.

ACL application

ACLs can be applied as follows:

ACL type Direction	IPv4+6 In	MAC In
L2 interface (port)	Yes	Yes
L2 LAG	Yes	Yes
VLAN	Yes	Yes
Control plane (default VRF)	Yes	



The following match criteria is not supported. If this match criteria is attempted to be configured, an error message will be displayed and the action will not be completed.

TTL on IP ACLs

access-list copy

Syntax

```
access-list {ip|ipv6|mac} <ACL-NAME> copy <DESTINATION-ACL>
```

Description

Copies an IPv4, IPv6, or MAC ACL to a new destination ACL or overwrites an existing ACL.

Command context

config

Parameters

{ip|ipv6|mac}

Specifies the type of ACL.

<ACL-NAME>

Specifies the name of the ACL to be copied.

<DESTINATION-ACL>

Specifies the name of the destination ACL.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Copying MY_IP_ACL to MY_IP_ACL2:

```
switch(config)# access-list ip MY_IP_ACL copy MY_IP_ACL2
switch(config-acl-ip)# exit

switch(config)# do show access-list
Type      Name
Sequence  Comment
          Action
          Source IP Address
          Destination IP Address
          Additional Parameters
-----
IPv4      MY_IP_ACL
1 permit
  any
  172.16.1.0/255.255.255.0
2 permit
  172.16.2.0/255.255.0.0
  any
3 permit
  172.26.1.0/255.255.255.0
  any
  dscp: AF11
  ack
  syn
4 deny
  any
  any
  Hit-counts: enabled
-----
IPv4      MY_IP_ACL2
1 permit
  any
  172.16.1.0/255.255.255.0
2 permit
  172.16.2.0/255.255.0.0
  any
3 permit
  172.26.1.0/255.255.255.0
  any
  dscp: AF11
  ack
  syn
4 deny
  any
  any
  Hit-counts: enabled
```

Copying MY_IPV6_ACL to MY_IPV6_ACL2:

```
switch(config)# access-list ipv6 MY_IPV6_ACL copy MY_IPV6_ACL2
switch(config-acl-ip)# exit
```

```
switch(config)# do show access-list
```

Type	Name	Sequence	Comment	Action	L3 Protocol	Source IP Address	Source L4 Port(s)	Destination IP Address	Destination L4 Port(s)	Additional Parameters
------	------	----------	---------	--------	-------------	-------------------	-------------------	------------------------	------------------------	-----------------------

IPv6	MY_IPV6_ACL	1	permit	any	udp	2001::1/64				
		2	Permit all TCP ephemeral ports	permit	tcp	2001:2001::2:1	> 1023			
		3	permit	any	tcp	2001:2011::1/64				
		4	deny	any	any					
										Hit-counts: enabled

IPv6	MY_IPV6_ACL2	1	permit	any	udp	2001::1/64				
		2	Permit all TCP ephemeral ports	permit	tcp	2001:2001::2:1	> 1023			
		3	permit	any	tcp	2001:2011::1/64				
		4	deny	any	any					
										Hit-counts: enabled

Copying MY_MAC_ACL to MY_MAC_ACL2:

```
switch(config)# access-list mac MY_MAC_ACL copy MY_MAC_ACL2
switch(config-acl-mac)# exit
```

```
switch(config)# do show access-list
```

Type	Name	Sequence	Comment	Action	EtherType	Source MAC Address	Destination MAC Address	Additional Parameters
------	------	----------	---------	--------	-----------	--------------------	-------------------------	-----------------------

MAC	MY_MAC_ACL	1	permit	1122.3344.5566/ffff.ffff.0000	ipv6			
		2	permit	any	any			


```

      aaaa.bbbb.cccc
      1111.2222.3333
      QoS Priority Code Point: 4
3 Permit all vlan-1 tagged Appletalk traffic
  permit                                appletalk
  any
  any
  VLAN: 1
4 deny                                any
  any
  any
  Hit-counts: enabled
-----
MAC      MY_MAC_ACL2
1 permit                                ipv6
  1122.3344.5566/ffff.ffff.0000
  any
2 permit                                any
  aaaa.bbbb.cccc
  1111.2222.3333
  QoS Priority Code Point: 4
3 Permit all vlan-1 tagged Appletalk traffic
  permit                                appletalk
  any
  any
  VLAN: 1
4 deny                                any
  any
  any
  Hit-counts: enabled

```

access-list ip

Syntax

Syntax to create an IPv4 ACL and enter its context. Plus syntax to remove an ACL:

```

access-list ip <ACL-NAME>
no access-list ip <ACL-NAME>

```

Syntax (within the ACL context) for creating or removing ACEs for protocols ah, gre, esp, igmp, ospf, pim (ip is available as an alias for any):

```

[<SEQUENCE-NUMBER>]
{permit|deny}
{any|ip|ah|gre|esp|igmp|ospf|pim|<IP-PROTOCOL-NUM>}
{any|<SRC-IP-ADDRESS>[/<PREFIX-LENGTH>|<SUBNET-MASK>]]}
{any|<DST-IP-ADDRESS>[/<PREFIX-LENGTH>|<SUBNET-MASK>]]}
[dscp <DSCP-SPECIFIER>] [ip-precedence <IP-PRECEDENCE-VALUE>]
[tos <TOS-VALUE>] [fragment] [vlan <VLAN-ID>] [count] [log]

no <SEQUENCE-NUMBER>

```

Syntax (within the ACL context) for creating or removing ACEs for protocols sctp, tcp, udp:

```

[<SEQUENCE-NUMBER>]
{permit|deny}
{sctp|tcp|udp}
{any|<SRC-IP-ADDRESS>[/<PREFIX-LENGTH>|<SUBNET-MASK>]]}
[{eq|gt|lt} <PORT>|range <MIN-PORT> <MAX-PORT>]
{any|<DST-IP-ADDRESS>[/<PREFIX-LENGTH>|<SUBNET-MASK>]]}
[{eq|gt|lt} <PORT>|range <MIN-PORT> <MAX-PORT>]
[urg] [ack] [psh] [rst] [syn] [fin] [established]
[dscp <DSCP-SPECIFIER>] [ip-precedence <IP-PRECEDENCE-VALUE>]

```

```
[tos <TOS-VALUE>] [fragment] [vlan <VLAN-ID>] [count] [log]
```

```
no <SEQUENCE-NUMBER>
```

Syntax (within the ACL context) for creating or removing ACEs for protocol icmp:

```
[<SEQUENCE-NUMBER>]
{permit|deny}
{icmp}
{any|<SRC-IP-ADDRESS>[/ {<PREFIX-LENGTH>|<SUBNET-MASK>}]}
{any|<DST-IP-ADDRESS>[/ {<PREFIX-LENGTH>|<SUBNET-MASK>}]}
[icmp-type {echo|echo-reply|<ICMP-TYPE-VALUE>}] [icmp-code <ICMP-CODE-VALUE>]
[dscp <DSCP-SPECIFIER>] [ip-precedence <IP-PRECEDENCE-VALUE>]
[tos <TOS-VALUE>] [fragment] [vlan <VLAN-ID>] [count] [log]
```

```
no <SEQUENCE-NUMBER>
```

Syntax (within the ACL context) for ACE comments:

```
[<SEQUENCE-NUMBER>] comment <TEXT-STRING>
```

```
no <SEQUENCE-NUMBER> comment
```

Description

Creates an IPv4 Access Control List (ACL) comprised of one or more Access Control Entries (ACEs) ordered and prioritized by sequence number. The lowest sequence number is the highest prioritized ACE.

The `no` form of this command deletes the entire ACL, or deletes an ACE identified by sequence number, or deletes only the comment from the ACE identified by sequence number.

Command context

config

The `access-list ip <ACL-NAME>` command takes you into the named ACL context where you enter the ACEs.

Parameters

<ACL-NAME>

Specifies the name of this ACL.

<SEQUENCE-NUMBER>

Specifies a sequence number for the ACE. Range: 1 to 4294967295.

{permit|deny}

Specifies whether to permit or deny traffic matching this ACE.

<IP-PROTOCOL-NUM>

Specifies the protocol as its Internet Protocol number. For example, 2 corresponds to the IGMP protocol. Range: 0 to 255.

{any|<SRC-IP-ADDRESS>[/ {<PREFIX-LENGTH>|<SUBNET-MASK>}]} }

Specifies the source IPv4 address.

- any - specifies any source IPv4 address.
- <SRC-IP-ADDRESS> - specifies the source IPv4 host address.
 - <PREFIX-LENGTH> - specifies the address bits to mask (CIDR subnet mask notation). Range: 1 to 32.
 - <SUBNET-MASK> - specifies the address bits to mask (dotted decimal notation).

{any|<DST-IP-ADDRESS>[/ {<PREFIX-LENGTH>|<SUBNET-MASK>}]} }

Specifies the destination IPv4 address.

- any - specifies any destination IPv4 address.

- `<DST-IP-ADDRESS>` - specifies the destination IPv4 host address.
 - `<PREFIX-LENGTH>` - specifies the address bits to mask (CIDR subnet mask notation). Range: 1 to 32.
 - `<SUBNET-MASK>` - specifies the address bits to mask (dotted decimal notation).

`[{eq|gt|lt} <PORT>|range <MIN-PORT> <MAX-PORT>]`

Specifies the port, or port range. Port numbers are in the range of 0 to 65535.

- `eq <PORT>` - specifies the Layer 4 port.
- `gt <PORT>` - specifies any Layer 4 port greater than the indicated port.
- `lt <PORT>` - specifies any Layer 4 port less than the indicated port.
- `range <MIN-PORT> <MAX-PORT>` - specifies the Layer 4 port range.



Upon application of the ACL, ACEs with L4 port ranges may consume more than one hardware entry.

`urg`

Specifies matching on the TCP Flag: Urgent.

`ack`

Specifies matching on the TCP Flag: Acknowledgment.

`psh`

Specifies matching on the TCP Flag: Push buffered data to receiving application.

`rst`

Specifies matching on the TCP Flag: Reset the connection.

`syn`

Specifies matching on the TCP Flag: Synchronize sequence numbers.

`fin`

Specifies matching on the TCP Flag: Finish connection.

`established`

Specifies matching on the TCP Flag: Established connection.

`[icmp-type {echo|echo-reply|<ICMP-TYPE-VALUE>}]`

Specifies the ICMP type.

- `echo` - specifies an ICMP echo request packet.
- `echo-reply` - specifies an ICMP echo reply packet.
- `<ICMP-TYPE-VALUE>` - specifies an ICMP type value. Range: 0 to 255.

`[icmp-code <ICMP-CODE-VALUE>]`

Specifies the ICMP code value. Range: 0 to 255.

`dscp DSCP-SPECIFIER`

Specifies the Differentiated Services Code Point (DSCP), either a numeric `<DSCP-VALUE>` (0 to 63) or one of these keywords:

- `AF11` - DSCP 10 (Assured Forwarding Class 1, low drop probability)
- `AF12` - DSCP 12 (Assured Forwarding Class 1, medium drop probability)
- `AF13` - DSCP 14 (Assured Forwarding Class 1, high drop probability)
- `AF21` - DSCP 18 (Assured Forwarding Class 2, low drop probability)
- `AF22` - DSCP 20 (Assured Forwarding Class 2, medium drop probability)
- `AF23` - DSCP 22 (Assured Forwarding Class 2, high drop probability)

- AF31 - DSCP 26 (Assured Forwarding Class 3, low drop probability)
- AF32 - DSCP 28 (Assured Forwarding Class 3, medium drop probability)
- AF33 - DSCP 30 (Assured Forwarding Class 3, high drop probability)
- AF41 - DSCP 34 (Assured Forwarding Class 4, low drop probability)
- AF42 - DSCP 36 (Assured Forwarding Class 4, medium drop probability)
- AF43 - DSCP 38 (Assured Forwarding Class 4, high drop probability)
- CS0 - DSCP 0 (Class Selector 0: Default)
- CS1 - DSCP 8 (Class Selector 1: Scavenger)
- CS2 - DSCP 16 (Class Selector 2: OAM)
- CS3 - DSCP 24 (Class Selector 3: Signaling)
- CS4 - DSCP 32 (Class Selector 4: Real time)
- CS5 - DSCP 40 (Class Selector 5: Broadcast video)
- CS6 - DSCP 48 (Class Selector 6: Network control)
- CS7 - DSCP 56 (Class Selector 7)
- EF - DSCP 46 (Expedited Forwarding)

`ip-precedence <IP-PRECEDENCE-VALUE>`

Specifies an IP precedence value. Range: 0 to 7.

`tos <TOS-VALUE>`

Specifies the Type of Service value. Range: 0 to 31.

`fragment`

Specifies a fragment packet.

`vlan <VLAN-ID>`

Specifies VLAN tag to match on. 802.1Q VLAN ID.



This parameter cannot be used in any ACL that will be applied to a VLAN.

`count`

Keeps the hit counts of the number of packets matching this ACE.

`log`

`[<SEQUENCE-NUMBER>] comment <TEXT-STRING>`

Adds a comment to an ACE. The `no` form removes only the comment from the ACE.

Authority

Administrators or local user group members with execution rights for this command.

Usage

- If the `<IP-PROTOCOL-NUM>` parameter is used instead of a protocol name, ensure that any needed ACE-definition parameters specific to the selected protocol are also provided.
- When using multiple ACL types (IPv4, IPv6, or MAC) with logging on the same interface, the first packet that matches an ACE with `log` option is logged. Until the log-timer wait-period is over, any packets matching other ACL types do not create a log. At the end of the wait-period, the switch creates a summary log for all the ACLs that were matched, regardless of type.

Examples

Creating an IPv4 ACL with four entries:

```

switch(config)# access-list ip MY_IP_ACL
switch(config-acl-ip)# 10 permit udp any 172.16.1.0/24
switch(config-acl-ip)# 20 permit tcp 172.16.2.0/16 gt 1023 any
switch(config-acl-ip)# 30 permit tcp 172.26.1.0/24 any syn ack dscp 10
switch(config-acl-ip)# 40 deny any any any count
switch(config-acl-ip)# exit

switch(config)# show access-list
Type      Name
Sequence  Comment
          Action                    L3 Protocol
          Source IP Address         Source L4 Port(s)
          Destination IP Address     Destination L4 Port(s)
          Additional Parameters
-----
IPv4      MY_IP_ACL
10 permit                                udp
   any
   172.16.1.0/255.255.255.0
20 permit                                tcp
   172.16.2.0/255.255.0.0             > 1023
   any
30 permit                                tcp
   172.26.1.0/255.255.255.0
   any
   dscp: AF11
   ack
   syn
40 deny                                any
   any
   any
Hit-counts: enabled

```

Adding a comment to an existing IPv4 ACE:

```

switch(config)# access-list ip MY_IP_ACL
switch(config-acl-ip)# 20 comment Permit all TCP ephemeral ports
switch(config-acl-ip)# exit

switch(config)# show access-list
Type      Name
Sequence  Comment
          Action                    L3 Protocol
          Source IP Address         Source L4 Port(s)
          Destination IP Address     Destination L4 Port(s)
          Additional Parameters
-----
IPv4      MY_IP_ACL
10 permit                                udp
   any
   172.16.1.0/255.255.255.0
20 Permit all TCP ephemeral ports
   permit                                tcp
   172.16.2.0/255.255.0.0             > 1023
   any
30 permit                                tcp
   172.26.1.0/255.255.255.0
   any
   dscp: AF11
   ack

```

```

    syn
40 deny
    any
    any
    Hit-counts: enabled
    any

```

Removing a comment from an existing IPv4 ACE:

```

switch(config)# access-list ip MY_IP_ACL
switch(config-acl-ip)# no 20 comment
switch(config-acl-ip)# exit

switch(config)# show access-list
Type      Name
Sequence  Comment
          Action
          Source IP Address
          Destination IP Address
          Additional Parameters
-----
IPv4      MY_IP_ACL
10 permit
   any
   172.16.1.0/255.255.255.0
20 permit
   172.16.2.0/255.255.0.0
   any
30 permit
   172.26.1.0/255.255.255.0
   any
   dscp: AF11
   ack
   syn
40 deny
   any
   any
   Hit-counts: enabled
          L3 Protocol
          Source L4 Port(s)
          Destination L4 Port(s)

```

Adding an ACE (insert line 25) to an existing IPv4 ACL:

```

switch(config)# access-list ip MY_IP_ACL
switch(config-acl-ip)# 25 permit icmp 172.16.2.0/16 any
switch(config-acl-ip)# exit

switch(config)# show access-list
Type      Name
Sequence  Comment
          Action
          Source IP Address
          Destination IP Address
          Additional Parameters
-----
IPv4      MY_IP_ACL
10 permit
   any
   172.16.1.0/255.255.255.0
20 permit
   172.16.2.0/255.255.0.0
25 permit
   172.16.2.0/16
   any
          L3 Protocol
          Source L4 Port(s)
          Destination L4 Port(s)

```

```

    any
25 permit 172.16.2.0/255.255.0.0 any icmp
30 permit 172.26.1.0/255.255.255.0 tcp
    any
    dscp: AF11
    ack
    syn
40 deny any any
    any
    Hit-counts: enabled

```

Replacing an ACE in an existing IPv4 ACL:

```

switch(config)# access-list ip MY_IP_ACL
switch(config-acl-ip)# 25 permit icmp 172.17.1.0/16 any
switch(config-acl-ip)# exit

switch(config)# show access-list
Type      Name
Sequence  Comment
          Action          L3 Protocol
          Source IP Address Source L4 Port(s)
          Destination IP Address Destination L4 Port(s)
          Additional Parameters
-----
IPv4      MY_IP_ACL
10 permit any udp
          172.16.1.0/255.255.255.0
20 permit 172.16.2.0/255.255.0.0 tcp
          any > 1023
25 permit 172.17.1.0/255.255.0.0 icmp
30 permit 172.26.1.0/255.255.255.0 tcp
    any
    dscp: AF11
    ack
    syn
40 deny any any
    any
    Hit-counts: enabled

```

Removing an ACE from an IPv4 ACL:

```

switch(config)# access-list ip MY_IP_ACL
switch(config-acl-ip)# no 25
switch(config-acl-ip)# exit

switch(config)# show access-list
Type      Name
Sequence  Comment
          Action          L3 Protocol

```

	Source IP Address	Source L4 Port(s)
	Destination IP Address	Destination L4 Port(s)
	Additional Parameters	

IPv4	MY_IP_ACL	
10	permit	udp
	any	
	172.16.1.0/255.255.255.0	
20	permit	tcp
	172.16.2.0/255.255.0.0	> 1023
	any	
30	permit	tcp
	172.26.1.0/255.255.255.0	
	any	
	dscp: AF11	
	ack	
	syn	
40	deny	any
	any	
	any	
	Hit-counts: enabled	

Copy an IPv4 ACL:

```
switch(config)# access-list ip MY_IP_ACL copy MY_IP_ACL2
switch(config)# show access-list
```

Type	Name		
Sequence	Comment		
	Action	L3 Protocol	
	Source IP Address	Source L4 Port(s)	
	Destination IP Address	Destination L4 Port(s)	
	Additional Parameters		

IPv4	MY_IP_ACL		
10	permit	udp	
	any		
	172.16.1.0/255.255.255.0		
20	permit	tcp	
	172.16.2.0/255.255.0.0	> 1023	
	any		
30	permit	tcp	
	172.26.1.0/255.255.255.0		
	any		
	dscp: AF11		
	ack		
	syn		
40	deny	any	
	any		
	any		
	Hit-counts: enabled		

IPv4	MY_IP_ACL2		
10	permit	udp	
	any		
	172.16.1.0/255.255.255.0		


```

20
  permit
  172.16.2.0/255.255.0.0      tcp
  any                        > 1023
30
  permit
  172.26.1.0/255.255.255.0    tcp
  any
  dscp: AF11
  ack
  syn
40
  deny                        any
  any
  any
  Hit-counts: enabled

```

Removing an IPv4 ACL:

```

switch(config)# no access-list ip MY_IP_ACL

switch(config)# show access-list
Type      Name
Sequence  Comment
          Action
          Source IP Address      L3 Protocol
          Destination IP Address  Source L4 Port(s)
          Additional Parameters   Destination L4 Port(s)
-----
IPv4      MY_IP_ACL2
1 permit
  any      udp
  172.16.1.0/255.255.255.0
2 permit
  172.16.2.0/255.255.0.0      tcp
  any                        > 1023
3 permit
  172.26.1.0/255.255.255.0    tcp
  any
  dscp: AF11
  ack
  syn
4 deny      any
  any
  any
  Hit-counts: enabled

```

access-list ipv6

Syntax

Syntax to create an IPv6 ACL and enter its context. Plus syntax to remove an ACL:

```

access-list ipv6 <ACL-NAME>
no access-list ipv6 <ACL-NAME>

```

Syntax (within the ACL context) for creating or removing ACEs for protocols ah, gre, esp, ospf, pim (ipv6 is available as an alias for any):

```

[<SEQUENCE-NUMBER>]
{permit|deny}
{any|ipv6|ah|gre|esp|ospf|pim|<IP-PROTOCOL-NUM>}

```

```

{any|<SRC-IP-ADDRESS>[/<PREFIX-LENGTH>]}
{any|<DST-IP-ADDRESS>[/<PREFIX-LENGTH>]}
[dscp <DSCP-SPECIFIER>] [ip-precedence <IP-PRECEDENCE-VALUE>]
[tos <TOS-VALUE>] [fragment] [vlan <VLAN-ID>] [count] [log]

no <SEQUENCE-NUMBER>

```

Syntax (within the ACL context) for creating or removing ACEs for protocols sctp, tcp, udp:

```

[<SEQUENCE-NUMBER>]
{permit|deny}
{sctp|tcp|udp}
{any|<SRC-IP-ADDRESS>[/<PREFIX-LENGTH>]}
[{eq|gt|lt} <PORT>|range <MIN-PORT> <MAX-PORT>]
{any|<DST-IP-ADDRESS>[/<PREFIX-LENGTH>]}
[{eq|gt|lt} <PORT>|range <MIN-PORT> <MAX-PORT>]
[urg] [ack] [psh] [rst] [syn] [fin] [established]
[dscp <DSCP-SPECIFIER>] [ip-precedence <IP-PRECEDENCE-VALUE>]
[tos <TOS-VALUE>] [fragment] [vlan <VLAN-ID>] [count] [log]

no <SEQUENCE-NUMBER>

```

Syntax (within the ACL context) for creating or removing ACEs for protocol icmpv6:

```

[<SEQUENCE-NUMBER>]
{permit|deny}
{icmpv6}
{any|<SRC-IP-ADDRESS>[/<PREFIX-LENGTH>]}
{any|<DST-IP-ADDRESS>[/<PREFIX-LENGTH>]}
[icmp-type {echo|echo-reply|<ICMP-TYPE-VALUE>}] [icmp-code <ICMP-CODE-VALUE>]
[dscp <DSCP-SPECIFIER>] [ip-precedence <IP-PRECEDENCE-VALUE>]
[tos <TOS-VALUE>] [fragment] [vlan <VLAN-ID>] [count] [log]

no <SEQUENCE-NUMBER>

```

Syntax (within the ACL context) for ACE comments:

```

[<SEQUENCE-NUMBER>] comment <TEXT-STRING>

no <SEQUENCE-NUMBER> comment

```

Description

Creates an IPv6 Access Control List (ACL). The ACL is made of one or more Access Control Entries (ACEs) ordered and prioritized by sequence number. The lowest sequence number is the highest prioritized ACE. The `no` form of this command deletes the entire ACL, or deletes an ACE identified by sequence number, or deletes only the comment from the ACE identified by sequence number.

Command context

config

The `access-list ipv6 <ACL-NAME>` command takes you into the named ACL context where you enter the ACEs.

Parameters

<ACL-NAME>

Specifies the name of this ACL.

<SEQUENCE-NUMBER>

Specifies a sequence number for the ACE. Range: 1 to 4294967295.

{permit|deny}

Specifies whether to permit or deny traffic matching this ACE.

<IP-PROTOCOL-NUM>

Specifies the protocol as its Internet Protocol number. For example, 2 corresponds to the IGMP protocol. Range: 0 to 255.

{any|<SRC-IP-ADDRESS>[/<PREFIX-LENGTH>]}

Specifies the source IPv6 address.

- any - specifies any source IPv6 address.
- <SRC-IP-ADDRESS> - specifies the source IPv6 host address.
 - <PREFIX-LENGTH> - specifies the address bits to mask (CIDR subnet mask notation). Range: 1 to 128.

```
{any|<DST-IP-ADDRESS>[/<PREFIX-LENGTH>]}
```

Specifies the destination IPv6 address.

- any - specifies any destination IPv6 address.
- <DST-IP-ADDRESS> - specifies the destination IPv6 host address.
 - <PREFIX-LENGTH> - specifies the address bits to mask (CIDR subnet mask notation). Range: 1 to 128.

```
[{eq|gt|lt} <PORT>|range <MIN-PORT> <MAX-PORT>]
```

Specifies the port, or port range. Port numbers are in the range of 0 to 65535.

- eq <PORT> - specifies the Layer 4 port.
- gt <PORT> - specifies any Layer 4 port greater than the indicated port.
- lt <PORT> - specifies any Layer 4 port less than the indicated port.
- range <MIN-PORT> <MAX-PORT> - specifies the Layer 4 port range.



Upon application of the ACL, ACEs with L4 port ranges may consume more than one hardware entry.

```
[icmp-type {echo|echo-reply|<ICMP-TYPE-VALUE>}]
```

Specifies the ICMP type.

- echo - specifies an ICMP echo request packet.
- echo-reply - specifies an ICMP echo reply packet.
- <ICMP-TYPE-VALUE> - specifies an ICMP type value. Range: 0 to 255.

```
[icmp-code <ICMP-CODE-VALUE>]
```

Specifies the ICMP code value. Range: 0 to 255.

```
dscp <DSCP-SPECIFIER>
```

Specifies the Differentiated Services Code Point (DSCP), either a numeric <DSCP-VALUE> (0 to 63) or one of these keywords:

- AF11 - DSCP 10 (Assured Forwarding Class 1, low drop probability)
- AF12 - DSCP 12 (Assured Forwarding Class 1, medium drop probability)
- AF13 - DSCP 14 (Assured Forwarding Class 1, high drop probability)
- AF21 - DSCP 18 (Assured Forwarding Class 2, low drop probability)
- AF22 - DSCP 20 (Assured Forwarding Class 2, medium drop probability)
- AF23 - DSCP 22 (Assured Forwarding Class 2, high drop probability)
- AF31 - DSCP 26 (Assured Forwarding Class 3, low drop probability)
- AF32 - DSCP 28 (Assured Forwarding Class 3, medium drop probability)
- AF33 - DSCP 30 (Assured Forwarding Class 3, high drop probability)
- AF41 - DSCP 34 (Assured Forwarding Class 4, low drop probability)
- AF42 - DSCP 36 (Assured Forwarding Class 4, medium drop probability)
- AF43 - DSCP 38 (Assured Forwarding Class 4, high drop probability)

- CS0 - DSCP 0 (Class Selector 0: Default)
- CS1 - DSCP 8 (Class Selector 1: Scavenger)
- CS2 - DSCP 16 (Class Selector 2: OAM)
- CS3 - DSCP 24 (Class Selector 3: Signaling)
- CS4 - DSCP 32 (Class Selector 4: Real time)
- CS5 - DSCP 40 (Class Selector 5: Broadcast video)
- CS6 - DSCP 48 (Class Selector 6: Network control)
- CS7 - DSCP 56 (Class Selector 7)
- EF - DSCP 46 (Expedited Forwarding)

`ip-precedence <IP-PRECEDENCE-VALUE>`

Specifies an IP precedence value. Range: 0-7.

`tos <TOS-VALUE>`

Specifies the Type of Service value. Range: 0-31.

`vlan <VLAN-ID>`

Specifies VLAN tag to match on. 802.1Q VLAN ID.



This parameter cannot be used in any ACL that will be applied to a VLAN.

`count`

Keeps the hit counts of the number of packets matching this ACE.

`log`

`[<SEQUENCE-NUMBER>] comment <TEXT-STRING>`

Adds a comment to an ACE. The `no` form removes only the comment from the ACE.

Authority

Administrators or local user group members with execution rights for this command.

Usage

- If the `<IP-PROTOCOL-NUM>` parameter is used instead of a protocol name, ensure that any needed ACE-definition parameters specific to the selected protocol are also provided.
- When using multiple ACL types (IPv4, IPv6, or MAC) with logging on the same interface, the first packet that matches an ACE with `log` option is logged. Until the log-timer wait-period is over, any packets matching other ACL types do not create a log. At the end of the wait-period, the switch creates a summary log all the ACLs that were matched, regardless of type.

Examples

Creating an IPv6 ACL with four entries:

```
switch(config)# access-list ipv6 MY_IPV6_ACL
switch(config-acl-ipv6)# 10 permit udp any 2001::1/64
switch(config-acl-ipv6)# 20 permit tcp 2001:2001::2:1/128 gt 1023 any
switch(config-acl-ipv6)# 30 permit tcp 2001:2011::1/64 any
switch(config-acl-ipv6)# 40 deny any any any count
switch(config-acl-ipv6)# exit

switch(config)# do show access-list
Type          Name
Sequence      Comment
              Action
              L3 Protocol
```

	Source IP Address	Destination IP Address	Source L4 Port(s)	Destination L4 Port(s)	Additional Parameters
IPv6	MY_IPV6_ACL				
10	permit	any	udp		2001::1/64
20	permit	2001:2001::2:1	tcp	> 1023	any
30	permit	2001:2011::1/64	tcp		any
40	deny	any	any		any
					Hit-counts: enabled

Adding a comment to an existing IPv6 ACE:

```
switch(config)# access-list ipv6 MY_IPV6_ACL
switch(config-acl-ipv6)# 20 comment Permit all TCP ephemeral ports
switch(config-acl-ipv6)# exit

switch(config)# do show access-list
Type      Name
Sequence  Comment
          Action
          Source IP Address
          Destination IP Address
          Additional Parameters
-----
IPv6      MY_IPV6_ACL
10 permit
   any
   2001::1/64
20 Permit all TCP ephemeral ports
   permit
   2001:2001::2:1
   any
30 permit
   2001:2011::1/64
   any
40 deny
   any
   any
   Hit-counts: enabled
```

Removing a comment from an existing IPv6 ACE:

```
switch(config)# access-list ipv6 MY_IPV6_ACL
switch(config-acl-ipv6)# no 20 comment
switch(config-acl-ipv6)# exit

switch(config)# do show access-list
Type      Name
Sequence  Comment
          Action
          Source IP Address
          Destination IP Address
          Additional Parameters
-----
IPv6      MY_IPV6_ACL
10 permit
   any
   2001::1/64
20 permit
   2001:2001::2:1
   any
30 permit
   2001:2011::1/64
   any
40 deny
   any
   any
   Hit-counts: enabled
```

	Source IP Address	Source L4 Port(s)
	Destination IP Address	Destination L4 Port(s)
	Additional Parameters	

IPv6	MY_IPV6_ACL	
10	permit	udp
	any	
	2001::1/64	
20	permit	tcp
	2001:2001::2:1	> 1023
	any	
30	permit	tcp
	2001:2011::1/64	
	any	
40	deny	any
	any	
	any	
	Hit-counts: enabled	

Adding an ACE to an existing IPv6 ACL:

```
switch(config)# access-list ipv6 MY_IPV6_ACL
switch(config-acl-ipv6)# 25 permit icmpv6 2001::1/64 any
switch(config-acl-ipv6)# exit

switch(config)# do show access-list
Type      Name
Sequence  Comment
          Action
          Source IP Address
          Destination IP Address
          Additional Parameters
-----
IPv6      MY_IPV6_ACL
10 permit
   any
   2001::1/64
20 permit
   2001:2001::2:1
   any
25 permit
   2001::1/64
   any
30 permit
   2001:2011::1/64
   any
40 deny
   any
   any
   Hit-counts: enabled
```

Replacing an ACE in an existing IPv6 ACL:

```
switch(config)# access-list ipv6 MY_IPV6_ACL
switch(config-acl-ipv6)# 25 permit icmpv6 2001::2:1/64 any
switch(config-acl-ipv6)# exit

switch(config)# do show access-list
Type      Name
```

Sequence	Comment		
	Action		L3 Protocol
	Source IP Address		Source L4 Port(s)
	Destination IP Address		Destination L4 Port(s)
	Additional Parameters		

IPv6	MY_IPV6_ACL		
10	permit		udp
	any		
	2001::1/64		
20	permit		tcp
	2001:2001::2:1		> 1023
	any		
25	permit		icmpv6
	2001::2:1/64		
	any		
30	permit		tcp
	2001:2011::1/64		
	any		
40	deny		any
	any		
	any		
	Hit-counts: enabled		

Removing an ACE from an IPv6 ACL:

```
switch(config)# access-list ipv6 MY_IPV6_ACL
switch(config-acl-ipv6)# no 25
switch(config-acl-ipv6)# exit

switch(config)# do show access-list
```

Type	Name		
Sequence	Comment		
	Action		L3 Protocol
	Source IP Address		Source L4 Port(s)
	Destination IP Address		Destination L4 Port(s)
	Additional Parameters		

IPv6	MY_IPV6_ACL		
10	permit		udp
	any		
	2001::1/64		
20	permit		tcp
	2001:2001::2:1		> 1023
	any		
30	permit		tcp
	2001:2011::1/64		
	any		
40	deny		any
	any		
	any		
	Hit-counts: enabled		

Removing an IPv6 ACL:

```
switch(config)# no access-list ipv6 MY_IPV6_ACL

switch(config)# do show access-list
```

Type	Name
------	------

Sequence	Comment				
	Action			L3 Protocol	
	Source IP Address			Source L4 Port(s)	
	Destination IP Address			Destination L4 Port(s)	
	Additional Parameters				

IPv6	MY_IPV6_ACL2				
1	permit			udp	
	any				
	2001::1/64				
2	Permit all TCP ephemeral ports				
	permit			tcp	
	2001:2001::2:1			> 1023	
	any				
3	permit			tcp	
	2001:2011::1/64				
	any				
4	deny			any	
	any				
	any				
	Hit-counts: enabled				

access-list log-timer

Syntax

```
access-list log-timer {default|<INTERVAL>}
```

Description

Sets the log timer interval for all ACEs that have the `log` parameter configured.

Command context

```
config
```

Parameters

default

Resets the log timer to its default 300 seconds.

<INTERVAL>

Specifies the log timer interval in seconds. Range: 30 to 300.

Authority

Administrators or local user group members with execution rights for this command.

Usage

- The first packet that matches an ACE with the `log` parameter within an ACL log timer window (configured with the `access-list log-timer` command) has its header contents extracted and sent to the configured logging destination, such as the console and syslog server. Each time the ACL log timer expires, a summary of all ACEs with `log` configured are sent to the logging destination. This capability allows throttling of logging ACL hits.
- If no further log messages are generated in the wait-period, the switch suspends the timer and resets itself to log as soon as a new match occurs.

- When using multiple ACL types (IPv4, IPv6, or MAC) with logging on the same interface, the first packet that matches an ACE with the `log` option is logged. Any packets, matching other ACL types, do not create a log until the log-timer wait-period is over. At the end of the wait-period, a summary log is made of all the ACLs that were matched, regardless of type.
- You may see a minor discrepancy between the ACL logging statistics and the hit counts statistics due to the time required to record the log message.

Examples



Although these examples use debug logging, you can alternatively use event logging.

Enabling debug logging for the ACL logging module:

```
switch# debug acl log severity info
switch# show debug
-----
module sub_module severity vlan port ip mac instance vrf
-----
acl acl_log info -----
```

Setting the debug destination to console with the minimum security level of info:

```
switch# debug destination console severity info
switch# show debug destination
-----
show debug destination
-----
CONSOLE:info
```

Setting the access list log-timer to 30 seconds:

```
switch(config)# access-list log-timer 30
switch(config)# do show access-list log-timer
ACL log timer length (frequency): 30 seconds
```

Creating an IPv4 ACL with one entry with the log parameter:

```
switch(config)# access-list ip MY_IP_ACL
switch(config-acl-ip)# deny icmp 1.1.1.1 1.1.1.2 log
switch(config-acl-ip)# do show access-list
Type      Name
Sequence  Comment
          Action          L3 Protocol
          Source IP Address Source L4 Port(s)
          Destination IP Address Destination L4 Port(s)
          Additional Parameters
-----
IPv4      MY_IP_ACL
10 deny          icmp
1.1.1.1
1.1.1.2
Logging: enabled
Hit-counts: enabled
```

Enabling interface 1/1/1 and applying the ACL:

Sending packets that will match the ACE and observe the ACL logging message on the console:

```
2017-10-10T20:13:36.044+00:00 ops-switchd[875]: debug|LOG_INFO|AMM|1/5|ACL|ACL_LOG|
List MY_IP_ACL, seq# 10 denied icmp 1.1.1.1 -> 1.1.1.2 type 8 code 0,
on vlan 1, port 1/1/1, direction in
```

When the access list log-timer expires, the summary message is printed on the console. The number 30 is the number of packets received during the last access list log-timer window.

```
2017-10-10T20:14:06.051+00:00 ops-switchd[875]: debug|LOG_INFO|AMM|1/5|ACL|ACL_LOG|
MY_IP_ACL on 1/1/1 (in): 30 10 deny icmp 1.1.1.1 1.1.1.2 log count
```

Resetting the ACL log timer to the default value:

```
switch(config)# access-list log-timer default
```

access-list mac

Syntax

```
access-list mac <ACL-NAME>
no access-list mac <ACL-NAME>
```

```
[<SEQUENCE-NUMBER>]
{permit|deny}
{any|<SRC-MAC-ADDRESS>[/<ETHERNET-MASK>]}
{any|<DST-MAC-ADDRESS>[/<ETHERNET-MASK>]}
{any|aarp|appletalk|arp|fcoe|fcoe-init|ip|ipv6|
    ipx-arpa|ipx-non-arpa|is-is|lldp|mpls-multicast|mpls-unicast|q-in-q|
    rbridge|trill|wake-on-lan|<NUMERIC-ETHERTYPE>}
[pcp <PCP-VALUE>] [vlan <VLAN-ID>] [count] [log]
no <SEQUENCE-NUMBER>
```

```
[<SEQUENCE-NUMBER>] comment <TEXT-STRING>
no <SEQUENCE-NUMBER> comment
```

Description

Creates a MAC Access Control List (ACL). The ACL is made of one or more Access Control Entries (ACEs) ordered and prioritized by sequence numbers. The lowest sequence number is the highest prioritized ACE. The `no` form of this command deletes the entire ACL, or deletes an ACE identified by sequence number, or deletes only the comment from the ACE identified by sequence number.

Command context

config

The `access-list mac <ACL-NAME>` command takes you into the named ACL context where you enter the ACEs.

Parameters

<ACL-NAME>

Specifies the name of this ACL.

<SEQUENCE-NUMBER>

Specifies a sequence number for the ACE. Range: 1 to 4294967295.

{permit|deny}

Specifies whether to permit or deny traffic matching this ACE.

comment

Specifies storing the remaining entered text as an ACE comment.

{any|<SRC-MAC-ADDRESS>[/<ETHERNET-MASK>]}

Specifies the source host MAC address (xxxx.xxxx.xxxx), OUI, or the keyword `any`. You can optionally include the following:

<ETHERNET-MASK> - The address bits to mask (xxxx.xxxx.xxxx).

{any|<DST-MAC-ADDRESS>[/<ETHERNET-MASK>]}

Specifies the destination host MAC address (xxxx.xxxx.xxxx), OUI, or the keyword `any`. You can optionally include the following:

<ETHERNET-MASK> - The address bits to mask (xxxx.xxxx.xxxx).

{any|arp|appletalk|...|wake-on-lan|<NUMERIC-ETHERTYPE>}

Specifies the protocol encapsulated in the Ethernet frame. The encapsulated protocol is identified by the EtherType Ethernet field. The EtherType is specified in one of the following three ways:

- `any` - any EtherType.
- <NUMERIC-ETHERTYPE> - the numerical EtherType protocol number. Range: 0x600 to 0xffff.
- One of these EtherType protocol name keywords:
 - `arp`
 - `appletalk`
 - `arp`
 - `fcoe`
 - `fcoe-init`
 - `ip`
 - `ipv6`
 - `ipx-arpa`
 - `ipx-non-arpa`
 - `is-is`
 - `lldp`
 - `mpls-multicast`
 - `mpls-unicast`
 - `q-in-q`
 - `rbridge`
 - `trill`
 - `wake-on-lan`

pcp <PCP-VALUE>

Specifies 802.1Q QoS Priority Code Point value. Range: 0 to 7.

vlan <VID>

Specifies a VLAN ID. The VLAN ID must exist.



This parameter cannot be used in any ACL that will be applied to a VLAN.

count

Keeps the hit counts of the number of packets matching this ACE.

log

Authority

Administrators or local user group members with execution rights for this command.

Usage

When using multiple ACL types (IPv4, IPv6, or MAC) with logging on the same interface, the first packet that matches an ACE with `log` option is logged. Until the log-timer wait-period is over, any packets matching other ACL types do not create a log. At the end of the wait-period, the switch creates a summary log all the ACLs that were matched, regardless of type.

Examples

Creating a MAC ACL with four entries:

```
switch(config)# access-list mac MY_MAC_ACL
switch(config-acl-ip)# 10 permit 1122.3344.5566/ffff.ffff.0000 any ipv6
switch(config-acl-ip)# 20 permit aaaa.bbbb.cccc 1111.2222.3333 any pcp 4
switch(config-acl-ip)# 30 permit any any appletalk vlan 40
switch(config-acl-ip)# 40 deny any any any count
switch(config-acl-ip)# exit

switch(config)# do show access-list
Type      Name
Sequence  Comment
          Action                      EtherType
          Source MAC Address
          Destination MAC Address
          Additional Parameters
-----
MAC       MY_MAC_ACL
10 permit 1122.3344.5566/ffff.ffff.0000  ipv6
   any
20 permit aaaa.bbbb.cccc 1111.2222.3333    any
   QoS Priority Code Point: 4
30 permit any any appletalk               appletalk
   any
   any
   VLAN: 40
40 deny  any any                          any
   any
   any
Hit-counts: enabled
```

Adding a comment to an existing MAC ACE:

```
switch(config)# access-list mac MY_MAC_ACL
switch(config-acl-ip)# 30 comment Permit all vlan-40 tagged Appletalk traffic
switch(config-acl-ip)# exit

switch(config)# do show access-list
Type      Name
Sequence  Comment
          Action                      EtherType
          Source MAC Address
          Destination MAC Address
          Additional Parameters
```

```

-----
MAC      MY_MAC_ACL
10 permit 1122.3344.5566/ffff.ffff.0000 ipv6
   any
20 permit                                any
   aaaa.bbbb.cccc
   1111.2222.3333
   QoS Priority Code Point: 4
30 Permit all vlan-40 tagged Appletalk traffic
   permit                                appletalk
   any
   any
   VLAN: 40
40 deny                                any
   any
   any
   Hit-counts: enabled

```

Removing a comment from an existing MAC ACE:

```

switch(config)# access-list mac MY_MAC_ACL
switch(config-acl-mac)# no 30 comment
switch(config-acl-mac)# exit

switch(config)# do show access-list
Type      Name
Sequence  Comment
          Action
          Source MAC Address
          Destination MAC Address
          Additional Parameters
-----
MAC      MY_MAC_ACL
10 permit 1122.3344.5566/ffff.ffff.0000 ipv6
   any
20 permit                                any
   aaaa.bbbb.cccc
   1111.2222.3333
   QoS Priority Code Point: 4
30 permit                                appletalk
   any
   any
   VLAN: 1
40 deny                                any
   any
   any
   Hit-counts: enabled

```

Adding an ACE to an existing MAC ACL:

```

switch(config)# access-list mac MY_MAC_ACL
switch(config-acl-ip)# 35 permit any aabb.cc11.1234 0xffee
switch(config-acl-ip)# exit

switch(config)# do show access-list
Type      Name
Sequence  Comment

```

	Action	EtherType
	Source MAC Address	
	Destination MAC Address	
	Additional Parameters	

MAC	MY_MAC_ACL	
10	permit 1122.3344.5566/ffff.ffff.0000 any	ipv6
20	permit aaaa.bbbb.cccc 1111.2222.3333 QoS Priority Code Point: 4	any
30	permit any any VLAN: 1	appletalk
35	permit any aabb.cc11.1234	0xffee
40	deny any any Hit-counts: enabled	any

Replacing an ACE in an existing MAC ACL:

```
switch(config)# access-list mac MY_MAC_ACL
switch(config-acl-ip)# 35 permit any aabb.cc11.1234 0xeeee
switch(config-acl-ip)# exit
```

```
switch(config)# do show access-list
```

Type	Name	
Sequence	Comment	
	Action	EtherType
	Source MAC Address	
	Destination MAC Address	
	Additional Parameters	

MAC	MY_MAC_ACL	
10	permit 1122.3344.5566/ffff.ffff.0000 any	ipv6
20	permit aaaa.bbbb.cccc 1111.2222.3333 QoS Priority Code Point: 4	any
30	permit any any VLAN: 1	appletalk
35	permit any aabb.cc11.1234	0xeeee
40	deny any any Hit-counts: enabled	any

Removing an ACE from an MAC ACL:

```

switch(config)# access-list mac MY_MAC_ACL
switch(config-acl-ip)# no 35
switch(config-acl-ip)# exit

switch(config)# do show access-list
Type          Name
Sequence      Comment
              Action                      EtherType
              Source MAC Address
              Destination MAC Address
              Additional Parameters
-----
MAC           MY_MAC_ACL
10 permit
   1122.3344.5566/ffff.ffff.0000
   any                                ipv6
20 permit
   aaaa.bbbb.cccc
   1111.2222.3333
   QoS Priority Code Point: 4
30 permit
   any                                appletalk
   any
   VLAN: 1
40 deny
   any                                any
   any
   Hit-counts: enabled

```

Removing a MAC ACL:

```

switch(config)# no access-list mac MY_MAC_ACL

switch(config)# do show access-list
Type          Name
Sequence      Comment
              Action                      EtherType
              Source MAC Address
              Destination MAC Address
              Additional Parameters
-----
MAC           MY_MAC_ACL2
1 permit
   1122.3344.5566/ffff.ffff.0000
   any                                ipv6
2 permit
   aaaa.bbbb.cccc
   1111.2222.3333
   QoS Priority Code Point: 4
3 Permit all vlan-40 tagged Appletalk traffic
   permit
   any                                appletalk
   any
   VLAN: 1
4 deny
   any                                any
   any
   Hit-counts: enabled

```

access-list resequence

Syntax

```
access-list {ip|ipv6|mac} <ACL-NAME> resequence <STARTING-SEQUENCE-NUMBER> <INCREMENT>
```

Description

Resequences the ACE sequence numbers in an ACL.

Command context

config

Parameters

{ip|ipv6|mac}

Specifies the ACL type.

<ACL-NAME>

Specifies the ACL name.

<STARTING-SEQUENCE-NUMBER>

Specifies the starting sequence number.

<INCREMENT>

Specifies the sequence number increment.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Resequencing an IPv4 ACL to start at 1 with an increment of 1:

```
switch(config)# access-list ip MY_IP_ACL resequence 1 1
switch(config-acl-ip)# exit

switch(config)# do show access-list
Type      Name
Sequence  Comment
          Action          L3 Protocol
          Source IP Address Source L4 Port(s)
          Destination IP Address Destination L4 Port(s)
          Additional Parameters
-----
IPv4      MY_IP_ACL
1 permit          udp
  any
  172.16.1.0/255.255.255.0
2 permit          tcp
  172.16.2.0/255.255.0.0    > 1023
  any
3 permit          tcp
  172.26.1.0/255.255.255.0
  any
  dscp: AF11
  ack
  syn
4 deny          any
  any
  any
Hit-counts: enabled
```

Resequencing an IPv6 ACL to start at 1 with an increment of 1:


```

switch(config)# access-list ipv6 MY_IPV6_ACL resequence 1 1
switch(config-acl-ip)# exit

switch(config)# do show access-list
Type          Name
Sequence      Comment
              Action                      L3 Protocol
              Source IP Address          Source L4 Port(s)
              Destination IP Address      Destination L4 Port(s)
              Additional Parameters
-----
IPv6          MY_IPV6_ACL
1 permit      any                          udp
  2001::1/64
2 Permit all TCP ephemeral ports
  permit      2001:2001::2:1                tcp
  > 1023
  any
3 permit      2001:2011::1/64                tcp
  any
4 deny        any                          any
  any
  any
Hit-counts: enabled

```

Resequencing a MAC ACL to start at 1 with an increment of 1:

```

switch(config)# access-list mac MY_MAC_ACL resequence 1 1
switch(config-acl-mac)# exit

switch(config)# do show access-list
Type          Name
Sequence      Comment
              Action                      EtherType
              Source MAC Address
              Destination MAC Address
              Additional Parameters
-----
MAC          MY_MAC_ACL
1 permit      1122.3344.5566/ffff.ffff.0000  ipv6
  any
2 permit      aaaa.bbbb.cccc                any
  1111.2222.3333
  QoS Priority Code Point: 4
3 Permit all vlan-40 tagged Appletalk traffic
  permit      any                          appletalk
  any
  any
  VLAN: 1
4 deny        any                          any
  any
  any
Hit-counts: enabled

```

access-list reset

Syntax

```
access-list {all|ip <ACL-NAME>|ipv6 <ACL-NAME>|mac <ACL-NAME>} reset
```

Description

Changes the user-specified ACL configuration to match the active ACL configuration. Use this command when a discrepancy exists between what the user configured and what is active and accepted by the system.

Command context

```
config
```

Parameters

```
all|ip <ACL-NAME>|ipv6 <ACL-NAME>|mac <ACL-NAME>
```

Specifies **one** of the following:

- a reset of all ACLs.
- a reset of a named IPv4 ACL.
- a reset of a named IPv6 ACL.
- a reset of a named MAC ACL.

Authority

Administrators or local user group members with execution rights for this command.

Usage

The output of the `show access-list` command displays the active configuration of the product. The active configuration is the ACLs that have been configured and accepted by the system. The output of the `show access-list` command with the `configuration` parameter, displays the ACLs that have been configured. The output of this command may not be the same as what was programmed in hardware or what is active on the product.

If the active ACLs and user-configured ACLs are not the same, a warning message is displayed in the output of the `show` command. Modify the user-configured ACL until the warning message is no longer displayed or run the `access-list reset` command to change the user-specified configuration to match the active configuration.

Examples

Apply an ACL with TCP acknowledgments (ACKs) on ingress, which is unsupported by hardware:

```
switch(config-acl)# 10 permit tcp 172.16.2.0/16 any ack
```

Displaying the user-specified configuration:

```
switch(config)# do show access-list commands
! access-list ip TEST_ACL user configuration does not match active configuration.
! run 'access-list TYPE NAME reset' to reset access-list to match active
configuration.
access-list ip TEST_ACL
! access-list ip TEST_ACL user configuration does not match active configuration.
! run 'access-list TYPE NAME reset' to reset access-list to match active
configuration.
interface 1/1/1
  apply access-list ip TEST_ACL in
```

```

switch(config)# do show access-list commands configuration
! access-list ip TEST_ACL user configuration does not match active configuration.
! run 'access-list TYPE NAME reset' to reset access-list to match active
configuration.
access-list ip TEST_ACL
  10 permit tcp 172.16.2.0/255.255.0.0 any ack
! access-list ip TEST_ACL user configuration does not match active configuration.
! run 'access-list TYPE NAME reset' to reset access-list to match active
configuration.
interface 1/1/1
  apply access-list ip TEST_ACL in

switch(config)# do show access-list
Type      Name
Sequence Comment
          Action                L3 Protocol
          Source IP Address      Source L4 Port(s)
          Destination IP Address Destination L4 Port(s)
          Additional Parameters
-----
% Warning: TEST_ACL user configuration does not match active configuration.
% run 'access-list TYPE NAME reset' to reset access-list to match active
configuration.
IPv4      TEST_ACL

switch(config)# do show access-list configuration
Type      Name
Sequence Comment
          Action                L3 Protocol
          Source IP Address      Source L4 Port(s)
          Destination IP Address Destination L4 Port(s)
          Additional Parameters
-----
% Warning: TEST_ACL user configuration does not match active configuration.
% run 'access-list TYPE NAME reset' to reset access-list to match active
configuration.
IPv4      TEST_ACL
          10
            permit                tcp
            172.16.2.0/255.255.0.0
            any
            ack

```

Resetting the user-specified configuration to match the active configuration.

```

switch(config)# access-list ip TEST_ACL reset

```

Displaying the updated user-specified configuration.

```

switch(config)# do show access-list commands
access-list ip TEST_ACL
interface 1/1/1
  apply access-list ip TEST_ACL in

switch(config)# do show access-list commands configuration
access-list ip TEST_ACL
interface 1/1/1
  apply access-list ip TEST_ACL in

```

```

switch(config)# do show access-list
Type          Name
Sequence      Comment
              Action          L3 Protocol
              Source IP Address Source L4 Port(s)
              Destination IP Address Destination L4 Port(s)
              Additional Parameters
-----
IPv4          TEST_ACL

switch(config)# do show access-list configuration
Type          Name
Sequence      Comment
              Action          L3 Protocol
              Source IP Address Source L4 Port(s)
              Destination IP Address Destination L4 Port(s)
              Additional Parameters
-----
IPv4          TEST_ACL

```

apply access-list control-plane

Syntax

```

apply access-list {ip|ipv6} <ACL-NAME> control-plane vrf <VRF-NAME>
no apply access-list {ip|ipv6} <ACL-NAME> control-plane vrf <VRF-NAME>

```

Description

Applies an ACL to the specified VRF.

The `no` form of this command removes application of the ACL from the specified VRF.

Command context

config

Parameters

`ip|ipv6`

Specifies the ACL type: `ip` for IPv4, or `ipv6` for IPv6.

`<ACL-NAME>`

Specifies the ACL name.

`vrf <VRF-NAME>`

Specifies the VRF name.

Authority

Administrators or local user group members with execution rights for this command.

Usage

Only one ACL per type (`ip`, or `ipv6`) may be applied to a control plane VRF at a time. Therefore, using the `apply access-list control-plane` command on a VRF with an already-applied ACL of the same type, will replace the applied ACL.

Examples

Applying `My_ip_ACL` to control plane traffic on the default VRF:

```
switch(config)# apply access-list ip My_ip_ACL control-plane vrf default
```

Replacing My_ip_ACL with My_Replacement_ACL on the default VRF:

```
switch(config)# apply access-list ip My_Replacement_ACL control-plane vrf default
```

Remove (unapply) the My_Replacement_ACL from the default VRF. Any other interfaces or VLANs with My_Replacement_ACL applied are unaffected.

```
switch(config)# no apply access-list ip My_Replacement_ACL control-plane vrf default
```

apply access-list (to interface or LAG)

Syntax

```
apply access-list {ip | ipv6 | mac} <ACL-NAME> {in}  
no apply access-list {ip | ipv6 | mac} <ACL-NAME> {in}
```

Description

Applies an ACL to the interface (Individual front plane port) or Link Aggregation Group (LAG) identified by the current interface or LAG context.

The `no` form of this command removes application of the ACL from the current interface or LAG identified by the current interface or LAG context.

Command context

```
config-if  
config-lag-if
```

Parameters

`ip|ipv6|mac`

Specifies the ACL type: `ip` for IPv4, `ipv6` for IPv6, or `mac` for MAC ACL.

`<ACL-NAME>`

Specifies the ACL name.

`in`

Selects the inbound (ingress) traffic direction.

Authority

Administrators or local user group members with execution rights for this command.

Usage

- Each ACL of a given type can be applied to the same interface or LAG once. Therefore, using the `apply access-list` command on an interface or LAG with an already-applied ACL of the same type will replace the currently applied ACL.
- An ACL can be applied to an individual front plane port or to a Link Aggregation Group (LAG).
- A port that is a member of a LAG with an applied ACL cannot have a different ACL applied to that member port.

- When the port membership of a LAG with an applied ACL is changed, the LAG ACL is automatically applied or removed from that port depending on the modification type.

Examples

Applying My_IP_ACL to ingress traffic on interface range 1/1/10 to 1/1/12:

```
switch(config)# int 1/1/10-1/1/12
switch(config-if-<1/1/10-1/1/12>)# apply access-list ip My_IP_ACL in
switch(config-if-<1/1/10-1/1/12>)# exit
```

Applying MY_IPV6_ACL to ingress traffic on interface 1/1/1 and to ingress traffic on LAG 100:

```
switch(config)# interface 1/1/1
switch(config-if)# apply access-list ipv6 MY_IPV6_ACL in
switch(config-if)# exit

switch(config)# interface lag 100
switch(config-lag-if)# apply access-list ipv6 MY_IPV6_ACL in
switch(config-lag-if)# exit
switch(config)#
```

Applying MY_MAC_ACL to ingress traffic on interface 1/1/1 and ingress traffic on interface 1/1/2:

```
switch(config)# interface 1/1/1
switch(config-if)# apply access-list mac MY_MAC_ACL in
switch(config-if)# exit

switch(config)# interface 1/1/2
switch(config-if)# apply access-list mac MY_MAC_ACL in
switch(config-if)# exit
switch(config)#
```

apply access-list (to VLAN)

Syntax

```
apply access-list {ip|ipv6|mac} <ACL-NAME> in
no apply access-list {ip|ipv6|mac} <ACL-NAME> in
```

Description

Applies an ACL to the VLAN identified by the current VLAN context.

The `no` form of this command removes application of the ACL from the VLAN identified by the current VLAN context.

Command context

config-vlan

Parameters

ip|ipv6|mac

Specifies the ACL type: `ip` for IPv4, `ipv6` for IPv6, or `mac` for MAC ACL.

<ACL-NAME>

Specifies the ACL name.

in

Selects the inbound (ingress) traffic direction.

Authority

Administrators or local user group members with execution rights for this command.

Usage

- Each ACL of a given type can be applied to the same VLAN once. Therefore, using the `apply access-list` command on a VLAN with an already-applied ACL of the same type, will replace the applied ACL.
- When an ACL is applied to a VLAN, it will create hardware entries on all stack members regardless of whether a VLAN member exists on any specific stack member.

Examples

Applying `My_ip_ACL` to ingress traffic on VLAN range 20 to 25:

```
switch(config)# vlan 20-25  
switch(config-vlan-<20-25>)# apply access-list ip My_ip_ACL in
```

Applying `My_ip_ACL` to ingress traffic on VLAN 10:

```
switch(config)# vlan 10  
switch(config-vlan-10)# apply access-list ip My_ip_ACL in
```

Applying `My_ipv6_ACL` to ingress traffic on VLAN 10:

```
switch(config)# vlan 10  
switch(config-vlan-10)# apply access-list ipv6 My_ipv6_ACL in
```

Applying `My_mac_ACL` to ingress traffic on VLAN 10:

```
switch(config)# vlan 10  
switch(config-vlan-10)# apply access-list mac My_mac_ACL in
```

Replacing `My_ipv6_ACL` with `My_Replacement_ACL` on VLAN 10 (following the preceding examples):

```
switch(config)# vlan 10  
switch(config-vlan-10)# apply access-list ipv6 My_Replacement_ACL in
```

Removing (unapplying) several ACLs on VLAN 10:

```
switch(config)# vlan 10  
switch(config-vlan-10)# no apply access-list ipv6 My_Replacement_ACL in  
switch(config-vlan-10)# no apply access-list mac My_mac_ACL in
```

clear access-list hitcounts

Syntax

```
clear access-list hitcounts { all | [{ip|ipv6|mac} <ACL-NAME>] [interface <IF-NAME>| vlan <VLAN-ID>] [in] }
```

Description

Clears the hit counts for ACLs with ACEs that include the `count` keyword.

Command context

Operator (>) or Manager (#)

Parameters

`all`

Selects all ACLs.

`ip|ipv6|mac`

Specifies the ACL type: `ip` for IPv4, `ipv6` for IPv6, or `mac` for MAC.

`<ACL-NAME>`

Specifies the ACL name.

`interface <IF-NAME>`

Specifies the interface name (port or LAG).

`vlan <VLAN-ID>`

Specifies the VLAN.

`in`

Selects the inbound (ingress) traffic direction.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Clearing the hit counts for `My_ip_ACL` applied to VLAN 10 (ingress):

```
switch# clear access-list hitcounts ip My_ip_ACL vlan 10 in
```

Clearing the hit counts for all ACLs:

```
switch# clear access-list hitcounts all
```

clear access-list hitcounts control-plane

Syntax

```
clear access-list hitcounts [{ip|ipv6} <ACL-NAME>] control-plane vrf <VRF-NAME>
```

Description

Clears the hit counts for ACLs applied to the Control Plane VRF.

Command context

Operator (>) or Manager (#)

Parameters

ip|ipv6

Specifies the ACL type: ip for IPv4, or ipv6 for IPv6.

<ACL-NAME>

Specifies the ACL name.

vrf <VRF-NAME>

Specifies the VRF name.

Authority

Operators or Administrators or local user group members with execution rights for this command.

Operators can execute this command from the operator context (>) only.

Examples

Clearing the hit counts for an IPv4 ACL applied to the Control Plane default VRF:

```
switch# clear access-list hitcounts ip My_ipv4_ACL control-plane vrf default
```

Clearing the hit counts for an IPv6 ACL applied to the Control Plane default VRF:

```
switch# clear access-list hitcounts ipv6 My_ipv6_ACL control-plane vrf default
```

show access-list

Syntax

Syntax that filters by ACLs applied to an interface or VLAN:

```
show access-list [interface <IF-NAME>|vlan <VLAN-ID>] [ip|ipv6|mac]
                  [in] [commands] [configuration]
```

Syntax that filters by the named ACL:

```
show access-list [ip|ipv6|mac] [<ACL-NAME>] [commands] [configuration]
```

Description

Shows information about your defined ACLs and where they have been applied. When `show access-list` is entered without parameters, information for all ACLs is shown. The parameters filter the list of ACLs for which information is shown.

Available filtering includes:

- The content of a specific ACL.
- All ACLs of a specific type.
- All ACLs applied to a specific interface (port or LAG).
- All ACLs applied to a specific VLAN.
- All IPv4 or IPv6 ACLs applied to interface VLANs.

Command context

Operator (>) or Manager (#)

Parameters

interface <IF-NAME>

Specifies the interface name (port or LAG).

vlan <VLAN-ID>

Specifies the VLAN.

ip|ipv6|mac

Specifies the ACL type:

ip for IPv4,
ipv6 for IPv6, or
mac for MAC.

in

Selects the inbound (ingress) traffic direction.

<ACL-NAME>

Specifies the ACL name.

commands

Specifies that the ACL definition is to be shown as the commands and parameters used to create it rather than in tabular form.

configuration

Specifies that the user-configured ACLs be shown as entered, even if the ACLs are not active due to ACE-definition command issues or hardware issues. This parameter is useful if there is a mismatch between the entered configuration and the previous successfully programmed (active) ACLs configuration.

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Examples

Showing an IPv4 ACL:

```
switch# show access-list ip MY_ACL
Type      Name
Sequence  Comment
          Action
          Source IP Address
          Destination IP Address
          Additional Parameters
-----
IPv4      MY_ACL
10 permit                                udp
   any
   172.16.1.0/255.255.255.0
20 permit                                tcp
   172.16.2.0/255.255.0.0
   any
   > 1023
30 permit                                tcp
   172.26.1.0//255.255.255.0
   any
   syn
   ack
   dscp 10
40 deny                                    any
   any
   any
Hit-counts: enabled
-----
```

Showing an IPv4 ACL as commands:

```

switch# show access-list ip MY_ACL commands
access-list ip MY_ACL
  10 permit udp any 172.16.1.0/255.255.255.0
  20 permit tcp 172.16.2.0/255.255.0.0 gt 1023 any
  30 permit tcp 172.26.1.0/255.255.255.0 any syn ack dscp 10
  40 deny any any any count

```

Showing IPv4 ACLs applied to VLAN 10, inbound:

```

switch# show access-list vlan 10 ip in
Type      Name
Sequence  Comment
          Action                L3 Protocol
          Source IP Address      Source L4 Port(s)
          Destination IP Address  Destination L4 Port(s)
          Additional Parameters
-----
IPv4      My_ip_ACL
  10 permit                                udp
    any
    172.16.1.0/255.255.255.0
  20 permit                                tcp
    172.16.2.0/255.255.0.0      > 1023
    any
  30 permit                                tcp
    172.26.1.0//255.255.255.0
    any
    syn
    ack
    dscp 10
  40 deny                                any
    any
    any
    Hit-counts: enabled
-----

```

Showing IPv6 ACLs applied to LAG 128, inbound:

```

switch# show access-list interface lag128 ipv6 in
Type      Name
Sequence  Comment
          Action                L3 Protocol
          Source IP Address      Source L4 Port(s)
          Destination IP Address  Destination L4 Port(s)
          Additional Parameters
-----
IPv6      MY_IPV6_ACL
  10 permit                                udp
    any
    2001::1/64
  20 permit                                tcp
    2001:2001::2:1/128        > 1023
    any
  30 permit                                tcp
    2001:2011::1/64
  40 deny                                any
    any
    any

```

```
Hit-counts: enabled
```

Showing an IPv6 ACL as commands:

```
switch# show access-list ipv6 MY_IPV6_ACL commands
access-list ipv6 MY_IPV6_ACL
  10 permit udp any 2001::1/64
  20 permit tcp 2001:2001::2:1/128 gt 1023 any
  40 deny any any any count
```

Showing a MAC ACL:

```
switch# show access-list mac MY_MAC_ACL
Type      Name
Sequence  Comment
          Action
          Source MAC Address
          Destination MAC Address
          Additional Parameters
-----
MAC       MY_MAC_ACL
  10 permit
    1122.3344.5566/ffff.ffff.0000
    any
    ipv6
  20 permit
    aaaa.bbbb.cccc
    1111.2222.3333
    QoS Priority Code Point: 4
    any
    any
    any
    Hit-counts: enabled
-----
```

Showing a MAC ACL as commands:

```
switch# show access-list mac MY_MAC_ACL commands
access-list mac MY_MAC_ACL
  10 permit 1122.3344.5566/ffff.ffff.0000 any ipv6
  20 permit aaaa.bbbb.cccc 1111.2222.3333 any pcp 4
  30 deny any any any count
```

show access-list control-plane

Syntax

```
show access-list [ip|ipv6] [<ACL-NAME>] control-plane [vrf <VRF-NAME>]
                  [commands] [configuration]
```

Description

Shows information about your defined ACLs that have been applied to the Control Plane. When `show access-list control-plane` is entered without parameters, information for all ACLs applied to the Control Plane is shown. The parameters filter the list of ACLs for which information is shown.

Available filtering includes:

- The content of a specific ACL that has been applied to the Control Plane.
- All ACLs of a specific type that have been applied to the Control Plane.
- All ACLs applied to the Control Plane for a specific VRF.

Command context

Operator (>) or Manager (#)

Parameters

ip|ipv6

Specifies the ACL type: ip for IPv4, or ipv6 for IPv6.

<ACL-NAME>

Specifies the ACL name.

vrf <VRF-NAME>

Specifies the VRF name.

[commands]

Specifies that the ACL definition is to be shown as the commands and parameters used to create it rather than in tabular form.

[configuration]

Specifies that the user-configured ACLs be shown as entered, even if the ACLs are not active due to ACE-definition command issues or hardware issues. This parameter is useful if there is a mismatch between the entered configuration and the previous successfully programmed (active) ACLs configuration.

Authority

Operators or Administrators or local user group members with execution rights for this command.

Operators can execute this command from the operator context (>) only.

Examples

Showing an IPv4 ACL applied to the Control Plane default VRF:

```
switch# show access-list ip My_ipv4_ACL control-plane vrf default
Type      Name
Sequence  Comment
          Action                      L3 Protocol
          Source IP Address           Source L4 Port(s)
          Destination IP Address       Destination L4 Port(s)
          Additional Parameters
-----
IPv4      My_ipv4_ACL
10 permit                                udp
   any
   172.16.1.0/24
20 permit                                tcp
   172.16.2.0/16
   any                                   > 1023
30 permit                                tcp
   172.26.1.0/24
   any
   syn
   ack
   dscp 10
40 deny                                    any
   any
   any
```

```
Hit-counts: enabled
```

Showing an IPv6 ACL applied to the Control Plane default VRF:

```
switch# show access-list ipv6 My_ipv6_ACL control-plane vrf default
Type      Name
Sequence  Comment
          Action
          Source IP Address      L3 Protocol
          Destination IP Address Source L4 Port(s)
          Additional Parameters  Destination L4 Port(s)
-----
IPv6      My_ipv6_ACL
10 permit                                udp
   any
   2001::1/64
20 permit                                tcp
   2001:2001::2:1/128                > 1023
   any
30 permit                                tcp
   2001:2011::1/64
40 deny                                    any
   any
   any
Hit-counts: enabled
-----
```

show access-list hitcounts

Syntax

```
show access-list hitcounts { [{ip|ipv6|mac} <ACL-NAME>] [interface <IF-NAME> |
                             vlan <VLAN-ID>] [in]]}
```

Description

Shows the hit count of the number of times an ACL has matched a packet or frame for ACEs with the `count` keyword. For ACEs without the `count` keyword, a dash is shown in place of a hit count.

Command context

Operator (>) or Manager (#)

Parameters

`ip|ipv6|mac`

Specifies the ACL type: `ip` for IPv4, `ipv6` for IPv6, or `mac` for MAC.

`<ACL-NAME>`

Specifies the ACL name.

`interface <IF-NAME>`

Specifies the interface name (port or LAG).

`vlan <VLAN-ID>`

Specifies the VLAN.

`in`

Selects the inbound (ingress) traffic direction.

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Usage

- ACL hit counts are aggregated across all:
 - physical interfaces to which the ACL is applied to on ingress,
 - VLANs to which the ACL is applied to on ingress.
- If an ACL with an ACE with the `count` keyword is applied to multiple physical interfaces or VLANs, the hit counts are aggregated. There is one aggregation for physical interfaces and another for VLANs.
- Accumulated hit counts for an applied ACL are cleared upon any modification of the ACL.

Examples

Showing the hit counts for `My_ip_ACL` applied to port 1/1/2:

```
switch# show access-list hitcounts ip My_ip_ACL interface 1/1/2
Statistics for ACL My_ip_ACL (ipv4):
interface 1/1/2* (in):
    Hit Count  Configuration
              - 10 permit udp any 172.16.1.0/24
              - 20 permit tcp 172.16.2.0/16 gt 1023 any
              - 30 permit tcp 172.26.1.0/24 any syn ack dscp 10
              0 40 deny any any any count
* access-list statistics are shared among each combination of context type
(interface, VLAN, VRF) and direction (in, control-plane). Use 'access-list
TYPE NAME copy' to create a new access-list for separate statistics.
```

Showing the hit counts for `My_ip_ACL` applied to VLAN 10:

```
switch# show access-list hitcounts ip My_ip_ACL vlan 10
Statistics for ACL My_ip_ACL (ipv4):
vlan 10* (in):
    Hit Count  Configuration
              - 10 permit udp any 172.16.1.0/24
              - 20 permit tcp 172.16.2.0/16 gt 1023 any
              - 30 permit tcp 172.26.1.0/24 any syn ack dscp 10
              0 40 deny any any any count
* access-list statistics are shared among each combination of context type
(interface, VLAN, VRF) and direction (in, control-plane). Use 'access-list
TYPE NAME copy' to create a new access-list for separate statistics.
```

show access-list hitcounts control-plane

Syntax

```
show access-list hitcounts [{ip|ipv6} <ACL-NAME>] control-plane vrf <VRF-NAME>
```

Description

Shows the hit count of the number of times an ACL (applied to the Control Plane) has matched a packet for ACEs with the `count` keyword. For ACEs without the `count` keyword, a dash is shown in place of a hit count.

Command context

Operator (>) or Manager (#)

Parameters

ip|ipv6

Specifies the ACL type: **ip** for IPv4, or **ipv6** for IPv6.

<ACL-NAME>

Specifies the ACL name.

vrf **<VRF-NAME>**

Specifies the VRF name.

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Usage

- ACL hit counts are aggregated across all VRFs to which the ACL is applied to on ingress.
- Accumulated hit counts for an applied ACL are cleared upon any modification of the ACL.

Examples

Showing the hit counts for an IPv4 ACL applied to the Control Plane default VRF:

```
switch# show access-list hitcounts ip My_ipv4_ACL control-plane vrf default
Statistics for ACL My_ipv4_ACL (ipv4):
VRF default* (control-plane):
    Hit Count  Configuration
    - 10 permit udp any 172.16.1.0/255.255.255.0
    - 20 permit tcp 172.16.2.0/255.255.0.0 gt 1023 any
    - 30 permit tcp 172.26.1.0/255.255.255.0 any syn ack dscp 10
    8 40 deny any any any count
* access-list statistics are shared among each combination of
context type (interface, VLAN, VRF) and direction (in, control-plane).
use 'access-list TYPE NAME copy' to create a uniquely-named access-list.
```

show capacities

Syntax

show capacities **<FEATURE>**

Description

Shows system capacities and their values for all features or a specific feature.

Command context

Manager (#)

Parameters

<FEATURE>

Specifies a feature. For example, **aaa**.

Authority

Administrators or local user group members with execution rights for this command.

Usage

Capacities are expressed in user-understandable terms. Thus they may not map to a specific hardware or software resource or component. They are not intended to define a feature exhaustively.

Examples

Showing classifier-related capacities on the 6100:

```
switch# show capacities classifier

System Capacities: Filter Classifier
Capacities Name                                     Value
-----
Maximum number of Access Control Entries configurable in a system      4096
Maximum number of Access Control Lists configurable in a system         512
Maximum number of class entries configurable in a system                4096
Maximum number of classes configurable in a system                     512
Maximum number of entries in an Access Control List                    1024
Maximum number of entries in a class                                   1024
Maximum number of entries in a policy                                   64
Maximum number of classifier policies configurable in a system          512
Maximum number of policy entries configurable in a system               4096
```

Showing all available capacities on the 6100:

```
switch# show capacities

System Capacities:
Capacities Name
Value
-----
Maximum number of Access Control Entries configurable in a system      4096
Maximum number of Access Control Lists configurable in a system         512
Maximum number of class entries configurable in a system                4096
Maximum number of classes configurable in a system                     512
Maximum number of entries in an Access Control List                    1024
Maximum number of entries in a class                                   1024
Maximum number of entries in a policy                                   64
Maximum number of classifier policies configurable in a system          512
Maximum number of policy entries configurable in a system               4096
Maximum number of clients supported for tracking the IP address in the system 128
Maximum number of dynamic VLANs that can be allowed using MVRP        256
Maximum number of nexthops per IP ECMP group                           1
Maximum number of IP neighbors (IPv4+IPv6) supported in the system     1024
Maximum number of IPv4 neighbors(# of ARP entries) supported in the system 1024
```

```

Maximum number of IPv6 neighbors(# of ND entries) supported in the system
512
Maximum number of L2 MAC addresses supported in the system
8192
Maximum number of L3 Groups for IP Tunnels and ECMP Groups
1
Maximum number of L3 Destinations for Routes, Nexthops in Tunnels and ECMP groups
1024
Maximum number of configurable LAG ports
8
Maximum number of members supported by a LAG port
8
Maximum number of VLANs across ports allowed in loop-protect
3328
Maximum number of IGMP/MLD groups supported
512
Maximum number of IGMP/MLD snooping groups supported
512
Maximum number of Mirror Sessions configurable in a system
4
Maximum number of enabled Mirror Sessions in a system
4
Maximum number of mstp instances configurable in a system
16
Maximum number of Clients that can be authenticated on a port
32
Maximum number of Device Profiles allowed to be created on the system
8
Maximum number of Port Access Roles allowed to be created on the system
32
Maximum number of MAC Address that can be authorized on a port
32
Maximum number of Port Access Role VLAN IDs allowed to be created on the system
50
Maximum number of Port Access Role VLAN names allowed to be created on the system
50
Maximum number of RBAC rules per user group
1024
Maximum number of RPVST VLANs configurable on the system
16
Maximum number of RPVST VPORTs supported in a system
512
Maximum number of SVIs supported in the system
16
Maximum number of routes (IPv4+IPv6) on the system
512
Maximum number of IPv4 routes on the system
512
Maximum number of IPv6 routes on the system
512
Maximum number of VLANs supported in the system
512

```

Showing all available capacities for mirroring:

```
switch# show capacities mirroring
```

```
System Capacities: Filter Mirroring
Capacities Name
```

```
Value
```

```
-----
Maximum number of Mirror Sessions configurable in a system
```

```
4
```

Showing all available capacities for MSTP:

```
switch# show capacities mstp

System Capacities: Filter MSTP
Capacities Name                                     Value
-----
Maximum number of mstp instances configurable in a system      64
```

Showing all available capacities for VLAN count:

```
switch# show capacities vlan-count

System Capacities: Filter VLAN Count
Capacities Name                                     Value
-----
Maximum number of VLANs supported in the system              4094
```

show capacities-status

Syntax

```
show capacities-status <FEATURE>
```

Description

Shows system capacities status and their values for all features or a specific feature.

Command context

Manager (#)

Parameters

<FEATURE>

Specifies the feature, for example `aaa` for which to display capacities, values, and status. Required.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Showing the system capacities status for all features:

```
switch# show capacities-status
System Capacities Status
Capacities Status Name                                     Value
Maximum
```

```

-----
--
Number of Access Control Entries currently configured      0
4096
Number of Access Control Lists currently configured      0
512
Number of class entries currently configured              0
4096
Number of classes currently configured                   0
512
Number of policies currently configured                  0
512
Number of policy entries currently configured             0
4096
Number of dynamic VLANs currently learnt using MVRP      0
256
Number of IP neighbor (IPv4+IPv6) entries                1
1024
Number of IPv4 neighbor(ARP) entries                    1
1024
Number of IPv6 neighbor(ND) entries                     0
512
Number of L3 Groups for IP Tunnels and ECMP Groups currently configured 0
1
Number of L3 Destinations for Routes, Nexthops in ECMP groups and Tunnels
currently configured                                     0
1024
Number of Mirror Sessions currently configured           0
4
Number of Mirror Sessions currently enabled              0
4
Number of mstp instances currently configured            0
16
Number of RPVST VLANs currently configured              0
16
Number of routes (IPv4+IPv6) currently configured       1
512
Number of IPv4 routes currently configured              1
512
Number of IPv6 routes currently configured              0
512
Number of VLANs currently configured                    1
512

```

show resources

Syntax

```
show resources
```

Description

Shows hardware resource consumption. Resource data is updated every 10 seconds.

Hardware resource consumption information is shown for:

- TCAM entries
- TCAM lookups
- Policers

Command context

Operator (>) or Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command.

Operators can execute this command from the operator context (>) only.

Usage

The widths for show resources can have features combined (IPv4 + IPv6) into one TCAM lookup. Therefore, the table widths for each ACL/classifier policy type are variable depending on what is applied. For example:

```
"Ingress IP Port ACL" = Ingress v4 Port ACLs + Ingress v6 Port ACLs
                       = 1 TCAM entry + 4 TCAM entries
                       = 5 TCAM entries
```

Widths per feature are as follows:

```
IPv4 ACL    2
MAC ACL     2
IPv6 ACL    4
IPv4 Class  2
IPv6 Class  4
```

Examples

Showing hardware resource consumption on a 6100 switch:

```
switch# show resources
```

```
Resource Usage:
```

Mod	Description Resource	Used	Reserved	Free
1/1	Ingress IPv4 VLAN ACL Lookup			
	Ingress TCAM Entries	4	128	
	Ingress IPv6 VLAN ACL Lookup			
	Ingress TCAM Entries	8	128	
	Ingress IP CPURX Lookup			
	Ingress TCAM Entries	126	128	
	Ingress Policers	19		
	Ingress IP Port Policy Lookup			
	Ingress TCAM Entries	2	128	
	Ingress IP VLAN Policy Lookup			
	Ingress TCAM Entries	12	128	
	Total			
	Ingress TCAM Entries	152	640	3448
	Ingress Lookups	5		27
	Ingress Policers	19		2029

arp inspection

Syntax

```
arp inspection
```

Description

Enables Dynamic ARP Inspection on the current VLAN, forcing all ARP packets from untrusted ports to be subjected to a MAC-IP association check against a binding table.

The `no` form of this command disables Dynamic ARP Inspection on the VLAN.

Command context

```
config-vlan-<VLAN-ID>
```

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling dynamic ARP inspection:

```
switch# configure terminal
switch(config)# vlan 1
switch(config-vlan)# arp inspection
```

Disabling dynamic ARP inspection:

```
switch# configure terminal
switch(config)# vlan 1
switch(config-vlan)# no arp inspection
```

arp inspection trust

Syntax

```
arp inspection trust
no arp inspection trust
```

Description

Configures the interface as a trusted. All interfaces are untrusted by default.

The `no` form of this command returns the interface to the default state (untrusted).

Command context

config-if

Authority

Administrators or local user group members with execution rights for this command.

Example

Setting an interface as trusted:

```
switch(config-if)# arp inspection trust
```

arp ipv4 mac

Syntax

```
arp ipv4 <IPV4_ADDR> mac <MAC_ADDR>  
no arp ipv4 <IPV4_ADDR> mac <MAC_ADDR>
```

Description

Specifies a permanent static neighbor entry in the ARP table (for IPv4 neighbors).

The `no` form of this command deletes a permanent static neighbor entry from the ARP table.

Command context

config-if
config-if-vlan

Parameters

ipv4 <IPV4-ADDR>

Specifies the IP address of the neighbor or the virtual IP address of the cluster in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255. . Range: 4096 to 131072. Default: 131072.

mac <MAC-ADDR>

Specifies the MAC address of the neighbor or the multicast MAC address in IANA format (xx:xx:xx:xx:xx:xx), where x is a hexadecimal number from 0 to F. Range: 4096 to 131072. Default: 131072.

Authority

Administrators or local user group members with execution rights for this command.

Example

Configuring a static ARP entry on a interface VLAN 10:

```
switch(config)# interface vlan 10  
switch(config-if-vlan)# arp ipv4 2.2.2.2 mac 01:00:5e:00:00:01
```

Removing a static ARP entry on interface VLAN10:

```
switch(config)# interface vlan 10  
switch(config-if-vlan)# no arp ipv4 2.2.2.2 mac 01:00:5e:00:00:01
```


clear arp

Syntax

```
clear arp [port <PORT-ID> | vrf {all-vrfs | <VRF-NAME>}]
```

Description

Clears IPv4 and IPv6 neighbor entries from the ARP table. If you do not specify any parameters, ARP table entries are cleared for the default VRF.

Command context

Manager (#)

Parameters

port <PORT-ID>

Specifies a physical port on the switch. Format: member/slot/port. For example: vlan 2. .
all-vrfs

Selects all VRFs.

<VRF-NAME>

Specifies the name of a VRF. Default: default.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Clearing all IPv4 and IPv6 neighbor ARP entries for the default VRF:

```
switch# clear arp
```

Clearing all ARP neighbor entries for a port:

```
switch# clear arp vlan 2
```

ip local-proxy-arp

Syntax

```
ip local-proxy-arp  
no ip local-proxy-arp
```

Description

Enables local proxy ARP on the specified interface. Local proxy ARP is supported on Layer 3 physical interfaces and on VLAN interfaces. To enable local proxy ARP on an interface, routing must be enabled on that interface.

The no form of this command disables local proxy ARP on the specified interface.

Command context

```
config-if  
config-if-vlan
```

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling local proxy ARP on interface **1/1/1**:

```
switch# interface 1/1/1
switch(config-if)# ip local proxy-arp
```

Enabling local proxy ARP on interface VLAN **3**:

```
switch# interface vlan 3
switch(config-if-vlan)# ip local-proxy-arp
```

Disabling local proxy ARP on on interface **1/1/1**.

```
switch# interface 1/1/1
switch(config-if)# no ip local-proxy-arp
```

ipv6 neighbor mac

Syntax

```
ipv6 neighbor <IPV6-ADDR> mac <MAC-ADDR>
no ipv6 neighbor <IPV6-ADDR> mac <MAC-ADDR>
```

Description

Specifies a permanent static neighbor entry in the ARP table (for IPv6 neighbors).

The **no** form of this command deletes a permanent static neighbor entry from the ARP table.

Command context

config-if

Parameters

<IPV6-ADDR>>

Specifies an IP address in IPv6 format (xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx), where x is a hexadecimal number from 0 to F. Range: 4096 to 131072. Default: 131072.

mac <MAC-ADDR>>

Specifies the MAC address of the neighbor (xx:xx:xx:xx:xx:xx), where x is a hexadecimal number from 0 to F. Range: 4096 to 131072. Default: 131072.

Authority

Administrators or local user group members with execution rights for this command.

Example

Creates a static ARP entry on interface **vlan 2**.

```
switch(config)# interface vlan 2
switch(config-if-vlan)# arp ipv6 neighbor 2001:0db8:85a3::8a2e:0370:7334 mac
00:50:56:96:df:c8
```

ip proxy-arp

Syntax

```
ip proxy-arp
no ip proxy-arp
```

Description

Enables proxy ARP for the specified Layer 3 interface. Proxy ARP is supported on Layer 3 physical interfaces, LAG interfaces, and VLAN interfaces. It is disabled by default. To enable proxy ARP on an interface, routing must be enabled on that interface.

The `no` form of this command disables proxy ARP for the specified interface.

Command context

```
config-if
config-if-vlan
config-lag-vlan
```

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling proxy ARP on interface **1/1/1**:

```
switch# interface 1/1/1
switch(config-if)# ip proxy-arp
```

Enabling proxy ARP on VLAN **3**:

```
switch# interface vlan 3
switch(config-if-vlan)# ip proxy-arp
```

Enabling proxy ARP on a LAG **11**:

```
switch(config)# int lag 11
switch(config-lag-if)# ip proxy-arp
```

Disabling proxy ARP on interface 1/1/1:

```
switch# interface 1/1/1
switch(config-if)# no ip proxy-arp
```

show arp

Syntax

```
show arp
```

Description

Shows the entries in the ARP (Address Resolution Protocol) table.

Command context

Manager (#)

Authority

Administrators or local user group members with execution rights for this command.

Usage

This command displays information about ARP entries, including the IP address, MAC address, port, and state.

When no parameters are specified, the `show arp` command shows all ARP entries for the default VRF (Virtual Router Forwarding) instance.

Examples

```
switch# show arp
```

IPv4 Address	MAC	Port	Physical Port	
192.168.1.2	00:50:56:96:7b:e0	vlan10	1/1/29	stale
192.168.1.3	00:50:56:96:7b:ac	vlan10	1/1/1	reachable

Total Number Of ARP Entries Listed- 2.

show arp inspection interface

Syntax

```
show arp inspection interface
```

Description

Displays the current configuration of dynamic ARP inspection on a VLAN or interface.

Command context

Operator (>) or Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

```
switch# show arp inspection interface
```

```

-----
Interface          Trust-State
-----
1/1/1              Untrusted
-----

```

```
switch# show arp inspection interface vsx-peer
```

```

-----
Interface          Trust-State
-----
1/1/1              Untrusted
lag100             Trusted
-----

```

```
switch# show arp inspection interface 1/1/1
```

```

-----
Interface          Trust-State
-----
1/1/1              Untrusted
-----

```

show arp inspection statistics

Syntax

```
show arp inspection statistics
```

Description

Displays statistics about forwarded and dropped ARP packets.

Command context

Operator (>) or Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Examples

```
switch# show arp inspection statistics vlan 1-200
```

```

-----
VLAN   Name                Forwarded    Dropped
-----
1      DEFAULT_VLAN_1      0           0
-----

```

```
switch# show arp inspection statistics vlan
```

VLAN	Name	Forwarded	Dropped
1	DEFAULT_VLAN_1	0	0
200	VLAN200	0	0

show arp state

Syntax

```
show arp state {all | failed | incomplete | permanent | reachable | stale}
```

Description

Shows ARP (Address Resolution Protocol) cache entries that are in the specified state.

Command context

Operator (>) or Manager (#)

Parameters

all

Shows the ARP cache entries for all VRF (Virtual Router Forwarding) instances.

failed

Shows the ARP cache entries that are in `failed` state. The neighbor might have been deleted.

incomplete

Shows the ARP cache entries that are in `incomplete` state.

An incomplete state means that address resolution is in progress and the link-layer address of the neighbor has not yet been determined. A solicitation request was sent, and the switch is waiting for a solicitation reply or a timeout.

permanent

Shows the ARP cache entries that are in `permanent` state. ARP entries that are in a permanent state can be removed by administrative action only.

reachable

Shows the ARP cache entries that are in `reachable` state, meaning that the neighbor is known to have been reachable recently.

stale

Shows ARP cache entries that are in `stale` state.

ARP cache entries are in the `stale` state if the elapsed time is in excess of the ARP timeout in seconds since the last positive confirmation that the forwarding path was functioning properly.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

```
switch# show arp state failed
```

IPv4 Address	MAC	Port	Physical Port	State

192.168.1.4

vlan10

failed

show arp summary

Syntax

```
show arp summary [all-vrfs | vrf <VRF-NAME>]
```

Description

Shows a summary of the IPv4 and IPv6 neighbor entries on the switch for all VRFs or a specific VRF.

Command context

Operator (>) or Manager (#)

Parameters

all-vrfs

Selects all VRFs.

vrf <VRF-NAME>

Specifies the name of a VRF.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Showing summary ARP information for all VRFs:

```
switch# show arp summary all-vrfs

ARP Entry's State           : IPv4      IPv6
-----
Number of Reachable ARP entries : 2          0
Number of Stale ARP entries    : 0          0
Number of Failed ARP entries   : 2          2
Number of Incomplete ARP entries : 0          0
Number of Permanent ARP entries : 0          0
-----
Total ARP Entries: 6          : 4          2
-----
```

Showing a summary of all IPv4 and IPv6 neighbor entries on the primary and secondary (peer) switches:

```
vsx-primary# show arp summary
ARP Entry's State      IPv4      IPv6
-----
Number of Reachable ARP entries  25858    32231
Number of Stale ARP entries     0         1
Number of Failed ARP entries    0        257
Number of Incomplete ARP entries 0         0
Number of Permanent ARP entries 0         0
-----
Total ARP Entries- 58347      25858    32489

vsx-primary# show arp summary vsx-peer
ARP Entry's State      IPv4      IPv6
-----
Number of Reachable ARP entries  25858    32168
Number of Stale ARP entries     0         3
Number of Failed ARP entries    0        317
Number of Incomplete ARP entries 0         0
Number of Permanent ARP entries 0         0
-----
Total ARP Entries- 58346      25858    32488
-----
```

show arp timeout

Syntax

```
show arp timeout [<INTERFACE>]
```

Description

Shows the age-out period for each ARP (Address Resolution Protocol) entry for a port, LAG, or VLAN interface.

Command context

Operator (>) or Manager (#)

Parameters

<INTERFACE>

Specifies a physical port, VLAN, or LAG on the switch. For physical ports, use the format (for example, 1/3/1).

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Showing ARP timeout information for a VLAN:

```
switch# show arp timeout vlan2
Port      VRF      Timeout
-----
vlan2     default  1800
```


show arp vrf

Syntax

```
show arp {all-vrfs | vrf <VRF-NAME>}
```

Description

Shows the ARP table for all VRF instances, or for the named VRF.

Command context

Operator (>) or Manager (#)

Parameters

all-vrfs

Specifies all VRFs.

vrf <VRF-NAME>

Specifies the name of a VRF. Length: 1 to 32 alphanumeric characters.

Authority

Operators or Administrators or local user group members with execution rights for this command.

Operators can execute this command from the operator context (>) only.

Examples

Showing ARP entries for VRF **test**.

```
switch# show arp vrf test
ARP IPv4 Entries:
-----
IPv4 Address      MAC                Port    Physical Port  State      VRF
10.20.30.40       00:50:56:bd:6a:c5  1/1/29  1/1/29         reachable  test
-----
Total Number Of ARP Entries Listed: 1.
-----

switch# show arp all-vrfs
ARP IPv4 Entries:
-----
IPv4 Address      MAC                Port    Physical Port  State      VRF
192.168.120.10    00:50:56:bd:10:be  1/1/32  1/1/32         reachable  red
10.20.30.40       00:50:56:bd:6a:c5  1/1/29  1/1/29         reachable  test
-----
Total Number Of ARP Entries Listed: 2.
-----
```

Showing ARP entries for all VRFs.

```
switch# show arp all-vrfs
ARP IPv4 Entries:
-----
IPv4 Address      MAC                Port    Physical Port  State      VRF
192.168.120.10    00:50:56:bd:10:be  1/1/32  1/1/32         reachable  red
10.20.30.40       00:50:56:bd:6a:c5  1/1/29  1/1/29         reachable  test
-----
Total Number Of ARP Entries Listed: 2.
-----
```

show ipv6 neighbors

Syntax

```
show ipv6 neighbors {all-vrfs | vrf <VRF-NAME>}
```

Description

Shows entries in the ARP table for all IPv6 neighbors for all VRFs or for a specific VRF.

When no parameters are specified, this command shows all ARP entries for the default VRF, and state information for `reachable` and `stale` entries only.

Command context

Operator (>) or Manager (#)

Authority

Administrators or local user group members with execution rights for this command.

Parameters

`all-vrfs`

Specifies all VRFs.

`vrf <VRF-NAME>`

Specifies a VRF name. Length: 1 to 32 alphanumeric characters.

Examples

```
switch# show ipv6 neighbors
IPv6 Entries:

-----

IPv6 Address          MAC                Port      Physical Port  State
-----
fe80::a21d:48ff:fe8f:2700  a0:1d:48:8f:27:00  vlan2300  1/1/31         reachable
fe80::f603:43ff:fe80:a600  f4:03:43:80:a6:00  vlan2300  1/1/30         reachable
-----

Total Number Of IPv6 Neighbors Entries Listed: 2.

-----
```

show ipv6 neighbors state

Syntax

```
show ipv6 neighbors state {all | failed | incomplete | permanent | reachable | stale}
```

Description

Shows all IPv6 neighbor ARP (Address Resolution Protocol) cache entries, or those cache entries that are in the specified state.

Command context

Operator (>) or Manager (#)

Parameters

`all`

Shows all ARP cache entries.

`failed`

Shows ARP cache entries that are in `failed` state. The neighbor might have been deleted. Set the neighbor to be unreachable.

`incomplete`

Shows ARP cache entries that are in `incomplete` state.

An incomplete state means that address resolution is in progress and the link-layer address of the neighbor has not yet been determined. This means that a solicitation request was sent, and you are waiting for a solicitation reply or a timeout.

`permanent`

Shows ARP cache entries that are in `permanent` state.

`reachable`

Shows ARP cache entries that are in `reachable` state, meaning that the neighbor is known to have been reachable recently.

`stale`

Shows ARP cache entries that are in `stale` state.

ARP cache entries are in the `stale` state if the elapsed time is in excess of the ARP timeout in seconds since the last positive confirmation that the forwarding path was functioning properly.

Authority

Operators or Administrators or local user group members with execution rights for this command.

Operators can execute this command from the operator context (>) only.

Example

```
switch# show ipv6 neighbors state all
```

IPv6 Address	MAC	Port	Physical Port	State
100::2	48:0f:cf:af:f1:cc	lag1	lag1	reachable
300::3	48:0f:cf:af:33:be	vlan3	1/4/20	reachable
fe80::4a0f:cfff:feaf:f1cc	48:0f:cf:af:f1:cc	lag1	lag1	reachable
200::3	48:0f:cf:af:33:be	1/4/11	1/4/11	reachable
fe80::4a0f:cfff:feaf:33be	48:0f:cf:af:33:be	vlan3	1/4/20	reachable

```
Total Number Of IPv6 Neighbors Entries Listed- 5.
```

aruba-central

Syntax

```
aruba-central
```

```
no aruba-central
```

Description

Creates or enters the Aruba Central configuration context (`config-aruba-central`).

Command context

```
config
```

Authority

Administrators or local user group members with execution rights for this command.

Example

Creating the Aruba Central configuration context:

```
switch(config)# aruba-central
switch(config-aruba-central)#
```

aruba-central support-mode

Syntax

```
aruba-central support-mode
```

```
no aruba-central support-mode
```

Description

Allows the device to be writable for all operations in Aruba Central lockout mode for troubleshooting. The no form of this command disables this activity.



Support-mode is disabled by default when the switch is managed by Aruba Central. This command is only effective in the CLI session where it is executed.

Command context

```
Manager (#)
```

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring the device to be writable for all operations in Aruba Central lockout mode:

```
switch# aruba-central support-mode
switch#
```

Removing the configuration that allows the device to be writable for all operations in Aruba Central lockout mode:

```
switch# no aruba-central support-mode
switch#
```

configuration-lockout central managed

Syntax

```
configuration-lockout central managed
```

```
no configuration-lockout central managed
```

Description

Configures the device to only be writable from Aruba Central. Aruba Central will be the only agent that can add, modify, or delete configurations on the device. The no form of this command disables this feature.



The no form of this command is only available when the device is disconnected from Aruba Central.

Command context

```
config
```

Authority

Administrators or local user group members with execution rights for this command.

Usage

The AOS-CX switch connects to Aruba Central in either of two modes: monitor or managed. When the device is connected in monitor mode, Aruba Central monitors the configurations on the switch. When the device is connected in managed mode, the `configuration-lockout central managed` command does not allow configuration changes from other interfaces such as CLI or Web UI.

Examples

Configuring the device to only be writable from Aruba Central :

```
switch(config)# configuration-lockout central managed
switch# show configuration-lockout
configuration lockout
-----
central: managed
switch# sh aruba-central
```

```

Central admin state           :enable
Central location              :20.0.0.2:8083
VRF for connection            :default
Central connection status     :connected

Central source                :cli
Central source connection status :connected
Central source last connected on :Tue Feb 9 17:53:13 UTC 2021

Activate Server URL           :devices-v2.arubanetworks.com
CLI location                   :20.0.2:8083
CLI VRF                        :default
switch(config)# end

```

disable

Syntax

```
disable
```

Description

Disables connection to Aruba Central server.

When the connection is disabled, the switch does not attempt to connect to the Aruba Central server or fetch central location from any of the three sources (CLI/Aruba Activate/DHCP). It also disconnects any active connection to the Aruba Central server.

Command context

```
config-aruba-central
```

Authority

Administrators or local user group members with execution rights for this command.

Example

```

switch(config-aruba-central)# disable
switch(config-aruba-central)#

```

enable

Syntax

```
enable
```

Description

Enables connection to Aruba Central server. When the connection is enabled, the switch attempts to download the location of the Aruba Central server in one of the following ways at startup and after the connection is lost:

- Using command-line interface (CLI).
- Connecting to Aruba Activate server.
- Using DHCP options provided during ZTP.

DHCP servers provide the options requested by the device to connect to Central, Central On-premise management, or the TFTP server.

Command context

config-aruba-central

Authority

Administrators or local user group members with execution rights for this command.

Examples

```
switch(config-aruba-central) # enable
switch(config-aruba-central) #
```

location-override

Syntax

```
location-override <location> [vrf <VRF default>]
no location-override
```

Description

When `location` and `vrf` are configured, the switch overrides existing connections to Aruba Central. The switch attempts to establish connection to Aruba Central with the specified location and VRF with highest priority.

The `no` form of this command removes location override values from the Aruba Central configuration context.

Command context

config-aruba-central

Authority

Administrators or local user group members with execution rights for this command.

Parameters

<location>

Specifies one of these values:

- <FQDN>: a fully qualified domain name.
- <IPv4>: an IPv4 address.
- <IPv6>: an IPv6 address.

vrf <VRF-NAME>

Specifies the VRF name to be used for communicating with the server. If no VRF name is provided, the default VRF named `default` is used.

Examples

Configuring location override with location and VRF:

```
switch(config-aruba-central)# location-override aruba-central.com vrf default
switch(config-aruba-central)#
```

Configuring location override with location only:

```
switch(config-aruba-central)# location-override aruba-central.com
switch(config-aruba-central)#
```

Removing location override values from the Aruba Central configuration context:

```
switch(config-aruba-central)# no location-override
switch(config-aruba-central)#
```

show aruba-central

Syntax

```
show aruba-central
```

Description

Shows information about Aruba Central connection and the status of the Activate server connection.

Command context

Operator (>) or Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Example of a switch that has the Aruba Central connection:

```
switch# show aruba-central
Central admin state           :enabled
Central location              : N/A
VRF for connection           : N/A
Central connection status     : N/A
Central source                : dhcp
Central source connection status : connection_failure
Central source last connected on : N/A
System time synchronized from Activate : True
Activate server URL           : 172.17.0.1
CLI location                  : N/A
CLI VRF                       : N/A
Source IP                     : N/A
Source IP Overridden          : false
Central support mode          : disabled
```

show running-config current-context

Syntax


```
show running-config current-context
```

Description

Shows the running configuration for the current-context. If user is in the context of Aruba-Central(config-aruba-central), then Aruba Central running configuration is displayed.

Command context

Operator (>) or Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Shows the running configuration of Aruba Central:

```
switch(config-aruba-central)# show running-config current-context
aruba-central
    disable
```

banner

Syntax

```
banner {motd | exec} <DELIMITER>  
no banner {motd | exec}
```

Description

Enables the customization of the MOTD or the EXEC banner.

The `no` form of this command disables the MOTD or the EXEC banner.

Command context

`config`

Parameters

`motd`

Configures the banner shown before the login prompt.

`exec`

Configures the banner shown after a successful login.

`<DELIMITER>`

Specifies the character used to terminate the input string.

Authority

Administrators or local user group members with execution rights for this command.

Usage

This command enables the customization of two types of banners:

- The MOTD banner. The banner displayed on attempting to connect to a management interface.
- The EXEC banner. The banner displayed upon successful authentication.

You can create a banner that spans multiple lines. The maximum length of a banner is 4,095 characters. This requirement includes any non-visible characters. The minimum number of characters allowed is an empty string, which displays no banner.

End the banner text with a chosen delimiter character. A delimiter character can be any non-whitespace character that does not have special meaning to the CLI, such as the caret (^). A question mark (?) is not permitted. Question marks can however be included as part of the banner text.

Examples

Configuring the banner displayed before login:

```
switch(config)# banner motd ^
Enter a new banner. Terminate the banner with the delimiter you have chosen.
(banner-motd)# This is an example of a banner text which a connecting user
(banner-motd)# will see before they are prompted for their password.
(banner-motd)#
(banner-motd)# As you can see it may span multiple lines and the input
(banner-motd)# will be terminated when the delimiter character is
(banner-motd)# encountered.^
```

Configuring the banner displayed after a successful login:

```
switch(config)# banner exec &
Enter a new banner. Terminate the banner with the delimiter you have chosen.
(banner-motd)# This is an example of different banner text. This time
(banner-motd)# the banner entered will be displayed after a user has
(banner-motd)# authenticated.
(banner-motd)#
(banner-motd)# & This text will not be included because it comes after the &
```

Disabling the MOTD banner:

```
switch(config)# no banner motd
```

show banner

Syntax

```
show banner {motd | exec} [vsx-peer]}
```

Description

Shows the MOTD or EXEC banner message.

Command context

Operator (>) or Manager (#)

Parameters

motd

Shows the banner displayed before the login prompt.

exec

Shows the banner displayed after a successful login.

[vsx-peer]

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Showing the MOTD banner displayed before the login prompt:

```
switch(config)# show banner motd
This is an example of a banner text which a connecting user
will see before they are prompted for their password.
```

As you can see it may span multiple lines and the input will be terminated when the delimiter character is encountered.

boot set-default

Syntax

```
boot set-default {primary | secondary}
```

Description

Sets the default operating system image to use when the system is booted.

Command context

Manager (#)

Parameters

`primary`

Selects the primary network operating system image.

`secondary`

Selects the secondary network operating system image.

Authority

Administrators or local user group members with execution rights for this command.

Example

Selecting the primary image as the default boot image:

```
switch# boot set-default primary
Default boot image set to primary.
```

boot system

Syntax

```
boot system [primary | secondary | serviceos]
```

Description

Reboots all modules on the switch. By default, the configured default operating system image is used. Optional parameters enable you to specify which system image to use for the reboot operation and for future reboot operations.

Command context

Manager (#)

Parameters

`primary`

Selects the primary operating system image for this reboot and sets the configured default operating system image to `primary` for future reboots.

`secondary`

Selects the secondary operating system image for this reboot and sets the configured default operating system image to `secondary` for future reboots.

`serviceos`

Selects the service operating system for this reboot. Does not change the configured default operating system image. The service operating system acts as a standalone bootloader and recovery OS for switches running the AOS-CX operating system and is used in rare cases when troubleshooting a switch.

Authority

Administrators or local user group members with execution rights for this command.

Usage

This command reboots the entire system. If you do not select one of the optional parameters, the system reboots from the configured default boot image.

You can use the `show images` command to show information about the primary and secondary system images.

Choosing one of the optional parameters affects the setting for the default boot image:

- If you select the `primary` or `secondary` optional parameter, that image becomes the configured default boot image for future system reboots. The command fails if the switch is not able to set the operating system image to the image you selected.

You can use the `boot set-default` command to change the configured default operating system image.

- If you select `serviceos` as the optional parameter, the configured default boot image remains the same, and the system reboots all management modules with the service operating system.

If the configuration of the switch has changed since the last reboot, when you execute the `boot system` command you are prompted to save the configuration and you are prompted to confirm the reboot operation.

Saving the configuration is not required. However, if you attempt to save the configuration and there is an error during the save operation, the `boot system` command is aborted.

Examples

Rebooting the system from the configured default operating system image:

```
switch# boot system
Do you want to save the current configuration (y/n)? y
The running configuration was saved to the startup configuration.

This will reboot the entire switch and render it unavailable
until the process is complete.
Continue (y/n)? y
The system is going down for reboot.
```

Rebooting the system from the secondary operating system image, setting the secondary operating system image as the configured default boot image:

```
switch# boot system secondary
Default boot image set to secondary.

Do you want to save the current configuration (y/n)? n

This will reboot the entire switch and render it unavailable
until the process is complete.
Continue (y/n)? y
The system is going down for reboot.
```

Canceling a system reboot:

```
switch# boot system

Do you want to save the current configuration (y/n)? n

This will reboot the entire switch and render it unavailable
until the process is complete.
Continue (y/n)? n
Reboot aborted.
switch#
```

show boot-history

Syntax

```
show boot-history [all]
```

Description

Shows boot information. When no parameters are specified, shows the most recent information about the boot operation, and the three previous boot operations for the active management module. When the `all` parameter is specified, shows the boot information for the active management module.

Command context

Manager (#)

Parameters

`all`

Shows boot information for the active management module and all available line modules.

Authority

Administrators or local user group members with execution rights for this command.

Usage

This command displays the boot-index, boot-ID, and up time in seconds for the current boot. If there is a previous boot, it displays boot-index, boot-ID, reboot time (based on the time zone configured in the system) and reboot reasons. Previous boot information is displayed in reverse chronological order.

Index

The position of the boot in the history file. Range: 0 to 3.

Boot ID

A unique ID for the boot . A system-generated 128-bit string.

Current Boot, up for <SECONDS> seconds

For the current boot, the `show boot-history` command shows the number of seconds the module has been running on the current software.

Timestamp boot reason

For previous boot operations, the `show boot-history` command shows the time at which the operation occurred and the reason for the boot. The reason for the boot is one of the following values:

`<DAEMON-NAME> crash`

The daemon identified by `<DAEMON-NAME>` caused the module to boot.

`Kernel crash`

The operating system software associated with the module caused the module to boot.

`Reboot requested through database`

The reboot occurred because of a request made through the CLI or other API.

`Uncontrolled reboot`

The reason for the reboot is not known.

Examples

Showing the boot history of the active management module:

```
switch# show boot-history
Management module
=====

Index : 3
Boot ID : f1bf071bdd04492bbf8439c6e479d612
Current Boot, up for 22 hrs 12 mins 22 secs

Index : 2
Boot ID : edfa2d6598d24e989668306c4a56a06d
07 Aug 18 16:28:01 : Reboot requested through database

Index : 1
Boot ID : 0bda8d0361df4a7e8e3acdc1dba5caad
07 Aug 18 14:08:46 : Reboot requested through database

Index : 0
Boot ID : 23da2b0e26d048d7b3f4b6721b69c110
07 Aug 18 13:00:46 : Reboot requested through database
switch#
```


diag cable-diagnostic

Syntax

```
diag cable-diagnostic <IF-NAME>
```

Description

Runs a cable diagnostic test on an interface.

Command context

Manager (#)

Parameters

IF-NAME

Specifies the name of the interface.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Running a cable diagnostic test on interfaces:

```
switch# diag cable-diagnostic 1/1/1
This command will cause a loss of link on the port under test and will take
several seconds to complete.
Continue (y/n)? y
```

Interface	MDI Pair	Cable Status	Distance to Fault (Meters)	MDI Mode
1/1/1	1-2	good	5	mdi
	3-6	good	5	mdi
	4-5	good	5	mdi
	7-8	open	3	

```
switch# diag cable-diagnostic 1/1/2
This command will cause a loss of link on the port under test and will take
several seconds to complete.
Continue (y/n)? y
```

Interface	MDI Pair	Cable Status	Distance to Fault (Meters)	MDI Mode
1/1/2	1-2	good	5	mdix

3-6	good	5	mdix
4-5	short	1	
7-8	good	5	mdix



Running a cable diagnostic test will result in a brief interruption in connectivity on all tested ports.

cdp

Syntax

cdp

Description

Configures CDP support globally on all active interfaces or on a specific interface. By default, CDP is enabled on all active interfaces.

When CDP is enabled, the switch adds entries to its CDP Neighbors table for any CDP packets it receives from neighboring CDP devices.

When CDP is disabled, the CDP Neighbors table is cleared and the switch drops all inbound CDP packets without entering the data in the CDP Neighbors table.

The `no` form of this command disables CDP support globally on all active interfaces or on a specific interface.

Command context

config

config-if

Authority

Administrators or local user group members with execution rights for this command.

Examples

Not supported on the 6100 Switch Series.

Enabling CDP globally:

```
switch(config) # cdp
```

Disabling CDP globally:

```
switch(config) # no cdp
```

Enabling CDP on interface **1/1/1**:

```
switch(config) # interface 1/1/1
switch(config-if) # cdp
```

Disabling CDP on interface **1/1/1**:

```
switch(config)# interface 1/1/1
switch(config-if)# no cdp
```

clear cdp counters

Syntax

```
clear cdp counters
```

Description

Clears CDP counters.

Command context

```
config
```

Authority

Administrators or local user group members with execution rights for this command.

Examples

Clearing CDP counters:

```
switch(config) clear cdp counters
```

clear cdp neighbor-info

Syntax

```
clear cdp neighbor-info
```

Description

Clears CDP neighbor information.

Command context

```
config
```

Authority

Administrators or local user group members with execution rights for this command.

Examples

Clearing CDP neighbor information:

```
switch(config) clear neighbor-info
```

show cdp

Syntax

```
show cdp
```

Description

Shows CDP information for all interfaces.

Command context

config

Authority

Administrators or local user group members with execution rights for this command.

Examples

Showing CDP information:

```
switch(config)# show cdp
CDP Global Information
=====
CDP           : Enabled
CDP Mode      : Rx only
CDP Hold Time : 180 seconds

Port          CDP
-----
1/1/1         Enabled
1/1/2         Enabled
1/1/3         Enabled
1/1/4         Enabled
1/1/5         Enabled
1/1/6         Enabled
1/1/7         Enabled
1/1/8         Enabled
1/1/9         Enabled
1/1/10        Enabled
```

show cdp neighbor-info

Syntax

```
show cdp neighbor-info <INTERFACE-ID>
```

Description

Shows CDP information for all neighbors or for CDP information on a specific interface.

Command context

config

Parameters

<INTERFACE-ID>

Specifies an interface. Format: member/slot/port.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Showing all CDP neighbor information:

```
switch(config)# show cdp neighbor-info
```

Port	Device ID	Platform	Capability
1/1/1	myswitch	cisco WS-C2950-12	SI

Showing CDP information for interface **1/1/1**:

```
switch(config)# show cdp neighbor-info 1/1/1
Local Port : 1/1/1
MAC          : 3c:a8:2a:7b:6b:2b
Device ID    : SEPd4adbd2a30d6
Address      : 2.71.0.230
Platform     : Cisco IP Phone 3905
Duplex       : full
Capability    : host
Voice VLAN Support : Yes
Neighbor Port-ID : Port 1
```

show cdp traffic

Syntax

```
show cdp neighbor-info
```

Description

Shows CDP statistics for each interface.

Command context

```
config
```

Authority

Administrators or local user group members with execution rights for this command.

Examples

Showing CDP traffic statistics:

```
switch(config)# show cdp traffic
```

CDP Statistics			
Port	Transmitted Frames	Received Frames	Discarded Frames
1/1/1	0	4	0
1/1/2	0	0	0
1/1/3	0	2	0
1/1/4	0	0	0
1/1/5	0	0	0

checkpoint auto

Syntax

```
checkpoint auto <TIME-LAPSE-INTERVAL>
```

Description

Starts auto checkpoint mode. In auto checkpoint mode, the switch temporarily saves the runtime configuration as a checkpoint only for the specified time lapse interval. Configuration changes must be saved before the interval expires, otherwise the runtime configuration is restored from the temporary checkpoint.

Command context

Manager (#)

Parameters

<TIME-LAPSE-INTERVAL>

Specifies the time lapse interval in minutes. Range: 1 to 60.

Authority

Administrators or local user group members with execution rights for this command.

Usage

To save the runtime checkpoint permanently, run the `checkpoint auto confirm` command during the time lapse interval. The filename for the saved checkpoint is named `AUTO<YYYYMMDDHHMMSS>`. If the `checkpoint auto confirm` command is not entered during the specified time lapse interval, the previous runtime configuration is restored.

Examples

Confirming the auto checkpoint:

```
switch# checkpoint auto 20
Auto checkpoint mode expires in 20 minute(s)
switch# WARNING Please "checkpoint auto confirm" within 2 minutes
switch# checkpoint auto confirm
checkpoint AUTO20170801011154 created
```

In this example, the runtime checkpoint was saved because the `checkpoint auto confirm` command was entered within the value set by the `time-lapse-interval` parameter, which was 20 minutes.

Not confirming the auto checkpoint:

```
switch# checkpoint auto 20  
Auto checkpoint mode expires in 20 minute(s)  
switch# WARNING Please "checkpoint auto confirm" within 2 minutes  
WARNING: Restoring configuration. Do NOT add any new configuration.  
Restoration successful
```

In this example, the runtime checkpoint was reverted because the `checkpoint auto confirm` command was not entered within the value set by the `time-lapse-interval` parameter, which was 20 minutes.

checkpoint auto confirm

Syntax

```
checkpoint auto confirm
```

Description

Signals to the switch to save the running configuration used during the auto checkpoint mode. This command also ends the auto checkpoint mode.

Command context

Manager (#)

Authority

Administrators or local user group members with execution rights for this command.

Usage

To save the runtime checkpoint permanently, run the `checkpoint auto confirm` command during the time lapse value set by the `checkpoint auto <TIME-LAPSE-INTERVAL>` command. The generated checkpoint name will be in the format `AUTO<YYYYMMDDHHMMSS>`. If the `checkpoint auto confirm` command is not entered during the specified time lapse interval, the previous runtime configuration is restored.

Examples

Confirming the auto checkpoint:

```
switch# checkpoint auto confirm
```

checkpoint diff

Syntax

```
checkpoint diff {<CHECKPOINT-NAME1> | running-config | startup-config}  
               {<CHECKPOINT-NAME2> | running-config | startup-config}
```

Description

Shows the difference in configuration between two configurations. Compare checkpoints, the running configuration, or the startup configuration.

Command context

Manager (#)

Parameters

{<CHECKPOINT-NAME1> | running-config | startup-config}

Selects either a checkpoint, the running configuration, or the startup configuration as the baseline.

{<CHECKPOINT-NAME2> | running-config | startup-config}

Selects either a checkpoint, the running configuration, or the startup configuration to compare.

Authority

Administrators or local user group members with execution rights for this command.

Usability

The output of the `checkpoint diff` command has several symbols:

- The plus sign (+) at the beginning of a line indicates that the line exists in the comparison but not in the baseline.
- The minus sign (-) at the beginning of a line indicates that the line exists in the baseline but not in the comparison.

Examples

In the following example, the configurations of checkpoints `cp1` and `cp2` are displayed before the `checkpoint diff` command, so that you can see the context of the `checkpoint diff` command.

```
Switch# checkpoint diff chkpt01 chkpt02
--- /tmp/chkpt011607564301327
+++ /tmp/chkpt021607564301353
@@ -1,7 +1,7 @@
!
!Version ArubaOS-CX PL.10.06.0100V
!export-password: default
-hostname Switch
+hostname Switch1
+user admin group administrators password ciphertext
AQBapTyg9tpaiAaTfSVV5eNdFzOORRvZ6CMpglh1P+LQUHQLYgAAAGAhmRqFbkNvrgy2SBVk7H8C5hvg/Iib
8rWYFZLEaSCrobNP9EwMu+hLNM0xmsh45yG8dncP7WkxjwrW4p4Qra6dVfr0EW8xh/lpQf8F/2Wki20Lc9JL
XiYge7ti0H6cVn+G
+radius-server tracking interval 60
+no usb

Switch1#
```

checkpoint post-configuration

Syntax

`checkpoint post-configuration`

`no checkpoint post-configuration`

Description

Enables creation of system generated checkpoints when configuration changes occur. This feature is enabled by default.

The `no` form of this command disables system generated checkpoints.

Command context

Manager (#)

Authority

Administrators or local user group members with execution rights for this command.

Usage

System generated checkpoints are automatically created by default. Whenever a configuration change occurs, the switch starts a timeout counter (300 seconds by default). For each additional configuration change, the timeout counter is restarted. If the timeout expires with no additional configuration changes being made, the switch generates a new checkpoint.

System generated checkpoints are named with the prefix `CPC` followed by a time stamp in the format `<YYYYMMDDHHMMSS>`. For example: `CPC20170630073127`.

System checkpoints can be applied using the checkpoint rollback feature or copy command.

A maximum of 32 system checkpoints can be created. Beyond this limit, the newest system checkpoint replaces the oldest system checkpoint.

Examples

Enabling system checkpoints:

```
switch(config)# checkpoint post-configuration
```

Disabling system checkpoints:

```
switch(config)# no checkpoint post-configuration
```

checkpoint post-configuration timeout

Syntax

```
checkpoint post-configuration timeout <TIMEOUT>
```

```
no checkpoint post-configuration timeout <TIMEOUT>
```

Description

Sets the timeout for the creation of system checkpoints. The timeout specifies the amount of time since the latest configuration for the switch to create a system checkpoint.

The `no` form of this command resets the timeout to 300 seconds, regardless of the value of the `<TIMEOUT>` parameter.

Command context

Manager (#)

Parameters

```
timeout <TIMEOUT>
```

Specifies the timeout in seconds. Range: 5 to 600. Default: 300.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the timeout for system checkpoints to 60 seconds:

```
switch(config)# checkpoint post-configuration timeout 60
```

Resetting the timeout for system checkpoints to 300 seconds:

```
switch(config)# no checkpoint post-configuration timeout 1
```

checkpoint rename

Syntax

```
checkpoint rename <OLD-CHECKPOINT-NAME> <NEW-CHECKPOINT-NAME>
```

Description

Renames an existing checkpoint.

Command context

Manager (#)

Parameters

<OLD-CHECKPOINT-NAME>

Specifies the name of an existing checkpoint to be renamed.

<NEW-CHECKPOINT-NAME>

Specifies the new name for the checkpoint. The checkpoint name can be alphanumeric. It can also contain underscores (_) and dashes (-).



Do not start the checkpoint name with `CPC` because it is used for system-generated checkpoints.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Renaming checkpoint **ckpt1** to **cfg001**:

```
switch# checkpoint rename ckpt1 cfg001
```

checkpoint rollback

Syntax

```
checkpoint rollback {<CHECKPOINT-NAME> | startup-config}
```

Description

Applies the configuration from a pre-existing checkpoint or the startup configuration to the running configuration.

Command context

Manager (#)

Parameters

<CHECKPOINT-NAME>

Specifies a checkpoint name.

startup-config

Specifies the startup configuration.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Applying a checkpoint named `ckpt1` to the running configuration:

```
switch# checkpoint rollback ckpt1  
Success
```

Applying a startup checkpoint to the running configuration:

```
switch# checkpoint rollback startup-config  
Success
```

copy checkpoint *<CHECKPOINT-NAME>* *<REMOTE-URL>*

Syntax

```
copy checkpoint <CHECKPOINT-NAME> <REMOTE-URL> [vrf <VRF-NAME>]
```

Description

Copies a checkpoint configuration to a remote location as a file. The configuration is exported in checkpoint format, which includes switch configuration and relevant metadata.

Command context

Manager (#)

Parameters

<CHECKPOINT-NAME>

Specifies the name of a checkpoint.

<REMOTE-URL>

Specifies the remote destination and filename using the syntax: `{tftp | sftp}://<IP-ADDRESS>`

`[:<PORT-NUMBER>] [;blocksize=<BLOCKSIZE-VALUE>] /<FILE-NAME>`

vrf <VRF-NAME>

Specifies a VRF name.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Copying checkpoint configuration to remote file through TFTP:

```
switch# copy checkpoint ckpt1 tftp://192.168.1.10/ckptmeta vrf default
##### 100.0%
Success
```

Copying checkpoint configuration to remote file through SFTP:

```
switch# copy checkpoint ckpt1 sftp://root@192.168.1.10/ckptmeta vrf default
The authenticity of host '192.168.1.10 (192.168.1.10)' can't be established.
ECDSA key fingerprint is SHA256:FtOm6Uxuxumil7VCwLnhz92H9LkjY+eURbdddOETy50.
Are you sure you want to continue connecting (yes/no)? yes
root@192.168.1.10's password:
sftp> put /tmp/ckptmeta ckptmeta
Uploading /tmp/ckptmeta to /root/ckptmeta
Warning: Permanently added '192.168.1.10' (ECDSA) to the list of known hosts.
Connected to 192.168.1.10.
Success
```

copy checkpoint <CHECKPOINT-NAME> {running-config | startup-config}

Syntax

```
copy checkpoint <CHECKPOINT-NAME> {running-config | startup-config}
```

Description

Copies an existing checkpoint configuration to the running configuration or to the startup configuration.

Command context

Manager (#)

Parameters

<CHECKPOINT-NAME>

Specifies the name of an existing checkpoint.

{running-config | startup-config}

Selects whether the running configuration or the startup configuration receives the copied checkpoint configuration. If the startup configuration is already present, the command overwrites the startup configuration.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Copying **ckpt1** checkpoint to the running configuration:

```
switch# copy checkpoint ckpt1 running-config
Success
```

Copying **ckpt1** checkpoint to the startup configuration:

```
switch# copy checkpoint ckpt1 startup-config
Success
```

copy checkpoint <CHECKPOINT-NAME> <STORAGE-URL>

Syntax

```
copy checkpoint <CHECKPOINT-NAME> <STORAGE-URL>
```

Description

Copies an existing checkpoint configuration to a USB drive. The file format is defined when the checkpoint was created.

Command context

Manager (#)

Parameters

<CHECKPOINT-NAME>

Specifies the name of the checkpoint to copy. The checkpoint name can be alphanumeric. It can also contain underscores (_) and dashes (-).

<STORAGE-URL>>

Specifies the name of the target file on the USB drive using the following syntax: `usb:/<FILE>`

The USB drive must be formatted with the FAT file system.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Copying the `test` checkpoint to the `testCheck` file on the USB drive:

```
switch# copy checkpoint test usb:/testCheck
Success
```

copy <REMOTE-URL> checkpoint <CHECKPOINT-NAME>

Syntax

```
copy <REMOTE-URL> checkpoint <CHECKPOINT-NAME> [vrf <VRF-NAME>]
```

Description

Copies a remote configuration file to a checkpoint. The remote configuration file must be in checkpoint format.

Command context

Manager (#)

Parameters

<REMOTE-URL>

Specifies a remote file using the following syntax: {tftp | sftp}://<IP-ADDRESS>[:<PORT-NUMBER>]
[;blocksize=<BLOCKSIZE-VALUE>]/<FILE-NAME>
<CHECKPOINT-NAME>

Specifies the name of the target checkpoint. The checkpoint name can be alphanumeric. It can also contain underscores (_) and dashes (-). Required.



Do not start the checkpoint name with CPC because it is used for system-generated checkpoints.

vrf <VRF-NAME>

Specifies a VRF name. Default: default.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Copying a checkpoint format file to checkpoint **ckpt5** on the default VRF:

```
switch# copy tftp://192.168.1.10/ckptmeta checkpoint ckpt5
##### 100.0%
100.0%
Success
```

copy <REMOTE-URL> {running-config | startup-config}

Syntax

copy <REMOTE-URL> {running-config | startup-config } [vrf <VRF-NAME>]

Description

Copies a remote file containing a switch configuration to the running configuration or to the startup configuration.

Command context

Manager (#)

Parameters

<REMOTE-URL>

Specifies a remote file with the following syntax: {tftp | sftp}://<IP-ADDRESS>[:<PORT-NUMBER>]
[;blocksize=<BLOCKSIZE-VALUE>]/<FILE-NAME>

{running-config | startup-config}

Selects whether the running configuration or the startup configuration receives the copied checkpoint configuration. If the startup configuration is already present, the command overwrites the startup configuration.

vrf <VRF-NAME>

Specifies the name of a VRF. Default: default.

Authority

Administrators or local user group members with execution rights for this command.

Usage

The switch copies only certain file types. The format of the file is automatically detected from contents of the file. The `startup-config` option only supports the JSON file format and checkpoints, but the `running-config` option supports the JSON and CLI file formats and checkpoints.

When a file of the CLI format is copied, it overwrites the running configuration. The CLI command does not clear the running configuration before applying the CLI commands. All of the CLI commands in the file are applied line-by-line. If a particular CLI command fails, the switch logs the failure and it continues to the next line in the CLI configuration. The event log (`show events -d hpe-config`) provides information as to which command failed.

Examples

Copying a JSON format file to the running configuration:

```
switch# copy tftp://192.168.1.10/runjson running-config
##### 100.0%
Configuration may take several minutes to complete according to configuration file
size
--0%----10%----20%----30%----40%----50%----60%----70%----80%----90%----100%--
Success
```

Copying a CLI format file to the running configuration with an error in the file:

```
switch# copy tftp://192.168.1.10/runcli running-config
##### 100.0%
Configuration may take several minutes to complete according to configuration file
size
--0%----10%----20%----30%----40%----50%----60%----70%----80%----90%----100%--
Some of the configuration lines from the file were NOT applied. Use 'show
events -d hpe-config' for more info.
```

Copying a CLI format file to the startup configuration:

```
switch# copy tftp://192.168.1.10/startjson startup-config
##### 100.0%
100.0%
Success
```

Copying an unsupported file format to the startup configuration:

```
switch# copy tftp://192.168.1.10/startfile startup-config
##### 100.0%
100.0%
unsupported file format
```

copy running-config {startup-config | checkpoint <CHECKPOINT-NAME>}

Syntax

```
copy running-config {startup-config | checkpoint <CHECKPOINT-NAME>}
```

Description

Copies the running configuration to the startup configuration or to a new checkpoint. If the startup configuration is already present, the command overwrites the existing startup configuration.

Command context

Manager (#)

Parameters

startup-config

Specifies that the startup configuration receives a copy of the running configuration.

checkpoint <CHECKPOINT-NAME>

Specifies the name of a new checkpoint to receive a copy of the running configuration. The checkpoint name can be alphanumeric. It can also contain underscores (_) and dashes (-).



Do not start the checkpoint name with `CPC` because it is used for system-generated checkpoints.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Copying the running configuration to the startup configuration:

```
switch# copy running-config startup-config
Success
```

Copying the running configuration to a new checkpoint named **ckpt1**:

```
switch# copy running-config checkpoint ckpt1
Success
```

copy {running-config | startup-config} <REMOTE-URL>

Syntax

```
copy {running-config | startup-config} <REMOTE-URL> {cli | json} [vrf <VRF-NAME>]
```

Description

Copies the running configuration or the startup configuration to a remote file in either CLI or JSON format.

Command context

Manager (#)

Parameters

{running-config | startup-config}

Selects whether the running configuration or the startup configuration is copied to a remote file.

<REMOTE-URL>

Specifies the remote file using the syntax: {tftp | sftp}://<IP-ADDRESS>[:<PORT-NUMBER>]

[;blocksize=<BLOCKSIZE-VALUE>]/<FILE-NAME>

{cli | json}

Selects the remote file format: P: CLI or JSON.

vrf <VRF-NAME>

Specifies the name of a VRF. Default: default.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Copying a running configuration to a remote file in CLI format:

```
switch# copy running-config tftp://192.168.1.10/runcli cli
##### 100.0%
Success
```

Copying a running configuration to a remote file in JSON format:

```
switch# copy running-config tftp://192.168.1.10/runjson json
##### 100.0%
Success
```

Copying a startup configuration to a remote file in CLI format:

```
switch# copy startup-config sftp://root@192.168.1.10/startcli cli
root@192.168.1.10's password:
sftp> put /tmp/startcli startcli
Uploading /tmp/startcli to /root/startcli
Connected to 192.168.1.10.
Success
```

Copying a startup configuration to a remote file in JSON format:

```
switch# copy startup-config sftp://root@192.168.1.10/startjson json
root@192.168.1.10's password:
sftp> put /tmp/startjson startjson
Uploading /tmp/startjson to /root/startjson
Connected to 192.168.1.10.
Success
```

copy {running-config | startup-config} <STORAGE-URL>

Syntax

copy {running-config | startup-config} <STORAGE-URL> {cli | json}

Description

Copies the running configuration or a startup configuration to a USB drive.

Command context

Manager (#)

Parameters

{running-config | startup-config}

Selects the running configuration or the startup configuration to be copied to the switch USB drive.
<STORAGE-URL>

Specifies a remote file with the following syntax: `usb:/<file>`
{cli | json}

Selects the format of the remote file: CLI or JSON.

Authority

Administrators or local user group members with execution rights for this command.

Usage

The switch supports JSON and CLI file formats when copying the running or starting configuration to the USB drive. The USB drive must be formatted with the FAT file system.

The USB drive must be enabled and mounted with the following commands:

```
switch(config)# usb
switch(config)# end
switch# usb mount
```

Examples

Copying a running configuration to a file named `runCLI` on the USB drive:

```
switch# copy running-config usb:/runCLI cli
Success
```

Copying a startup configuration to a file named `startCLI` on the USB drive:

```
switch# copy startup-config usb:/startCLI cli
Success
```

copy startup-config running-config

Syntax

```
copy startup-config running-config
```

Description

Copies the startup configuration to the running configuration.

Command context

Manager (#)

Authority

Administrators or local user group members with execution rights for this command.

Examples

```
switch# copy startup-config running-config
Success
```

copy <STORAGE-URL> running-config

Syntax

```
copy <STORAGE-URL> {running-config | startup-config | checkpoint <CHECKPOINT-NAME>}
```

Description

This command copies a specified configuration from the USB drive to the running configuration, to a startup configuration, or to a checkpoint.

Command context

Manager (#)

Parameters

<STORAGE-URL>

Specifies the name of a configuration file on the USB drive with the syntax: `usb:/<FILE>`

running-config

Specifies that the configuration file is copied to the running configuration. The file must be in CLI, JSON, or checkpoint format or the copy will fail. the copy will not work.

startup-config

Specifies that the configuration file is copied to the startup configuration. The switch stores this configuration between reboots. The startup configuration is used as the operating configuration following a reboot of the switch. The file must be in JSON or checkpoint format or the copy will fail.

checkpoint <CHECKPOINT-NAME>

Specifies the name of a new checkpoint file to receive a copy of the configuration. The configuration file on the USB drive must be in checkpoint format.



Do not start the checkpoint name with `CPC` because it is used for system-generated checkpoints.

Authority

Administrators or local user group members with execution rights for this command.

Usage

This command requires that the USB drive is formatted with the FAT file system and that the file be in the appropriate format as follows:

- `running-config`: This option requires the file on the USB drive be in CLI, JSON, or checkpoint format.
- `startup-config`: This option requires the file on the USB drive be in JSON or checkpoint format.
- `checkpoint <checkpoint-name>`: This option requires the file on the USB drive be in checkpoint format.

Examples

Copying the file **runCli** from the USB drive to the running configuration:

```
switch# copy usb:/runCli running-config
Configuration may take several minutes to complete according to configuration
file size
--0%-----10%-----20%-----30%-----40%-----50%-----60%-----70%-----80%-----90%-----100%--
Success
```

Copying the file **startUp** from the USB drive to the startup configuration:

```
switch# copy usb:/startUp startup-config
Success
```

Copying the file **testCheck** from the USB drive to the **abc** checkpoint:

```
switch# copy usb:/testCheck checkpoint abc
Success
```

erase {checkpoint <CHECKPOINT-NAME> | startup-config | all}

Syntax

```
erase {checkpoint <CHECKPOINT-NAME> | startup-config | all}
```

Description

Deletes an existing checkpoint, startup configuration, or all checkpoints.

Command context

Manager (#)

Parameters

checkpoint <CHECKPOINT-NAME>

Specifies the name of a checkpoint.

startup-config

Specifies the startup configuration.

all

Specifies all checkpoints.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Erasing checkpoint **ckpt1**:

```
switch# erase checkpoint ckpt1
```

Erasing the startup configuration:

```
switch# erase startup-config
```

Erasing all checkpoints:

```
switch# erase checkpoint all
```

show checkpoint <CHECKPOINT-NAME>

Syntax

```
show checkpoint <CHECKPOINT-NAME> [json]
```

Description

Shows the configuration of a checkpoint.

Command context

Manager (#)

Parameters

```
checkpoint <CHECKPOINT-NAME>
```

Specifies the name of a checkpoint.

```
[json]
```

Specifies that the output is displayed in JSON format.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Showing the configuration of the `ckpt1` checkpoint in non-JSON format:

```
switch# show checkpoint ckpt1
Checkpoint configuration:
!
!Version ArubaOS-CX PL.10.07.0000K-75-g55e5193
!export-password: default
lacp system-priority 65535
user admin group administrators password ciphertext
AQBapQjwipebv36io0jFfde7ZzrHckncal1D+3n8XFTZKQdmYgAAADetYOeHSme93xzdD0uz6Vr9Kl+XBzB+
2GB0UBxSF7rvGN2x8KSgkqv7iqXVQ0Te6LkSMnH4BdNaT3Bf25qyvOqmr4YakO1V3rg8zAOADkPktQD8joTH
XflzwomoIzcmv/uX
cli-session
    timeout 0
!
!
!
!
ssh server vrf default
vlan 1
spanning-tree
interface lag 1
    no shutdown
    vlan access 1
interface lag 128
    no shutdown
    vlan access 1
interface lag 129
    shutdown
    vlan access 1
    lacp mode active
interface 1/1/1
    no shutdown
    lag 128
    lacp port-id 65535
interface 1/1/2
    no shutdown
```

```

        vlan access 1
interface 1/1/3
    no shutdown
    vlan access 1
interface 1/1/4
    no shutdown
    vlan access 1
interface 1/1/5
    no shutdown
    vlan access 1
interface 1/1/6
    no shutdown
    vlan access 1
interface 1/1/7
    no shutdown
    vlan access 1
interface 1/1/8
    no shutdown
    vlan access 1
interface 1/1/9
    no shutdown
    vlan access 1
interface 1/1/10
    no shutdown
    vlan access 1
interface 1/1/11
    no shutdown
    vlan access 1
interface 1/1/12
    no shutdown
    vlan access 1
interface 1/1/13
    no shutdown
    vlan access 1
interface 1/1/14
    no shutdown
    vlan access 1
interface 1/1/15
    no shutdown
    vlan access 1
interface 1/1/16
    no shutdown
    vlan access 1
interface vlan 1
    ip dhcp
snmp-server vrf default
!
!
!
!
!
https-server vrf default

```

Showing the configuration of the `ckpt1` checkpoint in JSON format:

```

switch# show checkpoint ckpt1 json
Checkpoint configuration:
{
    "AAA_Server_Group": {
        "local": {
            "group_name": "local"

```

```

    },
    "none": {
        "group_name": "none"
    }
},
...
...
...
...

```

show checkpoint post-configuration

Syntax

```
show checkpoint post-configuration
```

Description

Shows the configuration settings for creating system checkpoints.

Command context

Manager (#)

Authority

Administrators or local user group members with execution rights for this command.

Examples

```

switch# show checkpoint post-configuration

Checkpoint Post-Configuration feature
-----

Status           : enabled
Timeout (sec)    : 300

```

show checkpoint list

Syntax

```
show checkpoint list {all | <START-DATE> <END-DATE>}
```

Description

Shows a list of saved checkpoints.

Command context

Manager (#)

Parameters

all

Shows a detailed list of all saved checkpoints.

<START-DATE>

Specifies the starting date for the range of saved checkpoints to show. Format: YYYY-MM-DD.

<END-DATE>

Specifies the ending date for the range of saved checkpoints to show. Format: YYYY-MM-DD.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Showing the list of saved checkpoints:

```
switch# show checkpoint list
ckpt1
ckpt2
ckpt3
startup-config
AUTO20170308214100
```

Showing a detailed list of all saved system and user checkpoints:

```
switch# show checkpoint list all
|NAME          |TYPE          |WRITER  |DATE (UTC)          |HARDWARE
|IMAGE VERSION |              |        |                    |
|CPC20171017195548 |checkpoint    |System  |2020-10-17T19:55:48Z|000000
|xx.10.0x.xxxxX-1-g7691be0|
|ckpt23         |checkpoint    |User    |2020-10-17T20:09:00Z|000000
|xx.10.0x.xxxxX-1-g7691be0|
|ckpt24         |checkpoint    |User    |2020-10-17T20:09:09Z|000000
|xx.10.0x.xxxxX-1-g7691be0|
|ckpt30         |checkpoint    |User    |2020-10-17T20:10:36Z|000000
|xx.10.0x.xxxxX-1-g7691be0|
|startup-config-backup|checkpoint    |System  |2020-10-17T20:17:56Z|000000
|xx.10.0x.xxxxX-1-g7691be0|
|CPC20171017201712 |checkpoint    |System  |2020-10-17T20:19:12Z|000000
|xx.10.0x.xxxxAZ-10-g4c6b4446bd6|
|ckpt31         |checkpoint    |User    |2020-10-17T20:19:24Z|000000
|xx.10.0x.xxxxAZ-10-g4c6b4446bd6|
|startup-config  |startup      |User    |2020-10-17T20:47:11Z|000000
|xx.10.0x.xxxxAZ-10-g4c6b4446bd6|
|ckpt32         |checkpoint    |User    |2020-10-17T20:50:24Z|000000
|xx.10.0x.xxxxAZ-10-g4c6b4446bd6|
|CPC20171017205110 |checkpoint    |System  |2020-10-17T20:51:10Z|000000
|xx.10.0x.xxxxAZ-10-g4c6b4446bd6|
```

Showing a detailed list of saved checkpoints for a specific date range:

```
switch# show checkpoint list date 2020-03-08 2020-03-12
|NAME |TYPE    |WRITER |DATE          |HARDWARE |IMAGE VERSION |
|ckpt2 |checkpoint |User   |2020-03-08 18:10:01 |          |0.0.0         |
|ckpt3 |checkpoint |User   |2020-03-09 23:11:02 |          |0.0.0         |
|ckpt4 |checkpoint |User   |2020-03-11 00:00:03 |          |0.0.0         |
```

write memory

Syntax

write memory

Description

Saves the running configuration to the startup configuration. It is an alias of the command `copy running-config startup-config`. If the startup configuration is already present, this command overwrites the startup configuration.

Command context

Manager (#)

Authority

Administrators or local user group members with execution rights for this command.

Examples

```
switch# write memory  
Success
```

Classifier policy application

Classifier policies can be applied as follows:

Policy type Direction	IPv4 In	IPv6 In
L2 interface (port)	Yes	Yes
L2 LAG	Yes	Yes
VLAN	Yes	Yes



Port policies and port-access client policies cannot be configured at the same time.

apply policy (Context: config)

Syntax

```
apply policy <POLICY-NAME> in
no apply policy <POLICY-NAME> in
```

Description

Applies a policy to the global config context.

Only one policy can be globally applied at a time. Applying a policy globally again, replaces the previous globally applied policy.

The `no` form of this command removes application of the global policy.

Command context

config

Parameters

<POLICY-NAME>

Specifies the policy to apply.

in

Selects the inbound (ingress) traffic direction.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Applying policy global1 to the global config context:

```
switch(config)# apply policy global1 in
```

Removing application of policy global1 from the global config context:

```
switch(config)# no apply policy global1 in
```

apply policy (Contexts: config-if, config-vlan)

Syntax

Context: config-if:

```
apply policy <POLICY-NAME> {in}  
no apply policy <POLICY-NAME> {in}
```

Context: config-vlan:

```
apply policy <POLICY-NAME> in  
  
no apply policy <POLICY-NAME> in
```

Description

Applies a policy to the current interface or VLAN context.

The `no` form of this command removes a policy from the interface or VLAN specified by the current context.

Command context

```
config-if  
config-vlan
```

Parameters

<POLICY-NAME>

Specifies the policy to apply.

in

Selects the inbound (ingress) traffic direction.

Authority

Administrators or local user group members with execution rights for this command.

Usage (applies to `config-vlan` context)

Only one policy may be applied to a VLAN at a time. Therefore, using the `apply policy` command on a VLAN with an already-applied policy of the same type, will replace the applied policy.

Examples

Applying a policy to an interface (ingress):

```
switch(config)# interface 1/1/1  
switch(config-if)# apply policy MY_POLICY in
```

Applying a policy to a VLAN (ingress):

```
switch(config)# vlan 10  
switch(config-vlan-10)# apply policy MY_POLICY in
```

Removing a policy from a VLAN (ingress):

```
switch(config)# vlan 10  
switch(config-vlan-10)# no apply policy MY_POLICY in
```

class copy

Syntax

```
class {ip|ipv6} <CLASS-NAME> copy <DESTINATION-CLASS>
```

Description

Copies a class to a new destination class or overwrites an existing class. Copying a class copies all entries as well.

Command context

config

Parameters

{ip|ipv6} <CLASS-NAME>

Specifies the type and name of the class to be copied.

<DESTINATION-CLASS>

Specifies the name of the destination class.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Copying an IPv4 class. Copying a class with entries copies all its entries as well:

```
switch(config)# class ip MY_IP_CLASS copy MY_IP_CLASS2  
switch(config)# do show class
```

Type	Sequence	Name	Comment	Action	Source IP Address	Destination IP Address	Additional Parameters	L3 Protocol
IPv4		MY_IP_CLASS						
	11	ignore						udp
		any						
	21	match						tcp
					192.168.0.1			
					192.168.0.2			
IPv4		MY_IP_CLASS2						
	11	ignore						udp

```

any
any
21 match                                tcp
192.168.0.1
192.168.0.2

```

Copying an IPv6 class:

```

switch(config)# class ipv6 MY_IPV6_CLASS copy MY_IPV6_CLASS2
switch(config)# do show class
Type          Name
Sequence      Comment
              Action                      L3 Protocol
              Source IP Address           Source L4 Port(s)
              Destination IP Address       Destination L4 Port(s)
              Additional Parameters
-----
IPv6          MY_IPV6_CLASS
2 ignore                                udp
any
any
-----
IPv6          MY_IPV6_CLASS2
2 ignore                                udp
any
any

```

class ip

Syntax

Syntax to create an IPv4 class and enter its context. Plus syntax to remove a class:

```
class ip <CLASS-NAME>
```

```
no class ip <CLASS-NAME>
```

Syntax (within the class context) for creating or removing class entries for protocols ah, gre, esp, igmp, ospf,

pim (ip is available as an alias for any):

```

[<SEQUENCE-NUMBER>]
{match|ignore}
{any|ip|ah|gre|esp|igmp|ospf|pim|<IP-PROTOCOL-NUM>}
{any|<SRC-IP-ADDRESS>[/<PREFIX-LENGTH>|<SUBNET-MASK>]]}
{any|<DST-IP-ADDRESS>[/<PREFIX-LENGTH>|<SUBNET-MASK>]]}
[dscp <DSCP-SPECIFIER>] [ip-precedence <IP-PRECEDENCE-VALUE>]
[tos <TOS-VALUE>] [fragment] [vlan <VLAN-ID>] [count]

```

```
no <SEQUENCE-NUMBER>
```

Syntax (within the class context) for creating or removing class entries for protocols sctp, tcp, udp:

```

[<SEQUENCE-NUMBER>]
{match|ignore}
{sctp|tcp|udp}
{any|<SRC-IP-ADDRESS>[/<PREFIX-LENGTH>|<SUBNET-MASK>]]}
[{eq|gt|lt} <PORT>|range <MIN-PORT> <MAX-PORT>]
{any|<DST-IP-ADDRESS>[/<PREFIX-LENGTH>|<SUBNET-MASK>]]}
[{eq|gt|lt} <PORT>|range <MIN-PORT> <MAX-PORT>]
[urg] [ack] [psh] [rst] [syn] [fin] [established]
[dscp <DSCP-SPECIFIER>] [ip-precedence <IP-PRECEDENCE-VALUE>]
[tos <TOS-VALUE>] [fragment] [vlan <VLAN-ID>] [count]

```

```
no <SEQUENCE-NUMBER>
```

Syntax (within the class context) for creating or removing class entries for protocol icmp:

```
[<SEQUENCE-NUMBER>]
{match|ignore}
icmp
{any|<SRC-IP-ADDRESS>[/ {<PREFIX-LENGTH>|<SUBNET-MASK>}] }
{any|<DST-IP-ADDRESS>[/ {<PREFIX-LENGTH>|<SUBNET-MASK>}] }
[icmp-type {echo|echo-reply|<ICMP-TYPE-VALUE>}] [icmp-code <ICMP-CODE-VALUE>]
[dscp <DSCP-SPECIFIER>] [ip-precedence <IP-PRECEDENCE-VALUE>]
[tos <TOS-VALUE>] [fragment] [vlan <VLAN-ID>] [count]

no <SEQUENCE-NUMBER>
```

Syntax (within the class context) for class entry comments:

```
[<SEQUENCE-NUMBER>] comment <TEXT-STRING>

no <SEQUENCE-NUMBER> comment
```

Description

Creates or modifies an IPv4 traffic class to match specified packets. Class is composed of one or more class entries ordered and prioritized by sequence numbers. With this command, the class can classify traffic based on IPv4 header information.

The `no` form of the command can be used to delete either an IPv4 traffic class (use `no` with the class command) or an individual IPv4 traffic class entry (use `no` with the sequence number).

Command context

config

The `class ip <CLASS-NAME>` command takes you into the `config-class-ip` context where you enter the class entries.

Parameters

ip

Specifies create or modify an IPv4 class.

<CLASS-NAME>

Specifies the name of this class.

<SEQUENCE-NUMBER>

Specifies a sequence number for the class entry. Optional. Range: 1-4294967295.

{match|ignore}

Creates a rule to match or ignore specified packets.

<IP-PROTOCOL-NUM>

Specifies the protocol as its Internet Protocol number. For example, 2 corresponds to the IGMP protocol. Range: 0 to 255.

{any|<SRC-IP-ADDRESS>[/ {<PREFIX-LENGTH>|<SUBNET-MASK>}] }

Specifies the source IPv4 address.

- any - specifies any source IPv4 address.
- <SRC-IP-ADDRESS> - specifies the source IPv4 host address.
 - <PREFIX-LENGTH> - specifies the address bits to mask (CIDR subnet mask notation). Range: 1 to 32.
 - <SUBNET-MASK> - specifies the address bits to mask (dotted decimal notation).

{any|<DST-IP-ADDRESS>[/ {<PREFIX-LENGTH>|<SUBNET-MASK>}] }

Specifies the destination IPv4 address.

- any - specifies any destination IPv4 address.

- `<DST-IP-ADDRESS>` - specifies the destination IPv4 host address.
 - `<PREFIX-LENGTH>` - specifies the address bits to mask (CIDR subnet mask notation). Range: 1 to 32.
 - `<SUBNET-MASK>` - specifies the address bits to mask (dotted decimal notation).

`[{eq|gt|lt} <PORT>|range <MIN-PORT> <MAX-PORT>]`

Specifies the port or port range. Port numbers are in the range of 0 to 65535.

- `eq <PORT>` - specifies the Layer 4 port.
- `gt <PORT>` - specifies any Layer 4 port greater than the indicated port.
- `lt <PORT>` - specifies any Layer 4 port less than the indicated port.
- `range <MIN-PORT> <MAX-PORT>` - specifies the Layer 4 port range.

`urg`

Specifies matching on the TCP Flag: Urgent.

`ack`

Specifies matching on the TCP Flag: Acknowledgment.

`psh`

Specifies matching on the TCP Flag: Push buffered data to receiving application.

`rst`

Specifies matching on the TCP Flag: Reset the connection.

`syn`

Specifies matching on the TCP Flag: Synchronize sequence numbers.

`fin`

Specifies matching on the TCP Flag: Finish connection.

`established`

Specifies matching on the TCP Flag: Established connection.

`dscp <DSCP-SPECIFIER>`

Specifies the Differentiated Services Code Point (DSCP), either a numeric `<DSCP-VALUE>` (0 to 63) or one of these keywords:

- `AF11` - DSCP 10 (Assured Forwarding Class 1, low drop probability)
- `AF12` - DSCP 12 (Assured Forwarding Class 1, medium drop probability)
- `AF13` - DSCP 14 (Assured Forwarding Class 1, high drop probability)
- `AF21` - DSCP 18 (Assured Forwarding Class 2, low drop probability)
- `AF22` - DSCP 20 (Assured Forwarding Class 2, medium drop probability)
- `AF23` - DSCP 22 (Assured Forwarding Class 2, high drop probability)
- `AF31` - DSCP 26 (Assured Forwarding Class 3, low drop probability)
- `AF32` - DSCP 28 (Assured Forwarding Class 3, medium drop probability)
- `AF33` - DSCP 30 (Assured Forwarding Class 3, high drop probability)
- `AF41` - DSCP 34 (Assured Forwarding Class 4, low drop probability)
- `AF42` - DSCP 36 (Assured Forwarding Class 4, medium drop probability)
- `AF43` - DSCP 38 (Assured Forwarding Class 4, high drop probability)
- `CS0` - DSCP 0 (Class Selector 0: Default)
- `CS1` - DSCP 8 (Class Selector 1: Scavenger)
- `CS2` - DSCP 16 (Class Selector 2: OAM)
- `CS3` - DSCP 24 (Class Selector 3: Signaling)

- CS4 - DSCP 32 (Class Selector 4: Realtime)
- CS5 - DSCP 40 (Class Selector 5: Broadcast video)
- CS6 - DSCP 48 (Class Selector 6: Network control)
- CS7 - DSCP 56 (Class Selector 7)
- EF - DSCP 46 (Expedited Forwarding)

`ip-precedence <IP-PRECEDENCE-VALUE>`

Specifies an IP precedence value. Range: 0 to 7.

`tos <TOS-VALUE>`

Specifies the Type of Service value. Range: 0 to 31.

`fragment`

Specifies a fragment packet.

`vlan <VLAN-ID>`

Specifies VLAN tag to match on. 802.1Q VLAN ID.



This parameter cannot be used in any class that will be applied to a VLAN.

`count`

Keeps the hit counts of the number of packets matching this class entry.

`[<SEQUENCE-NUMBER>] comment <TEXT-STRING>`

Adds a comment to a class entry. The `no` form removes only the comment from the class entry.

Authority

Administrators or local user group members with execution rights for this command.

Usage

- Entering an existing `<CLASS-NAME>` value will cause the existing class to be modified, with any new `<SEQUENCE-NUMBER>` value creating an additional class entry, and any existing `<SEQUENCE-NUMBER>` value replacing the existing class entry with the same sequence number.
- If no sequence number is specified, a new class entry will be appended to the end of the class with a sequence number equal to the highest class entry currently in the list plus 10.
- If the `<IP-PROTOCOL-NUM>` parameter is used instead of a protocol name, ensure that any needed class entry-definition parameters specific to the selected protocol are also provided.

Examples

Creating an IPv4 class with three entries:

```
switch(config)# class ip MY_IP_CLASS
switch(config-class-ip)# 10 match icmp any any
switch(config-class-ip)# 20 ignore udp any any
switch(config-class-ip)# 30 match tcp 192.168.0.1 192.168.0.2
switch(config-class-ip)# exit

switch(config)# do show class
Type          Name
Sequence      Comment
              Action          L3 Protocol
              Source IP Address Source L4 Port(s)
              Destination IP Address Destination L4 Port(s)
              Additional Parameters
-----
```

```

IPv4      MY_IP_CLASS
10 match                                     icmp
   any
   any
20 ignore                                     udp
   any
   any
30 match                                     tcp
   192.168.0.1
   192.168.0.2

```

Adding a comment to an existing IPv4 class entry:

```

switch(config)# class ip MY_IP_CLASS
switch(config-class-ip)# 30 comment myipClass
switch(config-class-ip)# exit

switch(config)# do show class
Type      Name
Sequence  Comment
          Action          L3 Protocol
          Source IP Address Source L4 Port(s)
          Destination IP Address Destination L4 Port(s)
          Additional Parameters
-----
IPv4      MY_IP_CLASS
10 match                                     icmp
   any
   any
20 ignore                                     udp
   any
   any
30 myipClass
   match                                     tcp
   192.168.0.1
   192.168.0.2

```

Removing a comment from an existing IPv4 class entry:

```

switch(config)# class ip MY_IP_CLASS
switch(config-class-ip)# no 30 comment
switch(config-class-ip)# exit

switch(config)# do show class
Type      Name
Sequence  Comment
          Action          L3 Protocol
          Source IP Address Source L4 Port(s)
          Destination IP Address Destination L4 Port(s)
          Additional Parameters
-----
IPv4      MY_IP_CLASS
10 match                                     icmp
   any
   any
20 ignore                                     udp
   any
   any
30 match                                     tcp

```

```
192.168.0.1
192.168.0.2
```

Replacing an IPv4 class entry in an existing class:

```
switch(config)# class ip MY_IP_CLASS
switch(config-class-ip)# 10 match igmp any any
switch(config-class-ip)# exit

switch(config)# do show class
Type      Name
Sequence Comment
          Action          L3 Protocol
          Source IP Address Source L4 Port(s)
          Destination IP Address Destination L4 Port(s)
          Additional Parameters
-----
IPv4      MY_IP_CLASS
          10 match          igmp
             any
             any
          20 ignore          udp
             any
             any
          30 match          tcp
             192.168.0.1
             192.168.0.2
```

Removing an IPv4 class entry:

```
switch(config)# class ip MY_IP_CLASS
switch(config-class-ip)# no 10
switch(config-class-ip)# exit

switch(config)# do show class
Type      Name
Sequence Comment
          Action          L3 Protocol
          Source IP Address Source L4 Port(s)
          Destination IP Address Destination L4 Port(s)
          Additional Parameters
-----
IPv4      MY_IP_CLASS
          20 ignore          udp
             any
             any
          30 match          tcp
             192.168.0.1
             192.168.0.2
```

Removing an IPv4 class. Removing a class with entries removes all its entries as well. If a class associated with a policy entry (or multiple policy entries) is removed, the corresponding entries are also removed.



The corresponding entries are only removed if the class is unused by all policy entries.

```
switch(config)# no class ip MY_IP_CLASS

switch(config)# do show class
No Class found.
```

class ipv6

Syntax

Syntax to create an IPv6 class and enter its context. Plus syntax to remove a class:

```
class ipv6 <CLASS-NAME>
no class ipv6 <CLASS-NAME>
```

Syntax (within the class context) for creating or removing class entries for protocols ah, gre, esp, igmp, ospf, pim (ipv6 is available as an alias for any):

```
[<SEQUENCE-NUMBER>]
{match|ignore}
{any|ipv6|ah|gre|esp|igmp|ospf|pim|<IP-PROTOCOL-NUM>}
{any|<SRC-IP-ADDRESS>[/ {<PREFIX-LENGTH>|<SUBNET-MASK>}]}
{any|<DST-IP-ADDRESS>[/ {<PREFIX-LENGTH>|<SUBNET-MASK>}]}
[dscp <DSCP-SPECIFIER>] [ip-precedence <IP-PRECEDENCE-VALUE>]
[tos <TOS-VALUE>] [fragment] [vlan <VLAN-ID>] [count]

no <SEQUENCE-NUMBER>
```

Syntax (within the class context) for creating or removing class entries for protocols sctp, tcp, udp:

```
[<SEQUENCE-NUMBER>]
{match|ignore}
{sctp|tcp|udp}
{any|<SRC-IP-ADDRESS>[/ {<PREFIX-LENGTH>|<SUBNET-MASK>}]}
[{eq|gt|lt} <PORT>|range <MIN-PORT> <MAX-PORT>]
{any|<DST-IP-ADDRESS>[/ {<PREFIX-LENGTH>|<SUBNET-MASK>}]}
[{eq|gt|lt} <PORT>|range <MIN-PORT> <MAX-PORT>]
[urg] [ack] [psh] [rst] [syn] [fin] [established]
[dscp <DSCP-SPECIFIER>] [ip-precedence <IP-PRECEDENCE-VALUE>]
[tos <TOS-VALUE>] [fragment] [vlan <VLAN-ID>] [count]

no <SEQUENCE-NUMBER>
```

Syntax (within the class context) for creating or removing class entries for protocol icmpv6:

```
[<SEQUENCE-NUMBER>]
{permit|deny}
{icmpv6}
{any|<SRC-IP-ADDRESS>[/ {<PREFIX-LENGTH>|<SUBNET-MASK>}]}
{any|<DST-IP-ADDRESS>[/ {<PREFIX-LENGTH>|<SUBNET-MASK>}]}
[icmp-type {echo|echo-reply|<ICMP-TYPE-VALUE>}] [icmp-code <ICMP-CODE-VALUE>]
[dscp <DSCP-SPECIFIER>] [ip-precedence <IP-PRECEDENCE-VALUE>]
[tos <TOS-VALUE>] [fragment] [vlan <VLAN-ID>] [count]

no <SEQUENCE-NUMBER>
```

Syntax (within the class context) for class entry comments:

```
[<SEQUENCE-NUMBER>] comment <TEXT-STRING>

no <SEQUENCE-NUMBER> comment
```

Description

Creates or modifies an IPv6 traffic class to match specified packets. Class is composed of one or more class entries ordered and prioritized by sequence numbers. With this command, each class can classify traffic based on IPv6 header information.

The `no` form of the command deletes either an IPv6 traffic class (use `no` with the `class` command) or an individual IPv6 traffic class entry (use `no` with the sequence number).

Command context

`config`

The `class ipv6 <CLASS-NAME>` command takes you into the `config-class-ipv6` command context where you enter the class entries.

Parameters

`ipv6`

Specifies create or modify an IPv6 class.

`<CLASS-NAME>`

Specifies the name of this class.

`<SEQUENCE-NUMBER>`

Specifies a sequence number for the class entry. Optional. Range: 1-4294967295.

`{match|ignore}`

Creates a rule to match or ignore specified packets.

`<IP-PROTOCOL-NUM>`

Specifies the protocol as its Internet Protocol number. For example, 2 corresponds to the IGMP protocol. Range: 0 to 255.

`{any|<SRC-IP-ADDRESS>[/ {<PREFIX-LENGTH>|<SUBNET-MASK>}]}`

Specifies the source IPv6 address.

- `any` - specifies any source IPv6 address.
- `<SRC-IP-ADDRESS>` - specifies the source IPv4 host address.
 - `<PREFIX-LENGTH>` - specifies the address bits to mask (CIDR subnet mask notation). Range: 1 to 32.
 - `<SUBNET-MASK>` - specifies the address bits to mask (dotted decimal notation).

`{any|<DST-IP-ADDRESS>[/ {<PREFIX-LENGTH>|<SUBNET-MASK>}]}`

Specifies the destination IPv4 address.

- `any` - specifies any destination IPv6 address.
- `<DST-IP-ADDRESS>` - specifies the destination IPv6 host address.
 - `<PREFIX-LENGTH>` - specifies the address bits to mask (CIDR subnet mask notation). Range: 1 to 32.
 - `<SUBNET-MASK>` - specifies the address bits to mask (dotted decimal notation).

`[(eq|gt|lt) <PORT>|range <MIN-PORT> <MAX-PORT>]`

Specifies the port or port range. Port numbers are in the range of 0 to 65535.

- `eq <PORT>` - specifies the Layer 4 port.
- `gt <PORT>` - specifies any Layer 4 port greater than the indicated port.
- `lt <PORT>` - specifies any Layer 4 port less than the indicated port.
- `range <MIN-PORT> <MAX-PORT>` - specifies the Layer 4 port range.

`urg, ack, psh, rst, syn, fin, established`

These TCP flag matching parameters are not supported.

`dscp <DSCP-SPECIFIER>`

Specifies the Differentiated Services Code Point (DSCP), either a numeric `<DSCP-VALUE>` (0 to 63) or one of these keywords:

- `AF11` - DSCP 10 (Assured Forwarding Class 1, low drop probability)

- AF12 - DSCP 12 (Assured Forwarding Class 1, medium drop probability)
- AF13 - DSCP 14 (Assured Forwarding Class 1, high drop probability)
- AF21 - DSCP 18 (Assured Forwarding Class 2, low drop probability)
- AF22 - DSCP 20 (Assured Forwarding Class 2, medium drop probability)
- AF23 - DSCP 22 (Assured Forwarding Class 2, high drop probability)
- AF31 - DSCP 26 (Assured Forwarding Class 3, low drop probability)
- AF32 - DSCP 28 (Assured Forwarding Class 3, medium drop probability)
- AF33 - DSCP 30 (Assured Forwarding Class 3, high drop probability)
- AF41 - DSCP 34 (Assured Forwarding Class 4, low drop probability)
- AF42 - DSCP 36 (Assured Forwarding Class 4, medium drop probability)
- AF43 - DSCP 38 (Assured Forwarding Class 4, high drop probability)
- CS0 - DSCP 0 (Class Selector 0: Default)
- CS1 - DSCP 8 (Class Selector 1: Scavenger)
- CS2 - DSCP 16 (Class Selector 2: OAM)
- CS3 - DSCP 24 (Class Selector 3: Signaling)
- CS4 - DSCP 32 (Class Selector 4: Real time)
- CS5 - DSCP 40 (Class Selector 5: Broadcast video)
- CS6 - DSCP 48 (Class Selector 6: Network control)
- CS7 - DSCP 56 (Class Selector 7)
- EF - DSCP 46 (Expedited Forwarding)

`ip-precedence <IP-PRECEDENCE-VALUE>`

Specifies an IP precedence value. Range: 0 to 7.

`tos <TOS-VALUE>`

Specifies the Type of Service value. Range: 0 to 31.

`fragment`

Specifies a fragment packet.

`vlan <VLAN-ID>`

Specifies VLAN tag to match on. 802.1Q VLAN ID.



This parameter cannot be used in any class that will be applied to a VLAN.

`count`

Keeps the hit counts of the number of packets matching this class entry.

`[<SEQUENCE-NUMBER>] comment <TEXT-STRING>`

Adds a comment to a class entry. The `no` form removes only the comment from the class entry.

Authority

Administrators or local user group members with execution rights for this command.

Usage

- If you enter an existing `<CLASS-NAME>` value, the existing class is modified with any new `<SEQUENCE-NUMBER>` value. This action creates an additional class entry. Any existing `<SEQUENCE-NUMBER>` value replaces the existing class entry with the same sequence number.
- If no sequence number is specified, a new class entry is appended to the end of the class with a sequence number equal to the highest class entry currently in the list plus 10.

- If the `<IP-PROTOCOL-NUM>` parameter is used instead of a protocol name, ensure that any needed class entry-definition parameters specific to the selected protocol are also provided.

Examples

Creating an IPv6 class with two entries:

```
switch(config)# class ipv6 MY_IPV6_CLASS
switch(config-class-ipv6)# 10 match icmpv6 any any
switch(config-class-ipv6)# 20 ignore udp any any
switch(config-class-ipv6)# exit

switch(config)# do show class
Type      Name
Sequence  Comment
          Action          L3 Protocol
          Source IP Address Source L4 Port(s)
          Destination IP Address Destination L4 Port(s)
          Additional Parameters
-----
IPv6      MY_IPV6_CLASS
          10 match          icmpv6
          any
          any
          20 ignore          udp
          any
          any
```

Adding a comment to an existing IPv6 class entry:

```
switch(config)# class ipv6 MY_IPV6_CLASS
switch(config-class-ipv6)# 10 match icmpv6 any any
switch(config-class-ipv6)# 20 ignore udp any any
switch(config-class-ipv6)# 20 comment myipv6class
switch(config-class-ipv6)# exit

switch(config)# do show class
Type      Name
Sequence  Comment
          Action          L3 Protocol
          Source IP Address Source L4 Port(s)
          Destination IP Address Destination L4 Port(s)
          Additional Parameters
-----
IPv6      MY_IPV6_CLASS
          10 match          icmpv6
          any
          any
          20 myipv6class
          ignore          udp
          any
          any
```

Removing a comment from an existing IPv6 class entry:

```
switch(config)# class ipv6 MY_IPV6_CLASS
switch(config-class-ipv6)# no 20 comment
switch(config-class-ipv6)# exit
```

```

switch(config)# do show class
Type      Name
Sequence  Comment
          Action                L3 Protocol
          Source IP Address      Source L4 Port(s)
          Destination IP Address Destination L4 Port(s)
          Additional Parameters
-----
IPv6      MY_IPV6_CLASS
          10 match                icmpv6
             any
             any
          20 ignore                udp
             any
             any

```

Replacing an IPv6 class entry in an existing IPv6 class:

```

switch(config)# class ipv6 MY_IPV6_CLASS
switch(config-class-ipv6)# 10 match any any 1020::
switch(config-class-ipv6)# exit

switch(config)# do show class
Type      Name
Sequence  Comment
          Action                L3 Protocol
          Source IP Address      Source L4 Port(s)
          Destination IP Address Destination L4 Port(s)
          Additional Parameters
-----
IPv6      MY_IPV6_CLASS
          10 match                any
             any
             1020::
          20 ignore                udp
             any
             any

```

Removing an IPv6 class entry:

```

switch(config)# class ipv6 MY_IPV6_CLASS
switch(config-class-ipv6)# no 10
switch(config-class-ipv6)# exit

switch(config)# do show class
Type      Name
Sequence  Comment
          Action                L3 Protocol
          Source IP Address      Source L4 Port(s)
          Destination IP Address Destination L4 Port(s)
          Additional Parameters
-----
IPv6      MY_IPV6_CLASS
          20 ignore                udp
             any
             any

```

Removing an IPv6 class. Removing a class with entries removes all its entries as well. If a class associated with a policy entry (or multiple policy entries) is removed, the corresponding entries are also removed.



The corresponding entries are only removed if the class is unused by all policy entries.

```
switch(config)# no class ipv6 MY_IPV6_CLASS

switch(config)# do show class
No Class found.
```

class resequence

Syntax

```
class {ip|ipv6} <CLASS-NAME> resequence <STARTING-SEQUENCE-NUMBER> <INCREMENT>
```

Description

Resequencing numbering in an IPv4, or IPv6 class.

Command context

config

Parameters

{ip|ipv6} <CLASS-NAME>

Specifies the class where you want to resequence class entries.

<STARTING-SEQUENCE-NUMBER>

Specifies the sequence number to start resequencing from.

<INCREMENT>

Specifies how much to increment the sequence numbers by.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Resequencing an IPv4 class:

```
switch(config)# class ip MY_IP_CLASS resequence 1 10
switch(config)# do show class
Type      Name
Sequence  Comment
          Action          L3 Protocol
          Source IP Address Source L4 Port(s)
          Destination IP Address Destination L4 Port(s)
          Additional Parameters
-----
IPv4      MY_IP_CLASS
          1 match          igmp
          any
          any
          11 ignore        udp
          any
          any
          21 match          tcp
```

```
192.168.0.1
192.168.0.2
```

Resequencing an IPv6 class:

```
switch(config)# class ipv6 MY_IPV6_CLASS resequence 1 1
switch(config-class-ipv6)# exit
switch(config)# do show class
```

Type	Name	Sequence	Comment	Action	L3 Protocol	Source IP Address	Source L4 Port(s)	Destination IP Address	Destination L4 Port(s)	Additional Parameters
IPv6	MY_IPV6_CLASS									
		1	match		any					
					1020::					
		2	ignore		udp					
					any					
					any					

class reset

Syntax

```
class { all | ip <CLASS-NAME> | ipv6 <CLASS-NAME>} reset
```

Description

Changes the user-specified class configuration to match the active class configuration. Use this command when there is a discrepancy between what the user configured and what is active and accepted by the system.

Command context

config

Parameters

```
{ all | ip <CLASS-NAME> | ipv6 <CLASS-NAME>}
```

Specifies either `all` classes be reset or specifies the type (`ip` for IPv4, or `ipv6` for IPv6) and name of the class to be reset.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Resetting the user-specified configuration to the active configuration:

```
switch(config)# class all reset
```

clear policy hitcounts

Syntax

Description

Clears the policy hit count statistics.

Command context

Operator (>) or Manager (#)

Parameters

`all`

Selects all policies.

`<POLICY-NAME>`

Specifies the policy name.

`interface <IF-NAME>`

Specifies the interface name.

`vlan <VLAN-ID>`

Specifies the VLAN.

`in`

Specifies the inbound (ingress) traffic direction.

`global`

Selects the globally applied policy.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Clearing hit counts for policy `MY_IPv6_Policy` applied to VLAN 10 (ingress):

```
switch# clear policy hitcounts My_IPv6_Policy vlan 10 in
```

Clearing hit counts for all policies:

```
switch# clear policy hitcounts all
```

policy

Syntax

`policy <POLICY-NAME>`

```
[<SEQUENCE-NUMBER>]
class {ip|ipv6} <CLASS-NAME>
    action {<REMARK-ACTIONS> | <POLICE-ACTIONS> | <OTHER-ACTIONS>}
    [{<REMARK-ACTIONS> | <POLICE-ACTIONS> | <OTHER-ACTIONS>}]

[<SEQUENCE-NUMBER>]
comment ...
```

Description

Creates or modifies classifier policy and policy entries. A policy is made up of one or more policy entries ordered and prioritized by sequence numbers. Each entry has an IPv4/IPv6 class and zero or more policy actions associated with it.

A policy must be applied using the `apply` command.

The `no` form of the command can be used to delete either a policy (use `no` with the `policy` command) or an individual policy entry (use `no` with the sequence number).

Command context

`config`

The `policy` command takes you into the `config-policy` context where you enter the policy entries.

Parameters

`<POLICY-NAME>`

Specifies the name of the policy.

`<SEQUENCE-NUMBER>`

Specifies a sequence number for the policy entry. Optional. Range: 1 to 4294967295.

`comment`

Stores the remaining entered text as a policy entry comment.

`class {ip|ipv6} <CLASS-NAME>`

Specifies a type of class, `ip` for IPv4, `ipv6` for IPv6.

`<REMARK-ACTIONS>`

Remark actions can be any of the following options: `{cos <COS-VALUE> | ip-precedence <IP-PRECEDENCE-VALUE> | dscp <DSCP-VALUE>}` where:

`cos <COS-VALUE>`

Specifies the Class of Service (CoS) value.

`ip-precedence <IP-PRECEDENCE-VALUE>`

Specifies the numeric IP precedence value. Range: 0 to 7.

`dscp <DSCP-VALUE>`

Specifies a Differentiated Services Code Point (DSCP) value. Enter either a numeric value (0 to 63) or a keyword as follows:

- AF11 - DSCP 10 (Assured Forwarding Class 1, low drop probability)
- AF12 - DSCP 12 (Assured Forwarding Class 1, medium drop probability)
- AF13 - DSCP 14 (Assured Forwarding Class 1, high drop probability)
- AF21 - DSCP 18 (Assured Forwarding Class 2, low drop probability)
- AF22 - DSCP 20 (Assured Forwarding Class 2, medium drop probability)
- AF23 - DSCP 22 (Assured Forwarding Class 2, high drop probability)
- AF31 - DSCP 26 (Assured Forwarding Class 3, low drop probability)
- AF32 - DSCP 28 (Assured Forwarding Class 3, medium drop probability)
- AF33 - DSCP 30 (Assured Forwarding Class 3, high drop probability)
- AF41 - DSCP 34 (Assured Forwarding Class 4, low drop probability)
- AF42 - DSCP 36 (Assured Forwarding Class 4, medium drop probability)
- AF43 - DSCP 38 (Assured Forwarding Class 4, high drop probability)
- CS0 - DSCP 0 (Class Selector 0: Default)
- CS1 - DSCP 8 (Class Selector 1: Scavenger)
- CS2 - DSCP 16 (Class Selector 2: OAM)
- CS3 - DSCP 24 (Class Selector 3: Signaling)

- CS4 - DSCP 32 (Class Selector 4: Real time)
- CS5 - DSCP 40 (Class Selector 5: Broadcast video)
- CS6 - DSCP 48 (Class Selector 6: Network control)
- CS7 - DSCP 56 (Class Selector 7)
- EF - DSCP 46 (Expedited Forwarding)

<POLICE-ACTIONS>

Police actions can be the following {cir <RATE-BPS> exceed} where:

cir kbps <RATE-KBPS>

Specifies a Committed Information Rate value in Kilobits per second. Range: 1 to 4294967295.
exceed

Specifies action to take on packets that exceed the rate limit.

<OTHER-ACTIONS>

Other actions can be the following:

drop

Specifies drop traffic.

Authority

Administrators or local user group members with execution rights for this command.

Usage

- An applied policy will process a packet sequentially against policy entries in the list until the last policy entry in the list has been evaluated or the packet matches an entry.
- Entering an existing <POLICY-NAME> value will cause the existing policy to be modified, with any new <SEQUENCE-NUMBER> value creating an additional policy entry, and any existing <SEQUENCE-NUMBER> value replacing the existing policy entry with the same sequence number.
- If no sequence number is specified, a new policy entry will be appended to the end of the entry list with a sequence number equal to the highest policy entry currently in the list plus 10.

Examples

Creating a policy with several entries:

```
switch(config)# policy MY_POLICY
switch(config-policy)# 10 class ipv6 MY_CLASS1 action dscp af21 action drop
switch(config-policy)# 20 class ip MY_CLASS3 action mirror 1
switch(config-policy)# exit
switch(config)# do show policy
```

Sequence	Name	Comment	Class	Type	action

	MY_POLICY				
10	MY_CLASS1	ipv6		drop	dscp AF21
20	MY_CLASS3	ipv4		mirror	1

Adding a comment to an existing policy entry:

```

switch(config)# policy MY_POLICY
switch(config-policy)# 20 comment MY_TEST_POLICY
switch(config-policy)# exit
switch(config)# do show policy
      Name
Sequence Comment
      Class Type
              action
-----
      MY_POLICY
10      MY_CLASS1 ipv6
              drop
              dscp AF21

20 MY_TEST_POLICY
      MY_CLASS3 ipv4
              mirror 1

```

Removing a comment from an existing policy entry:

```

switch(config)# policy MY_POLICY
switch(config-policy)# no 20 comment
switch(config-policy)# exit
switch(config)# do show policy
      Name
Sequence Comment
      Class Type
              action
-----
      MY_POLICY
10      MY_CLASS1 ipv6
              drop
              dscp AF21

20
      MY_CLASS3 ipv4
              mirror 1

```

Adding/Replacing a policy entry in an existing policy:

```

switch(config)# policy MY_POLICY
switch(config-policy)# 10 class ip MY_CLASS3 action drop action dscp af21
switch(config-policy)# exit
switch(config)# do show policy
      Name
Sequence Comment
      Class Type
              action
-----
      MY_POLICY
10      MY_CLASS3 ipv4
              drop
              dscp AF21

20

```

```
MY_CLASS3 ipv4
    mirror 1
```

Removing a policy entry:

```
switch(config)# policy MY_POLICY
switch(config-policy)# no 10
switch(config-policy)# exit
switch(config)# do show policy
      Name
Sequence Comment
      Class Type
              action
-----
      MY_POLICY
20
      MY_CLASS3 ipv4
              mirror 1
```

Removing a policy:

```
switch(config)# no policy MY_POLICY
switch(config)# do show policy
      Name
Sequence Comment
      Class Type
              action
-----
      MY_POLICY2
2
      MY_CLASS3 ipv4
              mirror 1
```

policy copy

Syntax

```
policy <POLICY-NAME> copy <DESTINATION-POLICY>
```

Description

Copies a policy to a new destination policy or overwrites an existing policy. Copying a policy copies all its entries as well.

Command context

config

Parameters

<POLICY-NAME>

Specifies the policy to be copied.

<DESTINATION-POLICY>

Specifies the name of the destination policy.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Copying a policy:

```
switch(config)# policy MY_POLICY copy MY_POLICY2
switch(config)# do show policy
      Name
Sequence Comment
      Class Type
              action
-----
      MY_POLICY
      2
      my_class3 ipv4
              mirror 1
-----
      MY_POLICY2
      2
      my_class3 ipv4
              mirror 1
```

policy resequence

Syntax

```
policy <POLICY-NAME> resequence <STARTING-SEQ-NUM> <INCREMENT>
```

Description

Resequences numbering in a policy.

Command context

config

Parameters

<POLICY-NAME>

Specifies the policy where you want to resequence policy entries.

<STARTING-SEQ-NUM>

Specifies the sequence number to start resequencing from.

<INCREMENT>

Specifies how much to increment the sequence numbers by.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Resequencing a policy:

```
switch(config)# policy MY_POLICY resequence 1 1
switch(config)# do show policy
      Name
Sequence Comment
```



```

      Class Type
      action
-----
      MY_POLICY
1      MY_CLASS3 ipv4
      drop
      dscp AF21

2
      MY_CLASS3 ipv4
      mirror 1

```

policy reset

Syntax

```
policy <POLICY-NAME> reset
```

Description

Changes the user-specified policy configuration to match the active policy configuration. Use this command when a discrepancy exists between what the user configured and what is active and accepted by the system.

Command context

```
config
```

Parameters

<POLICY-NAME>

Specifies the policy to be reset.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Resetting a policy:

```
switch(config)# policy MY_POLICY reset
```

show class

Syntax

```
show class [ip | ipv6] [<CLASS-NAME>] [commands] [configuration]
```

Description

Shows class configuration information.

Command context

Operator (>) or Manager (#)

Parameters

All parameters are optional.

[ip | ipv6]

Selects the class type for the display: ip for IPv4, ipv6 for IPv6.

<CLASS-NAME>

Specifies the class name.

commands

Specifies whether to display output as the CLI commands showing the configured class entries.

configuration

Specifies whether to display classes that have been configured by the user, even if they are not active due to issues with the command parameters or hardware issues. This parameter is useful during a mismatch between the entered configuration and the previous successfully programmed (active) classes.

Authority

Operators or Administrators or local user group members with execution rights for this command.

Operators can execute this command from the operator context (>) only.

Examples

Showing all class configuration:

```
switch# show class
Type Name
Sequence Comment
          action          L3 Protocol
          Source IP address Source L4 Port(s)
          Destination IP address Destination L4 Port(s)
          Additional Parameters
-----
ipv4 MY_IPV4_CLASS
  10 my first class entry comment
    match          icmp
    192.168.0.1/255.255.255.0
    192.168.1.1/255.255.255.0
    VLAN: 1
  20 my second class entry comment
    ignore          tcp
    10.100.0.10/255.255.255.0 < 3000
    10.100.1.10/255.255.255.0 > 2000
    VLAN: 1
-----
```

Showing class configuration for the IPv4 class MY_IPV4_CLASS as CLI commands:

```
switch# show class ip MY_IPV4_CLASS commands
class ip "MY_IPV4_CLASS"
  10 match icmp 192.168.0.1/255.255.255.0 192.168.1.1/255.255.255.0 vlan 1
  10 comment my first class entry comment
  20 ignore tcp 10.100.0.10/255.255.255.0 lt 3000 10.100.1.10/255.255.255.0 gt
    2000 vlan 1
  20 comment my second class entry comment
```

show policy

Syntax

Syntax that shows information for all policies:

```
show policy [commands] [configuration]
```

Syntax that filters by policies applied to an interface or VLAN:

```
show policy [interface <IF-NAME> [in] | vlan <VLAN-ID> [in]]  
           [commands] [configuration]
```

Syntax that filters by the named policy:

```
show policy <POLICY-NAME> [commands] [configuration]
```

Syntax that filters by the globally applied policy:

```
show policy global [commands] [configuration]
```

Syntax that shows statistical information in the form of hit counts:

```
show policy hitcounts <POLICY-NAME> [interface <IF-NAME> [in] |  
                                   vlan <VLAN-ID> [in]]
```

Syntax that shows statistical information in the form of hit counts for the globally applied policy:

```
show policy hitcounts global
```

Description

Shows information about your defined policies and where they have been applied. When `show policy` is entered without parameters, information for all policies is shown. The parameters filter the list of policies for which information is shown.

Available filtering includes:

- The content of a specific policy.
- All policies applied to a specific interface.
- All policies applied to a specific VLAN.
- The globally applied policy.

To display policy statistics, use the `show policy hitcounts` form of this command.

Command context

Operator (>) or Manager (#)

Parameters

`interface <IF-NAME>`

Specifies the interface name.

`vlan <VLAN-ID>`

Specifies the VLAN.

`in`

Selects the inbound (ingress) traffic direction.

`<POLICY-NAME>`

Specifies the policy name.

`commands`

Causes the policy definition to be shown as the commands and parameters used to create it rather than in tabular form.

`configuration`

Causes the user-configured policies be shown as entered, even if the policies are not active due to policy-definition command issues or hardware issues. This parameter is useful if there is a mismatch between the entered configuration and the previous successfully programmed (active) policies configuration.

`global`

Selects the globally applied policy.

`hitcounts`

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Examples

Showing information for all policies:

```
switch# show policy
      Name
Sequence Comment
      Class Type
              action
-----
      my_policy
10 This is my QOS class.
   class1 ipv4
              dscp af21
              drop
-----
```

Showing a policy as commands:

```
switch# show policy commands
policy my_policy
      10 class ip my_class1 action dscp af21 action drop
```

Showing the globally applied policy:

```
switch# show policy global commands
policy global1
      10 class ip my_class1 action drop
apply policy my_policy in
```

Showing policy hit counts (statistics):

```
switch# show policy hitcounts my_policy
Statistics for Policy my_policy:
Interface 1/1/1* (in):
      Hit Count Configuration
10 class ip my_class1 action dscp af21 action drop
      20 10 match any any any count
* policy statistics are shared among each context type (interface, VLAN).
  Use 'policy NAME copy' to create a new policy for separate statistics.
```

Showing policy hit counts (statistics) for the globally applied policy:

```
switch# show policy hitcounts global
Statistics for Policy global1:
Global Policy:
      Hit Count Configuration
10 class ip my_class1 action mirror
      20 10 match any any any count
* policy statistics are shared among each context type (interface, vlan, VRF).
  use 'policy NAME copy' to create a uniquely-named policy
```

Showing policy hit counts with conform rate from my_policy applied inbound:

```
switch# show policy hitcounts my_policy
Statistics for Policy my_policy:
Interface 1/1/1* (in):
    Hit Count  Configuration
10 class ipv6 my_class1 action cir kbps 1024 exceed drop [1024 kbps conform ]
    - 10 match any any any
* policy statistics are shared among each context type (interface, VLAN).
  Use 'policy NAME copy' to create a new policy for separate statistics.
```

alias

Syntax

```
alias <ALIAS-NAME> <COMMAND-STRING>
no alias <ALIAS-NAME>
```

Description

Defines an alias for one or more CLI commands. The alias and its definition are valid only for the user that creates the alias.



The alias name cannot be an existing token name.

The `no` form of this command removes the specified alias.

Command context

config

Parameters

<ALIAS-NAME>

Specifies the name of the alias you are defining.

<COMMAND-STRING>

Specifies one or more commands and their parameters. Separate commands with a semicolon (;).
Length: 1 to 400 characters.

For commands that require user-supplied parameters, use `$1` through `$n`, in order, as placeholders. These parameters are replaced by the corresponding arguments from the command line, and must match the number of parameters required by the original command. For alias definitions that include multiple commands, continue numbering parameters through all commands. Do not restart numbering for each command.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Defining an alias:

```
switch(config)# alias srci show running-config interface $1
switch(config)# srci?
  shv  Execute "show alias" to list the command list. Arguments to replace $1,
       $2 etc.
switch(config)# show alias
Alias Name           Alias Definition
```

```
-----
srci                               show running-config interface $1
-----
```

Using `alias` in config context:

```
switch(config)# srci 1/1/1
interface 1/1/1
  no shutdown
  ip address 1.1.1.1/24
exit
```

Using `alias` in enable context:

```
switch# srci 1/1/1
interface 1/1/1
  no shutdown
  ip address 1.1.1.1/24
exit
```

Using `alias` in operator context:

```
switch> srci 1/1/1
interface 1/1/1
  no shutdown
  ip address 1.1.1.1/24
exit
```

Removing an alias:

```
switch(config)# no alias srci
switch(config)# show alias
Alias Name                               Alias Definition
-----
```

auto-confirm

Syntax

```
auto-confirm
no auto-confirm
```

Description

Specifies that the CLI automatically enters the affirmative response (`y`) to all confirmation prompts, enabling commands to execute without waiting for user confirmation.

The `no` form of this command sets auto-confirmation to the default value disabled.

Command context

Manager (#)

Authority

Administrators or local user group members with execution rights for this command.

Usage

Some commands, such as `boot` command, prompt to confirm execution of the command or to save the current configuration. Typically, such commands display a confirmation message similar to the following:

```
Continue (y/n)?
```

This command is useful for automating switch configuration, but Hewlett Packard Enterprise recommends that you use the REST API instead of using CLI scripts to automate configuration operations.

When the switch reboots, auto-confirmation is set to the default (disabled).

Example

```
switch# auto-confirm
```

configure terminal

Syntax

```
configure [terminal]
```

Description

Enters the global configuration (`config`) context.

Command context

Manager (#)

Parameters

terminal

Configure from the terminal. This is the default parameter.

Authority

Administrators or local user group members with execution rights for this command.

Example

```
switch# configure terminal
switch(config)#
```

disable

Syntax

```
disable
```

Description

Exits the manager context (#) and enters the operator context (>).

Command context

Manager (#)

Authority

Administrators or local user group members with execution rights for this command.

Example

```
switch# disable  
switch>
```

do

Syntax

do <COMMAND>

Description

Executes a manager context (#) or operator context (>) command from a configuration (config) context.



You can execute `exec` commands from a configuration context with or without using the `do` command.

Command context

A configuration context such as `config` or `config-vlan-10`

Parameters

<COMMAND>

Specifies the manager context (#) or the operator context (>) command to execute.

Authority

Administrators or local user group members with execution rights for this command.

Usage

You can execute `exec` commands from a configuration context with or without using the `do` command.

Use the `do` command to execute commands such as `clear`, `checkpoint`, `auto-confirm` and `show` commands while you are in a configuration context (such as `config` or `config-vlan-10`).

For all `exec` commands you can use with the `do` command, from the global configuration context (`config`), enter `do`, followed by a space, and then press the tab key twice.

Examples

Clearing LLDP neighbors from the global configuration context:

```
switch(config)# clear lldp neighbors  
switch(config)# do clear lldp neighbors  
switch(config)#
```

enable

Syntax

enable

Description

Exits the operator context (>) and enters the manager context (#).

Command context

Operator (>)

Authority

Administrators or local user group members with execution rights for this command.

Example

```
switch> enable
switch#
```

end

Syntax

end

Description

Exits the current context and enters the manager context (#).

Command context

Any context

Authority

Administrators or local user group members with execution rights for this command.

Example

```
switch# configure terminal
switch(config)# vlan 10
switch(config-vlan-10)# vlan 22
switch(config-vlan-22)# end
switch#
```

exit

Syntax

exit

Description

Exits the current context and enters its parent context.

Command context

Any context

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Example

```
switch# configure terminal
switch(config)# vlan 10
switch(config-vlan-10)# vlan 22
switch(config-vlan-22)# exit
switch(config)# exit
switch#
```

list

Syntax

list

Description

Shows a list of commands available from the current context.

Command context

Any context

Command context

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Example

```
switch> list
list
enable
exit
show session-timeout
...
```

page

Syntax

page [<LINES>]

no page

Description

Specifies the number of output lines the CLI displays before pausing to wait for a user to press a key. This value is the number of lines supported by the terminal session. This setting is not persistent and applies to the current session only.

The `page` command is enabled by default on the switch with the number of lines supported by the terminal. Change this default by using the `page` command to specify a different number of output lines.

The `no` form of this command sets the number of lines that are displayed to the default, which is the number of lines supported by the current terminal session.

Command context

Manager (#)

Parameters

<LINES>

Specifies the number to display before pausing. If not specified, the number of lines supported by the current terminal session is used. Range: 2-1000 lines.

Default: The number of lines supported by the current terminal session

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the page size to an unlimited number of lines:

```
switch# no page
switch#
```

Example output of a command after setting the page size to 10 lines:

```
switch# page 10
switch# list
  show hostname
  show domain-name
  list
  configure { terminal }
  disable
  exit
  end
  page
  page <2-1000>
-- MORE --, next page: Space, next line: Enter, quit: q
```

Pipe (|) command

Syntax

```
show running-config | begin 2 "vlan" | redirect "abc.txt"
show running-config | include "vlan" | exclude "vlan2" | count
show vlan | include "up" | include "VLAN100"
```

```
list | include "show" | exclude "show" | count
```

Description

The pipe (|) command filters the output of `show` or `list` commands using the options `include`, `exclude`, `begin`, `count`, or `redirect`.

Command context

Manager (#)

Authority

Administrators or local user group members with execution rights for this command.

Usage

```
show {<SHOW-COMMAND-OPTIONS>}... [ | {include <PATTERN-STRING> |
                                     exclude <PATTERN-STRING> |
                                     begin {<LINES-TO-DISPLAY>} <PATTERN-STRING>}}]...
                                     [ | {count [<PATTERN-STRING>] |
                                     redirect [<FILE-NAME>}}]

list [ | {include <PATTERN-STRING> |
          exclude <PATTERN-STRING> |
          begin {<LINES-TO-DISPLAY>} <PATTERN-STRING>}}]...
      [ | {count [<PATTERN-STRING>] |
          redirect [<FILE-NAME>}}]
```

- The pipe (|) command is supported for use with the `show` and `list` commands only.
- You can use multiple pipe commands with a single `show` or `list` command.

Examples

```
show running-config | redirect "abc.txt"
show running-config | begin 2 "vlan" | begin -2 "vlan" | begin "vlan"
show running-config | include "vlan" | exclude "vlan2" | count
show vlan | include "up" | include "VLAN100"
list | include "show" | exclude "show" | count
```

repeat

Syntax

```
repeat [id <POSITION>] [<COUNT>] [<DELAY>]
```

Description

Repeatedly executes one or more commands. By default, the most recent command in the history is executed until you press Ctrl+C.

Command context

Manager (#)

Parameters

<POSITION>

Specifies the position of a command, or range of positions of multiple commands, in the history list as shown in the output of the `show history` command. <POSITION> can be a single number, a comma-separated list of numbers, or a range of numbers specified by the beginning and end of the range, separated by a hyphen.

If the `id` parameter is not specified, the `repeat` command executes the command that was entered most recently. Default: 1.

<COUNT>

Specifies number of times to execute the command or commands. Default: The command repeats an infinite number of times.

<DELAY>

Specifies the number of seconds to delay before executing the command. Default: 2

Authority

Administrators or local user group members with execution rights for this command.

Example

```
switch# repeat id 1-4,7-8,10 count 2 delay 3
```

session-timeout

Syntax

```
session-timeout <MINUTES>  
no session-timeout
```

Description

Specifies the number of minutes a CLI session can be idle before the session is automatically terminated and the user is logged out.

The `no` form of this command sets the timeout to the default value of 30 minutes.

Command context

config

Parameters

<MINUTES>

Specifies the number of minutes the CLI session can remain idle. Specify 0 to configure CLI sessions to never time out. Range: 0 to 4320. Default: 30

Authority

Administrators or local user group members with execution rights for this command.

Example

```
switch(config)# session-timeout 15S
```

show session-timeout

Syntax

```
show session-timeout
```

Description

Shows the configured session timeout value.

Command context

Manager (#)

Authority

Administrators or local user group members with execution rights for this command.

Example

```
switch# show session-timeout
session-timeout: 30 minutes (Default)
```

show alias

Syntax

```
show alias
```

Description

Shows the command aliases that are defined on the switch.

Command context

Manager (#)

Authority

Administrators or local user group members with execution rights for this command.

Example

```
switch# show alias
Alias Name      Alias Definition
-----
hst             hostname $1
int_config     interface $1; no shutdown; ip address $2; lldp receive; mtu $3;
exit

switch# show hst
Alias Name      Alias Definition
-----
hst            hostname $1
```

show history

Syntax

```
show history [timestamp]
```

Description

Shows a numbered list of the commands that have been executed during this CLI session. Commands are displayed in descending order, with the most recent command having the lowest number

Command context

Manager (#)

Parameters

timestamp

Specifies that output include the time of execution of each command in the command history.

Authority

Administrators or local user group members with execution rights for this command.

Usage

Use the `show history` command to show the commands executed during this session. You use the command numbers to specify commands to repeat using the `repeat` command.

A command is not saved in the history if the command is one of the following:

- The same command as the command that was entered most recently.
- The `show history` command.
- The `repeat` command.

The commands saved in the history command buffer are in the same format in which you entered the commands. For example, if you execute the `show startup-config` command repeatedly, the system saves only one command in the history command buffer. If you execute the command in the format of `show start` and `show startup` respectively, the system saves them as two commands.

Example

```
switch# show history
6      show vers
5      conf
4      int mgmt
3      ip static 10.1.1.2/24
2      no shut
1      end
switch# show history timestamp
6      Tue Jan 30 08:42:24 2018      show vers
5      Tue Jan 30 08:45:52 2018      conf
4      Tue Jan 30 08:45:54 2018      int mgmt
3      Tue Jan 30 08:46:00 2018      ip static 10.1.1.2/24
2      Tue Jan 30 08:46:02 2018      no shut
1      Tue Jan 30 08:46:02 2018      end
switch#
```


cli-session

Syntax

```
cli-session  
no cli-session
```

Description

Enters the CLI session context (shown in the switch prompt as `config-cli-session`) for the purpose of configuring CLI user session management. Session management enhances security by enforcing specific CLI user session requirements. The following information is provided at time of successful login:

- When applicable, the number of failed login attempts since the most recent successful login.
- The date, time, and location (console or IP address or hostname) of the most recent previous successful login.
- The count of successful logins within the past (configurable) time period.

For example:

```
switch login: admin  
Password:
```

```
There were 3 failed login attempts since the last successful login  
Last login: 2019-04-20 08:51:33 from the console  
User "admin" has logged in 73 times in the past 30 days
```

The no form of this command disables concurrent CLI user session restrictions and reverts `timeout` and `tracking-range` to their default values.



To ensure that enhanced security is maintained, it is recommended that you keep CLI user session management fully enabled by setting `max-per-user` to a nondefault value.



The `cli-session` command applies only to SSH/console login connection types. It does not apply to other connection types such as REST.

Command context

```
config
```

Subcommands

These subcommands are available within the CLI session context.

```
[no] max-per-user <SESSIONS>
```

Specifies the maximum number of concurrent CLI sessions per user. The no form of this subcommand disables concurrent CLI user session restrictions. Default: Disabled (no value). Range: 1 to 5.



When the same user name is configured for both local and remote authentication, both users, regardless of privilege level, are considered to be the same user for the purpose of counting concurrent CLI sessions. For example, with `max-per-user` set to 1 and user `admin1` configured for local and remote authentication, only the local user `admin1` or the remote user `admin1` can be logged in at any given moment. Both `admin1` users cannot be logged in simultaneously unless `max-per-user` is increased to at least 2.

[no] `timeout` <MINUTES>

Specifies the number of minutes a CLI session can be idle before the session is automatically terminated and the user is logged out. A value of 0 minutes disables the session timeout. The `no` form of this subcommand sets the timeout value to the default. Default 30: Range 0 to 4320.



This subcommand is the recommended replacement for the `session-timeout` command.

[no] `tracking-range` <DAYS>

Specifies the maximum number of days to track CLI user session logins. The `no` form of this subcommand resets the value to its default. Default 30: Range 1 to 30.

`exit`

Exits the CLI session context.

`end`

Exits the CLI session context and then the config context.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring CLI user session settings for a maximum of one concurrent session, a 20-minute timeout, and tracking for a maximum of 25 days.

```
switch(config)# cli-session
switch(config-cli-session)# max-per-user 1
switch(config-cli-session)# timeout 20
switch(config-cli-session)# tracking-range 25
switch# exit
```

After successful earlier logins, logging in from the console without any intervening unsuccessful logins.

```
switch login: admin1
Password:

Last login: 2019-04-15 14:10:21 from the console
User 'admin1' has logged in 65 times in the past 25 days
```

Attempting to log in as `admin1` when already logged in as `admin1` from elsewhere.

```
switch login: admin1
Password:
Too many logins for 'admin1'
```

After successful earlier logins, attempting to log in twice with an invalid password, followed by a successful login.

```
switch login: admin1  
Password:
```

```
Login incorrect  
switch login: admin1  
Password:
```

```
Login incorrect  
switch login: admin1  
Password:
```

```
There were 2 failed login attempts since the last successful login  
Last login: 2019-04-15 17:22:45 from 192.168.1.1  
User 'admin1' has logged in 72 times in the past 25 days
```

clock date

Syntax

```
clock date <DATE>
```

Description

Sets the switch date.

Command context

```
config
```

Parameters

<DATE>

Specifies the date. Format: YYYY-MM-DD.

Authority

Administrators or local user group members with execution rights for this command.

Examples

This example sets the date to Dec 14, 2017.

```
switch(config)# clock date 2017-12-14
```

clock datetime

Syntax

```
clock datetime <DATE> <TIME>
```

Description

Sets the switch date and time.

Command context

```
config
```

Parameters

<DATE>

Specifies the date. Format: YYYY-MM-DD.

<TIME>

Specifies the time in 24-hour clock format. Seconds are optional. Format: HH:MM or HH:MM:SS.

Authority

Administrators or local user group members with execution rights for this command.

Examples

This example sets the date and time to Dec 13, 2017 at 15:00.

```
switch(config)# clock datetime 2017-12-13 14:15:00
```

clock time

Syntax

```
clock time <TIME>
```

Description

Sets the switch time.

Command context

```
config
```

Parameters

<TIME>

Specifies the time in 24-hour clock format. Seconds are optional. Format: HH:MM or HH:MM:SS.

Authority

Administrators or local user group members with execution rights for this command.

Examples

This example sets the time to 15:01:23.

```
switch(config)# clock time 15:01:23
```

clock timezone

Syntax

```
clock timezone <TIME-ZONE>
```

```
no clock timezone
```

Description

Sets the time zone and its associated daylight savings time rule.

The `no` form of this command sets the time zone to the default value of UTC.

Command context

```
config
```

Parameters

timezone <TIME-ZONE>

Specifies the time zone, <TIME-ZONE>, using a name defined in the IANA time zone database. See https://en.wikipedia.org/wiki/List_of_tz_database_time_zones.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the time zone to Eastern Standard Time (EST):

```
switch(config)# clock timezone EST
```

show clock

Syntax

```
show clock
```

Description

This command displays the current date, time, and time zone.

Command context

Operator (>) or Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

```
switch# show clock
Wed Nov 22 23:29:10 PDT 2017
System is configured for timezone : US/Pacific
```

Classes of traffic

The different classes of traffic that can be individually configured are:

- `arp-broadcast`: Address Resolution Protocol packets with a broadcast destination MAC address.
- `arp-unicast`: Address Resolution Protocol packets with a switch system destination MAC address.
- `dhcp`: Dynamic Host Configuration Protocol packets. Also includes snooped DHCP packets if DHCP snooping is enabled.
- `icmp-broadcast-ipv4`: Internet Control Message Protocol packets with a broadcast or multicast destination IPv4 address.
- `icmp-multicast-ipv6`: Internet Control Message Protocol packets with a well-known multicast destination IPv6 address.
- `icmp-security-ipv6`: IPv6 Internet Control Message Protocol packets intercepted and inspected.
- `icmp-unicast-ipv4`: Internet Control Message Protocol packets with a destination IPv4 address owned by the switch
- `icmp-unicast-ipv6`: Internet Control Message Protocol packets with a destination IPv6 address owned by the switch.
- `igmp`: Internet Group Management Protocol packets.
- `lacp`: Link Aggregation Control Protocol packets with the destination MAC address 01:80:c2:00:00:02.
- `lldp`: Link Layer Discovery Protocol packets with the destination MAC address 01:80:c2:00:00:0e.
- `loop-protect`: Loop Protection packets with the destination MAC address 09:00:09:09:13:a6.
- `manageability`: Unicast IP packets addressed to the switch for specific protocols that do not have a dedicated CoPP class like HTTP, SSH, RADIUS.
- `mvrp`: Multiple VLAN Registration Protocol packets with the destination MAC address 01:80:c2:00:00:20 or 01:80:c2:00:00:21
- `ntp`: Network Time Protocol packets with a destination IP address owned by the switch.
- `stp`: Spanning Tree Protocol (STP) packets with the destination MAC address 01:80:c2:00:00:00 or Per-VLAN Spanning Tree (PVST) packets with the destination MAC address 01:00:0c:cc:cc:cd.
- `udld`: Unidirectional Link Detection packets with the destination MAC address 01:00:0c:cc:cc:cc or 00:e0:52:00:00:00, or Cisco Discovery Protocol packets with the destination MAC address 01:00:0c:cc:cc:cc.
- `unresolved-ip-unicast`: Packets to be software forwarded by the management processor.

To regulate any other traffic destined for the CPU, every CoPP policy has a class named `default` that can also be configured to regulate other traffic to the CPU or prevent other traffic from being delivered.



All IPsec traffic received by the CPU will be regulated by the `ipsec` class regardless of the encapsulated protocol.

apply copp-policy

Syntax

```
apply copp-policy { <NAME> | default }  
no apply copp-policy <NAME>
```

Description

Applies a CoPP policy to the switch, replacing the policy that is in effect. There may be a brief interruption in traffic flow to the management processor while the switch implements the change.

Enter the `no apply copp-policy <NAME>` command with the name of a CoPP policy to unapply a CoPP policy and apply the default CoPP policy. This will only take effect if the specified policy is actively applied. Since there must always be a CoPP policy applied, this command effectively attempts to replace the applied CoPP policy with the default CoPP policy. The default CoPP policy cannot be unapplied using this command.

Command context

config

Parameters

<NAME>

Specifies the name of the policy to apply. Length: 1 to 64 characters.

default

Applies the default policy.

Authority

Administrators or local user group members with execution rights for this command.

Usage

If the new policy cannot be applied (for example, due to a lack of hardware resources), the previous policy remains in effect. Use the [show copp-policy](#) command to determine which policy is in effect.

Examples

Applying a policy named My_CoppPolicy:

```
switch(config)# apply copp-policy My_CoppPolicy
```

Applying the default policy:

```
switch(config)# apply copp-policy default
```

Unapplying a policy named My_CoppPolicy:

```
switch(config)# no apply copp-policy My_CoppPolicy
```

class

Syntax

```
class <CLASS> {drop | priority <PRIORITY> rate <RATE>}  
  
no class <CLASS> {drop | priority <PRIORITY> rate <RATE>}
```


Description

Adds a class to a CoPP policy. If the class exists, the existing class is modified. Changes made to an active (applied) policy take effect immediately.

When adding or modifying a class in an active policy, CoPP immediately activates the change on the switch. In cases where insufficient hardware resources exist to support a class or its action, CoPP fails to activate the changed class on the switch. When this failure occurs, the active configuration on the switch will be out of sync with its definition. To diagnose and remedy this situation:

- Use the [show copp-policy](#) command to determine which classes are out of sync between the active policy and its definition.
- Use the [reset copp-policy](#) command to synchronize the active policy with its definition. This synchronization changes the classes in the definition to match the classes in the active policy.

The `no` form of this command removes the configuration for the class. Traffic for the class will be prioritized and regulated using the factory default configuration for the class. Use the `show copp-policy factory-default` command to display the factory default CoPP policy. To stop a class of traffic from reaching the processor, set the class action to drop.

Command context

config-copp

Parameters

<CLASS>

Specifies the class to add or edit.

drop

Drop packets matching the selected class.

priority <PRIORITY>

Specifies the priority for packets matching the selected class. Range: 0 to 6.

rate <RATE>

Specifies the maximum rate, in packets per second (pps), for packets matching the selected class. Range: 25 to 99999.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Adding a class to handle LACP traffic with priority of 2 and rate of 2000:

```
switch(config-copp)# class lacp priority 2 rate 2000
```

Modifying the class to drop LLDP packets:

```
switch(config-copp)# class lldp drop
```

Removing the class that handles LLDP packets.

```
switch(config-copp)# no class lldp
```

clear copp-policy statistics

Syntax

```
clear copp-policy statistics
```

Description

Resets statistics for all CoPP classes to zero.

Command context

Operator (>) or Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Examples

Displaying and then resetting statistics for all classes in the active policy:

copp-policy

Syntax

```
copp-policy {<NAME> | default [revert]}  
no copp-policy <NAME>
```

Description

Creates a CoPP policy and switches to the `config-copp` context for the policy. Or, if the specified policy exists, switches to the `config-copp` context for the policy. A predefined policy, named `default`, contains factory default classes and is applied to the switch at first startup. This policy cannot be deleted, but its configuration can be changed.

The `no` form of this command removes a CoPP policy. If a policy is active (applied), it cannot be removed. It must be replaced with another policy before it can be removed.

Command context

```
config
```

Parameters

<NAME>

Specifies the name of the policy to add or edit. Length: 1 to 64 characters. The name must not be a substring of any of the following reserved words: `default`, `factory-default`, `commands`, `configuration`, or `statistics`.

`default`

Specifies the default CoPP policy. Use this default policy to configure the default policy.

`revert`

Sets the default CoPP policy to its factory settings.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Creating a policy named My_CoppPolicy:

```
switch(config)# copp-policy My_CoppPolicy
switch(config-copp)#
```

Removing a policy named My_CoppPolicy:

```
switch(config)# no copp-policy My_CoppPolicy
```

Setting the default policy to its factory settings:

```
switch(config)# copp-policy default revert
```

Unapplying the policy named My_CoppPolicy:

```
switch(config)# no apply copp-policy My_CoppPolicy
```

default-class

Syntax

```
default-class priority <PRIORITY> rate <RATE>
```

Description

Configures the default class that is automatically defined for all CoPP policies. The default class cannot be removed, but its configuration can be changed. The default class is applied to traffic that does not match any other class defined for a policy.

Command context

config-copp

Parameters

priority <PRIORITY>

Specifies the priority for packets matching the selected class. Range: 0 to 7.

rate <RATE>

Specifies the maximum rate, in packets per second (pps), for packets matching the selected class. Range: 25 to 99999.

Authority

Administrators or local user group members with execution rights for this command.

Example

Setting the default class to a priority of **2** and rate of **2000**:

```
switch(config-copp)# default-class priority 2 rate 2000
```

reset copp-policy

Syntax

```
reset copp-policy { <NAME> | default }
```

Description

Resets an active CoPP policy to match the settings that are currently in effect for the active policy on the switch. Changes made to the active policy that could not be activated are removed from the active policy. When the switch fails to add or modify a class in an active CoPP policy, it is possible the active policy settings on the switch may be out of sync with those defined in the policy.

Command context

config

Parameters

<NAME>

Specifies the name of the policy to reset. Length: 1 to 64 characters.

default

Resets the default policy to match its active settings.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Resetting a policy named My_CoppPolicy:

```
switch# show copp-policy My_CoppPolicy
class          drop priority rate pps hardware rate pps
-----
igmp            6          5000    5000
lacp            2          2000    2000
default         1          6000    6000
switch# config terminal
switch(config)# copp-policy My_CoppPolicy
switch(config-copp)# class stp priority 4 rate 4000
switch(config-copp)# do show copp-policy My_CoppPolicy
class          drop priority rate pps hardware rate pps
-----
igmp            6          5000    5000
lacp            2          2000    2000
default         1          6000    6000
% Warning: user-specified classes in CoPP policy My_CoppPolicy do not match
active configuration.
switch(config-copp)# do show copp-policy My_CoppPolicy configuration
class          drop priority rate pps applied
-----
igmp            6          5000    yes
lacp            2          2000    yes
stp             4          4000    no
default         1          6000    yes
% Warning: user-specified classes in CoPP policy My_CoppPolicy do not match
active configuration.
switch(config-copp)# exit
switch(config)# reset copp-policy My_CoppPolicy
switch(config)# do show copp-policy My_CoppPolicy
```

class	drop	priority	rate pps	hardware	rate pps
igmp		6	5000		5000
lcp		2	2000		2000
default		1	6000		6000

Resetting the default policy:

```
switch(config)# reset copp-policy default
```

show copp-policy

Syntax

```
show copp-policy [<NAME> | default] [commands] [configuration]
```

Description

Shows CoPP policy settings for a specific CoPP policy. When entered without specifying either a name or the `default` parameter, shows all the CoPP policy settings that are active on the switch and have successfully been programmed into the hardware.

A warning is displayed if:

- The active and user-specified applications of a policy do not match.
- The active and user-specified configurations of a policy do not match.

Command context

Operator (>) or Manager (#)

Parameters

<NAME>

Specifies the name of the policy for which to display settings. Length: 1 to 64 characters.

default

Displays CoPP settings for the default policy.

commands

Displays output as CLI commands.

configuration

Displays user-specified CoPP settings and **not** the active settings.

Authority

Operators or Administrators or local user group members with execution rights for this command.

Operators can execute this command from the operator context (>) only.

Example

Displaying the CoPP policies defined in the configuration and the active application:

```
switch# show copp-policy
applied copp_policy_name
-----
```

```
My_CoppPolicy
applied default
switch#
```

Displaying the active configuration of all CoPP policies as CLI commands:

Displaying the default policy:

```
switch# show copp-policy default
class                drop priority rate pps hardware rate pps
-----
arp-broadcast        2          1250    1250
arp-unicast          3          825     825
<---OUTPUT OMITTED FOR BREVITY-->
default              2          4225    4225
```

show copp-policy factory-default

Syntax

```
show copp-policy factory-default [commands]
```

Description

Display the configuration for the factory-default CoPP policy.

Command context

Operator (>) or Manager (#)

Parameters

commands

Displays output as CLI commands.

Authority

Operators or Administrators or local user group members with execution rights for this command.

Operators can execute this command from the operator context (>) only.

Example

Displaying the factory-default policy:

```
switch# show copp-policy factory-default
class                drop priority rate pps
-----
arp-broadcast        2          1250
arp-unicast          3          825
<---OUTPUT OMITTED FOR BREVITY-->
default              2          4225
```

Displaying the active configuration of My_CoppPolicy (My_CoppPolicy is applied):

```
switch# config terminal
switch(config)# apply copp-policy My_CoppPolicy
```

```
switch(config)# do show copp-policy My_CoppPolicy
class                drop priority rate pps hardware rate pps
-----
igmp                  6          5000    5000
larp                  2          2000    2000
default               1          6000    6000
```

Displaying the active configuration of `My_CoppPolicy` as CLI commands:

```
switch# show copp-policy My_CoppPolicy commands
copp-policy My_CoppPolicy
  class igmp priority 6 rate 5000
  class larp priority 2 rate 2000
  default-class priority 1 rate 6000
apply copp-policy My_CoppPolicy
```

Displaying the user-specified configuration of `My_CoppPolicy`:

```
switch# show copp-policy My_CoppPolicy configuration
class                drop priority rate pps applied
-----
igmp                  6          5000    yes
larp                  2          2000    yes
default               1          6000    yes
```

Displaying the user-specified configuration of `My_CoppPolicy` as CLI commands:

```
switch# show copp-policy My_CoppPolicy commands configuration
copp-policy My_CoppPolicy
  class igmp priority 6 rate 5000
  class larp priority 2 rate 2000
  default-class priority 1 rate 6000
apply copp-policy My_CoppPolicy
```

show copp-policy statistics

Syntax

```
show copp-policy statistics [class <CLASS> | default-class | non-zero]
```

Description

Displays statistics for all classes, a single class, or all classes with non-zero statistics in the active CoPP policy.

Command context

Operator (>) or Manager (#)

Parameters

<CLASS>

Specifies the [class](#) for which to display statistics.

default-class

Displays statistics for the default class.

non-zero

Displays statistics for all classes with non-zero statistics.

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Usage

If a single class is specified, the priority and rate that has been programmed in hardware for that class will be shown.

Examples

Applying the `default` CoPP policy and displaying statistics for all classes in the actively applied policy:



The rate displayed is the actual rate in hardware.

```
switch# config terminal
switch(config)# apply copp-policy default
switch(config)# exit
switch# show copp-policy statistics
Statistics for CoPP policy 'default':
Totals:
  packets passed   : 1000           packets dropped   : 1500
Class: default
  packets passed   : 400           packets dropped   : 600
Class: arp-broadcast
  packets passed   : 500           packets dropped   : 800
  <--OUTPUT OMITTED FOR BREVITY-->
```

Displaying statistics for the default class in the active policy:

```
switch(config)# show copp-policy statistics default-class
Statistics for CoPP policy 'default':
Class: default
Description: Default
  priority          : 2
  rate (pps)        : 4225

  packets passed    : 400           packets dropped    : 600
```

Displaying statistics for the class `arp-broadcast` in the actively applied policy:

```
switch# show copp-policy statistics class arp-broadcast
Statistics for CoPP policy 'default':
Class: arp-broadcast
Description: Address Resolution Protocol broadcast
  priority          : 2
  rate (pps)        : 1250

  packets passed    : 500           packets dropped    : 800
```

show tech copp

Syntax

```
show tech copp
```

Description

Displays the output of all show commands supported by CoPP.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Capturing the command output into a local file:

```
switch# show tech copp local-file
Show Tech output stored in local-file. Please use 'copy show-tech local-file' to
copy-out this file.

switch# copy show-tech local-file ?
  REMOTE_URL    URL of syntax
                 {tftp://|sftp://USER@}{IP|HOST}[:PORT] [;blocksize=VAL]/FILE
  STORAGE_URL   URL of syntax usb:/file
switch# copy show-tech local-file
```

clear debug buffer

Syntax

```
clear debug buffer
```

Description

Clears all debug logs. Using the `show debug buffer` command will only display the logs generated after the `clear debug buffer` command.

Command context

Manager (#)

Authority

Administrators or local user group members with execution rights for this command.

Examples

Clearing all generated debug logs:

```
switch# show debug buffer
-----
show debug buffer
-----
2018-10-14:09:10:58.558710|lldpd|LOG_DEBUG|MSTR||LLDP|LLDP_CONFIG|No Port cfg changes
2018-10-14:09:10:58.558737|lldpd|LOG_DEBUG|MSTR||LLDP|LLDP_EVENT|lldpd_stats_run
entered at time 8257199
2018-10-14:09:10:58.569317|lldpd|LOG_DEBUG|MSTR||LLDP|LLDP_CONFIG|No Port cfg changes
2018-10-14:09:11:21.881907|hpe-sysmond|LOG_INFO|MSTR||SYSMON|SYSMON_CONFIG|Sysmon
poll interval changed to 32

switch# clear debug buffer
switch# show debug buffer
-----
show debug buffer
-----
2018-10-14:09:13:24.481407|hpe-sysmond|LOG_INFO|MSTR||SYSMON|SYSMON_CONFIG|Sysmon
poll interval changed to 51
```

debug {all | <MODULE-NAME>}

Syntax

```

debug {all | <MODULE-NAME>} [<SUBMODULE-NAME>] [severity
    (emer|crit|alert|err|notice|warning|info|debug)] {port <PORT-NAME> |
    vlan <VLAN-ID> | ip <IP-ADDRESS> | mac <MAC-ADDRESS> |
    vrf <VRF-NAME> | instance <INSTANCE-ID>}
no debug {all | <MODULE-NAME>} [<SUBMODULE-NAME>] {port | vlan | ip | mac |
    vrf | instance}

```

Description

Enables debug logging for modules or submodules by name, with optional filtering by specific criteria.

The `no` form of this command disables debug logging.

Command context

Manager (#)

Parameters

`all`

Enables debug logging for all modules.

`<MODULE-NAME>`

Enables debug logging for a specific module. For a list of supported modules, enter the `debug` command followed by a space and a question mark (?).

`<SUBMODULE-NAME>`

Enables debug logging for a specific submodule. For a list of supported submodules, enter the `debug`

`<MODULE-NAME>` command followed by a space and a question mark (?).

`severity (emer|crit|alert|err|notice|warning|info|debug)`

Selects the minimum severity log level for the destination. If a severity is not provided, the default log level is `debug`. Optional.

`emer`

Specifies storage of debug logs with a severity level of `emergency` only.

`crit`

Specifies storage of debug logs with severity level of `critical` and above.

`alert`

Specifies storage of debug logs with severity level of `alert` and above.

`err`

Specifies storage of debug logs with severity level of `error` and above.

`notice`

Specifies storage of debug logs with severity level of `notice` and above.

`warning`

Specifies storage of debug logs with severity level of `warning` and above.

`info`

Specifies storage of debug logs with severity level of `info` and above.

`debug`

Specifies storage of debug logs with severity level of `debug` (default).

`port`

Displays debug logs for the specified port, for example 1/1/1.

`vlan <VLAN-ID>`

Displays debug logs for the specified VLAN. Provide a VLAN from 1 to 4094.

`ip <IP-ADDRESS>`

Displays debug logs for the specified IP Address.

`mac <MAC-ADDRESS>`

Displays debug logs for the specified MAC Address, for example A:B:C:D:E:F.

`vrf <VRF-NAME>`

Displays debug logs for the specified VRF.

instance <INSTANCE-ID>

Displays debug logs for the specified instance. Provide an instance ID from 1 to 255.

Authority

Administrators or local user group members with execution rights for this command.

Examples

```
switch# debug all
```

debug db

Syntax

```
debug db {all | sub-module} [level <MINIMUM-SEVERITY>] [filter]
```

```
no debug db {all | sub-module} [level <MINIMUM-SEVERITY>] [filter]
```

Description

Enables or disables debug logging for a db module or submodules, with an option to filter by specific criteria. The `no` form of this command disables debug logging for the db module or submodule.

Command context

Manager (#)

Parameters

`all`

Enables all submodules for the db log.

`sub-module`

Enables debug logging for supported submodules. Specify `rx` or `tx` debug logs.

`filter`

Specifies supported filters for the db log. Specify `table`, `column`, or `client`. Optional

`severity` (`emer`|`crit`|`alert`|`err`|`notice`|`warning`|`info`|`debug`)

Selects the minimum severity log level for the destination. If a severity is not provided, the default log level is `debug`. Optional.

`emer`

Specifies storage of debug logs with a severity level of `emergency` only.

`crit`

Specifies storage of debug logs with severity level of `critical` and above.

`alert`

Specifies storage of debug logs with severity level of `alert` and above.

`err`

Specifies storage of debug logs with severity level of `error` and above.

`notice`

Specifies storage of debug logs with severity level of `notice` and above.

`warning`

Specifies storage of debug logs with severity level of `warning` and above.

`info`

Specifies storage of debug logs with severity level of `info` and above.

`debug`

Specifies storage of debug logs with severity level of `debug` (default).

Authority

Administrators or local user group members with execution rights for this command.

Usage

DBlog is a high performance, configuration, and state database server logging infrastructure where a user can log the transactions which are sent or received by clients to the configuration and state database server. It can be enabled through the CLI and REST, and also supports filters where a user can filter out logs on the basis of table, column, or client. It is helpful for debugging when the user wants to debug an issue with a particular client, table, or column combination. It is not enabled by default. A combination of filters can also be applied to filter out messages based on table, column, and client.

There are three submodules for the "db" module:

1. `all`: When All is enabled, no filters are applied to any of the debug logs, even if other submodules are configured with filters.
2. `tx`: If enabled, only the replies and notifications sent out for the initial and incremental updates are logged.
3. `rx`: If enabled, only the transactions sent to the configuration and state database server are logged.

The keyword `all` may be used to enable or disable debug logging for all sub-modules. Also a combination of filters can be used to filter the message types.

If the table or client filter is applied, then the messages belonging to this specific table or client will be logged. The column filter can also be applied to further filter messages on a table, providing a mechanism to filter messages on a column. The table and client filter can be used in combination or separately, but column can only be used in conjunction with table.

Examples

Configuring all submodules with severity `debug`:

```
switch# debug db all severity debug
```

Configuring the `tx` submodule with `table Interface` filter and severity `debug`:

```
switch# debug db tx table Interface severity debug
```

Configuring the `rx` submodule with `table Interface column statistics` filter and severity `debug`:

```
switch# debug db rx table Interface column statistics severity debug
```

Disabling the `rx` submodule:

```
switch# no debug db rx
```

Disabling the `tx` submodule `table Interface`:

```
switch# no debug db tx table Interface
```

debug destination

Syntax

```
debug destination {syslog | file | console | buffer} [severity  
(emer|crit|alert|err|notice|warning|info|debug)]
```

```
no debug destination {syslog | file | console}
```

Description

Sets the destination for debug logs and the minimum severity level for each destination

The `no` form of this command unsets the destination for debug logs.

Command context

Manager (#)

Parameters

`{syslog | file | console | buffer}`

Selects the destination to store debug logs. Required.

`syslog`

Specifies that the debug logs are stored in the `syslog`.

`file`

Specifies that debug logs are stored in `file`.

`console`

Specifies that debug logs are stored in `console`.

`buffer`

Specifies that debug logs are stored in `buffer` (default).

`severity (emer|crit|alert|err|notice|warning|info|debug)`

Selects the minimum severity log level for the destination. If a severity is not provided, the default log level is `debug`. Optional.

`emer`

Specifies storage of debug logs with a severity level of `emergency` only.

`crit`

Specifies storage of debug logs with severity level of `critical` and above.

`alert`

Specifies storage of debug logs with severity level of `alert` and above.

`err`

Specifies storage of debug logs with severity level of `error` and above.

`notice`

Specifies storage of debug logs with severity level of `notice` and above.

`warning`

Specifies storage of debug logs with severity level of `warning` and above.

`info`

Specifies storage of debug logs with severity level of `info` and above.

`debug`

Specifies storage of debug logs with severity level of `debug` (default).

Authority

Administrators or local user group members with execution rights for this command.

Usage

Events that have a severity equal to or higher than the configured severity level are stored in the designated destination. The product defaults to `buffer` for destination and `debug` as a severity level.

Examples

```
switch# debug destination syslog severity alert
switch# debug destination console severity info
switch# debug destination file severity warning
switch# debug destination buffer severity err
```

show debug

Syntax

```
show debug
```

Description

Displays the enabled debug types.

Command context

Manager (#)

Authority

Administrators or local user group members with execution rights for this command.

Examples

```
switch# show debug
-----
module sub_module severity vlan port   ip       mac           instance vrf
-----
all      all          err      1    1/1/1  10.0.0.1  1a:2b:3c:4d:5e:6f  2      default
```

show debug buffer

Syntax

```
show debug buffer [module <MODULE-NAME> | severity
(emer|crit|alert|err|notice|warning|info|debug)]
```

Description

Displays debug logs stored in the specified debug buffer with optional filtering by module or severity.

Command context

Manager (#)

Parameters

<MODULE-NAME>

Filters debug logs displayed by the specified module name.

severity (emer|crit|alert|err|notice|warning|info|debug)

Displays debug logs with a specified severity level. Defaults to debug. Optional.

emer

Displays debug logs with a severity level of emergency only.

crit

Displays debug logs with a severity level of critical and above.

alert

Displays debug logs with a severity level of alert and above.

err

Specifies storage of debug logs with severity level of error and above.

notice

Specifies storage of debug logs with severity level of notice and above.

warning

Displays debug logs with a severity level of warning and above.

info

Displays debug logs with a severity level of info and above.

debug

Displays debug logs with a severity level of debug (default).

Authority

Administrators or local user group members with execution rights for this command.

Examples

```
switch# show debug buffer

-----
show debug buffer
-----
2017-03-06:06:51:15.089967|hpe-sysmond|SYSMON|SYSMON_CONFIG|LOG_INFO|Sysmon poll
interval changed to 20
```

show debug destination

Syntax

show debug destination

Description

Displays the configured debug destination and severity.

Command context

Manager (#)

Authority

Administrators or local user group members with execution rights for this command.

Examples


```
switch# show debug destination
```

```
-----
```

```
show debug destination
```

```
-----
```

```
CONSOLE:info
```

```
FILE:warning
```

aaa authentication port-access allow-cdp-bpdu

Syntax

```
aaa authentication port-access allow-cdp-bpdu
```

```
no aaa authentication port-access allow-cdp-bpdu
```

Description

Allows all packets related to the CDP (Cisco Discovery Protocol) BPDU (Bridge Protocol Data Unit) on a secure port.

The `no` form of this command blocks the CDP BPDU on a secure port. On a nonsecure port, the command has no effect.

Command context

```
config-if
```

Authority

Administrators or local user group members with execution rights for this command.

Examples

Allowing a CDP BPDU on secure port **1/1/1**:

```
switch(config)# interface 1/1/1
switch(config-if)# aaa authentication port-access allow-cdp-bpdu
switch(config-if)# do show running-config
Current configuration:
!
!Version ArubaOS-CX 10.0X.0000
led locator on
!
!
vlan 1
aaa authentication port-access mac-auth
    enable
aaa authentication port-access dot1x authenticator
    enable
interface 1/1/1
    no shutdown
    vlan access 1
    aaa authentication port-access allow-cdp-bpdu
    aaa authentication port-access mac-auth
        enable
    aaa authentication port-access dot1x authenticator
        enable

switch(config-if)# do show port-access device-profile interface all
```

```
Port 1/1/1, Neighbor-Mac 00:0c:29:9e:d1:20
  Profile Name      : access_switches
  LLDP Group       :
  CDP Group        : aruba-ap_cdp
  Role             : test_ap_role
  Status           : In Progress
  Failure Reason    :
```

Blocking LLDP packet on secure port **1/1/1**:

```
switch(config)# interface 1/1/1
switch(config-if)# no aaa authentication port-access allow-cdp-bpdu
switch(config-if)# do show running-config
Current configuration:
!
!Version ArubaOS-CX 10.0X.0000
led locator on
!
!
vlan 1
aaa authentication port-access mac-auth
    enable
interface 1/1/1
    no shutdown
    vlan access 1
    aaa authentication port-access mac-auth
        enable
```

aaa authentication port-access allow-lldp-bpdu

Syntax

```
aaa authentication port-access allow-lldp-bpdu
```

```
no aaa authentication port-access allow-lldp-bpdu
```

Description

Allows all packets related to the LLDP BPDU (Bridge Protocol Data Unit) on a secure port.

The `no` form of this command blocks the LLDP BPDU on a secure port. On a nonsecure port, the command has no effect.

Command context

```
config-if
```

Authority

Administrators or local user group members with execution rights for this command.

Examples

Allowing an LLDP BPDU on secure port **1/1/1**:

```
switch(config)# interface 1/1/1
switch(config-if)# aaa authentication port-access allow-lldp-bpdu
```

```

switch(config-if)# do show running-config
Current configuration:
!
!Version ArubaOS-CX 10.0X.0000
led locator on
!
!
vlan 1
aaa authentication port-access mac-auth
    enable
interface 1/1/1
    no shutdown
    vlan access 1
    aaa authentication port-access allow-lldp-bpdu
    aaa authentication port-access mac-auth
        enable

switch(config-if)# do show port-access device-profile interface all
Port 1/1/1, Neighbor-Mac 00:0c:29:9e:d1:20
  Profile Name      : access_switches
  LLDP Group        : 2920-grp
  CDP Group         :
  Role              : local_2920_role
  Status            : Profile Applied
  Failure Reason    :

```

Blocking LLDP BPDU on secure port **1/1/1**:

```

switch(config)# interface 1/1/1
switch(config-if)# no aaa authentication port-access allow-lldp-bpdu
switch(config-if)# do show running-config
Current configuration:
!
!Version ArubaOS-CX 10.0X.0000led locator on
!
!
vlan 1
aaa authentication port-access mac-auth
    enable
interface 1/1/1
    no shutdown
    vlan access 1
    aaa authentication port-access mac-auth
        enable

```

associate cdp-group

Syntax

```
associate cdp-group <GROUP-NAME>
```

```
no associate cdp-group <GROUP-NAME>
```

Description

Associates a CDP (Cisco Discovery Protocol) group with a device profile. A maximum of two CDP groups can be associated with a device profile.

The `no` form of this command removes a CDP group from a device profile.

Command context

config-device-profile

Parameters

<GROUP-NAME>

Specifies the name of the CDP group to associate with this device profile. Range: 1 to 32 alphanumeric characters.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Associating the CDP group **my-cdp-group** with the device profile **profile01**:

```
switch(config)# port-access device-profile profile01
switch(config-device-profile)# associate cdp-group my-cdp-group
```

Removing the CDP group **my-cdp-group** from the device profile **profile01**:

```
switch(config)# port-access device-profile profile01
switch(config-device-profile)# no associate cdp-group my-cdp-group
```

associate lldp-group

Syntax

```
associate lldp-group <GROUP-NAME>
```

```
no associate lldp-group <GROUP-NAME>
```

Description

Associates an LLDP group with a device profile. A maximum of two LLDP groups can be associated with a device profile

The **no** form of this command removes an LLDP group from a device profile.

Command context

config-device-profile

Parameters

<GROUP-NAME>

Specifies the name of the LLDP group to associate with the device profile. Range: 1 to 32 alphanumeric characters.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Associating the LLDP group **my-lldp-group** with the device profile **profile01**:

```
switch(config)# port-access device-profile profile01
switch(config-device-profile)# associate lldp-group my-lldp-group
```

Removing the LLDP group **my-lldp-group** from the device profile **profile01**:

```
switch(config)# port-access device-profile profile01
switch(config-device-profile)# no associate lldp-group my-lldp-group
```

associate mac-group

Syntax

```
associate mac-group <GROUP-NAME>
```

```
no associate mac-group <GROUP-NAME>
```

Description

Associates a MAC group with a device profile. A maximum of two MAC groups can be associated with a device profile.

The **no** form of this command removes a MAC group from a device profile.

Command context

config-device-profile

Parameters

<GROUP-NAME>

Specifies the name of the MAC group to associate with this device profile. Range: 1 to 32 alphanumeric characters.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Associating the MAC group **mac01-group** with the device profile **profile01**:

```
switch(config)# port-access device-profile profile01
switch(config-device-profile)# associate mac-group mac01-group
```

Removing the MAC group **mac01-group** from the device profile **profile01**:

```
switch(config)# port-access device-profile profile01
switch(config-device-profile)# no associate mac-group mac01-group
```

associate role

Syntax

```
associate role <ROLE-NAME>
```

```
no associate role <ROLE-NAME>
```

Description

Associates a role with a device profile. Only one role can be associated with a device profile. For information on how to configure a role, see the port access role information in the *Security Guide*.

The `no` form of this command removes a role from a device profile.

Command context

```
config-device-profile
```

Parameters

<ROLE-NAME>

Specifies the name of the role to associate with the device profile. Range: 1 to 64 alphanumeric characters.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Associating the role **my-role** with the device profile **profile01**:

```
switch(config)# port-access device-profile profile01
switch(config-device-profile)# associate role my-role
```

Removing the role **my-role** from the device profile **profile01**:

```
switch(config)# port-access device-profile profile01
switch(config-device-profile)# no associate role my-role
```

disable

Syntax

```
disable
```

```
no disable
```

Description

Disables a device profile.

The `no` form of this command enables a device profile.

Command context

```
config-device-profile
```

Authority

Administrators or local user group members with execution rights for this command.

Examples

Disabling a device profile:

```
switch(config)# port-access device-profile profile01
switch(config-device-profile)# disable
```

Enabling a device profile named profile01:

```
switch(config)# port-access device-profile profile01
switch(config-device-profile)# no disable
```

enable

Syntax

enable

no enable

Description

Enables a device profile.

The `no` form of this command disables a device profile.

Command context

config-device-profile

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling a device profile:

```
switch(config)# port-access device-profile profile01
switch(config-device-profile)# enable
```

Disabling a device profile named profile01:

```
switch(config)# port-access device-profile profile01
switch(config-device-profile)# no enable
```

ignore (for CDP groups)

Syntax

```
ignore [seq <SEQ-NUM>] {platform <PLATFORM> | sw-version <SWVERSION> |
voice-vlan-query <VLAN-ID>}
```

```
no ignore [seq <SEQ-ID>] {platform <PLATFORM> | sw-version <SWVERSION> |
voice-vlan-query <VLAN-ID>}
```

Description

Defines a rule to ignore devices for a CDP (Cisco Discovery Protocol) group. Up to 64 match/ignore rules can be defined for a group.

The `no` form of this command removes a rule for ignoring devices from a CDP group.

Command context

`config-cdp-group`

Parameters

`seq <SEQ-ID>`

Specifies the ID of the rule to create or modify. If no ID is specified when adding a rule, an ID is automatically assigned in increments of 10 in the order in which rules are added. When more than one rule matches the command entered, the rule with the lowest ID takes precedence.

`platform <PLATFORM>`

Specifies the hardware or model details of the neighbor. Range: 1 to 128 alphanumeric characters.

`sw-version <SWVERSION>`

Specifies the software version of the neighbor. Range: 1 to 128 alphanumeric characters.

`voice-vlan-query <VLAN-ID>`

Specifies the VLAN query value of the neighbor. Range: 1 to 65535.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Adding a rule to the CDP group **grp01** that ignores a device that transmits **PLATFORM01** in the platform TLV:

```
switch(config)# port-access cdp-group grp01
switch(config-cdp-group)# ignore platform PLATFORM01
```

Adding a rule to the CDP group **grp01** that ignores a device that transmits **SWVERSION** in software version TLV:

```
switch(config)# port-access cdp-group grp01
switch(config-cdp-group)# ignore sw-version SWVERSION
```

Removing the rule that matches the sequence number **25** from the CDP group named **grp01**.

```
switch(config)# port-access cdp-group grp01
switch(config-cdp-group)# no ignore seq 25
```

ignore (for LLDP groups)

Syntax

```
ignore [seq <SEQ-ID>] {sys-desc <SYS-DESC> | sysname <SYS-NAME> |
    vendor-oui <VENDOR-OUI> [type <KEY> [value <VALUE>]]}
```

```
no ignore [seq <SEQ-ID>] {sys-desc <SYS-DESC> | sysname <SYS-NAME> |
    vendor-oui <VENDOR-OUI> [type <KEY> [value <VALUE>]]}
```

Description

Defines a rule to ignore devices for an LLDP group. Up to 64 match/ignore rules can be defined for a group. The `no` form of this command removes a rule for ignoring devices from an LLDP group.

Command context

`config-lddp-group`

Parameters

`seq` *<SEQ-ID>*

Specifies the ID of the rule to create or modify. If no ID is specified when adding a rule, an ID is automatically assigned in increments of 10 in the order in which rules are added. When more than one rule matches the command entered, the rule with the lowest ID takes precedence.

`sys-desc` *<SYS-DESC>*

Specifies the LLDP system description type-length-value (TLV). Range: 1 to 256 alphanumeric characters.

`sysname` *<SYS-NAME>*

Specifies the LLDP system name TLV. Range: 1 to 64 alphanumeric characters.

`vendor-oui` *<VENDOR-OUI>*

Specifies the LLDP system vendor OUI TLV. Range: 1 to 6 alphanumeric characters.

`type` *<KEY>*

Specifies the vendor OUI subtype key. Optional.

`value` *<VALUE>*

Specifies the vendor OUI subtype value. Range: 1 to 256 alphanumeric characters.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Adding a rule to the LLDP group **grp01** that ignores a device that transmits **PLATFORM01** in the system description TLV:

```
switch(config)# port-access lldp-group grp01
switch(config-lddp-group)# ignore sys-desc PLATFORM01
```

Removing the rule that matches the sequence number **25** from the LLDP group named **grp01**.

```
switch(config)# port-access lldp-group grp01
switch(config-lddp-group)# no match seq 25
```

ignore (for MAC groups)

Syntax

```
[seq <SEQ-ID>] ignore {mac <MAC-ADDR> | mac-mask <MAC-MASK> | mac-oui <MAC-OUI>}
```

```
no [seq <SEQ-ID>] ignore {mac <MAC-ADDR> | mac-mask <MAC-MASK> | mac-oui <MAC-OUI>}
```

Description

Defines a rule to ignore devices for a MAC group based on the criteria of MAC address, MAC address mask, or MAC Organizational Unique Identifier (OUI). Up to 64 ignore rules can be defined for a group.

The `no` form of this command removes a rule for ignoring devices from a MAC group.

Command context

config-mac-group

Parameters

seq <SEQ-ID>

Specifies the entry sequence ID of the rule to create or modify a MAC group. If no ID is specified when adding a rule, an ID is automatically assigned in increments of 10 in the order in which rules are added. When more than one rule matches the command entered, the rule with the lowest ID takes precedence.

Range: 1 to 4294967295.

mac <MAC-ADDR>

Specifies the MAC address of the device to ignore.

mac-mask <MAC-MASK>

Specifies the MAC address mask to ignore devices in that range. Supported MAC address masks: /32 and /40.

mac-oui <MAC-OUI>

Specifies the MAC OUI to ignore devices in that range. Supports MAC OUI address of maximum length of 24 bits.

Authority

Administrators or local user group members with execution rights for this command.

Usage

To achieve the required configuration of matches for devices, it is recommended to first ignore the devices that you do not want to add. Then match the criteria for the rest of the devices that you want to add to the MAC group.

For example, if you want to ignore a specific device but add all the other devices that belong to a MAC OUI, then you must first configure the ignore criteria with a lower sequence number. And then configure match criteria with a higher sequence number.

Examples

Adding a rule to the MAC group **grp01** to ignore a device based on MAC address, but match all other devices belonging to a MAC OUI:

```
switch(config)# mac-group grp01
switch(config-mac-group)# ignore mac 1a:2b:3c:4d:5e:6f
switch(config-mac-group)# match mac-oui 1a:2b:3c
switch(config-mac-group)# exit
switch(config)# do show running-config
Current configuration:
!
!Version ArubaOS-CX Virtual.10.0X.0001
!export-password: default
led locator on
!
!
!
!
ssh server vrf
!
!
!
!
```

```

vlan 1
interface vlan 1
    no shutdown
    ip dhcp
mac-group grp01
    seq 10 ignore mac 1a:2b:3c:4d:5e:6f
    seq 20 match mac-oui 1a:2b:3c
...

```

Adding a rule to the MAC group **grp01** to ignore devices based on MAC address mask, but match all other devices belonging to a MAC OUI:

```

switch(config)# mac-group grp01
switch(config-mac-group)# ignore mac-mask 1a:2b:3c:4d/32
switch(config-mac-group)# match mac-oui 1a:2b:3c
switch(config-mac-group)# exit
switch(config)# do show running-config
Current configuration:
!
!Version ArubaOS-CX Virtual.10.0X.0001
!export-password: default
led locator on
!
!
!
!
ssh server vrf
!
!
!
!
vlan 1
interface vlan 1
    no shutdown
    ip dhcp
mac-group grp01
    seq 10 ignore mac-mask 1a:2b:3c:4d/32
    seq 20 match mac-oui 1a:2b:3c
...

```

Adding a rule to the MAC group **grp01** that ignores a device based on complete MAC address:

```

switch(config)# mac-group grp01
switch(config-mac-group)# ignore mac 1a:2b:3c:4d:5e:6f

```

Adding a rule to the MAC group **grp02** that ignores devices based on MAC mask:

```

switch(config)# mac-group grp01
switch(config-mac-group)# ignore mac-mask 1a:2b:3c:4d:5e/40
switch(config-mac-group)# ignore mac-mask 18:e3:ab:73/32

```

Adding a rule to the MAC group **grp03** that ignores devices based on MAC OUI:

```
switch(config)# mac-group grp03
switch(config-mac-group)# ignore mac-oui 81:cd:93
```

Adding a rule to the MAC group **grp01** that ignores devices with a sequence number and based on MAC address:

```
switch(config)# mac-group grp01
switch(config-mac-group)# seq 10 ignore mac b2:c3:44:12:78:11
switch(config-mac-group)# exit
switch(config)# do show running-config
Current configuration:
!
!Version ArubaOS-CX Virtual.10.0X.0001
!export-password: default
led locator on
!
!
vlan 1
interface vlan 1
    no shutdown
    ip dhcp
mac-group grp01
    seq 10 ignore mac b2:c3:44:12:78:11
...

```

Removing the rule from the MAC group **grp01** based on sequence number:

```
switch(config)# mac-group grp01
switch(config-mac-group)# no ignore seq 10
switch(config-mac-group)# exit
switch(config)# do show running-config
Current configuration:
!
!Version ArubaOS-CX Virtual.10.0X.0001
!export-password: default
led locator on
!
!
vlan 1
interface vlan 1
    no shutdown
    ip dhcp
mac-group grp01
...

```

Adding a rule to the MAC group **grp01** that ignores devices with MAC entry sequence number and based on MAC OUI:

```
switch(config)# mac-group grp01
switch(config)# mac-group grp01
switch(config-mac-group)# seq 10 ignore mac b2:c3:44:12:78:11
switch(config-mac-group)# seq 20 ignore mac-oui 1a:2b:3c
switch(config-mac-group)# seq 30 ignore mac-mask 71:14:89:42/32
switch(config-mac-group)# exit
switch(config)#

```

```
switch(config)# ^Z

switch#
switch#
switch# show running-config
Current configuration:
!
!Version ArubaOS-CX PL.10.06.0002
!export-password: default
!
!
!
!
ssh server vrf default
vlan 1
spanning-tree
mac-group grp01
    seq 10 ignore mac b2:c3:44:12:78:11
    seq 20 ignore mac-oui 1a:2b:3c
    seq 30 ignore mac-mask 71:14:89:42/32
interface 1/1/1
    no shutdown
    vlan access 1
interface 1/1/2
    no shutdown
    vlan access 1
interface 1/1/3
    no shutdown
    vlan access 1
interface 1/1/4
    no shutdown
    vlan access 1
interface 1/1/5
    no shutdown
    vlan access 1
interface 1/1/6
    no shutdown
    vlan access 1
interface 1/1/7
    no shutdown
    vlan access 1
interface 1/1/8
    no shutdown
    vlan access 1
interface 1/1/9
    no shutdown
    vlan access 1
interface 1/1/10
    no shutdown
    vlan access 1
interface 1/1/11
    no shutdown
    vlan access 1
interface 1/1/12
    no shutdown
    vlan access 1
interface 1/1/13
    no shutdown
    vlan access 1
interface 1/1/14
    no shutdown
    vlan access 1
```

```

interface 1/1/15
  no shutdown
  vlan access 1
interface 1/1/16
  no shutdown
  vlan access 1
interface vlan 1
  ip dhcp
!
!
!
!
!
https-server vrf default

```

Removing the rule from the MAC group **grp01** based on sequence number and MAC OUI:

```

switch(config)# mac-group grp01
switch(config-mac-group)# no seq 20 ignore mac-oui 1a:2b:3c
switch(config-mac-group)# exit
switch(config)# do show running-config
Current configuration:
!
!Version ArubaOS-CX Virtual.10.0X.0001
!export-password: default
led locator on
!
!
vlan 1
interface vlan 1
  no shutdown
  ip dhcp
mac-group grp01
  seq 10 ignore mac b2:c3:44:12:78:11
  seq 30 ignore mac-mask 71:14:89:f3/32
...

```

Removing the rule that matches the sequence number **25** from the MAC group named **grp01**.

```

switch(config)# mac-group grp01
switch(config-mac-group)# no ignore seq 25

```

mac-group

Syntax

```
mac-group <MAC-GROUP-NAME>
```

```
no mac-group <MAC-GROUP-NAME>
```

Description

Creates a MAC group or modifies an existing MAC group. A MAC group is used to classify connected devices based on the MAC address details, such as mask or OUI.

A maximum of 32 MAC groups can be configured on the switch. A maximum of 2 MAC groups can be associated with a device profile. Each group accepts 64 match or ignore commands.

The `no` form of this command removes a MAC group.

Command context

`config`

Parameters

`<MAC-GROUP-NAME>`

Specifies the name of the MAC group to create or modify. The maximum number of characters supported is 32.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Creating a MAC group named `grp01`:

```
switch(config)# mac-group grp01
switch(config-mac-group)# exit
```

Removing a MAC group named `grp01`:

```
switch(config)# no mac-group grp01
```

match (for CDP groups)

Syntax

```
match [seq <SEQ-ID>] {platform <PLATFORM> | sw-version <SWVERSION> |
    voice-vlan-query <VLAN-ID>}

no match [seq <SEQ-ID>] {platform <PLATFORM> | sw-version <SWVERSION> |
    voice-vlan-query <VLAN-ID>}
```

Description

Defines a rule to match devices for a CDP group. A maximum of 32 CDP groups can be configured on the switch. Up to 64 match or ignore rules can be defined for each group.

The `no` form of this command removes a rule for adding devices to a CDP group.

Command context

`config-cdp-group`

Parameters

`seq <SEQ-ID>`

Specifies the ID of the rule to create or modify. If no ID is specified when adding a rule, an ID is automatically assigned in increments of 10 in the order in which rules are added. When more than one rule matches the command entered, the rule with the lowest ID takes precedence.

`platform <PLATFORM>`

Specifies the hardware or model details of the neighbor. Range: 1 to 128 alphanumeric characters.

sw-version <SWVERSION>

Specifies the software version of the neighbor. Range: 1 to 128 alphanumeric characters.

voice-vlan-query <VLAN-ID>

Specifies the VLAN query value of the neighbor. Range: 1 to 65535.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Adding rules to match a Cisco device with a specific software version on VLAN **512** to the CDP group **grp01**:

```
switch(config)# port-access cdp-group grp01
switch(config-cdp-group)# match platform CISCO
switch(config-cdp-group)# match sw-version 11.2(12)P
switch(config-cdp-group)# match voice-vlan-query 512
switch(config-cdp-group)# match seq 50 platform cisco sw-version 11.2(12)P voice-
vlan-query 512
switch(config-cdp-group)# exit
switch(config)# do show running-config

Current configuration:
!
!Version ArubaOS-CX Virtual.10.0X.000
!export-password: default
led locator on
!
!
vlan 1
port-access cdp-group grp01
    seq 10 match platform CISCO
    seq 20 match sw-version 11.2(12)P
    seq 30 match voice-vlan-query 512
    seq 50 match platform cisco sw-version 11.2(12)P voice-vlan-query 512
```

Removing a rule that matches the sequence number **25** from the CDP group named **grp01**:

```
switch(config)# port-access cdp-group grp01
switch(config-cdp-group)# no match seq 25
```

Adding a rule that matches the value of vendor-OUI **000b86** to the CDP group named **grp01**:

```
switch(config)# port-access cdp-group grp01
switch(config-cdp-group)# match vendor-oui 000b86
```

match (for LLDP groups)

Syntax

```
match [seq <SEQ-ID>] {sys-desc <SYS-DESC> | sysname <SYS-NAME> |
    vendor-oui <VENDOR-OUI> [type <KEY> [value <VALUE>]]}
```

```
no match [seq <SEQ-ID>] {sys-desc <SYS-DESC> | sysname <SYS-NAME> |
    vendor-oui <VENDOR-OUI> [type <KEY> [value <VALUE>]]}
```

Description

Defines a rule to match devices for an LLDP group. Up to 64 match/ignore rules can be defined for a group. The `no` form of this command removes a rule.

Command context

`config-lldp-group`

Parameters

`seq <SEQ-ID>`

Specifies the ID of the rule to create or modify. If no ID is specified when adding a rule, an ID is automatically assigned in increments of 10 in the order in which rules are added. When more than one rule matches the command entered, the rule with the lowest ID takes precedence.

`sys-desc <SYS-DESC>`

Specifies the LLDP system description type-length-value (TLV). Range: 1 to 256 alphanumeric characters.

`sysname <SYS-NAME>`

Specifies the LLDP system name TLV. Range: 1 to 64 alphanumeric characters.

`vendor-oui <VENDOR-OUI>`

Specifies the LLDP system vendor OUI TLV. Range: 1 to 6 alphanumeric characters.

`type <KEY>`

Specifies the vendor OUI subtype key.

`value <VALUE>`

Specifies the vendor OUI subtype value. Range: 1 to 256 alphanumeric characters.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Adding rules that match the LLDP system description **ArubaSwitch** and system name **Aruba** to the LLDP group named **grp01**:

```
switch(config)# port-access lldp-group grp01
switch(config-lldp-group)# match sys-desc ArubaSwitch
switch(config-lldp-group)# match sysname Aruba
switch(config)# do show running-config
```

Current configuration:

```
!
!Version ArubaOS-CX Virtual.10.0X.000
!export-password: default
led locator on
!
!
vlan 1
port-access lldp-group grp01
    seq 10 match sys-desc ArubaSwitch
    seq 20 match sysname Aruba
```

Removing a rule that matches the sequence number **25** from an LLDP group named **grp01**:

```
switch(config)# port-access lldp-group grp01
switch(config-lldp-group)# no match seq 25
```

Adding a rule that matches the value of vendor-OUI **000b86** with type of **1** to the LLDP group named **grp01**:

```
switch(config)# port-access lldp-group grp01
switch(config-lldp-group)# match vendor-oui 000b86 type 1
```

Adding a rule that matches the value of vendor-OUI **000c34** to the LLDP group named **grp01**:

```
switch(config)# port-access lldp-group grp01
switch(config-lldp-group)# match vendor-oui 000c34
```

match (for MAC groups)

Syntax

```
[seq <SEQ-ID>] match {mac <MAC-ADDR> | mac-mask <MAC-MASK> | mac-oui <MAC-OUI>}
no [seq <SEQ-ID>] match {mac <MAC-ADDR> | mac-mask <MAC-MASK> | mac-oui <MAC-OUI>}
```

Description

Defines a rule to match devices for a MAC group based on the criteria of MAC address, MAC address mask, or MAC Organizational Unique Identifier (OUI). Up to 64 match rules can be defined for a group.

You must not configure the following special MAC addresses:

- Null MAC—For example, 00:00:00:00:00:00 or 00:00:00/32
- Multicast MAC
- Broadcast MAC—For example, ff:ff:ff:ff:ff:ff
- System MAC

Although the switch accepts these addresses, it will not process these addresses for the local MAC match feature.

The `no` form of this command removes a rule for adding devices to a MAC group.

The number of clients that can onboard based on the match criteria is configured in the `aaa authentication port-access client-limit` command. For information about this command, see the *Security Guide* for your switch.

Command context

config-mac-group

Parameters

seq <SEQ-ID>

Specifies the entry sequence ID of the rule to create or modify a MAC group. If no ID is specified when adding a rule, an ID is automatically assigned in increments of 10 in the order in which rules are added. When more than one rule matches the command entered, the rule with the lowest ID takes precedence.

Range: 1 to 4294967295.

mac <MAC-ADDR>

Specifies the MAC address of the device.

mac-mask <MAC-MASK>

Specifies the MAC address mask to add devices in that range. Supported MAC address masks: /32 and /40.

mac-oui <MAC-OUI>

Specifies the MAC OUI to add devices in that range. Supports MAC OUI address of maximum length of 24 bits.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Adding a device to the MAC group **grp01** based on complete MAC address:

```
switch(config)# mac-group grp01
switch(config-mac-group)# match mac 1a:2b:3c:4d:5e:6f
switch(config-mac-group)# exit
```

Adding devices to the MAC group **grp02** based on MAC mask:

```
switch(config)# mac-group grp01
switch(config-mac-group)# match mac-mask 1a:2b:3c:4d:5e/40
switch(config-mac-group)# match mac-mask 18:e3:ab:73/32
switch(config-mac-group)# exit
```

Adding devices to the MAC group **grp03** based on MAC OUI:

```
switch(config)# mac-group grp03
switch(config-mac-group)# match mac-oui 81:cd:93
switch(config-mac-group)# exit
```

Adding devices to the MAC group **grp01** with MAC entry sequence number and based on MAC address:

```
switch(config)# mac-group grp01
switch(config-mac-group)# seq 10 match mac b2:c3:44:12:78:11
switch(config-mac-group)# exit
switch(config)# do show running-config
Current configuration:
!
!Version ArubaOS-CX Virtual.10.0X.0001
!export-password: default
led locator on
!
!
vlan 1
interface vlan 1
    no shutdown
    ip dhcp
    mac-group grp01
        seq 10 match mac b2:c3:44:12:78:11
...

```

Removing devices from the MAC group **grp01** based on sequence number:

```
switch(config)# mac-group grp01
switch(config-mac-group)# no match seq 10
switch(config-mac-group)# exit
```

```

switch(config)# do show running-config
Current configuration:
!
!Version ArubaOS-CX Virtual.10.0X.0001
!export-password: default
led locator on
!
!
vlan 1
interface vlan 1
    no shutdown
    ip dhcp
mac-group grp01
...

```

Adding devices to the MAC group **grp01** with MAC entry sequence number and based on MAC address, MAC address mask, and MAC OUI:

```

switch(config)# mac-group grp01
switch(config-mac-group)# seq 10 match mac b2:c3:44:12:78:11
switch(config-mac-group)# seq 20 match mac-oui 1a:2b:3c
switch(config-mac-group)# seq 30 match mac-mask 71:14:89:f3/32
switch(config-mac-group)# exit
switch(config)# do show running-config
Current configuration:
!
!Version ArubaOS-CX Virtual.10.0X.0001
!export-password: default
led locator on
!
!
vlan 1
interface vlan 1
    no shutdown
    ip dhcp
mac-group grp01
    seq 10 match mac b2:c3:44:12:78:11
    seq 20 match mac-oui 1a:2b:3c
    seq 30 match mac-mask 71:14:89:f3/32
...

```

Removing devices from the MAC group **grp01** based on MAC OUI:

```

switch(config)# mac-group grp01
switch(config-mac-group)# no seq 20 match mac-oui 1a:2b:3c
switch(config-mac-group)# exit
switch(config)# do show running-config
Current configuration:
!
!Version ArubaOS-CX Virtual.10.0X.0001
!export-password: default
led locator on
!
!
vlan 1
interface vlan 1
    no shutdown

```

```

    ip dhcp
mac-group grp01
    seq 10 match mac b2:c3:44:12:78:11
    seq 30 match mac-mask 71:14:89:f3/32

...

```

Adding devices to the MAC group **grp03** with MAC entry sequence number and based on MAC address mask:

```

switch(config)# mac-group grp03
switch(config-mac-group)# seq 10 match mac-mask 10:14:a3:b7:55/40
switch(config-mac-group)# exit
switch(config)# do show running-config
Current configuration:
!
!Version ArubaOS-CX Virtual.10.0X.0001
!export-password: default
led locator on
!
!
vlan 1
interface vlan 1
    no shutdown
    ip dhcp
mac-group grp03
    seq 10 match mac-mask 10:14:a3:b7:55/40

...

```

Removing devices from the MAC group **grp03** based on MAC address mask:

```

switch(config)# mac-group grp03
switch(config-mac-group)# no seq 10 match mac-mask 10:14:a3:b7:55/40
switch(config-mac-group)# exit
switch(config)# do show running-config
Current configuration:
!
!Version ArubaOS-CX Virtual.10.0X.0001
!export-password: default
led locator on
!
!
vlan 1
interface vlan 1
    no shutdown
    ip dhcp
mac-group grp03

...

```

port-access cdp-group

Syntax

```
port-access cdp-group <CDP-GROUP-NAME>
```

```
no port-access cdp-group <CDP-GROUP-NAME>
```

Description

Creates a CDP (Cisco Discovery Protocol) group or modifies an existing CDP group. A CDP Group is used to classify connected devices based on the CDP packet details advertised by the device. A maximum of 32 CDP groups can be configured on the switch. Each group accepts 64 match/ignore commands.

The `no` form of this command removes a CDP group.

Command context

config

Parameters

<CDP-GROUP-NAME>

Specifies the name of the CDP group to create or modify. The maximum number of characters supported is 32. Required.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Creating a CDP group named grp01:

```
switch(config)# port-access cdp-group grp01
switch(config-cdp-group)# match platform CISCO
switch(config-cdp-group)# match sw-version 11.2(12)P
switch(config-cdp-group)# match voice-vlan-query 512
switch(config-cdp-group)# seq 50 match platform cisco sw-version 11.2(12)P voice-
vlan-query 512
switch(config-cdp-group)# exit
switch(config)# do show running-config
```

Current configuration:

```
!
!Version ArubaOS-CX Virtual.10.0X.000
!export-password: default
led locator on
!
!
vlan 1
port-access cdp-group grp01
    seq 10 match platform CISCO
    seq 20 match sw-version 11.2(12)P
    seq 30 match voice-vlan-query 512
    seq 50 match platform cisco sw-version 11.2(12)P voice-vlan-query 512
```

Removing a CDP group named grp01:

```
switch(config)# no port-access cdp-group grp01
```

port-access device-profile

Syntax

```
port-access device-profile <DEVICE-PROFILE-NAME>

no port-access device-profile <DEVICE-PROFILE-NAME>
```

Description

Creates a new device profile and switches to the `config-device-profile` context. A maximum of 32 device profiles can be created.

The `no` form of this command removes a device profile.

Command context

`config`

Parameters

`<DEVICE-PROFILE-NAME>`

Specifies the name of a device profile. Range: 1 to 32 alphanumeric characters.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Creating a device profile named **profile01**:

```
switch(config)# port-access device-profile profile01
switch(config-device-profile)#
```

Removing a device profile named **profile01**:

```
switch(config)# no port-access device-profile profile01
```

port-access device-profile mode block-until-profile-applied



You must configure this mode in device profile only on standalone ports where there is no security configured and when you not want the port to be offline until one client is onboarded.

Syntax

```
port-access device-profile mode block-until-profile-applied

no port-access device-profile mode block-until-profile-applied
```

Description

Configures the switch to block the port until a profile match occurs for a device. This configuration is required when no security feature is enabled on the port.

You must enable this mode or security on the port for local MAC match feature to operate. You must not enable both features on the same port at the same time.



You must not combine any other AAA configurations with the block-until-profile-applied mode.

The `no` form of this command removes a rule for adding devices to a MAC group.

Command context

config-if
config-if-deviceprofile

Authority

Administrators or local user group members with execution rights for this command.

Example

Configuring block-until-profile applied mode on port 1/1/1:

```
switch(config)# interface 1/1/1
switch(config-if)# port-access device-profile
switch(config-if-deviceprofile)# mode block-until-profile-applied
switch(config-if-deviceprofile)# end
```

port-access lldp-group

Syntax

```
port-access lldp-group <LLDP-GROUP-NAME>
no port-access lldp-group <LLDP-GROUP-NAME>
```

Description

Creates an LLDP group or modifies an existing LLDP group. An LLDP group is used to classify connected devices based on the LLDP type-length-values (TLVs) advertised by the device. A maximum of 32 LLDP groups can be configured on the switch. Each group accepts 64 match/ignore commands.

The `no` form of this command removes an LLDP group.

Command context

config

Parameters

<LLDP-GROUP-NAME>

Specifies the name of the LLDP group to create or modify. The maximum number of characters supported is 32. Required.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Creating an LLDP group named grp01:

```
switch(config)# port-access lldp-group grp01
switch(config-lldp-group)#
```

Removing an LLDP group named grp01:

```
switch(config)# no port-access lldp-group grp01
```

show port-access device-profile

Syntax

```
show port-access device-profile [[interface {all | <INTERFACE-ID>}  
    [client-status <MAC-ADDR>]] | name <DEVICE-PROFILE-NAME>]
```

Description

Shows the client status for a specific MAC address or profile name.

Command context

Manager (#)

Parameters

interface {all | <INTERFACE-ID>}

Select **all** for all interfaces or specify the name of an interface in the format: member/slot/port.

client-status <MAC-ADDR>

Specifies a MAC address (xx:xx:xx:xx:xx:xx), where x is a hexadecimal number from 0 to F.

name <DEVICE-PROFILE-NAME>

Specifies the name of the device profile.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Showing the applied state of the device profiles:

```
switch# show port-access device-profile

Profile Name      : accesspoints
LLDP Groups       : 2920-grp
CDP Groups        :
MAC Groups        : 2920-mac-grp1,2920-iot-grp2
Role              : local_role_1
State             : Enabled

Profile Name      : access_switches
LLDP Groups       : 2920-grp
CDP Groups        :
MAC Groups        :
Role              : local_2920_role
State             : Enabled

Profile Name      : iot_devices
LLDP Groups       :
CDP Groups        :
MAC Groups        : iot_camera-grp1,iot_sensors-grp1
Role              : local_2920_role
State             : Enabled

Profile Name      : lobbyaps
LLDP Groups       :
CDP Groups        : lobby_ap_cdp_grp
MAC Groups        :
Role              : test_ap_role
State             : Disabled
```

Showing the applied state of the device profile on interface 1/1/3:

```
switch# show port-access device-profile interface 1/1/3 client-status
00:0c:29:9e:d1:20

Port 1/1/3, Neighbor-Mac 00:0c:29:9e:d1:20
  Profile Name      : lobbyaps
  LLDP Group        :
  CDP Group         : aruba-ap_cdp
  MAC Group         :
  Role              : test_ap_role
  Status            : Failed
  Failure Reason    : Failed to apply MAC based VLAN
```

Showing the applied state of a specific device profile:

```
switch# show port-access device-profile name lldp-group

  Profile Name      : lldp-group
  LLDP Groups       :
  CDP Groups        :
  MAC Groups        : pc-behind-phone, lldp
  Role              : auth_role
  State             : Enabled
```

ip dhcp

Syntax

```
ip dhcp
```

Description

Enables the DHCP client on the management interface enabling the interface to automatically obtain an IP address from a DHCP server on the network. By default, the DHCP client is enabled.

Command context

Authority

Administrators or local user group members with execution rights for this command.

Examples

This example enables the DHCP client on the management interface.

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ip dhcp
switch(config-if-vlan)# no shutdown
```

If the interface is not enabled, you can enable it by entering the `no shutdown` command.

http proxy

Syntax

```
http-proxy {<FQDN | IPV4-ADDR>} [vrf <VRF-NAME>]
no http-proxy
```

Description

Specifies HTTP proxy location and VRF.

The `no` form of this command removes a specified HTTP proxy location.

Command context

```
config
```

Parameters

<FQDN>

Specifies FQDN for HTTP proxy location.

<IPV4-ADDR>

Specifies IPV4 address for HTTP proxy location.

<VRF-NAME>

Specifies VRF for HTTP proxy.

Authority

Administrators or local user group members with execution rights for this command.

Usage

- HTTP proxy location can be configured using the CLI/REST interface or auto-configured through the DHCP server connected to the switch.
- There are three sources for HTTP proxy location:
 - User configured HTTP proxy via CLI or REST interface.
 - DHCP options received via management/OOBM port.
 - DHCP options received via VLAN 1 on supported switch platforms.
- Operational configuration for HTTP proxy location is determined by the source with the highest priority. Source priority:
 1. User configured.
 2. DHCP options received via management/OOBM port.
 3. DHCP options received via VLAN 1.



-
- HTTP proxy location can only be a FQDN or an IPV4 address.
 - When HTTP proxy location and VRF are configured, they override any existing HTTP proxy location and VRF.
 - If this command is executed without the VRF parameter, the default VRF will be used.
 - Port number may need to be specified at the end of the IP address for FQDN to connect via HTTP proxy.
 - For example, 8088 is the TCP port number: http-proxy 192.168.248.248:**8088**
-

Examples

Specifying a FQDN for HTTP proxy location and MGMT VRF:

```
switch(config)# http-proxy http-proxy.aruba.com vrf mgmt
```

Removing HTTP proxy location

```
switch(config)# no http-proxy
```

ip dns domain-list

Syntax

```
ip dns domain-list <DOMAIN-NAME> [vrf default]
no ip dns domain-list <DOMAIN-NAME> [vrf default]
```

Description

Configures one or more domain names that are appended to the DNS request. The DNS client appends each name in succession until the DNS server replies. Domains can be either IPv4 or IPv6. By default, requests are forwarded on the default VRF.

The `no` form of this command removes a domain from the list.

Command context

config

Parameters

`list <DOMAIN-NAME>`

Specifies a domain name. Up to six domains can be added to the list. Length: 1 to 256 characters.

Authority

Administrators or local user group members with execution rights for this command.

Examples

This example defines a list with two entries: **domain1.com** and **domain2.com**.

```
switch(config)# ip dns domain-list domain1.com
switch(config)# ip dns domain-list domain2.com
```

This example removes the entry **domain1.com**.

```
switch(config)# no ip dns domain-list domain1.com
```

ip dns domain-name

Syntax

```
ip dns domain-name <DOMAIN-NAME> [ vrf <VRF-NAME> ]
no ip dns domain-name <DOMAIN-NAME> [ vrf <VRF-NAME> ]
```

Description

Configures a domain name that is appended to the DNS request. The domain can be either IPv4 or IPv6. By default, requests are forwarded on the default VRF. If a domain list is defined with the command `ip dns domain-list`, the domain name defined with this command is ignored.

The `no` form of this command removes the domain name.

Command context

config

Parameters

`<DOMAIN-NAME>`

Specifies the domain name to append to DNS requests. Length: 1 to 256 characters.

`vrf <VRF-NAME>`

Specifies a VRF name. Default: default.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the default domain name to `domain.com`:

```
switch(config)# ip dns domain-name domain.com
```

Removing the default domain name `domain.com`:

```
switch(config)# no ip dns domain-name domain.com
```

ip dns host

Syntax

```
ip dns host <HOST-NAME> <IP-ADDR> [ vrf <VRF-NAME> ]
```

```
no ip dns host <HOST-NAME> <IP-ADDR> [ vrf <VRF-NAME> ]
```

Description

Associates a static IP address with a hostname. The DNS client returns this IP address instead of querying a DNS server for an IP address for the hostname. Up to six hosts can be defined. If no VRF is defined, the default VRF is used.

The `no` form of this command removes a static IP address associated with a hostname.

Command context

config

Parameters

`host <HOST-NAME>`

Specifies the name of a host. Length: 1 to 256 characters.

`<IP-ADDR>`

Specifies an IP address in IPv4 format (`x.x.x.x`), where `x` is a decimal number from 0 to 255, or IPv6 format (`xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx`), where `x` is a hexadecimal number from 0 to F.

vrf <VRF-NAME>

Specifies a VRF name. Default: default.

Authority

Administrators or local user group members with execution rights for this command.

Examples

This example defines an IPv4 address of **3.3.3.3** for **host1**.

```
switch(config)# ip dns host host1 3.3.3.3
```

This example defines an IPv6 address of **b::5** for **host 1**.

```
switch(config)# ip dns host host1 b::5
```

This example defines removes the entry for **host 1** with address **b::5**.

```
switch(config)# no ip dns host host1 b::5
```

ip dns server address

Syntax

```
ip dns server-address <IP-ADDR> [ vrf <VRF-NAME> ]
```

```
no ip dns server-address <IP-ADDR> [ vrf <VRF-NAME> ]
```

Description

Configures the DNS name servers that the DNS client queries to resolve DNS queries. Up to six name servers can be defined. The DNS client queries the servers in the order that they are defined. If no VRF is defined, the default VRF is used.

The **no** form of this command removes a name server from the list.

Command context

config

Parameters

<IP-ADDR>

Specifies an IP address in IPv4 format (**x.x.x.x**), where **x** is a decimal number from 0 to 255, or IPv6 format (**xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx**), where **x** is a hexadecimal number from 0 to F.

vrf <VRF-NAME>

Specifies a VRF name. Default: default.

Authority

Administrators or local user group members with execution rights for this command.

Examples

This example defines a name server at **1.1.1.1**.


```
switch(config)# ip dns server-address 1.1.1.1
```

This example defines a name server at **a::1**.

```
switch(config)# ip dns server-address a::1
```

This example removes a name server at **a::1**.

```
switch(config)# no ip dns server-address a::1
```

show ip dns

Syntax

```
show ip dns [vrf <VRF-NAME>]
```

Description

Shows all DNS client configuration settings or the settings for a specific VRF.

Command context

Manager (#)

Parameters

vrf <VRF-NAME>

Specifies the VRF for which to show information. If no VRF is defined, the default VRF is used.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

(enabling, disabling faults)

```
{all | <FAULT>}  
no {all | <FAULT>}
```

Description

Within the selected fault monitor profile context, enables all faults or specific faults for monitoring.



Faults enabled with this command use default actions and thresholds unless the actions and thresholds are configured with respective commands `action` and `threshold`.

By default, all faults are disabled in a profile and remain disabled until enabled as described here. Configuring the action and threshold does not enable the fault.

The `no` form of this command disables faults for monitoring.

Parameter	Description
all	Selects all faults.
<FAULT>	Selects a specific fault. Available fault names: excessive-jabbers excessive-crc-errors excessive-oversize-packets excessive-fragments excessive-tx-drops excessive-collisions excessive-late-collisions excessive-alignment-errors excessive-link-flaps excessive-broadcasts excessive-multicasts

Examples

Enabling faults:

```
switch(config-fault-monitor-profile)# all  
switch(config-fault-monitor-profile)# excessive-oversize-packets  
switch(config-fault-monitor-profile)# excessive-jabbers  
switch(config-fault-monitor-profile)# excessive-fragments  
switch(config-fault-monitor-profile)# excessive-crc-errors  
switch(config-fault-monitor-profile)# excessive-tx-drops
```

```

switch(config-fault-monitor-profile) # excessive-link-flaps
switch(config-fault-monitor-profile) # excessive-broadcasts
switch(config-fault-monitor-profile) # excessive-multicasts
switch(config-fault-monitor-profile) # excessive-collisions
switch(config-fault-monitor-profile) # excessive-late-collisions
switch(config-fault-monitor-profile) # excessive-alignment-errors

```

Disabling faults:

```

switch(config-fault-monitor-profile) # no all
switch(config-fault-monitor-profile) # no excessive-oversize-packets
switch(config-fault-monitor-profile) # no excessive-jabbers
switch(config-fault-monitor-profile) # no excessive-fragments
switch(config-fault-monitor-profile) # no excessive-crc-errors
switch(config-fault-monitor-profile) # no excessive-tx-drops
switch(config-fault-monitor-profile) # no excessive-link-flaps
switch(config-fault-monitor-profile) # no excessive-broadcasts
switch(config-fault-monitor-profile) # no excessive-multicasts
switch(config-fault-monitor-profile) # no excessive-collisions
switch(config-fault-monitor-profile) # no excessive-late-collisions
switch(config-fault-monitor-profile) # no excessive-alignment-errors

```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config-fault-monitor-profile	Administrators or local user group members with execution rights for this command.

action

```

{all | <FAULT>} action {notify | notify-and-disable [auto-enable <TIMEOUT>]}
no {all | <FAULT>} action {notify | notify-and-disable [auto-enable <TIMEOUT>]}

```

Description

Within the selected fault monitor profile context, configures the fault monitoring action for the specified fault. Default action: `notify` with `auto-enable` disabled.

The `no` form of this command removes the action and disables `auto-enable`.

Parameter	Description
<code>all</code>	Selects all faults.
<code><FAULT></code>	Selects a specific fault. Available fault names: <code>excessive-jabbers</code>

Parameter	Description
	excessive-crc-errors excessive-oversize-packets excessive-fragments excessive-tx-drops excessive-collisions excessive-late-collisions excessive-alignment-errors excessive-link-flaps excessive-broadcasts excessive-multicasts
notify	Selects the notify action. Notifies through events, DLOGs, and SNMP trap. This action is enabled by default.
notify-and-disable	Selects the action as notify-and-disable. Notifies through events, DLOGs, and SNMP trap, and then disables the port.
auto-enable <TIMEOUT>	Sets the number of seconds after which a port disabled by the notify-and-disable action is automatically re-enabled. Range: 1 to 604800 seconds.



The fault parameter values are saved even after a fault is disabled in the profile. The saved values will be used if the fault is later re-enabled in the profile again.

Examples

Configuring the notify fault action:

```
switch(config-fault-monitor-profile)# all action notify
switch(config-fault-monitor-profile)# excessive-oversize-packets action notify
switch(config-fault-monitor-profile)# excessive-jabbers action notify
switch(config-fault-monitor-profile)# excessive-collisions action notify
```

Configuring the notify-and-disable fault action:

```
switch(config-fault-monitor-profile)# all action notify-and-disable
switch(config-fault-monitor-profile)# excessive-oversize-packets action notify-and-disable
switch(config-fault-monitor-profile)# excessive-jabbers action notify-and-disable
switch(config-fault-monitor-profile)# excessive-late-collisions action notify-and-disable
switch(config-fault-monitor-profile)# excessive-alignment-errors action notify-and-disable
```

Configuring the notify-and-disable action with auto-enable:

```
switch(config-fault-monitor-profile)# excessive-oversize-packets action notify-and-disable auto-enable 80
switch(config-fault-monitor-profile)# excessive-jabbers action notify-and-disable auto-enable 100
```

```
switch(config-fault-monitor-profile) # excessive-collisions action notify-and-disable auto-enable 70
```

Resetting the fault action to default:

```
switch(config-fault-monitor-profile) # no excessive-oversize-packets action
switch(config-fault-monitor-profile) # no excessive-jabbers action
switch(config-fault-monitor-profile) # no excessive-alignment-errors action
```

Disabling auto-enable:

```
switch(config-fault-monitor-profile) # no all action notify-and-disable auto-enable
switch(config-fault-monitor-profile) # no excessive-jabbers action notify-and-disable auto-enable
switch(config-fault-monitor-profile) # no excessive-alignment-errors action notify-and-disable auto-enable
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config-fault-monitor-profile	Administrators or local user group members with execution rights for this command.

apply fault-monitor profile

```
apply fault-monitor profile <PROFILE-NAME>
no apply fault-monitor profile [<PROFILE-NAME>]
```

Description

Applies a fault monitoring profile to the selected interface or interface range.

The **no** form of this command removes the fault monitoring profile from the selected interface or interface range.

Parameter	Description
<PROFILE-NAME>	Specifies the fault monitor profile name. Range: Up to 64 alphanumeric and special characters.

Examples

Applying the fault monitoring profile to a interface:

```
switch(config)# interface 1/1/1
switch(config-if)# apply fault-monitor profile noisy-ports
```

Applying the fault monitoring profile to a interface range:

```
switch(config)# interface 1/1/2-1/1/24
switch(config-if)# apply fault-monitor profile quiet-ports
```

Command History

Release	Modification
10.08	Made the <code><PROFILE-NAME></code> parameter optional in the <code>no</code> form of the command.
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config-if	Administrators or local user group members with execution rights for this command.

fault-monitor profile

```
fault-monitor profile <PROFILE-NAME>
no fault-monitor profile <PROFILE-NAME>
```

Description

Creates a fault monitoring profile and enters its context. If the profile already exists, this command enters the profile context. A maximum of 16 fault monitoring profiles are supported.

The `no` form of this command deletes the fault monitoring profile.

Parameter	Description
<code><PROFILE-NAME></code>	Specifies the fault monitor profile name. Range: Up to 64 alphanumeric and special characters.

Examples

Creating a fault monitor profile:

```
switch(config)# fault-monitor profile noisy-ports
switch(config-fault-monitor-profile)#
```

Deleting a fault monitor profile:

```
switch(config)# no fault-monitor profile noisy-ports
switch(config)#
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config	Administrators or local user group members with execution rights for this command.

show fault-monitor profile

show fault-monitor profile <PROFILE-NAME>

Description

Shows fault monitoring profile information for all profiles or a specific profile.

Parameter	Description
<PROFILE-NAME>	Specifies the fault monitor profile name. Range: Up to 64 alphanumeric and special characters.

Example

Showing information for all fault monitoring profiles:

```
switch# show fault-monitor profile
-----
Fault monitor profile: noisy-ports
-----
Auto
Fault                               Enabled  Threshold  Action                Enable
-----
excessive-broadcasts                yes      5%          notify-and-disable    --
excessive-multicasts                yes      1000 pps    notify-and-disable    --
excessive-link-flaps                yes      7           notify-and-disable    --
excessive-oversize-packets          yes      25          notify-and-disable    --
excessive-jabbers                   yes      25          notify-and-disable    --
excessive-fragments                 yes      25          notify-and-disable    --
excessive-crc-errors                yes      25          notify-and-disable    --
excessive-late-collisions            yes      25          notify-and-disable    --
excessive-collisions                yes      25          notify-and-disable    --
excessive-tx-drops                  yes      25          notify-and-disable    --
excessive-alignment-errors          yes      25          notify-and-disable    --
-----
Fault monitor profile: quiet-ports
```

Auto Fault	Enabled	Threshold	Action	Enable
excessive-broadcasts	yes	20%	notify-and-disable	--
excessive-multicasts	yes	25000 pps	notify-and-disable	40
excessive-link-flaps	yes	7	notify	--
excessive-oversize-packets	yes	30	notify-and-disable	--
excessive-jabbers	no	30	notify-and-disable	100
excessive-fragments	yes	30	notify-and-disable	--
excessive-crc-errors	yes	30	notify-and-disable	--
excessive-late-collisions	yes	30	notify-and-disable	--
excessive-collisions	yes	30	notify-and-disable	--
excessive-tx-drops	yes	30	notify-and-disable	--
excessive-alignment-errors	yes	30	notify-and-disable	--

Showing information for a particular fault monitoring profile:

```
switch# show fault-monitor profile noisy-ports
```

Auto Fault	Enabled	Threshold	Action	Enable
excessive-broadcasts	yes	5%	notify-and-disable	--
excessive-multicasts	yes	1000 pps	notify-and-disable	--
excessive-link-flaps	yes	7	notify-and-disable	--
excessive-oversize-packets	yes	25	notify-and-disable	--
excessive-jabbers	yes	25	notify-and-disable	--
excessive-fragments	yes	25	notify-and-disable	--
excessive-crc-errors	yes	25	notify-and-disable	--
excessive-late-collisions	yes	25	notify-and-disable	--
excessive-collisions	yes	25	notify-and-disable	--
excessive-tx-drops	yes	25	notify-and-disable	--
excessive-alignment-errors	yes	25	notify-and-disable	--

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
4100i 6000 6100	Manager (#)	Administrators or local user group members with execution rights for this command.

show interface fault-monitor profile

show interface [<INTERFACE>|<IF-RANGE>] fault-monitor profile

Description

Shows fault monitoring profile configuration information for all or specific interfaces.

Parameter	Description
<INTERFACE>	Specifies a single interface.
<IF-RANGE>	Specifies a interface range,

Example

Showing all interfaces with applied fault monitoring profiles:

```
switch# show interface fault-monitor profile
```

```
-----  
Port      Fault Monitor Profile  
-----
```

```
1/1/1     noisy-ports  
1/1/2     quiet-ports  
1/1/4     quiet-ports  
1/1/5     noisy-ports  
1/1/6     noisy-ports  
1/1/7     quiet-ports
```

Showing a range of interfaces with applied fault monitoring profiles:

```
switch# show interface 1/1/1-1/1/2,1/1/6 fault-monitor profile
```

```
-----  
Port      Fault Monitor Profile  
-----
```

```
1/1/1     noisy-ports  
1/1/2     quiet-ports  
1/1/6     noisy-ports
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
4100i 6000 6100	Manager (#)	Administrators or local user group members with execution rights for this command.

show interface fault-monitor status

```
show interface [<INTERFACE>|<IF-RANGE>] fault-monitor status
```

Description

Shows active fault information for all or specific interfaces.

Parameter	Description
<INTERFACE>	Specifies a single interface.
<IF-RANGE>	Specifies a interface range,

Example

Showing active fault information for all interfaces with applied fault monitoring profiles:

```
switch# show interface fault-monitor status
```

Port	Fault	Fault Elapsed Time	Port State	Time Left
1/1/1	excessive-broadcasts	Tue Apr 14 14:29:09 UTC 2020	down	60
	excessive-jabbers	Tue Apr 15 14:29:09 UTC 2020	--	--
1/1/2	excessive-oversize-packets	Tue Apr 16 14:29:09 UTC 2020	down	--

Showing active fault information for a range of interfaces with applied fault monitoring profiles:

```
switch# show interface 1/3/1,1/3/3 fault-monitor status
```

Port	Fault	Occurring Since	Port State	Time Left
1/1/4	excessive-broadcasts	Tue Apr 14 14:29:09 UTC 2020	down	60
	excessive-jabbers	Tue Apr 15 14:29:09 UTC 2020	--	100

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
4100i 6000 6100	Manager (#)	Administrators or local user group members with execution rights for this command.

show running-config

```
show running-config [interface <IFNAME> | current-context | all]
```

Description

Displays the fault-monitor profile configurations and profile-name applied to an interface.

Parameter	Description
interface <IFNAME>	Specifies a single interface.
current-context	Displays only current context information.
all	Displays all options in the running config.

Example

Showing the running configuration for the fault monitoring profiles:

```
switch# show running-config
fault-monitor profile noisy-ports
  excessive-broadcasts
  excessive-broadcasts threshold pps 10000
  excessive-broadcasts action notify-and-disable auto-enable 2000
  excessive-multicasts
  excessive-multicasts threshold pps 10000
  excessive-link-flaps
  excessive-link-flaps action notify-and-disable auto-enable 2000
interface 1/1/1
  apply fault-monitor profile noisy-ports
```

Showing the running configuration with the all option:

```
switch# show running-config all
fault-monitor profile noisy-ports
  excessive-broadcasts
  excessive-broadcasts threshold pps 10000
  excessive-broadcasts action notify-and-disable auto-enable 2000
  excessive-multicasts
  excessive-multicasts threshold pps 10000
  excessive-multicasts action notify
  excessive-link-flaps
  excessive-link-flaps threshold count 7
  excessive-link-flaps action notify-and-disable auto-enable 2000
  no excessive-oversize-packets
  excessive-oversize-packets threshold value 25
  excessive-oversize-packets action notify
  no excessive-jabbers
  excessive-jabbers threshold value 25
  excessive-jabbers action notify
  no excessive-fragments
  excessive-fragments threshold value 25
  excessive-fragments action notify
  no excessive-crc-errors
  excessive-crc-errors threshold value 25
  excessive-crc-errors action notify
  no excessive-late-collisions
  excessive-late-collisions threshold value 25
  excessive-late-collisions action notify
  no excessive-collisions
  excessive-collisions threshold value 25
  excessive-collisions action notify
  no excessive-tx-drops
  excessive-tx-drops threshold value 25
  excessive-tx-drops action notify
  no excessive-alignment-errors
```

```
excessive-alignment-errors threshold value 25
excessive-alignment-errors action notify
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
4100i 6000 6100	Manager (#)	Administrators or local user group members with execution rights for this command.

threshold

```
<FAULT> threshold value <VALUE>
no <FAULT> threshold
```

```
excessive-link-flaps threshold count <COUNT>
no excessive-link-flaps threshold
```

```
{excessive-broadcasts | excessive-multicasts}
  threshold {percent <BW-PERCENT> | pps <PPS>}
no {excessive-broadcasts | excessive-multicasts} threshold
```

Description

Within the selected fault monitor profile context, configures the fault threshold value for the profile.

The `no` form of this command sets the threshold to its default value.

Parameter	Description
<FAULT>	Available fault names: excessive-jabbers excessive-crc-errors excessive-oversize-packets excessive-fragments excessive-tx-drops excessive-collisions excessive-late-collisions excessive-alignment-errors excessive-alignment-errors
threshold value <VALUE>	Specifies the fault threshold value. Default: 25.
threshold count <COUNT>	Specifies the fault threshold count. Default threshold count: 7.
threshold percent <BW-PERCENT>	Specifies the fault threshold bandwidth percentage. Range: 1 to 100. Default: 5.

Parameter	Description
threshold pps <PPS>	Specifies the fault threshold PPS (packets per second). Range: 1 to 195312500.

If excessive-broadcast or excessive-multicast faults are configured with the threshold higher than the `rate-limit` threshold, the following occurs:



- Fault reporting still happens as the port has actually received packets at a rate that violated its threshold.
- Traffic gets shaped as per `rate-limit` configuration and any packet exceeding the `rate-limit` threshold gets dropped.

Examples

Configuring with threshold values:

```
switch(config-fault-monitor-profile) # excessive-jabbers threshold value 30
switch(config-fault-monitor-profile) # excessive-oversize-packets threshold value 40
switch(config-fault-monitor-profile) # excessive-crc-errors threshold value 35
switch(config-fault-monitor-profile) # excessive-fragments threshold value 50
switch(config-fault-monitor-profile) # excessive-tx-drops threshold value 20
switch(config-fault-monitor-profile) # excessive-collisions threshold value 40
switch(config-fault-monitor-profile) # excessive-late-collisions threshold value 30
switch(config-fault-monitor-profile) # excessive-alignment-errors threshold value 50
```

Configuring with a threshold count:

```
switch(config-fault-monitor-profile) # excessive-link-flaps threshold count 10
```

Configuring with threshold percents and PPS:

```
switch(config-fault-monitor-profile) # excessive-broadcasts threshold percent 40
switch(config-fault-monitor-profile) # excessive-multicasts threshold pps 10000
```

Removing the configured threshold value for the faults:

```
switch(config-fault-monitor-profile) # no excessive-jabbers threshold
switch(config-fault-monitor-profile) # no excessive-oversize-packets threshold
switch(config-fault-monitor-profile) # no excessive-crc-errors threshold
switch(config-fault-monitor-profile) # no excessive-fragments threshold
switch(config-fault-monitor-profile) # no excessive-tx-drops threshold
switch(config-fault-monitor-profile) # no excessive-collisions threshold
switch(config-fault-monitor-profile) # no excessive-late-collisions threshold
switch(config-fault-monitor-profile) # no excessive-alignment-errors threshold
switch(config-fault-monitor-profile) # no excessive-link-flaps threshold
switch(config-fault-monitor-profile) # no excessive-broadcasts threshold
switch(config-fault-monitor-profile) # no excessive-multicasts threshold
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config-fault-monitor-profile	Administrators or local user group members with execution rights for this command.

vsx-sync (fault monitor)

vsx-sync
no vsx-sync

Description

Within the selected fault monitor profile context, configures VSX synchronization for the selected fault monitoring profile.

The `no` form of this command removes the VSX synchronization for a fault monitoring profile.

Example

Configuring VSX synchronization for a fault monitoring profile:

```
switch(config-fault-monitor-profile)# vsx-sync
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config-fault-monitor-profile	Administrators or local user group members with execution rights for this command.

copy {primary | secondary} <REMOTE-URL>

Syntax

```
copy {primary | secondary} <REMOTE-URL> [vrf <VRF-NAME>]
```

Description

Uploads a firmware image to a TFTP or SFTP server.

Command context

Manager (#)

Parameters

{primary | secondary}

Selects the primary or secondary image profile to upload. Required

<REMOTE-URL>

Specifies the URL to receive the uploaded firmware using SFTP or TFTP. For information on how to format the remote URL, see URL formatting for copy commands.

vrf <VRF-NAME>

Specifies a VRF name. Default: default.

Authority

Administrators or local user group members with execution rights for this command.

Examples

TFTP upload:

```
switch# copy primary tftp://192.0.2.0/00_10_00_0002.swi
##### 100.0%
Verifying and writing system firmware...
```

SFTP upload:

```
switch# copy primary sftp://swuser@192.0.2.0/00_10_00_0002.swi
swuser@192.0.2.0's password:
Connected to 192.0.2.0.
sftp> put primary.swi XL_10_00_0002.swi
Uploading primary.swi to /users/swuser/00_10_00_0002.swi
primary.swi          100% 179MB 35.8MB/s 00:05
```

copy {primary | secondary} <FIRMWARE-FILENAME>

Syntax

```
copy {primary | secondary} <FIRMWARE-FILENAME>
```

Description

Copies a firmware image to USB storage.

Command context

Manager (#)

Parameters

```
{primary | secondary}
```

Selects the primary or secondary image from which to copy the firmware. Required

```
<FIRMWARE-FILENAME>
```

Specifies the name of the firmware file to create on the USB storage device. Prefix the filename with `usb:/`. For example: `usb:/firmware_v1.2.3.swi`

For information on how to format the path to a firmware file on a USB drive, see USB URL.

Authority

Administrators or local user group members with execution rights for this command.

Examples

```
switch# copy primary usb:/11.10.00.0002.swi
```

copy primary secondary

Syntax

```
copy primary secondary
```

Description

Copies the firmware image from the primary to the secondary location.

Command context

Manager (#)

Authority

Administrators or local user group members with execution rights for this command.

Examples

```
switch# copy primary secondary
The secondary image will be deleted.

Continue (y/n)? y
Verifying and writing system firmware...
```

copy <REMOTE-URL>

Syntax

```
copy <REMOTE-URL> {primary | secondary} [vrf <VRF-NAME>]
```

Description

Downloads and installs a firmware image from a TFTP or SFTP server.

Command context

Manager (#)

Parameters

<REMOTE-URL>

Specifies the URL from which to download the firmware using SFTP or TFTP. For information on how to format the remote URL, see URL formatting for copy commands.

{primary | secondary}

Selects the primary or secondary image profile for receiving the downloaded firmware. Required.

vrf <VRF-NAME>

Specifies the name of a VRF. Default: default.

Authority

Administrators or local user group members with execution rights for this command.

Examples

TFTP download:

```
switch# copy tftp://192.10.12.0/ss.10.00.0002.swi primary
The primary image will be deleted.

Continue (y/n)? y
##### 100.0%
Verifying and writing system firmware...
```

SFTP download:

```
switch# copy sftp://swuser@192.10.12.0/ss.10.00.0002.swi primary
The primary image will be deleted.

Continue (y/n)? y
The authenticity of host '192.10.12.0 (192.10.12.0)' can't be established.
ECDSA key fingerprint is SHA256:L64khLwlyLgXlARKRMiwcAAK8oRaQ8C0oWP+PkGBXHY.
Are you sure you want to continue connecting (yes/no)? yes
Warning: Permanently added '192.10.12.0' (ECDSA) to the list of known hosts.
swuser@192.10.12.0's password:
Connected to 192.10.12.0.
Fetching /users/swuser/ss.10.00.0002.swi to ss.10.00.0002.swi.dnld
/users/swuser/ss.10.00.0002.swi          100% 179MB 25.6MB/s   00:07
Verifying and writing system firmware...
```

copy secondary primary

Syntax

```
copy secondary primary
```

Description

Copies the firmware image from the secondary to the primary location.

Command context

Manager (#)

Authority

Administrators or local user group members with execution rights for this command.

Examples

```
switch# copy secondary primary
The primary image will be deleted.

Continue (y/n)? y
Verifying and writing system firmware...
```

```
switch# copy sftp://stor@192.22.1.0/im-switch.swi primary vrf default
The primary image will be deleted.

Continue (y/n)? y
The authenticity of host '192.22.1.0 (192.22.1.0)' can't be established.
ECDSA key fingerprint is SHA256:MyI1xbdKnehYut0NLfL69gDpNzCmZqBVvBaRR46m7o8.
Are you sure you want to continue connecting (yes/no)? yes
Warning: Permanently added '192.22.1.0' (ECDSA) to the list of known hosts.
stor@192.22.1.0's password:
Connected to 192.22.1.0.
sftp> get c8d5b9f-topflite.swi c8d5b9f-topflite.swi.dnld
Fetching /home/dr/im-switch.swi to c8d5b9f-topflite.swi.dnld
/home/dr/im-switch.swi          100% 226MB 56.6MB/s 00:04

Verifying and writing system firmware...
```

copy <STORAGE-URL>

Syntax

```
copy <STORAGE-URL> {primary | secondary}
```

Description

Copies, verifies, and installs a firmware image from a USB storage device connected to the active management module.

Command context

Manager (#)

Parameters

<STORAGE-URL>

Specifies the name of the firmware file to copy from the USB storage device. Required. Prefix the filename with `usb:/`. For example, `usb:/firmware_v1.2.3.swi` For information on how to format the path to a firmware file on a USB drive, see [USB URL](#).

{primary | secondary}

Selects the primary or secondary image profile for receiving the copied firmware.

Authority

Administrators or local user group members with execution rights for this command.

Examples

```
switch# copy usb:/11.10.00.0002.swi primary
The primary image will be deleted.

Continue (y/n)? y

Verifying and writing system firmware...
```

URL formatting for copy commands

TFTP URL

Syntax

tftp://<IP-ADDR>[:<PORT-NUM>][;blocksize=<Value>]/<FILENAME>

Examples

To specify a URL with:

- an IPv4 address: tftp://1.1.1.1/a.txt
- an IPv6 address: tftp://[2000::2]/a.txt
- a hostname: tftp://hpe.com/a.txt

To specify TFTP with:

- the port number of the server in the URL: tftp://1.1.1.1:12/a.txt
- the blocksize in the URL: tftp://1.1.1.1;blocksize=1462/a.txt

The valid blocksize range is 8 to 65464.

- the port number of the server and blocksize in the URL: tftp://1.1.1.1:12;blocksize=1462/a.txt

To specify a file in a directory of URL: tftp://1.1.1.1/dir/a.txt

SFTP URL

Syntax

sftp://<USERNAME>@<IP-ADDR>[:<PORT-NUM>]/<FILENAME>

Examples

To specify:

- A URL with an IPv4 address: sftp://user@1.1.1.1/a.txt
- A URL with an IPv6 address: sftp://user@[2000::2]/a.txt
- A URL with a hostname: sftp://user@hpe.com/a.txt
- SFTP port number of a server in the URL: sftp://user@1.1.1.1:12/a.txt
- A file in a directory of URL: sftp://user@1.1.1.1/dir/a.txt
- To specify a file with absolute path in the URL: sftp://user@1.1.1.1//home/user/a.txt

USB URL

Syntax

`usb: /<FILENAME>`

Examples

To specify a file:

- In a USB storage device: `usb: /a.txt`
- In a directory of a USB storage device: `usb: /dir/a.txt`

https-server rest access-mode

Syntax

```
https-server rest access-mode {read-only | read-write}
```

Description

Changes the REST API access mode. The default mode is read-write.

Command context

config

Parameters

read-write

Selects the read/write mode. Allows POST, PUT, and DELETE methods to be called on all configurable elements in the switch database.

read-only

Selects the read-only mode. Write access to most switch resources through the REST API is disabled.

Authority

Administrators or local user group members with execution rights for this command.

Usage

Setting the mode to `read-write` on the REST API allows POST, PUT, and DELETE methods to be called on all configurable elements in the switch database.

By default, REST APIs in the device are in the read-write mode. Some switch resources allow POST, PUT, and DELETE regardless of REST API mode. REST APIs that are required to support the Web UI or the Network Analytics Engine expose POST, PUT, or DELETE operations, even if the REST API access mode is set to read-only.

The REST API in read/write mode is intended for use by advanced programmers who have a good understanding of the system schema and data relationships in the switch database.



Because the REST API in read/write mode can access every configurable element in the database, it is powerful but must be used with extreme caution: No semantic validation is performed on the data you write to the database, and configuration errors can destabilize the switch.

Example

```
switch(config)# https-server rest access-mode read-only
```

https-server session close all

Syntax

```
https-server session close all
```

Description

Invalidates and closes all HTTPS sessions. All existing Web UI and REST sessions are logged out and all real-time notification feature WebSocket connections are closed.

Command context

Manager (#)

Authority

Administrators or local user group members with execution rights for this command.

Usage

Typically, a user that has consumed the allowed concurrent HTTPS sessions and is unable to access the session cookie to log out manually must wait for the session idle timeout to start another session. This command is intended as a workaround to waiting for the idle timeout to close an HTTPS session. This command stops and starts the `hpe-restd` service, so using this command affects all existing REST sessions, Web UI sessions, and real-time notification subscriptions.

Example

```
switch# https-server session close all
```

https-server vrf



Since the Aruba 6100 Switch Series only supports the `default` VRF, this command does not apply to the 6100.

Syntax

```
https-server vrf <VRF-NAME>  
no https-server vrf <VRF-NAME>
```

Description

Configures and starts the HTTPS server on the specified VRF. HTTPS server features include the REST API and the web user interfaces.

The `no` form of the command stops any HTTPS servers running on the specified VRF and removes the HTTPS server configuration.

Command context

config

Parameters

<VRF-NAME>

Specifies the VRF name. Required. Length: Up to 32 alpha numeric characters.

Authority

Administrators or local user group members with execution rights for this command.

Usage

By using this command, you enable access to both the Web UI and to the REST API on the specified VRF. You can enable access on multiple VRFs.

When the HTTPS server is not configured and running, attempts to access the Web UI or REST API result in 404 Not Found errors.

The VRF you select determines from which network the Web UI and REST API can be accessed.

For example:

- If you want to enable access to the REST API and Web UI through the OOBM port (management IP address), specify the built-in management VRF (`mgmt`).
- If you want to enable access to the REST API and Web UI through the data ports (for "inband management"), specify the built-in default VRF (`default`).
- If you want to enable access to the REST API and Web UI through only a subset of data ports on the switch, specify other VRFs you have created.

Aruba Network Analytics Engine scripts run in the default VRF, but you do not have to enable HTTPS server access on the default VRF for the scripts to run. If the switch has custom Aruba Network Analytics Engine scripts that require access to the Internet, then for those scripts to perform their functions, you must configure a DNS name server on the default VRF.

Examples

Enabling access on all ports on the switch, specify the default VRF:

```
switch(config)# https-server vrf default
```

Enabling access on the OOBM port (management interface IP address), specify the management VRF:

```
switch(config)# https-server vrf mgmt
```

Enabling access on ports that are members of the VRF named `vrfprogs`, specify `vrfprogs`:

```
switch(config)# https-server vrf vrfprogs
```

Enabling access on the management port and ports that are members of the VRF named `vrfprogs`, enter two commands:

```
switch(config)# https-server vrf mgmt
switch(config)# https-server vrf vrfprogs
```

show https-server

Syntax

```
show https-server
```

Description

Shows the status and configuration of the HTTPS server. The REST API and web user interface are accessible only on VRFs that have the HTTPS server features configured.

Command context

Manager (#)

Authority

Administrators or local user group members with execution rights for this command.

Usage

Shows the configuration of the HTTPS server features.

VRF

Shows the VRFs, if any, for which HTTPS server features are configured.

REST Access Mode

Shows the configuration of the REST access mode:

read-write

POST, PUT, and DELETE methods can be called on all configurable elements in the switch database. This is the default value.

read-only

Write access to most switch resources through the REST API is disabled.

Examples

```
switch# show https-server

HTTPS Server Configuration
-----

VRF                : default, mgmt
REST Access Mode   : read-write

Max sessions per user : 6

Session timeout      : 20
```


ip icmp redirect

Syntax

```
ip icmp redirect  
no ip icmp redirect
```

Description

Enables the sending of ICMPv4 and ICMPv6 redirect messages to the source host. Enabled by default. The `no` form of this command disables ICMPv4 and ICMPv6 redirect messages to the source host.

Command context

config

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling ICMP redirect messages:

```
switch(config)# ip icmp redirect
```

Disabling ICMP redirect messages:

```
switch(config)# no ip icmp redirect
```

ip icmp throttle

Syntax

```
ip icmp throttle <packet-interval>  
no ip icmp throttle
```

Description

Used to configure the throttle parameter for both ICMPv4 and ICMPv6 error messages and redirect messages.

The `no` form of this command disables the throttle parameter for both ICMPv4 and ICMPv6 error messages and redirect messages.

Command context

config

Parameters

<packet-interval>

Specifies the ICMPv4/v6 packet interval in seconds. Default: 1 second. Range: 1-86400.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling the throttle parameter for both ICMPv4 and ICMPv6 error messages and redirect messages:

```
switch(config)# ip icmp throttle 3000
```

Disabling the throttle parameter for both ICMPv4 and ICMPv6 error messages and redirect messages:

```
switch(config)# no ip icmp throttle
```

ip icmp unreachable

Syntax

```
ip icmp unreachable  
no ip icmp unreachable
```

Description

Enables the sending of ICMPv4 and ICMPv6 destination unreachable messages on the switch to a source host when a specific host is unreachable. The unreachable host address originates from the failed packet. Default setting.

The `no` form of this command disables the sending of ICMPv4 and ICMPv6 destination unreachable messages from the switch to a source host when a specific host is unreachable. This command does not prevent other hosts from sending an ICMP unreachable message.

Command context

config

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling ICMPv4 and ICMPv6 destination unreachable messages to a source host:

```
switch(config)# ip icmp unreachable
```

Disabling ICMPv4 and ICMPv6 destination unreachable messages to a source host:

```
switch(config)# no ip icmp unreachable
```

For commands in the interface configuration context, the interface must be an L3 interface. The supported contexts includes: `config-if-vlan`.

ip igmp

Syntax

```
ip igmp {enable | disable}
```

Description

Enables or disables IGMP on the current interface. IGMP is disabled by default.

Command context

```
config-if-vlan
```

Parameters

`enable`

Enable IGMP.

`disable`

Disable IGMP.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling IGMP on interface VLAN 2:

```
switch(config)# interface vlan 2  
switch(config-if-vlan)# ip igmp enable
```

Disabling IGMP on interface VLAN 2:

```
switch(config)# interface vlan 2  
switch(config-if-vlan)# ip igmp disable
```

ip igmp apply access-list

Syntax

```
ip igmp apply access-list <ACL-NAME>  
no ip igmp apply access-list <ACL-NAME>
```

Description

Configures the ACL on a particular interface to filter the IGMP join or leave packets based on rules set in the particular ACL name.

The `no` form of this command unconfigures the rules set for the ACL.



This configuration will override the ACL associated with IGMP snooping on the corresponding L2 VLAN.

Command context

`config-if-vlan`

Parameters

`access-list`

Associates an ACL with the IGMP.

`<ACL-NAME>`

Specifies the name of the ACL.

Authority

Administrators or local user group members with execution rights for this command.

Usage

Existing classifier commands are used to configure the ACL. In case an IGMPv3 packet with multiple group addresses is received, it will only process the permitted group addresses based on the ACL rule set, and any existing joins will time out. If there is no match or if there is a deny rule match, the packet is dropped.

Examples

Configuring the ACL on a VLAN to filter IGMP packets based on rules set in access list `mygroup`:

```
switch(config)# access-list ip mygroup
switch(config-acl-ip)# permit igmp any 239.1.1.1
switch(config-acl-ip)# exit
switch(config)# interface vlan 2
switch(config-if-vlan)# ip igmp apply access-list mygroup
```

Configuring the ACL to remove the rules set in access list `mygroup`:

```
switch(config-if-vlan)# no ip igmp apply access-list mygroup
```

ip igmp last-member-query-interval

Syntax

```
ip igmp last-member-query-interval <INTERVAL-VALUE>
no ip igmp last-member-query-interval <INTERVAL-VALUE>
```

Description

Configures an IGMP last member query interval value in seconds on an interface, depending on the command context you are in.

The `no` form of this command sets the value to a default of 1 second on an interface.

Command context

config-if-vlan

Parameters

<INTERVAL-VALUE>

Specifies an IGMP last-member-query-interval on the interface. Default: 1 second. Range: 1-2 seconds.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring an IGMP last member query interval of 2 on interface VLAN 2:

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ip igmp last-member-query-interval 2
switch(config-if-vlan)# no ip igmp last-member-query-interval
```

ip igmp querier

Syntax

```
ip igmp querier
no ip igmp querier
```

Description

Configures an IGMP querier on an interface, depending on the command context you are in. This functionality will allow an interface to join in the querier-election process.

The `no` form of this command disables IGMP querier on an interface.

Command context

config-if-vlan

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring an IGMP querier on interface VLAN 2:

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ip igmp querier
switch(config-if-vlan)# no ip igmp querier
```

ip igmp querier interval

Syntax

```
ip igmp querier interval <INTERVAL-VALUE>
no ip igmp querier interval
```

Description

Configures the interval between IGMP queries on an interface, depending on the command context you are in.

The `no` form of this command sets the IGMP querier interval to the default value of 125 seconds on an interface.

Command context

`config-if-vlan`

Parameters

`<INTERVAL-VALUE>`

Specifies the IGMP querier interval in seconds on the interface. Default: 125 seconds. Range: 5-300.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring an IGMP querier interface interval of 100 on interface VLAN 2:

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ip igmp querier interval 100
```

Resetting an IGMP querier interval to the default value:

```
switch(config-if-vlan)# no ip igmp querier interval
```

ip igmp querier query-max-response-time

Syntax

```
ip igmp querier query-max-response-time <RESPONSE-TIME>
no ip igmp querier query-max-response-time <RESPONSE-TIME>
```

Description

Configures the IGMP querier max response time value in seconds on an interface, depending on the command context you are in.

The `no` form of this command sets the querier max response time value to the default of 10 seconds on an interface.

Command context

`config-if-vlan`

Parameters

`<RESPONSE-TIME>`

Specifies the IGMP querier max response time value on the interface. Default: 10 seconds. Range: 10-128 seconds.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring the IGMP querier maximum response time of 50 for interface VLAN 2:

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ip igmp query-max-response-time 50
```

Resetting an IGMP querier interval to the default value:

```
switch(config-if-vlan)# no ip igmp query-max-response-time
```

ip igmp robustness

Syntax

```
ip igmp robustness <VALUE>
no ip igmp robustness <VALUE>
```

Description

Configures IGMP robustness on an interface, depending on the command context. The robustness parameter allows tuning for the expected packet loss on a subnet.

The `no` form of this command sets the robustness value to the default of 2 on an interface.

Command context

config-if-vlan

Parameters

<VALUE>

Specifies an IGMP robustness value on the interface. Default: 2. Range: 1-7.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring an IGMP robustness of 5 on interface VLAN 2:

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ip igmp robustness 5
```

Resetting the IGMP robustness to the default:

```
switch(config-if-vlan)# no ip igmp robustness
```

ip igmp static-group

Syntax

```
ip igmp static-group <MULTICAST-GROUP-IP>
no ip igmp static-group <MULTICAST-GROUP-IP>
```

Description

Configures an IGMP static multicast group on an interface, depending on the command context you are in. You can configure a maximum of 32 IGMP static groups.

The `no` form of the command unconfigures IGMP static multicast group on an interface.

Command context

config-if-vlan

Parameters

<MULTICAST-GROUP-IP>

Specifies an IGMP static multicast group IP address on the interface. Format: A.B.C.D

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring an IGMP static group on interface VLAN 2:

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ip igmp static-group 239.1.1.1
```

Resetting an IGMP static group on an interface to the default (none):

```
switch(config-if)# no ip igmp static-group 239.1.1.10
```

ip igmp version

Syntax

ip igmp version <VERSION>

Description

Configures the IGMP version on an interface, depending on the command context you are in.

Command context

config-if-vlan

Parameters

<VERSION>

Specifies the IGMP version on the interface. Select 2 for IGMPv2 (RFC2236). Select 3 for IGMPv3 (RFC3376). Values: 2 or 3.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring an IGMP version on interface VLAN 2:


```
switch(config)# interface vlan 2  
switch(config-if-vlan)# ip igmp version 2
```

ip igmp version strict

Syntax

```
ip igmp version <VERSION> strict  
no ip igmp version <VERSION> strict
```

Description

Configures an IGMP strict version on an interface, depending on the command context you are in. Drops packets that do not match the configured version.

The `no` form of the command removes the strict version configuration from the interface.

Command context

config-if-vlan

Parameters

<VERSION>

Specifies the IGMP version on the interface. Select 2 for IGMPv2 (RFC2236). Select 3 for IGMPv3 (RFC3376). Values: 2 or 3.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring the IGMP strict version to 2 on interface VLAN 2:

```
switch(config)# interface vlan 2  
switch(config-if-vlan)# ip igmp version 2 strict
```

Resetting the IGMP strict version to the default (none):

```
switch(config-if)# no ip igmp version 2 strict
```

no ip igmp

Syntax

```
no ip igmp
```

Description

Disables all IGMP configurations on an interface, depending on the command context you are in.

Command context

config-if-vlan

Authority

Administrators or local user group members with execution rights for this command.

Examples

Removes IGMP on interface VLAN 2:

```
switch(config)# interface vlan 2
switch(config-if-vlan)# no ip igmp
```

show ip igmp

Syntax

```
show ip igmp [all-vrfs]
```

Description

Shows IGMP configuration information and status, or shows information by VRF.

Command context

Operator (>) or Manager (#)

Parameters

all-vrfs

To show information for all VRFs, specify `all-vrfs`.

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Examples

Showing IGMP configuration and status:

```
switch# show ip igmp

VRF Name   : default
Interface  : vlan2
IGMP Configured Version   : 3
IGMP Operating Version    : 3
Querier State              : Querier
Querier IP [this switch]  : 20.1.1.1
Querier Uptime             : 1m 4s
Querier Expiration Time   : 0m 1s
IGMP Snoop Enabled on VLAN : True
```

Showing IGMP information for all VRFs:

```
switch# show ip igmp all-vrfs

VRF Name   : default
Interface  : vlan5
IGMP Configured Version   : 3
IGMP Operating Version    : 2
Querier State              : Querier
```

```
Querier IP [this switch] : 50.1.1.1
Querier Uptime           : 1m 1s
Querier Expiration Time  : 0m 4s
IGMP Snoop Enabled on VLAN : False
```

show ip igmp counters

Syntax

```
show ip igmp counters [all-vrfs]
```

Description

Shows IGMP counter details, or shows counters by VRF.

Command context

Operator (>) or Manager (#)

Parameters

`all-vrfs`

Specify `all-vrfs` to show information for all VRFs.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Showing IGMP counters:

```
switch# show ip igmp counters

IGMP Counters

Interface Name      : vlan2
VRF Name            : default
Membership Timeout  : 0

                                     Rx          Tx
                                     -----
V1 All Hosts Queries      0          0
V2 All Hosts Queries      0          12
V3 All Hosts Queries      0          0
V2 Group Specific Queries  0          0
V3 Group Specific Queries  0          0
Group And Source Specific Queries  0          0
V3 Member Reports         0          N/A
V2 Member Reports         0          N/A
V1 Member Reports         0          N/A
V2 Member Leaves          0          N/A
Packets dropped by ACL     0          N/A
```

Showing IGMP counters for the default VRF:

```
switch# show ip igmp counters vrf default
```

IGMP Counters

```
Interface Name      : vlan2
VRF Name            : default
Membership Timeout   : 0
```

	Rx	Tx
	-----	-----
V1 All Hosts Queries	0	0
V2 All Hosts Queries	0	12
V3 All Hosts Queries	0	0
V2 Group Specific Queries	0	0
V3 Group Specific Queries	0	0
Group And Source Specific Queries	0	0
V3 Member Reports	0	N/A
V2 Member Reports	0	N/A
V1 Member Reports	0	N/A
V2 Member Leaves	0	N/A
Packets dropped by ACL	0	N/A

show ip igmp group

Syntax

```
show ip igmp group <GROUP-IP> [source <SOURCE-IP>] [all-vrfs]
```

Description

Shows IGMP joined group information for the specified group, or shows joined group source and display information by VRF.

Command context

Operator (>) or Manager (#)

Parameters

<GROUP-IP>

Specifies the IP address of the group. Format: A.B.C.D

source <SOURCE-IP>

Specifies the IP address of the source. Format: A.B.C.D

all-vrfs

Specify `all-vrfs` to show information for all VRFs.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Showing IGMP joined group details for group 239.1.1.10:

```
switch# show ip igmp group 239.1.1.10
```

```
IGMP group information for group 239.1.1.10
```

```

Interface Name      : vlan2
VRF Name            : default

Group Address       : 239.1.1.10
Last Reporter       : 100.1.1.10

Vers  Mode  Uptime    Expires    V1      V2      Sources  Sources
-----  -
3      EXC   16m 34s    2m 27s    Timer   Timer   Forwarded Blocked

```

Showing IGMP joined group details for group 239.1.1.10 and source 10.1.1.10:

```

switch# show ip igmp group 239.1.1.10 source 10.1.1.10

Interface Name      : vlan2
VRF Name            : default
Group Address       : 239.1.1.10
Source Address      : 10.1.1.10

Mode  Uptime    Expire
-----
      0m 13s    4m 7s

```

Showing IGMP joined group details for group 239.1.1.10 for all VRFs:

```

switch# show ip igmp group 239.1.1.10 all-vrfs

IGMP group information for group 239.1.1.10

Interface Name      : vlan10
VRF Name            : default

Group Address       : 239.1.1.10
Last Reporter       : 100.1.1.10

Vers  Mode  Uptime    Expires    V1      V2      Sources  Sources
-----  -
3      EXC   17m 5s    4m 2s    Timer   Timer   Forwarded Blocked

```

Showing IGMP joined group details for group 239.1.1.10 source 10.1.1.10 for all VRFs:

```

switch# show ip igmp group 239.1.1.10 source 10.1.1.10 all-vrfs

Interface Name      : vlan10
VRF Name            : default
Group Address       : 239.1.1.10
Source Address      : 10.1.1.10

Mode  Uptime    Expire
-----
      0m 39s    3m 41s

```

Showing IGMP joined group details group 239.1.1.10 for the default VRF:

```
switch# show ip igmp group 239.1.1.10 vrf default
```

IGMP group information for group 239.1.1.10

Interface Name : vlan2
VRF Name : default

Group Address : 239.1.1.10
Last Reporter : 100.1.1.10

Vers	Mode	Uptime	Expires	V1 Timer	V2 Timer	Sources Forwarded	Sources Blocked
3	EXC	17m 35s	3m 32s				

Showing IGMP joined group details group 239.1.1.10 source 10.1.1.10 for the default VRF:

```
switch# show ip igmp group 239.1.1.10 source 10.1.1.10 vrf default
```

Interface Name : vlan10
VRF Name : default
Group Address : 239.1.1.10
Source Address : 10.1.1.10

Mode	Uptime	Expire
	0m 59s	3m 21s

show ip igmp groups

Syntax

```
show ip igmp groups [all-vrfs]
```

Description

Shows IGMP group information, or you can display group information by VRF.

Command context

Operator (>) or Manager (#)

Parameters

all-vrfs

Specify `all-vrfs` to show information for all VRFs.

Authority

Operators or Administrators or local user group members with execution rights for this command.

Operators can execute this command from the operator context (>) only.

Examples

Showing IGMP group information:

```
switch# show ip igmp groups
```

```
IGMP group information for group 239.1.1.10
```

```
Interface Name   : vlan2  
VRF Name         : default
```

```
Group Address    : 239.1.1.10  
Last Reporter    : 100.1.1.10
```

Vers	Mode	Uptime	Expires	V1 Timer	V2 Timer	Sources Forwarded	Sources Blocked
3	EXC	0m 36s	3m 44s				

```
IGMP group information for group 239.1.1.11
```

```
Interface Name   : vlan2  
VRF Name         : default
```

```
Group Address    : 239.1.1.11  
Last Reporter    : 100.1.1.10
```

Vers	Mode	Uptime	Expires	V1 Timer	V2 Timer	Sources Forwarded	Sources Blocked
3	EXC	0m 36s	3m 44s				

Showing IGMP groups for all VRFs:

```
switch# show ip igmp groups all-vrfs
```

```
IGMP group information for group 239.1.1.1
```

```
Interface Name   : vlan20  
VRF Name         : default
```

```
Group Address    : 239.1.1.1  
Last Reporter    : 200.1.1.10
```

Vers	Mode	Uptime	Expires	V1 Timer	V2 Timer	Sources Forwarded	Sources Blocked
3	EXC	0m 13s	4m 7s				

```
IGMP group information for group 239.1.1.2
```

```
Interface Name   : vlan20  
VRF Name         : default
```

```
Group Address    : 239.1.1.2  
Last Reporter    : 200.1.1.10
```

Vers	Mode	Uptime	Expires	V1 Timer	V2 Timer	Sources Forwarded	Sources Blocked
3	EXC	0m 13s	4m 7s				

Showing IGMP groups for the default VRF:

```
switch# show ip igmp groups vrf default

IGMP group information for group 239.1.1.10

Interface Name      : vlan2
VRF Name            : default

Group Address       : 239.1.1.10
Last Reporter       : 100.1.1.10

Vers  Mode  Uptime      Expires      V1          V2          Sources     Sources
-----  -
3      EXC    9m 23s      3m 20s      Timer       Timer       Forwarded   Blocked

IGMP group information for group 239.1.1.11

Interface Name      : vlan2
VRF Name            : default

Group Address       : 239.1.1.11
Last Reporter       : 100.1.1.10

Vers  Mode  Uptime      Expires      V1          V2          Sources     Sources
-----  -
3      EXC    9m 23s      3m 20s      Timer       Timer       Forwarded   Blocked
```

show ip igmp interface

Syntax

```
show ip igmp interface {vlan <VLAN-ID>}
```

Description

Shows IGMP configuration information for a specific interface (VLAN).

Command context

Operator (>) or Manager (#)

Parameters

vlan <VLAN-ID>

Specifies a VLAN. Values: 1-4094.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Showing IGMP configuration information for interface VLAN 2:

```
switch# show ip igmp interface vlan 2

IGMP Configured Version : 3
```



```
IGMP Operating Version   : 3
Querier State            : Querier
Querier IP [this switch] : 20.1.1.1
Querier Uptime           : 1m 46s
Querier Expiration Time  : 0m 1s
Snoop Enabled on VLAN    : True
```

show ip igmp interface counters

Syntax

```
show ip igmp interface {vlan <VLAN-ID>} counters
```

Description

Shows IGMP counter details for a specific interface or VLAN interface.

Command context

Operator (>) or Manager (#)

Parameters

vlan <VLAN-ID>

Specifies a VLAN. Values: 1-4094.

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Examples

Showing IGMP counters for interface VLAN 2:

```
switch# show ip igmp interface vlan 2 counters
```

IGMP Counters

```
Interface Name      : vlan2
VRF Name            : default
Membership Timeout   : 0
```

	Rx	Tx
	-----	-----
V1 All Hosts Queries	0	0
V2 All Hosts Queries	0	0
V3 All Hosts Queries	0	29
V2 Group Specific Queries	0	0
V3 Group Specific Queries	0	2
Group And Source Specific Queries	0	2
V3 Member Reports	0	N/A
V2 Member Reports	0	N/A
V1 Member Reports	0	N/A
V2 Member Leaves	0	N/A
Packets dropped by ACL	0	N/A

show ip igmp interface group

Syntax

```
show ip igmp interface {vlan <VLAN-ID>} group <GROUP-ID> [source <SOURCE-IP>]
```

Description

Shows IGMP joined group information for a specific interface or VLAN interface, or specify a source IP.

Command context

Operator (>) or Manager (#)

Parameters

vlan <VLAN-ID>

Specifies a VLAN. Values: 1-4094.

<GROUP-ID>

Specifies the IP address of the group. Format: A.B.C.D

source <SOURCE-IP>

Specifies the IP address of the source. Format: A.B.C.D

Authority

Operators or Administrators or local user group members with execution rights for this command.

Operators can execute this command from the operator context (>) only.

Examples

Showing IGMP joined group details for group 239.1.1.1 for interface VLAN 10:

```
switch# show ip igmp interface vlan 10 group 239.1.1.1

IGMP group information for group 239.1.1.1

Interface Name   : vlan10
VRF Name        : default

Group Address    : 239.1.1.1
Last Reporter    : 100.1.1.10

Vers  Mode  Uptime    Expires    V1      V2      Sources  Sources
-----  -
3      INC    8m 10s    2m 21s    Timer   Timer   Forwarded Blocked
-----  -

Group Address   : 239.1.1.1
Source Address  : 10.1.1.1

Mode  Uptime    Expire
-----  -
INC   8m 10s    2m 21s
```

Showing IGMP joined group details for group 239.1.1.1 for interface VLAN 10 with source details for 10.1.1.1:

```
switch# show ip igmp interface vlan 10 group 239.1.1.1 source 10.1.1.1

Interface Name   : vlan10
VRF Name        : default
```

```
Group Address : 239.1.1.1
Source Address : 10.1.1.1
```

Mode	Uptime	Expire
INC	8m 52s	3m 51s

show ip igmp interface groups

Syntax

```
show ip igmp interface {vlan <VLAN-ID>} groups
```

Description

Shows IGMP group information for a specific interface or VLAN interface.

Command context

Operator (>) or Manager (#)

Parameters

vlan <VLAN-ID>

Specifies a VLAN. Values: 1-4094.

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Examples

Showing IGMP groups for interface VLAN 2:

```
switch# show ip igmp interface vlan 2 groups

IGMP group information for group 239.1.1.1

Interface Name      : vlan2
VRF Name            : default

Group Address       : 239.1.1.1
Last Reporter       : 100.1.1.10

Vers  Mode  Uptime    Expires    V1          V2          Sources    Sources
-----  ---  -----  -
3      INC   4m 40s    3m 51s    Timer       Timer       Forwarded  Blocked
-----  ---  -----  -
1

Group Address       : 239.1.1.1
Source Address      : 10.1.1.1

Mode  Uptime    Expire
-----
INC   4m 40s    3m 51s

IGMP group information for group 239.1.1.2
```

```

Interface Name      : vlan2
VRF Name            : default

Group Address       : 239.1.1.2
Last Reporter       : 100.1.1.10

Vers  Mode  Uptime      Expires    V1      V2      Sources  Sources
-----  -
3      INC   4m 40s      3m 51s    Timer   Timer   Forwarded Blocked
-----  -

Group Address       : 239.1.1.2
Source Address      : 10.1.1.1

Mode  Uptime      Expire
-----  -
INC   4m 40s      3m 51s

```

show ip igmp interface statistics

Syntax

```
show ip igmp interface {vlan <VLAN-ID>} statistics
```

Description

Shows IGMP statistics for a specific interface or VLAN interface, including groups joined.

Command context

Operator (>) or Manager (#)

Parameters

vlan <VLAN-ID>

Specifies a VLAN. Values: 1-4094.

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Examples

Showing IGMP statistics for interface VLAN 2:

```

switch# show ip igmp interface vlan 2 statistics

IGMP statistics

Interface Name : vlan2
VRF Name       : default

Number of Include Groups      : 2
Number of Exclude Groups     : 0
Number of Static Groups      : 0
Total Multicast Groups Joined : 2

```

show ip igmp static-groups

Syntax

```
show ip igmp static-groups [all-vrfs]
```

Description

Shows IGMP static groups, or shows information by VRF.

Command context

Operator (>) or Manager (#)

Parameters

`all-vrfs`

Specify `all-vrfs` to show information for all VRFs.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Showing IGMP static-group information:

```
switch# show ip igmp static-groups

IGMP Static Group Address Information

VRF Name      default
Interface Name Group Address
-----
vlan10        238.1.1.1
```

Showing IGMP statics-group information for all VRFs:

```
switch# show ip igmp static-groups all-vrfs

IGMP Static Group Address Information

VRF Name      :default
Interface Name Group Address
-----
vlan10        238.1.1.1
```

show ip igmp statistics

Syntax

```
show ip igmp statistics [all-vrfs]
```

Description

Shows IGMP statistics, including groups joined, or shows statistics by VRF.

Command context

Operator (>) or Manager (#)

Parameters

`all-vrfs`

Specify `all-vrfs` to show information for all VRFs.

Authority

Operators or Administrators or local user group members with execution rights for this command.

Operators can execute this command from the operator context (>) only.

Examples

Showing IGMP statistics:

```
switch# show ip igmp statistics
IGMP statistics

VRF Name      : default

Number of Include Groups      : 1
Number of Exclude Groups     : 0
Number of Static Groups      : 0
Total Multicast Groups Joined : 1
```

Showing IGMP statistics for all VRFs:

```
switch# show ip igmp statistics all-vrfs
IGMP statistics

VRF Name      : default

Number of Include Groups      : 1
Number of Exclude Groups     : 0
Number of Static Groups      : 0
Total Multicast Groups Joined : 1
```

clear update-log

Syntax

```
clear update-log
```

Description

Clears stored log files of any In-System Programming updates on the system.

Command context

Manager (#)

Authority

Administrators or local user group members with execution rights for this command.

show needed-updates

Syntax

```
show needed-updates [next-boot [primary|secondary]]
```

Description

Displays whether any programmable devices are in need of an update.

Without the `next-boot` parameter, this command displays needed updates relative to the currently running AOS-CX image.

With the `next-boot` parameter, this command displays needed updates relative to an AOS-CX image file in the persistent storage of the switch, which might be different from the currently running image. If either the `primary` or `secondary` parameter is specified, this command queries that specific AOS-CX image file. Otherwise, it queries the default AOS-CX image file as set by the most recent `boot system` or `boot set-default` command.

Command context

Manager (#)

Authority

Administrators or local user group members with execution rights for this command.

allow-unsupported-transceiver

Syntax

```
allow-unsupported-transceiver [confirm | log-interval {none | <INTERVAL>}]  
  
no allow-unsupported-transceiver
```

Description

Allows unsupported transceivers to be enabled or establish connections. Only 1G and 10G transceivers are enabled by this command and unsupported transceivers of other speeds will remain disabled.

The `no` form of this command disallows using unsupported transceivers. This is the default.

Command context

config

Parameters

`confirm`

Specifies that unsupported transceiver warnings are to be automatically confirmed.

`log-interval none`

Disables unsupported transceiver logging.

`log-interval <INTERVAL>`

Sets the unsupported transceiver logging interval in minutes. Default: 1440 minutes. Range: 1440 to 10080 minutes.

Authority

Administrators or local user group members with execution rights for this command.

Usage

When none of the parameters are specified it will display a warning message to accept the warranty terms. With `confirm` option the warning message is displayed but the user is not prompted to `(y/n)` answering. Warranty terms must be agreed to as part of enablement and the support is on best effort basis.

Examples

Allowing unsupported transceivers with follow-up confirmation:

```
switch(config)# allow-unsupported-transceiver  
Warning: The use of unsupported transceivers, DACs, and AOCs is at your  
own risk and may void support and warranty. Please see HPE Warranty terms  
and conditions.  
  
Do you agree and do you want to continue (y/n)? y
```

Allowing unsupported transceivers with confirmation in command syntax:


```
switch(config)# allow-unsupported-transceiver confirm  
Warning: The use of unsupported transceivers, DACs, and AOCs is at your  
own risk and may void support and warranty. Please see HPE Warranty terms  
and conditions.
```

Configuring unsupported transceiver logging with an interval of every 48 hours:

```
switch(config)# allow-unsupported-transceiver log-interval 2880
```

Disabling unsupported transceiver logging:

```
switch(config)# allow-unsupported-transceiver log-interval none
```

Disallowing unsupported transceivers with follow-up confirmation:

```
switch(config)# no allow-unsupported-transceiver  
Warning: Unsupported transceivers, DACs, and AOCs will be disabled,  
which could impact network connectivity. Use 'show allow-unsupported-transceiver'  
to identify unsupported transceivers, DACs, and AOCs.  
  
Continue (y/n)? y
```

Disallowing unsupported transceivers with confirmation in command syntax:

```
switch(config)# no allow-unsupported-transceiver confirm  
Warning: Unsupported transceivers, DACs, and AOCs will be disabled,  
which could impact network connectivity. Use 'show allow unsupported-transceiver'  
to identify unsupported transceivers, DACs, and AOCs.  
  
switch(config)#
```

default interface

Syntax

```
default interface <INTERFACE-ID>
```

Description

Sets an interface (or a range of interfaces) to factory default values.

Command context

config

Parameters

<INTERFACE-ID>

Specifies the ID of a single interface or range of interfaces. Format: member/slot/port or member/slot/port-member/slot/port to specify a range.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Resetting an interface:

```
switch(config)# default default interface 1/1/1
```

Resetting an range of interfaces:

```
switch(config)# default default interface 1/1/1-1/1/10
```

description

Syntax

```
description <DESCRIPTION>  
no description
```

Description

Associates descriptive information with an interface to help administrators and operators identify the purpose or role of an interface.

The `no` form of this command removes a description from an interface.

Command context

config-if

Parameters

<DESCRIPTION>

Specify a description for the interface. Range: 1 to 64 ASCII characters (including space, excluding question mark).

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the description for an interface to **DataLink 01**:

```
switch(config-if)# description DataLink 01
```

Removing the description for an interface.

```
switch(config-if)# no description
```

energy-efficient-ethernet

Syntax

```
energy-efficient-ethernet
```

Description

Enables auto-negotiation of Energy-Efficient Ethernet (EEE) on an interface. EEE Negotiation is established only on auto-link negotiation with supported link partners.

Command context

config

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring an interface:

```
switch(config)# interface 1/1/1
switch(config-if)# energy-efficient-ethernet
```

Disabling Energy Efficient Ethernet on an interface:

```
switch(config)# interface 1/1/1
switch(config-if)# no energy-efficient-ethernet
```

flow-control

Syntax

Description

The no form disables flow control support on the current interface.

Command context

config-if

Authority

Administrators or local user group members with execution rights for this command.

Parameters

Examples

interface

Syntax

```
interface <PORT-NUM>
```

Description

Switches to the config-if context for a physical port. This is where you define the configuration settings for the logical interface associated with the physical port.

Command context

config

Parameters

<PORT-NUM>

Specifies a physical port number. Format: member/slot/port.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring an interface:

```
switch(config)# interface 1/1/1
switch(config-if)#
```

interface vlan

Syntax

```
interface vlan <VLAN-ID>
```

```
no interface vlan <VLAN-ID>
```

Description

Creates an interface VLAN also know as an SVI (switched virtual interface) and changes to the `config-if-vlan` context. The specified VLAN must already be defined on the switch.

The `no` form of this command deletes an interface VLAN.

Command context

config

Parameters

<VLAN-ID>

Specifies the interface ID. Range: 2 to 4094

Authority

Administrators or local user group members with execution rights for this command.

Examples

```
switch# config
switch(config)# vlan 10
switch(config-vlan-10)# exit
switch(config)# interface vlan 10
switch(config-if-vlan)#
```

ip address

Syntax

```
ip address <IPV4-ADDR>/<MASK> [secondary]
```

```
no ip address <IPV4-ADDR>/<MASK> [secondary]
```

Description

Sets an IP/IPv6 address on the interface VLAN.

The `no` form of this command removes the IP/IPv6 address from the interface.

Command context

`config-if-vlan`

Parameters

`<IPv4-ADDR>`

Specifies an IP address in IPv4 format (`x.x.x.x`), where `x` is a decimal number from 0 to 255. You can remove leading zeros. For example, the address `192.169.005.100` becomes `192.168.5.100`.

`<MASK>`

Specifies the number of bits in the address mask in CIDR format (`x`), where `x` is a decimal number from 0 to 128.

`secondary`

Specifies a secondary IP address.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Assigning the IP address **192.168.199.1** with a mask of **24** bits to interface VLAN **10**:

```
switch(config)# interface vlan 10
switch(config-if-vlan)# ip address 192.168.199.1/24
```

Removing the IP address **192.168.199.1** with a mask of **24** bits from interface VLAN **10**:

```
switch(config)# interface vlan 10
switch(config-if-vlan)# no ip address 192.168.199.1/24
```

ip mtu

Syntax

`ip mtu <VALUE>`

`ip no mtu`

Description

Sets the IP MTU (maximum transmission unit) for an interface. This defines the largest IP packet that can be sent or received by the interface.

The `no` form of this command sets the IP MTU to the default value 1500.

Command context

`config-if-vlan`

Parameters

`<VALUE>`

Specifies the IP MTU in bytes. Range: 68 to 9198. Default: 1500.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the IP MTU to 576 bytes:

```
switch(config-if-vlan)# ip mtu 576
```

ip source-interface

Syntax

```
ip source-interface {sflow | tftp | radius | tacacs | ntp | syslog | dhcp-relay |  
simplivity | dns | all} {interface <IFNAME> | <IPV4-ADDR>} [vrf <VRF-NAME>]
```

```
no ip source-interface {sflow | tftp | radius | tacacs | ntp | syslog | dhcp-relay |  
simplivity | dns | all} [interface <IFNAME> | <IPV4-ADDR>] [vrf <VRF-NAME>]
```

Description

Sets a single source IP address for a feature on the switch. This ensures that all traffic sent the feature has the same source IP address regardless of how it egresses the switch. You can define a single global address that applies to all supported features, or an individual address for each feature.

This command provides two ways to set the source IP addresses: either by specifying a static IP address, or by using the address assigned to a switch interface. If you define both options, then the static IP address takes precedence.

The `no` form of this command deletes the single source IP address for all supported services, or a specific service.

Command context

config

Parameters

sflow | tftp | radius | tacacs | ntp | syslog | dhcp-relay | simplivity | dns | all

Sets a single source IP address for a specific service. The `all` option sets a global address that applies to all protocols that do not have an address set. For DHCP relay, the address is used as both the source IP and GIADDR.

interface <IFNAME>

Specifies the name of the interface from which the specified service obtains its source IP address. The interface must have a valid IP address assigned to it. If the interface has both a primary and secondary IP address, the primary IP address is used.

<IPV4-ADDR>

Specifies the source IP address to use for the specified service. The IP address must be defined on the switch, and it must exist on the specified VRF (which is the `default` VRF, if the `vrf` option is not used). Specify the address in IPv4 format (`x.x.x.x`), where `x` is a decimal number from 0 to 255.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the IPv4 address 10.10.10.5 as the global single source address:

```
switch# config  
switch(config)# ip source-interface all 10.10.10.5
```

Clearing the global single source IP address **10.10.10.5**:

```
switch(config)# no ip source-interface all 10.10.10.5
```

ipv6 address

Syntax

```
ipv6 address <IPv6-ADDR>/<MASK>{eui64 | [tag <ID>]}
```

```
no ipv6 address <IPv6-ADDR>/<MASK>
```

Description

Sets an IPv6 address on the interface.

The `no` form of this command removes the IPv6 address on the interface.



This command automatically creates an IPv6 link-local address on the interface. However, it does not add the `ipv6 address link-local` command to the running configuration. If you remove the IPv6 address, the link-local address is also removed. To maintain the link-local address, you must manually execute the `ipv6 address link-local` command.

Command context

config-if

Parameters

<IPv6-ADDR>

Specifies the IP address in IPv6 format (xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx), where x is a hexadecimal number from 0 to F. You can use two colons (::) to represent consecutive zeros (but only once), remove leading zeros, and collapse a hextet of four zeros to a single 0. For example, this address 2222:0000:3333:0000:0000:0000:4444:0055 becomes 2222:0:3333::4444:55.

<MASK>

Specifies the number of bits in the address mask in CIDR format (x), where x is a decimal number from 0 to 128.

eui64

Configure the IPv6 address in the EUI-64 bit format.

tag <ID>

Configure route tag for connected routes. Range: 0 to 4294967295. Default: 0.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the IPv6 address **2001:0db8:85a3::8a2e:0370:7334** with a mask of 24 bits:

```
switch(config-if)# ipv6 address 2001:0db8:85a3::8a2e:0370:7334/24
```

Removing the IP address 2001:0db8:85a3::8a2e:0370:7334 with mask of 24 bits:

```
switch(config-if)# no ipv6 address 2001:0db8:85a3::8a2e:0370:7334/24
```

ipv6 source-interface

Syntax

```
ipv6 source-interface {sflow | tftp | radius | tacacs | ntp | syslog | dhcp-relay |  
simplivity | dns | all} [interface <IFNAME> | <IPV6-ADDR>]
```

```
no ipv6 source-interface {sflow | tftp | radius | tacacs | ntp | syslog | dhcp-relay |  
simplivity | dns | all} [interface <IFNAME> | <IPV6-ADDR>]
```

Description

Sets a single source IP address for a feature on the switch. This ensures that all traffic sent the feature has the same source IP address regardless of how it egresses the switch. You can define a single global address that applies to all supported features, or an individual address for each feature.

This command provides two ways to set the source IP addresses: either by specifying a static IP address, or by using the address assigned to a switch interface. If you define both options, then the static IP address takes precedence.

The **no** form of this command deletes the single source IP address for all supported protocols, or a specific protocol.

Command context

config

Parameters

sflow | tftp | radius | tacacs | ntp | syslog | dhcp-relay | simplivity | dns | all

Sets a single source IP address for a specific protocol. The **all** option sets a global address that applies to all protocols that do not have an address set.

interface <IFNAME>

Specifies the name of the interface from which the specified protocol obtains its source IP address.

<IPV6-ADDR>

Specifies the source IP address to use for the specified protocol. The IP address must be defined on the switch, and it must exist on the specified VRF (which is the **default** VRF). Specify the IP address in IPv6 format (xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx), where x is a hexadecimal number from 0 to F.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring the IPv6 address 2001:DB8::1 as the global single source address:

```
switch# config  
switch(config)# ip source-interface all 2001:DB8::1/32
```


Stop the source IP address from using the IP address on interface **1/1/1** on VRF `default`.

```
switch(config)# no ip source-interface all interface 1/1/1 vrf default
```

Clear the source IP address `2001:DB8::1`.

```
switch(config)# no ip source-interface all 2001:DB8::1
```

mtu

Syntax

```
mtu <VALUE>
```

```
no mtu
```

Description

Sets the MTU (maximum transmission unit) for an interface. This defines the maximum size of a layer 2 (Ethernet) frame. Frames larger than the MTU (1500 bytes by default) are dropped.

To support jumbo frames (frames larger than 1522 bytes), increase the MTU as required by your network. A frame size of up to 9198 bytes is supported.

The largest possible layer 1 frame will be 18 bytes larger than the MTU value to allow for link layer headers and trailers.

The `no` form of this command sets the MTU to the default value 1500.

Command context

```
config-if
```

Parameters

<VALUE>

Specifies the MTU in bytes. Range: 46 to 9198. Default: 1500.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the MTU on interface **1/1/1** to 1000 bytes:

```
switch(config)# interface 1/1/1
switch(config-if)# no routing
switch(config-if)# mtu 1000
```

Setting the MTU on interface **1/1/1** to the default value:

```
switch(config)# interface 1/1/1
switch(config-if)# no routing
switch(config-if)# no mtu
```

show allow-unsupported-transceiver

Syntax

```
show allow-unsupported-transceiver
```

Description

Displays configuration and status of unsupported transceivers.

Command context

```
config
```

Authority

Administrators or local user group members with execution rights for this command.

Examples

Showing unallowed unsupported transceivers:

```
switch(config)# show allow-unsupported-transceiver

Allow unsupported transceivers : no
Logging interval               : 1440 minutes

-----
Port        Type           Status
-----
1/1/31      SFP-SX              unsupported
1/1/32      SFP-1G-BXD          unsupported
1/1/2       SFP28DAC3           unsupported
```

Showing allowed unsupported transceivers:

```
switch# show allow-unsupported-transceiver

Allow unsupported transceivers : yes
Logging interval               : 1440 minutes

-----
Port        Type           Status
-----
1/1/31      SFP-SX              unsupported-allowed
1/1/32      SFP-1G-BXD          unsupported-allowed
1/1/2       SFP28DAC3           unsupported
```

show interface

Syntax

```
show interface [<IFNAME>|<IFRANGE>] [brief | physical | extended [non-zero]]
show interface [lag | vlan ] [<ID>] [brief | physical]
show interface [lag | vlan ] [<ID>] [extended [non-zero]]
```

Description

Displays active configurations and operational status information for interfaces.

Command context

Operator (>) or Manager (#)

Parameters

<IFNAME>

Specifies a interface name.

<IFRANGE>

Specifies the port identifier range.

brief

Shows brief info in tabular format.

physical

Shows the physical connection info in tabular format.

extended

Shows additional statistics.

non-zero

Shows only non zero statistics.

LAG

Shows LAG interface information.

VLAN

Shows VLAN interface information.

<LAG-ID>

Specifies the LAG number. Range: 1-256

<VLAN-ID>

Specifies the VLAN ID. Range: 1-4094

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Examples

The following example shows when interface 1/1/1 is configured:

```
switch# show interface 1/1/1
Interface 1/1/1 is up
Admin state is up
Link state: up for 1 minute (since Thu Nov 26 10:26:34 UTC 2020)
Link transitions: 3
Description:
Hardware: Ethernet, MAC Address: 88:3a:30:47:d1:df
MTU 1500
Type 1GbT
Full-duplex
qos trust cos
Speed 1000 Mb/s
Auto-negotiation is on
Energy-Efficient Ethernet is disabled
Flow-control: off
Error-control: off
MDI mode: MDIX
VLAN Mode: native-untagged
Native VLAN: 1
Allowed VLAN List: all
```

Rate collection interval: 300 seconds

Rates	RX	TX	Total (RX+TX)
Mbits / sec	0.00	0.00	0.00
KPkts / sec	0.00	0.00	0.00
Unicast	0.00	0.00	0.00
Multicast	0.00	0.00	0.00
Broadcast	0.00	0.00	0.00
Utilization %	0.00	0.00	0.00
Statistics	RX	TX	Total
Packets	0	0	0
Unicast	0	0	0
Multicast	0	0	0
Broadcast	0	0	0
Bytes	0	0	0
Jumbos	0	0	0
Dropped	0	0	0
Filtered	0	0	0
Pause Frames	0	0	0
Errors	0	0	0
CRC/FCS	0	n/a	0
Collision	n/a	0	0
Runts	0	n/a	0
Giants	0	n/a	0

When the interface is currently linked at a downshifted speed:

```
switch(config-if)# show interface 1/1/1

Interface 1/1/1 is up
...
Auto-negotiation is on with downshift active
```

When the interface is currently linked with energy-efficient-ethernet negotiated:

```
switch(config-if)# show interface 1/1/1

Interface 1/1/1 is up
...
Energy-Efficient Ethernet is enabled and active
```

When the interface is configured with EEE and the EEE has auto-negotiated:

```
switch(config-if)# show interface 1/1/1 physical

-----
-----
----
EEE      PoE Power      Link      Admin      Speed      Flow-Control
Port      Type      Status      Config      Status | Config      Port
Status | Config      (Watts)      State Information      Status | Config      Description
-----
```

```

-----
-----
1/1/1      1GbT      up      up      1G      auto      off      off      on
      on      --      10M/100M/1G

```

show interface dom

Syntax

```
show interface [<INTERFACE-ID>] dom [detail]
```

Description

Shows diagnostics information and alarm/warning flags for the optical transceivers (SFP, SFP+). This information is known as DOM (Digital Optical Monitoring). DOM information also consists of vendor determined thresholds which trigger high/low alarms and warning flags.

Command context

Manager (#)

Parameters

<INTERFACE-ID>

Specifies an interface. Format: member/slot/port.

detail

Show detailed information.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

```
switch# show interface dom
```

```

-----
Port      Type      Channel      Temperature      Voltage      Tx Bias      Rx Power      Tx Power
          (Celsius)    (Volts)      (mA)          (mW/dBm)      (mW/dBm)
-----
1/1/1     SFP+SR      47.65        3.31           8.40         0.08, -10.96  0.63, -2.49
1/1/2     SFP+SR      n/a          n/a            n/a          n/a           n/a
1/1/3     SFP+DA3     42.10        3.24           n/a          n/a           n/a

```

show interface energy-efficient ethernet

Syntax

```
show interface [<IFNAME>|<IFRANGE>] energy-efficient-ethernet
```

Description

Displays Energy-Efficient Ethernet information for the interface.

Command context

config

Parameters

<IFNAME>

Specifies the name of an interface on the switch. Use the format `member/slot/port` (for example, 1/1/1).

<IFRANGE>

Specifies the port identifier range of an interface on the switch. Use the format `member/slot/port` (for example, 1/1/1).

Authority

Operators or Administrators or local user group members with execution rights for this command.

Operators can execute this command from the operator context (>) only.

Example

The following example shows when the interfaces are Energy-Efficient Ethernet capable:

```
switch# show interface energy-efficient-ethernet
```

Port	Enabled	Negotiated	Speed (MB/s)	TX Wake Time (us)	RX Wake Time (us)
1/1/1	no	no	--	--	--
1/1/2	yes	yes	100	36	36
1/1/3	yes	yes	1000	17	17
1/1/4	no	no	--	--	--
1/1/5	yes	no	1000	--	--

```
switch# show interface 1/1/1 energy-efficient-ethernet
```

Port	Enabled	Negotiated	Speed (Mb/s)	TX Wake Time (us)	RX Wake Time (us)
1/1/1	no	no	1000	--	--

```
switch#
```

show interface transceiver

Syntax

```
show interface [<INTERFACE-ID>] transceiver [detail | threshold-violations]
```

Description

Displays information about transceivers present in the switch. The information shown varies for different transceiver types and manufacturers. Only basic information is shown for unsupported HPE and third-party transceivers installed in the switch and they are also identified with an asterisk in the output.

Command context

Manager (#)

Parameters

<INTERFACE-ID>

Specifies the name or range of an interface on the switch. Use the format `member/slot/port` (for example, `1/3/1`).

detail

Show detailed information for the interfaces.

threshold-violations

Show threshold violations for transceivers.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

Showing summary transceiver information with identification of unsupported transceivers:

```
switch(config)# show interface transceiver
```

Port	Type	Product Number	Serial Number	Part Number
1/1/15	SFP+SR	J9150D	CN92KJW05P	1990-4634
1/1/16	SFP+SR	J9150D	CN92KJW1RG	1990-4634

Showing detailed transceiver information:

```
switch(config)# show interface transceiver detail
```

Transceiver in 1/1/15

Interface Name : 1/1/15
Type : SFP+SR
Connector Type : LC
Wavelength : 850nm
Transfer Distance : 0.00km (SMF), 20m (OM1), 80m (OM2), 300m (OM3)
Diagnostic Support : DOM
Product Number : J9150D
Serial Number : CN92KJW05P
Part Number : 1990-4634

Status

Temperature : 30.38C
Voltage : 3.26V
Tx Bias : 5.54mA
Rx Power : 0.56mW, -2.52dBm
Tx Power : 0.62mW, -2.08dBm

Recent Alarms:

Recent Errors:

Transceiver in 1/1/16

Interface Name : 1/1/16
Type : SFP+SR
Connector Type : LC
Wavelength : 850nm
Transfer Distance : 0.00km (SMF), 20m (OM1), 80m (OM2), 300m (OM3)
Diagnostic Support : DOM
Product Number : J9150D
Serial Number : CN92KJW1RG

Part Number : 1990-4634

Status

Temperature : 30.62C
Voltage : 3.28V
Tx Bias : 5.64mA
Rx Power : 0.61mW, -2.15dBm
Tx Power : 0.59mW, -2.29dBm

Recent Alarms:

Recent Errors:

Showing detailed transceiver information with identification of unsupported transceivers:

```
switch# show interface transceiver detail
```

Transceiver in 1/1/2

Interface Name : 1/1/2
Type : SFP+ER (unsupported)
Connector Type : LC
Wavelength : 3590nm
Transfer Distance : 80m (SMF), 0m (OM1), 0m (OM2), 0m (OM3)
Diagnostic Support : DOM
Vendor Name : INNOLIGHT
Vendor Part Number : TR-PX15Z-NHP
Vendor Part Revision: 1A
Vendor Serial number: MYxxxxxxx

Status

Temperature : 28.88C
Voltage : 3.30V
Tx Bias : 65.53mA
Rx Power : 0.00mW, -inf
Tx Power : 1.47mW, 1.67dBm

Recent Alarms:

Rx Power low alarm
Rx Power low warning

Recent Errors:

Rx loss of signal

Showing transceiver threshold-violations:

```
switch(config)# show interface transceiver threshold-violations
```

Port	Type	Channel#	Recent Threshold Violations
1/1/15	SFP+SR		none
1/1/16	SFP+SR		none

switch#

show ip interface

Syntax

show ip interface <INTERFACE-ID>

Description

Shows status and configuration information for an IPv4 interface.

Command context

Manager (#)

Parameters

<INTERFACE-ID>

Specifies the name of an interface. Format: member/slot/port.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

```
switch# show ip interface vlan1

Interface vlan1 is up
Admin state is up
Hardware: Ethernet, MAC Address: f8:60:f0:c9:11:60
IP MTU 1500
IPv4 address 10.120.3.8/26
```

show ip source-interface

Syntax

```
show ip source-interface {sflow | tftp | radius | tacacs | all} [vrf <VRF-NAME>]
```

Description

Shows single source IP address configuration settings.

Command context

Manager (#)

Parameters

sflow | tftp | radius | tacacs | all

Shows single source IP address configuration settings for a specific protocol. The `all` option shows the global setting that applies to all protocols that do not have an address set.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Showing single source IP address configuration settings for sFlow:

```
switch# show ip source-interface sflow

Source-interface Configuration Information
-----
Protocol      Source Interface
-----
sflow         10.10.10.1
```

Showing single source IP address configuration settings for all protocols:

```
switch# show ip source-interface all

Source-interface Configuration Information
-----
Protocol      Src-Interface      Src-IP              VRF
-----
all           vlan2              2.2.2.2             default
switch#
```

show ipv6 interface

Syntax

```
show ipv6 interface <INTERFACE-ID>
```

Description

Shows status and configuration information for an IPv6 interface.

Command context

Manager (#)

Parameters

<INTERFACE-ID>

Specifies an interface ID. Format: member/slot/port.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

```
switch# show ipv6 interface vlan2

Interface vlan2 is up
Admin state is up
IPv6 address:
  2001::1/64 [VALID]
IPv6 link-local address: fe80::883a:3080:247:c1c0/64 [VALID]
IPv6 Forwarding feature: enabled
IPv6 multicast groups locally joined:
  ff02::1 ff02::1:ff00:1 ff02::1:ff47:c1c0 ff02::1:ff00:0
  ff02::2
```

```
IPv6 MTU 1500
IPv6 unicast reverse path forwarding: none
IPv6 load sharing: none
switch#
```

show ipv6 source-interface

Syntax

```
show ipv6 source-interface {sflow | tftp | radius | tacacs | all} [vrf <VRF-NAME>]
```

Description

Shows single source IP address configuration settings.

Command context

Manager (#)

Parameters

sflow | tftp | radius | tacacs | all

Shows single source IP address configuration settings for a specific protocol. The `all` option shows the global setting that applies to all protocols that do not have an address set.

vrf <VRF-NAME>

Specifies the name of a VRF.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Showing single source IP address configuration settings for sFlow:

```
switch# show ipv6 source-interface sflow

Source-interface Configuration Information
-----
Protocol          Source Interface
-----
sflow             2001:DB8::1
```

Showing single source IP address configuration settings for all protocols:

```
switch# show ipv6 source-interface all

Source-interface Configuration Information
-----
Protocol          Src-Interface          Src-IP          VRF
-----
all               vlan2                  2001::1         default

switch#
```

shutdown

Syntax

shutdown

no shutdown

Description

Disables an interface. Interfaces are disabled by default when created.

The `no` form of this command enables an interface.

Command context

config-if

Authority

Administrators or local user group members with execution rights for this command.

Examples

Disabling an interface:

```
switch(config-if) # shutdown
```

Enabling an interface:

```
switch(config-if) # no shutdown
```

client track ip

Syntax

```
client track ip
```

Description

Enables client IP address tracking on the switch. The default is disabled on global and VLAN levels. Admin users can enable client IP address tracking at the VLAN level.



Tracking enabling will take effect only if the client IP address tracking is enabled at system and VLAN level.

The `no` form of the command disables client IP address tracking. If tracking is disabled at switch level, it will be stopped even if it is enabled at VLAN or port level.

Command context

```
config
```

Parameters

None

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

Enable client IP address tracking at switch level:

```
switch(config)# client track ip
```

Enable client IP address tracking on VLAN 100:

```
switch(config)# vlan 100
switch(config-vlan-100)# client track ip
Enable client IP address tracking on VLANs 10 to 100:
switch(config)# vlan 10-100
switch(config-vlan-<10-100>)# client track ip
```

client track ip { enable | disable | auto }

Syntax

```
client track ip { enable | disable | auto }
```

Description

Enables client IP address tracking on the specified set of interfaces. Tracking will take effect only if client IP address tracking is enabled at both the system level and for the VLAN to which the port belongs. Default: auto.

The `no` form of the command disables client IP address tracking on the specified set of interfaces.

Command context

config-if

Parameters

enable

Specifies that all client IP addresses will be tracked in the port.

disable

Specifies that client IP addresses will not be tracked in the port.

auto

Specifies the following:

- For LLDP devices: Only the specified client IP address will be tracked in the port and other client IP addresses will not be tracked.
- For non-LLDP devices: All client IP addresses will be tracked in the port.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

Enable client IP address tracking on interface 1/1/1:

```
switch(config)# interface 1/1/1  
switch(config-if)# client track ip enable
```

Enable client IP address tracking on interfaces 1/1/1 to 1/1/5:

```
switch(config)# interface 1/1/1-1/1/5  
switch(config-if-<1/1/1-1/1/5>)# client track ip enable
```

client track ip client-limit

Syntax

```
client track ip client-limit <CLIENT-LIMIT>
```

Description

Configures the maximum number of clients to be tracked on the specified set of interfaces.

For the 6100 Switch Series the `no` form of the command resets the client limit to the default value of 128.

Command context

config-if

Parameters

CLIENT-LIMIT

Specifies the maximum number of clients tracked on a port. Required. For the 6100 Switch Series, Range: 1-128. Default: 128.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

Configure the maximum number of clients to be tracked on interface 1/1/5:

```
switch(config)# interface 1/1/5
switch(config-if)# client track ip client-limit 32
```

client track ip update-interval

Syntax

client track ip update-interval <INTERVAL>

Description

Configures how often client IP addresses are updated.

The `no` form of the command resets the update interval to the default of 1800 seconds.

Command context

config-if

Parameters

INTERVAL

Specifies the update interval in seconds. Required. Range: 60-28000. Default: 1800.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

Configure the update interval for an interface:

```
switch(config)# interface 1/1/1
switch(config-if)# client track ip update-interval 600
```

client track ip update-method probe

Syntax

client track ip update-method probe

Description

Enables probing the client to update the IP address.

The probe is sent to all clients on the tracking list that have an IP address in the following scenarios:

1. IP packets are not received from the clients during the IP address update cycle.
2. There is no IP packet from a learned IP address. In this case, a probe will be sent for the IP address to confirm if it is still owned by that client.

The `no` form of the command disables probing.

Command context

`config`

Parameters

None

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

Disable probing to update the client IP address:

```
switch(config)# no client track ip update-method probe
```

show capacities

Syntax

`show capacities`

Description

Shows the capacities configured on the switch.

Command context

Operator (>)

Parameters

None

Command context

Administrators or local user group members with execution rights for this command.

Example

```
switch# show capacities

System Capacities:
```



```

Capacities Name
                Value
-----
Maximum number of Access Control Entries configurable in a system
                14336
Maximum number of Access Control Lists configurable in a system
                1024
Maximum number of class entries configurable in a system
                1024
Maximum number of classes configurable in a system
                512
Maximum number of entries in an Access Control List
                1024
Maximum number of entries in a class
                1024
Maximum number of entries in a policy
                1024
Maximum number of classifier policies configurable in a system
                512
Maximum number of policy entries configurable in a system
                1024
Maximum number of clients supported for tracking the IP address in the system
                128
switch# show capacities client-track-ip-client-limit

System Capacities: Filter Client Track IP Client Limit
Capacities Name
                Value
-----
Maximum number of clients supported for tracking the IP address in the system

```

show client ip { count | port | vlan }

Syntax

```
show client ip { count | port | vlan }
```

Description

Shows number of client IP addresses or information about client IP addresses tracked on ports and VLANs.

Command context

Operator (>)

Parameters

count

Displays number of clients tracked.

port

Displays client IP addresses tracked on the ports.

vlan

Displays client IP addresses tracked on the VLANs.

Authority

Administrators or local user group members with execution rights for this command.

ipv6 address <global-unicast-address>

Syntax

```
ipv6 address <global-unicast-address>  
no ipv6 address <global-unicast-address>
```

Description

Sets a global unicast address on the interface.

The `no` form of this command removes the global unicast address on the interface.



This command automatically creates an IPv6 link-local address on the interface. However, it does not add the `ipv6 address link-local` command to the running configuration. If you remove the IPv6 address, the link-local address is also removed. To maintain the link-local address, you must manually execute the `ipv6 address link-local` command.

Command context

config-if

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

Enabling a global unicast address:

```
switch(config)# interface vlan 2  
switch(config-if-vlan)# ipv6 address 3731:54:65fe:2::a7
```

Disabling a global unicast address:

```
switch(config)# interface vlan 2  
switch(config-if-vlan)# no ipv6 address 3731:54:65fe:2::a7
```

ipv6 address autoconfig

Syntax

```
ipv6 address autoconfig  
no ipv6 address autoconfig
```

Description

Enables the interface to automatically obtain an IPv6 address using router advertisement information and the EUI-64 identifier.

The `no` form of this command disables address auto-configuration.



- A maximum of 15 autoconfigured addresses are supported.
- This command automatically creates an IPv6 link-local address on the interface. However, it does not add the `ipv6 address link-local` command to the running configuration. If you remove the IPv6 address, the link-local address is also removed. To maintain the link-local address, you must manually execute the `ipv6 address link-local` command.

Command context

`config-if`

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Usage

The IPv6 SLAAC feature lets the router obtain the IPv6 address for the interface it is configured through the SLAAC method. This feature is not available on the `mgmt` VRF.

Example

Enabling unicast autoconfiguring:

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 address autoconfig
```

Disabling unicast autoconfiguring:

```
switch(config)# interface vlan 2
switch(config-if-vlan)# no ipv6 address autoconfig
```

ipv6 address link-local

Syntax

`ipv6 address link-local [<IPV6-ADDR>/<MASK>]`

Description

Enables IPv6 on the current interface. If no address is specified, an IPv6 link-local address is auto-generated for the interface. If an address is specified, auto-configuration is disabled and the specified address/mask is assigned to the interface.

To disable IPv6 link-local on the interface, remove `ipv6 address link-local`, `ipv6 address <global-ipv6-address>`, and `ipv6 address autoconfig` from the interface.



This feature is not available on the management VRF.

Command context

config-if

Parameters

<IPv6-ADDR>

Specifies the IP address in IPv6 format (xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx), where x is a hexadecimal number from 0 to F. You can use two colons (::) to represent consecutive zeros (but only once), remove leading zeros, and collapse a hextet of four zeros to a single 0. For example, this address

2222:0000:3333:0000:0000:0000:4444:0055 becomes 2222:0:3333::4444:55.

<MASK>

Specifies the number of bits in the address mask in CIDR format (x), where x is a decimal number from 0 to 128.

Authority

Operators or Administrators or local user group members with execution rights for this command.

Operators can execute this command from the operator context (>) only.

Example

Enabling IPv6 link-local on the interface:

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 address link-local
```

ipv6 nd cache-limit

Syntax

```
ipv6 nd cache-limit <CACHELIMIT>
no ipv6 nd cache-limit [<CACHELIMIT>]
```

Description

Configures the limit on the number of neighbor entries in the ND cache.

The no form of this command sets the cache limit to the default value.

Command context

config

Parameters

<CACHELIMIT>

Specifies the neighbor cache entries limit. Range: 1-131072. Default: 131072.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the cache limit to 20.

```
switch(config)# ipv6 nd cache-limit 20
```

ipv6 nd dad attempts

Syntax

```
ipv6 nd dad attempts <NUM-ATTEMPTS>  
no ipv6 nd dad attempts [<NUM-ATTEMPTS>]
```

Description

Configures the number of neighbor solicitations to be sent when performing duplicate address detection (DAD) for a unicast address configured on an interface.

The `no` form of this command sets the number of attempts to the default value.

Command context

config-if

Parameters

dad attempts <NUM-ATTEMPTS>

Specifies the number of neighbor solicitations to send. Range: 0-15. Default: 1.

Authority

Administrators or local user group members with execution rights for this command.

Examples

```
switch(config)# interface vlan 2  
switch(config-if-vlan)# ipv6 nd dad attempts 5
```

ipv6 nd hop-limit

Syntax

```
ipv6 nd hop-limit <HOPLIMIT>  
no ipv6 nd hop-limit [<HOPLIMIT>]
```

Description

Configures the hop limit to be sent in RAs.

The `no` form of this command resets the hop limit to 0. This reset eliminates the hop limit from the RAs that originate on the interface, so the host determines the hop limit.

Command context

config-if

Parameters

hop-limit <HOPLIMIT>

Specifies the hop limit. Range: 0-255. Default: 64.

Authority

Administrators or local user group members with execution rights for this command.

Examples

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 nd hop-limit 64
```

ipv6 nd mtu

Syntax

```
ipv6 nd mtu <MTU-VALUE>
no ipv6 nd mtu [<MTU-VALUE>]
```

Description

Configures the MTU size to be sent in the RA messages.

The `no` form of this command sets hop limit to the default value.

Command context

config-if

Parameters

<MTU-VALUE>

Specifies the MTU size. Range: 1280-65535 bytes. Default: 1500 bytes.

Authority

Administrators or local user group members with execution rights for this command.

Examples

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 nd mtu 1300
```

ipv6 nd ns-interval

Syntax

```
ipv6 nd ns-interval <TIME>
no ipv6 nd ns-interval [<TIME>]
```

Description

Configures the ND time between DAD neighbor solicitations sent for an unresolved destination, or between duplicate address detection neighbor solicitation requests. Increase this setting when neighbor solicitation retries or failures are occurring, or in a slow (WAN) network.

The `no` form of this command sets the ns-interval to the default value.

Command context

config-if

Parameters

<TIME>

Specifies the neighbor solicitation interval. Range: 1000-3600000 milliseconds. Default: 1000 milliseconds.

Authority

Administrators or local user group members with execution rights for this command.

Examples

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 nd ns-interval 1200
```

ipv6 nd prefix

Syntax

```
ipv6 nd prefix <IPV6-ADDR>/<PREFIX-LEN>
    [no-advertise | [valid <LIFETIME-VALUE> preferred
    <LIFETIME-VALUE>] | no-autoconfig | no-onlink]

no ipv6 nd prefix <IPV6-ADDR>/<PREFIX-LEN> [no-advertise
    | [valid <LIFETIME-VALUE> preferred <LIFETIME-VALUE>
    ] | no-autoconfig | no-onlink]

ipv6 nd prefix default [no-advertise | [valid <LIFETIME-VALUE>
    preferred <LIFETIME-VALUE>] | no-autoconfig | no-onlink]}

no ipv6 nd prefix default [no-advertise | [valid <LIFETIME-VALUE>
    preferred <LIFETIME-VALUE>] | no-autoconfig | no-onlink]}
```

Description

Specifies prefixes for the routing switch to include in RAs transmitted on the interface. IPv6 hosts use the prefixes in RAs to autoconfigure themselves with global unicast addresses. The autoconfigured address of a host is composed of the advertised prefix and the interface identifier in the current link-local address of the host.

By default, advertise, autoconfig, and onlink are set.

The `no` form of this command removes the configuration on the interface.

Command context

config-if

Parameters

<IPV6-ADDR>/<PREFIX-LEN>

Specifies the IPv6 prefix to advertise in RA. Format: X:X::X:X/M

default

Specifies apply configuration to all on-link prefixes that are not individually set by the `ipv6 ra prefix <IPV6-ADDR>/<PREFIX-LEN>` command. It applies the same valid and preferred lifetimes, link state, autoconfiguration state, and advertise options to the advertisements sent for all on-link prefixes that are not individually configured with a unique lifetime. This also applies to the prefixes for any global unicast addresses configured later on the same interface.

Using default once, and then using it again with any new parameter values results in the new values replacing the former values in advertisements. If default is used without the `no-advertise`, `no-autoconfig`, or `no-onlink` parameter, the advertisement setting for the absent parameter is returned to its default setting.

no-advertise

Specifies do not advertise prefix in RA.

valid <LIFETIME-VALUE>

Specifies the total time, in seconds, the prefix remains available before becoming unusable. After preferred-lifetime expiration, any autoconfigured address is deprecated and used only for transactions only before preferred-lifetime expires. If the valid lifetime expires, the address becomes invalid.

You can enter a value in seconds or enter `valid infinite` which sets infinite lifetime. Default: 2,592,000 seconds which is 30 days. Range: 0–4294967294 seconds.

`preferred <LIFETIME-VALUE>`

Specifies the span of time during which the address can be freely used as a source and destination for traffic. This setting must be less than or equal to the corresponding valid-lifetime setting.

You can enter a value in seconds or enter `preferred infinite` which sets infinite lifetime. Default: 604,800 seconds which is seven days. Range: 0–4294967294 seconds.

`no-autoconfig`

Specifies do not use prefix for autoconfiguration.

`no-onlink`

Specifies do not use prefix for onlink determination.

Authority

Administrators or local user group members with execution rights for this command.

Examples

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 nd prefix 4001::1/64 valid 30 preferred 10 no-
autoconfig no-onlink
```

ipv6 nd ra dns search-list

Syntax

```
ipv6 nd ra dns search-list <DOMAIN-NAME> [lifetime <TIME>]
no ipv6 nd ra dns search-list <DOMAIN-NAME>
```

Description

Configures the DNS Search List (DNSSL) to include in Router Advertisements (RAs) transmitted on the interface.

The `no` form of this command removes the DNS Search List from the RAs transmitted on the interface.

Command context

`config-if`

Parameters

`<DOMAIN-NAME>`

Specifies the domain names for DNS queries.

`lifetime <TIME>`

Specifies lifetime in seconds. Range: 4-4294967295 seconds. Default: 1800 seconds.

Authority

Administrators or local user group members with execution rights for this command.

Usage

- DNSSL contains the domain names of DNS suffixes or IPv6 hosts to append to short, unqualified domain names for DNS queries.
- Multiple DNS domain names can be added to the DNSSL by using the command repeatedly.
- A maximum of eight server addresses are allowed.

Examples

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 nd ra dns search-list test.com lifetime 500
```

ipv6 nd ra dns server

Syntax

```
ipv6 nd ra dns server <IPv6-ADDR> [lifetime <TIME>]
no ipv6 nd ra dns server <IPv6-ADDR>
```

Description

Configures the IPv6 address of a preferred Recursive DNS Server (RDNSS) to be included in Router Advertisements (RAs) transmitted on the interface.

The `no` form of this command removes the configured DNS server from the RAs transmitted on the interface.

Command context

config-if

Parameters

<IPv6-ADDR>

Specifies the RDNSS address in IPv6 format (xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx), where x is a hexadecimal number from 0 to F. You can use two colons (::) to represent consecutive zeros (but only once), remove leading zeros, and collapse a hextet of four zeros to a single 0. For example, this address 2222:0000:3333:0000:0000:0000:4444:0055 becomes 2222:0:3333::4444:55.

lifetime <TIME>

Specifies IPv6 DNS server lifetime in seconds. Range: 4-4294967295 seconds. Default: 1800 seconds.

Authority

Administrators or local user group members with execution rights for this command.

Usage

- Including RDNSS information in RAs provides DNS server configuration for connected IPv6 hosts without requiring DHCPv6.
- Multiple servers can be configured on the interface by using the command repeatedly.
- A maximum of eight server addresses are allowed.

Examples

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 nd ra dns server 2001::1 lifetime 400
```

ipv6 nd ra lifetime

Syntax

```
ipv6 nd ra lifetime <TIME>  
no ipv6 nd ra lifetime [<TIME>]
```

Description

Configures the lifetime, in seconds, for the routing switch to be used as a default router by hosts on the current interface.

The `no` form of this command sets lifetime to the default of 1800 seconds.

Command context

config-if

Parameters

<TIME>

Specifies lifetime in seconds of a default router. A setting of 0 for default router lifetime in an RA indicates that the routing switch is not a default router on the interface. Range: 0-9000 seconds. Default: 1800 seconds.

Authority

Administrators or local user group members with execution rights for this command.

Usage

- A given host on an interface refreshes the default router lifetime for a specific router each time the host receives an RA from that router.
- A specific router ceases to be a default router candidate for a given host if the default router lifetime expires before the host is updated with a new RA from the router.

Examples

```
switch(config)# interface vlan 2  
switch(config-if-vlan)# ipv6 nd ra lifetime 1200
```

ipv6 nd ra managed-config-flag

Syntax

```
ipv6 nd ra managed-config-flag  
no ipv6 nd ra managed-config-flag
```

Description

Controls the M flag setting in RAs the router transmits on the current interface. Enable the M flag to indicate that hosts can obtain IP address through DHCPv6. The M flag is disabled by default.

The `no` form of this command turns off (disables) the M flag.

Command context

config-if

Authority

Administrators or local user group members with execution rights for this command.

Usage

- Enabling the M flag directs hosts to acquire their IPv6 addressing for the current interface from a DHCPv6 server.
- When the M-bit is enabled, receiving hosts ignore the O flag setting, which is configured using the command `ipv6 nd ra other-config-flag`.
- When the M-bit is disabled (the default), receiving hosts expect to receive their IPv6 addresses from RA.

M flag	O flag	Description
0	0	Indicates that no information is available via DHCPv6.
0	1	Indicates that other configuration information is available via DHCPv6. Examples of such information are DNS-related information or information on other servers within the network.
1	0	Indicates that addresses are available via Dynamic Host Configuration Protocol (DHCPv6).
1	1	If the M flag is set, the O flag is redundant and can be ignored because DHCPv6 will return all available configuration information.

Examples

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 nd ra managed-config-flag
```

ipv6 nd ra max-interval

Syntax

```
ipv6 nd ra max-interval <TIME>
no ipv6 nd ra max-interval [<TIME>]
```

Description

Configures the maximum interval between transmissions of IPv6 RAs on the interface. The interval between RA transmissions on an interface is a random value that changes every time an RA is sent. The interval is calculated to be a value between the current max-interval and min-interval settings.

The `no` form of this command returns the setting to its default, provided the default value is less than the default lifetime value.

Command context

config-if

Parameters

<TIME>

Specifies the maximum advertisement time in seconds. Range: 4-1800. Default: 600 seconds.

Authority

Administrators or local user group members with execution rights for this command.

Usage

- This value has one setting per interface. The setting does not apply to RAs sent in response to a router solicitation received from another device.
- Attempting to set max-interval to a value that is not sufficiently larger than the current min-interval also results in an error message.

Examples

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 nd ra max-interval 30
```

ipv6 nd ra min-interval

Syntax

```
ipv6 nd ra min-interval <TIME>
no ipv6 nd ra min-interval [<TIME>]
```

Description

Configures the minimum interval between transmissions of IPv6 RAs on the interface. The interval between RA transmissions on an interface is a random value that changes every time an RA is sent. The interval is calculated to be a value between the current max-interval and min-interval settings.

The `no` form of this command returns the setting to its default, provided the default value is less than the current max-interval setting.

Command context

config-if

Parameters

<TIME>

Specifies a minimum advertisement time in seconds. Range: 3-1350. Default: 200 seconds.

Authority

Administrators or local user group members with execution rights for this command.

Usage

- This value has one setting per interface and does not apply to RAs sent in response to a router solicitation received from another device.
- The min-interval must be less than the max-interval. Attempting to set min-interval to a higher value results in an error message.

Examples

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 nd ra min-interval 25
```

ipv6 nd ra other-config-flag

Syntax

```
ipv6 nd ra other-config-flag  
no ipv6 nd ra other-config-flag
```

Description

Controls the O-bit in RAs the router transmits on the current interface; but is ignored unless the M-bit is disabled in RAs. Configure to set the O-bit in RA messages for host to obtain network parameters through DHCPv6. The other-config-flag is disabled by default.

For more information on configuring the M-bit, see `ipv6 nd ra managed-config-flag`.

The `no` form of this command turns off (disables) the setting for this command in RAs.

Command context

```
config-if
```

Authority

Administrators or local user group members with execution rights for this command.

Usage

Enabling the O-bit while the M-bit is disabled directs hosts on the interface to acquire their other configuration information from DHCPv6. Examples of such information are DNS-related information or information on other servers within the network.

Examples

```
switch(config)# interface vlan 2  
switch(config-if-vlan)# ipv6 nd ra other-config-flag
```

ipv6 nd ra reachable-time

Syntax

```
ipv6 nd ra reachable-time <TIME>  
no ipv6 nd ra reachable-time [<TIME>]
```

Description

Sets the amount of time that the interface considers a device to be reachable after receiving a reachability confirmation from the device.

The `no` form of this command sets the reachable time to the default value of 0. (no limit).

Command context

```
config-if
```

Parameters

<TIME>

Specifies the reachable time in milliseconds. Range: 1000-3600000. Default: 0 (no limit).

Authority

Administrators or local user group members with execution rights for this command.

Examples

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 nd ra reachable-time 2000
```

ipv6 nd ra retrans-timer

Syntax

```
ipv6 nd ra retrans-timer <TIME>
no ipv6 nd ra retrans-timer [<TIME>]
```

Description

Configures the period (retransmit timer) between ND solicitations sent by a host for an unresolved destination, or between DAD neighbor solicitation requests. By default, hosts on the interface use their own locally configured NS-interval settings instead of using the value received in the RAs.

Increase this timer when neighbor solicitation retries or failures are occur, or in a "slow" (WAN) network.

The `no` form of this command sets the value to the default of 0.

Command context

config-if

Parameters

<TIME>

Specifies the retransmit timer value in milliseconds. Range: 0 - 4294967295 milliseconds. Default: 0 (Use locally configured NS-interval).

Authority

Administrators or local user group members with execution rights for this command.

Examples

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 nd ra retrans-timer 400
```

ipv6 nd router-preference

Syntax

```
ipv6 nd router-preference {high | medium | low}
no ipv6 nd router-preference [high | medium | low]
```

Description

Specifies the value that is set in the Default Router Preference (DRP) field of Router Advertisements (RAs) that the switch sends from an interface. An interface with a DRP value of high will be preferred by other devices on the network over interfaces with an RA value of medium or low.

The `no` form of this command set the value to the default of medium.

Command context

config-if

Parameters

high

Sets DRP to high.

medium

Sets DRP to medium. Default.

low

Sets DRP to low.

Authority

Administrators or local user group members with execution rights for this command.

Examples

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 nd router-preference high
```

ipv6 nd suppress-ra

Syntax

```
ipv6 nd suppress-ra [<SUPPRESS-OPTION>]
no ipv6 nd ra suppress-ra [<SUPPRESS-OPTION>]
```

Description

Configures suppression of IPv6 Router Advertisement transmissions on an interface.

The `no` form of this command restores transmission of IPv6 Router Advertisement and options.

Command context

config-if

Parameters

suppress-ra [<SUPPRESS-OPTION>]

Specifies suppressing RA transmissions. Entering `suppress-ra` without any options, suppresses all RA messages (default). Or you can enter one of the following options.

dnssl

Specifies suppressing DNSSL options in RA messages.

mtu

Specifies suppressing MTU options in RA messages.

rdnss

Specifies suppressing RDNSS options in RA messages.

Authority

Administrators or local user group members with execution rights for this command.

Examples

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 nd suppress-ra mtu dnssl rdnss
switch(config-if-vlan)# no ipv6 nd suppress-ra mtu dnssl rdnss
```

show ipv6 nd global traffic

Syntax

```
show ipv6 nd global traffic
```

Description

Displays IPV6 Neighbor Discovery traffic details on a device.

Command context

Operator (>) or Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Examples

```
switch# show ipv6 nd global traffic
ICMPv6 packet Statistics (sent/received)
  Total Messages           :      18/0
  Error Messages           :       0/0
  Destination Unreachables :       0/0
  Time Exceeded            :       0/0
  Parameter Problems       :       0/0
  Echo Request             :       0/0
  Echo Replies             :       0/0
  Redirects                :       0/0
  Packet Too Big           :       0/0
  Router Advertisements    :       4/0
  Router Solicitations     :       0/0
  Neighbor Advertisements  :       0/0
  Neighbor Solicitations   :       3/0
  Duplicate router RA received :    0/0
ICMPv6 MLD Statistics (sent/received)
  V1 Queries :      0/0
  V2 Queries :      0/0
  V1 Reports  :      0/0
  V2 Reports  :     11/0
  V1 Leaves  :      0/0
```

show ipv6 nd interface

Syntax

```
show ipv6 nd interface [<IF-NAME> | all-vrfs | vrf <VRF-NAME>]
```

Description

Displays neighbor discovery information for an interface. If no options are specified, displays information for the default VRF.

Command context

Operator (>) or Manager (#)

Parameters

<IF-NAME>

Displays information about the specified IPv6 enabled interface.

all-vrfs

Displays information about interfaces in all VRFs.

vrf <VRF-NAME>

Displays information about interfaces in a particular VRF. Or, if <VRF-NAME> is not specified, information for the default VRF is displayed.

Authority

Operators or Administrators or local user group members with execution rights for this command.

Operators can execute this command from the operator context (>) only.

Examples

Showing information for all VRFs:

```
switch# show ipv6 nd interface all-vrfs

List of IPv6 Interfaces for VRF default
Interface vlan2 is up
  Admin state is up
  IPv6 address:
  IPv6 link-local address: fe80::7272:cfff:fee7:a8b9/64 [VALID]
  ICMPv6 active timers:
    Last Router-Advertisement sent:
    Next Router-Advertisement sent in:
  Router-Advertisement parameters:
    Periodic interval: 200 to 600 secs
    Router Preference: medium
    Send "Managed Address Configuration" flag: false
    Send "Other Stateful Configuration" flag: false
    Send "Current Hop Limit" field: 64
    Send "MTU" option value: 1500
    Send "Router Lifetime" field: 1800
    Send "Reachable Time" field: 0
    Send "Retrans Timer" field: 0
    Suppress RA: true
    Suppress MTU in RA: true
  ICMPv6 error message parameters:
    Send redirects: false
  ICMPv6 DAD parameters:
    Current DAD attempt: 1

List of IPv6 Interfaces for VRF red
Interface vlan3 is up
  Admin state is up
  IPv6 address:
    2001::1/64 [VALID]
  IPv6 link-local address: fe80::7272:cfff:fee7:a8b9/64 [VALID]
  ICMPv6 active timers:
    Last Router-Advertisement sent:
    Next Router-Advertisement sent in:
  Router-Advertisement parameters:
```

```
Periodic interval: 200 to 600 secs
Router Preference: medium
Send "Managed Address Configuration" flag: false
Send "Other Stateful Configuration" flag: false
Send "Current Hop Limit" field: 64
Send "MTU" option value: 1500
Send "Router Lifetime" field: 1800
Send "Reachable Time" field: 0
Send "Retrans Timer" field: 0
Suppress RA: true
Suppress MTU in RA: true
ICMPv6 error message parameters:
  Send redirects: false
ICMPv6 DAD parameters:
  Current DAD attempt: 1
```

Showing information for interface vlan 2:

```
switch# show ipv6 nd interface vlan 2
Interface vlan2 is up
Admin state is up
IPv6 address:
IPv6 link-local address: fe80::7272:cfff:fee7:a8b9/64 [VALID]
ICMPv6 active timers:
  Last Router-Advertisement sent:
  Next Router-Advertisement sent in:
Router-Advertisement parameters:
  Periodic interval: 200 to 600 secs
  Router Preference: high
  Send "Managed Address Configuration" flag: false
  Send "Other Stateful Configuration" flag: false
  Send "Current Hop Limit" field: 64
  Send "MTU" option value: 1500
  Send "Router Lifetime" field: 1800
  Send "Reachable Time" field: 0
  Send "Retrans Timer" field: 0
  Suppress RA: true
  Suppress MTU in RA: true
ICMPv6 error message parameters:
  Send redirects: false
ICMPv6 DAD parameters:
  Current DAD attempt: 1
```

Showing information for the default VRF:

```
switch# show ipv6 nd interface

List of IPv6 Interfaces for VRF default
Interface vlan2 is up
Admin state is up
IPv6 address:
  2001::1/64 [VALID]
IPv6 link-local address: fe80::7272:cfff:fee7:a8b9/64 [VALID]
ICMPv6 active timers:
  Last Router-Advertisement sent: 6 Secs
  Next Router-Advertisement sent in: 7 Secs
Router-Advertisement parameters:
  Periodic interval: 3 to 13 secs
  Router Preference: medium
```

```
Send "Managed Address Configuration" flag: false
Send "Other Stateful Configuration" flag: false
Send "Current Hop Limit" field: 64
Send "MTU" option value: 1500
Send "Router Lifetime" field: 1900
Send "Reachable Time" field: 0
Send "Retrans Timer" field: 0
Suppress RA: true
Suppress MTU in RA: true
ICMPv6 error message parameters:
  Send redirects: false
ICMPv6 DAD parameters:
  Current DAD attempt: 1
```

show ipv6 nd interface prefix

Syntax

```
show ipv6 nd interface prefix [all-vrfs | vrf <VRF-NAME>]
```

Description

Shows IPv6 prefix information for all VRFs or a specific VRF. If no options are specified, shows information for the default VRF.

Command context

Operator (>) or Manager (#)

Parameters

all-vrfs

Shows prefix information for all VRFs.

vrf <VRF-NAME>

Name of a VRF.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Showing prefix information for the default VRF:

```
switch# show ipv6 nd interface prefix

List of IPv6 Interfaces for VRF default
List of IPv6 Prefix advertised on vlan2
  Prefix : 4545::/65
  Enabled : Yes
  Validlife time : 2592000
  Preferred lifetime : 604800
  On-link : Yes
  Autonomous : Yes
```

Showing information for VRF red:

```
switch# show ipv6 nd interface prefix vrf red
```

```
List of IPv6 Interfaces for VRF red
List of IPv6 Prefix advertised on vlan3
  Prefix : 2001::/64
  Enabled : Yes
  Validlife time : 2592000
  Preferred lifetime : 604800
  On-link : Yes
  Autonomous : Yes
```

show ipv6 nd ra dns search-list

Syntax

```
show ipv6 nd ra dns search-list
```

Description

Displays domain name information on all interfaces.

Command context

Operator (>) or Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Examples

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 nd ra dns search-list test.com
switch# show ipv6 nd ra dns search-list
Recursive DNS Search List on: 1
  Suppress DNS Search List: Yes
  DNS Search 1: test.com    lifetime 1800
```

show ipv6 nd ra dns server

Syntax

```
show ipv6 nd ra dns server
```

Description

Displays DNS server information on all interfaces.

Command context

Operator (>) or Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Examples

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 nd ra dns server 2001::1
switch# show ipv6 nd ra dns server
Recursive DNS Server List on: 1
    Suppress DNS Server List: Yes
    DNS Server 1: 2001::1    lifetime 1800
```

diag-dump irdp basic

Syntax

```
diag-dump irdp basic
```

Description

Displays diagnostic information for IRDP.

Command context

Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

```
switch# diag-dump irdp basic
=====
[Start] Feature irdp Time : Thu Jan  7 04:46:25 2021
=====
-----
[Start] Daemon hpe-rdiscd
-----
Interface: vlan2 (state : Down)
rdisc ipv4 (enabled: 1, max:600, min:450, hold:1800, pref:0, isBcast:0)
No advertisable IPv4 addresses on the interface
Interface: vlan1 (state : Down)
rdisc ipv4 (enabled: 0, max:600, min:450, hold:1800, pref:0, isBcast:0)
No advertisable IPv4 addresses on the interface
-----
[End] Daemon hpe-rdiscd
-----
[End] Feature irdp
=====
Diagnostic-dump captured for feature irdp
```

ip irdp

Syntax

```
ip irdp [broadcast | multicast]
no ip irdp
```

Description

Enables IRDP on an interface and specifies the packet type that is used to send advertisements. By default, the packet type is set to `multicast`. IRDP is only supported on layer 3 interfaces.

The `no` form of this command disables IRDP on an interface.

Command context

`config-if-vlan`

Parameters

`broadcast`

Advertisements are sent as broadcast packets to IP address 255.255.255.255.

`multicast`

Advertisements are sent as multicast packets to the multicast group with IP address 24.0.0.1. Default.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling IRDP on interface vlan 2 with packet type set to the default value (multicast).

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ip irdp
```

Enabling IRDP on interface vlan 2 with packet type set to broadcast.

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ip irdp broadcast
```

Disabling IRDP.

```
switch(config)# interface vlan 2
switch(config-if-vlan)# no ip irdp
```

ip irdp holdtime

Syntax

`ip irdp holdtime <TIME>`

Description

Specifies the maximum amount of time the host will consider an advertisement to be valid until a newer advertisement arrives. When a new advertisement arrives, hold time is reset. Hold time must be greater than or equal to the maximum advertisement interval. Therefore, if the hold time for an advertisement expires, the host can reasonably conclude that the router interface that sent the advertisement is no longer available. The default hold time is three times the maximum advertisement interval.

Command context

`config-if-vlan`

Parameters

<TIME>

Specifies the lifetime of router advertisements sent from this interface. Range: 4 to 9000 seconds.
Default: 1800 seconds.

Authority

Administrators or local user group members with execution rights for this command.

Example

Setting the hold time for VLAN interface 2 to 5000 seconds:

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ip irdp holdtime 5000
```

ip irdp maxadvertinterval

Syntax

ip irdp maxadvertinterval <TIME>

Description

Specifies the maximum router advertisement interval.

Command context

config-if-vlan

Parameters

<TIME>

Specifies the maximum time allowed between the sending of unsolicited router advertisements. Range: 4 to 1800 seconds. Default: 600 seconds.

Authority

Administrators or local user group members with execution rights for this command.

Example

Setting the advertisement interval for VLAN interface 2 to 30 seconds:

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ip irdp maxadvertinterval 30
```

ip irdp minadvertinterval

Syntax

ip irdp minadvertinterval <TIME>

Description

Specifies the minimum amount of time the switch waits between sending router advertisements. By default, this value is automatically set by the switch to be 75% of the value configured for maximum router advertisement interval. Use this command to override the automatically configured value.

Command context

config-if-vlan

Parameters

<TIME>

Specifies the minimum time allowed between the sending of unsolicited router advertisements. Range: 3 to 1800 seconds. Default: 450 seconds (75% of the default value for maximum router advertisement interval).

Authority

Administrators or local user group members with execution rights for this command.

Example

Setting the minimum advertisement interval for VLAN interface 2 to 25 seconds:

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ip irdp minadvertinterval 25
```

ip irdp preference

Syntax

ip irdp preference <LEVEL>

Description

Specifies the IRDP preference level. If a host receives multiple router advertisement messages from different routers, the host selects the router that sent the message with the highest preference as the default gateway.

Command context

config-if-vlan

Parameters

<LEVEL>

Specifies the IRDP preference level. Range: -2147483648 to 2147483647. Default: 0.

Authority

Administrators or local user group members with execution rights for this command.

Example

Setting the IRDP preference level for VLAN interface 2 to 25.

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ip irdp preference 25
```

show ip irdp

Syntax

```
show ip irdp
```

Description

Displays IRDP configuration settings.

Command context

Manager (#)

Authority

Administrators or local user group members with execution rights for this command.

Example

```
switch# sh ip irdp
```

ICMP Router Discovery Protocol

Interface	Status	Advertising Address	Minimum Interval	Maximum Interval	Holdtime	Preference
vlan1	Disabled	multicast	450	600	1800	0
bridge_normal	Disabled	multicast	450	600	1800	0

accept-lifetime

Syntax

```
accept-lifetime [start-time <time> <month>/<day>/<year>] {duration {<seconds> | infinite} |  
end-time <time> <month>/<day>/<year>}
```

Description

Configures the duration for which the key is valid for receiving packets.

The `no` form of this command configures the key packet receiving duration to the default value of an infinite time.

Command context

config-keychain-key

Parameters

start-time

Time at which the key chain lifetime starts. Required. Format: HH:MM:SS

end-time

Time at which the key chain lifetime expires. Required. Format: HH:MM:SS

day

Day of the month. Required. Range: 1-31.

month

Month of the year. Required.

year

Year. Required. Range: 2020-2050

duration

Time in seconds. Optional. Range: 1-2147483646.

infinite

Specifies infinite time for the key. Optional.

Authority

Operators or Administrators or local user group members with execution rights for this command.

Operators can execute this command from the operator context (>) only.

Examples

Configuring the duration for which the key is valid for receiving packets:

```
switch# configure terminal  
switch(config)# keychain ospf_keys  
switch(config-keychain)# key 1  
switch(config-keychain-key)# accept-lifetime start-time 10:10:10 10/25/2020 end-time
```

```
10:10:10 11/25/2020
switch(config-keychain-key)# accept-lifetime start-time 10:10:10 10/25/2020 duration
1000
switch(config-keychain-key)# accept-lifetime start-time 10:10:10 10/25/2020 duration
infinite
switch(config-keychain-key)# accept-lifetime end-time 10:10:10 11/25/2020
switch(config-keychain-key)# accept-lifetime duration 1000
switch(config-keychain-key)# accept-lifetime duration infinite
```

Configuring the key packet receiving duration to the default value of an infinite time:

```
switch# configure terminal
switch(config)# keychain ospf_keys
switch(config-keychain)# key 1
switch(config-keychain-key)# no accept-lifetime
```

key

Syntax

key <key-id>

Description

Creates the key for a key chain and enters the key chain key context. A maximum of 64 keys can be configured per key chain.

The `no` form of this command deletes the key from the key chain.

Command context

config-keychain

Parameters

key-id

ID of the key. Required. Range: 1-255.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Creating a key for a key chain:

```
switch# configure terminal
switch(config)# keychain ospf_keys
switch(config-keychain)# key 1
```

Deleting a key from a key chain:

```
switch# configure terminal
switch(config)# keychain ospf_keys
switch(config-keychain)# no key 1
```

keychain

Syntax

keychain <keychain-name>

Description

Creates the key chain and enters the key chain context. A maximum of 64 key chains can be configured in the system.

The `no` form of this command removes the key chain if it is not used by any subscribers.

Command context

config

Parameters

keychain-name

Name of the key chain. Required.

Command context

Operators or Administrators or local user group members with execution rights for this command.

Operators can execute this command from the operator context (>) only.

Examples

Creating a key chain:

```
switch# configure terminal
switch(config)# keychain ospf_keys
```

Removing a key chain:

```
switch# configure terminal
switch(config)# keychain ospf_keys
```

key-string

Syntax

key-string [{ciphertext | plaintext} <PASSWORD>]

Description

Sets the key password. The password is internally stored in encrypted form. The key is not valid until its password has been set.

The `no` form of this command deletes the password used for the key.

Command context

config-keychain-key

Parameters

ciphertext

Specifies that the key password is provided as ciphertext.

plaintext

Specifies that the key password is provided as plaintext.

<PASSWORD>

Specifies the key password.



When the key password is not provided on the command line, plaintext password prompting occurs upon pressing Enter. The entered password characters are masked with asterisks.

Authority

Operators or Administrators or local user group members with execution rights for this command.

Operators can execute this command from the operator context (>) only.

Examples

Setting the key password with plaintext:

```
switch(config)# keychain ospf_keys
switch(config-keychain)# key 1
switch(config-keychain-key)# key-string plaintext F82#450bHP
```

Setting the key password with plaintext prompting:

```
switch(config)# keychain ospf_keys
switch(config-keychain)# key 1
switch(config-keychain-key)# key-string
Enter the key password: *****
Re-Enter the key password: *****
```

Setting the key password with ciphertext:

```
switch(config)# keychain ospf_keys
switch(config-keychain)# key 1
switch(config-keychain-key)# key-string ciphertext AQBpfciFZ/P...biAAAOjc0a8=
```

Deleting the password for the key:

```
switch(config)# keychain ospf_keys
switch(config-keychain)# key 1
switch(config-keychain-key)# no key-string
```

send-lifetime

Syntax

```
send-lifetime [start-time <time> <month>/<day>/<year>] {duration {<seconds> | infinite} |
end-time <time> <month>/<day>/<year>}
```

Description

Configures the duration for which the key is valid for sending packets.

The `no` form of this command configures the key packet sending duration to the default value of an infinite time.

Command context

config-keychain-key

Parameters

start-time

Time at which the key chain lifetime starts. Required. Format: HH:MM:SS

end-time

Time at which the key chain lifetime expires. Required. Format: HH:MM:SS

day

Day of the month. Required. Range: 1-31.

month

Month of the year. Required.

year

Year. Required. Range: 2020-2050

duration

Time in seconds. Optional. Range: 1-2147483646.

infinite

Specifies infinite time for the key. Optional.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Configuring the duration for which the key is valid for sending packets:

```
switch# configure terminal
switch(config)# keychain ospf_keys
switch(config-keychain)# key 1
switch(config-keychain-key)# send-lifetime start-time 10:10:10 10/25/2020 end-time
10:10:10 11/25/2020
switch(config-keychain-key)# send-lifetime start-time 10:10:10 10/25/2020 duration
1000
switch(config-keychain-key)# send-lifetime start-time 10:10:10 10/25/2020 duration
infinite
switch(config-keychain-key)# send-lifetime end-time 10:10:10 11/25/2020
switch(config-keychain-key)# send-lifetime duration 1000
switch(config-keychain-key)# send-lifetime duration infinite
```

Configuring the key packet sending duration to the default value of an infinite time:

```
switch# configure terminal
switch(config)# keychain ospf_keys
switch(config-keychain)# key 1
switch(config-keychain-key)# no send-lifetime
```

show capacities keychain

Syntax

show capacities keychain

Description

Shows the maximum number of key chains and keys configurable in a key chain.

Command context

Operator (>)

Parameters

None

Authority

Administrators or local user group members with execution rights for this command.

Example

```
switch# show capacities keychain

System Capacities: Filter Keychain
Capacities Name      Value
-----
Maximum number of keychains supported in the system      64
Maximum number of Keys supported in a single Keychain    64
```

show keychain

Syntax

```
show keychain [<keychain-name>]
```

Description

Shows information about configured and active keys of a named key chain or (if `keychain-name` is not specified) all configured key chains.

Command context

Operator (>)

Parameters

`keychain-name`

Name of the key chain. Optional.

Command context

Administrators or local user group members with execution rights for this command.

Example

```
switch# show keychain
Keychain Name : ospf_keys
  Number of Keys      : 2
  Active Send Key ID  : 7
```



```

Active Recv Key IDs : 7, 200

Key ID          : 7
Key string      : AQBapZ1OHio9W3JwRqnjtLfbV73BPLS1S6TGVg+Lzl7N4e5eBAAAAPWaPBE=
Send Key Validity : 00:00:01 10/1/2020 to 23:59:01 10/1/2021
Recv Key Validity : 00:00:01 10/1/2020 to infinite
Key ID          : 200
Key string      : AQBapZ1OHio9W3JwRqnjtLfbV73BPLS1S6TGVg+Lzl7N4e5eBAAAAPWaPBE=
Send Key Validity : 00:00:01 10/1/2020 to 23:59:01 10/1/2021
Recv Key Validity : 00:00:01 10/1/2020 to 23:59:01 10/1/2021

Keychain Name : bgp_keys
Number of Keys : 2
Active Send Key ID : 7
Active Recv Key IDs : 7

Key ID          : 7
Key string      : AQBapZ1OHio9W3JwRqnjtLfbV73BPLS1S6TGVg+Lzl7N4e5eBAAAAPWaPBE=
Send Key Validity : 00:00:01 10/26/2020 to 23:59:01 10/1/2021
Recv Key Validity : 00:00:01 10/22/2020 to infinite
Key ID          : 8
Key string      : AQBapZ1OHio9W3JwRqnjtLfbV73BPLS1S6TGVg+Lzl7N4e5eBAAAAPWaPBE=
Send Key Validity : 00:00:01 10/1/2021 to 23:59:01 10/1/2021
Recv Key Validity : 00:00:01 10/1/2021 to 23:59:01 10/1/2021

Keychain Name : ospf_keys
Number of Keys : 2
Active Send Key ID : 7
Active Recv Key IDs : 7, 200

Key ID          : 7
Key string      : AQBapZ1OHio9W3JwRqnjtLfbV73BPLS1S6TGVg+Lzl7N4e5eBAAAAPWaPBE=
Send Key Validity : 00:00:01 10/1/2020 to 23:59:01 10/1/2021
Recv Key Validity : 00:00:01 10/1/2020 to infinite
Key ID          : 200
Key string      : AQBapZ1OHio9W3JwRqnjtLfbV73BPLS1S6TGVg+Lzl7N4e5eBAAAAPWaPBE=
Send Key Validity : 00:00:01 10/1/2020 to 23:59:01 10/1/2021
Recv Key Validity : 00:00:01 10/1/2020 to 23:59:01 10/1/2021

```

show running-config keychain

Syntax

show running-config keychain

Description

Shows the configurations for key chain protocol.

Command context

Operator (>)

Parameters

None

Authority

Administrators or local user group members with execution rights for this command.

Example

```
switch# show running-config keychain
keychain ospf_keys
  key 1
    key-string ciphertext
    AQBapZlOHio9W3JwRqnjtLfbV73BPLS1S6TGVg+Lzl7N4e5eBAAAAPWaPBE=
    accept-lifetime start-time 10:10:10 10/25/2020 end-time 10:10:10 11/25/2020
    send-lifetime start-time 10:10:10 10/25/2020 end-time 10:10:10 11/25/2020
  key 45
    key-string ciphertext
    AQBapZlOHio9W3JwRqnjtLfbV73BPLS1S6TGVg+Lzl7N4e5eBAAAAPWaPBE=
    accept-lifetime start-time 10:10:10 10/25/2020 end-time 10:10:10 11/25/2020
  key 33
switch#
```

downshift enable

Syntax

```
downshift-enable  
no downshift-enable
```

Description

Enables/disables automatic speed downshift on an interface that supports downshift, generally 1GBASE-T ports. When enabled, downshift allows an interface to link at a lower advertised speed when unable to establish a stable link at the maximum speed. Downshifting only applies to physical interfaces that are not members of a LAG and is only available when auto-negotiation is enabled. When only one speed is advertised, downshift will not be triggered.

Command context

```
config-if
```

Authority

Administrators or local user group members with execution rights for this command.

Examples

```
switch(config-if)# interface 1/1/1  
switch(config-if)# downshift-enable
```

Warning: this is a non-standard mode for use only when standards-based auto-negotiation is not able to establish a stable link. Enabling this may cause the port to link at a lower than expected speed and should not be used on ports that are members of a LAG. Support calls may require this feature to be disabled

```
Continue (y/n)?
```

```
switch(config-if)#
```

When automatic downshift is enabled:

```
switch(config-if)# show running-config interface  
interface 1/1/1  
    downshift-enable
```

Disabling automatic speed downshift:

```
switch(config-if)# interface 1/1/1  
switch(config-if)# no downshift-enable
```

show interface

Syntax

```
show interface [<IFNAME>|<IFRANGE>] [brief | physical | extended [non-zero]]  
show interface [lag | loopback | tunnel | vlan ] [<ID>] [brief | physical]  
show interface [lag | loopback | tunnel | vlan ] [<ID>] [extended | non-zero]
```

Description

Displays active configurations and operational status information for interfaces.

Command context

config

Parameters

<IFNAME>

Specifies a interface name.

<IFRANGE>

Specifies the port identifier range.

brief

Displays brief info in tabular format.

extended

Displays the physical connection info in tabular format.

non-zero

Displays only non zero statistics.

LAG

Displays LAG interface information.

LOOPBACK

Displays loopback interface information.

TUNNEL

Displays tunnel interface information.

VLAN

Displays VLAN interface information.

<LAG-ID>

Specifies the LAG number. Range: 1-256

<LOOPBACK-ID>

Specifies the LOOPBACK number. Range: 0-255

<TUNNEL-ID>

Specifies the tunnel ID. Range: 1-255

<VLAN-ID>

Specifies the VLAN ID. Range: 1-4094

VXLAN

Displays the VXLAN interface information.

<VXLAN-ID>

Specifies the VXLAN interface identifier. Default: 1

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Examples

Showing interfaces configured for L2 forwarding:

```
switch(config-if)# show interface
Interface 1/1/1 is down (Administratively down)
Admin state is down
State information: Administratively down
Link transitions: 2
Description: Backup data center link
Hardware: Ethernet, MAC Address: 70:72:cf:fd:e7:b4
MTU 9198
Type SFP-SX
Full-duplex
qos trust none
Speed 0 Mb/s
Auto-negotiation is on
Flow-control: off
Error-control: off
VLAN Mode: access
Access VLAN: 1
Rx
    1386055 total packets          586397526 total bytes
    1374287 unicast packets
    11764  multicast packets
    4      broadcast packets
    0 errors                          0 dropped
    0 CRC/FCS
Tx
    2475612 total packets          1003506711 total bytes
    2311806 unicast packets
    100369  multicast packets
    63437   broadcast packets
    0 errors                          2462773 dropped
    0 collision

Interface 1/2/1 is down (Administratively down)
Admin state is down
State information: Administratively down
Link transitions: 0
Description:
Hardware: Ethernet, MAC Address: 70:72:cf:fd:e7:b4
MTU 9198
Type QSFP+SR4
Full-duplex
qos trust none
Speed 0 Mb/s
Auto-negotiation is off
Flow-control: off
Error-control: off
VLAN Mode: access
Access VLAN: 1
Rx
    0 total packets          0 total bytes
    0 unicast packets
    0 multicast packets
    0 broadcast packets
    0 errors                  0 dropped
```

```

Tx          0 CRC/FCS
            0 total packets          0 total bytes
            0 unicast packets
            0 multicast packets
            0 broadcast packets
            0 errors                  0 dropped
            0 collision

```

When the interface is currently linked at a downshifted speed:

```

switch(config-if)# show interface 1/1/1

Interface 1/1/1 is up
...
Auto-negotiation is on with downshift active

```

show interface downshift-enable

Syntax

```
show interface [<IFNAME>|<IFRANGE>] downshift-enable
```

Description

Displays speed downshift information, including the interface speed status and configuration.

Command context

config

Parameters

<IFNAME>

Specifies a interface name.

<IFRANGE>

Specifies the port identifier range.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Showing automatic downshift information:

```

switch(config-if)# show interface downshift-enable
-----
Port          Downshift          Speed
              Enabled | Active   Status   | Config
-----
1/1/1         yes      yes      100M-FDx  auto
1/1/2         yes      no       1G       auto
1/1/3         yes      no       100M-FDx 100M-FDx
1/1/4         no       no       --       auto

```

Showing automatic downshift information on per interface:

```
switch(config-if)# show interface 1/1/2 downshift-enable
```

Port	Downshift		Status	Speed	
	Enabled	Active			Config
1/1/2	yes	no	1G		auto

show running-config interface

Syntax

```
show running-config interface [<IFNAME>|<IFRANGE>]  
show running-config interface [lag | loopback | tunnel | vlan ] [<ID>]
```

Description

Displays active configurations of various switch interfaces.

Command context

config

Parameters

<IFNAME>

Specifies a interface name.

<IFRANGE>

Specifies the port identifier range.

LAG

Specifies LAG interface information

LOOPBACK

Specifies loopback interface information.

TUNNEL

Specifies tunnel interface information.

VLAN

Specifies VLAN interface information.

<LAG-ID>

Specifies the LAG number. Range: 1-256.

<LOOPBACK-ID>

Specifies the LOOPBACK number. Range: 0-255.

<TUNNEL-ID>

Specifies the tunnel ID. Range: 1-255.

<VLAN-ID>

Specifies the VLAN ID. Range: 1-4094.

VXLAN

Specifies the VXLAN interface information.

<VXLAN-ID>

Specifies the VXLAN interface identifier. Default: 1.

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Examples

Showing 1/1/2 interface configuration:

```
switch(config-if)# show running-config interface 1/1/2

interface 1/1/2
  no shutdown
  description DC-23
  exit
```

Showing loopback interfaces configured:

```
switch(config-if)# show running-config interface loopback

interface loopback 1
  description lb interface 1
  exit
interface loopback 2
  description lb interface 2
  exit
```

Showing loopback interfaces not configured:

```
switch(config-if)# show running-config interface loopback

No loopback interfaces configured.
```


description

Syntax

```
description <TEXT>
```

```
no description <TEXT>
```

Description

Provides a brief description of the LAG interface. The description text is saved in the configuration of the LAG. It is available even after a reboot.

The `no` form of this command removes the description of the LAG interface from the configuration.

Command context

```
config-lag-if
```

Parameters

<TEXT>

Specifies the description of the LAG interface.

Authority

Administrators or local user group members with execution rights for this command.

Example

```
switch(config)# interface lag 10
switch(config-lag-if)# description This LAG is used for an example.
switch(config-lag-if)# show running-config
...
vlan 1
interface lag 10
    description This LAG is used for an example.
interface lag 60
switch(config-lag-if)#
```

hash

Syntax

```
hash [l2-src-dst | l3-src-dst | l4-src-dst]
```

Description

This command controls the selection of an interface in a group of aggregate interfaces. The hash type value helps transmit a frame. This configuration must be done at the LAG interface level.

Command context

config-lag-if

Parameters

l2-src-dst

Specifies the load-balancing calculation to include only Layer 2 items, such as source and destination MAC addresses.

l3-src-dst

Specifies the load-balancing calculation to include only Layer 3 items, such as source and destination IP addresses. Default setting.

l4-src-dst

Specifies the load-balancing calculation to include only Layer 4 items, such as source and destination UDP/TCP ports.

Authority

Administrators or local user group members with execution rights for this command.

Example

```
switch(config-lag-if)# hash l2-src-dst
```

interface lag

Syntax

```
interface lag <ID>  
no interface lag <ID>
```

Description

Creates a Link Aggregation Group (LAG) interface represented by an ID.

The `no` form of this command deletes a LAG interface represented by an ID.

Command context

config

Parameters

<ID>

Specifies a LAG interface ID.

Authority

Administrators or local user group members with execution rights for this command.

Usage

Keep in mind the following requirements when adding interfaces to a LAG:

- To determine the maximum number of LAG interfaces for your type of switch, look at the output from the `show capacities lag` command; however, the number of LAGs that can be created depends on the availability of the physical interface since each LAG interface needs at least one physical interface as a member link.

- After the maximum limit of members is reached in a LAG, an additional port cannot be added to the aggregation group. If a port belongs to a card type with a different speed than the other aggregation members, the port can still be added to the aggregation group. If dynamic LAG is enabled, any port member with a speed different than other aggregation members is blocked or ineligible from the same aggregation group. Any operational keys/attributes or configuration changes might affect the aggregation states of the member ports.
- The nondefaults configuration on an interface is removed automatically when the interface is added to a link aggregation. For example: Assume that you remove a member interface from an existing LAG and add it to another LAG. The software removes the nondefault configurations on the interface when it is added to the new LAG.

Examples

Creating a Link Aggregation Group (LAG) interface represented by an ID of 100:

```
switch(config)# interface lag 100
```

Deleting a Link Aggregation Group (LAG) interface represented by an ID of 100:

```
switch(config)# no interface lag 100
```

ip address

Not supported on the 6100 and 6200 Switch Series.

Syntax

```
ip address <IPv4-ADDR>/<MASK> [secondary]
no ip address <IPv4-ADDR>/<MASK> [secondary]
```

Description

Sets an IPv4 address and subnet mask to a LAG interface. One primary and up to 31 secondary address can be configured per interface.

The `no` form of this command removes the IPv4 address from the interface.

Command context

config-lag-if

Parameters

<IPv4-ADDR>

Specifies an IP address in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255. You can remove leading zeros. For example, the address 192.169.005.100 becomes 192.168.5.100.

<MASK>

Specifies the number of bits in the address mask in CIDR format (x), where x is a decimal number from 0 to 32.

secondary

Specifies a secondary IP address.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting an IP address on the LAG interface 1 to 198.51.100.1 with a mask of 24 bits:

```
switch(config)# interface lag 1
switch(config-lag-if)# ip address 198.51.100.1/24
```

Removing the IP address 198.51.100.1 with a mask of 24 bits from LAG interface 1:

```
switch(config)# interface lag 1
switch(config-lag-if)# no ip address 198.51.100.1/24
```

ipv6 address

Not supported on the 6100 and 6200 Switch Series.

Syntax

```
ipv6 address <IPV6-ADDR>/<MASK>
no ipv6 address <IPV6-ADDR>/<MASK>
```

Description

Sets an IPv6 address and subnet mask to a LAG interface.

The `no` form of this command removes the IPv6 address from the interface.

Command context

config-lag-if

Parameters

<IPV6-ADDR>

Specifies the IP address in IPv6 format (xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx), where x is a hexadecimal number from 0 to F. You can use two colons (::) to represent consecutive zeros (but only once), remove leading zeros, and collapse a quartet of four zeros to a single 0. For example, this address 2222:0000:3333:0000:0000:0000:4444:0055 becomes 2222:0:3333::4444:55.

<MASK>

Specifies the number of bits in the address mask in CIDR format (x), where x is a decimal number from 0 to 128.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the IPv6 address on LAG interface 1 to 2001:0db8:85a3::8a2e:0370:7334 with a mask of 24 bits:

```
switch(config)# interface lag 1
switch(config-lag-if)# ipv6 address 2001:0db8:85a3::8a2e:0370:7334/24
```

Removing the IP address 2001:0db8:85a3::8a2e:0370:7334 with mask of 24 bits with a mask of 24 bits from LAG interface 1:

```
switch(config)# interface lag 1
switch(config-lag-if)# no ipv6 address 2001:0db8:85a3::8a2e:0370:7334/24
```

lacp hash

Supported only on the 6100 and 8400 Switch Series.

Syntax

```
lacp hash [l2-src-dst | l3-src-dst | l4-src-dst]
```

Description

Controls the selection of an interface in a group of aggregate interfaces. The hash type value helps transmit a frame. This configuration must be done at the global level.

Command context

config

Parameters

l2-src-dst

Specifies the load-balancing calculation to include only Layer 2 items, such as source and destination MAC addresses.

l3-src-dst

Specifies the load-balancing calculation to include only Layer 3 items, such as source and destination IP addresses.

l4-src-dst

Specifies the load-balancing calculation to include only Layer 4 items, such as source and destination UDP/TCP ports.

Authority

Administrators or local user group members with execution rights for this command.

Example

```
switch(config)# lacp hash l2-src-dst
```

lacp mode

Syntax

```
lacp mode {active | passive}
no lacp mode {active | passive}
```

Description

Sets an LACP mode to active or passive.

The `no` form of this command sets the LACP mode to `off`, returning the LAG to a static mode aggregation.

Command context

config-lag-if

Parameters

active

Specifies that the local switch will transmit LACP Data Units (LACPDUs) to attempt to negotiate with the remote device.

passive

Specifies that the local switch will listen for LACPDUs from the remote device for LACP negotiation.



A momentary traffic drop occurs because LACP partners reconverge when changing the mode from active to passive or from passive to active.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the LACP mode to `active`:

```
switch(config)# interface lag 1
switch(config-lag-if)# lacp mode active
```

Setting the LACP mode to `off`:

```
switch(config)# interface lag 1
switch(config-lag-if)# no lacp mode active
```

lacp port-id

Syntax

lacp port-id <PORT-ID>

no lacp port-id

Description

Sets the LACP port ID value of the member interface of the LAG.

The `no` form of this command removes the LACP port ID value from the interface.

Command context

config-if

Parameters

<PORT-ID>

Specifies a port ID value. Range: 1 to 65535.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting an LACP port ID to a value of 10:

```
switch(config-if)# lacp port-id 10
```

Removing the LACP port ID value:

```
switch(config-if)# no lacp port-id
```

lacp port-priority

Syntax

```
lacp port-priority <PORT-PRIORITY>  
no lacp port-priority
```

Description

Sets an LACP port priority value for the member interface of the LAG.

The `no` form of this command reverts the LACP port priority to the default, which is 1.

Command context

config-if

Parameters

<PORT-PRIORITY>

Specifies a port priority value. Range: 1 to 65535.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting a LACP port priority value of 10:

```
switch(config-if)# lacp port-priority 10
```

Reverting the LACP port ID to the default:

```
switch(config-if)# no lacp port-priority
```

lacp rate

Syntax

```
lacp rate {fast | slow}  
no lacp rate
```

Description

Sets an LACP heartbeat request time to fast or slow.

The `no` form of the command sets an LACP rate to `slow`.

Command context

config-lag-if

Parameters

fast

Specifies the heartbeat request to every second, and the timeout period is a three-consecutive heartbeat loss that is 3 seconds.

slow

Specifies the heartbeat request to every 30 seconds. The timeout period is three-consecutive heartbeat loss that is 90 seconds. Default setting.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the LACP heartbeat request time to `fast`:

```
switch(config)# interface lag 1
switch(config-lag-if)# lacp rate fast
```

Resetting the LACP heartbeat request time to the default, which is `slow`:

```
switch(config)# interface lag 1
switch(config-lag-if)# no lacp rate
```

Another way to set the LACP heartbeat request time to the default, which is `slow`:

```
switch(config)# interface lag 1
switch(config-lag-if)# lacp rate slow
```

lacp system-priority

Syntax

```
lacp system-priority <SYSTEM-PRIORITY-VALUE>
no lacp system-priority
```

Description

Sets a Link Aggregation Control Protocol (LACP) system priority.

The `no` form of this command sets an LACP system priority to the default, which is 65534.

Command context

config

Parameters

<SYSTEM-PRIORITY-VALUE>

Specifies a system priority value. Range: 0 to 65535.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting a Link Aggregation Control Protocol (LACP) system priority to 100:

```
switch(config)# lacp system-priority 100
```

Setting an LACP system priority to the default (65534):

```
switch(config)# no lacp system-priority
```

A momentary traffic drop can be seen in case the LACP state machine must renegotiate.

lag

Syntax

```
lag <ID>  
no lag <ID>
```

Description

Adds an interface to a specified LAG interface ID.

The `no` form of this command removes an interface from a specified LAG interface ID. The member loses its LACP configuration when removed from the LAG. The member also reaches the default state with an administrative shutdown. For 6300 and 6400 series switches, the administrative state is enabled. Configurations, such as MTU and UDLD, are retained.

Command context

```
config-if
```

Parameters

<ID>

Specifies a LAG interface ID. Range: 1 to 256.

Authority

Administrators or local user group members with execution rights for this command.

Usage

- All members of the LAG must have the same speed. If a member comes up late with a different speed, it will not participate in the LAG/LACP. The hardware restriction is applied before adding an interface to LAG. The member belongs to the card type that has the same maximum speed as the reference port card type.
- To move an interface from LagA to LagB, first remove the interface from LagA and then add it to LagB. When a member is attached to a LAG, the nondefault configurations on the member are removed silently.
- After removing a physical interface from a LAG, the interface associated with the LAG becomes L3 ports with default L3 configurations and administrative down. For example, suppose interface 1/1/1 was part of LAG 3 and you had administratively enabled the interface. If you later remove interface 1/1/1 from LAG 3, the administrative status automatically changes to down. If you want to use the interface again, you must administratively enable it again.

Examples

Adding an interface to a Link Aggregation Group (LAG) represented by an ID of 100:

```
switch(config)# interface 1/1/1
switch(config-if)# lag 100
```

Deleting an interface from a Link Aggregation Group (LAG) represented by an ID of 100:

```
switch(config)# interface 1/1/1
switch(config-if)# no lag 100
```

show interface

Syntax

```
show interfaces <LAG-NAME> [vsx-peer]
```

Description

Displays information about a specific LAG.

Command context

Operator (>) or Manager (#)

Parameters

<LAG-NAME>

Specifies a LAG name.

[vsx-peer]

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Displaying information about LAG 100:

```
switch# show interface lag100
Aggregate lag100 is up
Admin state is up
Description :
MAC Address           : 48:0f:cf:af:43:9c
Aggregated-interfaces : 1/1/2
Aggregation-key       : 100
Aggregate mode        : active
Speed                 : 2000 Mb/s
L3 Counters: Rx Disabled, Tx Disabled
qos trust none
VLAN Mode: access
```

Access VLAN: 1

Statistics	RX	TX	Total
Packets	20	45	65
Unicast	5	5	10
Multicast	5	15	20
Broadcast	10	25	35
Bytes	5658	2584	8242
Jumbos	0	0	0
Dropped	0	0	0
Filtered	0	0	0
Pause Frames	0	0	0
Errors	0	0	0
CRC/FCS	0	n/a	0
Collision	n/a	0	0
Runts	0	n/a	0
Giants	0	n/a	0

show lacp aggregates

Syntax

```
show lacp aggregates [<LAG-NAME>] [vsx-peer]
```

Description

Displays all LACP aggregate information configured for all LAGs, or for a specific LAG.

Command context

Operator (>) or Manager (#)

Parameters

<LAG-NAME>

Optional: Specifies a lag name.

[vsx-peer]

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Displaying LACP aggregate information configured for lag10:

```
switch# show lacp aggregates lag10

Aggregate-name       : lag10
Aggregated-interfaces : 1/1/1 1/1/2
Heartbeat rate       : slow
Hash                  : 13-src-dst
Aggregate mode        : active
```

Displaying LACP aggregates:

```
switch# show lacp aggregates

Aggregate-name      : lag1
Aggregated-interfaces : 1/1/27 1/1/28 1/1/29
Heartbeat rate      : slow
Hash                : l3-src-dst
Aggregate mode      : active

Aggregate-name      : lag2
Aggregated-interfaces : 1/1/48
Heartbeat rate      : slow
Hash                : l2-src-dst
Aggregate mode      : passive
```

show lacp configuration

Syntax

```
show lacp configuration [vsx-peer]
```

Description

Displays global LACP configuration.

Command context

Operator (>) or Manager (#)

Parameters

[vsx-peer]

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

show lacp interfaces

Syntax

```
show lacp interfaces [<IFNAME>]
```

Description

Displays an LACP configuration of the physical interfaces, including VSXs. If an interface name is passed as argument, it only displays an LACP configuration of a specified interface.

Command context

Operator (>) or Manager (#)

Parameters

<IFNAME>

Optional: Specifies an interface name.

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Examples

This example displays an LACP configuration of the physical interfaces. One of the interfaces has the `lacp-block` forwarding state. If a VSX switch has loop protect enabled on an interface and a loop occurs, VSX blocks the interface to stop the loop. The forwarding state of the blocked interface is set to `lacp-block`.

```
switch# show lacp interfaces
State abbreviations :
A - Active          P - Passive          F - Aggregable I - Individual
S - Short-timeout  L - Long-timeout  N - InSync      O - OutofSync
C - Collecting     D - Distributing
X - State m/c expired      E - Default neighbor state

Actor details of all interfaces:
-----
Intf   Aggr   Port   Port   State   System-id           System   Aggr   Forwarding
      name  id     Pri    State   System-id           Pri     Key   State
-----
1/1/1  lag10  17     1      ALFOE   70:72:cf:37:a3:5c   20      10    lacp-block
1/1/2  lag128 69     1      ALFNCD  70:72:cf:37:a3:5c   20      128   up
1/1/3  lag128 14     1      ALFNCD  70:72:cf:37:a3:5c   20      128   up
1/1/4  lag128                down
1/1/5  lag20                up

Partner details of all interfaces:
-----
Intf   Aggr   Partner Port   State   System-id           System   Aggr
      name  Port-id Pri    State   System-id           Priority Key
-----
1/1/1  lag10  0       65534   PLFOEX  00:00:00:00:00:00  65534   0
1/1/2  lag128 69       1       PLFNCD  70:72:cf:8c:60:a7  65534   128
1/1/3  lag128 14       1       PLFNCD  70:72:cf:8c:60:a7  65534   128
1/1/4  lag128
1/1/5  lag20
```

Displaying static LAG:

```
switch# show lacp interfaces
State abbreviations :
A - Active          P - Passive          F - Aggregable I - Individual
S - Short-timeout  L - Long-timeout  N - InSync      O - OutofSync
C - Collecting     D - Distributing
X - State m/c expired      E - Default neighbor state

Actor details of all interfaces:
-----
Intf   Aggr   Port   Port   State   System-id           System   Aggr   Forwarding
      Name  Id     Pri    State   System-id           Pri     Key   State
-----
1/1/1  lag10                up
```

```
1/1/2 lag10 up
```

Partner details of all interfaces:

Intf	Aggr Name	Port Id	Port Pri	State	System-id	System Pri	Aggr Key
1/1/1	lag10						
1/1/2	lag10						

Displaying an LACP configuration of the 1/1/1 interface:

```
switch# show lacp interfaces 1/1/1
```

State abbreviations :

A - Active P - Passive F - Aggregable I - Individual
S - Short-timeout L - Long-timeout N - InSync O - OutofSync
C - Collecting D - Distributing
X - State m/c expired E - Default neighbor state

Aggregate-name : lag1

	Actor	Partner
Port-id	28	31
Port-priority	1	1
Key	1	1
State	ALFNCD	ALFNCD
System-id	98:f2:b3:68:40:a0	98:f2:b3:68:60:a6
System-priority	65534	65534

Displaying an LACP configuration after loop-protect is enabled on the primary VSX switch:

```
switch# show lacp interfaces
```

State abbreviations :

A - Active P - Passive F - Aggregable I - Individual
S - Short-timeout L - Long-timeout N - InSync O - OutofSync
C - Collecting D - Distributing
X - State m/c expired E - Default neighbor state

Actor details of all interfaces:

Intf	Aggr Name	Port Id	Port Pri	State	System-ID	System Pri	Aggr Key	Forwarding State
1/4/14	lag1 (mc)	206	1	ALFNCD	f8:60:f0:06:49:00	65534	1	up
1/5/15	lag2 (mc)							down

Partner details of all interfaces:

Intf	Aggr Name	Port Id	Port Pri	State	System-ID	System Pri	Aggr Key
1/4/14	lag1 (mc)	130	1	ALFNCD	f8:60:f0:06:87:00	65534	1
1/5/15	lag2 (mc)						

Displaying an LACP configuration after loop-protect is enabled on the secondary VSX switch:

```
switch# show lacp interfaces
```

State abbreviations :

A - Active P - Passive F - Aggregable I - Individual
S - Short-timeout L - Long-timeout N - InSync O - OutofSync
C - Collecting D - Distributing
X - State m/c expired E - Default neighbor state

Actor details of all interfaces:

Intf	Aggr Name	Port Id	Port Pri	State	System-ID	System Pri	Aggr Key	Forwarding State
1/3/2	lag1 (mc)	1130	1	ALFNCD	f8:60:f0:06:49:00	65534	1	up
1/9/3	lag2 (mc)							down

Partner details of all interfaces:

Intf	Aggr Name	Port Id	Port Pri	State	System-ID	System Pri	Aggr Key
1/3/2	lag1 (mc)	131	1	ALFNCD	f8:60:f0:06:87:00	65534	1
1/9/3	lag2 (mc)						

shutdown

Syntax

```
shutdown  
no shutdown
```

Description

Sets every interface in the LAG operationally down.

The `no` form of this command sets every interface operationally up.

Command context

```
config-lag-if
```

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting every interface in the LAG to shutdown:

```
switch(config)# interface lag 1  
switch(config-lag-if)# shutdown
```

Resetting every interface in the LAG to the default (up):

```
switch(config)# interface lag 1  
switch(config-lag-if)# no shutdown
```

vlan trunk native

Syntax

```
vlan trunk native <VLAN-ID>  
no vlan trunk native [<VLAN-ID>]
```

Description

Assigns a native VLAN ID to a LAG interface.

The `no` form of this command removes a native VLAN from a LAG interface and assigns VLAN ID 1 as its native VLAN.

Command context

```
config-if  
config-lag-if
```

Parameters

<VLAN-ID>

Specifies the number of the VLAN ID to assign. The VLAN ID must exist.

Maximum number of VLANs supported: 512 (6100); ; 4096 (6300, 6400, 8320, 8325, 8360, 8400).

VLAN ID range: 2 to 4094.

Authority

Administrators or local user group members with execution rights for this command.

Usage

By default, VLAN ID 1 is assigned as the LAG VLAN ID for all LAG interfaces. VLANs can only be assigned to a nonrouted (layer 2) interface or LAG interface.

Only one VLAN ID can be assigned as the native VLAN. For the interface to forward the native VLAN traffic, the interface has to be allowed explicitly by entering `vlan trunk allowed <ID>` where the ID is the native VLAN ID. This setting is also applicable to the physical interface.

Examples

Configuring a Layer 2 dynamic aggregation group with native VLAN ID **1** assigned to LAG **1**:

For 6100 and 6200 switch series:

```
switch(config)# interface lag 1  
switch(config-lag-if)# no shutdown  
switch(config-lag-if)# lacp mode active  
switch(config-lag-if)# vlan trunk native 1  
switch(config-lag-if)# vlan trunk allowed 1
```

Configuring a Layer 2 dynamic aggregation group with native VLAN ID **20** assigned to LAG **1**:

For 6100 and 6200 switch series:

```
switch(config)# interface lag 1  
switch(config-lag-if)# no shutdown  
switch(config-lag-if)# lacp mode active  
switch(config-lag-if)# vlan trunk native 20  
switch(config-lag-if)# vlan trunk allowed 20
```


Removing a native VLAN from LAG 1:

```
switch(config)# interface lag 1  
switch(config-lag-if)# no vlan trunk native
```

clear lldp neighbors

Syntax

```
clear lldp neighbors
```

Description

Clears all LLDP neighbor details.

Command context

Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Clearing all LLDP neighbor details:

```
switch# clear lldp neighbors
```

clear lldp statistics

Syntax

```
clear lldp statistics
```

Description

Clears all LLDP neighbor statistics.

Command context

Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Clearing all LLDP neighbor statistics:

```
switch# clear lldp statistics
```

lldp

Syntax

```
lldp
```

```
no lldp
```

Description

Enables LLDP support globally on all active interfaces. By default, LLDP is enabled.

The `no` form of this command disables LLDP support globally on all active interfaces. It does not remove any LLDP configuration settings.

Command context

```
config
```

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling LLDP:

```
switch(config)# lldp
```

Disabling LLDP:

```
switch(config)# no lldp
```

lldp dot3

Syntax

```
lldp dot3 {poe | macphy}
```

```
no lldp dot3 {poe | macphy}
```

Description

Sets the 802.3 TLVs to be advertised. By default, advertisement of both POE and MAC/PHY TLVs is enabled.

The `no` form of this command disables advertisement of 802.3 TLVs.

Command context

```
config-if
```

Parameters

`poe`

Specifies advertisement of power over Ethernet data link classification.

`macphy`

Specifies advertisement of media access control and physical layer information.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling advertisement of the POE TLV:

```
switch(config-if)# lldp dot3 poe
```

Disabling advertisement of the POE TLV:

```
switch(config-if)# no lldp dot3 poe
```

lldp holdtime

Syntax

```
lldp holdtime <TIME>
```

```
no lldp holdtime
```

Description

Sets the holdtime that is used to calculate the LLDP Time-to-Live value. Time-to-Live defines the length of time that neighbors consider LLDP information sent by this agent as valid. When Time-to-Live expires, the information is deleted by the neighbor. Time-to-live is calculated by multiplying holdtime by the value of `lldp timer`.

The `no` form of this command sets the holdtime to its default value of 4.

Command context

```
config
```

Parameters

<TIME>

Specifies the holdtime in seconds. Range: 2 to 10. Default: 4.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the holdtime to 8 seconds:

```
switch(config)# lldp holdtime 8
```

Setting the holdtime to the default value of 4 seconds:

```
switch(config)# no lldp holdtime
```

lldp management-ipv4-address

Syntax

```
lldp management-ipv4-address <IPv4-ADDR>  
no lldp management-ipv4-address
```

Description

Defines the IPv4 management address of the switch which is sent in the management address TLV. One IPv4 and one IPv6 management address can be configured.

If you do not define an LLDP management address, then LLDP uses one of the following (in order):

- IP address of SVI [interface VLAN <vid>]
- Base MAC address of the switch

The `no` form of this command removes the IPv4 management address of the switch.

Command context

config

Parameters

<IPv4-ADDR>

Specifies the management address of the switch as an IPv4 format (x.x.x.x), where x is a decimal value from 0 to 255.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the management address to **10.10.10.2**:

```
switch(config)# lldp management-ipv4-address 10.10.10.2
```

Removing the management address:

```
switch(config)# no lldp management-ipv4-address
```

lldp management-ipv6-address

Syntax

```
lldp management-ipv6-address <IPv6-ADDR>  
no lldp management-ipv6-address
```

Description

Defines the IPv6 management address of the switch. The management address is encapsulated in the management address TLV.

If you do not define an LLDP management address, then LLDP uses one of the following (in order):

- IP address of SVI [interface VLAN <vid>]
- Base MAC address of the switch

The `no` form of this command removes the IPv6 management address of the switch.

Command context

config

Parameters

<IPv6-ADDR>

Specifies an IP address in IPv6 format (`xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx`), where `x` is a hexadecimal number from 0 to F.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the management address to 2001:db8:85a3::8a2e:370:7334:

```
switch(config)# lldp management-ipv6-address 2001:0db8:85a3::8a2e:0370:7334
```

Removing the management address:

```
switch(config)# no lldp management-ipv6-address
```

lldp med

Syntax

```
lldp med [poe [priority-override] | capability | network-policy]
```

```
no med [poe [priority-override] | capability | network-policy]
```

Description

Configures support for the LLDP-MED TLV. LLDP-MED (media endpoint devices) is an extension to LLDP developed by TIA to support interoperability between VoIP endpoint devices and other networking end-devices. The switch only sends the LLDP MED TLV after receiving a MED TLV from and connected endpoint device.

The `no` form of this command disables support for the LLDP MED TLV.

Command context

config-if

Parameters

poe [priority-override]

Specifies advertisement of power over Ethernet data link classification. The `priority-override` option overrides user-configured port priority for Power over Ethernet. When both `lldp dot3 poe` and `lldp med poe` are enabled, the `lldp dot3 poe3` setting takes precedence. Default: enabled.

capability

Specifies advertisement of supported LLDP MED TLVs. The capability TLV is always sent with other MED TLVs, therefore it cannot be disabled when other MED TLVs are enabled. Default: enabled.

network-policy

Network policy discovery lets endpoints and network devices advertise their VLAN IDs, and IEEE 802.1p (PCP and DSCP) values for voice applications. This TLV is only sent when a voice VLAN policy is present. Default: enabled.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling advertisement of the network policy TLV:

```
switch(config-if)# lldp med network-policy
```

Disabling advertisement of the network policy TLV:

```
switch(config-if)# no lldp med network-policy
```

lldp med-location

Syntax

```
lldp med-location {civic-addr | elin-addr }
```

```
no med-location {civic-addr | elin-addr }
```

Description

Configures support for the LLDP-MED TLV. Supports only civic address and emergency location information number (ELIN). Coordinate-based location is not supported.

The `no` form of this command disables support for the LLDP MED TLV.

Command context

config-if

Parameters

civic-addr

Configures the LLDP MED civic location TLV.

elin-addr

Configures support for the LLDP MED emergency location TLV.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling support for the LLDP MED emergency location TLV:

```
switch(config-if)# lldp med-location elin-addr gher
```

Disabling support for the LLDP MED emergency location TLV:

```
switch(config-if)# no lldp med-location elin-addr gher
```

Enabling support for the LLDP MED civic address TLV:

```
switch(config-if)# lldp med-location civic-addr US 1 4 ret 6 tyu 7 tiyuo
```

Disabling support for the LLDP MED civic address TLV:

```
switch(config-if)# no lldp med-location civic-addr US 1 4 ret 6 tyu 7 tiyuo
```

lldp receive

Syntax

```
lldp receive
```

```
no lldp receive
```

Description

Enables reception of LLDP information on an interface. By default, LLDP reception is enabled on all active interfaces.

The `no` form of this command disables reception of LLDP information on an interface.

Command context

```
config-if
```

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling LLDP reception on interface **1/1/1**:

```
switch(config)# interface 1/1/1
switch(config-if)# lldp receive
```

Disabling LLDP reception on interface **1/1/1**:

```
switch(config)# interface 1/1/1
switch(config-if)# no lldp receive
```

lldp reinit

Syntax

```
lldp reinit <TIME>
```

```
no lldp reinit
```


Description

Sets the amount of time (in seconds) to wait before performing LLDP initialization on an interface. The `no` form of this command sets the reinitialization time to its default value of 2 seconds.

Command context

config

Parameters

<TIME>

Specifies the reinitialization time in seconds. Range: 1 to 10. Default: 2 seconds.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the reinitialization time to 5 seconds:

```
switch(config)# lldp reinit 5
```

Setting the reinitialization time to the default value of 2 seconds:

```
switch(config)# no lldp reinit
```

lldp select-tlv

Syntax

```
lldp select-tlv <TLV-NAME>
```

```
no lldp select-tlv <TLV-NAME>
```

Description

Selects a TLV that the LLDP agent will send and receive. By default, all supported TLVs are sent and received. The `no` form of this command stops the LLDP agent from sending and receiving a specific TLV.

Command context

config

Parameters

select-tlv <TLV-NAME>

Specifies the TLV name to send. The following TLV names are supported:

- `management-address`: Selected as follows:
 1. IPv4 or IPV6 management address.
 2. If layer 2, the IP address of the SVI.
 3. Base MAC address of the switch.
- `port-description`: A description of the port.
- `port-vlan-id`: VLAN ID assigned to the port.

- **system-capabilities**: Identifies the primary switch functions that are enabled, such as routing.
- **system-description**: Description of the system, comprised of the following information: hardware serial number, hardware revision number, and firmware version.
- **system-name**: Host name assigned to the switch.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Stopping the LLDP agent from sending the **port-description** TLV:

```
switch(config)# no lldp select-tlv port-description
```

Enabling the LLDP agent to send the **port-description** TLV:

```
switch(config)# lldp select-tlv port-description
```

lldp timer

Syntax

```
lldp timer <TIME>
```

```
no lldp timer
```

Description

Sets the interval (in seconds) at which local LLDP information is updated and TLVs are sent to neighboring network devices by the LLDP agent. The minimum setting for this timer must be four times the value of `lldp txdelay`.

For example, this is a valid configuration:

- `lldp timer = 16`
- `lldp txdelay = 4`

And, this is an invalid configuration:

- `lldp timer = 5`
- `lldp txdelay = 2`

When copying a saved configuration to the running configuration, the value for `lldp timer` is applied before the value of `lldp txdelay`. This can result in a configuration error if the saved configuration has a value of `lldp timer` that is not four times the value of `lldp txdelay` in the running configuration.

For example, if the saved configuration has the settings:

- `lldp timer = 16`
- `lldp txdelay = 4`



And the running configuration has the settings:

- `lldp timer = 30`
- `lldp txdelay = 7`

Then you will see an error indicating that certain configuration settings could not be applied, and you will have to manually adjust the value of `lldp txdelay` in the running configuration.

The `no` form of this command sets the update interval to its default value of 30 seconds.

Command context

`config`

Parameters

<TIME>

Specifies the update interval (in seconds). Range: 5 to 32768. Default: 30.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the update interval to 7 seconds:

```
switch(config)# lldp timer 7
```

Setting the update interval to the default value of 30 seconds:

```
switch(config)# no lldp timer
```

lldp transmit

Syntax

`lldp transmit`

`no lldp transmit`

Description

Enables transmission of LLDP information on specific interface. By default, LLDP transmission is enabled on all active interfaces.

The `no` form of this command disables transmission of LLDP information on an interface.

Command context

config-if

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling LLDP transmission on interface **1/1/1**:

```
switch(config)# interface 1/1/1
switch(config-if)# lldp transsmmit
```

Disabling LLDP transmission on interface **1/1/1**:

```
switch(config)# interface 1/1/1
switch(config-if)# no lldp transsmmit
```

lldp txdelay

Syntax

```
lldp txdelay <TIME>
```

```
no lldp txdelay
```

Description

Sets the amount of time (in seconds) to wait before sending LLDP information from any interface. The maximum value for `txdelay` is 25% of the value of `lldp tx timer`.

The `no` form of this command sets the delay time to its default value of 2 seconds.

Command context

config

Parameters

<TIME>

Specifies the delay time in seconds. Range: 0 to 10. Default: 2.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the delay time to 8 seconds:

```
switch(config)# lldp txdelay 8
```

Setting the delay time to the default value of 2 seconds:

```
switch(config)# no lldp txdelay
```

lldp trap enable

Syntax

```
lldp trap enable
```

```
no lldp trap enable
```

Description

Enables sending SNMP traps for LLDP related events from a particular interface. LLDP trap generation is enabled by default on all the interfaces and has to be disabled for interfaces on which traps are not required to be generated.

The `no` form of this command disables the LLDP trap generation.



LLDP trap generation is disabled by default at the global level and must be enabled before any LLDP traps are sent.

Command context

`config` and `config-if`

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling LLDP trap generation on global level:

```
switch(config)# lldp trap enable
```

Enabling LLDP trap generation on interface level:

```
switch(config-if)# lldp trap enable
```

Disabling LLDP trap generation on global level:

```
switch(config)# no lldp trap enable
```

Disabling LLDP trap generation on interface level:

```
switch(config-if)# no lldp trap enable
```

Displaying LLDP global configuration:

```
switch# show lldp configuration
```

```
LLDP Global Configuration
```

```
=====
```

```
LLDP Enabled : No
```

```

LLDP Transmit Interval      : 30
LLDP Hold Time Multiplier   : 4
LLDP Transmit Delay Interval : 2
LLDP Reinit Timer Interval  : 2
LLDP Trap Enabled           : No

```

TLVs Advertised

=====

Management Address

Port Description

Port VLAN-ID

System Description

System Name

LLDP Port Configuration

=====

PORT	TX-ENABLED	RX-ENABLED	INTF-TRAP-ENABLED
1/1/1	Yes	Yes	Yes
1/1/2	Yes	Yes	Yes
1/1/3	Yes	Yes	Yes
1/1/4	Yes	Yes	Yes
1/1/5	Yes	Yes	Yes
1/1/6	Yes	Yes	Yes
.....			
.....			
mgmt	Yes	Yes	Yes

Displaying LLDP Configuration for the interface:

```
switch# show lldp configuration 1/1/1
```

LLDP Global Configuration

=====

```

LLDP Enabled           : Yes
LLDP Transmit Interval : 30
LLDP Hold Time Multiplier : 4
LLDP Transmit Delay Interval : 2
LLDP Reinit Timer Interval : 2
LLDP Trap Enabled      : No

```

LLDP Port Configuration

=====

PORT	TX-ENABLED	RX-ENABLED	INTF-TRAP-ENABLED
1/1/1	Yes	Yes	Yes

Displaying LLDP Configuration for the management interface:

```
switch# show lldp configuration mgmt
```

LLDP Global Configuration

=====

```

LLDP Enabled           : Yes
LLDP Transmit Interval : 30
LLDP Hold Time Multiplier : 4
LLDP Transmit Delay Interval : 2

```

```
LLDP Reinit Timer Interval : 2
LLDP Trap Enabled          : Yes
```

LLDP Port Configuration

```
=====
PORT          TX-ENABLED      RX-ENABLED      INTF-TRAP-ENABLED
-----
mgmt          Yes             Yes             Yes
```

show lldp configuration

Syntax

```
show lldp configuration [<INTERFACE-ID>]
```

Description

Shows LLDP configuration settings for all interfaces or a specific interface.

Command context

Manager (#)

Parameters

<INTERFACE-ID>

Specifies an interface. Format: member/slot/port.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

Showing configuration settings for all interfaces:

This example shows configuration settings for interface **1/1/1**.

```
switch# show lldp configuration 1/1/1
```

LLDP Global Configuration

```
=====
LLDP Enabled          : Yes
LLDP Transmit Interval : 30
LLDP Hold Time Multiplier : 4
LLDP Transmit Delay Interval : 2
LLDP Reinit Timer Interval : 2
LLDP Trap Enabled      : No
```

LLDP Port Configuration

```
=====
PORT          TX-ENABLED      RX-ENABLED      INTF-TRAP-ENABLED
-----
1/1/1          Yes             Yes             Yes
```

show lldp local-device

Syntax

```
show lldp local-device
```

Description

Shows global LLDP information advertised by the switch, as well as port-based data. If VLANs are configured on any active interfaces, the VLAN ID is only shown for trunk native or untagged VLAN IDs on access interfaces.

Command context

Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

Showing global LLDP information only:

```
switch# show lldp local-device

Global Data
=====

Chassis-ID           : 88:3a:30:47:c1:c0
System Name          : 6100
System Description    : Aruba JL679A  PL.10.06.0001-346-g56a12a8f4cf15
Management Address   : 88:3a:30:47:c1:c0
Capabilities Available : Bridge, Router
Capabilities Enabled   : Bridge, Router
TTL                   : 120
```

Showing all ports except 1/1/11 as administratively down:

```
switch# show lldp local-device

Global Data
=====

Chassis-ID           : 88:3a:30:47:c1:c0
System Name          : 6100
System Description    : Aruba JL679A  PL.10.06.0001-346-g56a12a8f4cf15
Management Address   : 11.11.11.11
Capabilities Available : Bridge, Router
Capabilities Enabled   : Bridge, Router
TTL                   : 120

Port Based Data
=====

Port-ID              : 1/1/11
Port-Desc             : "1/1/11"
Port Mgmt-Address    : 11.11.11.11
```



```
Port VLAN ID      : 1
Parent Interface  : interface 1/1/11

switch#
```

In this example, all the ports except **1/1/11** are administratively down, and VLAN ID 100 is configured on this access interface.

```
switch# show lldp local-device

Global Data
=====

Chassis-ID          : 1c:98:ec:e3:45:00
System Name         : switch
System Description   : Aruba
Management Address   : 192.168.10.1
Capabilities Available : Bridge, Router
Capabilities Enabled  : Bridge, Router
TTL                  : 120

Port Based Data
=====

Port-ID             : 1/1/11
Port-Desc           : "1/1/11"
Port VLAN ID        : 100
Parent Interface     : interface 1/1/11
```

show lldp neighbor-info

Syntax

```
show lldp neighbor-info [<INTERFACE-NAME>]
```

Description

Displays information about neighboring devices for all interfaces or for a specific interface. The information displayed varies depending on the type of neighbor connected and the type of TLVs sent by the neighbor.

Command context

Manager (#)

Parameters

<INTERFACE-NAME>

Specifies the interface for which to show information for neighboring devices. Use the format member/slot/port (for example, 1/3/1).

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Showing LLDP information for all interfaces:

```
switch# show lldp neighbor-info
```

```
LLDP Neighbor Information
=====
```

```
Total Neighbor Entries      : 1
Total Neighbor Entries Deleted : 5
Total Neighbor Entries Dropped : 0
Total Neighbor Entries Aged-Out : 3
```

LOCAL-PORT	CHASSIS-ID	PORT-ID	PORT-DESC	TTL	SYS-NAME
1/1/2	38:21:c7:5c:df:40	1/1/2	1/1/2	120	switch
1/1/3	f8:60:f0:c9:e0:a0	1/1/3	1/1/3	120	switch

Showing information for interface **1/1/3** when it has only one switch connected as a neighbor:

```
Aruba-6100-Switch2# show lldp neighbor-info 1/1/3
```

```
Port : 1/1/3
Neighbor Entries : 1
Neighbor Entries Deleted : 1
Neighbor Entries Dropped : 0
Neighbor Entries Aged-Out : 1
Neighbor Chassis-Name : 6100
Neighbor Chassis-Description : Aruba JL679A PL.10.06.0001
Neighbor Chassis-ID : 88:3a:30:47:d1:c0
Neighbor Management-Address : 88:3a:30:47:d1:c0
Chassis Capabilities Available : Bridge, Router
Chassis Capabilities Enabled : Bridge, Router
Neighbor Port-ID : 1/1/3
Neighbor Port-Desc : 1/1/3
Neighbor Port VLAN ID : 1
TTL : 120
Neighbor Mac-Phy details
Neighbor Auto-neg Supported : True
Neighbor Auto-neg Enabled : True
Neighbor Auto-neg Advertised : 1000 BASE_TFD, 100 BASE_T4, 10 BASE_TFD
Neighbor MAU Type : 1000 BASE_TFD
```

Showing neighbor information for interface 1/3/2 when it has EEE enabled and successfully auto-negotiated:

```
switch# show lldp neighbor-info 1/3/2
```

```
Port : 1/3/2
Neighbor Entries : 1
Neighbor Entries Deleted : 1
Neighbor Entries Dropped : 0
Neighbor Entries Aged-Out : 1
Neighbor Chassis-Name : BLDG01-F1-6300
Neighbor Chassis-Description : Aruba JL668A FL.10.07.0001BN
Neighbor Chassis-ID : 88:3a:30:92:a5:c0
Neighbor Management-Address : 10.6.9.15
Chassis Capabilities Available : Bridge, Router
Chassis Capabilities Enabled : Bridge, Router
Neighbor Port-ID : 1/1/1
Neighbor Port-Desc : 1/1/1
Neighbor Port VLAN ID : 1
```

```

TTL                                : 120

Neighbor Mac-Phy details
Neighbor Auto-neg Supported        : true
Neighbor Auto-Neg Enabled         : true
Neighbor Auto-Neg Advertised       : 1000 BASE_TFD, 100 BASE_T4, 10 BASET_FD
Neighbor MAU type                  : 1000 BASETFD

Neighbor EEE information           : DOT3
Neighbor TX Wake time              : 17 us
Neighbor RX Wake time              : 17 us
Neighbor Fallback time             : 17 us
Neighbor TX Echo time              : 17 us
Neighbor RX Echo time              : 17 us

```

show lldp neighbor-info detail

Syntax

```
show lldp neighbor-info detail
```

Description

Shows detailed LLDP neighbor information for all LLDP neighbor connected interfaces.

Command context

Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Examples

Showing detailed LLDP information for all interfaces:

```

switch# show lldp neighbor-info detail

LLDP Neighbor Information
=====

Total Neighbor Entries      : 6
Total Neighbor Entries Deleted : 2
Total Neighbor Entries Dropped : 0
Total Neighbor Entries Aged-Out : 2

-----

Port                        : 1/1/1
Neighbor Entries           : 1
Neighbor Entries Deleted   : 0
Neighbor Entries Dropped   : 0
Neighbor Entries Aged-Out  : 0
Neighbor Chassis-Name      : 6300
Neighbor Chassis-Description : Aruba ...
Neighbor Chassis-ID        : 38:11:17:1a:d5:00
Neighbor Management-Address : 38:11:17:1a:d5:00

```

Chassis Capabilities Available : Bridge, Router
Chassis Capabilities Enabled : Bridge, Router
Neighbor Port-ID : 1/1/4
Neighbor Port-Desc : 1/1/4
Neighbor Port VLAN ID : 1
TTL : 120

Neighbor Mac-Phy details
Neighbor Auto-neg Supported : true
Neighbor Auto-Neg Enabled : true
Neighbor Auto-Neg Advertised : 1000 BASE_TFD, 100 BASE_T4, 10 BASET_FD
Neighbor MAU type : 1000 BASETFD

Port : 1/1/2
Neighbor Entries : 1
Neighbor Entries Deleted : 0
Neighbor Entries Dropped : 0
Neighbor Entries Aged-Out : 0
Neighbor Chassis-Name : 6300
Neighbor Chassis-Description : Aruba ...
Neighbor Chassis-ID : 38:11:17:1a:d5:00
Neighbor Management-Address : 38:11:17:1a:d5:00
Chassis Capabilities Available : Bridge, Router
Chassis Capabilities Enabled : Bridge, Router
Neighbor Port-ID : 1/1/5
Neighbor Port-Desc : 1/1/5
Neighbor Port VLAN ID : 1
TTL : 120

Neighbor Mac-Phy details
Neighbor Auto-neg Supported : true
Neighbor Auto-Neg Enabled : true
Neighbor Auto-Neg Advertised : 1000 BASE_TFD, 100 BASE_T4, 10 BASET_FD
Neighbor MAU type : 1000 BASETFD

Port : 1/1/3
Neighbor Entries : 1
Neighbor Entries Deleted : 0
Neighbor Entries Dropped : 0
Neighbor Entries Aged-Out : 0
Neighbor Chassis-Name : 6300
Neighbor Chassis-Description : Aruba ...
Neighbor Chassis-ID : 38:11:17:1a:d5:00
Neighbor Management-Address : 38:11:17:1a:d5:00
Chassis Capabilities Available : Bridge, Router
Chassis Capabilities Enabled : Bridge, Router
Neighbor Port-ID : 1/1/6
Neighbor Port-Desc : 1/1/6
Neighbor Port VLAN ID : 1
TTL : 120

Neighbor Mac-Phy details
Neighbor Auto-neg Supported : true
Neighbor Auto-Neg Enabled : true
Neighbor Auto-Neg Advertised : 1000 BASE_TFD, 100 BASE_T4, 10 BASET_FD
Neighbor MAU type : 1000 BASETFD

```

Port : 1/1/46
Neighbor Entries : 1
Neighbor Entries Deleted : 0
Neighbor Entries Dropped : 0
Neighbor Entries Aged-Out : 0
Neighbor Chassis-Name : 6300
Neighbor Chassis-Description : Aruba ...
Neighbor Chassis-ID : 38:11:17:1a:d5:00
Neighbor Management-Address : 38:11:17:1a:d5:00
Chassis Capabilities Available : Bridge, Router
Chassis Capabilities Enabled : Bridge, Router
Neighbor Port-ID : 1/1/19
Neighbor Port-Desc : 1/1/19
Neighbor Port VLAN ID : 1
TTL : 120

Neighbor Mac-Phy details
Neighbor Auto-neg Supported : true
Neighbor Auto-Neg Enabled : true
Neighbor Auto-Neg Advertised : 1000 BASE_TFD, 100 BASE_T4, 10 BASET_FD
Neighbor MAU type : 1000 BASETFD

```

```

-----
Port : 1/1/47
Neighbor Entries : 1
Neighbor Entries Deleted : 0
Neighbor Entries Dropped : 0
Neighbor Entries Aged-Out : 0
Neighbor Chassis-Name : 6300
Neighbor Chassis-Description : Aruba ...
Neighbor Chassis-ID : 38:11:17:1a:d5:00
Neighbor Management-Address : 38:11:17:1a:d5:00
Chassis Cap

```

show lldp statistics

Syntax

```
show lldp statistics [<INTERFACE-ID>]
```

Description

Shows global LLDP statistics or statistics for a specific interface.

Command context

Manager (#)

Parameters

<INTERFACE-ID>

Specifies an interface. Format: member/slot/port.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

Showing global statistics for all interfaces:

```
switch# show copp-policy statistics
switch# show lldp statistics
LLDP Global Statistics
=====

Total Packets Transmitted      : 25
Total Packets Received        : 20
Total Packets Received And Discarded : 0
Total TLVs Unrecognized       : 0

LLDP Port Statistics
=====
```

PORT-ID	TX-PACKETS	RX-PACKETS	RX-DISCARDED	TLVS-UNKNOWN
1/1/1	25	20	0	0
1/1/2	0	0	0	0
1/1/3	0	0	0	0
1/1/4	0	0	0	0
1/1/5	0	0	0	0
1/1/6	0	0	0	0
1/1/7	0	0	0	0
1/1/8	0	0	0	0
1/1/9	0	0	0	0
1/1/10	0	0	0	0
1/1/11	0	0	0	0
1/1/12	0	0	0	0
1/1/13	0	0	0	0
1/1/14	0	0	0	0
1/1/15	0	0	0	0
1/1/16	0	0	0	0
1/1/17	0	0	0	0
1/1/18	0	0	0	0
1/1/19	0	0	0	0
1/1/20	0	0	0	0
1/1/21	0	0	0	0
1/1/22	0	0	0	0
1/1/23	0	0	0	0
1/1/24	0	0	0	0
1/1/25	0	0	0	0
1/1/26	0	0	0	0
1/1/27	0	0	0	0
1/1/28	0	0	0	0

Showing statistics for interface **1/1/1**:

```
switch# show lldp statistics 1/1/1

LLDP Statistics
=====

Port Name      : 1/1/1
Packets Transmitted : 159
Packets Received   : 163
Packets Received And Discarded : 0
Packets Received And Unrecognized : 0
```

show lldp tlv

Syntax

```
show lldp tlv
```

Description

Shows the LLDP TLVs that are configured for send and receive.

Command context

Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Example

```
switch# show lldp tlv
```

```
TLVs Advertised
```

```
=====
```

```
Management Address  
Port Description  
Port VLAN-ID  
System Capabilities  
System Description  
System Name
```

aaa accounting all-mgmt

Syntax

```
aaa accounting all-mgmt <CONNECTION-TYPE> start-stop {local | group <GROUP-LIST>}  
no aaa accounting all-mgmt <CONNECTION-TYPE>
```

Description

Defines accounting as being local (with the name `local`) (the default). Or defines a sequence of remote AAA server groups to be accessed for accounting purposes.

For remote accounting, the information is sent to the first reachable remote server that was configured with this command for remote accounting. If no remote server is reachable, local accounting remains available. Each available connection type (channel) can be configured individually as either local or using remote AAA server groups. All server groups named in your command, must exist. This command can be issued multiple times, once for each connection type. Local is always available for any connection type not configured for remote accounting.



The system accounting log is not associated with any connection type (channel) and is therefore sent to the accounting method configured on the default connection type (channel) only.

The `no` form of this command removes for the specified connection type, any defined remote AAA server group accounting sequence. Local accounting is available for connection types without a configured remote AAA server group list (whether default or for the specific connection type).

Command context

`config`

Parameters

`<CONNECTION-TYPE>`

One of these connection types (channels):

`default`

Defines a list of accounting server groups to be used for the `default` connection type. This configuration applies to all other connection types (`console`, `https-server`, `ssh`) that are not explicitly configured with this command. For example, if you do not use `aaa accounting all-mgmt console...` to define the console accounting list, then this default configuration is used for console.

`console`

Defines a list of accounting server groups to be used for the `console` connection type.

`https-server`

Defines a list of accounting server groups to be used for the `https-server` (REST, Web UI) connection type.

`ssh`

Defines a list of accounting server groups to be used for the `ssh` connection type.

`start-stop`

Selects accounting information capture at both the beginning and end of a process.

`local`

Selects local-only accounting when used without the `group` parameter.

`group <GROUP-LIST>`

Specifies the list of remote AAA server group names. Each name can be specified one time. Predefined remote AAA group names `tacacs` and `radius` are available. Although not a group name, predefined name `local` is available. User-defined TACACS+ and RADIUS server group names may also be used. The remote AAA server groups are accessed in the order that the group names are listed in this command. Within each group, the servers are accessed in the order in which the servers were added to the group. Server groups are defined using command `aaa group server` and servers are added to a server group with the command `server`.

Authority

Administrators or local user group members with execution rights for this command.

Usage

Local accounting is always active. It cannot be turned off.

Examples

Setting local accounting for the default connection type:

```
switch(config)# aaa accounting all-mgmt default start-stop local
```

Setting local accounting for the console connection type:

```
switch(config)# aaa accounting all-mgmt console start-stop local
```

aaa authentication console-login-attempts

Syntax

```
aaa authentication console-login-attempts <ATTEMPTS> console-lockout-time <LOCKOUT-TIME>  
no aaa authentication console-login-attempts
```

Description

For the console interface only (not SSH or REST), enables console login attempt limiting. If the number of failed console login attempts equals the configured threshold, the user is locked out for the configured duration..

The `no` form of this command disables console login attempt limits.



Important: If you enable the lockout using this command and also enable the SSH and REST lockout using command `aaa authentication limit-login-attempts`, and then enter too many consecutive wrong passwords, you will become locked out, and will have to wait for the configured lockout time to elapse before logging in on any interface.



This console login attempt limiting feature is only available when not using remote authentication through AAA servers (TACACS+ or RADIUS) on any interface. Remote authentication through AAA servers (TACACS+ or RADIUS) is not possible when limit login attempts is configured on any interface.

Command context

config

Parameters

<ATTEMPTS>

Specifies the threshold of failed console login attempts that triggers user lockout. Range: 1 to 10. For example, if <ATTEMPTS> is set to 1, a single failed login attempt triggers immediate user lockout.

<LOCKOUT-TIME>

Specifies the amount of time a user is locked out. Range: 1 to 3600 seconds.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling console login attempt failure limiting with a 60 second lockout being triggered upon the third consecutive login attempt failure.

```
switch(config)# aaa authentication console-login-attempts 3 console-lockout-time 60
```

Disabling console login attempt failure limiting:

```
switch(config)# no aaa authentication console-login-attempts
```

aaa authentication limit-login-attempts

Syntax

```
aaa authentication limit-login-attempts <ATTEMPTS> lockout-time <LOCKOUT-TIME>  
no aaa authentication limit-login-attempts
```

Description

For the SSH and REST interface, enables local login attempt limiting. If the number of failed local login attempts equals the configured threshold, the user is locked out for the configured duration.

The `no` form of this command disables local login attempt limits.



Important: If you enable the lockout using this command and also enable the console lockout using command `aaa authentication console-login-attempts`, and then enter too many consecutive wrong passwords, you will become locked out, and will have to wait for the configured lockout time to elapse before logging in on any interface.



This local login attempt limiting feature is only available when not using remote authentication through AAA servers (TACACS+ or RADIUS) on any interface. Remote authentication through AAA servers (TACACS+ or RADIUS) is not possible when limit login attempts is configured on any interface.

Command context

config

Parameters

<ATTEMPTS>

Specifies the threshold of failed local login attempts that triggers user logout. Range: 1 to 10. For example, if *<ATTEMPTS>* is set to 1, a single failed login attempt triggers immediate user logout.

<LOCKOUT-TIME>

Specifies the amount of time a user is locked out. Range: 1 to 3600 seconds.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling local login attempt failure limiting with a 20 second logout being triggered upon the fourth consecutive login attempt failure.

```
switch(config)# aaa authentication limit-login-attempts 4 lockout-time 20
```

Disabling login attempt failure limiting:

```
switch(config)# no aaa authentication limit-login-attempts
```

aaa authentication login

Syntax

```
aaa authentication login <CONNECTION-TYPE> {local | group <GROUP-LIST>}  
no aaa authentication login <CONNECTION-TYPE>
```

Description

Defines authentication as being local (with the name `local`) (the default). Or defines a sequence of remote AAA server groups to be accessed for authentication purposes. Each available connection type (channel) can be configured individually as either local or using remote AAA server groups. All server groups named in your command, must exist. This command can be issued multiple times, once for each connection type. Local is always available for any connection type not configured for remote AAA authentication.

The `no` form of this command removes for the specified connection type, any defined remote AAA server group authentication sequence. Local authentication is available for connection types without a configured remote AAA server group list (whether default or for the specific connection type).

Command context

config

Parameters

<CONNECTION-TYPE>

One of these connection types (channels):

`default`

Defines a list of authentication server groups to be used for the `default` connection type. This configuration applies to all other connection types (`console`, `https-server`, `ssh`) that are not explicitly configured with this command. For example, if you do not use `aaa authentication login console...` to define the console authentication list, then this default configuration is used for `console`.

`console`

Defines a list of authentication server groups to be used for the `console` connection type.

`https-server`

Defines a list of authentication server groups to be used for the `https-server` (REST, Web UI) connection type.

`ssh`

Defines a list of authentication server groups to be used for the `ssh` connection type.

`local`

Selects local-only authentication when used without the `group` parameter.

`group` <GROUP-LIST>

Specifies the list of remote AAA server group names. Each name can be specified one time. Predefined remote AAA group names `tacacs` and `radius` are available. Although not a group name, predefined name `local` is available. User-defined TACACS+ and RADIUS server group names may also be used. The remote AAA server groups are accessed in the order that the group names are listed in this command. Within each group, the servers are accessed in the order in which the servers were added to the group. Server groups are defined using command `aaa group server` and servers are added to a server group with the command `server`.

If no AAA server is reachable, local authentication is attempted.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting local authentication for the default connection type:

```
switch(config)# aaa authentication login default local
```

Setting local authentication for the console connection type:

```
switch(config)# aaa authentication login console local
```

aaa authentication minimum-password-length

Syntax

`aaa authentication minimum-password-length <LENGTH>`

`no aaa authentication minimum-password-length`

Description

Enables minimum password length checking. Existing passwords shorter than the minimum length are unaffected. Length checking does not apply to ciphertext passwords. Length checking applies both to local and remote authentication.

The `no` form of this command disables minimum password length checking.

Command context

config

Parameters

<LENGTH>

Specifies the minimum password length. Range: 1 to 32.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling password length checking, with a minimum length of 12.

```
switch(config)# aaa authentication minimum-password-length 12
```

Disabling minimum password length checking:

```
switch(config)# no aaa authentication minimum-password-length
```

aaa authorization commands

Syntax

```
aaa authorization commands <CONNECTION-TYPE> {local | none}
aaa authorization commands <CONNECTION-TYPE> group <GROUP-LIST>

no aaa authorization commands <CONNECTION-TYPE>
```

Description

Defines authorization as being basic local RBAC (specified as `none`), or as full-fledged local RBAC specified as `local` (the default), or as remote TACACS+ (specified with `group <GROUP-LIST>`). Each available connection type (channel) can be configured individually. All server groups named in the command, must exist. This command can be issued multiple times, once for each connection type.

The `no` form of this command unconfigures authorization for the specified connection type, reverting to the default of `local`.



Although only TACACS+ servers are supported for remote authorization, local authorization (basic or full-fledged) can be used with remote RADIUS authentication.

Command context

config

Parameters

<CONNECTION-TYPE>

One of these connection types (channels):

`default`

Selects the `default` connection type for configuration. This configuration applies to all other connection types (`console`, `ssh`) that are not explicitly configured with this command. For example, if you do not use `aaa authorization commands console...` to define the console authorization list, then this default configuration is used for console.

`console`

Selects the `console` connection type for configuration.

`ssh`

Selects the `ssh` connection type for configuration.

`local` (the default)

When used alone without `group <GROUP-LIST>`, selects local authorization which can be used to provide authorization for a purely local setup without any remote AAA servers and also for when RADIUS is used for remote Authentication and Accounting but Authorization is local.

When used after `group`, provides for fallback (to full-fledged local authorization) when every server in every specified TACACS+ server group cannot be reached.



If any TACACS+ server in the specified groups is reachable, but the command fails to be authorized by that server, the command is rejected and local authorization is never attempted. Local authorization is only attempted if every TACACS+ server cannot be reached.

`none`

When used alone without `group <GROUP-LIST>`, selects basic local RBAC authorization, for use with the built-in user groups (`administrators`, `operators`, `auditors`).

When used after `group`, provides for fallback (to basic local RBAC authorization) when every server in every specified TACACS+ server group cannot be reached.



With `none`, for users belonging to user-defined user groups, all commands can be executed regardless of what authorization rules are defined in such groups. For per-command local authorization, use `local` instead.

`group <GROUP-LIST>`

Specifies the list of remote AAA server group names. Predefined remote AAA group name `tacacs` is available. User-defined TACACS+ server group names may also be used. The remote AAA server groups are accessed in the order that the group names are listed in this command. Within each group, the servers are accessed in the order in which the servers were added to the group. Server groups are defined using command `aaa server group` and servers are added to a server group using command `server`.

It is recommended to always include either the special name `local` or `none` as the last name in the group list. If both `local` and `none` are omitted, and no remote AAA server is reachable (or the first reachable server cannot authorize the command), command execution for the current user will not be possible.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the authorization for default to `local`:

```
switch(config)# aaa authorization commands default local
```

Setting the authorization for the SSH interface to `none`:

```
switch(config)# aaa authorization commands ssh none
```

show aaa accounting

Syntax

```
show aaa accounting
```

Description

Shows the accounting configuration per connection type (channel).

Command context

Operator (>) or Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Example

Configuring and then showing local accounting for the default and console connection types:

```
switch(config)# aaa accounting all default start-stop local
switch(config)# aaa accounting all console start-stop local
switch(config)# exit
switch# show aaa accounting
AAA Accounting:
  Accounting Type           : all
  Accounting Mode           : start-stop
```

Accounting for default channel:

GROUP NAME	GROUP PRIORITY
local	0

Accounting for console channel:

GROUP NAME	GROUP PRIORITY
local	0

show aaa authentication

Syntax

```
show aaa authentication
```

Description

Shows the authentication configuration per connection type (channel).

Command context

Operator (>) or Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Example

Configuring and then showing local authentication for the default and console connection types (channels):

```
switch(config)# aaa authentication login default local
switch(config)# aaa authentication login console local
switch(config)# exit
switch# show aaa authentication
```

```
AAA Authentication:
  Fail-through           : Disabled
  Limit Login Attempts   : Not set
  Lockout Time           : 300
  Minimum Password Length : Not set
```

Authentication for default channel:

GROUP NAME	GROUP PRIORITY
local	0

Authentication for console channel:

GROUP NAME	GROUP PRIORITY
local	0

show aaa authorization

Syntax

```
show aaa authorization
```

Description

Shows the authorization configuration per connection type (channel).

Command context

Operator (>) or Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Example

Configuring and then showing full-fledged local RBAC authorization for the default and console connection types (channels):

```
switch(config)# aaa authorization commands default none
switch(config)#
switch(config)# aaa authorization commands console none
```



```
switch(config)# exit
switch#
switch# show aaa authorization
Authorization for default channel:
```

GROUP NAME	GROUP PRIORITY
none	0

```
Authorization for console channel:
```

GROUP NAME	GROUP PRIORITY
none	0

show ssh authentication-method

Syntax

```
show ssh authentication-method
```

Description

Shows the status of the SSH public key method and the local password-based (through SSH client) authentication method.

Command context

Operator (>) or Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

Showing the authentication methods.

```
switch# show ssh authentication-method
SSH publickey authentication : Enabled
SSH password authentication : Enabled
```

show user

Syntax

```
show user <USERNAME> authorized-key
```

Description

Shows the SSH client public key list for a specified user.

Command context

Operator (>) or Manager (#)

Parameters

<USERNAME>

Specifies the username for which you want to show the SSH client public key list.

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Usage

Any user can show their own public key list; however, administrators can also show a public key list of other users.

Examples

Showing a client public key:

```
switch# show user admin authorized-key

1. Key Type : RSA      Key size : 2048
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQDMtyMBmmAaF6r1zxf3DZNHSYVHBjhlbBlyAIqQ8DSHK
...
U+aEl4UW/ifIukmK67sIHwK+FhhRYwPztQc5pjyOPk128a4pgKQaHCcOF169Z admin@switch
```

Showing two client public keys:

```
switch# show user admin authorized-key

1. Key Type : ECDSA    Curve : nistp256
ecdsa-sha2-nistp256 AAAAE2VjZHNhLXNoYTItbmlzdHAyNTYAAAAIbmlzdHAyNTYAAABBBEqEFevZ0
...
176V+D0svdCJ9Wo32zqI9OeAdTJw/eZYp5qknhNgS81HjAI6J/4/kAqdZAJbqQUiCAk= admin@switch

2. Key Type : RSA      Key size : 2048
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQDXQHrQv7+/GcMdOhr//IRjJkX7TQKupW89j80bL7xq8
...
j8qKuHWSN0/h/HxjzQJuYDVmZN5vG3DhpXbBZU1ZNnchVod13QLCesqA3VLKN admin@switch
```

ssh password-authentication

Syntax

```
ssh password-authentication
```

```
no ssh password-authentication
```

Description

Enables the password-based authentication method for use with SSH clients.

The `no` form of this command disables the password-based authentication method for use with SSH clients.

Command context

```
config
```

Authority

Administrators or local user group members with execution rights for this command.

Usage

The switch ships with password-based authentication (for SSH clients) enabled. The maximum number of password retries is three.

Examples

Enabling password authentication for use with SSH clients:

```
switch(config)# ssh password-authentication
```

Disabling password authentication for use with SSH clients:

```
switch(config)# no ssh password-authentication
```

ssh public-key-authentication

Syntax

```
ssh public-key-authentication
```

```
no ssh public-key-authentication
```

Description

Enables the SSH public key authentication method. The switch ships with SSH public key authentication enabled.

The `no` form of this command disables the SSH public key authentication method.



Although SSH public key authentication is enabled by default, it cannot be used until SSH public keys are added with the `user authorized-key` command.

Command context

```
config
```

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling SSH public key authentication:

```
switch(config)# ssh public-key-authentication
```

Disabling SSH public key authentication:

```
switch(config)# no ssh public-key-authentication
```

user authorized-key

Syntax

```
user <USERNAME> authorized-key <PUBKEY>
no user <USERNAME> authorized-key [<KEYNUM>]
```

Description

Copies an SSH client public key into the key list. If the key list and the public key do not exist, it creates a list with the public key. If the SSH client public key exists, the command appends the new key to the existing list. The client public key list holds a maximum of 32 client keys.

The `no` form of the command removes either one or all SSH public keys from the key list.

Command context

config

Parameters

<USERNAME>

Specifies the name of the user.

<PUBKEY>

Specifies the SSH client public key to be copied into the key list.

<KEYNUM>

Specifies the key number. The range is 1 to 32. Use the `show user <USERNAME> authorized-key` command to find the key number associated with the key.

Authority

Operators or Administrators or local user group members with execution rights for this command.

Operators can execute this command from the operator context (>) only.

Usage

Each key on the key list has a key identifier. The `show user <USERNAME> authorized-key` command displays the key identifier associated with the key.

Administrators can add and remove the public keys of themselves and other users. Operators can add and remove only their own public keys. If the public key authentication method is enabled, the client public key present is used by the SSH server to authenticate the client. The authentication method reverts to the password authentication method and prompts for a client password when one of the following occurs:

- The client public keys are not present.
- The server does not have the keys enabled.
- The public key method is disabled.

You can either remove all keys or a specific key. Each key on the key list has a key identifier. If you provide the key identifier in this command, the command removes the corresponding key from the list. If you provide no key identifier, the command removes all keys from the key list.

Examples

Adding a public key:

```
switch(config)#user admin authorized-key ecdsa-sha2-nistp256 AAAAE2VjZHNhLXNoYTItbmlzdHAyNTYAAAAIbmlzdHAyNTYAAABBBEgEFevZ0176V+D0svdCJ9Wo32zqI9OeAIdTJwT/eZYp50qkAnhZNgS81HBjAI6QJ/4/kAyqdZ9oAjbiqQUiCAk= root@switch
```

Removing all SSH public keys from the list:

```
switch(config)# no user admin authorized-key
```

Removing the specified SSH public key from the list:

```
switch(config)# no user admin authorized-key 2
```

logrotate maxsize

Syntax

```
logrotate maxsize <MAX-SIZE>

no logrotate maxsize
```

Description

Specifies the maximum allowed log file size.

The `no` form of this command resets the size of the log file to the default (100 MB).

Command context

config

Parameters

<MAX-SIZE>

Specifies the allowed size the log file can reach before it is compressed and stored locally or transferred to a remote host. The size is a value in the range of 10- 200 MB, and it cannot exceed 32 MB for transferred files.

Authority

Administrators or local user group members with execution rights for this command.

Usage

A log file that exceeds either the configured <MAX-SIZE> value or the `logrotate period`, triggers rotation for that log file. Log rotation occurs during the next hourly maintenance cycle.

Logs are stored locally (event logs in the `/var/log/event.log` file, and authentication logs in the `/var/log/auth.log` file) or transferred to the configured remote destination target using TFTP.

Examples

```
switch(config)# logrotate maxsize 32
```

```
switch(config)# no logrotate maxsize
```

logrotate period

Syntax

```
logrotate period {daily | hourly | monthly | weekly}
```

```
no logrotate period
```

Description

Sets the rotate period for the event logs, stored in the `/var/log/event.log` file, and authentication logs, stored in the `/var/log/auth.log` file. Defaults to `daily`.

A log file that exceeds either the `logrotate <MAX-SIZE>` value or the `logrotate period` (whichever happens first), triggers rotation for that log file.

The `no` form of this command resets the log rotation period to the default.

Command context

`config`

Parameters

`daily`

Rotates log files on a daily basis (default) at 1:00 am local time.

`hourly`

Rotates log files every hour at the first second of the hour.

`monthly`

Rotates log files monthly on the first day of the month.

`weekly`

Rotates log files once a week on Sunday.

Authority

Administrators or local user group members with execution rights for this command.

Examples

```
switch(config)# logrotate period weekly
```

logrotate target

Syntax

```
logrotate target {tftp://<IPV4_ADDR> | tftp://<IPV6_ADDR>}
```

```
no logrotate target
```

Description

Specifies the target remote host Universal Resource Identifier (URI) using TFTP protocol to allow transfer of rotated and compressed files to a remote target. Rotated log files are stored locally (event logs in the `/var/log/event.log` file, and authentication logs in the `/var/log/auth.log` file) or transferred to the configured remote destination target.

The `no` form of this command resets the target to the default, which stores the rotated and compressed log files locally in `/var/log/`.

Command context

`config`

Parameters

<IPv4_ADDR>

Specifies an IPv4 IP Address location for log file storage.

<IPv6_ADDR>

Specifies an IPv6 IP Address location for log file storage.

Authority

Administrators or local user group members with execution rights for this command.

Usage

To transfer rotated log files remotely, use the TFTP protocol only, and make sure that the rotated log file is less than 32 MB in size. Use the Linux TFTP command to transfer the file. The rotated log file is removed from the local path `/var/log/` when the log file is moved to a TFTP server.

Examples

Setting an IPv4 target:

```
switch(config)# logrotate target tftp://192.168.1.132
```

Setting an IPv6 target:

```
switch(config)# logrotate target tftp://2001:db8:0:1::128
```

Removing a logrotate target :

```
switch(config)# logrotate target tftp://1.1.1.1
switch(config)# do show logrotate
Logrotate configurations :
Period : daily
Maxsize : 10MB
Target : tftp://1.1.1.1

switch(config)# no logrotate target
switch(config)# do show logrotate
Logrotate configurations :
Period : daily
Maxsize : 10MB

switch(config)#
```

show logrotate

Syntax

```
show logrotate
```

Description

Displays `logrotate` configuration details.

Command context

Manager (#)

Authority

Administrators or local user group members with execution rights for this command.

Examples

```
switch# show logrotate
Logrotate configurations :
Period           : daily
Maxsize          : 100MB
Target           : local
switch#
```

loop-protect

Syntax

```
loop-protect
no loop-protect
```

Description

Enables loop protection on a layer 2 interface or LAG. Loop protection packets are sent/received on the LAG and not the interface which are members of the LAG. Loop protection only works on layer 2 interfaces. If a layer 2 interface is changed to a layer 3 interface, all loop protection configuration settings are lost for that interface.

The `no` form of this command disables loop protection on a layer 2 interface or LAG.

Command context

```
config-if
config-lag-if
```

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling loop protection on interface **1/1/1**:

```
switch# config
switch(config)# interface 1/1/1
switch(config-if)# loop-protect
```

Enabling loop protection on LAG **25**:

```
switch# config
switch(config)# interface lag 25
switch(config-lag-if)# loop-protect
```

loop-protect action

Syntax

```
loop-protect action {do-not-disable | tx-disable | tx-rx-disable}
no loop-protect action {do-not-disable | tx-disable | tx-rx-disable}
```

Description

Sets the action to be taken when a loop protection packet is received on a port.

If an action is configured after a loop is detected, then the new action only takes effect after the re-enable timer expires. To have the action take effect immediately, disable and then re-enable loop protect.

The `no` form of this command resets the action to the default (`tx-disable`).

Command context

`config-if`

Parameters

`do-not-disable`

No ports are disabled. On every transmit interval, the loop will be detected and the detection will be reported via an SNMP trap and an event log message.

`tx-disable`

The port that transmitted the loop detection packet is disabled. When this setting is enabled, environments with N loops, must have loop protection be configured on at least N-1 ports to have a loop free topology. Default.

`tx-rx-disable`

The ports that transmitted and received the loop detection packet are disabled.

Authority

Administrators or local user group members with execution rights for this command.

Example

```
switch(config-if)# loop-protect action do-not-disable
switch(config-if)# no loop-protect action do-not-disable
```

loop-protect re-enable-timer

Syntax

```
loop-protect re-enable-timer <TIME>
no loop-protect re-enable-timer <TIME>
```

Description

Configures the time interval after which an interface disabled by loop protection is re-enabled. The loop protection timer is disabled by default.

The `no` form of this command disables the loop protect timer.

Command context

`config`

Parameters

`<TIME>`

Specify the number of seconds after which a disabled interface is re-enabled. Range: 15 to 604800.

Authority

Administrators or local user group members with execution rights for this command.

Example

```
switch# config  
switch(config)# loop-protect re-enable-timer 60
```

loop-protect transmit-interval

Syntax

```
loop-protect transmit-interval <TIME>  
no loop-protect transmit-interval [<TIME>]
```

Description

Configures the time interval between successive loop protect packets sent on an interface. The **no** form of this command sets the time interval to the default value of 5 seconds.

Command context

config

Parameters

<TIME>

Configures the transmit interval in seconds. Range: 5 to 10. Default: 5.

Authority

Administrators or local user group members with execution rights for this command.

Examples

```
switch(config)# loop-protect transmit-interval 10  
switch(config)# no loop-protect transmit-interval
```

loop-protect trap loop-detected

Syntax

```
loop-protect trap loop-detected  
no loop-protect trap loop-detected
```

Description

Enables sending SNMP traps for loop-protect related events. The **no** form of this command disables sending SNMP traps for loop-protect related events.

Command context

config

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling the sending of SNMP traps:

```
switch# loop-protect trap loop-detected
```

Disabling the sending of SNMP traps:

```
switch# no loop-protect trap loop-detected
```

loop-protect vlan

Syntax

```
loop-protect vlan <VLAN-LIST>  
no loop-protect vlan
```

Description

Specifies the trunk allowed VLANs on which loop protection packets are sent. By default, loop protection packets are only sent on access VLANs and native VLANs on a port. To send loop protection packets on trunk allowed VLANs, the VLANs must be explicitly added using this command.

Loop protection can be configured on a maximum of 40942048 VLANs across all interfaces.

The `no` form of this command removes loop protection from all VLANs on the interface.

Command context

```
config-if
```

Parameters

<VLAN-LIST>

Specifies the number of a single VLAN, or a series of numbers for a range of VLANs, separated by commas (1, 2, 3, 4), dashes (1-4), or both (1-4, 6).

Authority

Administrators or local user group members with execution rights for this command.

Example

```
switch(config-if)# loop-protect vlan 2-6,10,15-20
```

show loop-protect

Syntax

```
show loop-protect [<INTERFACE-NAME>]
```

Description

Shows loop protection information for all interfaces or for a specific interface.

Command context

Operator (>) or Manager (#)

Parameters

<INTERFACE-NAME>

Specifies the name of a logical interface on the switch. This can be one of the following:

- An Ethernet interface associated with a physical port. Format: member/slot/port.
- A LAG (link aggregation group). Specify the ID of LAG . For example: lag100.

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Examples

```
switch# show loop-protect

Transmit Interval (sec)           : 5
Port Re-enable Timer (sec)       : Disabled
Loop Detected Trap                : Enabled

Interface 1/1/1
  Loop-protect enabled           : Yes
  Loop-Protect enabled VLANs    :
  Action on loop detection       : TX disable
  Loop detected count            : 0
  Loop detected                  : No
  Interface status               : up

Interface 1/1/2
  Loop-protect enabled           : Yes
  Loop-Protect enabled VLANs    :
  Action on loop detection       : TX disable
  Loop detected count            : 0
  Loop detected                  : No
  Interface status               : up
```

```
switch# show loop-protect 1/1/3

Status and Counters - Loop Protection Information

Transmit Interval (sec)           : 5
Port Re-enable Timer (sec)       : 0
Loop Detected Trap                : Disabled

Interface 1
  Loop-protect enabled           : Yes
  Loop-Protect enabled VLANs    :
  Action on loop detection       : TX disable
  Loop detected count            : 0
  Loop detected                  : No
  Interface status               : up
```

interface loopback

Syntax

```
interface loopback <INSTANCE>  
no interface loopback <INSTANCE>
```

Description

Creates a loopback interface and enters loopback configuration mode.
The `no` form of this command deletes a loopback interface.

Command context

config

Parameters

<INSTANCE>

Selects the loopback interface ID. Range: 0 to 4

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Examples

```
switch(config)# interface loopback 1  
switch(config-loopback-if)#
```

ip address

Syntax

```
ip address <IPv4-ADDR/MASK> [secondary]  
no ip address <IPv4-ADDR/MASK> [secondary]
```

Description

Sets the IPv4 address for a loopback interface.
The `no` form of this command reverses the set of the IPv4 address for a loopback interface.

Command context

config

Parameters

<IPv4-ADDR>

Specifies an IP address in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255.

<MASK>

Specifies the number of bits in the address mask in CIDR format (x), where x is a decimal number from 0 to 128.

secondary

Indicates that the IPv4 address is a secondary address.

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Examples

```
switch(config)# interface loopback 1  
switch(config-loopback-if)# ip address 16.93.50.2/24  
switch(config-loopback-if)# ip address 20.1.1.1/24 secondary
```

ipv6 address

Syntax

ipv6 address <IPv6-ADDR/MASK>

Description

Sets the IPv6 address for a loopback interface.

Command context

config

Parameters

<IPv6-ADDR>

Specifies an IP address in IPv6 format (xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx), where x is a hexadecimal number from 0 to F.

<MASK>

Specifies the number of bits in the address mask in CIDR format (x), where x is a decimal number from 0 to 128.

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Examples

```
switch(config)# interface loopback 1  
switch(config-loopback-if)# ipv6 address fd00:5708::f02d:4df6/64
```

show interface loopback

Syntax

show interface loopback [brief | instance <ID>]

Description

This command displays the configuration and status of loopback interfaces.

Command context

Operator (>) or Manager (#)

Parameters

brief

Displays brief information about all configured loopback interfaces.

instance <ID>

Displays the configuration and status of a loopback interface ID. Range: 1-255

Authority

Operators or Administrators or local user group members with execution rights for this command.

Operators can execute this command from the operator context (>) only.

Examples

```
switch# show interface loopback
```

```
Interface loopback1 is up
IPv4 address 192.168.1.1/24
```

```
Interface loopback2 is up
IPv4 address 182.168.1.1/24
```

```
switch# show interface loopback brief
```

```
-----
Loopback      IP Address                               Status
Interface
-----
```

```
loopback1    10.1.1.1/24                               up
loopback1    1111:2222:3333:4444::6666/128            up
```

```
switch# show interface loopback 1
```

```
Interface loopback1 is up
IPv4 address 192.168.1.1/24
```

clear mac-address

Syntax

```
clear mac-address {port <PORT-NUM> [vlan <VLAN-ID>] | vlan <VLAN-ID> [port <PORT-NUM>]}
```

Description

Clears the dynamic learned MAC addresses on the specified port, VLAN, or combination of port and VLAN.

Command context

Manager (#)

Parameters

<PORT-NUM>

Specifies a physical port on the switch. Format: member/slot/port.

<VLAN-ID>

Specifies the number of a VLAN.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Clearing the learned MAC addresses on a port:

```
switch# clear mac-address port 1/1/1
```

Clearing the learned MAC addresses on a combination of a VLAN and a port:

```
switch# clear mac-address port 1/1/1 vlan 20
```

```
switch# clear mac-address vlan 2 port 1/1/3
```

mac-address-table age-time

Syntax

```
mac-address-table age-time <SECONDS>  
no mac-address-table age-time [<SECONDS>]
```

Description

Sets the maximum amount of time a MAC address remains in the MAC address table. When this time expires, the MAC address is removed.

The `no` form of this command resets the MAC aging timer to the default value (300 seconds).

Command context

`config`

Parameters

`age-time` *<SECONDS>*

Specifies the MAC address aging time in seconds. Range: 60 to 3600. Default: 300.

Authority

Administrators or local user group members with execution rights for this command.

Example

```
switch(config)# mac-address-table age-time 120
```

mac-lockout

Syntax

`mac-lockout` *<MAC-ADDR>*
`no mac-lockout` *<MAC-ADDR>*

Description

Locks a MAC address globally on the switch and all VLANs. The switch drops all data packets addressed to or from the given address.

The `no` form of this command unlocks the MAC address globally on the switch and all VLANs.

Command context

`config`

Parameters

<MAC-ADDR>

Specifies the MAC address.

Authority

Administrators or local user group members with execution rights for this command.

Usage

MAC lockout is implemented on each switch individually. MAC lockout overrides MAC lockdown, port security (secure MAC), and 802.1X authentication. The MAC lockout feature is not intended to lock broadcast/multicast MAC addresses and switch agent MACs.

A maximum of 128 MAC lockouts can be configured on a switch.

Example

Enabling MAC lockout:

```
switch(config)# mac-lockout 00:00:00:00:00:01
```

Disabling MAC lockout:

```
switch(config)# no mac-lockout 00:00:00:00:00:01
```

show mac-address-table

Syntax

```
show mac-address-table
```

Description

Shows MAC address table information.

Command context

Operator (>) or Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Showing output when table entries exist:

```
switch# show mac-address-table
MAC age-time           : 300 seconds
Number of MAC addresses : 5

MAC Address            VLAN    Type      Port
-----
00:00:00:00:00:05      1      dynamic   1/1/2
00:00:00:00:00:06      2      dynamic   1/1/1
```

Showing output when there are no MAC table entries:

```
switch# show mac-address-table
No MAC entries found.
```

show mac-address-table address

Syntax

```
show mac-address-table address <MAC-ADDR>
```

Description

Shows MAC address table information for a specific MAC address.

Command context

Operator (>) or Manager (#)

Parameters

<MAC-ADDR>

Specifies the MAC address.

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Example

```
switch# show mac-address-table address 00:00:00:00:00:01
MAC age-time           : 300 seconds
Number of MAC addresses : 2

MAC Address            VLAN    Type      Port
-----
00:00:00:00:00:01     2      dynamic  1/1/1
00:00:00:00:00:01     1      dynamic  1/1/1
```

show mac-address-table count

Syntax

```
show mac-address-table count
    [dynamic | port <PORT-NUM> | vlan <VLAN-ID>]
```

Description

Displays the number of MAC addresses.

Command context

Operator (>) or Manager (#)

Parameters

dynamic

Show the count of dynamically learned MAC addresses.

<PORT-NUM>

Specifies a physical port on the switch. Format: member/slot/port.

vlan <VLAN-ID>

Specifies the number of a VLAN.

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Examples

Showing the number of MAC addresses:

```
switch# show mac-address-table count
Number of MAC addresses : 8
```

Showing the number of dynamically learned MAC addresses:

```
switch# show mac-address-table count dynamic
Number of MAC addresses : 8
```

Showing the number of MAC addresses per physical port on the switch:

```
switch# show mac-address-table count port 1/1/1
Number of MAC addresses : 2
```

Showing the number of MAC addresses per VLAN:

```
switch# show mac-address-table count vlan 100
Number of MAC addresses : 5
```

show mac-address-table dynamic

Syntax

```
show mac-address-table dynamic [port <PORT-NUM> | vlan <VLAN-ID>]
```

Description

Shows MAC address table information about dynamically learned MAC addresses.

Command context

Operator (>) or Manager (#)

Parameters

<PORT-NUM>

Specifies a physical port on the switch. Format: member/slot/port.

<VLAN-ID>

Specifies the number of a VLAN.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Showing all dynamic MAC address table entries:

```
switch# show mac-address-table dynamic
MAC age-time           : 300 seconds
Number of MAC addresses : 2

MAC Address            VLAN    Type    Port
```

```

-----
00:00:00:00:00:05    1      dynamic    1/1/2
00:00:00:00:00:06    2      dynamic    1/1/1

```

Showing dynamic MAC address table entries for VLAN 1:

```

switch# show mac-address-table dynamic vlan 1
MAC age-time          : 300 seconds
Number of MAC addresses : 1

MAC Address          VLAN    Type      Port
-----
00:00:00:00:00:05    1      dynamic    1/1/2

```

Showing dynamic MAC address table entries for port 1/1/1:

```

switch# show mac-address-table dynamic port 1/1/1
MAC age-time          : 300 seconds
Number of MAC addresses : 1

MAC Address          VLAN    Type      Port
-----
00:00:00:00:00:06    2      dynamic    1/1/1

```

show mac-address-table lockout

Syntax

```
show mac-address-table lockout
```

Description

Shows MAC lockout table information.

Command context

Operator (>) or Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Examples

```

switch# show mac-address-table lockout
Number of MAC lockout addresses :

2MAC Address          Type
-----
00:00:00:00:01:10    static
00:00:00:00:10:03    static

```

show mac-address-table port

Syntax

```
show mac-address-table port <PORT-NUM>
```

Description

Shows the MAC address table entries for the specified port.

Command context

Operator (>) or Manager (#)

Parameters

<PORT-NUM>

Specifies a physical port on the switch. Format: member/slot/port.

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Examples

Showing the MAC address table entries for port **1/1/1**:

```
switch# show mac-address-table port 1/1/1
MAC age-time           : 300 seconds
Number of MAC addresses : 1

MAC Address           VLAN    Type      Port
-----
00:00:00:00:00:01    2      dynamic   1/1/1
```

show mac-address-table static

Syntax

```
show mac-address-table static
```

Description

Shows all statically configured MAC addresses.

Command context

Operator (>) or Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Examples

```
switch# show mac-address-table static
Number of MAC addresses : 2

MAC Address           VLAN    Port
```



```
-----
00:00:00:00:10:02    1        1/1/1
00:00:00:00:10:03    1        1/1/1
```

show mac-address-table vlan

Syntax

```
show mac-address-table vlan <VLAN-ID>
```

Description

Shows MAC addresses learned by or configured on the specified VLAN.

Command context

Operator (>) or Manager (#)

Parameters

```
vlan <VLAN-ID>
```

Specifies the VLAN ID.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

```
switch# show mac-address-table vlan 1
MAC age-time           : 300 seconds
Number of MAC addresses : 1

MAC Address           VLAN    Type      Port
-----
00:00:00:00:00:01     1      dynamic  1/1/1
```

static-mac

Syntax

```
static-mac <MAC-ADDR> vlan <VLAN-ID> port <PORT-NUM>
no static-mac <MAC-ADDR> vlan <VLAN-ID> port <PORT-NUM>
```

Description

Adds a static MAC address to the MAC address table and associates it with a port or existing VLAN. Static MAC addresses can only be assigned to layer 2 (non-routed) interfaces. Static MAC addresses are not affected by the MAC address aging time.

The `no` form of this command deletes a static MAC address.

Command context

```
config
```

Parameters

<MAC-ADDR>

Specifies a MAC address (xx:xx:xx:xx:xx:xx), where x is a hexadecimal number from 0 to F.

vlan <VLAN-ID>

Specifies number of an existing VLAN.

port <PORT-NUM>

Specifies a physical port on the switch. Format: member/slot/port.

Authority

Administrators or local user group members with execution rights for this command.

Examples

```
switch(config)# static-mac 00:00:00:00:00:01 vlan 1 port 1/1/1
switch(config)# no static-mac 00:00:00:00:00:01 vlan 1 port 1/1/1

switch(config)# static-mac 00:00:00:00:00:01 vlan 1 port 1/1/2
1/1/2 is not an L2 port

switch(config)# static-mac 00:00:00:00:00:01 vlan 2 port 1/1/1
VLAN 2 not found
```

clear mirror

Syntax

```
clear mirror [all | <SESSION-ID>]
```

Description

Clears the mirror statistics for all configured mirror sessions or a specified session

Command context

Manager (#)

Parameters

all

Specifies all configured sessions.

<SESSION-ID>

Specifies a numeric identifier for the session. Range: 1 to 4

Authority

Administrators or local user group members with execution rights for this command.

Examples

Clearing mirror statistics for all configured mirror sessions:

```
switch# clear mirror all
```

Clearing mirror statistics for mirror session 1:

```
switch# clear mirror 1
```

comment

Syntax

```
comment <COMMENT>
```

```
no comment
```

Description

Specifies a comment for the mirroring session.

When used in mirror endpoint command context, specifies a comment for the mirror endpoint.

The `no` form of this command removes the comment.

Command context

```
config-mirror-<SESSION-ID>  
config-mirror-endpoint
```

Parameters

<COMMENT>

A comment string of up to 64 characters composed of letters, numbers, underscores, dashes, spaces, and periods.

Authority

Administrators or local user group members with execution rights for this command.

Usage

Comments are optional and can be added or removed at any time without affecting the state of the mirroring session.

Adding a comment to a session that already has a comment replaces the existing comment.

Examples

Adding a comment to a mirror session:

```
switch(config-mirror-3) # comment This Mirror will be removed during next maintenance window
```

Removing the comment from mirror session 3:

```
switch(config-mirror-3) # no comment
```

Adding a comment to a mirror endpoint:

```
switch(config-mirror-endpoint-test) # comment Monitor endpoint traffic
```

Replacing the existing comment for mirror endpoint:

```
switch(config-mirror-endpoint-test) # comment Monitor statistics on each endpoint interfaces
```

destination interface

Syntax

```
destination interface {<INTERFACE-ID>|<LAG-NAME>}  
no destination interface
```

Description

Configures the specified interface as the destination of the mirrored traffic.

The `no` form of this command immediately disables the mirroring session and removes the specified destination interface from the configuration.

Command context

config-mirror-<SESSION-ID>

Parameters

<INTERFACE-ID>

Specifies a interface. Format: member/slot/port.

<LAG-NAME>

Specifies a LAG (link aggregation group) identifier.

Authority

Administrators or local user group members with execution rights for this command.

Usage

Only one destination is allowed per session.

Configuring a different destination interface in an enabled mirroring session causes all mirrored traffic to use the new destination interface. This action might cause a temporary suspension of mirrored source traffic during the reconfiguration.

Examples

Configuring a mirroring session and adding an interface as a destination:

```
switch(config)# mirror session 1
switch(config-mirror-1)# destination interface 1/1/1
```

Replacing the existing destination with different interface:

```
switch(config-mirror-1)# destination interface 1/1/12
```

Removing a destination:

```
switch(config-mirror-1)# no destination interface
```

disable

Syntax

disable

Description

Disables the mirroring session specified by the current command context.

Command context

config-mirror-<SESSION-ID>

Authority

Administrators or local user group members with execution rights for this command.

Usage

By default, mirroring sessions are disabled.

When a mirroring session is disabled, the `show mirror` command for that session ID shows an `Admin Status of disable` and an `Operation Status of disabled`.

Example

Disabling a mirroring session:

```
switch(config)# mirror session 3
switch(config-mirror-3)# disable
```

enable

Syntax

`enable`

Description

Enables the mirroring session for the current command context.

Command context

`config-mirror-<SESSION-ID>`

Authority

Administrators or local user group members with execution rights for this command.

Usage

By default, mirroring sessions are disabled.

When a mirroring session is enabled, the `show mirror` command for that session ID shows an `Admin Status of enable` and an `Operation Status of enabled`.

If sFlow is enabled on an interface and a mirroring session specifies the same interface as the source of received traffic (the source is configured with a direction of `rx` or `both`):

- The attempt to enable the mirroring session fails and an error is returned.



When adding, removing, or changing the configuration of a source interface in an enabled mirroring session, packets from other mirror sources using the same destination interface might be interrupted.

Example

Configuring and enabling a mirroring session:

```
switch(config)# mirror session 3
switch(config-mirror-3)# source interface 1/1/2 rx
switch(config-mirror-3)# destination interface 1/1/3
switch(config-mirror-3)# comment Monitor router port ingress-only traffic
switch(config-mirror-3)# enable
```

mirror session

Syntax

```
mirror session <SESSION-ID>
no mirror session <SESSION-ID>
```

Description

Creates a mirroring session configuration context or enters an existing mirroring session configuration context.

From this context, you can enter commands to configure and enable or disable the mirroring session.

The `no` form of this command removes an existing mirroring session from the configuration.

Command context

config

Parameters

<SESSION-ID>

Specifies the session identifier. Range: 1 to 4

Authority

Administrators or local user group members with execution rights for this command.

Examples

```
switch(config)# mirror session 1
switch(config-mirror-1)#

switch(config)# mirror session 3
switch(config-mirror-3)#

switch(config)# no mirror session 1
switch(config)#
```

show mirror

Syntax

```
show mirror [<SESSION-ID>]
```

Description

Shows information about mirroring sessions. If <SESSION-ID> is not specified, then the command shows a summary of all configured mirroring sessions. If <SESSION-ID> is specified, then the command shows detailed information about the specified mirroring session.

Command context

Operator (>) or Manager (#)

Parameters

<SESSION-ID>

Specifies the session identifier. Range: 1 to 4

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Usage

Admin Status indicates the configured status. Admin Status is one of the following values:

`enable`

The mirroring session is enabled.

`disable`

The mirroring session has been configured but not yet enabled, or has been disabled.

Operation Status indicates the status of the mirroring session. Operation Status is one of the following values:

`dest_doesnt_exist`

The configured destination interface is not found in the system. The mirroring session cannot be enabled.

`destination_shutdown`

The mirroring session is enabled, but the destination interface is shut down. No traffic can be monitored.

`disabled`

The mirroring session is disabled and is not in an error condition.

`enabled`

The mirroring session is enabled.

`external/driver_error`

An internal ASIC hardware error occurred.

`hit_active_sessions_capacity`

The mirroring session could not be enabled because the maximum number of supported mirroring sessions are already enabled.

`internal_error`

An invalid parameter was passed to the ASIC software layer.

`no_dest_configured`

The mirroring session does not have a destination interface configured.

`no_name_configured`

A software error occurred. The mirroring session does not have a session ID in its configuration.

`null_mirror`

A software error occurred. The session object reference is invalid.

`out_of_memory`

The system is out of memory, reboot recommended.

`tunnel_route_resolution_not_populated`

If the destination tunnel IP address is not reachable.

`unknown_error`

An unexpected error occurred.

Examples

Showing summary information about all configured mirroring sessions:

```
switch# show mirror
ID  Admin Status  Operation Status
---  -
1   enable       enabled
2   disable      disabled
3   disable      disabled
4   enable       internal_error
```


Showing detailed information about a single mirroring session:

```
switch# show mirror 3
Mirror Session: 3
Admin Status: disable
Operation Status: disabled
Comment: Monitor router port ingress-only traffic
Source: interface 1/1/2 rx
Destination: interface 1/1/3
Output Packets: 0
Output Bytes: 0
switch#
```

source interface

Syntax

```
source interface {<PORT-NUM> | <LAG-NAME>} [<DIRECTION>]
no source interface {<PORT-NUM> | <LAG-NAME>} [<DIRECTION>]
```

Description

Configures the specified interface (either an Ethernet port or a LAG) as a source of traffic to be mirrored. The `no` form of this command ceases mirroring traffic from the specified source interface and removes the source interface from the mirroring session configuration.

Command context

config-mirror-<SESSION-ID>

Parameters

<PORT-NUM>

Specifies a physical port on the switch. Use the format `member/slot/port` (for example, `1/3/1`).

<LAG-NAME>

Specifies the identifier for the LAG (link aggregation group).

<DIRECTION>

Selects the direction of traffic to be mirrored from this source interface. There is no default for this parameter. Valid values are the following:

`both`

Mirror both transmitted and received packets.

`rx`

Mirror only received packets.

`tx`

Mirror only transmitted packets.

Authority

Administrators or local user group members with execution rights for this command.

Usage

There is a limit of four source interfaces in each direction of a given mirror session. However, there is a practical limit to the amount of traffic that a mirror destination can transmit. For example, mirroring session with multiple 10G sources can overwhelm a single 10G destination.



When adding, removing, or changing the configuration of a source port in an enabled mirroring session, packets from other mirror sources using the same destination port might be interrupted.

Examples

Configuring a mirrored traffic source interface:

```
switch(config-mirror-1)# source interface  
LAG-NAME      Enter a LAG name. For example, lag10  
PORT-NUM      Enter a port number
```

Creating a mirroring session and configuring a source interface to mirror both transmitted and received packets:

```
switch(config)# mirror session 1  
switch(config-mirror-1)# source interface 1/1/1 both
```

Creating a second mirroring session and configuring two source interfaces. One port mirroring only transmitted packets and the other mirroring both transmitted and received packets:

```
switch(config)# mirror session 2  
switch(config-mirror-2)# source interface 1/1/3 tx  
switch(config-mirror-2)# source interface 1/2/1 both
```

Removing the first source interface:

```
switch(config-mirror-2)# no source interface 1/2/3
```

Configuring a source interface to mirror received packets only:

```
switch(config-mirror-3)# source interface 1/1/2 rx
```

Configuring a source interface to mirror both transmitted and received packets:

```
switch(config-mirror-1)# source interface 1/1/1 both
```

Configuring a LAG as source interface to mirror both transmitted and received packets:

```
switch(config-mirror-4)# source interface lag1 both
```

Stopping the mirroring of received packets from a configured source interface:

```
switch(config-mirror-4)# no source interface lag1 rx
```

source vlan

Syntax

```
source vlan <VLAN-NUM> {rx | tx | both}
no source vlan <VLAN-NUM> [rx | tx | both]
```

Description

Adds or removes VLAN as a source of traffic to be mirrored. More than one source VLAN can be configured in a mirror session. Each VLAN may specify its own direction.

The `no` version of the command ceases mirroring traffic from the specified source VLAN and removes the source from the mirror configuration.

There is a limit of 1024 source VLANs in each direction of a given mirror session. The same VLAN can be configured as a mirror source for multiple sessions.

Command context

config

Parameters

<VLAN-NUM>

Configured VLAN number.

rx

Mirror only received traffic.

tx

Mirror only transmitted traffic.

both

Mirror both received and transmitted traffic.

Authority

Administrators or local user group members with execution rights for this command.

Example

Create a mirror session and add VLAN 10 as a source of traffic in both directions on that port.

```
switch(config)# mirror session 1
switch(config-mirror-1)# source vlan 10 both
```

Create a second mirror session and add VLAN 10 as a transmit sources of traffic and VLAN 20 in both receive and transmit directions.

```
switch(config)# mirror session 2
switch(config-mirror-2)# source vlan 10 tx
switch(config-mirror-2)# source vlan 20 both
```

Reconfigure the source in session 2 to be receive only by respecifying the source interface configuration.

```
switch(config-mirror-2)# source vlan 10 rx
```

From the second session, remove the first source interface entirely and remove the transmit direction from the other so that mirroring only occurs in the receive direction.

```
switch(config-mirror-2)# no source vlan 10  
switch(config-mirror-2)# no source vlan 20 tx
```

Message received when trying to add more than 1024 mirror source VLANs

```
switch(config-mirror-2)# source vlan 2000 rx  
The maximum number of source VLANs per mirror session is 1024 in each direction
```

ipv6 mld snooping

Syntax

```
ipv6 mld snooping [drop-unknown [vlan-shared | vlan-exclusive]]  
[no] ipv6 mld snooping
```

Description

This command configures the drop unknown mode. While MLD snooping is enabled, the traffic will be forwarded only to ports that initiate an MLD request for multicast. Drop unknown mode can be a filter across all VLANs (vlan-shared) or per VLAN (exclusive-vlan). The default configuration is vlan-shared.

The `no` form of this command disables drop unknown mode on the switch.

Command context

config

Parameters

vlan-shared

Required: Enable shared VLAN filter on the switch.

vlan-exclusive

Required: Enable exclusive drop unknown filter per VLAN.

Authority

Administrators or local user group members with execution rights for this command.

Example

```
switch(config)# ipv6 mld snooping drop-unknown vlan-shared  
switch(config)# ipv6 mld snooping drop-unknown vlan-exclusive  
switch(config)# no ipv6 mld snooping drop-unknown
```

ipv6 mld snooping

Syntax

```
ipv6 mld snooping {enable | disable}
```

Description

This command enables or disables MLD snooping on the VLAN.

The `no` form of this command disables all MLD snooping configurations on the VLAN.

Command context

```
config-vlan-<VLAN-ID>
```

Parameters

`enable`

Required: Enable MLD snooping on the VLAN.

`disable`

Required: Disable MLD snooping on the VLAN.

Authority

Administrators or local user group members with execution rights for this command.

Example

Enable MLD snooping on VLAN 2:

```
switch(config)# vlan 2  
switch(config-vlan)# ipv6 mld snooping enable  
switch(config-vlan)# ipv6 mld snooping disable
```

Remove all MLD snooping configurations on VLAN 2:

```
switch(config)# vlan 2  
switch(config-vlan)# no ipv6 mld snooping
```

ipv6 mld snooping fastlearn

Syntax

```
ipv6 mld snooping fastlearn <port-list>
```

Description

This command enables the port to learn group information on receiving topology change notification.

The `no` form of this command disables fastlearn on the ports.

Command context

config

Parameters

port-list

Required: 1/1/1-1/1/2, ports to be configured as fastlearn ports.

Authority

Administrators or local user group members with execution rights for this command.

Example

```
switch(config)# ipv6 mld snooping fastlearn 1/1/3
switch(config)# ipv6 mld snooping fastlearn 1/1/1-1/1/2
switch(config)# ipv6 mld snooping fastlearn 1/1/5,1/1/6
```

ipv6 mld snooping fastleave vlan

Syntax

```
ipv6 mld snooping [fastleave vlan <VLAN-LIST>]
no ipv6 mld snooping [fastleave vlan <VLAN-LIST>]
```

Description

Configures the specified ports as fastleave ports. Enables the switch to immediately remove an interface from the bridge table upon receiving the leave group message.

The `no` form of this command disables fastleave configuration on the ports.

Command context

config-vlan-<VLAN-ID>

Parameters

<VLAN-LIST>

Specifies a list of VLANs on which the port should be configured as a fastleave port. Specifies the number of a single VLAN or a series of numbers for a range of VLANs, separated by commas (10, 20, 30, 40), dashes (10-40), or both (10-40,60).

Authority

Administrators or local user group members with execution rights for this command.

Usage

MLD fastleave is configured for ports on a per-VLAN basis. By default, the querier sends a MLD Group-Specific Query message out of the interface, upon which the leave group message is received to ensure that no other receivers are connected to the interface. If receivers are directly attached to the switch, it is inefficient to send the membership query as the receiver wanting to leave is the only connected host. Fastleave processing eliminates the MLD Group-Specific Query message. Thus, it allows the switch to immediately remove an interface from the bridge table upon receiving the leave Group message. This

processing speeds up the overall leave process and also eliminates the CPU overhead of having to generate an MLD Group-Specific Query message.

Example

Configure fastleave ports for the VLAN:

```
switch# configure terminal
switch(config)# int 1/1/1
switch(config-vlan)# no shut
switch(config-vlan)# no routing
switch(config-vlan)# ipv6 mld snooping fastleave vlan 10
switch(config-vlan)# ipv6 mld snooping fastleave vlan 10-20
switch(config-vlan)# no ipv6 mld snooping fastleave vlan 10
switch(config-vlan)# no ipv6 mld snooping fastleave vlan 10-20
```

ipv6 mld snooping forced fastleave

Syntax

```
ipv6 mld snooping [forced-fastleave <VLAN-LIST>]
```

Description

Configures the given ports in forced fastleave mode.

The `no` form of this command disables forced fastleave configuration on the ports.

Command context

```
config-vlan-<VLAN-ID>
```

Parameters

<VLAN-LIST>

Specifies a list of VLANs on which the port should be configured as a forced fastleave port. Specifies the number of a single VLAN or a series of numbers for a range of VLANs, separated by commas (10, 20, 30, 40), dashes (10-40), or both (10-40,60).

Authority

Administrators or local user group members with execution rights for this command.

Usage

With forced fastleave enabled, MLD speeds up the process of blocking unnecessary multicast traffic to a switch port that is connected to multiple end nodes. When a port having multiple end nodes receives a leave group request from one end node for a given multicast group, forced fastleave activates and waits a small amount of time to receive a join request from any other member of the same group on that port. If the port does not receive a join request for that group within the forced fastleave interval, the switch then blocks any further traffic to that group on that port.

Example

Configure forced-fastleave ports for the VLAN:

```
switch# configure terminal
switch(config)# int 1/1/1
```



```
switch(config-vlan)# no shut
switch(config-vlan)# no routing
switch(config-vlan)# ipv6 mld snooping forced-fastleave vlan 10
switch(config-vlan)# ipv6 mld snooping forced-fastleave vlan 10-20
switch(config-vlan)# no ipv6 mld snooping forced-fastleave vlan 10
switch(config-vlan)# no ipv6 mld snooping forced-fastleave vlan 10-20
```

ipv6 mld snooping [version <ver>]

Syntax

```
ipv6 mld snooping [version <ver>]
```

Description

This command configures the MLD snooping version on the VLAN. MLD version 2 is the default.

Command context

```
config-vlan-<VLAN-ID>
```

Parameters

ver

Required: 1-2, MLD snooping version.

Authority

Administrators or local user group members with execution rights for this command.

Example

```
switch(config)# vlan 2
switch(config-vlan)# ipv6 mld snooping version 2
```

ipv6 mld snooping apply access-list

Syntax

```
ipv6 mld snooping apply access-list <ACL-NAME>
no ipv6 mld snooping apply access-list <ACL-NAME>
```

Description

Configures the ACL on a particular interface to filter the MLD join or leave packets based on rules set in the particular ACL name.

The `no` form of this command disables the rules set for the ACL.



This configuration will override the ACL associated with IGMP snooping on the corresponding L2 VLAN.

Command context

```
config-vlan
```

Parameters

```
access-list
```

Associates an ACL with the IGMP.

<ACL-NAME>

Specifies the name of the ACL.

Authority

Administrators or local user group members with execution rights for this command.

Usage

Existing classifier commands are used to configure the ACL. In case an MLDv2 packet with multiple group addresses is received, it will only process the permitted group addresses based on the ACL rule set, and any existing joins will time out. If there is no match or if there is a deny rule match, the packet is dropped.



If the access list is configured for both L2 VLAN and L3 VLAN, the L3 VLAN configuration will be applied.

Examples

Configuring the ACL to filter MLD packets based on rules set in access list `mygroup`:

```
switch(config)# access-list ipv6 mygroup
switch(config-acl-ip)# permit icmpv6 any ff55::1
switch(config-acl-ip)# exit
switch(config)# interface vlan 2
switch(config-vlan)# ipv6 mld snooping apply access-list mygroup
```

Configuring the ACL to remove the rules set in access list `mygroup`:

```
switch(config-vlan)# no ipv6 mld snooping apply access-list mygroup
```

ipv6 mld snooping auto vlan

Syntax

```
ipv6 mld snooping [auto vlan <VLAN-LIST>]
no ipv6 mld snooping [auto vlan <VLAN-LIST>]
```

Description

This command configures the given ports in auto mode, which is the default port mode.

The `no` form of this command disables auto ports.

Command context

config-vlan-<VLAN-ID>

Parameters

<VLAN-LIST>

Required: Specifies a list of VLANs on which the port should be configured as an auto port. Specifies the number of a single VLAN or a series of numbers for a range of VLANs, separated by commas (10, 20, 30, 40), dashes (10-40), or both (10-40,60).

Authority

Administrators or local user group members with execution rights for this command.

Example

Configure auto ports for VLANs on the interface:

```
switch# configure terminal
switch(config)# int 1/1/1
switch(config-vlan)# no shut
switch(config-vlan)# no routing
switch(config-vlan)# ipv6 mld snooping auto vlan 10
switch(config-vlan)# ipv6 mld snooping auto vlan 10-20
```

ipv6 mld snooping blocked vlan

Syntax

```
ipv6 mld snooping [blocked vlan <VLAN-LIST>]
```

Description

By default ports are configured in auto mode. This command configures the given ports in blocked mode. The `no` form of this command removes blocked ports.

Command context

config-vlan-<VLAN-ID>

Parameters

<VLAN-LIST>

Required: Specifies a list of VLANs on which the port should be configured as a blocked port. Specifies the number of a single VLAN or a series of numbers for a range of VLANs, separated by commas (10, 20, 30, 40), dashes (10-40), or both (10-40,60).

Authority

Administrators or local user group members with execution rights for this command.

Example

Configuring blocked ports for the VLANs on the interface:

```
switch# configure terminal
switch(config)# int 1/1/1
switch(config-vlan)# no shut
switch(config-vlan)# no routing
switch(config-vlan)# ipv6 mld snooping blocked vlan 10
switch(config-vlan)# ipv6 mld snooping blocked vlan 10-20
switch(config-vlan)# no ipv6 mld snooping blocked vlan 10
switch(config-vlan)# no ipv6 mld snooping blocked vlan 10-20
```

ipv6 mld snooping forward vlan

Syntax

```
ipv6 mld snooping [forward vlan <VLAN-LIST>]
```

Description

By default ports are configured in auto mode. This command configures the given ports in forward mode. The `no` form of this command disables forward ports.

Command context

`config-vlan-<VLAN-ID>`

Parameters

`<VLAN-LIST>`

Required: Specifies a list of VLANs on which ports should be configured as forward ports. Specifies the number of a single VLAN or a series of numbers for a range of VLANs, separated by commas (10, 20, 30, 40), dashes (10-40), or both (10-40,60).

Authority

Administrators or local user group members with execution rights for this command.

Example

Configure forward ports for VLANs on the interface:

```
switch# configureterminal
switch(config)# int 1/1/1
switch(config-vlan)# no shut
switch(config-vlan)# no routing
switch(config-vlan)# ipv6 mld snooping forward vlan 10
switch(config-vlan)# ipv6 mld snooping forward vlan 10-20
switch(config-vlan)# no ipv6 mld snooping forward vlan 10
switch(config-vlan)# no ipv6 mld snooping forward vlan 10-20
```

ipv6 mld snooping [static-group <X:X::X:X>]

Syntax

`ipv6 mld snooping [static-group <X:X::X:X>]`

Description

This command configures static multicast group.

The `no` form of this command disables static multicast group.

Command context

`config-vlan-<VLAN-ID>`

Parameters

`static-group`

Required: `<x:x::x:x>`, MLD static multicast group.

Authority

Administrators or local user group members with execution rights for this command.

Example

```
switch(config)# vlan 2  
switch(config-vlan)# ipv6 mld snooping static-group ff12::c  
switch(config-vlan)# no ipv6 mld snooping static-group ff12::c
```

show ipv6 mld snooping

Syntax

```
show ipv6 mld snooping
```

Description

This command shows MLD snooping configuration details for all VLANs.

Command context

Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

```
switch# show ipv6 mld snooping

MLD Snooping Protocol Info

Total VLANs with MLD enabled           : 1
Current count of multicast groups joined : 0

MLD Drop Unknown Multicast             : Global

VLAN ID                               : 1
VLAN Name                             : DEFAULT_VLAN_1
MLD Snooping is not enabled

VLAN ID                               : 2
VLAN Name                             : VLAN2
MLD Configured Version                 : 2
MLD Operating Version                   : 2
Querier Address [this switch]          : fe80::218:71ff:fec4:2f00
Querier Port                           :
Querier UpTime                         :0m 21s
Querier Expiration Time                 :0m 2s
```

show ipv6 mld snooping [counters]

Syntax

```
show ipv6 mld snooping [counters]
```

Description

This command shows MLD snooping query packet Tx, Rx, and Error packet counter details.

Command context

Manager (#)

Parameters

counters

Optional, show MLD snooping counters.

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Example

```
switch# show ipv6 mld snooping counters
MLD Snooping VLAN Counters

Rx Counters :

V1 All Hosts Queries           0
V2 All Hosts Queries           0
V2 Group Specific Queries      0
Group And Source Specific Queries 0
V1 Member Reports              0
V2 Member Reports              0
V1 Member Leaves               0

Tx Counters :

Flood on vlan                   44
V1 Group Specific Queries       0
V2 Group Specific Queries       0

Errors:

Unknown Message Type           0
Malformed Packets              0
Bad Checksum                    0
Packet received on MLD-disabled Interface 0
Interface Wrong Version Queries 0
Packets dropped by ACL          0

Port Counters:

Membership Timeout              0
```

show ipv6 mld snooping [groups]

Syntax

```
show ipv6 mld snooping [groups]
```

Description

This command shows MLD snooping group details for the specified VLAN.

Command context

Manager (#)

Parameters

groups

Optional, show MLD snooping groups information.

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Example

```
switch# show ipv6 mld snooping groups
```

MLD Group Address Information

VLAN ID	Group Address	Expires	UpTime	Last Reporter	Type
10	ff12::c	3m 54s	0m 26s	2001::1	Filter
10	ff12::d	4m 17s	0m 3s	2001::1	

show ipv6 mld snooping [statistics]

Syntax

```
show ipv6 mld snooping [statistics]
```

Description

This command shows MLD snooping statistics information.

Command context

Manager (#)

Parameters

statistics

Optional, show MLD snooping statistics.

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Example

```
switch# show ipv6 mld snooping statistics
```

MLD Snooping Protocol Info

Total VLANs with MLD enabled : 1

Current count of multicast groups joined : 2

MLD Drop Unknown Multicast : Global

MLD Snooping Joined Groups Statistics

VLAN ID	VLAN Name	Total	Static	INCLUDE	EXCLUDE
1	DEFAULT_VLAN_1	0	0	0	0
2	VLAN2	2	2	0	0

show ipv6 mld snooping [vlan <vlan-id> [counters]]

Syntax

```
show ipv6 mld snooping [vlan <vlan-id> [counters]]
```

Description

This command shows MLD snooping protocol information and number of different groups joined for the VLAN.

Command context

Manager (#)

Parameters

vlan-id

Required, 1-4094, shows MLD snooping information.

counters

Optional, shows MLD query packet Tx, Rx, Error packet counters on a specified VLAN.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

```
switch# show ipv6 mld snooping vlan 2 counters
MLD Snooping VLAN Counters
```

```
VLAN ID   : 2
VLAN Name : VLAN2
```

Rx Counters :

```
V1 All Hosts Queries      0
V2 All Hosts Queries      0
V1 Group Specific Queries 0
V2 Group Specific Queries 0
Group And Source Specific Queries 0
V1 Member Reports         0
V2 Member Reports         0
V1 Member Leaves          0
```

Tx Counters :

```
Flood on vlan             71
V1 Group Specific Queries 0
```

```

V2 Group Specific Queries                                0

Errors:

Unknown Message Type                                    0
Malformed Packets                                       0
Bad Checksum                                            0
Packet received on MLD-disabled Interface              0
Interface Wrong Version Queries                        0
Packets dropped by ACL                                  0

Port Counters:

Membership Timeout                                      0
switch#

```

show ipv6 mld snooping [vlan <vlan-id> [statistics]]

Syntax

```
show ipv6 mld snooping [vlan <vlan-id> [statistics]]
```

Description

This command shows MLD snooping statistics details for the specified VLAN, including the number of different groups joined for the VLAN.

Command context

Manager (#)

Parameters

vlan-id

Required, 1-4094, shows MLD query packet Tx, Rx, error packet counters on VLAN.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

```

switch# show ipv6 mld snooping vlan 2 statistics
MLD Snooping statistics

VLAN ID      :    2
VLAN Name    :   VLAN2

Number of Include Groups      :    1
Number of Exclude Groups     :    0
Number of Static Groups      :    1
Total Multicast Groups Joined :    2

```

show ipv6 mld snooping [vlan <vlan-id> [group [<group-ip>] [source <source-ip>]]]

Syntax

```
show ipv6 mld snooping [vlan <vlan-id> [group [<group-ip>] [source <source-ip>]]]
```

Description

This command shows MLD snooping details for the specified VLAN, including the number of different groups joined for the VLAN.

Command context

Manager (#)

Parameters

vlan-id

Required: 1-4094, shows MLD protocol information for the specified VLAN.

group-ip

Optional: X:X::X:X, MLD source information for the specified group.

source-ip

Optional: X:X::X:X, MLD source information for the specified group.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

```
switch# show ipv6 mld snooping vlan 2

MLD Snooping Protocol Info

Total VLANs with MLD enabled           : 2
Current count of multicast groups joined : 0

MLD Drop Unknown Multicast             : Global

VLAN ID                               : 2
VLAN Name                             : VLAN2
MLD Configured Version                 : 2
MLD Operating Version                 : 2
Querier Address [this switch]         : fe80::218:71ff:fec4:2f00
Querier Port                           :
Querier UpTime                         :0m 21s
Querier Expiration Time                :0m 2s

Active Group Address                   Tracking Vers Mode  Uptime    Expires
-----
ff05::2:1                             Filter    2    EXC    0m 17s    4m 3s

switch# show ipv6 mld snooping vlan 2 group

MLD ports and group information for group ff05::2:1

VLAN ID                               : 2
VLAN Name                             : VLAN2

Group Address                         : ff05::2:1
Last Reporter                         : 2001::1
```

```

Group Type                               : Filter

Port      Vers Mode Uptime    Expires    V1      Sources Sources
-----
1/1/1     2    EXC 0m 5s    4m 15s    4m 15s    0        0

```

```
switch# show ipv6 mld snooping vlan 2 group ff05::2:1
```

```
MLD ports and group information for group ff05::2:1
```

```

VLAN ID           : 2
VLAN Name         : VLAN2

Group Address      : ff05::2:1
Last Reporter      : 2001::1
Group Type         : Filter

```

```

Port      Vers Mode Uptime    Expires    V1      Sources Sources
-----
1/1/1     2    EXC 0m 5s    4m 15s    4m 15s    0        0

```

```
switch# show ipv mld snooping vlan 2 group ff05::2:1 source 3000::3
```

```

VLAN ID           : 2
VLAN Name         : VLAN2
Group Address      : ff05::2:1
Source Address     : 3000::3
Source Type        : Filter

```

```

Port      Mode Uptime    Expires    Configured Mode
-----
1/1/1     INC  0m 27s    3m 53s    Auto

```

show ipv6 mld snooping [vlan <vlan-id> [group [port <port_id>]]

Syntax

```
show ipv6 mld snooping [vlan <vlan-id> [group [port <port_id>]]
```

Description

This command shows MLD snooping details for the specified VLAN, including the number of different groups joined for the VLAN.

Command context

Manager (#)

Parameters

port-id

Required: <PORT>, shows MLD protocol information for the specified port of a VLAN.

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Example

```
switch# show ipv mld snooping vlan 2 group port 1/1/1

VLAN ID    : 2
VLAN Name  : VLAN2

Group Address : ff05::2:1
Last Reporter : fe80::1
Group Type   : Filter

Port      Vers Mode Uptime    Expires    V1      Sources  Sources
-----  -
1/1/1     2    INC  1m 46s    2m 34s     Timer   Forwarded Blocked
-----  -
Group Address : ff05::2:1
Source Address : 3000::1
Source Type   : Filter

Port      Mode Uptime    Expires    Configured Mode
-----  -
1/1/1     INC  1m 46s    2m 34s     Auto

Group Address : ff05::2:1
Source Address : 3000::2
Source Type   : Filter

Port      Mode Uptime    Expires    Configured Mode
-----  -
1/1/1     INC  1m 46s    2m 34s     Auto

Group Address : ff05::2:1
Source Address : 3000::3
Source Type   : Filter

Port      Mode Uptime    Expires    Configured Mode
-----  -
1/1/1     INC  1m 46s    2m 34s     Auto
```

show ipv6 mld snooping [static-groups]

Syntax

```
show ipv6 mld snooping [static-groups]
```

Description

This command shows MLD snooping static group details, including the number of static groups joined.

Command context

Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Example

```
switch# show ipv6 mld snooping static-groups
```

MLD Static Group Address Information

VLAN ID	Group Address
---------	---------------

10	ff12::1
10	ff12::2

ipv6 mld {enable | disable}

Syntax

```
ipv6 mld {enable | disable}
```

Description

This command enables or disables MLD on the interface VLAN.

Command context

```
config-if-vlan
```

Parameters

enable

Required: Enable MLD on the interface VLAN.

disable

Required: Disable MLD on the interface VLAN.

Authority

Administrators or local user group members with execution rights for this command.

Example

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 mld enable
switch(config-if-vlan)# ipv6 mld disable
```

ipv6 mld apply access-list

Syntax

```
ipv6 mld apply access-list <ACL-NAME>
no ipv6 mld apply access-list <ACL-NAME>
```

Description

Configures the ACL on a particular interface to filter the MLD join or leave packets based on rules set in the particular ACL name.

The **no** form of this command disables the rules set for the ACL.

Command context

```
config-vlan
```

Parameters

access-list

Associates an ACL with the IGMP.

<ACL-NAME>

Specifies the name of the ACL.

Authority

Administrators or local user group members with execution rights for this command.

Usage

Existing classifier commands are used to configure the ACL. In case an MLDv2 packet with multiple group addresses is received, it will only process the permitted group addresses based on the ACL rule set, and any existing joins will time out. If there is no match or if there is a deny rule match, the packet is dropped.

Examples

Configuring the ACL to filter MLD packets based on rules set in access list `mygroup`:

```
switch(config)# access-list ipv6 mygroup
switch(config-acl-ip)# permit icmpv6 any ff55::1
switch(config-acl-ip)# exit
switch(config)# interface vlan 2
switch(config-vlan)# ipv6 mld apply access-list mygroup
```

Configuring the ACL to remove the rules set in access list `mygroup`:

```
switch(config-vlan)# no ipv6 mld apply access-list mygroup
```

no ipv6 mld

Syntax

no ipv6 mld

Description

This command removes all MLD configurations on the interface.

Authority

Administrators or local user group members with execution rights for this command.

Command context

config-if

Example

```
switch(config)# interface vlan 1
switch(config-if)# no ipv6 mld
```

ipv6 mld querier

Syntax


```
ipv6 mld querier
```

Description

This command configures MLD querier.

The `no` form of this command disables MLD querier.

Command context

```
config-if-vlan
```

Authority

Administrators or local user group members with execution rights for this command.

Example

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 mld querier
switch(config-if-vlan)# no ipv6 mld querier
```

ipv6 mld querier [interval <interval-value>]

Syntax

```
ipv6 mld querier [interval <interval-value>]
```

Description

This command configures MLD querier interval. The default interval-value is 125.

Command context

```
config-if-vlan
```

Parameters

```
interval-value
```

Required: 5-300, configures MLD querier interval.



Default interval-value is 125. Use the `no ipv6 mld querier interval` command to set interval-value to the default.

Authority

Administrators or local user group members with execution rights for this command.

Example

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 mld querier interval 100
switch(config-if-vlan)# no ipv6 mld querier interval
```

ipv6 mld last-member-query-interval <interval-value>

Syntax

```
ipv6 mld last-member-query-interval <interval-value>
```

Description

This command configures MLD last member query interval value in seconds. The default interval-value is 1 second.

Command context

config-if-vlan

Parameters

interval-value

Required: 1-2, configures MLD last-member-query-interval.



Default interval-value is 1 second. Use the `no ipv6 mld last-member-query-interval` command to set interval-value to the default.

Authority

Administrators or local user group members with execution rights for this command.

Example

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 mld last-member-query-interval 2
switch(config-if-vlan)# no ipv6 mld last-member-query-interval
```

ipv6 mld querier query-max-response-time <response-time>

Syntax

```
ipv6 mld querier query-max-response-time <response-time>
```

Description

This command configures MLD max response time value in seconds. The default max-response-time-value is 10 seconds.

Command context

config-if-vlan

Parameters

max-response-time-value

Required: 10-128, configures MLD querier max-response-time.



Default max-response-time-value is 10 seconds. Use the `no ipv6 mld querier query-max-response-time` command to set max-response-time-value to the default.

Authority

Administrators or local user group members with execution rights for this command.

Example

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 mld query-max-response-time 50
switch(config-if-vlan)# no ipv6 mld query-max-response-time
```

ipv6 mld robustness

Syntax

```
ipv6 mld robustness <VALUE>
```

Description

This command configures MLD robustness. The robustness value represents the number of times the querier retries queries on the connected subnets. The default robustness-value is 2 seconds.

Command context

config-if-vlan

Parameters

<VALUE>

Required: 1-7, configures MLD robustness.



Default robustness-value is 2 seconds. Use the `no ipv6 mld robustness` command to set robustness-value to the default.

Authority

Administrators or local user group members with execution rights for this command.

Example

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 mld robustness 5
switch(config-if-vlan)# no ipv6 mld robustness
```

ipv6 mld static-group <MULTICAST-GROUP-IP>

Syntax

```
ipv6 mld static-group <MULTICAST-GROUP-IP>
```

Description

This command configures MLD static group.

Command context

config-if-vlan

Parameters

<MULTICAST-GROUP-IP>

Required: X:X::X:X, configures MLD static group.

Authority

Administrators or local user group members with execution rights for this command.

Example

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 mld static-group ff12::c
switch(config-if-vlan)# no ipv6 mld static-group ff12::c
```

ipv6 mld version <VERSION>

Syntax

```
ipv6 mld version <VERSION>
```

Description

This command configures MLD version.

Command context

config-if-vlan

Parameters

<VERSION>

Required: 1-2, configures MLD version.

Authority

Administrators or local user group members with execution rights for this command.

Example

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 mld version 2
```

ipv6 mld version <VERSION> [strict]

Syntax

```
ipv6 mld version <VERSION> [strict]
```

Description

This command configures MLD strict version. Packets that do not match the configured version will be dropped. By default, strict option is not enabled.

Command context

config-if-vlan

Parameters

<VERSION>

Required: 1-2, configures MLD version.

Authority

Administrators or local user group members with execution rights for this command.

Example

```
switch(config)# interface vlan 2  
switch(config-if-vlan)# ipv6 mld version 2 strict  
switch(config-if-vlan)# no ipv6 mld version 2 strict
```

show ipv6 mld

Syntax

```
show ipv6 mld
```

Description

This command shows MLD configuration on VLAN.

Command context

Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

```
switch# show ipv6 mld

VRF Name           : default
Interface          : vlan10
MLD Configured Version : 2
MLD Operating Version : 2
Querier State      : Querier
Querier IP [this switch] : fe80::7272:cfff:fe96:d3ec
Querier Uptime     : 39m 44s
Querier Expiration Time : 0m 31s
MLD Snoop Enabled on VLAN : True
```

show ipv6 mld [interface <intf_id> | vlan <vlan-id>]

Syntax

```
show ipv6 mld [interface <IFNAME> | vlan <VLAN-ID>]
```

Description

This command shows MLD configuration on a specific VLAN.

Command context

Manager (#)

Parameters

<VLAN-ID>

Required: 1-4094, shows MLD configuration on a specified VLAN.
<IFNAME>

Required: Shows MLD configuration on a specified interface.

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Example

```
switch# show ipv6 mld interface vlan 10

MLD Configured Version   : 2
MLD Operating Version    : 2
Querier State            : Querier
Querier IP [this switch] : fe80::7272:cfff:fe96:d3ec
Querier Uptime           : 40m 42s
Querier Expiration Time  : 1m 39s
MLD Snoop Enabled on VLAN : True

switch# show ipv6 mld interface 1/1/2

MLD Configured Version   : 2
MLD Operating Version    : 2
Querier State            : Querier
Querier IP [this switch] : fe80::7272:cfff:fe96:d3ec
Querier Uptime           : 40m 42s
Querier Expiration Time  : 1m 39s
MLD Snoop Enabled on VLAN : True
```

show ipv6 mld [all-vrfs]

Syntax

```
show ipv6 mld [all-vrfs ]
```

Description

This command shows MLD information for the specified VRF.

Command context

Manager (#)

Parameters

all-vrfs

Optional: shows MLD information status for all VRFs.

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Example

```

switch(config)# show ipv6 mld all-vrfs
VRF Name      : default
Interface     : vlan2
MLD Configured Version : 2
MLD Operating Version  : 2
Querier State      : Querier
Querier IP [this switch] : fe80::a00:9ff:fe06:67cd
Querier Uptime      : 23m 53s
Querier Expiration Time : 0m 17s
MLD Snoop Enabled on VLAN : True

Active Group Address      Vers Mode Uptime      Expires
-----
ff05::2:1                2    INC   3m 56s    1m 47s

```

show ipv6 mld [interface <intf-id> | vlan <vlan-id>] [counters]]

Syntax

```
show ipv6 mld [interface <intf-id> | vlan <vlan-id>] [counters]]
```

Description

This command shows MLD query packet Tx and Rx on a specific VLAN.

Command context

Manager (#)

Parameters

vlan-id

Required: 1-4094, shows MLD configuration on a specified VLAN.

intf-id

Required: IFNAME, shows MLD configuration on a specified interface.

counters

Optional: Shows MLD query packet counter Tx-Rx on a specified VLAN.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

```

switch# show ipv6 mld interface vlan 2 counters

MLD Counters

Interface Name      : vlan2
VRF Name            : default
Membership Timeout   : 0

Rx                  Tx
-----
V1 All Hosts Queries 0          0

```


V2 All Hosts Queries	0	0
V1 Group Specific Queries	0	0
V2 Group Specific Queries	0	2
Group And Source Specific Queries	0	2
V2 Member Reports	0	N/A
V1 Member Reports	0	N/A
V1 Member Leaves	0	N/A
Packets dropped by ACL	0	N/A

```
switch# show ipv6 mld interface 1/1/1 counters
```

MLD Counters

```
Interface Name      : 1/1/1
VRF Name            : default
Membership Timeout  : 0
```

	Rx	Tx
	-----	-----
V1 All Hosts Queries	0	0
V2 All Hosts Queries	0	0
V1 Group Specific Queries	0	0
V2 Group Specific Queries	0	0
Group And Source Specific Queries	0	0
V2 Member Reports	0	N/A
V1 Member Reports	0	N/A
V1 Member Leaves	0	N/A
Packets dropped by ACL	0	N/A

show ipv6 mld [interface <intf-id> | vlan <vlan-id>] [groups]]

Syntax

```
show ipv6 mld [interface <intf-id> | vlan <vlan-id>] [groups]]
```

Description

This command shows MLD groups joined details.

Command context

Manager (#)

Parameters

vlan-id

Required: 1-4094, shows MLD information on a specified VLAN.

intf-id

Required: IFNAME, shows MLD information on a specified interface.

groups

Optional: Shows MLD groups information on a specified interface.

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Example

```
switch# show ipv mld interface vlan 2 groups

MLD group information for group ff05::2:1

Interface Name      : vlan2
VRF Name            : default

Group Address       : ff05::2:1
Last Reporter       : fe80::1

Vers  Mode  Uptime      Expires      V1      Sources  Sources
-----  -
2      INC   6m 2s      0m 4s      Timer    Forwarded Blocked
-----  -

Group Address       : ff05::2:1
Source Address      : 3000::1

Mode  Uptime      Expire
-----  -
INC   6m 2s      0m 4s
```

show ipv6 mld [interface (<intf-id> | vlan <vlan-id>) [group <group_ip>] [source <source_ip>]]]

Syntax

```
show ipv6 mld [interface (<intf-id> | vlan <vlan-id>) [group <group_ip>] [source <source_ip>]]]
```

Description

This command shows MLD joined group details on a specified interface.

Command context

Manager (#)

Parameters

group_ip

Required: X:X::X:X, shows MLD joined group details.

source_ip

Required: X:X::X:X, shows MLD joined group details for a specified source.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

```
switch# show ipv mld interface vlan 2 group ff55::5

MLD group information for group ff55::5

Interface Name      : vlan2
VRF Name            : default
```

```
Group Address      : ff55::5
Last Reporter      : fe80::1
```

Vers	Mode	Uptime	Expires	V1 Timer	Sources Forwarded	Sources Blocked
2	INC	6m 2s	0m 4s		1	

```
Group Address      : ff55::5
Source Address     : 3000::1
```

Mode	Uptime	Expire
INC	6m 2s	0m 4s

```
switch# show ipv mld interface vlan 2 group ff55::5 source 3000::1
```

```
Interface Name     : vlan2
VRF Name           : default
Group Address      : ff55::5
Source Address     : 3000::1
```

Mode	Uptime	Expire
INC	9m 37s	2m 0s

show ipv6 mld [groups]

Syntax

```
show ipv6 mld [groups]
```

Description

This command shows MLD groups joined details.

Command context

Manager (#)

Parameters

groups

Options: shows MLD groups information.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

```
switch# show ipv6 mld groups

MLD group information for group ff05::2:11

Interface Name     : vlan2
VRF Name           : default
```

```

Group Address      : ff05::2:11
Last Reporter      : 2001::1

Vers Mode Uptime    Expires    V1      Sources  Sources
-----
1          2m 27s    1m 53s   1m 53s   Forwarded Blocked

MLD group information for group ff05::2:12

Interface Name     : vlan2
VRF Name           : default

Group Address      : ff05::2:12
Last Reporter      : 2001::1

Vers Mode Uptime    Expires    V1      Sources  Sources
-----
1          0m 3s     4m 18s   4m 18s   Forwarded Blocked

```

show ipv6 mld groups [all-vrfs]

Syntax

```
show ipv6 mld groups [all-vrfs]
```

Description

This command shows MLD groups joined details on VRFs.

Command context

Manager (#)

Parameters

all-vrfs

Optional: shows MLD groups joined in all VRFs.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

```

switch# show ipv6 mld groups all-vrfs

MLD group information for group ff05::2:11

Interface Name     : vlan1
VRF Name           : default

Group Address      : ff05::2:11
Last Reporter      : 2001::1

V1      Sources  Sources

```

```

Vers Mode Uptime      Expires      Timer      Forwarded Blocked
-----
1          4m 4s      2m 38s      2m 38s

switch# show ipv6 mld groups vrf default

MLD group information for group ff05::2:11

Interface Name      : vlan2
VRF Name            : default

Group Address       : ff05::2:11
Last Reporter       : 2001::1

V1      Sources      Sources
Vers Mode Uptime      Expires      Timer      Forwarded Blocked
-----
1          5m 25s      1m 17s      1m 17s

```

show ipv6 mld [interface <intf-id> [counters]]

Syntax

```
show ipv6 mld [interface <intf-id> [counters]]
```

Description

This command shows MLD query packet Tx and Rx on a specific interface.

Command context

Manager (#)

Parameters

intf-id

Required: shows MLD configuration on a specified interface.

counters

Optional: shows MLD query packet counter Tx-Rx on a specified interface.

Authority

Operators or Administrators or local user group members with execution rights for this command.

Operators can execute this command from the operator context (>) only.

Example

```

switch# show ipv6 mld interface 1/1/1 counters

MLD Counters

Interface Name      : 1/1/1
VRF Name            : default
Membership Timeout  : 0

V1      Rx      Tx
V1 All Hosts Queries      0      0
V2 All Hosts Queries      0      9

```

V1 Group Specific Queries	0	0
V2 Group Specific Queries	0	0
Group And Source Specific Queries	0	0
V2 Member Reports	0	N/A
V1 Member Reports	0	N/A
V1 Member Leaves	0	N/A
Packets dropped by ACL	0	N/A

show ipv6 mld [interface <intf-id> [statistics]]

Syntax

```
show ipv6 mld [interface <intf-id> [statistics]]
```

Description

This command shows MLD statistics on a specific interface.

Command context

Manager (#)

Parameters

intf-id

Required: shows MLD statistics on a specified interface.

statistics

Optional: shows MLD statistics on a specified interface.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

```
switch# show ipv6 mld interface 1/1/1 statistics

MLD statistics

Interface Name : 1/1/1
VRF Name       : default

Number of Include Groups      : 2
Number of Exclude Groups     : 0
Number of Static Groups      : 0
Total Multicast Groups Joined : 2
```

show ipv6 mld [interface <intf-id> [groups]]

Syntax

```
show ipv6 mld [interface <intf-id> [groups]]
```

Description

This command shows MLD groups joined details.

Command context

Manager (#)

Parameters

intf-id

Required: shows MLD configuration on a specified interface.

groups

Optional: shows MLD groups information.

Authority

Operators or Administrators or local user group members with execution rights for this command.

Operators can execute this command from the operator context (>) only.

Example

```
switch# show ipv6 mld interface 1/1/1 groups

MLD group information for group ff55::1

Interface Name      : 1/1/1
VRF Name            : default

Group Address       : ff55::1
Last Reporter       : fe80::a00:9ff:fe77:1062

Vers  Mode  Uptime      Expires   Vl      Sources  Sources
-----  -
2      EXC    0m 14s      4m 6s    Timer   Forwarded Blocked
```

show ipv6 mld [interface (<intf-id> | vlan <vlan-id>) [group <group_ip>] [source <source_ip>]]]

Syntax

show ipv6 mld [interface (<intf-id> | vlan <vlan-id>) [group <group_ip>] [source <source_ip>]]]

Description

This command shows MLD joined group details on a specified interface.

Command context

Manager (#)

Parameters

group_ip

Required: X:X::X:X, shows MLD joined group details.

source_ip

Required: X:X::X:X, shows MLD joined group details for a specified source.

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Example

```
switch# show ipv mld interface vlan 2 group ff55::5

MLD group information for group ff55::5

Interface Name      : vlan2
VRF Name            : default

Group Address       : ff55::5
Last Reporter       : fe80::1

Vers  Mode  Uptime      Expires      V1      Sources  Sources
-----  -
2      INC    6m 2s          0m 4s          Timer    Forwarded Blocked
-----  -

Group Address       : ff55::5
Source Address      : 3000::1

Mode  Uptime      Expire
-----  -
INC    6m 2s          0m 4s

switch# show ipv mld interface vlan 2 group ff55::5 source 3000::1

Interface Name      : vlan2
VRF Name            : default
Group Address       : ff55::5
Source Address      : 3000::1

Mode  Uptime      Expire
-----  -
INC    9m 37s          2m 0s
```

show ipv6 mld [group <group_ip> [all-vrfs]]

Syntax

```
show ipv6 mld [group <group_ip> [all-vrfs]]
```

Description

This command shows MLD joined group details on VRF.

Command context

Manager (#)

Parameters

group_ip

Required: X:X::X:X, shows MLD joined group details.

all-vrfs

Optional: shows MLD groups joined in all VRFs.

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Example

```
switch# show ipv6 mld group ff55::1

MLD group information for group ff55::1

Interface Name      : 1/1/1
VRF Name            : default

Group Address       : ff55::1
Last Reporter       : fe80::a00:9ff:fe77:1062

Vers Mode Uptime    Expires    V1      Sources  Sources
-----
2      EXC  3m 12s    3m 46s    Timer    Forwarded Blocked
-----

switch# show ipv6 mld group ff05::2:11 all-vrfs

MLD group information for group ff05::2:11

Interface Name      : vlan2
VRF Name            : default

Group Address       : ff05::2:11
Last Reporter       : 2001::1

Vers Mode Uptime    Expires    V1      Sources  Sources
-----
1              1m 16s    3m 4s    3m 4s    Forwarded Blocked
-----
```

show ipv6 mld [group <group_ip> [source <source_ip> [all-vrfs]]]

Syntax

show ipv6 mld [group <group_ip> [source <source_ip> [all-vrfs]]]

Description

This command shows MLD joined group details for a source on VRF.

Command context

Manager (#)

Parameters

group_ip

Required: X:X::X:X, shows MLD joined group details.

source_ip

Required: X:X::X:X, shows MLD joined group details for a source.

all-vrfs

Optional: shows MLD groups joined in all VRFs.

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Example

```
switch# show ipv6 mld group ff05::2:1 source 3000::1

Interface Name   : vlan2
VRF Name        : default
Group Address    : ff05::2:1
Source Address   : 3000::1

Mode Uptime      Expire
----
INC  0m 53s      3m 27s

switch# show ipv6 mld group ff05::2:1 source 3000::1 all-vrfs

Interface Name   : vlan2
VRF Name        : default
Group Address    : ff05::2:1
Source Address   : 3000::1

Mode Uptime      Expire
----
INC  1m 38s      4m 5s
```

show ipv6 mld [interface vlan <vlan-id> [statistics]]

Syntax

```
show ipv6 mld [interface vlan <vlan-id> [statistics]]
```

Description

This command shows MLD statistics on a specific interface VLAN.

Command context

Manager (#)

Parameters

vlan-id

Required: 1-4094, shows MLD information on a specified VLAN.

statistics

Optional: shows MLD query packet Tx, Rx, Error packet counters on a specified VLAN.

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Example

```
switch# show ipv6 mld interface vlan 2 statistics
```

```
MLD statistics

Interface Name : vlan2
VRF Name       : default

Number of Include Groups      : 2
Number of Exclude Groups     : 0
Number of Static Groups      : 0
Total Multicast Groups Joined : 2
```

show ipv6 mld [static-groups [all-vrfs]]

Syntax

```
show ipv6 mld [static-groups [all-vrfs]]
```

Description

This command shows MLD static groups.

Command context

Manager (#)

Parameters

all-vrfs

Optional: shows MLD groups joined in all VRFs.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

```
switch# show ipv6 mld static-groups

MLD Static Group Address Information

VRF Name      :default
Interface Name  Group Address
-----
vlan2          ff12::c
vlan2          ff12::d

switch# show ipv6 mld static-groups all-vrfs

MLD Static Group Address Information

VRF Name      :default
Interface Name  Group Address
-----
vlan2          ff12::c
vlan2          ff12::d
```

show ipv6 mld counters

Syntax

```
show ipv6 mld [counters]
```

Description

This command shows MLD counters.

Command context

Manager (#)

Parameters

vrf

Optional: shows MLD counter status in a specific VRF.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

```
switch# show ipv6 mld counters
```

MLD Counters

Interface Name : vlan2
VRF Name : default
Membership Timeout : 0

	Rx	Tx
	-----	-----
V1 All Hosts Queries	0	0
V2 All Hosts Queries	0	12
V1 Group Specific Queries	0	0
V2 Group Specific Queries	0	0
Group And Source Specific Queries	0	0
V2 Member Reports	0	N/A
V1 Member Reports	0	N/A
V1 Member Leaves	0	N/A
Packets dropped by ACL	0	N/A

```
switch# show ipv6 mld counters vrf default
```

MLD Counters

Interface Name : vlan2
VRF Name : default
Membership Timeout : 0

	Rx	Tx
	-----	-----
V1 All Hosts Queries	0	0
V2 All Hosts Queries	0	12
V1 Group Specific Queries	0	0
V2 Group Specific Queries	0	0
Group And Source Specific Queries	0	0
V2 Member Reports	0	N/A
V1 Member Reports	0	N/A

Vl Member Leaves	0	N/A
Packets dropped by ACL	0	N/A

ipv6 mld {enable | disable}

Syntax

```
ipv6 mld {enable | disable}
```

Description

This command enables or disables MLD on the interface.

Command context

```
config-if
```

Parameters

enable

Required: Enable MLD on the interface.

disable

Required: Disable MLD on the interface.

Authority

Administrators or local user group members with execution rights for this command.

Example

```
switch(config)# interface vlan 1
switch(config-if)# ipv6 mld enable
switch(config-if)# ipv6 mld disable
```

ipv6 mld apply access-list

Syntax

```
ipv6 mld apply access-list <ACL-NAME>
no ipv6 mld apply access-list <ACL-NAME>
```

Description

Configures the ACL on a particular interface to filter the MLD join or leave packets based on rules set in the particular ACL name.

The `no` form of this command removes the rules set for the ACL.

Command context

```
config-vlan
```

Parameters

access-list

Associates an ACL with the IGMP.

<ACL-NAME>

Specifies the name of the ACL.

Authority

Administrators or local user group members with execution rights for this command.

Usage

Existing classifier commands are used to configure the ACL. In case an MLDv2 packet with multiple group addresses is received, it will only process the permitted group addresses based on the ACL rule set, and any existing joins will time out. If there is no match or if there is a deny rule match, the packet is dropped.

Examples

Configuring the ACL to filter MLD packets based on rules set in access list `mygroup`:

```
switch(config)# access-list ipv6 mygroup
switch(config-acl-ip)# permit icmpv6 any ff55::1
switch(config-acl-ip)# exit
switch(config)# interface vlan 1
switch(config-vlan)# ipv6 mld apply access-list mygroup
```

Configuring the ACL to remove the rules set in access list `mygroup`:

```
switch(config-vlan)# no ipv6 mld apply access-list mygroup
```

no ipv6 mld

Syntax

no ipv6 mld

Description

This command removes all MLD configurations on the interface.

Authority

Administrators or local user group members with execution rights for this command.

Command context

config-if

Example

```
switch(config)# interface vlan 1
switch(config-if)# no ipv6 mld
```

ipv6 mld querier

Syntax

```
ipv6 mld querier
```

Description

This command configures MLD querier. This functionality will allow the interface to join in the querier-election process.

Command context

```
config-if
```

Authority

Administrators or local user group members with execution rights for this command.

Example

```
switch(config)# interface vlan 1
switch(config-if)# ipv6 mld querier
switch(config-if)# no ipv6 mld querier
```

ipv6 mld querier [interval <interval-value>]

Syntax

```
ipv6 mld querier [interval <interval-value>]
```

Description

This command configures MLD querier interval. The default interval-value is 125.

Command context

```
config-if
```

Parameters

```
interval-value
```

Required: 5-300, configures MLD querier interval.



Default interval-value is 125. Use the `no ipv6 mld querier interval` command to set interval-value to the default.

Authority

Administrators or local user group members with execution rights for this command.

Example

```
switch(config)# interface vlan 1
switch(config-if)# ipv6 mld querier interval 100
switch(config-if)# no ipv6 mld querier interval
```

ipv6 mld last-member-query-interval

Syntax


```
ipv6 mld last-member-query-interval <interval-value>
```

Description

This command configures MLD last member query interval value in seconds. The default interval-value is 1 second.

Command context

config-if

Parameters

interval-value

Required: 1-2, configures MLD last-member-query-interval.



Default interval-value is 1 second. Use the `no ipv6 mld last-member-query-interval` command to set interval-value to the default.

Authority

Administrators or local user group members with execution rights for this command.

Example

```
switch(config)# interface vlan 1
switch(config-if)# ipv6 mld last-member-query-interval 2
switch(config-if)# no ipv6 mld last-member-query-interval
```

ipv6 mld querier query-max-response-time

Syntax

```
ipv6 mld querier query-max-response-time <response-time>
```

Description

This command configures MLD max response time value in seconds. The default max-response-time-value is 10 seconds.

Command context

config-if

Parameters

max-response-time-value

Required: 10-128, configures MLD querier max-response-time.



Default max-response-time-value is 10 seconds. Use the `no ipv6 mld querier query-max-response-time` command to set max-response-time-value to the default.

Authority

Administrators or local user group members with execution rights for this command.

Example

```
switch(config)# interface vlan 1
switch(config-if)# ipv6 mld query-max-response-time 50
switch(config-if)# no ipv6 mld query-max-response-time
```

ipv6 mld robustness

Syntax

```
ipv6 mld robustness <value>
```

Description

This command configures MLD robustness. The robustness value represents the number of times the querier retries queries on the connected subnets. The default robustness-value is 2 seconds.

Command context

config-if

Parameters

robustness-value

Required: 1-7, configures MLD robustness.



Default robustness-value is 2 seconds. Use the `no ipv6 mld robustness` command to set robustness-value to the default.

Authority

Administrators or local user group members with execution rights for this command.

Example

```
switch(config)# interface vlan 1
switch(config-if)# ipv6 mld robustness 5
switch(config-if)# no ipv6 mld robustness
```

ipv6 mld static-group

Syntax

```
ipv6 mld static-group <multicast-group-ip>
```

Description

This command configures MLD static group.

Command context

config-if

Parameters

multicast-group-ip

Required: X:X::X:X, configures MLD static group.

Authority

Administrators or local user group members with execution rights for this command.

Example

```
switch(config)# interface vlan 1
switch(config-if)# ipv6 mld static-group ff12::c
switch(config-if)# no ipv6 mld static-group ff12::c
```

ipv6 mld version

Syntax

```
ipv6 mld version <version>
```

Description

This command configures MLD version.

Command context

config-if

Parameters

version

Required: 1-2, configures MLD version.

Authority

Administrators or local user group members with execution rights for this command.

Example

```
switch(config)# interface vlan 1
switch(config-if)# ipv6 mld version 2
```

ipv6 mld version [strict]

Syntax

```
ipv6 mld version <version> [strict]
```

Description

This command configures MLD strict version. Packets that do not match the configured version will be dropped. By default, strict option is not enabled.

Command context

config-if

Parameters

version

Required: 1-2, configures MLD version.

Authority

Administrators or local user group members with execution rights for this command.

Example

```
switch(config)# interface vlan 1
switch(config-if)# ipv6 mld version 2 strict
switch(config-if)# no ipv6 mld version 2 strict
```

show spanning-tree

Syntax

```
show spanning-tree
```

Description

Shows priority, address, Hello-time, Max-age, and Forward-delay for bridge and root node.

Command context

Operator (>) or Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Example

Showing spanning tree standard information:

```
switch# show spanning-tree
Spanning tree status      : Enabled Protocol: MSTP

MST0
  Root ID
    Priority      : 32768, Root
    MAC-Address   : 48:0F:CF:AF:04:76
    Hello time    : 2 seconds
    Max Age       : 20 seconds
    Forward Delay : 15 seconds

  Bridge ID
    Priority      : 32768
    MAC-Address   : 48:0F:CF:AF:04:76
    Hello time    : 2 seconds
    Max Age       : 20 seconds
    Forward Delay : 15 seconds
```

PORT	ROLE	STATE	COST	PRIORITY	TYPE	BPDU-Tx	BPDU-Rx	TCN-Tx	TCN-Rx
1/1/1	Designated	Forwarding	20000	128	P2P Edge	100	60	20	10
1/1/2	Designated	Forwarding	20000	128	P2P	100	60	20	10
1/1/3	Designated	Forwarding	20000	128	Shr	100	60	20	10
1/1/4	Designated	Forwarding	20000	128	Shr Edge	100	60	20	10
1/1/5	Alternate	Loop-Inc	20000	128	Shr Edge	100	60	20	10
1/1/6	Alternate	Root-Inc	20000	128	Shr Edge	100	60	20	10

```
Number of topology changes      : 4
Last topology change occurred   : 516 seconds ago
```

show spanning-tree detail

Syntax

```
show spanning-tree detail
```

Description

Shows spanning tree detail including CIST and corresponding port information.

Command context

Operator (>) or Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Example

Showing spanning tree detailed information:

```
switch# show spanning-tree detail
Spanning tree status      : Enabled Protocol: MSTP

MST0
  Root ID      Priority    : 32768
                MAC-Address: 48:0f:cf:af:04:76
                This bridge is the root
                Hello time(in seconds):2  Max Age(in seconds):20
                Forward Delay(in seconds):15

  Bridge ID    Priority    : 32768
                MAC-Address: 48:0f:cf:af:04:76
                Hello time(in seconds):2  Max Age(in seconds):20
                Forward Delay(in seconds):15

  PORT        ROLE        STATE        COST        PRIORITY    TYPE        BPDU-Tx    BPDU-Rx
  TCN-Tx      TCN-Rx
  -----
  1/1/1       Designated Forwarding  20000       128         P2P Edge    100         60
  20          10
  1/1/2       Designated Forwarding  20000       128         P2P         100         60
  20          10
  1/1/3       Designated Forwarding  20000       128         Shr         100         60
  20          10
  1/1/4       Designated Forwarding  20000       128         Shr Edge    100         60
  20          10
  1/1/5       Alternate  Loop-Inc    20000       128         Shr Edge    100         60
  20          10
  1/1/6       Alternate  Root-Inc    20000       128         Shr Edge    100         60
  20          10

Topology change flag      : True
```

```

Number of topology changes      : 4
Last topology change occurred  : 516 seconds ago
Timers:      Hello expiry  1 , Forward delay expiry 18

Port 1/1/1
Designated root has priority          :32768 Address: 48:0f:cf:af:04:76
Designated bridge has priority        :32768 Address: 48:0f:cf:af:04:76
Designated port                      :1/1/1
Number of transitions to forwarding state : 3
Bpdus sent 347, received 9
TCN_Tx: 20, TCN_Rx: 10

Port 1/1/2
Designated root has priority          :32768 Address: 48:0f:cf:af:04:76
Designated bridge has priority        :32768 Address: 48:0f:cf:af:04:76
Designated port                      :1/1/2
Number of transitions to forwarding state : 3
Bpdus sent 350, received 11
TCN_Tx: 20, TCN_Rx: 10

Port lag1 ID 321
Designated root has priority          : 32768          Address:
48:0F:CF:AF:04:76
Designated bridge has priority        : 32768          Address:
48:0F:CF:AF:04:76
Designated port id                   : 321
Multi-Chassis role                   : active
Number of transitions to forwarding state : 3
BPDUs sent                          : 340
BPDUs received                      : 5
TCN_Tx: 20, TCN_Rx: 10

```

show spanning-tree inconsistent-ports

Syntax

```
show spanning-tree inconsistent-ports [instance <INSTANCE-ID>]
```

Description

Shows ports blocked by STP protection functions such as Root guard, Loop guard, BPDU guard, and RPVST guard in addition to MSTI information.

Command context

Operator (>) or Manager (#)

Parameters

<INSTANCE-ID>

Specifies the MSTP instance ID. Range: 0 to 64.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

Showing spanning tree inconsistent ports:

```
switch# show spanning-tree inconsistent-ports
Instance ID Blocked Port Reason
-----
0           1/1/13      BPDU Guard
```

Showing inconsistent port information for instances 1-4:

```
switch# show spanning-tree inconsistent-ports instance 1-4
Instance ID Blocked Port Reason
-----
1           1/1/3      Root Guard
2           1/1/7      BPDU Guard
3           1/1/9      Loop Guard
4           1/1/37     RPVST Guard
```

show spanning-tree mst

Syntax

```
show spanning-tree mst
```

Description

Shows MSTP configuration and status information for each instance.

Command context

Operator (>) or Manager (#)

Parameters

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Examples

Showing MSTP configuration and status information:

```
switch# show spanning-tree mst
#### MST0
Vlans mapped   : 2,4-4094
Bridge Address : 48:0F:CF:AF:04:76
Priority       : 32768
Root
Regional Root
Operational    Hello time   : 2 seconds           Forward delay: 15 seconds
                Max-age     : 20 seconds          TxHoldCount  : 6 pps
Configured     Hello time   : 2 seconds           Forward delay: 15 seconds
                Max-age     : 20 seconds          Max-Hops     : 20
Root           Address     : 48:0F:CF:AF:04:76      Priority     : 32768
                Port        : 0                   Path cost    : 0
Regional Root  Address     : 48:0F:CF:AF:04:76      Priority     : 32768
                Internal cost: 0                   Rem Hops     : 20
```

PORT	ROLE	STATE	COST	PRIORITY	TYPE	BPDU-Tx	BPDU-Rx
------	------	-------	------	----------	------	---------	---------

TCN-Tx	TCN-Rx						
1/1/1	Designated	Forwarding	20000	128	P2P Edge	100	60
20	10						
1/1/2	Designated	Forwarding	20000	128	P2P	100	60
20	10						
1/1/3	Designated	Forwarding	20000	128	Shr	100	60
20	10						
1/1/4	Designated	Forwarding	20000	128	Shr Edge	100	60
20	10						
1/1/5	Alternate	Loop-Inc	20000	128	Shr Edge	100	60
20	10						
1/1/6	Alternate	Root-Inc	20000	128	Shr Edge	100	60
20	10						

Topology change flag : True
Number of topology changes : 4
Last topology change occurred : 516 seconds ago

MST1

Vlans mapped: 1
Bridge Address : 48:0F:CF:AF:04:76 Priority: 32768
Root Address : 48:0F:CF:AF:04:76 Priority: 32768
Port : 0 Cost : 0
Rem Hops: 20

PORT	ROLE	STATE	COST	PRIORITY	TYPE	BPDU-Tx	BPDU-Rx
TCN-Tx	TCN-Rx						
1/1/1	Designated	Forwarding	20000	128	P2P Edge	100	60
20	10						
1/1/2	Designated	Forwarding	20000	128	P2P	100	60
20	10						
1/1/3	Designated	Forwarding	20000	128	Shr	100	60
20	10						
1/1/4	Designated	Forwarding	20000	128	Shr Edge	100	60
20	10						
1/1/5	Alternate	Loop-Inc	20000	128	Shr Edge	100	60
20	10						
1/1/6	Alternate	Root-Inc	20000	128	Shr Edge	100	60
20	10						

Topology change flag : True
Number of topology changes : 4
Last topology change occurred : 516 seconds ago

MST2

Vlans mapped: 3
Bridge Address : 48:0F:CF:AF:04:76 Priority: 32768
Root Address : 48:0F:CF:AF:04:76 Priority: 32768
Port : 0 Cost : 0
Rem Hops: 20

PORT	ROLE	STATE	COST	PRIORITY	TYPE	BPDU-Tx	BPDU-Rx
TCN-Tx	TCN-Rx						
1/1/1	Designated	Forwarding	20000	128	P2P Edge	100	60
20	10						
1/1/2	Designated	Forwarding	20000	128	P2P	100	60

```

20          10
1/1/3      Designated Forwarding 20000      128      Shr      100      60
20          10
1/1/4      Designated Forwarding 20000      128      Shr Edge 100      60
20          10
1/1/5      Alternate  Loop-Inc   20000      128      Shr Edge 100      60
20          10
1/1/6      Alternate  Root-Inc   20000      128      Shr Edge 100      60
20          10

Topology change flag      : True
Number of topology changes : 4
Last topology change occurred : 516 seconds ago

```

show spanning-tree mst-config

Syntax

```
show spanning-tree mst-config
```

Description

Shows MSTP instance and corresponding VLAN information.

Command context

Operator (>) or Manager (#)

Parameters

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Examples

Showing configuration information for MST instances and corresponding VLANs:

```

switch# show spanning-tree mst-config
MST configuration information
  MST config ID      : reg
  MST config revision : 1
  MST config digest   : 2D2BC9A32097B463C48EE1817673FA2D
  Number of instances : 2

Instance ID      Member VLANs
-----
0                2,4-4094
1                1
2                3

```

show spanning-tree mst detail

Syntax

```
show spanning-tree mst detail
```

Description

Shows detailed information for all MST instances.

Command context

Operator (>) or Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Example

Showing detailed information for all MST instances:

```
switch# show spanning-tree mst detail
#### MST0
Vlans mapped: 2,4-4094
Bridge      Address: 48:0F:CF:AF:04:76      Priority: 32768
Root
Regional Root
Operational  Hello time   : 2 seconds           Forward delay: 15 seconds
              Max-age    : 20 seconds          TxHoldCount  : 6 pps
Configured   Hello time   : 2 seconds           Forward delay: 15 seconds
              Max-age    : 20 seconds          Max-Hops     : 20
Root         Address    : 48:0F:CF:AF:04:76   Priority      : 32768
              Port       : 0                  Path cost    : 0
Regional Root Address    : 48:0F:CF:AF:04:76   Priority      : 32768
              Internal cost: 0                Rem Hops     : 20
```

PORT	ROLE	STATE	COST	PRIORITY	TYPE	BPDU-Tx	BPDU-Rx
TCN-Tx	TCN-Rx						
1/1/1	Designated	Forwarding	20000	128	P2P Edge	100	60
20	10						
1/1/2	Designated	Forwarding	20000	128	P2P	100	60
20	10						
1/1/3	Designated	Forwarding	20000	128	Shr	100	60
20	10						
1/1/4	Designated	Forwarding	20000	128	Shr Edge	100	60
20	10						
1/1/5	Alternate	Loop-Inc	20000	128	Shr Edge	100	60
20	10						
1/1/6	Alternate	Root-Inc	20000	128	Shr Edge	100	60
20	10						

```
Topology change flag      : True
Number of topology changes : 4
Last topology change occurred : 516 seconds ago
```

```
Port 1/1/1
Designated root address      : 48:0F:CF:AF:04:76
Designated regional root address : 48:0F:CF:AF:04:76
Designated bridge address    : 48:0F:CF:AF:04:76
Priority                      : 32768
BPDUs sent                   : 638
BPDUs received               : 9
Message expiry               : 1 second
Forward delay expiry         : 18 seconds
Forward transitions           : 3
```

TCN_Tx: 10, TCN_Rx: 10

Port 1/1/2

Designated root address : 48:0F:CF:AF:04:76

Designated regional root address : 48:0F:CF:AF:04:76

Designated bridge address : 48:0F:CF:AF:04:76

Priority : 32768

BPDUs sent : 641

BPDUs received : 11

Message expiry : 1 second

Forward delay expiry : 18 seconds

Forward transitions : 3

TCN_Tx: 10, TCN_Rx: 10

MST1

Vlans mapped: 1

Bridge Address : 48:0F:CF:AF:04:76 Priority: 32768

Root Address : 48:0F:CF:AF:04:76 Priority: 32768

Port : 0 Cost : 0

Rem Hops: 20

PORT	ROLE	STATE	COST	PRIORITY	TYPE	BPDU-Tx	BPDU-Rx
TCN-Tx	TCN-Rx						
-----	-----	-----	-----	-----	-----	-----	-----
1/1/1	Designated	Forwarding	20000	128	P2P Edge	100	60
20	10						
1/1/2	Designated	Forwarding	20000	128	P2P	100	60
20	10						
1/1/3	Designated	Forwarding	20000	128	Shr	100	60
20	10						
1/1/4	Designated	Forwarding	20000	128	Shr Edge	100	60
20	10						
1/1/5	Alternate	Loop-Inc	20000	128	Shr Edge	100	60
20	10						
1/1/6	Alternate	Root-Inc	20000	128	Shr Edge	100	60
20	10						

Topology change flag : True

Number of topology changes : 4

Last topology change occurred : 516 seconds ago

Port 1/1/1

Designated root address : 48:0F:CF:AF:04:76

Designated bridge address : 48:0F:CF:AF:04:76

Priority : 32768

BPDUs sent : 638

BPDUs received : 9

Message expiry : 1 second

Forward delay expiry : 18 seconds

Forward transitions : 4

TCN_Tx: 10, TCN_Rx: 10

Port 1/1/2

Designated root address : 48:0F:CF:AF:04:76

Designated bridge address : 48:0F:CF:AF:04:76

Priority : 32768

BPDUs sent : 641

BPDUs received : 11

Message expiry : 1 second

Forward delay expiry : 18 seconds

Forward transitions : 4

TCN_Tx: 10, TCN_Rx: 10

MST2

Vlans mapped: 3

Bridge Address : 48:0F:CF:AF:04:76 Priority: 32768

Root Address : 48:0F:CF:AF:04:76 Priority: 32768

Port : 0 Cost : 0

Rem Hops: 20

PORT	ROLE	STATE	COST	PRIORITY	TYPE	BPDU-Tx	BPDU-Rx
TCN-Tx	TCN-Rx						
1/1/1	Designated	Forwarding	20000	128	P2P Edge	100	60
20	10						
1/1/2	Designated	Forwarding	20000	128	P2P	100	60
20	10						
1/1/3	Designated	Forwarding	20000	128	Shr	100	60
20	10						
1/1/4	Designated	Forwarding	20000	128	Shr Edge	100	60
20	10						
1/1/5	Alternate	Loop-Inc	20000	128	Shr Edge	100	60
20	10						
1/1/6	Alternate	Root-Inc	20000	128	Shr Edge	100	60
20	10						

Topology change flag : True

Number of topology changes : 4

Last topology change occurred : 516 seconds ago

Port 1/1/1

Designated root address : 48:0F:CF:AF:04:76

Designated bridge address : 48:0F:CF:AF:04:76

Priority : 32768

BPDUs sent : 638

BPDUs received : 9

Message expiry : 1 second

Forward delay expiry : 18 seconds

Forward transitions : 3

TCN_Tx: 10, TCN_Rx: 10

Port 1/1/2

Designated root address : 48:0F:CF:AF:04:76

Designated bridge address : 48:0F:CF:AF:04:76

Priority : 32768

BPDUs sent : 641

BPDUs received : 11

Message expiry : 1 second

Forward delay expiry : 18 seconds

Forward transitions : 3

TCN_Tx: 10, TCN_Rx: 10

show spanning-tree mst <INSTANCE-ID>

Syntax

show spanning-tree mst <INSTANCE-ID>

Description

Displays MSTP configurations for the given instance ID.

Command context

Operator (>) or Manager (#)

Parameters

<INSTANCE-ID>

Specifies the MSTP instance number. Range: 0 to 64.

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Example

```
switch# show spanning-tree mst 1

#### MST1
Vlans mapped: 1
Bridge        Address : 48:0F:CF:AF:04:76      Priority: 32768
Root          Address : 48:0F:CF:AF:04:76      Priority: 32768
              Port      : 0                  Cost      : 0
              Rem Hops: 20

PORT          ROLE          STATE      COST      PRIORITY  TYPE        BPDU-Tx    BPDU-Rx
TCN-Tx        TCN-Rx
-----
1/1/1        Designated Forwarding 20000      128        P2P Edge    100         60
20           10
1/1/2        Designated Forwarding 20000      128        P2P         100         60
20           10
1/1/3        Designated Forwarding 20000      128        Shr         100         60
20           10
1/1/4        Designated Forwarding 20000      128        Shr Edge    100         60
20           10
1/1/5        Alternate Loop-Inc  20000      128        Shr Edge    100         60
20           10
1/1/6        Alternate Root-Inc  20000      128        Shr Edge    100         60
20           10

Topology change flag      : True
Number of topology changes : 4
Last topology change occurred : 516 seconds ago
```

show spanning-tree mst <INSTANCE-ID> detail

Syntax

```
show spanning-tree mst <INSTANCE-ID> detail
```

Description

Displays MSTP configurations for the given instance ID with corresponding port details.

Command context

Operator (>) or Manager (#)

Parameters

<INSTANCE-ID>

Specifies the MSTP instance number. Range: 0 to 64.

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Example

```
switch# show spanning-tree mst 1 detail
```

```
#### MST1
```

```
Vlans mapped: 1
```

```
Bridge Address : 48:0F:CF:AF:04:76 Priority: 32768
```

```
Root Address : 48:0F:CF:AF:04:76 Priority: 32768
```

```
Port : 0 Cost : 0
```

```
Rem Hops: 20
```

PORT TCN-Tx	ROLE TCN-Rx	STATE	COST	PRIORITY	TYPE	BPDU-Tx	BPDU-Rx
1/1/1 20	Designated 10	Forwarding	20000	128	P2P Edge	100	60
1/1/2 20	Designated 10	Forwarding	20000	128	P2P	100	60
1/1/3 20	Designated 10	Forwarding	20000	128	Shr	100	60
1/1/4 20	Designated 10	Forwarding	20000	128	Shr Edge	100	60
1/1/5 20	Alternate 10	Loop-Inc	20000	128	Shr Edge	100	60
1/1/6 20	Alternate 10	Root-Inc	20000	128	Shr Edge	100	60

```
Topology change flag : True
```

```
Number of topology changes : 4
```

```
Last topology change occurred : 516 seconds ago
```

```
Port 1/1/1
```

```
Designated root address : 48:0F:CF:AF:04:76
```

```
Designated bridge address : 48:0F:CF:AF:04:76
```

```
Priority : 32768
```

```
BPDUs sent : 667
```

```
BPDUs received : 9
```

```
Message expiry : 0 second
```

```
Forward delay expiry : 18 seconds
```

```
Forward transitions : 4
```

```
TCN_Tx: 10, TCN_Rx: 10
```

```
Port 1/1/2
```

```
Designated root address : 48:0F:CF:AF:04:76
```

```
Designated bridge address : 48:0F:CF:AF:04:76
```

```
Priority : 32768
```

```
BPDUs sent : 670
```

```
BPDUs received : 11
```

```
Message expiry : 0 second
```

```
Forward delay expiry : 18 seconds
```

```
Forward transitions : 4
```

```
TCN_Tx: 10, TCN_Rx: 10
```

show spanning-tree mst interface

Syntax

```
show spanning-tree mst <INSTANCE-ID> interface <IFNAME>
```

Description

Shows MSTP configurations for the given instance ID with corresponding port details.

Command context

Operator (>) or Manager (#)

Parameters

<INSTANCE-ID>

Specifies the MSTP instance number. Range: 0 to 64.

<IFNAME>

Specifies an interface.

Authority

Operators or Administrators or local user group members with execution rights for this command.

Operators can execute this command from the operator context (>) only.

Examples

Showing MST configuration and port details:

```
switch# show spanning-tree mst 1 interface 1/1/1
Port 1/1/1
```

Instance	Role	State	Cost	Priority	Vlans mapped
1	Designated	Forwarding	20000	128	1

show spanning-tree summary port

Syntax

```
show spanning-tree summary port
```

Description

Shows spanning tree port summary information.

Command context

Operator (>) or Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command.

Operators can execute this command from the operator context (>) only.

Example

Showing summary of spanning tree ports:


```
switch# show spanning-tree summary port

STP status           : Enabled
Protocol             : MSTP
BPDU guard timeout value : None
BPDU guard enabled interfaces : 1/1/1-1/1/9,1/1/11,1/1/13,1/1/15,1/1/17,1/1/19,
                               1/1/21,lag1,lag2
BPDU filter enabled interfaces : None
Root guard enabled interfaces : 1/1/3
Loop guard enabled interfaces : 1/1/2
TCN guard enabled interfaces : 1/1/1-1/1/3
```

Interface count by state

Instance ID	Blocking	Listening	Learning	Forwarding
0	2	0	0	15
1	2	0	0	15
2	2	0	0	15
Total = 3	6	0	0	45

show spanning-tree summary root

Syntax

```
show spanning-tree summary root
```

Description

Shows spanning tree root summary information.

Command context

Operator (>) or Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Example

Showing spanning tree root summary:

```
switch# show spanning-tree summary root

STP status           : Enabled
Protocol             : MSTP
System ID            : 70:72:cf:32:50:f5

Root bridge for STP Instance : 0,1,2
```

Instance ID	Priority	Root ID	Root cost	Hello Time	Max Age	Fwd Dly	Root Port
0	32768	70:72:cf:32:50:f5	0	2	20	15	n/a
1	32768	70:72:cf:32:50:f5	0	2	20	15	n/a
2	32768	70:72:cf:32:50:f5	200	2	20	15	1/1/1

spanning-tree

Syntax

```
spanning-tree  
no spanning-tree
```

Description

Enables the spanning tree protocol on the switch.

The `no` form of this command disables the spanning tree protocol on the switch.

Command context

```
config
```

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling spanning tree:

```
switch(config)# spanning-tree
```

Disabling spanning tree:

```
switch(config)# no spanning-tree
```

spanning-tree bdpdu-filter

Syntax

```
spanning-tree bdpdu-filter  
no spanning-tree bdpdu-filter
```

Description

Enables the BDPU filter for the interface.

The BPDU filter feature allows control of spanning tree participation on a per-port basis. It can be used to exclude specific ports from becoming part of spanning tree operations. A port with the BPDU filter enabled will ignore incoming BPDU packets and stay locked in the spanning tree forwarding state. All other ports maintain their role. Typical uses for this parameter include:

- To have MSTP operations running on selected ports of the switch rather than every port of the switch at a time.
- To prevent the spread of errant BPDU frames.
- To eliminate the need for a topology change when a port's link status changes. For example, ports that connect to servers and workstations can be configured to remain outside of spanning tree operations.
- To protect the network from denial of service attacks that use spoofing BPDUs by dropping incoming BPDU frames. For this scenario, BPDU protection offers a more secure alternative, implementing port shut down and a detection alert when errant BPDU frames are received.



Ports configured with the BPDU filter mode remain active (learning and forward frames). However, spanning tree cannot receive or transmit BPDUs on the port. The port remains in a forwarding state, permitting all broadcast traffic. This can create a network storm if there are any loops (that is, redundant links) using these ports. If you suddenly have a high load, disconnect the link and disable the BPDU filter (using the `no` command.)

The `no` form of the command sets the BPDU filter status to the default of disabled on the interface.

Command context

`config-if`

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling the BPDU filter on interface 1/1/1:

```
switch(config)# interface 1/1/1
switch(config-if)# spanning-tree bdpu-filter
```

Disabling BPDU filter on interface 1/1/1:

```
switch(config)# interface 1/1/1
switch(config-if)# no spanning-tree bdpu-filter
```

spanning-tree bpduguard

Syntax

```
spanning-tree bpduguard
no spanning-tree bpduguard
```

Description

Enables the BPDU guard on the selected switch interface. When BPDU guard is enabled, interfaces receiving MSTP BPDUs become disabled.

BPDU protection is a security feature designed to protect the active MSTP topology by preventing spoofed BPDU packets from entering the MSTP domain. In a typical implementation, BPDU protection would be applied to edge ports connected to end user devices that do not run MSTP. If MSTP BPDU packets are received on a protected port, this feature disables that port and alerts the network manager using an SNMP trap.

Occasionally a hardware or software failure can cause MSTP to fail, creating forwarding loops that can cause network failures where unidirectional links are used. The non-designated port transitions in a faulty manner because the port is no longer receiving MSTP BPDUs.

The `no` form of the command disables BPDU guard on the selected interface.

Command context

`config-if`

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling the BPDU guard on interface 1/1/1:

```
switch(config)# interface 1/1/1
switch(config-if)# spanning-tree bpdu-guard
```

Disabling BPDU guard on interface 1/1/1:

```
switch(config)# interface 1/1/1
switch(config-if)# no spanning-tree bpdu-guard
```

spanning-tree bpdu-guard timeout

Syntax

```
spanning-tree bpdu-guard timeout <INTERVAL>
no spanning-tree bpdu-guard timeout
```

Description

Enables and configures the auto re-enable timeout in seconds for all interfaces with BPDU guard enabled. When an interface is disabled after receiving an unauthorized BPDU it will automatically be re-enabled after the timeout expires. The default is for the interface to stay disabled until manually re-enabled.

The `no` form of the command disables BPDU guard timeout on the interface. This is the default.

Command context

config

Parameters

<INTERVAL>

Specifies the re-enable timeout in seconds. Range: 1 to 65535.

Authority

Administrators or local user group members with execution rights for this command.

Example

Enabling the BPDU guard timeout on interface 1/1/1:

```
switch(config)# interface 1/1/1
switch(config-if)# spanning-tree bpdu-guard timeout 10
```

Disabling BPDU guard timeout on interface 1/1/1:

```
switch(config)# interface 1/1/1
switch(config-if)# no spanning-tree bpdu-guard
```

spanning-tree config-name

Syntax

```
spanning-tree config-name <CONFIG-NAME>  
no spanning-tree config-name [<CONFIG-NAME>]
```

Description

Sets the configuration name for the MST region in which the switch resides.

All switches within an MST region must have identical configuration names. For more than one MSTP switch in the same MST region, the identical region name must be configured on all such switches. If the default configuration name is retained on a switch, it cannot exist in the same MST region with another switch.

The `no` form of this command overwrites the currently configured name with the default name. The default name is a text string using the hexadecimal representation of the system MAC address.

Command context

config

Parameters

<CONFIG-NAME>

Specifies the configuration name for the MST region in which the switch resides. Default: text string using the hexadecimal representation of the MAC address of the switch. Range: 1 - 32 nonblank characters (case-sensitive).

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the configuration name to MST0:

```
switch(config)# spanning-tree config-name MST0
```

Setting the configuration name to the default value:

```
switch(config)# no spanning-tree config-name
```

spanning-tree config-revision

Syntax

```
spanning-tree config-revision <REVISION-NUMBER>  
no spanning-tree config-revision [<REVISION-NUMBER>]
```

Description

Configures the revision number for the MST region in which the switch resides. All switches within an MST region must have identical revision numbers. Use this setting to differentiate between region configurations. For example, when changing configuration settings within a region where you want to track the configuration versions you use, or when creating a new region from a subset of switches in a current region and you want to maintain the same region name.

The `no` form of this command overwrites the currently configured revision number of the MST region and sets it to the default value of 0.

Command context

`config`

Parameters

`<REVISION-NUMBER>`

Specifies the revision number for the MST region in which the switch resides. Range: 0 - 65535. Default: 0.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the revision to 40:

```
switch(config)# spanning-tree config-revision 40
```

Setting the revision to the default value:

```
switch(config)# no spanning-tree config-revision
```

spanning-tree cost

Syntax

```
spanning-tree cost <PORT-COST>  
no spanning-tree cost [<PORT-COST>]
```

Description

Sets individual port cost for MSTI 0.

For a given port, the path cost setting can be different for different MSTIs to which the port may belong.

The switch uses the path cost to determine which ports are the forwarding ports in the MSTI; that is, which links to use for the active topology of the MSTI and which ports to block.

The `no` form of the command sets the port cost for MSTI 0 instance to the default value.

Command context

`config-if`

Parameters

`<PORT-COST>`

Specifies the cost of the port for MSTI 0. Range: 1-200,000,000. Default is calculated from the port link speed:

- 10 Mbps link speed equals a path cost of 2,000,000.
- 100 Mbps link speed equals a path cost of 200,000.
- 1 Gbps link speed equals a path cost of 20,000.

- 2 Gbps link speed equals a path cost of 10,000.
- 10 Gbps link speed equals a path cost of 2,000.
- 100 Gbps link speed equals a path cost of 200.
- 1 Tbps link speed equals a path cost of 20.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the cost to **2000** on interface **1/1/1**:

```
switch(config)# interface 1/1/1
switch(config-if)# spanning-tree cost 2000
```

Setting the cost to the default on interface **1/1/1**:

```
switch(config)# interface 1/1/1
switch(config-if)# no spanning-tree cost
```

spanning-tree forward-delay

Syntax

```
spanning-tree forward-delay <DELAY-IN-SECS>
no spanning-tree forward-delay [<DELAY-IN-SECS>]
```

Description

Configures the time the switch waits between transitions from listening to learning and from learning to forwarding states.

The `no` form of this command sets forward delay time for the bridge to the default of 15 seconds.

Command context

config

Parameters

<DELAY-IN-SECS>

Specifies the forward delay time in seconds. Default: 15 seconds. Range: 4-30.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting forward delay to 6 seconds:

```
switch(config)# spanning-tree forward-delay 6
```

Setting forward delay to the default of 15 seconds:

```
switch(config)# no spanning-tree forward-delay
```

spanning-tree hello-time

Syntax

```
spanning-tree hello-time <HELLO-IN-SECS>  
no spanning-tree hello-time [<HELLO-IN-SECS>]
```

Description

Configures the transmission interval between consecutive Bridge Protocol Data Units (BPDU) that the switch sends as a root bridge. The hello time interval is inserted in outbound BPDUs.

The `no` form of this command sets hello time to the default of 2 seconds.

Command context

config

Parameters

<HELLO-IN-SECS>

Specifies the hello time interval in seconds. Default: 2 seconds. Range: 2-10.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the hello time interval to 6 seconds:

```
switch(config)# spanning-tree hello-time 6
```

Setting the hello time interval to the default of 2 seconds:

```
switch(config)# no spanning-tree hello-time
```

spanning-tree instance cost

Syntax

```
spanning-tree instance <INSTANCE-ID> cost <PORT-COST>  
no spanning-tree instance <INSTANCE-ID> cost [<PORT-COST>]
```

Description

Sets the individual port cost for an MSTI. The switch uses the path cost to determine which links to use for the active topology of the MSTI (forwarding ports) and which ports to block. The path cost setting for a port can be different on each MSTI to which the port belongs.

The `no` form of this command sets the port cost for an MSTI to the default value.

Command context

config-if

Parameters

<INSTANCE-ID>

Specifies the MSTI number. Range: 1-64.

<PORT-COST>

Specifies the cost of the port for the MSTI. Range: 1-200000000. Default value is calculated from the port link speed:

- 10 Mbps link speed equals a path cost of 2000000.
- 100 Mbps link speed equals a path cost of 200000.
- 1 Gbps link speed equals a path cost of 20000.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the port **1/1/1** cost for MSTI **1** to **2000**:

```
switch(config)# interface 1/1/1
switch(config-if)# spanning-tree instance 1 cost 2000
```

Setting the port **1/1/1** cost for MSTI **1** to the default:

```
switch(config)# interface 1/1/1
switch(config-if)# no spanning-tree instance 1 cost
```

spanning-tree instance port-priority

Syntax

```
spanning-tree instance <INSTANCE-ID> port-priority <PRIORITY-MULTIPLIER>
no spanning-tree instance <INSTANCE-ID> port-priority [<PRIORITY-MULTIPLIER>]
```

Description

Configures the priority as a priority multiplier for the specified ports in the specified MST instance.

For a given port, the priority setting can be different for different MST instances to which the port may belong.

The `no` form of this command sets the port priority to the default value of 8 for the MST instance. The default priority value is derived by multiplying 8 by 16.

Command context

config-if

Parameters

<INSTANCE-ID>

Specifies the MSTP instance number. Range: 1-64.

<PRIORITY-MULTIPLIER>

Specifies the priority as a multiplier. Default: 8. Range: 0 to 15.

The priority range for a port in a given MST instance is 0 to 255. However, this command specifies the priority as a multiplier (0 to 15) of 16. When you specify a priority multiplier of 0 to 15, the actual priority assigned to the switch is: (priority-multiplier) x 16.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the port **1/1/1** priority for instance **1** to **8**:

```
switch(config)# interface 1/1/1
switch(config-if)# spanning-tree instance 1 port-priority 8
```

Setting the port 1/1/1 priority for instance 1 to the default:

```
switch(config)# interface 1/1/1
switch(config-if)# no spanning-tree instance 1 port-priority
```

spanning-tree instance priority

Syntax

```
spanning-tree instance <INSTANCE-ID> priority <PRIORITY-MULTIPLIER>
no spanning-tree instance <INSTANCE-ID> priority [<PRIORITY-MULTIPLIER>]
```

Description

Sets the switch priority for the specified MST instance.

The **no** form of this command sets the priority for the specified instance to the default of 8.

Command context

config

Parameters

<INSTANCE-ID>

Specifies the MSTP instance number. Range: 1 to 64.

<PRIORITY-MULTIPLIER>

Specifies the priority as a multiplier. Default: 8. Range: 0 to 15.

The priority range for an MSTP switch is 0-61440. However, this command specifies the priority as a multiplier (0 - 15) of 4096. That is, when you specify a priority multiplier value of 0 - 15, the actual priority assigned to the switch is: (priority-multiplier) x 4096. For example, with 2 as the priority-multiplier on a given MSTP switch, the switch priority setting is 8,192.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the priority multiplier for instance 1 to 5:

```
switch(config)# spanning-tree instance 1 priority 5
```

Setting the priority multiplier for instance 1 to the default of 8:

```
switch(config)# no spanning-tree instance 1 priority
```

spanning-tree instance vlan

Syntax

```
spanning-tree instance <INSTANCE-ID> vlan <VLAN-ID>  
no spanning-tree instance <INSTANCE-ID> vlan <VLAN-ID>
```

Description

Creates a new instance with VLANs mapped or maps VLANs to an existing instance.

Each instance must have at least one VLAN mapped to it. When VLANs are mapped to an instance, they are automatically unmapped from the instance they were mapped to before. Any MSTP instance can have all the VLANs configured on the switch.

The `no` form of this command removes the specified VLAN from the MSTP instance.

Command context

config

Parameters

<INSTANCE-ID>

Specifies the MSTP instance number. Range: 1 to 64.

<VLAN-ID>

Specifies a VLAN ID number.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Mapping VLAN 1 to instance 1:

```
switch(config)# spanning-tree instance 1 vlan 1
```

Removing VLAN 1 from instance 1:

```
switch(config)# no spanning-tree instance 1 vlan 1
```

spanning-tree link-type

Syntax

```
spanning-tree link-type {point-to-point|shared}
```

Description

Specifies the link type of the interface, which is normally derived from the duplex setting of the port. The default setting depends on the duplex mode of the port: full-duplex ports are point-to-point, half-duplex ports are shared.

Command context

config-if

Parameters

point-to-point

Specifies the link type as point-to-point.

shared

Specifies the link type as shared.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the link type to point-to-point on interface **1/1/1**:

```
switch(config)# interface 1/1/1
switch(config-if)# spanning-tree link-type point-to-point
```

Setting the link type to shared on interface **1/1/1**:

```
switch(config)# interface 1/1/1
switch(config-if)# spanning-tree link-type shared
```

spanning-tree loop-guard

Syntax

```
spanning-tree loop-guard
no spanning-tree loop-guard
```

Description

Enables the loop guard on the interface. STP loop guard is best applied on blocking or forwarding ports. The `no` form of the command sets the loop guard status to the default of disabled on the interface.

Command context

config-if

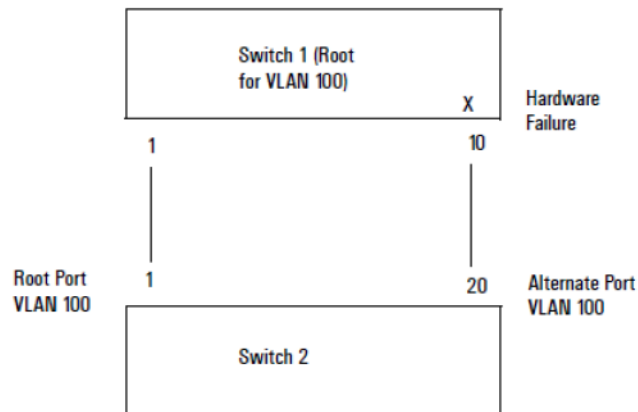
Authority

Administrators or local user group members with execution rights for this command.

Usage

Occasionally a hardware or software failure can cause MSTP to fail, creating forwarding loops that can cause network failures where unidirectional links are used. The non-designated port transitions in a faulty manner because the port is no longer receiving MSTP BPDUs.

Loop guard causes the non-designated port to go into the MSTP loop inconsistent state instead of the forwarding state. In the loop inconsistent state the port prevents data traffic and BPDU transmission through the link, therefore avoiding the loop creation. When BPDUs again are received on the inconsistent port, it resumes normal MSTP operation automatically.



In this example, the transmission from switch 1 port 10 to switch 2 port 20 is blocked due to a hardware failure. Switch 2 port 2 does not receive BPDUs and goes into a forwarding state, creating a loop.

When loop guard is configured for switch 2 port 20, this port goes from a forwarding state to an inconsistent state, and does not forward the traffic through the link, thus avoiding loop creation.

Examples

Enabling the loop guard on interface **1/1/1**:

```
switch(config)# interface 1/1/1
switch(config-if)# spanning-tree loop-guard
```

Disabling loop guard on interface **1/1/1**:

```
switch(config)# interface 1/1/1
switch(config-if)# no spanning-tree loop-guard
```

spanning-tree max-age

Syntax

```
spanning-tree max-age <AGE-IN-SECS>
no spanning-tree max-age [<AGE-IN-SECS>]
```

Description

Sets the maximum age timer, which specifies the maximum age value that the switch inserts in outbound BPDU packets it sends as a root bridge. Max-age is the interval, specified in the BPDU, that BPDU data remains valid after its reception.

The bridge recomputes the spanning tree topology if it does not receive a new BPDU before max-age expiry.

The `no` form of this command sets the max-age value to the default of 20 seconds.

Command context

config

Parameters

<AGE-IN-SECS>

Specifies the max-age in seconds. Range: 6 to 40. Default: 20.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the max-age to 10 seconds:

```
switch(config)# spanning-tree max-age 10
```

Setting the max-age to the default of 20 seconds:

```
switch(config)# no spanning-tree max-age
```

spanning-tree max-hops

Syntax

spanning-tree max-hops <HOP-COUNT>

no spanning-tree max-hops [<HOP-COUNT>]

Description

Configures the max hop setting that the switch inserts into BPDUs that it sends out as the root bridge. The max hop setting determines the number of bridges in an MST region that a BPDU can traverse before it is discarded.

The `no` form of this command sets the maximum number of hops to the default of 20.

Command context

config

Parameters

<HOP-COUNT>

Specifies the maximum number of hops. Range: 1 to 40. Default: 20.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the hop count to 10:

```
switch(config)# spanning-tree max-hops 10
```

Setting the max-age to the default of 20:

```
switch(config)# no spanning-tree max-hops
```

spanning-tree mode

Syntax

```
spanning-tree mode {mstp|rpvst}
```

Description

Sets the spanning tree mode to either MSTP mode (Multiple-instance Spanning Tree Protocol) or RPVST mode (Rapid Per VLAN Spanning Tree).

The `no` form of this command sets the spanning tree mode to the default value `mstp`.

Command context

config

Parameters

mstp

Sets the mode to MSTP (Multiple-instance Spanning Tree Protocol), which applies the STP (spanning tree protocol) separately for each set of VLANs (called an MSTI - multiple spanning tree instance).

rpvst

Sets the mode to RPVST (Rapid Per VLAN Spanning Tree).

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling MSTP mode:

```
switch(config)# spanning-tree mode mstp
```

Enabling RPVST mode:

```
switch(config)# spanning-tree mode rpvst
```

spanning-tree port-priority

Syntax

```
spanning-tree port-priority <PRIORITY-MULTIPLIER>  
no spanning-tree port-priority [<PRIORITY-MULTIPLIER>]
```

Description

Configures the port priority. The priority of a port can be different for each MST instance to which it belongs.

The `no` form of the command sets the port priority for MST instance 0 to the default of 8. The default priority value is derived by multiplying 8 by 8. For LAG interfaces the default is 4.

Command context

config-if

Parameters

<PRIORITY-MULTIPLIER>

Specifies the port priority as a multiplier. Default: 8, except for LAG interfaces where the default is 4.
Range: 0 to 15.

The priority range for a port in a given MSTI is 0 to 255. However, this command specifies the priority as a multiplier (0 to 15) of 16. When you specify a priority multiplier of 0 to 15, the actual priority assigned to the switch is: (priority-multiplier) x 16.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the port priority to 8 on interface **1/1/1**:

```
switch(config)# interface 1/1/1
switch(config-if)# spanning-tree port-priority 8
```

Setting the port priority to the default on interface **1/1/1**:

```
switch(config)# interface 1/1/1
switch(config-if)# no spanning-tree port-priority
```

spanning-tree port-type

Syntax

```
spanning-tree port-type {admin-edge|admin-network}
no spanning-tree port-type [admin-edge|admin-network]
```

Description

Sets the STP port type for the interface.

Port types include: admin-edge and admin-network.

The **no** form of the command sets the port type to the default of admin-network.

Command context

config-if

Parameters

admin-edge

Specifies the port type as administrative edge. During spanning tree establishment, ports with admin-edge enabled transition immediately to the forwarding state.

admin-network

Specifies the port type as administrative network. When this option is selected, the port looks for BPDUs for the first 3 seconds. If there are none, the port is classified as an edge port and immediately starts forwarding packets. If BPDUs are seen on the port, the port is classified as a non-edge port and normal STP operation commences on that port.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the port type to admin-edge on interface **1/1/1**:

```
switch(config)# interface 1/1/1
switch(config-if)# spanning-tree port-type admin-edge
```

Setting the port type to admin-network on interface **1/1/1**:

```
switch(config)# interface 1/1/1
switch(config-if)# spanning-tree port-type admin-network
```

Setting the port type to the default of admin-network on interface 1/1/1:

```
switch(config)# interface 1/1/1
switch(config-if)# no spanning-tree port-type
```

spanning-tree priority

Syntax

```
spanning-tree priority <PRIORITY-MULTIPLIER>
no spanning-tree priority [<PRIORITY-MULTIPLIER>]
```

Description

Configures the switch (bridge) priority for the designated region in which the switch resides.

The switch compares this priority with the priorities of other switches in the same region to determine the root switch for the region. The lower the priority value, the higher the priority.

The `no` form of this command sets the bridge priority to the default of 8. The default priority value is derived by multiplying 8 by 4096.

Command context

config

Parameters

<PRIORITY-MULTIPLIER>

Specifies the priority as a multiplier. Range: 0 to 15. Default: 8.

The priority range for an MSTP switch is 0-61440. However, this command specifies the priority as a multiplier (0 to 15) of 4096. That is, when you specify a priority multiplier value of 0 to 15, the actual priority assigned to the switch is: (priority-multiplier) x 4096. For example, with 2 as the priority-multiplier on a given MSTP switch, the switch priority setting is 8,192.

Authority

Administrators or local user group members with execution rights for this command.

Usage

Every switch running an instance of MSTP has a Bridge Identifier, which is a unique identifier that helps distinguish this switch from all others. The switch with the lowest Bridge Identifier is elected as the root for the tree. The Bridge Identifier is composed of a configurable priority component (2 bytes) and the bridge's MAC address (6 bytes). You can change the priority component provides flexibility in determining which switch will be the root for the tree, regardless of its MAC address.

Examples

Setting the priority multiplier to 12:

```
switch(config)# spanning-tree priority 12
```

Setting the priority multiplier to the default of 8:

```
switch(config)# no spanning-tree priority
```

spanning-tree root-guard

Syntax

```
spanning-tree root-guard  
no spanning-tree root-guard
```

Description

Enables the root guard on the interface.

When a port is enabled as root-guard, it cannot be selected as the root port even if it receives superior STP BPDUs. The port is assigned an "alternate" port role and enters a blocking state if it receives superior MSTP BPDUs.

A superior BPDU contains both "better" information on the root bridge and path cost to the root bridge, which would normally replace the current root bridge selection.

The `no` form of the command sets the root guard status to the default of disabled on the interface.

Command context

```
config-if
```

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling the root guard on interface **1/1/1**:

```
switch(config)# interface 1/1/1  
switch(config-if)# spanning-tree root-guard
```

Disabling root guard on interface **1/1/1**:

```
switch(config)# interface 1/1/1  
switch(config-if)# no spanning-tree root-guard
```

spanning-tree rpvst-filter

Syntax

```
spanning-tree rpvst-filter  
no spanning-tree rpvst-filter
```

Description

Enables the RPVST filter for the interface.

When the RPVST filter is enabled, the ingressing RPVST proprietary BPDUs are dropped after copying to CPU whereas the standard IEEE RPVST BPDUs are still allowed. This helps in preventing the flooding of RPVST proprietary BPDUs under an MSTP-RPVST interop environment.



If the neighboring switch is running RPVST then this pair of switches will not converge as RPVST BPDUs will not reach them.

If enabling RPVST filter causes a high traffic load, shutdown the port and reconfigure the BPDU filter with the CLI command: `no spanning tree rpvst-filter`.

RPVST filter is disabled by default.

Command context

config-if

Authority

Administrators or local user group members with execution rights for this command.

Example

Enabling the RPVST filter on interface 1/1/1:

```
switch# configure terminal
switch(config)# interface 1/1/1
switch(config-if)# spanning-tree rpvst-filter
```

Disabling RPVST filter on interface 1/1/1:

```
switch# configure terminal
switch(config)# interface 1/1/1
switch(config-if)# no spanning-tree rpvst-filter
```

spanning-tree rpvst-guard

Syntax

```
spanning-tree rpvst-guard
no spanning-tree rpvst-guard
```

Description

Enables RPVST guard on the switch interface. When RPVST guard is enabled on an interface, it will disable that interface if RPVST BPDUs are received on it.

The `no` form of the command sets the RPVST guard status to the default of disabled on the interface.

Command context

config-if

Authority

Administrators or local user group members with execution rights for this command.

Example

Enabling RPVST guard on interface 1/1/1:

```
switch# configure terminal
switch(config)# interface 1/1/1
switch(config-if)# spanning-tree rpvst-guard
```

Disabling RPVST guard on interface 1/1/1:

```
switch# configure terminal
switch(config)# interface 1/1/1
switch(config-if)# no spanning-tree rpvst-guard
```

spanning-tree rpvst-mstp interconnect vlan

Syntax

```
spanning-tree rpvst-mstp-interconnect-vlan <VLAN-ID>
no spanning-tree rpvst-mstp-interconnect-vlan
```

Description

Configures the VLAN that has to be used to interconnect RPVST and MSTP domains. VLAN 1 is used by default.

The `no` form of this command removes the VLAN configuration.

- It is required to create the interconnect VLAN and then configure RPVST spanning tree on it.
- The same interconnect VLAN must be kept on all the switches in the network.
- Adding or deleting the interconnect VLAN triggers a re-convergence in the network.
- Deleting a VLAN that is configured as the interconnect VLAN does not reset the value to the default.

Command context

config

Parameters

<VLAN-ID>

Specifies the number of a VLAN.

Authority

Administrators or local user group members with execution rights for this command.

Examples

This example configures VLAN 10 to used to interconnect RPVST and MSTP domains.

```
switch#(config)# spanning-tree rpvst-mstp-interconnect-vlan 10
```

spanning-tree tcn-guard

Syntax

```
spanning-tree tcn-guard
no spanning-tree tcn-guard
```

Description

Enables the TCN (Topology Change Notification) guard in the interface. When enabled for a port, the port stops propagating received topology change notifications and topology changes to other ports.

The `no` form of the command sets the TCN guard status to the default of disabled on the interface.

Command context

```
config-if
```

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling TCN guard on interface **1/1/1**:

```
switch(config)# interface 1/1/1
switch(config-if)# spanning-tree tcn-guard
```

Disabling TCN guard on interface **1/1/1**:

```
switch(config)# interface 1/1/1
switch(config-if)# no spanning-tree tcn-guard
```

spanning-tree transmit-hold-count

Syntax

```
spanning-tree transmit-hold-count <COUNT>
no spanning-tree transmit-hold-count [<COUNT>]
```

Description

Sets the maximum number of BPDUs per second that the switch can send from an interface.

The `no` form of this command sets the transmit-hold-count to the default of 6.

Command context

```
config
```

Parameters

<COUNT>

Specifies the number of BPDUs that can be sent per second. Range: 1 to 10. Default: 6.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the transmit-hold-count to 5:

```
switch(config)# spanning-tree transmit-hold-count 5
```

Setting the transmit-hold-count to the default of 6:

```
switch(config)# no spanning-tree transmit-hold-count
```

spanning-tree trap

Syntax

```
spanning-tree trap {new-root|topology-change [instance <0-64>] | errant-bpdu |  
    root-guard-inconsistency | loop-guard-inconsistency}  
  
no spanning-tree trap {new-root|topology-change [instance <0-64>] | errant-bpdu |  
    root-guard-inconsistency | loop-guard-inconsistency}
```

Description

This command enables SNMP traps for new root, topology change event, errant-bpdu received event, root-guard inconsistency, and loop-guard inconsistency notifications. It is disabled by default.

The `no` form of this command disables SNMP traps for new root and topology change event notifications.

Command context

config

Parameters

new-root

Enabling SNMP notification when a new root is elected on any MST instance on the switch.

topology-change

Enabling SNMP notification when a topology change event occurs in the specified MST instance on the switch.

errant-bpdu

Enabling SNMP notification when an errant bpdu is received by any MST instance on the switch.

root-guard-inconsistency

Enabling SNMP notification when the root-guard finds the port inconsistent for any MST instance on the switch.

loop-guard-inconsistency

Enabling SNMP notification when the loop-guard finds the port inconsistent for any MST instance on the switch.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling SNMP notification when a new root is elected:

```
switch(config)# spanning-tree trap new-root
```

Enabling SNMP notification when a topology change event occurs:

```
switch(config)# spanning-tree trap topology-change
```

Enabling SNMP notification when a topology change event occurs on instance 1:

```
switch(config)# spanning-tree trap topology-change instance 1
```

Disabling SNMP traps when a topology change event notification occurs on instance 1:

```
switch(config)# no spanning-tree trap topology-change instance 1
```

clear mvrp statistics

Syntax

```
clear mvrp statistics [<PORT-NUM> | <PORT-LIST> | LAG <LAG-NUM>]
```

Command context

Manager (#)

Description

Resets the MVRP statistic counters globally or for the specified ports or LAG.

Parameters

<PORT-NUM>

Specifies a port number.

<PORT-LIST>

Specifies a list of ports.

LAG <LAG-NUM>

Specifies a Link Aggregation number. Range: 1 to 128.

Authority

Administrators or local user group members with execution rights for this command.

Examples

```
switch# clear mvrp statistics 1/1/1
```

mvrp

Syntax

```
mvrp  
no mvrp
```

Description

Enables the MVRP feature globally or on a specific interface. By default, MVRP is disabled.

The `no` form of this command disables MVRP.



MVRP and VLAN translation cannot be enabled on the same interface.

Command context


```
config
config-if
```

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling MVRP globally:

```
switch(config)# mvrp
```

Enabling MVRP on an interface:

```
switch(config)# interface 1/1/1
switch(config-if)# mvrp
```

mvrp registration

Syntax

```
mvrp registration {normal | fixed | forbidden [<VLAN-LIST>]}
no mvrp registration forbidden {<VLAN-LIST>}
```

Description

Configures the MVRP registrar state which determines how an MVRP participant responds to MRP messages. The default registration mode is normal.

The `no` command removes the specified VLANs from the forbidden list.

Command context

```
config-if
```

Parameters

`normal`

Enables dynamic registration and deregistration of VLANs on the interface, and propagates VLAN information to other switches on the network. Default.

`fixed`

Disables dynamic deregistration of VLANs and drops received MVRP frames. The interface does not deregister dynamic VLANs or register new dynamic VLANs.

`forbidden`

Disables dynamic registration of VLANs and drops received MVRP frames. The MVRP participant does not register new dynamic VLANs or re-register a deregistered dynamic VLAN.

`<VLAN-LIST>`

Disables dynamic registration of VLANs and drops received MVRP frames for specific VLANs only. Normal behavior applies to all other VLANs. Specify the number of a single VLAN, or a series of numbers for a range of VLANs, separated by commas (1, 2, 3, 4), dashes (1-4), or both (1-4,6).

Authority

Administrators or local user group members with execution rights for this command.

Examples

```
switch(config)# switch(config-if)# mvrp registration forbidden 10
```

```
switch(config-if)# mvrp registration fixed
```

```
switch(config-if)# mvrp registration forbidden 1,2,10-20
```

mvrp timer

Syntax

```
mvrp timer {join | leave | leaveall | periodic} <TIME>  
no mvrp timer {join | leave | leaveall | periodic}
```

Description

Sets an MVRP timer.

The `no` form of this command sets the specified timer to its default value.

Command context

config-if

Parameters

join <TIME>

Sets the join timer. You can use the timer to space MVRP join messages. To ensure that join messages are transmitted to other participants, an MRP participant waits for the specified period of the join timer before sending a join message. The join timer must be less than half of the Leave Timer. Range: 20 to 100 in centiseconds. Default: 20.

leave <TIME>

Sets the leave timer for the port, specifying the time that the registrar state machine waits in the LV state before transiting to the MT state. The leave timer must be at least twice the join timer and must be less than the leave all timer. Range: 40 - 1000000 centiseconds. Default: 300 centiseconds.

leaveall <TIME>

Sets the leave all timer for the port, specifying the frequency with which the leave all state machine generates leave all PDUs. Range: 500 to 1000000 centiseconds. Default: 1000.

periodic <TIME>

Sets the periodic timer for the port, specifying the frequency with which the periodic transmission state machine generates periodic events. The periodic timer is set to 1 second when it is started. Range: 100 to 1000000 centiseconds. Default: 100.

Authority

Administrators or local user group members with execution rights for this command.

Examples

```
switch(config-if)# mvrp timer join 22
```

show mvrp config

Syntax

```
show mvrp config [<PORT-NUM> | <PORT-LIST> | LAG <LAG-NUM>]
```

Description

Displays the MVRP configuration for all L2 ports or optionally for the ports specified.

Command context

Operator (>) or Manager (#)

Parameters

<PORT-NUM>

Specifies displaying information for a particular port number.

<PORT-LIST>

Specifies displaying information for a list of ports.

LAG <LAG-NUM>

Specifies displaying information by LAG. Range: 1 to 128.

Authority

Operators or Administrators or local user group members with execution rights for this command.

Operators can execute this command from the operator context (>) only.

Examples

```
switch# show mvrp config
Configuration and Status - MVRP
Global MVRP status : Disabled
Port      Status   Registration Type   Join Timer   Leave Timer   LeaveAll Timer   Periodic Timer
-----
1/1/1     Disabled  Normal          20      300      1000      100
1/1/2     Disabled  Normal          20      300      1000      100
1/1/3     Disabled  Normal          20      300      1000      100
```

show mvrp state

Syntax

```
show mvrp state [<VLAN-ID> | <VLAN-ID> <PORT-NUM>]
```

Description

Displays the MVRP Registrar and Applicant state machine information for all ports on which MVRP is enabled, or for specific ports.

Command context

Operator (>) or Manager (#)

Parameters

<VLAN-ID>

<PORT-NUM>

Specifies a physical port on the switch. Format: member/slot/port.

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Examples

```
switch# show mvrp state 1
Configuration and Status - MVRP state for VLAN 1
Port   VLAN Registrar Applicant
      State      State
-----
1/1/1  1      MT          QA
```

```
switch# show mvrp state 10 1/1/1
Configuration and Status - MVRP state for VLAN 10
Port   VLAN Registrar Applicant Forbid
      State      State      Mode
-----
1/1/1  10      MT          LO          Yes
switch#
```

show mvrp statistics

Syntax

```
show mvrp statistics [<PORT-LIST>]
```

Description

Displays MVRP statistics for all ports or on the ports specified in the list.

Command context

Operator (>) or Manager (#)

Parameters

<PORT-LIST>

Specifies a list of ports. When specifying a list of ports, the ports for which there are no statistics will be listed in the output.

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Examples

```
switch# show mvrp statistics
Status and Counters - MVRP
MVRP statistics for port : 1/1/1
-----
Failed registration      : 0
Last PDU origin         : 48:0f:cf:af:b1:76
Total PDU Transmitted    : 13127
```

```

Total PDU Received      : 327
Frames Discarded        : 0
Message type      Transmitted      Received
-----
New                0                0
Empty              50029394         1264
In                 0                4
Join Empty         1425             48
Join In            563             555
Leave               0                0
Leaveall            12218           25

```

```
switch# show mvrp statistics 1/1/1
```

```

Status and Counters - MVRP
MVRP statistics for port : 1/1/1
-----
Failed registration      : 0
Last PDU origin          : 48:0f:cf:af:b1:76
Total PDU Transmitted    : 14874
Total PDU Received       : 327
Frames Discarded         : 0
Message type      Transmitted      Received
-----
New                0                0
Empty              57181612         1264
In                 0                4
Join Empty         1425             48
Join In            563             555
Leave               0                0
Leaveall            13965           25

```

ntp authentication

Syntax

```
ntp authentication
no ntp authentication
```

Description

Enables support for authentication when communicating with an NTP server. The `no` form of this command disables authentication support.

Command context

config

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling authentication support:

```
switch(config)# ntp authentication
```

Disabling authentication support:

```
switch(config)# no ntp authentication
```

ntp authentication-key

Syntax

```
ntp authentication-key <KEY-ID> {md5 | sha1}
    [{ <PLAINTEXT-KEY> [trusted] | ciphertext <ENCRYPTED-KEY> }]
no ntp authentication-key <KEY-ID>
```

Description

Defines an authentication key that is used to secure the exchange with an NTP time server. This command provides protection against accidentally synchronizing to a time source that is not trusted.

The `no` form of this command removes the authentication key.

Command context

config

Parameters

<KEY-ID>

Specifies the authentication key ID. Range: 1 to 65534.

md5

Selects MD5 key encryption.

sha1

Specifies SHA1 key encryption.

<PLAINTEXT-KEY>

Specifies the plaintext authentication key. Range: 8 to 40 characters. The key may contain printable ASCII characters excluding "#" or be entered in hex. Keys longer than 20 characters are assumed to be hex. To use an ASCII key longer than 20 characters, convert it to hex.

trusted

Specifies that this is a trusted key. When NTP authentication is enabled, the switch only synchronizes with time servers that transmit packets containing a trusted key.

ciphertext <ENCRYPTED-KEY>

Specifies the ciphertext authentication key in Base64 format. This is used to restore the NTP authentication key when copying configuration files between switches or when uploading a previously saved configuration.



When the key is not provided on the command line, plaintext key prompting occurs upon pressing Enter, followed by prompting as to whether the key is to be trusted. The entered key characters are masked with asterisks.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Defining key 10 with MD5 encryption and a provided plaintext trusted key:

```
switch(config)# ntp authentication-key 10 md5 F82#450b trusted
```

Defining key 5 with SHA1 encryption and a prompted plaintext trusted key:

```
switch(config)# ntp authentication-key 5 sha1
Enter the NTP authentication key: *****
Re-Enter the NTP authentication key: *****

Configure the key as trusted (y/n)? y
```

Removing key 10:

```
switch(config)# no ntp authentication-key 10
```

ntp disable

Syntax

ntp disable

Description

Disables the NTP client on the switch. The NTP client is disabled by default.

Command context

config

Authority

Administrators or local user group members with execution rights for this command.

Examples

Disabling the NTP client.

```
switch(config)# ntp disable
```

ntp enable

Syntax

ntp enable

no ntp enable

Description

Enables the NTP client on the switch to automatically adjust the local time and date on the switch. The NTP client is disabled by default.

The `no` form of this command disables the NTP client.

Command context

config

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling the NTP client.

```
switch(config)# ntp enable
```

Disabling the NTP client.

```
switch(config)# no ntp enable
```

ntp server

Syntax

```
ntp server <IP-ADDR> [key <KEY-NUM>] [minpoll <MIN-NUM>] [maxpoll <MAX-NUM>] [burst | iburst]
    [prefer] [version <VER-NUM>]
no ntp server <IP-ADDR>
no ntp server <IP-ADDR> [burst] [iburst] [prefer] [key-id <KEY-NUM>]
```


Description

Defines an NTP server to use for time synchronization, or updates the settings of an existing server with new values. Up to eight servers can be defined.

The `no` form of this command removes a configured NTP server.



The default NTP version is 4; it is backwards compatible with version 3.

Command context

`config`

Parameters

`server` <IP-ADDR>

Specifies the address of an NTP server as a DNS name, an IPv4 address (`x.x.x.x`), where `x` is a decimal number from 0 to 255, or an IPv6 address (`xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx`), where `x` is a hexadecimal number from 0 to F.

When specifying an IPv4 address, you can remove leading zeros. For example, the address `192.169.005.100` becomes `192.168.5.100`.

When specifying an IPv6 address, you can use two colons (`::`) to represent consecutive zeros (but only once), remove leading zeros, and collapse a hextet of four zeros to a single 0. For example, this address `2222:0000:3333:0000:0000:0000:4444:0055` becomes `2222:0:3333::4444:55`.

`key` <KEY-NUM>

Specifies the key to use when communicating with the server. A trusted key must be defined with the command `ntp authentication-key` and authentication must be enabled with the command `ntp authentication`. Range: 1 to 65534.

`minpoll` <MIN-NUM>

Specifies the minimum polling interval in seconds, as a power of 2. Range: 4 to 17. Default: 6 (64 seconds).

`maxpoll` <MAX-NUM>

Specifies the maximum polling interval in seconds, as a power of 2. Range: 4 to 17. Default: 10 (1024 seconds).

`burst`

Send a burst of packets instead of just one when connected to the server. Useful for reducing phase noise when the polling interval is long.

`iburst`

Send a burst of six packets when not connected to the server. Useful for reducing synchronization time at startup.

`prefer`

Make this the preferred server.

`version` <VER-NUM>

Specifies the version number to use for all outgoing NTP packets. Range: 3 or 4.

Authority

Administrators or local user group members with execution rights for this command.

Usage

For features such as Activate and ZTP, a switch that has a factory default configuration will automatically be configured with pool.ntp.org. NTP server configurations via DHCP options are supported. The DHCP server can be configured with maximum of two NTP server addresses which will be supported on the switch. Only IPV4 addresses are supported.

Examples

Defining the ntp server pool.ntp.org, using iburst, and NTP version 4.

```
switch(config)# ntp server pool.ntp.org iburst version 4
```

Removing the ntp server pool.ntp.org.

```
switch(config)# no ntp server pool.ntp.org
```

Defining the ntp server my-ntp.mydomain.com and makes it the preferred server.

```
switch(config)# ntp server my-ntp.mydomain.com prefer
```

ntp trusted-key

Syntax

```
ntp trusted-key <KEY-ID>
```

```
no ntp trusted-key <KEY-ID>
```

Description

Sets a key as trusted. When NTP authentication is enabled, the switch only synchronizes with time servers that transmit packets containing a trusted key.

The `no` form of this command removes the trusted designation from a key.

Command context

```
config
```

Parameters

<KEY-ID>

Specifies the identification number of the key to set as trusted. Range: 1 to 65534.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Defining key 10 as a trusted key.

```
switch(config)# ntp trusted-key 10
```

Removing trusted designation from key 10:

```
switch(config)# no ntp trusted-key 10
```

ntp vrf

Syntax

```
ntp vrf <VRF-NAME>
```

Description



6100 only supports default VRF.

Specifies the VRF on which the NTP client communicates with an NTP server.

Command context

config

Parameters

<VRF-NAME>

Specifies the name of a VRF.

Authority

Administrators or local user group members with execution rights for this command.

Example

Setting the switch to use the default VRF for NTP client traffic.

```
switch(config)# ntp vrf default
```

show ntp associations

Syntax

```
show ntp associations
```

Description

Shows the status of the connection to each NTP server. The following information is displayed for each server:

- Tally code : The first character is the Tally code:
 - (blank): No state information available (e.g. non-responding server)
 - x : Out of tolerance (discarded by intersection algorithm)
 - . : Discarded by table overflow (not used)
 - - : Out of tolerance (discarded by the cluster algorithm)
 - + : Good and a preferred remote peer or server (included by the combine algorithm)
 - # : Good remote peer or server, but not utilized (ready as a backup source)
 - * : Remote peer or server presently used as a primary reference
 - o : PPS peer (when the prefer peer is valid)
- ID: Server number.
- NAME: NTP server FQDN/IP address (Only the first 24 characters of the name are displayed).
- REMOTE: Remote server IP address.
- REF_ID: Reference ID for the remote server (Can be an IP address).

- ST: (Stratum) Number of hops between the NTP client and the reference clock.
- LAST: Time since the last packet was received in seconds unless another unit is indicated.
- POLL: Interval (in seconds) between NTP poll packets. Maximum (1024) reached as server and client sync.
- REACH: 8-bit octal number that displays status of the last eight NTP messages (377 = all messages received).

Command context

Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Example

```
switch# show ntp associations
-----
ID          NAME          REMOTE          REF-ID ST LAST POLL REACH
-----
  1         192.0.1.1        192.0.1.1        .INIT. 16   -   64    0
* 2  time.apple.com      17.253.2.253      .GPSs.  2   70  128   377
-----
```

show ntp authentication-keys

Syntax

```
show ntp authentication-keys
```

Description

Shows the currently defined authentication keys.

Command context

Manager (#)

Authority

Administrators or local user group members with execution rights for this command.

Examples

```
switch# show ntp authentication-keys
-----
Auth key  Trusted  MD5 password
-----
  10             No      *****
  20             Yes      *****
-----
```

show ntp servers

Syntax

```
show ntp servers
```

Description

Shows all configured NTP servers.

Command context

Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Example

```
switch# show ntp servers
-----
      NTP SERVER KEYID MINPOLL MAXPOLL OPTION VER
-----
      192.0.1.18    -       5      10 iburst  3
      192.0.1.19    -       6      10  none  4
      192.0.1.20    -       6       8  burst  3 prefer
-----
```

show ntp statistics

Syntax

```
show ntp statistics
```

Description

Shows global NTP statistics. The following information is displayed:

- Rx-pkts: Total NTP packets received.
- Current Version Rx-pkts: Number of NTP packets that match the current NTP version.
- Old Version Rx-pkts: Number of NTP packets that match the previous NTP version.
- Error pkts: Packets dropped due to all other error reasons.
- Auth-failed pkts: Packets dropped due to authentication failure.
- Declined pkts: Packets denied access for any reason.
- Restricted pkts: Packets dropped due to NTP access control.
- Rate-limited pkts: Number of packets discarded due to rate limitation.
- KOD pkts: Number of Kiss of Death packets sent.

Command context

```
config
```

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Examples

```
switch(config)# show ntp statistics
                Rx-pkts 100
Current Version Rx-pkts 80
Old Version Rx-pkts 20
                Err-pkts 2
Auth-failed-pkts 1
Declined-pkts 0
Restricted-pkts 0
Rate-limited-pkts 0
KoD-pkts 0
```

show ntp status

Syntax

```
show ntp status
```

Description

Shows the status of NTP on the switch.

Command context

Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Displaying the status information when the switch is not synced to an NTP server:

```
switch# show ntp status
NTP is enabled.
NTP authentication is enabled.
NTP is using the default VRF for NTP server connections.

Wed Nov 23 23:29:10 PDT 2016
NTP uptime: 187 days, 1 hours, 37 minutes, 48 seconds

Not synchronized with an NTP server.
```

Displaying the status information when the switch is synced to an NTP server:

```
switch# show ntp status
NTP is enabled.
NTP authentication is enabled.
NTP is using the default VRF for NTP server connections.

Wed Nov 23 23:29:10 PDT 2016
NTP uptime: 187 days, 1 hours, 37 minutes, 48 seconds

Synchronized to NTP Server 17.253.2.253 at stratum 2.
```

```
Poll interval = 1024 seconds.  
Time accuracy is within 0.994 seconds  
Reference time: Thu Jan 28 2016 0:57:06.647 (UTC)
```

ping

Syntax

```
ping <IPv4-address | hostname> [ data-fill <pattern> | datagram-size <size> | interval  
<time> | repetitions <number> | timeout <time> | tos <number> | ip-option (include-timestamp  
| include-timestamp-and-address | record-route ) | vrf <vrfname>]
```

Description

Pings the specified IPv4 address or hostname with or without optional parameters.

Command context

Operator (>) or Manager (#)

Parameters

`<IPv4-ADDR>`

Selects the IPv4 address to ping.

`<HOSTNAME>`

Selects the hostname to ping. Range: 1-256 characters

`data-fill <PATTERN>`

Specifies the data pattern in hexadecimal digits to send. A maximum of 16 "pad" bytes can be specified to fill out the ICMP packet. Default: AB

`datagram-size <SIZE>`

Specifies the ping datagram size. Range: 0-65399, default: 100.

`interval <TIME>`

Specifies the interval between successive ping requests in seconds. Range: 1-60 seconds, default: 1 second.

`repetitions <NUMBER>`

Specifies the number of packets to send. Range: 1-10000 packets, default: Five packets.

`timeout <TIME>`

Specifies the ping timeout in seconds. Range: 1-60 seconds, default: 2 seconds.

`tos <NUMBER>`

Specifies the IP Type of Service to be used in Ping request. Range: 0-255

`ip-option [include-timestamp | include-timestamp-and-address | record-route]`

Specifies an IP option (record-route or timestamp option).

`include-timestamp`

Specifies the intermediate router time stamp.

`include-timestamp-and-address`

Specifies the intermediate router time stamp and IP address.

`record-route`

Specifies the intermediate router addresses.

`vrf <VRF-NAME>`

Specifies the virtual routing and forwarding (VRF) to use. When VRF option is not given, the default VRF is used.

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Examples

Pinging an IPv4 address:

```
switch# ping 10.0.0.0
PING 10.0.0.0 (10.0.0.0) 100(128) bytes of data.
108 bytes from 10.0.0.0: icmp_seq=1 ttl=64 time=0.035 ms
108 bytes from 10.0.0.0: icmp_seq=2 ttl=64 time=0.034 ms
108 bytes from 10.0.0.0: icmp_seq=3 ttl=64 time=0.034 ms
108 bytes from 10.0.0.0: icmp_seq=4 ttl=64 time=0.034 ms
108 bytes from 10.0.0.0: icmp_seq=5 ttl=64 time=0.033 ms

--- 10.0.0.0 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 3999ms
rtt min/avg/max/mdev = 0.033/0.034/0.035/0.000 ms
```

Pinging the localhost:

```
switch# ping localhost
PING localhost (127.0.0.1) 100(128) bytes of data.
108 bytes from localhost (127.0.0.1): icmp_seq=1 ttl=64 time=0.060 ms
108 bytes from localhost (127.0.0.1): icmp_seq=2 ttl=64 time=0.035 ms
108 bytes from localhost (127.0.0.1): icmp_seq=3 ttl=64 time=0.043 ms
108 bytes from localhost (127.0.0.1): icmp_seq=4 ttl=64 time=0.041 ms
108 bytes from localhost (127.0.0.1): icmp_seq=5 ttl=64 time=0.034 ms

--- localhost ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 3998ms
rtt min/avg/max/mdev = 0.034/0.042/0.060/0.011 ms
```

Pinging a server with a data pattern:

```
switch# ping 10.0.0.2 data-fill 1234123412341234acde123456789012
PATTERN: 0x1234123412341234acde123456789012
PING 10.0.0.2 (10.0.0.2) 100(128) bytes of data.
108 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=0.207 ms
108 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.187 ms
108 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.225 ms
108 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.197 ms
108 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=0.210 ms

--- 10.0.0.2 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 3999ms
rtt min/avg/max/mdev = 0.187/0.205/0.225/0.015 ms
```

Pinging a server with a datagram size:

```
switch# ping 10.0.0.0 datagram-size 200
PING 10.0.0.0 (10.0.0.0) 200(228) bytes of data.
```

```
208 bytes from 10.0.0.0: icmp_seq=1 ttl=64 time=0.202 ms
208 bytes from 10.0.0.0: icmp_seq=2 ttl=64 time=0.194 ms
208 bytes from 10.0.0.0: icmp_seq=3 ttl=64 time=0.201 ms
208 bytes from 10.0.0.0: icmp_seq=4 ttl=64 time=0.200 ms
208 bytes from 10.0.0.0: icmp_seq=5 ttl=64 time=0.186 ms

--- 10.0.0.0 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4000ms
rtt min/avg/max/mdev = 0.186/0.196/0.202/0.016 ms
```

Pinging a server with an interval specified:

```
switch# ping 9.0.0.2 interval 2
PING 9.0.0.2 (9.0.0.2) 100(128) bytes of data.
108 bytes from 9.0.0.2: icmp_seq=1 ttl=64 time=0.199 ms
108 bytes from 9.0.0.2: icmp_seq=2 ttl=64 time=0.192 ms
108 bytes from 9.0.0.2: icmp_seq=3 ttl=64 time=0.208 ms
108 bytes from 9.0.0.2: icmp_seq=4 ttl=64 time=0.182 ms
108 bytes from 9.0.0.2: icmp_seq=5 ttl=64 time=0.194 ms

--- 9.0.0.2 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 7999ms
rtt min/avg/max/mdev = 0.182/0.195/0.208/0.008 ms
```

Pinging a server with a specified number of packets to send:

```
switch# ping 9.0.0.2 repetitions 10
PING 9.0.0.2 (9.0.0.2) 100(128) bytes of data.
108 bytes from 9.0.0.2: icmp_seq=1 ttl=64 time=0.213 ms
108 bytes from 9.0.0.2: icmp_seq=2 ttl=64 time=0.204 ms
108 bytes from 9.0.0.2: icmp_seq=3 ttl=64 time=0.201 ms
108 bytes from 9.0.0.2: icmp_seq=4 ttl=64 time=0.184 ms
108 bytes from 9.0.0.2: icmp_seq=5 ttl=64 time=0.202 ms
108 bytes from 9.0.0.2: icmp_seq=6 ttl=64 time=0.184 ms
108 bytes from 9.0.0.2: icmp_seq=7 ttl=64 time=0.193 ms
108 bytes from 9.0.0.2: icmp_seq=8 ttl=64 time=0.196 ms
108 bytes from 9.0.0.2: icmp_seq=9 ttl=64 time=0.193 ms
108 bytes from 9.0.0.2: icmp_seq=10 ttl=64 time=0.200 ms

--- 9.0.0.2 ping statistics ---
10 packets transmitted, 10 received, 0% packet loss, time 8999ms
rtt min/avg/max/mdev = 0.184/0.197/0.213/0.008 ms
```

Pinging a server with a specified timeout:

```
switch# ping 9.0.0.2 timeout 3
PING 9.0.0.2 (9.0.0.2) 100(128) bytes of data.
108 bytes from 9.0.0.2: icmp_seq=1 ttl=64 time=0.175 ms
108 bytes from 9.0.0.2: icmp_seq=2 ttl=64 time=0.192 ms
108 bytes from 9.0.0.2: icmp_seq=3 ttl=64 time=0.190 ms
108 bytes from 9.0.0.2: icmp_seq=4 ttl=64 time=0.181 ms
108 bytes from 9.0.0.2: icmp_seq=5 ttl=64 time=0.197 ms

--- 9.0.0.2 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4000ms
rtt min/avg/max/mdev = 0.175/0.187/0.197/0.007 ms
```

Pinging a server with the specified IP Type of Service:

```

switch# ping 9.0.0.2 tos 2
PING 9.0.0.2 (9.0.0.2) 100(128) bytes of data.
108 bytes from 9.0.0.2: icmp_seq=1 ttl=64 time=0.033 ms
108 bytes from 9.0.0.2: icmp_seq=2 ttl=64 time=0.034 ms
108 bytes from 9.0.0.2: icmp_seq=3 ttl=64 time=0.031 ms
108 bytes from 9.0.0.2: icmp_seq=4 ttl=64 time=0.034 ms
108 bytes from 9.0.0.2: icmp_seq=5 ttl=64 time=0.031 ms

--- 9.0.0.2 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 3999ms
rtt min/avg/max/mdev = 0.031/0.032/0.034/0.006 ms

```

Pinging a server with the intermediate router time stamp:

```

switch# ping 9.0.0.2 ip-option include-timestamp
PING 9.0.0.2 (9.0.0.2) 100(168) bytes of data.
108 bytes from 9.0.0.2: icmp_seq=1 ttl=64 time=0.031 ms
TS:      59909005 absolute
        0
        0
        0

108 bytes from 9.0.0.2: icmp_seq=2 ttl=64 time=0.034 ms
TS:      59910005 absolute
        0
        0
        0

108 bytes from 9.0.0.2: icmp_seq=3 ttl=64 time=0.038 ms
TS:      59911005 absolute
        0
        0
        0

108 bytes from 9.0.0.2: icmp_seq=4 ttl=64 time=0.035 ms
TS:      59912005 absolute
        0
        0
        0

108 bytes from 9.0.0.2: icmp_seq=5 ttl=64 time=0.037 ms
TS:      59913005 absolute
        0
        0
        0

--- 9.0.0.2 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 3999ms
rtt min/avg/max/mdev = 0.031/0.035/0.038/0.002 ms

```

Pinging a server with the intermediate router time stamp and address:

```

switch# ping 9.0.0.2 ip-option include-timestamp-and-address
PING 9.0.0.2 (9.0.0.2) 100(168) bytes of data.
108 bytes from 9.0.0.2: icmp_seq=1 ttl=64 time=0.030 ms
TS:      9.0.0.2 60007355 absolute
        9.0.0.2 0
        9.0.0.2 0

```

```

9.0.0.2 0

108 bytes from 9.0.0.2: icmp_seq=2 ttl=64 time=0.037 ms
TS: 9.0.0.2 60008355 absolute
    9.0.0.2 0
    9.0.0.2 0
    9.0.0.2 0

108 bytes from 9.0.0.2: icmp_seq=3 ttl=64 time=0.037 ms
TS: 9.0.0.2 60009355 absolute
    9.0.0.2 0
    9.0.0.2 0
    9.0.0.2 0

108 bytes from 9.0.0.2: icmp_seq=4 ttl=64 time=0.038 ms
TS: 9.0.0.2 60010355 absolute
    9.0.0.2 0
    9.0.0.2 0
    9.0.0.2 0

108 bytes from 9.0.0.2: icmp_seq=5 ttl=64 time=0.039 ms
TS: 9.0.0.2 60011355 absolute
    9.0.0.2 0
    9.0.0.2 0
    9.0.0.2 0

--- 9.0.0.2 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 3999ms
rtt min/avg/max/mdev = 0.030/0.036/0.039/0.005 ms

```

Pinging a server with the intermediate router address:

```

switch# ping 9.0.0.2 ip-option record-route
PING 9.0.0.2 (9.0.0.2) 100(168) bytes of data.
108 bytes from 9.0.0.2: icmp_seq=1 ttl=64 time=0.034 ms
RR: 9.0.0.2
    9.0.0.2
    9.0.0.2
    9.0.0.2

108 bytes from 9.0.0.2: icmp_seq=2 ttl=64 time=0.038 ms (same route)
108 bytes from 9.0.0.2: icmp_seq=3 ttl=64 time=0.036 ms (same route)
108 bytes from 9.0.0.2: icmp_seq=4 ttl=64 time=0.037 ms (same route)
108 bytes from 9.0.0.2: icmp_seq=5 ttl=64 time=0.035 ms (same route)

--- 9.0.0.2 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 3999ms
rtt min/avg/max/mdev = 0.034/0.036/0.038/0.001 ms

```

ping6

Syntax

```
ping6 {<IPv6-ADDR> | <HOSTNAME>} [data-fill <PATTERN> | datagram-size <SIZE> |
interval <TIME> | repetitions <NUMBER> | timeout <TIME> | vrf <VRF-NAME>]
```

Description

Pings the specified IPv6 address or hostname with or without optional parameters.

Command context

Operator (>) or Manager (#)

Parameters

IPv6-ADDR

Selects the IPv6 address to ping.

HOSTNAME

Selects the hostname to ping. Range: 1-256 characters

data-fill <PATTERN>

Specifies the data pattern in hexadecimal digits to send. A maximum of 16 "pad" bytes can be specified to fill out the ICMP packet. Default: AB

datagram-size <SIZE>

Specifies the ping datagram size. Range: 0-65399, default: 100.

interval <TIME>

Specifies the interval between successive ping requests in seconds. Range: 1-60 seconds, default: 1 second.

repetitions <NUMBER>

Specifies the number of packets to send. Range: 1-10000 packets, default: Five packets.

timeout <TIME>

Specifies the ping timeout in seconds. Range: 1-60 seconds, default: 2 seconds.

vrf <VRF-NAME>

Specifies the virtual routing and forwarding (VRF) to use. When this option is not provided, the default VRF is used.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Pinging an IPv6 address:

```
switch# ping6 2020::2
PING 2020::2(2020::2) 100 data bytes
108 bytes from 2020::2: icmp_seq=1 ttl=64 time=0.386 ms
108 bytes from 2020::2: icmp_seq=2 ttl=64 time=0.235 ms
108 bytes from 2020::2: icmp_seq=3 ttl=64 time=0.249 ms
108 bytes from 2020::2: icmp_seq=4 ttl=64 time=0.240 ms
108 bytes from 2020::2: icmp_seq=5 ttl=64 time=0.252 ms

--- 2020::2 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4000ms
rtt min/avg/max/mdev = 0.235/0.272/0.386/0.059 ms
```

Pinging the localhost:

```
switch# ping6 localhost
PING localhost(localhost) 100 data bytes
108 bytes from localhost: icmp_seq=1 ttl=64 time=0.093 ms
108 bytes from localhost: icmp_seq=2 ttl=64 time=0.051 ms
108 bytes from localhost: icmp_seq=3 ttl=64 time=0.055 ms
108 bytes from localhost: icmp_seq=4 ttl=64 time=0.046 ms
108 bytes from localhost: icmp_seq=5 ttl=64 time=0.048 ms
```

```
--- localhost ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 3998ms
rtt min/avg/max/mdev = 0.046/0.058/0.093/0.019 ms
```

Pinging a server with a data pattern:

```
switch# ping6 2020::2 data-fill ab
PATTERN: 0xab
PING 2020::2(2020::2) 100 data bytes
108 bytes from 2020::2: icmp_seq=1 ttl=64 time=0.038 ms
108 bytes from 2020::2: icmp_seq=2 ttl=64 time=0.074 ms
108 bytes from 2020::2: icmp_seq=3 ttl=64 time=0.076 ms
108 bytes from 2020::2: icmp_seq=4 ttl=64 time=0.075 ms
108 bytes from 2020::2: icmp_seq=5 ttl=64 time=0.077 ms

--- 2020::2 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 3999ms
rtt min/avg/max/mdev = 0.038/0.068/0.077/0.015 ms
```

Pinging a server with a datagram size:

```
switch# ping6 2020::2 datagram-size 200
PING 2020::2(2020::2) 200 data bytes
208 bytes from 2020::2: icmp_seq=1 ttl=64 time=0.037 ms
208 bytes from 2020::2: icmp_seq=2 ttl=64 time=0.076 ms
208 bytes from 2020::2: icmp_seq=3 ttl=64 time=0.076 ms
208 bytes from 2020::2: icmp_seq=4 ttl=64 time=0.077 ms
208 bytes from 2020::2: icmp_seq=5 ttl=64 time=0.066 ms

--- 2020::2 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 3999ms
rtt min/avg/max/mdev = 0.037/0.066/0.077/0.016 ms
```

Pinging a server with an interval specified:

```
switch# ping6 2020::2 interval 5
PING 2020::2(2020::2) 100 data bytes
108 bytes from 2020::2: icmp_seq=1 ttl=64 time=0.043 ms
108 bytes from 2020::2: icmp_seq=2 ttl=64 time=0.075 ms
108 bytes from 2020::2: icmp_seq=3 ttl=64 time=0.074 ms
108 bytes from 2020::2: icmp_seq=4 ttl=64 time=0.075 ms
108 bytes from 2020::2: icmp_seq=5 ttl=64 time=0.075 ms

--- 2020::2 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 19999ms
rtt min/avg/max/mdev = 0.043/0.068/0.075/0.014 ms
```

Pinging a server with a specified number of packets to send:

```
switch# ping6 2020::2 repetitions 6
PING 2020::2(2020::2) 100 data bytes
108 bytes from 2020::2: icmp_seq=1 ttl=64 time=0.039 ms
108 bytes from 2020::2: icmp_seq=2 ttl=64 time=0.070 ms
108 bytes from 2020::2: icmp_seq=3 ttl=64 time=0.076 ms
108 bytes from 2020::2: icmp_seq=4 ttl=64 time=0.076 ms
```

```
108 bytes from 2020::2: icmp_seq=5 ttl=64 time=0.071 ms
108 bytes from 2020::2: icmp_seq=6 ttl=64 time=0.078 ms

--- 2020::2 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 4999ms
rtt min/avg/max/mdev = 0.039/0.068/0.078/0.015 ms
```



On the 6100 Switch Series, only the vrf named `default` is available. Replace any references to the `mgmt` or other VRFs with `default`.

crypto pki application

Syntax

```
crypto pki application <APP-NAME> certificate <CERT-NAME>
no crypto pki application <APP-NAME> certificate
```

Description

Associates a leaf certificate with a feature (application) on the switch. By default, all features are associated with the default, self-signed certificate `local-cert`. This certificate is created by the switch the first time it starts.

The `no` form of this command associates the specified feature with the default certificate.

Command context

`config`

Parameters

`<APP-NAME>`

Specifies the name of a feature on the switch:

- `est-client`: EST client
- `hsc`: Hardware switch controller.
- `https-server`: HTTPS server.
- `syslog-client`: Syslog client.

`<CERT-NAME>`

Specifies the name of an installed leaf certificate.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Associating the EST client with leaf certificate **leaf-cert1**:

```
switch(config)# crypto pki application est-client certificate leaf-cert1
```

Associating the syslog client with leaf certificate **leaf-cert**:


```
switch(config)# crypto pki application syslog-client certificate leaf-cert
```



`syslog-client` communicates with syslog server over TLS.

You can associate a certificate with the `syslog-client` application by enrolling the certificate manually or through EST.

Setting the syslog client to use the default certificate:

```
switch(config)# no crypto pki application syslog-client certificate
```

Associating the HTTPS server with leaf certificate **leaf-cert2**:

```
switch(config)# crypto pki application https-server certificate leaf-cert2
```

crypto pki certificate

Syntax

```
crypto pki certificate <CERT-NAME>  
no crypto pki certificate <CERT-NAME>
```

Description

Creates a leaf certificate and changes to its context `config-cert-<CERT-NAME>`. If the specified leaf certificate exists, this command changes to its context.

The first time the switch starts it creates a self-signed, default leaf certificate called `local-cert`. This certificate is used by any switch application that does not have an associated leaf certificate.

The `no` form of this command deletes the specified leaf certificate. The default leaf certificate `local-cert` cannot be deleted.

Command context

`config`

Parameters

<CERT-NAME>

Specifies the name of a leaf certificate. Range: 1 to 32 alphanumeric characters (excluding ").

Authority

Administrators or local user group members with execution rights for this command.

Examples

Creating leaf certificate **leaf-cert**:

```
switch(config)# crypto pki certificate leaf-cert  
switch(config-cert-leaf-cert)#
```

Deleting leaf certificate **leaf-cert**:

```
switch(config)# no crypto pki certificate leaf-cert
The leaf certificate has associated applications. Deleting the certificate
will make the applications use the default certificate local-cert.
Continue (y/n)? y
switch(config)#
```

crypto pki ta-profile

Syntax

```
crypto pki ta-profile <TA-NAME>
no crypto pki ta-profile <TA-NAME>
```

Description

Creates a trust anchor (TA) profile and changes to the `config-ta-<TA-NAME>` context for the profile. Each TA profile stores the certificate for a trusted CA. Up to 64 profiles can be defined.

If the specified TA profile exists, this command changes to the `config-ta-<TA-NAME>` context for the profile.

The `no` form of this command removes the specified TA profile.



When creating a new profile, if you exit the `config-ta-<TA-NAME>` context without importing the TA certificate, the profile is discarded.

Command context

config

Parameters

<TA-NAME>

Specifies the TA profile name. Range: 1 to 48 alphanumeric characters excluding ".".



The TA profile name cannot end with `est-ta<nn>` where <nn> is 00 to 99. For example, `company-trust-anchor-est-ta01` is not allowed. This TA profile name suffix is reserved for TA profiles that are created for CA certificates from EST servers.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Creating the TA profile **root-cert**:

```
switch(config)# crypto pki ta-profile root-cert
switch(config-ta-root-cert)#
```

Removing TA profile **root-cert**:

```
switch(config)# no crypto pki ta-profile root-cert
```

enroll self-signed

Syntax

```
enroll self-signed
```

Description

Generates a key pair and generates a self-signed certificate with it.

The subject fields and key type of the current leaf certificate must be defined before running this command. If not, you are prompted to fill in the subject fields, and the key type is set to `RSA 2048`.

Command context

```
config-cert-<CERT-NAME>
```

Authority

Administrators or local user group members with execution rights for this command.

Example

Enrolling the leaf certificate **leaf-cert**:

```
switch(config-cert-leaf-cert)# enroll self-signed
You are enrolling a certificate with the following attributes:
Subject: C=US, ST=CA, L=Rocklin, OU=Site, O=Comp,
        CN=Leaf01
Key Type: RSA (2048)

Continue (y/n)? y
Self-signed certificate is created and enrolled successfully.

switch(config-cert-leaf-cert)#
```

enroll terminal

Syntax

```
enroll terminal
```

Description

Generates a key pair and certificate signing request (CSR) for the current leaf certificate. Use the CSR to obtain a signed certificate from a certificate authority (CA), and then import the certificate onto the switch with the command `import terminal`.

The key type, and the certificate common name in the subject fields of the current leaf certificate must be completed before running this command.

Command context

```
config-cert-<CERT-NAME>
```

Authority

Administrators or local user group members with execution rights for this command.

Example

Enrolling the leaf certificate **leaf-cert**:

```

switch(config-cert-leaf-cert)# enroll terminal
You are enrolling a certificate with the following attributes:
Subject: C=US, ST=CA, L=Rocklin, OU=Site, O=Comp,
        CN=Leaf01
Key Type: RSA (2048)

Continue (y/n)? y

-----BEGIN CERTIFICATE REQUEST-----
MIIBozCCAQwCAQAwYzEVMBMGAlUEAxMMcG9kMDEtODQwMC0xMQ4wDAYDVQQLEwV
nViYTEMMAoGAlUEChMDSFBFMRIwEAYDVQQHEw1Sb3Nldm1sbGUxCzAJBgNVBAGT
NBMQswCQYDVQQGEwJVUzCBnzANBgkqhkiG9w0BAQEFAAOBjQAwgYkCgYEAtKcLS
...
GBAJ4L3lFFFwBEL+KAKpOGjZcVmw1BMqSKFtOFNF9nzmUmONmU3SKy6dzQ+6ynR
7Au22mf3lWDxzrtCC/dj5RtWJeJekxp2LCIK/3eRXUwbYveQDKcxH7j9ZB+BAp2
ace+2tA68F2vlgRCQ/hcQH0YmNuaq4Ne3w0dhm7H1Urx
-----END CERTIFICATE REQUEST-----
switch(config-cert-leaf-cert)#

```

import (CA-signed leaf certificate)

Syntax

```

import terminal ta-profile <TA-NAME> [password <PW>]
import <REMOTE-URL> ta-profile <TA-NAME> [password <PW>] [vrf <VRF-NAME>]
import <STORAGE-URL> ta-profile <TA-NAME> [password <PW>]

```

Description

Imports a CA-signed leaf certificate and then validates the certificate against the specified TA profile. If the imported data includes a private key, the private key must match the leaf certificate being imported. If the imported data does not include a private key, the certificate must match a CSR that was previously generated with the command `enroll terminal` and must be signed by the CA whose root certificate is installed in the specified TA profile. The TA profile must exist and have a TA certificate configured.

Parameters

`terminal`

Import the certificate by pasting PEM-format data at the console. Upon execution, the `config-cert-import` context is entered for certificate pasting. To complete certificate data entry press Control-D in your terminal program. Alternatively, the pasted certificate data can include at its end the delimiter `END_OF_CERTIFICATE` (after the `-----END CERTIFICATE-----` line), making entry of Control-D unnecessary.

`ta-profile <TA-NAME>`

Specifies the TA profile name. Range: 1 to 48 alphanumeric characters excluding ".".

`<REMOTE-URL>`

Specifies a certificate data file on a remote TFTP or SFTP server. The URL syntax is:

`{tftp:// | sftp://<USER>@} {<IP> | <HOST>} [:<PORT>] [:blocksize=<SIZE>]/<FILE>`

`<STORAGE-URL>`

Available on switch families that provide USB device file import capability, specifies a certificate data file on a USB storage device inserted in the switch USB port. The URL syntax is:

`usb:/<FILE>`

`password <PW>`

Specifies the plaintext password used to decrypt the private key in the imported certificate data. When this parameter is omitted, the password is prompted for as required. Range: 1 to 32 alphanumeric characters.

vrf <VRF-NAME>

Specifies the name of the VRF to use for the remote URL file transfer. The default is `mgmt`.

Command context

config-cert-<CERT-NAME>

Authority

Administrators or local user group members with execution rights for this command.

Usage

- The imported data must include all the intermediate CA certificates in the certificate chain leading to the certificate imported into the specified TA profile.
- This command cannot be used with the default certificate `local-cert`.
- The PEM data format is supported for all import sources. The PKCS#12 data format is supported for <REMOTE-URL> and <STORAGE-URL>.
- The PEM data must be delimited with these lines for the certificate data:

```
-----BEGIN CERTIFICATE-----  
-----END CERTIFICATE-----
```

And the PEM data must be delimited with either of these line pairs for the private key data:

```
-----BEGIN PRIVATE KEY-----  
-----END PRIVATE KEY-----
```

```
-----BEGIN ENCRYPTED PRIVATE KEY-----  
-----END ENCRYPTED PRIVATE KEY-----
```

Examples

Importing a leaf certificate from the console:

```
switch(config)# crypto pki certificate leaf-cert  
switch(config-cert-leaf-cert1)# import terminal ta-profile root-cert  
Paste the certificate in PEM format below, then hit enter and ctrl-D:  
switch(config-cert-import)# -----BEGIN CERTIFICATE-----  
switch(config-cert-import)# MIIFRDCCAyYgAwIBAgQP8nS2Vp15u0xXMdkDJzANBgkqhkiG9w0Bv  
switch(config-cert-import)# MQswCQYDVQGEWJVUEOMAwGA1UCgwFXJlYmDAGNBAMM1Jvb3QgQ0Ew  
switch(config-cert-import)# HhcNMTkNDEwMjIwNT1WhecjIwMT0MjwNE1WjzQswQDVQGEWJVUzEL  
...  
switch(config-cert-import)# 1fIYZYGQyla0AwFuPTTxBXHYwRxTPbUYU5umJfRPmE4VY8S9DQgcr  
switch(config-cert-import)# 1NGNm3NG03GqPScs/TF9bVyFA5BOS5lmmkfRYK8D/kMTfRreSdxis  
switch(config-cert-import)# YQ1ul1NqShps=  
switch(config-cert-import)# -----END CERTIFICATE-----  
switch(config-cert-import)# -----BEGIN ENCRYPTED PRIVATE KEY-----  
switch(config-cert-import)# MIIFDjBABgkqhkiG9wBBQ0wMzAbBgqkw0QwwDQIpJMN7sVGwCaggA  
switch(config-cert-import)# MBQGCCqGSIB3DQMHAit+2qadNAASCgLYJ4Am3EfhH5p51Ggr86VqS  
switch(config-cert-import)# IJ6L/UhEtH523nUkdV6gvAgoYaD83PswToAGv5VS8OMFTPttrn5/K  
...  
switch(config-cert-import)# OgSecqZsG6arbx0ESaYBir1c/6rPspcjbx283iD1MWOpesS2aEmOX  
switch(config-cert-import)# iKnXnUMpVPfLc74ty2S41DtH0X9gf6aaljStg+7cND9XfGtjaV2+/  
switch(config-cert-import)# cb4=  
switch(config-cert-import)# -----END ENCRYPTED PRIVATE KEY-----  
switch(config-cert-import)#  
Enter import password: *****  
Leaf certificate is validated with root-cert and imported successfully.  
switch(config-cert-leaf-cert)#
```

Importing a leaf certificate from a remote file:

```

switch(config)# crypto pki certificate leaf-cert2
switch(config-cert-leaf-cert2)# import tftp://1.1.1.2/c2.p12 ta-profile root-cert
% Total      % Received % Xferd  Average Speed   Time    Time       Time  Current
             %                   Dload  Upload    Total     Spent    Left     Speed
100  3722  100  3722    0      0   391k      0  --:--:-- --:--:-- --:--:--   391k
100  3722  100  3722    0      0   376k      0  --:--:-- --:--:-- --:--:--   376k
Enter import password: *****
Leaf certificate is validated with root-cert and imported successfully.
switch(config-cert-leaf-cert2)#

```

import (self-signed leaf certificate)

Syntax

```

import terminal self-signed [password <PW>]
import <REMOTE-URL> self-signed [password <PW>] [vrf <VRF-NAME>]
import <STORAGE-URL> self-signed [password <PW>]

```

Description

Imports a self-signed leaf certificate including its matching private key.

Parameters

terminal

Import the certificate by pasting PEM-format data at the console. Upon execution, the `config-cert-import` context is entered for certificate pasting. To complete certificate data entry press Control-D in your terminal program. Alternatively, the pasted certificate data can include at its end the delimiter `END_OF_CERTIFICATE` (after the `-----END CERTIFICATE-----` line), making entry of Control-D unnecessary.

<REMOTE-URL>
Specifies a certificate data file on a remote TFTP or SFTP server. The URL syntax is:
{tftp:// | sftp://<USER>@} {<IP>|<HOST>} [:<PORT>] [;blocksize=<SIZE>]/<FILE>
<STORAGE-URL>

Available on switch families that provide USB device file import capability, specifies a certificate data file on a USB storage device inserted in the switch USB port. The URL syntax is:

usb:/<FILE>
password <PW>

Specifies the plaintext password used to decrypt the private key in the imported certificate data. When this parameter is omitted, the password is prompted for as required. Range: 1 to 32 alphanumeric characters.

vrf <VRF-NAME>

Specifies the name of the VRF to use for the remote URL file transfer. The default is `mgmt`.

Command context

config-cert-<CERT-NAME>

Authority

Administrators or local user group members with execution rights for this command.

Usage

- This command cannot be used with the default certificate `local-cert`.
- The PEM data format is supported for all import sources. The PKCS#12 data format is supported for <REMOTE-URL> and <STORAGE-URL>.

- The PEM data must be delimited with these lines for the certificate data:

```
-----BEGIN CERTIFICATE-----
-----END CERTIFICATE-----
```

And the PEM data must be delimited with either of these line pairs for the private key data:

```
-----BEGIN PRIVATE KEY-----
-----END PRIVATE KEY-----
```

```
-----BEGIN ENCRYPTED PRIVATE KEY-----
-----END ENCRYPTED PRIVATE KEY-----
```

Example

Importing a self-signed leaf certificate from the console:

```
switch(config)# crypto pki certificate ss-leaf-cert
switch(config-cert-ss-leaf-cert)# import terminal self-signed
Paste the certificate in PEM format below, then hit enter and ctrl-D:
switch(config-cert-import)# -----BEGIN CERTIFICATE-----
switch(config-cert-import)# MIID2TCCAsGgAwIBAgIJAKcrqokm6p9GMA0GCSqGSIb3DQEBCwUAM
switch(config-cert-import)# tDCCA5ygAwIBAgICEAEwDQYJKoZIhvcNAQELBQAwYgxCzABAYTA1
switch(config-cert-import)# VQQGEwJVUzELMAkGA1UECAwCQ0ExDTALBgNVBACMBFJvc2UxDDAKB
...
switch(config-cert-import)# +fWQLxhp+jKJGZGOZz/FENt2uSfZHxlXiu8n3g+EqqExenY1pBRJr
switch(config-cert-import)# VuEEoNb/YfkPXHHva4Zfx223q+f694wlVsHkENSzqr2goHpa2fOzq
switch(config-cert-import)# alewwdmVqCES+x8bvfh3C/6IB6ePkEsnMlHNTeM=
switch(config-cert-import)# -----END CERTIFICATE-----
switch(config-cert-import)# -----BEGIN ENCRYPTED PRIVATE KEY-----
switch(config-cert-import)# MIIFDjBABgkqhkiG9w0BBQ0wMzAbBgkqhkiG9w0BBQwwDgQIt8Ni3
switch(config-cert-import)# MBQGCCqGSib3DQMHBAlBHrejkcdpdASCBMjVxrrYYPnt3Vlabr9k8
switch(config-cert-import)# 5GE0U99awh9ys4360WR95xOFGThvjkTyRWG511nGwVeLZs/7TPXWI
...
switch(config-cert-import)# hzc5ZT/w2F08icRI5mFbGoTAAw9IIWMOXGweaWQJdyKGrhg89GrnV
switch(config-cert-import)# M2UuP/tYuuO328QcenKZEJmZKCbX78oFRR+pgma4oeMaFTIyXE6Pr
switch(config-cert-import)# GAdCK8tkDiJ9DKbqdm5W0/nTJfqwUQlfl127dNrBAodsHdrrw3UR99H
switch(config-cert-import)# SPo=
switch(config-cert-import)# -----END ENCRYPTED PRIVATE KEY-----
switch(config-cert-import)#
Enter import password: *****
Leaf certificate is validated as self-signed certificate and imported successfully.
switch(config-cert-ss-leaf-cert)#
```

Importing a leaf certificate from a remote file:

```
switch(config)# crypto pki certificate ss-leaf-cert2
switch(config-cert-ss-leaf-cert2)# import tftp://1.1.1.2/ss2.p12 self-signed
  % Total      % Received % Xferd  Average Speed   Time    Time       Time  Current
                                 Dload  Upload   Total     Spent    Left     Speed
100 3230 100 3230    0     0  875k      0  --:--:--  --:--:--  --:--:--  875k
100 3230 100 3230    0     0  831k      0  --:--:--  --:--:--  --:--:--  831k
Enter import password: *****
Leaf certificate is validated as self-signed certificate and imported successfully.
switch(config-cert-ss-leaf-cert2)#
```

key-type

Syntax

```
key-type {rsa [key-size <K-SIZE>] | ecdsa [curve-size <C-SIZE>]}
```

Description

Sets the key type and key size for the current leaf certificate. The key type of the default certificate `local-cert` cannot be changed.

Command context

`config-cert-<CERT-NAME>`

Parameters

`rsa`

Specifies the key type as RSA.

`key-size <K-SIZE>`

Specifies the RSA key size in bits. Supported values: 2048, 3072, 4096. Default: 2048

`ecdsa`

Specifies the key type as ECDSA.

`curve-size <C-SIZE>`

Specifies the ECDSA elliptic curve size in bits. Supported values: 256, 348, 521. Default: 256

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting RSA encryption on the leaf certificate **leaf-cert**:

```
switch(config)# crypto pki certificate leaf-cert
switch(config-cert-leaf-cert)# key-type rsa key-size 3072
```

Setting ECDSA encryption on the leaf certificate **leaf-cert**:

```
switch(config)# crypto pki certificate leaf-cert
switch(config-cert-leaf-cert)# key-type ecdsa curve-size 521
```

ocsp disable-nonce

Syntax

`ocsp disable-nonce`
`no ocsp disable-nonce`

Description

Configures exclusion of the nonce from OCSF requests. A nonce is a unique identifier that an OCSF client inserts in an OCSF request and expects the OCSF responder to include it in the corresponding OCSF response. The nonce mechanism helps prevent replay attacks in which a malicious player attempts to masquerade as the OCSF responder. Although the nonce is included by default, it can be excluded. Some OCSF responders choose to not support the use of the nonce due to performance considerations.

The `no` form of this command re-enables nonce inclusion in OCSF requests.

Command context

`config-ta-<TA-NAME>`

Authority

Administrators or local user group members with execution rights for this command.

Examples

Disable inclusion of the nonce in OCSF requests for TA profile `root-cert`:

```
switch(config)# crypto pki ta-profile root-cert
switch(config-ta-root-cert)# ocsf disable-nonce
```

Enable inclusion of the nonce in OCSF requests for TA profile `root-cert`:

```
switch(config)# crypto pki ta-profile root-cert
switch(config-ta-root-cert)# no ocsf disable-nonce
```

ocsf enforcement-level

Syntax

```
ocsf enforcement-level {strict | optional}
no enforcement-level
```

Description

Sets either strict or reduced enforcement of the OCSF check of certificates. Strict enforcement is enabled by default.

The `no` form of this command resets enforcement to its default of `strict`.

Command context

`config-ta-<TA-NAME>`

Parameters

`strict`

Sets strict OCSF checking of certificates. The certificate is accepted only if all possible checking (including validation failures, software system errors, configuration errors, transactional errors) is successful.

`optional`

Sets reduced OCSF checking of certificates. The certificate is accepted unless one or more of these validation errors occur:

- Response signature invalid.
- Nonce in response mismatch.
- Certificate revoked, but only when revocation checking is possible. If revocation check is not possible, the certificate is still accepted if there are no other validation errors.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting reduced OCSF checking of certificates:

```
switch(config)# crypto pki ta-profile root-cert  
switch(config-ta-root-cert)# ocsp enforcement-level optional
```

Setting strict OCSP checking of certificates:

```
switch(config)# crypto pki ta-profile root-cert  
switch(config-ta-root-cert)# ocsp enforcement-level strict
```

ocsp url

Syntax

```
ocsp url {primary | secondary} <URL>  
no ocsp url {primary | secondary}
```

Description

Configures the OCSP responder URLs that the current TA profile uses to verify the revocation status of an X.509 digital certificate. These URLs override the OCSP responder URL contained within the peer certificate being verified (as well as URLs defined in any intermediate CAs in the chain of trust).

If no OCSP responder URLs are defined for a TA profile (default setting), then the OCSP responder URL in the peer certificate is used for revocation status checking. (The OCSP responder URL is contained in a certificate's Authority Information Access field, which is an X.509 v3 certificate extension.)

The **no** form of this command deletes the specified OCSP responder URL (primary or secondary) from the current TA profile.

Command context

config-ta-<TA-NAME>

Parameters

{primary | secondary} <URL>

Specify the HTTP URL of the primary or secondary OCSP responder using either a fully qualified domain name or IPv4 address.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Defining the primary OCSP URL for the TA profile **root-cert**:

```
switch(config)# crypto pki ta-profile root-cert  
switch(config-ta-root-cert)# revocation-check ocsp  
switch(config-ta-root-cert)# ocsp url primary http://ocsp-server.site.com
```

Removing the primary OCSP URL from the TA profile **root-cert**:

```
switch(config)# crypto pki ta-profile oot-cert  
switch(config-ta-root-cert)# revocation-check ocsp  
switch(config-ta-root-cert)# no ocsp url primary
```

ocsp vrf

Syntax

```
ocsp vrf <VRF-NAME>  
no ocsp vrf
```

Description

Sets the VRF that the switch uses to communicate with OCSP responders for OCSP checking. VRF mgmt is used by default.

The `no` form of this command resets the VRF to its default `mgmt`.

Command context

```
config-ta-<TA-NAME>
```

Parameters

<VRF-NAME>

Specifies the name of the VRF the switch uses to communicate with OCSP responders. Default: `mgmt`.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Reverting the OCSP responder VRF to its default:

```
switch(config)# crypto pki ta-profile root-cert  
switch(config-ta-root-cert)# no ocsp vrf
```

revocation-check ocsp

Syntax

```
revocation-check ocsp  
no revocation-check
```

Description

Enables certificate revocation checking for the current profile using the online certificate status protocol (OCSP).

The `no` form of this command disables certificate revocation checking for the current profile.

Command context

```
config-ta-<TA-NAME>
```

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling revocation checking for the TA profile **root-cert**:

```
switch(config)# crypto pki ta-profile root-cert  
switch(config-ta-root-cert)# revocation-check oosp
```

Disabling revocation checking for the TA profile **root-cert**:

```
switch(config)# crypto pki ta-profile root-cert  
switch(config-ta-root-cert)# no revocation-check
```

show crypto pki application

Syntax

```
show crypto pki application
```

Description

Shows certificate information for all features (applications) using leaf certificates that are managed by PKI.

Command context

Manager (#)

Authority

Administrators or local user group members with execution rights for this command.

Examples

Showing certificate information for all features (applications) using leaf certificates:

```
switch# show crypto pki application1
```

Associated Applications	Certificate Name	Cert Status
https-server		not configured, using local-cert
syslog-client	local-cert	valid
hsc	xhscert	invalid, using local-cert

show crypto pki certificate

Syntax

```
show crypto pki certificate [<CERT-NAME> [plaintext | pem]]
```

Description

Shows a list of all configured leaf certificates, or detailed information for a specific leaf certificate.

Possible values for Cert Status are: CSR pending, expired, expires soon, installed, malformed, not yet known.

Possible values for EST Status are: enroll failed, enroll pending, enroll retrying, enroll success, n/a (certificate is not EST-enrolled), reenroll failed, reenroll pending, reenroll retrying.

Command context

Manager (#)

Parameters

<CERT-NAME>

Specifies the leaf certificate name. Range: 1 to 32 alphanumeric characters excluding ".
plaintext

Shows certificate information in plain text.

pem

Shows certificate information in PEM format.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Showing a list of all configured leaf certificates:

```
switch# show crypto pki certificate
Certificate Name      Cert Status      EST Status      Associated Applications
-----
device-identity      installed        n/a             none
pod01-99-1           installed        n/a             https-server, est-client
syslog-1             CSR pending      enroll retrying  syslog-client
leaf-cert1           installed        enroll success   none
leaf-cert2           CSR pending      enroll failed    none
```

Showing detailed information (in plaintext format) for leaf certificate pod01-99-1:

```
switch# show crypto pki certificate pod01-99-1 plaintext

Certificate Name: pod01-99-1
Associated Applications:
  https-server, est-client
Certificate Status: installed
EST Status: n/a
Certificate Type: regular
Intermediates:
  Subject: C = US, ST = CA, O = Company, OU = Lab-IT, CN = DeviceCA
  Issuer: C = US, ST = CA, O = Company, OU = Lab-IT, CN = Lab-CA
  Serial Number: 0x02
  Subject: C = US, ST = CA, O = Company, OU = Lab-IT, CN = Lab-CA
  Issuer: C = US, ST = CA, O = Company, OU = Lab-IT, CN = Lab-Root
  Serial Number: 0x01
Certificate:
  Data:
    Version: 1 (0x0)
    Serial Number: 14529416756121781768 (0xc9a2db8f3e3f4608)
  Signature Algorithm: sha256WithRSAEncryption
  Issuer: C=US, ST=CA, OU=Lab-IT, O=Company, CN=DeviceCA
  Validity
    Not Before: Jan 12 23:36:57 2018 GMT
    Not After : Nov 1 23:36:57 2020 GMT
  Subject: C=US, ST=CA, OU=Lab-IT, O=Company, CN=pod01-99-1
  Subject Public Key Info:
    Public Key Algorithm: rsaEncryption
    Public-Key: (2048 bit)
    Modulus:

      00:a0:cd:ef:1b:f9:b8:bd:39:fc:7a:0e:00:17:ff:
      2b:72:d8:4e:d4:df:49:36:ca:3a:f9:05:05:d7:e3:
```

```

d1:97:29:71:e6:33:b8:bb:8e:f0:ee:a6:e4:4a:f8:
...
fe:dd:d9:a0:af:59:47:25:b4:34:06:af:03:1d:33:
30:c3:85:fe:5c:e7:19:7f:ff:3a:b2:21:b8:e8:ed:
83:09
Exponent: 65537 (0x10001)
Signature Algorithm: sha256WithRSAEncryption
39:f6:03:86:03:d9:05:61:39:25:5f:0d:75:cc:05:ae:04:7e:
4c:a3:13:0b:f0:1e:af:68:0e:40:9f:ed:48:b6:5e:56:8c:53:
46:5b:c9:a4:e0:b0:bc:31:4b:a7:5d:0a:ed:7c:9c:f6:bf:1e:
...
39:f5:26:58:68:e2:13:ec:94:ac:60:8e:4b:b0:ba:45:cf:d6:
6a:4b:9f:7d:ae:3f:e5:2e:81:fe:ac:b3:65:44:35:47:a5:2f:
89:e7:58:a0

```

Showing detailed information (in PEM format) for leaf certificate `leaf-cert1` with a status of CSR pending:

```

switch# show crypto pki certificate leaf-cert1 pem

Certificate Name: leaf-cert1

Associated Applications:
  syslog-client
Certificate Status: CSR pending
EST Status: enroll retrying
Certificate Type: regular
-----BEGIN CERTIFICATE REQUEST-----

MIICtTCCA Z0CAQAwcDEWMBQGA1UEAxMNc3lzbG9nLTg0MBYGA1UECxMPQ
XJlYmEtUm9zZXZpbGx1MQ4wDAYDVQQKEyTESMBAGA1EBxMJUm9zZXZpbG
x1MQswCQYDVQQIEWJDQTELMAGA1UEBhMCVVMwggeiMSIb3DQEBAQUAA4I
...
cw2ytN6Idgh81k59x6DH7V/eORaKd5lq+oO7nkr6+QBf5L3f5Kb+TOFio
lei+EdCHMxxc07MK0n3dkziSW25HFUGsyEXVMK+BiD3zbKDoUe6XVhVqI
mamXyghigLYDcbsn6WVw==
-----END CERTIFICATE REQUEST-----

```

show crypto pki ta-profile

Syntax

```
show crypto pki ta-profile [<TA-NAME>]
```

Description

Shows a list of all configured TA profiles, or detailed information for a specific profile.



This command shows information for both directly-configured TA profiles and TA profiles that were dynamically downloaded from EST servers.

Command context

Manager (#)

Parameters

<TA-NAME>

Specifies the TA profile name. Range: 1 to 48 alphanumeric characters excluding ".".

Authority

Administrators or local user group members with execution rights for this command.

Examples

Showing a list of all configured TA profiles:

```
switch# show crypto pki ta-profile
```

Profile Name	TA Certificate	Revocation Check
BASE_CA	Installed,valid	disabled
BASE02_CA	Installed,expired	disabled
root-cert	Installed,valid	OCSP
ROOT-A_CA	Not Installed	OCSP
EST-Service1	Installed,valid	None
EST-Service2	Installed,valid	None

Showing detailed information for TA profile **root-cert**:

```
switch# show crypto pki ta-profile root-cert
```

TA Profile Name : root-cert
Revocation Check : OCSP
OCSP Primary URL : http://ocsp1.domain.com
OCSP Secondary URL : Not Configured
OCSP Disable-nonce : false
OCSP Enforcement Level: strict
OCSP VRF : mgmt
TA Certificate: Installed and valid
Version: 3 (0x2)
Serial Number:
74:e6:6d:22:3f:52:cc:94:43:41:ab:66:a8:8d:47:b1
Signature Algorithm: sha1withRSAEncryption
Issuer: OU=DeviceTrust, OU=Operations, O=Site, C=US,
CN=Site Trusted Computing Root CA 1.0
Validity
Not Before: Sep 14 03:12:06 2007 GMT
Not After : Sep 14 03:21:14 2032 GMT
Subject: OU=DeviceTrust, OU=Operations, O=Site, C=US,
CN=Site Trusted Computing Root CA 1.0
Subject Public Key Info:
Public Key Algorithm: rsaEncryption
RSA Public Key: (2048 bit)
Modulus (2048 bit):
30:0d:06:09:2a:86:48:86:f7:0d:01:01:01:05:33:
03:82:01:0f:00:30:82:01:3a:02:82:01:01:00:ac:
3d:60:3a:2e:ca:a4:34:db:5c:3b:6b:07:df:73:62:
...
20:c8:df:63:14:5a:e8:d3:ea:83:d8:47:a3:b5:2e:
bb:64:51:f0:be:13:b6:91:e4:32:45:58:5e:1f:0d:
02:03:01:00:01
Exponent: 65537 (0x10001)
X509v3 extensions:
X509v3 Key Usage:
Digital Signature, Certificate Signing, CRL Signing
X509v3 Basic Constraints:
CA:TRUE, pathlen:4
X509v3 Subject Key Identifier:
eb:d7:ec:db:8a:cb:f2:51:d5:06:e1:42:7b:39:a7:d0:1e:31:6e:bf

```
Signature Algorithm: sha1withRSAEncryption
1c:90:f3:a4:f0:0d:e2:e3:e9:ae:01:e1:7d:a7:13:e2:cc:0b:
17:31:26:92:a2:5d:1d:19:60:54:03:13:9b:e1:73:6c:e4:b3:
01:4f:4e:ae:61:bd:ae:b6:12:d3:ab:08:ae:8c:47:92:d7:0d:
...
ca:cf:11:78:55:6d:06:49:fa:d4:8d:f3:ef:7f:79:38:35:5d:
16:5a:57:7f:a8:dc:b0:f8:a2:04:0d:17:0b:bb:58:32:30:e0:
2d:a8:37:a2
```

ta-certificate

Syntax

```
ta-certificate { [import [terminal]] | import {<REMOTE-URL> | <STORAGE-URL>} }
```

Description

Imports a CA certificate for use in the current TA profile. The certificate must be in PEM format. The PEM data must be delimited with these lines:

```
-----BEGIN CERTIFICATE-----
-----END CERTIFICATE-----
```



Only the first certificate in the PEM data is imported. Any additional certificates are ignored.

Command context

```
config-ta-<TA-NAME>
```

Parameters

```
[import [terminal]]
```

Import the certificate by pasting at the console (the default). This form of importing is selected whether `ta-certificate` is entered without parameters or if only `import` is entered or if `import terminal` is entered. Upon execution, the `config-ta-cert` context is entered for certificate pasting. To complete certificate data entry press Control-D in your terminal program. Alternatively, the pasted certificate data can include at its end the delimiter `END_OF_CERTIFICATE` (after the `-----END CERTIFICATE-----` line), making entry of Control-D unnecessary..

```
import <REMOTE-URL>
```

Import the certificate from a file on a remote TFTP or SFTP server. The URL syntax is:

```
{tftp:// | sftp://<USER>@} {<IP> | <HOST>} [:<PORT>] [:blocksize=<SIZE>]/<FILE>
```

```
import <STORAGE-URL>
```

Available on switch families that provide USB device file import capability, import the certificate from a file on a USB storage device inserted in the switch USB port. The URL syntax is:

```
usb: /<FILE>
```

Authority

Administrators or local user group members with execution rights for this command.

Example

Importing a certificate into the TA profile **root-cert** by pasting PEM-format certificate data at the console:


```

switch(config)# crypto pki ta-profile root-cert
switch(config-ta-root-cert)# ta-certificate import terminal
Paste the certificate in PEM format below, then hit enter and ctrl-D:
switch(config-ta-cert)# -----BEGIN CERTIFICATE-----
switch(config-ta-cert)# MIIDuTCCAqECCQCuoXeJ2ZNYcjANBgkqhkiG9w0BAQsFADCBqzELMAEBh
switch(config-ta-cert)# VVMxEzARBgNVBAGMCKNhbGlmb3JuaWExEDAOBgNVBACMB1JvY2tsDAKBg
switch(config-ta-cert)# BAoMA0hQTjEVMBMGAlUECwwMSFBOUm9zZXZpbGx1MSowKAYDVQocG5zd
...
switch(config-ta-cert)# x3Wff3dFZ8o9sd5LVAHneH/ztb9MP34z+le1V346r12L2kpxmTOVJVyTO
switch(config-ta-cert)# BIzD/ST/HaWI+OS+S80rm93PSscEbb9GWk7vshh5EnW/moehBKcE40lzy
switch(config-ta-cert)# 3LvMLZcssSe5J2Ca2XIhfDme8UaNZ7syGYMsAW0nG7yYHWkEOQu9s
switch(config-ta-cert)# -----END CERTIFICATE-----
switch(config-ta-cert)#
The certificate you are importing has the following attributes:
Issuer: C=US, ST=CA, L=Rocklin, O=Company, OU=Site,
       CN=site.com/emailAddress=test.ca@site.com
Subject: C=US, ST=CA, L=Rocklin, O=Company, OU=Site,
        CN=9000/emailAddress=test.ca@site.com
Serial Number: 12121221634631568498 (0xae51217d5945772)

TA certificate import is allowed only once for a TA profile
Do you want to accept this certificate (y/n)? y
TA certificate accepted.
switch(config-ta-root-cert)#

```

Importing a certificate into the TA profile **root-cert2** from file `rcert2-data` on the USB device:

```

switch(config)# crypto pki ta-profile root-cert2
switch(config-ta-root-cert2)# ta-certificate import usb:/rcert2-data
The certificate you are importing has the following attributes:
Issuer: C=US, ST=California, L=Rocklin, O=Company, OU=Site,
       CN=site.com/emailAddress=test.ca@site.com
Subject: C=US, ST=California, L=Rocklin, O=Company, OU=Site,
        CN=9000/emailAddress=test.ca@site.com
Serial Number: 12121221634631568498 (0xae51217d5945772)

TA certificate import is allowed only once for a TA profile
Do you want to accept this certificate (y/n)? y
TA certificate accepted.
switch(config-ta-root-cert2)#

```

subject

Syntax

```

subject [common-name <COMMON-NAME>] [country <COUNTRY>] [locality <LOCALITY>]
[org <ORG-NAME>] [org-unit <ORG-UNIT>] [state <STATE>]

```

Description

Sets the subject fields for the current leaf certificate. If the `common-name` parameter is not specified, then you are prompted to define a value for each field. If a configured value exists for any field, it is presented as the default.

The subject fields of the default certificate `local-cert` cannot be changed.

Command context

```

config-cert-<CERT-NAME>

```

Parameters

`common-name <COMMON-NAME>`

Specifies the common name.

`country <COUNTRY>`

Specifies the region.

`locality <LOCALITY>`

Specifies the locality.

`org <ORG-NAME>`

Specifies the organization.

`org-unit <ORG-UNIT>`

Specifies the organizational unit.

`state <STATE>`

Specifies the state.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting subject fields for the leaf certificate **leaf-cert**:

```
switch(config-cert-leaf-cert)# subject common-name Leaf01 country US  
locality CA org Company org-unit Site state CA
```

Setting subject fields for the leaf certificate **leaf-cert** interactively:

```
switch(config-cert-leaf-cert)# subject  
Do you want to use the switch serial number as the common name (y/n)? n  
Enter Common Name : Leaf01  
Enter Org Unit : Site  
Enter Org Name : Company  
Enter Locality : Rocklin  
Enter State : CA  
Enter Country : US  
switch(config-cert-leaf-cert)#
```



On the 6100 Switch Series, only the vrf named `default` is available. Replace any references to the `mgmt` or other VRFs with `default`.

arbitrary-label

Syntax

```
arbitrary-label <LABEL>
no arbitrary-label
```

Description

Within the EST profile context, configures the generic optional label (also known as arbitrary label) to be concatenated to the EST server URL that is configured with the `url` command. There is no arbitrary label configured by default. Any existing arbitrary label is replaced by this command. The use of arbitrary labels is optional.

RFC 7030 allows the use of arbitrary labels so that one EST server may serve multiple CAs with the same server URL that gets concatenated with different arbitrary labels. The same label is used for every request made under a particular EST profile.

Some EST schemes use arbitrary labels in a more sophisticated way, defining different labels for different types of requests under the same EST profile. For example, the CA certificate request could use the generic label (configured with this `arbitrary-label` command), the certificate enrollment request could use the enrollment label (configured with the `arbitrary-label-enrollment` command), and the re-enrollment request could use the re-enrollment label (configured with the `arbitrary-label-reenrollment` command). Note that only one label of each of the three available types can be configured in any EST profile.

The `no` form of this command removes the generic arbitrary label.

Command context

```
config-est-<EST-NAME>
```

Parameters

<LABEL>

Specifies the generic arbitrary label. Range: Up to 64 characters.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring the URL and generic arbitrary label. Note that with the URL and arbitrary label configured in this example, the final URL the switch uses to request CA certificates from the EST server is `https://est-service999.com/.well-known/est/rsa2048/cacerts`.

```
switch(config)# crypto pki est-profile EST-service1  
switch(config)# url https://est-service999.com/.well-known/est  
switch(config-est-EST-service1)# arbitrary-label rsa2048
```

Removing the generic arbitrary label:

```
switch(config)# crypto pki est-profile EST-service1  
switch(config-est-EST-service1)# no arbitrary-label
```

arbitrary-label-enrollment

Syntax

```
arbitrary-label-enrollment <LABEL>  
no arbitrary-label-enrollment
```

Description

Within the EST profile context, configures the arbitrary enrollment label to be concatenated to the EST server URL that is configured with the `url` command. This label is specific to the enrollment operation. There is no arbitrary enrollment label configured by default. Any existing arbitrary enrollment label is replaced by this command. The use of arbitrary enrollment labels is optional.

When the enrollment label is not configured, the generic arbitrary label (created with the `arbitrary-label` command) is used (if configured) for enrollment.

RFC 7030 allows the use of arbitrary labels so that one EST server may serve multiple CAs with the same server URL that gets concatenated with different arbitrary labels. The same label is used for every request made under a particular EST profile.

Some EST schemes use arbitrary labels in a more sophisticated way, defining different labels for different types of requests under the same EST profile. For example, the CA certificate request could use the generic label (configured with the `arbitrary-label` command), the certificate enrollment request could use the enrollment label (configured with this `arbitrary-label-enrollment` command), and the re-enrollment request could use the re-enrollment label (configured with the `arbitrary-label-reenrollment` command). Note that only one label of each of the three available types can be configured in any EST profile.

The `no` form of this command removes the arbitrary enrollment label.

Command context

```
config-est-<EST-NAME>
```

Parameters

<LABEL>

Specifies the arbitrary enrollment label. Range: Up to 64 characters.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring the arbitrary enrollment label:

```
switch(config)# crypto pki est-profile EST-service1  
switch(config-est-EST-service1)# arbitrary-label-enrollment ipsec-v7
```

Removing the arbitrary enrollment label :

```
switch(config)# crypto pki est-profile EST-service1  
switch(config-est-EST-service1)# no arbitrary-label-enrollment
```

arbitrary-label-reenrollment

Syntax

```
arbitrary-label-reenrollment <LABEL>  
no arbitrary-label-reenrollment
```

Description

Within the EST profile context, configures the arbitrary re-enrollment label to be concatenated to the EST server URL that is configured with the `url` command. This label is specific to the re-enrollment operation. There is no arbitrary re-enrollment label configured by default. Any existing arbitrary re-enrollment label is replaced by this command. The use of arbitrary re-enrollment labels is optional.

When the re-enrollment label is not configured, the generic arbitrary label (created with the `arbitrary-label` command) is used (if configured) for re-enrollment.

RFC 7030 allows the use of arbitrary labels so that one EST server may serve multiple CAs with the same server URL that gets concatenated with different arbitrary labels. The same label is used for every request made under a particular EST profile.

Some EST schemes use arbitrary labels in a more sophisticated way, defining different labels for different types of requests under the same EST profile. For example, the CA certificate request could use the generic label (configured with the `arbitrary-label` command), the certificate enrollment request could use the enrollment label (configured with the `arbitrary-label-enrollment` command), and the re-enrollment request could use the re-enrollment label (configured with this `arbitrary-label-reenrollment` command). Note that only one label of each of the three available types can be configured in any EST profile.

The no form of this command removes the arbitrary re-enrollment label.

Command context

```
config-est-<EST-NAME>
```

Parameters

<LABEL>

Specifies the arbitrary re-enrollment label. Range: Up to 64 characters.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring the arbitrary re-enrollment label:

```
switch(config)# crypto pki est-profile EST-service1  
switch(config-est-EST-service1)# arbitrary-label-reenrollment ipsec-v7
```

Removing the arbitrary re-enrollment label :

```
switch(config)# crypto pki est-profile EST-service1  
switch(config-est-EST-service1)# no arbitrary-label-reenrollment
```

crypto pki est-profile

Syntax

```
crypto pki est-profile <EST-NAME>  
no crypto pki est-profile <EST-NAME>
```

Description

Creates a certificate Enrollment over Secure Transport (EST) profile and changes to the `config-est-<EST-NAME>` context for the profile. Each EST profile stores information about the EST service, including EST server URL. Up to 16 profiles can be created.

If the specified EST profile exists, this command changes to the `config-est-<EST-NAME>` context for the profile.

The `no` form of this command deletes the specified EST profile. It also deletes the TA profiles whose CA certificates were downloaded from the corresponding EST server, and the leaf certificates that were enrolled using this EST profile.



The deletion of the related TA profiles and enrolled certificates is permanent. If the EST profile is in the startup configuration and the EST profile is deleted but this deletion is not updated in the startup configuration before a switch reboot, the EST profile will still exist after the reboot but the related TA profiles and enrolled certificates will not exist.

Command context

config

Parameters

<EST-NAME>

Specifies the EST profile name. Range: Up to 32 alphanumeric characters (excluding ").

Authority

Administrators or local user group members with execution rights for this command.

Examples

Creating EST profile **EST-Service1**:

```
switch(config)# crypto pki est-profile EST-Service1  
switch(config-est-service1)#
```

Removing EST profile **service1**:

```
switch(config)# no crypto pki est-profile EST-Service1
```

enroll est-profile

Syntax

```
enroll est-profile <EST-NAME>
```

Description

Enrolls a leaf certificate through a remote EST (Enrollment over Secure Transport) server.

Per RFC 7030, EST enables clients to request certificate signing services over secure TLS connections. The switch generates a key pair and the corresponding CSR. The CSR is sent to the EST server to request signing, and the signed certificate is returned to the switch where it is validated. If the whole process succeeds, the certificate can be used as a leaf certificate on the switch. When the leaf certificate approaches its expiry date, it will be renewed automatically through the same EST server.

Each enrollment or re-enrollment attempt starts with a `/cacerts` request sent to the EST server to get the latest chain of CA certificates. After the enrollment or re-enrollment succeeds, this chain of CA certificates will be compared with those downloaded previously from the same EST server. Updates will be made as appropriate.

The subject fields of the current leaf certificate must be defined before running this command. If the common name subject field is not configured, this command is rejected.

This command cannot be used to enroll or renew the default certificate "local-cert."

Command context

```
config-cert-<CERT-NAME>
```

Parameters

<EST-NAME>

Specifies an existing EST profile name. Range: Up to 32 alphanumeric characters (excluding ").

Authority

Administrators or local user group members with execution rights for this command.

Example

Enrolling leaf certificate **leaf-cert1** through the EST server identified in EST profile `EST-service1`:

```
switch(config-cert-leaf-cert1)# enroll est-profile EST-service1
You are enrolling a certificate with the following attributes:
  Subject: C=US, ST=CA, L=Roseville, OU=Aruba-Roseville, O=Aruba,
          CN=leaf-cert1
  Key Type: RSA (2048 bits)

Continue (y/n)? y
Certificate enrollment via EST-service1 has been initiated.
Please use `show crypto pki certificate leaf-cert1` to check its status.

switch(config-cert-leaf-cert1)#
```

reenrollment-lead-time

Syntax

```
reenrollment-lead-time <LEAD-TIME>  
no reenrollment-lead-time
```

Description

Within the EST profile context, sets the certificate re-enrollment lead time which is the number of days before certificate expiry date that certificate re-enrollment will be initiated.

The no form of this command resets the EST server re-enrollment lead time to its default of 2 days.

Command context

```
config-est-<EST-NAME>
```

Parameters

<LEAD-TIME>

Specifies the certificate re-enrollment lead time in days. Range: 0 to 30 days. Default: 2 days.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the certificate re-enrollment lead time to 15 days:

```
switch(config)# crypto pki est-profile EST-service1  
switch(config-est-EST-service1)# reenrollment-lead-time 15
```

Resetting the certificate re-enrollment lead time to its default of 2 days :

```
switch(config)# crypto pki est-profile EST-service1  
switch(config-est-EST-service1)# no reenrollment-lead-time
```

retry-count

Syntax

```
retry-count <RETRIES>  
no retry-count
```

Description

Within the EST profile context, sets the maximum number of retries to be attempted after the initial certificate enrollment request fails.

The no form of this command resets the maximum number of certificate enrollment request retries to its default of 3.

Command context

```
config-est-<EST-NAME>
```

Parameters

<RETRIES>

Specifies the maximum number of certificate enrollment request retries. Range: 0 to 32 retries. Default: 3 retries.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the retry count to 5 retries:

```
switch(config)# crypto pki est-profile EST-service1  
switch(config-est-EST-service1)# retry-count 5
```

Resetting the retry count to its default of 3 retries:

```
switch(config)# crypto pki est-profile EST-service1  
switch(config-est-EST-service1)# no retry-count
```

retry-interval

Syntax

```
retry-interval <INTERVAL>  
no retry-interval
```

Description

Within the EST profile context, sets the interval at which a failed certificate enrollment request is retried. The no form of this command resets the enrollment request retry interval to its default of 30 seconds.

Command context

config-est-<EST-NAME>

Parameters

<INTERVAL>

Specifies the enrollment request retry interval in seconds. Range: 30 to 600 seconds. Default: 30 seconds.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the certificate enrollment request retry interval to 45 seconds:

```
switch(config)# crypto pki est-profile EST-service1  
switch(config-est-EST-service1)# retry-interval 45
```

Resetting the retry interval to its default of 30 seconds:

```
switch(config)# crypto pki est-profile EST-service1
switch(config-est-EST-service1)# no retry-interval
```

show crypto pki est-profile

Syntax

```
show crypto pki est-profile [<EST-NAME>]
```

Description

Shows a list of all configured EST profiles, or detailed information for a specific profile.

Command context

Manager (#)

Parameters

<EST-NAME>

Specifies the EST profile name. Range: Up to 32 alphanumeric characters excluding ".

Authority

Administrators or local user group members with execution rights for this command.

Examples

Showing a list of all configured EST profiles:

```
switch# show crypto pki est-profile
```

Profile Name	Downloaded TA Profiles	Enrolled Certificates
EST-service1	2	3
EST-service2	1	2
EST-service3	2	0

Showing detailed information for EST profile **EST-service1**:

```
switch# show crypto pki est-profile EST-service1
Profile Name           : EST-service1
Service VRF            : mgmt
Service URL             : https://est-service999.com
Arbitrary Label         : not configured
Arbitrary Label Enrollment : /ipsec-VP7
Arbitrary Label Reenrollment : not configured
Authentication Username : est1
Authentication Password :
  AQBapREALpWYm2z7LlLanOtr3vGkqhBN1hBUU2CuvQXUF/ggYgAAAnAnGTnKq49P4c
  dNQ6UqPbjHL4XzCO0T04djkhsUXPKGfnsWuFEONveh+JbEobqKImfwJjc3eWHiaUb
  eNpPx2zN2Q1DdyxAAQi4rmKr8LITMTTmd7qr
Retry Interval          : 45 seconds
Retry Count             : 5 times
Reenrollment Lead Time : 2 days
Downloaded TA Profiles  : 2
Enrolled Certificates   :
```

```
leaf-cert1  
leaf-cert2  
leaf-cert3
```

url

Syntax

```
url <URL>  
no url
```

Description

Within the EST profile context, configures the URL of the certificate enrollment EST server. This is not configured by default. Any existing URL is replaced by this command.

The no form of this command removes the EST server URL within the selected EST profile. The removal of the URL does not affect the TA profiles and enrolled certificates from the EST server.

Command context

```
config-est-<EST-NAME>
```

Parameters

<URL>

Specifies the EST server URL. Range: Up to 192 characters.

Authority

Administrators or local user group members with execution rights for this command.

Usage

- The configuration and update of the EST profile URL triggers the sending of a `/cacerts` request to the EST server. A successful request will result in a chain of trusted CA certificates being downloaded from the EST server. Each CA certificate, either root CA certificates or intermediate CA certificates, will be saved as a TA profile, with TA profile name `<est-name>-est-taNN` with NN representing two numerical digits. This TA profile naming scheme with the `-est-taNN` suffix is reserved for TA profiles downloaded from EST servers.
- Upon connection with an EST server, the switch authenticates the server by validating the server certificate. For this validation to succeed, a TA profile needs to pre-exist in the switch with a CA certificate from the issuer chain of the server certificate. Once the server is authenticated, all CA certificates in its `/cacerts` response will be trusted, with no further validation occurring for them.
- The TA profiles with CA certificates downloaded from an EST server will have their revocation check set to OCSP, enforcement set to optional, and the OCSP VRF set to the same as that of the EST profile.

Examples

Configuring the EST server URL :

```
switch(config)# crypto pki est-profile EST-service1  
switch(config-est-EST-service1)# url https://est-service999.com/.well-known/est
```

Removing the EST server URL :

```
switch(config)# crypto pki est-profile EST-service1  
switch(config-est-EST-service1)# no url
```

username

Syntax

```
username <USERNAME> password [ciphertext <CIPHERTEXT-PASSWORD> | plaintext <PLAINTEXT-PASSWORD>]
```

```
no username
```

Description

Within the EST profile context, configures the user account information for the EST server that is used to authenticate the switch before accepting requests from the switch. This is not configured by default. Any existing username and password is replaced by this command.

When entered without either optional `ciphertext` or `plaintext` parameters, the plaintext password is prompted for twice, with the characters entered masked with "*" symbols.

The no form of this command removes the user account information within the selected EST profile.

There are two ways the EST client on a CX switch can prove itself to an EST server: a certificate, and/or username and password. At least one of the two must be configured for the EST request to succeed. If both are configured, certificate authentication will be used. If a certificate is not configured or certificate authentication fails, and username and password is configured, the username and password will be sent to the EST server for authentication.

Command context

```
config-est-<EST-NAME>
```

Parameters

<USERNAME>

Specifies the EST server account user name. The exact user name requirements are set by the chosen EST service. Range: Up to 32 alphanumeric characters.

ciphertext <CIPHERTEXT-PASSWORD>

Specifies the EST server account password as Base64 ciphertext. No password prompts are provided and the ciphertext password is validated before the configuration is applied for the user.



The ciphertext password must be gotten from the EST service.

plaintext <PLAINTEXT-PASSWORD>

Specifies the password without prompting. The password is visible as cleartext when entered but is encrypted thereafter. The exact password requirements are set by the chosen EST service. Range: Up to 64 alphanumeric characters.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring an EST user with prompted cleartext password entry :

```
switch(config)# crypto pki est-profile EST-service1
switch(config-est-EST-service1)# username est1 password
Enter password: *****
Confirm password: *****
switch(config-est-EST-service1)#
```

Configuring an EST user with direct cleartext password entry:

```
switch(config)# crypto pki est-profile EST-service2
switch(config-est-EST-service2)# username est1 password plaintext concept_leap739
```

Configuring an EST user with ciphertext password entry :

```
switch(config)# crypto pki est-profile EST-service3
switch(config-est-EST-service3)# username est1 password ciphertext
AQBpRALpWYm2z7L1LanOtR3vGkqhN1hBU2CuvQXUF/ggYgAAAHWaPqxU6nAnGTnKq49P4cdNQ6U
qPbjHL4XzO0T04djKUPKGfnsWuFEONveh+JbEobq63+1k80qBKImfwJjc3eWHiaUbeNpPx2zN2Q
1DdyxAAQi4rmKr8LITMTTmd7qr
```

Removing the EST user account information for EST profile EST-service2:

```
switch(config)# crypto pki est-profile EST-service2
switch(config-est-EST-service2)# no username
```

vrf

Syntax

```
vrf <VRF-NAME>
no vrf
```

Description

Within the EST profile context, selects the VRF through which the EST server can be reached. Any existing VRF selection is replaced by this command. When this command is not used, VRF `mgmt` is used by default on switch families supporting the `mgmt` VRF, otherwise the default VRF named `default` is used.

The no form of this command selects the default VRF either `mgmt` or `default`.

Command context

```
config-est-<EST-NAME>
```

Parameters

```
<VRF-NAME>
```

Specifies the name of the VRF to use for EST server communication.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Selecting VRF `it-services` for EST server communications:

```
switch(config)# crypto pki est-profile EST-service1  
switch(config-est-EST-service1)# vrf it-services
```

Resetting the VRF to its default of `mgmt` for EST server communications:

```
switch(config)# crypto pki est-profile EST-service1  
switch(config-est-EST-service1)# no vrf
```

All PoE configuration commands except `threshold` configuration and `always-on poe` configuration are entered at the `config-if` context. The PoE threshold command is used at the system level whereas the `always-on poe` command is set at the slot level. These commands can only be configured in the global configuration context.

lldp dot3 poe

Syntax

```
lldp dot3 poe
no lldp dot3 poe
```

Description

Enables 802.3 TLV list in LLDP to advertise for Power over Ethernet Data Link Layer Classification. LLDP dot3 TLV is by default enabled for PoE.

The `no` form of this command disables 802.3 TLV list in LLDP.

Command context

```
config-if
```

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling 802.3 TLV list in LLDP:

```
switch(config)# interface 1/1/1
switch(config-if)# lldp dot3 poe
```

Disabling 802.3 TLV list in LLDP:

```
switch(config-if)# no lldp dot3 poe
```

lldp med poe

Syntax

```
lldp med poe [priority-override]
no lldp med poe [priority-override]
```

Description

Enables MED TLV list in LLDP to advertise for Power over Ethernet Data Link Layer Classification. Also enables the lldp-MED TLV priority to override user configured port priority for Power over Ethernet. When both dot3 and MED are enabled, dot 3 will take precedence. MED TLV is by default enabled for PoE. Priority over-ride is by default disabled.

The `no` form of this command disables MED TLV list in LLDP.

Command context

`config-if`

Parameters

[priority-override]

System defined name of the interface.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling and disabling LLDP MED PoE:

```
switch(config)# interface 1/1/1
switch(config-if)# lldp med poe
switch(config-if)# no lldp med poe
```

Enabling and disabling LLDP MED PoE priority override:

```
switch(config-if)# lldp med poe priority-override
```

power-over-ethernet

Syntax

`power-over-ethernet`
`no power-over-ethernet`

Description

Enables per-interface power distribution. Per-port power is enabled by default with priority low. PoE cannot be disabled for individual ports when Quick PoE is enabled for the entire switch or line module.

The `no` form of this command disables per-interface power distribution.

Command context

`config-if`

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling per-interface power distribution:


```
switch(config)# interface 1/1/1  
switch(config-if)# power-over-ethernet
```

Disabling per-interface power distribution:

```
switch(config-if)# no power-over-ethernet
```

Showing Quick PoE enabled:

```
switch(config-if)# power-over-ethernet quick-poe 1/1  
switch(config-if)# interface 1/1/1  
switch(config-if)# no power-over-ethernet  
Interface PoE cannot be disabled when Quick PoE is enabled.
```

power-over-ethernet allocate-by

Syntax

```
power-over-ethernet allocate-by {usage | class}  
no power-over-ethernet allocate-by {usage | class}
```

Description

Configures the power allocation method. Power allocation method is initially based on usage. PSE Allocated power value will change to LLDP negotiated power if and when LLDP exchange takes place between PSE and PD. When there is no LLDP negotiation, PSE Allocated Power Value will be the actual instantaneous power draw and reserve power based on actual consumption. In allocate-by class, power allocation is based on PD requested class and PSE allocated power value will be the LLDP negotiated power when LLDP exchange takes place between PSE and PD. . When there is no LLDP negotiation, PSE Allocate Power will be based on PD class. Reserve power is based on PD Class. By default, power allocation is by usage.

The `no` form of this command resets the action to default.

Command context

config-if

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring the power allocation method:

```
switch(config)# interface 1/1/1  
switch(config-if)# power-over-ethernet allocate-by usage  
switch(config-if)# power-over-ethernet allocate-by class
```

Resetting power allocation method:

```
switch(config-if)# no power-over-ethernet allocate-by class
```

power-over-ethernet assigned-class

Syntax

```
power-over-ethernet assigned-class {3 | 4 | 6}  
no power-over-ethernet assigned-class
```

Description

Limit PoE power based on the assigned class. When an user assigns a maximum class to an interface, the PSE will limit the maximum power delivered to the PD up to a total power draw not exceeding the PSE assigned-class power. Power demotion occurs when a PD requested class is higher than the PSE assigned class, permitting the PD to receive power and operate in a reduced power mode. PoE ports cannot set an assigned class when Quick PoE is enabled on the sybsystem. The default assigned class is 4 for 2-pair capable PSE and 6 for 4-pair capable PSE.

The `no` form of this command resets the action to default.

Command context

config-if

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting PoE assigned class:

```
switch(config)# interface 1/1/1  
switch(config-if)# power-over-ethernet assigned-class 4
```

Resetting PoE assigned class to default:

```
switch(config-if)# no power-over-ethernet assigned-class 4
```

Showing Quick PoE enabled:

```
switch(config)# power-over-ethernet quick-poe 1/1  
switch(config)# interface 1/1/1  
switch(config)# power-over-ethernet assigned-class 4  
Interface assigned class cannot be configured when Quick PoE is enabled.
```

power-over-ethernet pre-std-detect

Syntax

```
power-over-ethernet pre-std-detect  
no power-over-ethernet pre-std-detect
```

Description

Before IEEE 802.3 released the first Power over Ethernet standard (802.3af), vendors had shipped PoE capable switches and PD's. As we are backward compatible Aruba will support both IEEE standard and pre-standard 802.3af Power over Ethernet PD's concurrently. This CLI allows the user to enable or disable pre-

802.3af-standard device detection and powering on the specific port. When pre-std-detect is enabled, power will be delivered on PairA only. Default is disabled.

The `no` form of this command resets the action to default.

Command context

config-if

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling standard device detection:

```
switch(config)# interface 1/1/1
switch(config-if)# power-over-ethernet pre-std-detect
```

Disabling standard device detection:

```
switch(config-if)# no power-over-ethernet pre-std-detect
```

power-over-ethernet priority

Syntax

```
power-over-ethernet priority {critical | high | low}
no power-over-ethernet priority {critical | high | low}
```

Description

Sets PoE priority for an interface. Specifying critical, high, or low indicates the priority of the interface in the event of power over-subscription. Within the same priority level, higher power-priority line-module ports have higher precedence. With same PoE priority and same line-module priority, lower numbered line-module ports have higher precedence. Per-interface PoE priority is low by default.

The `no` form of this command resets the priority to default PoE priority "low".

Command context

config-if

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring PoE priority:

```
switch(config)# interface 1/1/1
switch(config-if)# power-over-ethernet priority critical
switch(config-if)# power-over-ethernet priority high
```

Resetting the PoE priority to default:

```
switch(config-if) # no power-over-ethernet priority high
```

power-over-ethernet threshold

Syntax

```
power-over-ethernet threshold <PERCENTAGE>  
no power-over-ethernet threshold <PERCENTAGE>
```

Description

Sets the threshold at which the system will send an excess power consumption notification trap. Default value is 80 percentage.

The `no` form of this command resets the action to default.

Command context

config

Parameters

<PERCENTAGE>

Excess power consumption trap threshold. Range 1-99.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the power-over-ethernet threshold:

```
switch(config) # power-over-ethernet threshold 75
```

Resetting the power-over-ethernet threshold to default:

```
switch(config-if) # no power-over-ethernet threshold 75
```

power-over-ethernet priority

Syntax

```
power-over-ethernet trap  
no power-over-ethernet trap
```

Description

This command enables/disables the SNMP trap generation for PoE related events at system level. PoE trap generation is enabled by default.

The `no` form of this command resets the priority to default PoE priority "low".

Command context

config-if

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling SNMP trap generation for PoE:

```
switch(config)# power-over-ethernet trap
```

Disabling SNMP trap generation for PoE:

```
switch(config-if)# no power-over-ethernet trap
```

show lldp local

Syntax

```
show lldp local-device [<INTERFACE-ID>]
```

Description

Displays information advertised by the switch if the LLDP feature is enabled by user.

Command context

Operator (>) or Manager (#)

Parameters

<INTERFACE-ID>

Specifies an interface. Format: member/slot/port

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Showing LLDP local device:

```
switch# show lldp local-device 1/1/10
Local Port Data
=====
Port-ID           : 1/1/10
Port-Desc         : "1/1/10"
Port VLAN ID      : 0

PoE Plus Information

PoE Device Type   : Type 2 PSE
Power Source      : Primary
Power Priority     : low
PSE Allocated Power: 25.0 W
PD Requested Power: 25.0 W
```

show lldp neighbor

Syntax

```
show lldp neighbor [<INTERFACE-ID>]
```

Description

Displays detailed information about a particular neighbor connected to a particular interface.

Command context

Operator (>) or Manager (#)

Parameters

<INTERFACE-ID>

Specifies an interface. Format: member/slot/port

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Showing LLDP neighbor information when there is only one neighbor:

```
switch# show lldp neighbor-info 1/1/10

Port                               : 1/1/10
Neighbor Entries                   : 1
Neighbor Entries Deleted           : 0
Neighbor Entries Dropped           : 0
Neighbor Entries Aged-Out          : 0
Neighbor Chassis-Name              : 84:d4:7e:ce:5d:68
Neighbor Chassis-Description       : ArubaOS (MODEL: 325), Version Aruba IAP
Neighbor Chassis-ID                : 84:d4:7e:ce:5d:68
Neighbor Management-Address        : 169.254.41.250
Chassis Capabilities Available     : Bridge, WLAN
Chassis Capabilities Enabled       :
Neighbor Port-ID                   : 84:d4:7e:ce:5d:68
Neighbor Port-Desc                  : eth0
TTL                                 : 120
Neighbor Port VLAN ID              :
Neighbor PoEplus information       : DOT3
Neighbor Device Type                : TYPE2 PD
Neighbor Power Priority              : Unkown
Neighbor Power Source               : Primary
Neighbor Power Requested            : 25.0 W
Neighbor Power Allocated            : 0.0 W
Neighbor Power Supported            : No
Neighbor Power Enabled              : No
Neighbor Power Class                : 5
Neighbor Power Paircontrol          : No
Neighbor Power Pairs                : SIGNAL
```

show power-over-ethernet

Syntax

```
show power-over-ethernet [<IFRANGE> [brief]
```

Description

Displays the status information of the full system. Displays the brief status of all port or given port if parameter brief is used. Displays the detailed status of given port.

Command context

Operator (>) or Manager (#)

Parameters

<IFNAME>

Display the detailed status of given port.

brief

Display the brief status of all ports or the given port.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

```
switch# show power-over-ethernet

System Power Status for member 1

Configured Power Status      : No redundancy
Operational Power Status     : No redundancy
Total Available Power        : 740 W
Total Failover Pwr Avl       : 0 W
Total Redundancy Power       : 0 W
Total Power Drawn            : 0 W +/- 6W
Total Power Reserved         : 0 W
Total Remaining Power        : 740 W
Trap Threshold               : 80 %
Trap Enabled                 : Yes
Always-on PoE Enabled        : 1/1
Quick PoE Enabled            : None
```

```
Internal Power
  PS      Total Power      Status
  ----      -
  1         0             Absent
  2        740             Ok
```

```
System Power Status for member 2

Configured Power Status      : No redundancy
Operational Power Status     : No redundancy
Total Available Power        : 600 W
Total Failover Pwr Avl       : 0 W
Total Redundancy Power       : 0 W
Total Power Drawn            : 0 W +/- 6W
Total Power Reserved         : 0 W
Total Remaining Power        : 600 W
Trap Threshold               : 80 %
Trap Enabled                 : Yes
```

Always-on PoE Enabled : None
Quick PoE Enabled : None

Internal Power

PS	Total Power (Watts)	Status
1	0	Absent
2	600	Ok

switch# **show power-over-ethernet member 1**

System Power Status for member 1

Configured Power Status : No redundancy
Operational Power Status : No redundancy
Total Available Power : 740 W
Total Failover Pwr Avl : 0 W
Total Redundancy Power : 0 W
Total Power Drawn : 0 W +/- 6W
Total Power Reserved : 0 W
Total Remaining Power : 740 W
Trap Threshold : 80 %
Trap Enabled : No
Always-on PoE Enabled : 1/1
Quick PoE Enabled : 1/1

Internal Power

PS	Total Power (Watts)	Status
1	0	Absent
2	740	Ok

switch# **show power-over-ethernet brief**

Status and Configuration Information for PoE

Member 1 Power Status

Available: 370 W Reserved: 55.60 W Remaining: 314.40 W
Always-on PoE Enabled: 1/1
Quick PoE Enabled: None

PoE Port	Pwr En	Power Priority	Pre-std Detect	Alloc Act	PSE Pwr Rsrvd	PD Pwr Draw	PoE Port Status	PD Sign	Cls	Type
1/1/1	Yes	Low	Off	Class	0.0 W	0.0 W	Denied	None	4	2
1/1/2	Yes	Critical	Off	Usage	1.6 W	1.5 W	Delivering*	Single	0	1
1/1/3	Yes	High	Off	Class	54.0 W	25.5 W	Delivering*^	Dual	1/3	3
1/1/4	No	Low	On	Usage	0.0 W	0.0 W	Disabled	None	N/A	N/A

Member 2 Power Status

Available: 600 W Reserved: 0.00 W Remaining: 600 W
Always-on PoE Enabled: None
Quick PoE Enabled: None

PoE Port	Pwr En	Power Priority	Pre-std Detect	Alloc Act	PSE Pwr Rsrvd	PD Pwr Draw	PoE Port Status	PD Sign	Cls	Type
----------	--------	----------------	----------------	-----------	---------------	-------------	-----------------	---------	-----	------

2/1/1	Yes	Low	Off	Class	0.0 W	0.0 W	Searching	None	N/A	N/A
2/1/2	Yes	Critical	Off	Usage	0.0 W	0.0 W	Searching	None	N/A	N/A
2/1/3	Yes	High	Off	Class	0.0 W	0.0 W	Searching	None	N/A	N/A
2/1/4	No	Low	On	Usage	0.0 W	0.0 W	Disabled	None	N/A	N/A

*This port may go down in the event of a PSU failure.

^This port is power demoted due to user config or power availability.

```
switch# show power-over-ethernet brief
```

Status and Configuration Information for PoE

Power Status

Available: 370 W Reserved: 55.60 W Remaining: 314.40 W

Always-on PoE Enabled: 1/1,1/3,1/4,1/7

Quick PoE Enabled: None

PoE Port	Pwr En	Power Priority	Pre-std Detect	Alloc Act	PSE Pwr Rsrvd	PD Pwr Draw	PoE Port Status	PD Sign	Cls	Type
1/1/1	Yes	Low	Off	Class	0.0 W	0.0 W	Denied	None	4	2
1/1/2	Yes	Critical	Off	Usage	1.6 W	1.5 W	Delivering*	Single	0	1
1/1/3	Yes	High	Off	Class	54.0 W	25.5 W	Delivering^	Dual	1/3	3
1/1/4	No	Low	On	Usage	0.0 W	0.0 W	Disabled	None	N/A	N/A

*This port may go down in the event of a PSU failure.

^This port is power demoted due to user config or power availability.

Showing sample output for power-over-ethernet brief per-port:

```
switch# show power-over-ethernet 1/1/1 brief
```

Status and Configuration Information for port 1/1/1

Member 1Power Status

Available: 370 W Reserved: 55.60 W Remaining: 314.40 W

Always-on PoE Enabled: 1/1

PoE Port	Pwr En	Power Priority	Pre-std Detect	Alloc Act	PSE Pwr Rsrvd	PD Pwr Draw	PoE Port Status	PD Sign	Cls	Type
1/1/1	Yes	Low	Off	Class	0.0 W	0.0 W	Denied	None	4	2

Showing sample output for power-over-ethernet brief for interface range:

```
switch# show power-over-ethernet 1/1/1-1/1/2 brief
```

Status and Configuration Information for port 1/1/1-1/1/2

Member 1Power Status

Available: 370 W Reserved: 55.60 W Remaining: 314.40 W

Always-on PoE Enabled: 1/1

PoE Port	Pwr En	Power Priority	Pre-std Detect	Alloc Act	PSE Pwr Rsrvd	PD Pwr Draw	PoE Port Status	PD Sign	Cls	Type
1/1/1	Yes	Low	Off	Class	0.0 W	0.0 W	Denied	None	4	2
1/1/2	Yes	Critical	Off	Usage	1.6 W	1.5 W	Delivering*	Single	0	1

Showing sample output for power-over-ethernet for a missing line card:

```
switch# show power-over-ethernet 1/3 brief
```

```
Module 1/3 is not physically present.
```

Showing sample output for power-over-ethernet brief for a missing member:

```
switch# show power-over-ethernet member 3 brief
```

```
Member 3 is not physically present.
```

Showing sample output for power-over-ethernet port when physical interface is not present:

```
switch# show power-over-ethernet 2/1/1
```

```
Interface 2/1/1 is not present.
```

aaa authentication port-access auth-mode

Syntax

```
aaa authentication port-access auth-mode {client-mode | device-mode}
```

Description

Configures the authentication mode for the port. By default, client mode is enabled.

Command context

config-if

Parameters

client-mode

Selects client mode. In this mode, all clients connecting to the port are sent for authentication.

The maximum number of clients allowed to connect to the port is limited by the client limit value configured with the `aaa authentication port-access client-limit` command.

device-mode

Selects device mode. In this mode, only the first client connecting to the port is sent for authentication. Once this client is authenticated, the port is considered as open and all subsequent clients trying to connect on that port are not sent for authentication.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring device mode authentication for a port:

```
switch(config-if)# aaa authentication port-access auth-mode device-mode
```

aaa authentication port-access auth-precedence

Syntax

```
aaa authentication port-access auth-precedence [dot1x mac-auth | mac-auth dot1x]
```

```
no aaa authentication port-access auth-precedence [dot1x mac-auth | mac-auth dot1x]
```

```
no aaa authentication port-access auth-precedence
```

Description

Configures the per port authentication precedence using the space separator.

By default, 802.1X authentication (`dot1x`) takes a higher precedence than MAC authentication (`mac-auth`).

The `no` form of the command resets the port access authentication precedence to the default, 802.1X authentication followed by MAC authentication.

Command context

`config-if`

Parameters

`dot1x mac-auth`

Specifies that the port access authentication precedence is 802.1X authentication followed by MAC authentication.

`mac-auth dot1x`

Specifies that the port access authentication precedence is MAC authentication followed by 802.1X authentication.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring MAC authentication precedence on a port:

```
switch(config-if)# aaa authentication port-access auth-precedence mac-auth dot1x
```

Resetting the authentication precedence to the default value:

```
switch(config-if)# no aaa authentication port-access auth-precedence mac-auth dot1x
```

aaa authentication port-access auth-priority

Syntax

```
aaa authentication port-access auth-priority [dot1x mac-auth | mac-auth dot1x]  
no aaa authentication port-access auth-priority [dot1x mac-auth | mac-auth dot1x]  
no aaa authentication port-access auth-priority
```

Description

Configures the authentication priority using the space separator to specific interface.

Default `auth-priority` with concurrent onboarding is 802.1x followed by MAC authentication. With authentication precedence, the default `auth-priority` follows the `auth-precedence` order.

The `no` form of the command resets the port access authentication priority to the default, is same as the configured `auth-precedence` order.

The authentication priority is useful in deployments where clients such as wireless access points (APs), IT-compliant-laptops or phones, or laptops without pre-loaded supplicant software must download the supplicant software or firmware patches before attempting 802.1X authentication. In such cases, configure the MAC authentication as the primary authentication method followed by 802.1X for the authentication order. Meanwhile, configure 802.1x as the primary authentication priority and MAC authentication as secondary to enforce access based on 802.1X. Thus the client (or end access device) will initially be authenticated by MAC authentication with the access required to onboard and install the software or patches, and subsequently attempt the 802.1X authentication.

Reauthentication will be triggered for all high priority methods and not just the final successful authentication method.

Command context

config-if

Parameters

dot1x mac-auth

Specifies that the port access authentication priority is 802.1X authentication followed by MAC authentication.

mac-auth dot1x

Specifies that the port access authentication priority is MAC authentication followed by 802.1X authentication.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring MAC authentication priority on a port:

```
switch(config-if)# aaa authentication port-access auth-priority mac-auth dot1x
```

Resetting the authentication priority to the default value:

```
switch(config-if)# no aaa authentication port-access auth-priority mac-auth dot1x  
switch(config-if)# no aaa authentication port-access auth-priority
```

Sample configuration:

```
interface 1/1/1  
  no shutdown  
  no routing  
  vlan access 1  
  aaa authentication port-access auth-precedence mac-auth dot1x  
  aaa authentication port-access auth-priority dot1x mac-auth
```

aaa authentication port-access client-limit

Syntax

```
aaa authentication port-access client-limit <CLIENTS>  
no aaa authentication port-access client-limit
```

Description

Configures the maximum number of clients that can simultaneously connect to a port.

The `no` form of this command resets the number of clients to the default.

Command context

config-if

Parameters

<CLIENTS>

Specifies the maximum number of clients. Default: 1. Range: 1 to 32 (6100).

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring the client limit for a port:

```
switch(config-if) # aaa authentication port-access client-limit 25
```

aaa authentication port-access (role)

Syntax

```
aaa authentication port-access [critical-role|preauth-role|reject-role|auth-role] <ROLE-NAME>  
no aaa authentication port-access [critical-role|preauth-role|reject-role|auth-role]
```

Description

Configures the role to assign to the clients depending on the client authentication state.

The `no` form of the command disassociates the roles that you assign to clients based on the authentication state.

Command context

config-if

Parameters

`critical-role`

Specifies the role that is applied when the RADIUS server is unreachable for authentication or when there is a request timeout.

`preauth-role`

Specifies the role that is applied when authentication is still in progress.

`reject-role`

Specifies the role that is applied when authentication has failed.

`auth-role`

Specifies the role that is applied to authenticated clients when a specific role is not assigned in the RADIUS server.

<ROLE-NAME>

Specifies the role name.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring critical role for clients:

```
switch(config-if)# aaa authentication port-access critical-role role1
```

port-access allow-flood-traffic

Syntax

```
port-access allow-flood-traffic {enable | disable}
```

Description

Enables or disables transmission of flood traffic, such as broadcast, multicast, and unknown unicast messages through a security enabled port on which no client has been authenticated.

By default, transmission of flood traffic is disabled.

Command context

```
config-if
```

Authority

Administrators or local user group members with execution rights for this command.

Usage

This command can be used to allow Wake-on-LAN packets on security enabled ports, before a client is authenticated.

Examples

Enabling flood traffic on a port:

```
switch(config-if)# port-access allow-flood-traffic enable
```

port-access client-move

Syntax

```
port-access client-move {enable | disable}
```

Description

When client move is enabled (the default), a port access client can move to other port access-enabled interfaces, at which time they will be re-authenticated on the new interface.

When client move is disabled, a client cannot move to other port access-enabled interfaces.

Command context

```
config
```

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling client move:

```
switch(config)# port-access client-move enable
```

Disabling client move:

```
switch(config)# port-access client-move disable
```

port-access fallback-role

Syntax

```
port-access fallback-role <ROLE-NAME>  
no port-access fallback-role <ROLE-NAME>
```

Description

Configures the fallback role to assign to the clients onboarding on a port. This role is applied only when no derived role is applied to the clients.

The `no` form of the command resets the fallback role.

Command context

config-if

Parameters

<ROLE-NAME>

Specifies the fallback role name. The maximum number of characters supported is 64.

Authority

Administrators or local user group members with execution rights for this command.

Usage

Following are the conditions for the fallback role to be applied on onboarding devices:

- The device profile local MAC match feature with block-until-profile-applied mode is configured.
- Device profile along with AAA is configured but no match was found for the device profile client.
- AAA method with no reject or critical role is configured, and the connection to RADIUS server failed.
- 802.1X authentication is enabled on the port, but the supplicant of the device timed out to respond to the authentication request.

Example

Configuring fallback role for a port:

```
switch(config)# interface 1/1/3  
switch(config-if)# port-access fallback-role fallback01
```

port-access log-off client

Syntax


```
port-access log-off client mac <MAC-ADDRESS>
port-access log-off client interface <INTERFACE-NAME>
port-access log-off client role <ROLE-NAME>
```

Description

Logs off the client connected to a port access-enabled interface.

Command context

Manager (#)

Parameters

mac <MAC-ADDRESS>

Specifies the client MAC address.

interface <INTERFACE-NAME>

Specifies the client interface.

role <ROLE-NAME>

Specifies the client role.

Authority

Administrators or local user group members with execution rights for this command.

Example

Logging a client off from the switch, specifying the MAC address:

```
switch# port-access log-off client mac 00:50:56:bd:04:2d
```

Logging a client off from the switch, specifying the interface:

```
switch# port-access log-off client interface 1/1/1
```

Logging a client off from the switch, specifying the role:

```
switch# port-access log-off client role r1
```

port-access onboarding-method precedence

Syntax

```
port-access onboarding-method precedence [aaa device-profile | device-profile aaa]
no port-access onboarding-method precedence [aaa device-profile | device-profile aaa]
```

Description

Configures the precedence for the method to be used to authenticate onboarding devices for each interface.

The `no` form of the command resets the authentication method precedence to the default precedence of AAA followed by device profile.

AAA includes the 802.1X and MAC authentication methods whose precedence can be configured using the `aaa authentication port-access auth-precedence` command. Here, the default precedence is 802.1X authentication.

For example, if you configure AAA (both 802.1X and MAC) authentication methods and device profile on a port, by default, the authentication precedence would be 802.1X, then MAC, and lastly device profile.

Command context

config-if

Parameters



aaa here refers to the authentication precedence configured using the `aaa authentication port-access auth-precedence` command.

aaa device-profile

Specifies that the precedence for per port onboarding authentication method is AAA followed by device profile.

device-profile aaa

Specifies that the precedence for per port onboarding authentication method is device profile followed by AAA.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring AAA method precedence on a port:

```
switch(config)# interface 1/1/1
switch(config-if)# port-access onboarding-method precedence device-profile aaa
```

Resetting the authentication method precedence:

```
switch(config)# interface 1/1/1
switch(config-if)# no port-access onboarding-method precedence device-profile aaa
```

port-access onboarding-method concurrent

Syntax

port-access onboarding-method concurrent <enable | disable>

Description

Configures all methods to start concurrently for faster onboarding process. If authentication priority is not configured when enabling concurrent onboarding, the priority will be 802.1x followed by `mac-auth` and `device-profile`.

Default priority for concurrent onboarding is 802.1x followed by `mac-auth` and `device-profile`.

When enabling concurrent onboarding on the port, existing clients will be de-authenticated and freshly onboarded concurrently.

When concurrent onboarding is enabled, then `auth-precedence` will be ignored.

If concurrent onboarding is configured, the client will stay in pre-auth role till it gets succeeded by one authentication method or gets failed by all the authentication methods.

When the authentication method with the highest priority fails, the profile of the next successful authentication method is applied.

If all methods fail, the reject or critical role is applied based on the 802.1X authentication failure reason and continues to reauthenticate with the 802.1X method.

Reauthentication will be triggered for all high priority methods and not just the final successful authentication method.

Some RADIUS server may block the client when it receives two requests, `mac-auth` and 802.1X, from the same client at the same time. This is because the RADIUS server allows only one authentication request. In such cases, concurrent onboarding is not feasible. To prevent such scenarios, configure `auth-precedence` with `auth-priority`.

Command context

`config-if`

Parameters

`enable`

Enable clients to be onboarded concurrently.

`disable`

Disable clients to be onboarded concurrently.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling concurrent onboarding on a port:

```
switch(config)# interface 1/1/1
switch(config-if)# port-access onboarding-method concurrent enable
```

Disabling concurrent onboarding on a port:

```
switch(config)# interface 1/1/1
switch(config-if)# port-access onboarding-method concurrent disable
```

Sample configuration:

```
interface 1/1/1
  no shutdown
  no routing
  vlan access 999
  !aaa authentication port-access auth-precedence mac-auth dot1x
  port-access onboarding-method concurrent enable
```

port-access reauthenticate interface

Syntax

`port-access reauthenticate interface <INTERFACE-NAME>`

Description



Forcefully reauthenticates all clients connected to an interface.

Clients that are in the `HELD` state are ignored.

Command context

Manager (#)

Parameters

<INTERFACE-NAME>

Specifies the interface name.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring reauthentication of all clients on a port (6200 and 6300 Switch Series):

```
switch# port-access reauthenticate interface 1/1/1
```

Configuring reauthentication of all clients on a port (6400 Switch Series):

```
switch# port-access reauthenticate interface 1/3/1
```

port-access security violation action

Syntax

```
port-access security violation action [notify | shutdown]  
no port-access security violation action
```

Description

Configures the action that the switch must take whenever a security violation occurs at a port, such as the number of clients exceeds the configured client limit.

The `no` form of the command resets the action to the default action, `notify`.

Command context

config-if

Parameters

`notify`

Specifies that the switch notifies any security violation as an event or log in the syslog server. This action is the default.

`shutdown`

Specifies that the switch shuts down the port where the client limit has exceeded.

A port that is shut down can be configured to auto-recover after a recovery period that can be configured with the `port-access security violation action shutdown auto-recovery` and `port-access security violation action shutdown recovery-timer` commands.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring the shutdown security violation action for a port:

```
sswitch(config-if)# port-access security violation action shutdown
```

Resetting the security violation action to the default value:

```
switch(config-if)# no port-access security violation action
```

port-access security violation action shutdown auto-recovery

Syntax

```
port-access security violation action shutdown auto-recovery [enable | disable]  
no port-access security violation action shutdown auto-recovery
```

Description

Configures auto-recovery of the port when the security violation action is configured as shutdown.

This configuration allows the port that is shut down when a security violation occurs to automatically enable after the recovery timer expires.

The `no` form of the command resets auto-recovery to the default, disable.

Command context

config-if

Parameters

enable

Enables auto-recovery of port when the security violation action is configured as shutdown.

disable

Disables auto-recovery of port when the security violation action is configured as shutdown.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling auto-recovery of port:

```
switch(config-if)# port-access security violation action shutdown auto-recovery  
enable
```

Disabling auto-recovery of port:

```
switch(config-if)# no port-access security violation action shutdown auto-recovery
```

port-access security violation action shutdown recovery-timer

Syntax

```
port-access security violation action shutdown recovery-timer <RECOVERY-TIME>
no port-access security violation action shutdown recovery-timer
```

Description

Configures security violation recovery timer for the port when the security violation action is configured as shutdown.

The `no` form of the command resets the shutdown recovery timer to the default, 10.

Command context

config-if

Parameters

<RECOVERY-TIME>

Specifies the recovery timer (in seconds) after which the port, which is shut down because of security violation, is automatically enabled. Default: 10. Range: 10 to 600.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring the shutdown recovery-timer on a port:

```
switch(config-if)# port-access security violation action shutdown recovery-timer 60
```

Resetting the shutdown recovery-timer to the default value:

```
switch(config-if)# no port-access security violation action shutdown recovery-timer
```

show aaa authentication port-access interface client-status

Syntax

```
show aaa authentication port-access interface {all|<INTERFACE-NAME>} client-status [mac <MAC-ADDRESS>]
```

Description

Shows information about active port access sessions. The output can be filtered by interface or MAC address.

Command context

Operator (>) or Manager (#)

Parameters

all

Specifies all interfaces.

<INTERFACE-NAME>

Specifies the interface name.

<MAC-ADDRESS>

Specifies the client MAC address.

Authority

Operators or Administrators or local user group members with execution rights for this command.

Operators can execute this command from the operator context (>) only.

Examples

Showing information for all ports.

```
switch# show aaa authentication port-access interface all client-status
```

```
Port Access Client Status Details
```

```
Client 00:50:56:96:93:d6, John Doe
```

```
=====
```

```
Session Details
```

```
-----
```

```
Port          : 1/1/13
```

```
Session Time  : 30s
```

```
Authentication Details
```

```
-----
```

```
Status          : dot1x Authenticated
```

```
Auth Precedence : dot1x - Authenticated, mac-auth - Not attempted
```

```
Authorization Details
```

```
-----
```

```
Role   : Employee
```

```
Status : Applied
```

```
Client 00:50:56:96:50:28
```

```
=====
```

```
Session Details
```

```
-----
```

```
Port          : 1/1/14
```

```
Session Time  : 10s
```

```
Authentication Details
```

```
-----
```

```
Status          : mac-auth Authenticated
```

```
Auth Precedence : dot1x - Unauthenticated, mac-auth - Authenticated
```

```
Authorization Details
```

```
-----
```

```
Role   : RADIUS_773420618
```

```
Status : Applied
```

show port-access clients

Syntax

```
show port-access clients [interface <INTERFACE-NAME>] [mac <MAC-ADDRESS>]
```

Description

Shows summarized active port access client information. The output can be filtered by interface or MAC address.

Command context

Operator (>) or Manager (#)

Parameters

<INTERFACE-NAME>

Specifies the interface name.

<MAC-ADDRESS>

Specifies the client MAC address.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Showing information for all clients.

```
switch# show port-access clients
```

Port Access Clients

Port	MAC Address	Onboarded Method	Status	Role
1/1/4	00:50:56:bd:04:c8	port-security	Success	
1/1/5	00:50:56:bd:32:07		Success	reject-role, Reject
1/1/5	00:50:56:bd:32:08		Fail	critical-..., Critical
1/1/6	00:50:56:bd:50:43	mac-auth	Success	auth-role, Auth
1/1/6	00:50:56:bd:50:45	dot1x	Success	RADIUS_773420618
1/1/19	08:97:34:ad:e4:00	device-profile	Success	ap-role
1/1/20	00:50:56:bd:32:08		In-Progress	preauth-role, Preauth
1/1/20	00:50:56:bd:32:06		In-Progress	
1/1/20	00:50:56:bd:32:09		Fail	
1/1/212	00:60:56:bd:50:43	mac-auth	Success	fallback-role, Fallback

Showing information for clients on a particular interface:

```
switch# show port-access clients interface 1/1/5
```

Port Access Clients

Port	MAC Address	Onboarded Method	Status	Role
1/1/5	00:50:56:bd:32:07		Success	reject-role, Reject
1/1/5	00:50:56:bd:32:08		Fail	critical-..., Critical

Showing information for a particular client MAC address:

```
switch# show port-access clients mac 00:50:56:bd:50:43
```


Port Access Clients

Port	MAC Address	Onboarded Method	Status	Role
1/1/6	00:50:56:bd:50:43	mac-auth	Success	auth-role, Auth

show port-access clients detail

Syntax

```
show port-access clients [interface <INTERFACE-NAME>] [mac <MAC-ADDRESS>] detail
```

Description

Shows detailed active port access clients information including the VLAN group and VLAN association for each of the authenticated clients. The output can be filtered by interface or MAC address.

Command context

Operator (>) or Manager (#)

Parameters

<INTERFACE-NAME>

Specifies the interface name.

<MAC-ADDRESS>

Specifies the client MAC address.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Showing detailed information for clients on a particular interface (one client):

```
switch# show port-access clients interface 1/1/7 detail
```

```
Port Access Client Status Details:
```

```
Client 2c:41:38:7f:35:b9, John Doe
```

```
=====
```

```
Session Details
```

```
-----
Port          : 1/1/7
Session Time  : 203s
IPv4 Address  : 10.10.10.10
IPv6 Address  :
```

```
VLAN Details
```

```
-----
VLAN Group Name : vlan_group1
VLANs Assigned  : 12
Access          : 12
Native Untagged :
```

Allowed Trunk :

Authentication Details

Status : mac-auth Authenticated
Auth Precedence : dot1x - Unauthenticated, mac-auth - Authenticated
Auth History : mac-auth - Authenticated, 203s ago

Authorization Details

Role : RADIUS_21963402
Status : Applied

Role Information:

Name : RADIUS_21963402
Type : radius

Reauthentication Period : 333 secs
Authentication Mode :
Session Timeout :
Client Inactivity Timeout :
Description :

Access VLAN :
Native VLAN :
Allowed Trunk VLANs :
Access VLAN Name :
Native VLAN Name :
Allowed Trunk VLAN Names :
VLAN Group Name : vlan_group1
MTU :
QOS Trust Mode :
STP Administrative Edge Port :
PoE Priority : low

Policy : PERMIT-ALL_87364653

Access Policy Details:

Policy Name : PERMIT-ALL_87364653
Policy Type : Local
Policy Status : Applied

SEQUENCE	CLASS	TYPE ACTION
10	dns	ipv4 permit
20	dhcp	ipv4 permit
30	clearpass-web	ipv4 cir kbps 1024 exceed drop

Class Details:

class ip dns
10 match tcp any any
class ip dhcp
20 match any any any
class ip clearpass-web

30 match any any any

Showing information for a particular client MAC address:

```
6300# show port-access clients mac 00:00:00:00:00:c8 detail
Port Access Client Status Details:
Client 00:00:00:00:00:c8, 00:00:00:00:00:c8
=====
Session Details
-----
Port          : 1/1/1
Session Time  : 888s
IPv4 Address  :
IPv6 Address  :
VLAN Details
-----
VLAN Group Name : test_group
VLANs Assigned  : 22
Access         : 22
Native Untagged :
Allowed Trunk   :
Authentication Details
-----
Status          : mac-auth Authenticated
Auth Precedence : dot1x - Unauthenticated, mac-auth - Authenticated
Auth History    : mac-auth - Authenticated, 288s ago
dot1x - Unauthenticated, Supplicant-Timeout, 703s ago
dot1x - Unauthenticated, 798s ago
mac-auth - Authenticated, 888s ago
Authorization Details
-----
Role   : RADIUS_2801090107
Status : Applied
Role Information:
Name   : RADIUS_2801090107
Type   : radius
-----
Reauthentication Period      : 600 secs
Cached Reauthentication Period :
Authentication Mode          :
Session Timeout              :
Client Inactivity Timeout    :
Description                   :
Access VLAN                   :
Native VLAN                   :
Allowed Trunk VLANs           :
Access VLAN Name              :
Native VLAN Name              :
Allowed Trunk VLAN Names      :
VLAN Group Name               : test_group
MTU                            :
QOS Trust Mode                :
STP Administrative Edge Port  :
PoE Priority                   :
Policy                         :
GBP                            :
6300#
```

show port-access clients onboarding-method

Syntax

```
show port-access clients onboarding-method <METHOD>
```

Description

Shows active port access client information for the specified onboarding method.

Command context

Operator (>) or Manager (#)

Parameters

<METHOD>

Selects the onboarding method. Available methods: device-profile, dot1x, mac-auth, port-security.

Authority

Operators or Administrators or local user group members with execution rights for this command.

Operators can execute this command from the operator context (>) only.

Examples

Showing information for clients onboarded using MAC authentication.

```
switch# show port-access clients onboarding-method mac-auth

Port Access Clients

Status codes: device-mode

-----
Port      MAC-Address      Onboarding      Status      Role
Method
-----
1/1/6     00:50:56:bd:50:43 mac-auth        Success     auth-role, auth
1/1/212   00:60:56:bd:50:43 mac-auth        Success     fallback-role, fallback
```

show port-access security violation client-limit-exceeded interface

Syntax

```
show port-access security violation client-limit-exceeded interface {all|<INTERFACE-NAME>}
```

Description

Shows information related to information on the number of client-limit-exceeded security violations that have occurred. The output can be filtered by interface.

Command context

Operator (>) or Manager (#)

Parameters

all

Specifies all interfaces.

<INTERFACE-NAME>

Specifies the interface name.

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Examples

Showing information for all ports.

```
switch# show port-access security violation client-limit-exceeded interface all

Client Limit Exceeded Violation Status Details

Port          : 1/1/1
Violation Count : 10
Violation State : Violation Occurred

Port          : 1/1/2
Violation Count : 10
Violation State : None
```

Showing information for a particular port.

```
switch# show port-access security violation client-limit-exceeded interface 1/1/1

Client Limit Exceeded Violation Status Details

Port          : 1/1/1
Violation Count : 10
Violation State : None
```

aaa authentication port-access dot1x authenticator

Syntax

```
aaa authentication port-access dot1x authenticator {enable | disable}  
no aaa authentication port-access dot1x authenticator
```

Description

Enables or disables 802.1X authentication globally or at the port-level.

The `no` form of the command deletes global 802.1X configuration details and disables 802.1x authentication.

Command context

```
config  
config-if
```

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling 802.1X authentication globally:

```
switch(config)# aaa authentication port-access dot1x authenticator enable
```

Disabling 802.1X authentication globally:

```
switch(config)# aaa authentication port-access dot1x authenticator disable
```

Deleting and disabling global 802.1X authentication:

```
switch(config)# no aaa authentication port-access dot1x authenticator
```

Enabling 802.1X authentication on a port:

```
switch(config-if)# aaa authentication port-access dot1x authenticator enable
```

Disabling 802.1X authentication on a port:

```
switch(config-if)# aaa authentication port-access dot1x authenticator disable
```

Deleting and disabling 802.1X authentication configuration on a port:

```
switch(config-if)# no aaa authentication port-access dot1x authenticator
```

aaa authentication port-access dot1x authenticator auth-method

Syntax

```
aaa authentication port-access dot1x authenticator auth-method eap-radius  
no aaa authentication port-access dot1x authenticator auth-method eap-radius
```

Description

Configures the authentication mechanism used to control access to the network. The configured authentication method will be used to authenticate 802.1X clients.

The `no` form of the command resets the authentication mechanism to the default, `eap-radius`.

Command context

config

Parameters

eap-radius

Specifies the EAP RADIUS as the 802.1X authentication method.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling the EAP RADIUS 802.1X authentication method on the switch:

```
switch(config)# aaa authentication port-access dot1x authenticator auth-method eap-radius
```

Resetting the EAP RADIUS 802.1X authentication method on the switch:

```
switch(config)# no aaa authentication port-access dot1x authenticator auth-method eap-radius
```

aaa authentication port-access dot1x authenticator cached-reauth

Syntax

```
aaa authentication port-access dot1x authenticator cached-reauth  
no aaa authentication port-access dot1x authenticator cached-reauth
```

Description

Enables cached reauthentication on a port. Cached reauthentication allows 802.1X reauthentications to succeed when the RADIUS server is unavailable. Users already authenticated retain their currently assigned RADIUS attributes.

The `no` form of the command disables the cached reauthentication on a port.

Command context

`config-if`

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling cached reauthentication on a port:

```
switch(config-if)# aaa authentication port-access dot1x authenticator cached-reauth
```

Disabling cached reauthentication on a port:

```
switch(config-if)# no aaa authentication port-access dot1x authenticator cached-reauth
```

aaa authentication port-access dot1x authenticator cached-reauth-period

Syntax

```
aaa authentication port-access dot1x authenticator cached-reauth-period <PERIOD>  
no aaa authentication port-access dot1x authenticator cached-reauth-period
```

Description

Configures the period during which an authenticated client, which has failed to reauthenticate because the RADIUS server is unreachable, remains authenticated.

The `no` form of the command resets the cached reauthentication period to the default, 30 seconds.

Command context

`config-if`

Parameters

<PERIOD>

Specifies the cached reauthentication period (in seconds). Default: 30. Range: 30 to 86400.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring the cached reauthentication period on a port:


```
switch(config-if)# aaa authentication port-access dot1x authenticator cached-reauth-period 300
```

Resetting the cached reauthentication period to the default value:

```
switch(config-if)# no aaa authentication port-access dot1x authenticator cached-reauth-period
```

aaa authentication port-access dot1x authenticator discovery-period

Syntax

```
aaa authentication port-access dot1x authenticator discovery-period <PERIOD>  
no aaa authentication port-access dot1x authenticator discovery-period
```

Description

Configures the period the port waits to retransmit the next EAPOL request identity frame on an 802.1X enabled port that has no authenticated clients.

The `no` form of the command resets the discovery period to the default, 30 seconds.

Command context

config-if

Parameters

<PERIOD>

Specifies the discovery period (in seconds). Default: 30. Range: 1 to 65535.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring the discovery period on a port:

```
switch(config-if)# aaa authentication port-access dot1x authenticator discovery-period 120
```

Resetting the discovery period to the default value:

```
switch(config-if)# no aaa authentication port-access dot1x authenticator discovery-period
```

aaa authentication port-access dot1x authenticator eapol-timeout

Syntax

```
aaa authentication port-access dot1x authenticator eapol-timeout <EAPOL-TIMEOUT>
```

```
no aaa authentication port-access dot1x authenticator eapol-timeout
```

Description

Configure the period the switch waits for a response from a client before retransmitting an EAPOL PDU. If the value is 0, the time period is calculated as per RFC 2988.



As per RFC 2988 2.1: Before Round-Trip Time (RTT) measurement, set Retransmission Timeout (RTO) to 3 seconds for initial retransmission and then double the RTO to provide back off as per section 5.5. Limit the maximum RTO (RTOmax) to 20 seconds as per section 4.3 of RFC 3748.

The `no` form of the command resets the timeout period to the default.

Command context

config-if

Parameters

<EAPOL-TIMEOUT>

Specifies the EAPOL timeout period (in seconds). Range: 1 to 65535.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring EAPOL timeout on a port:

```
switch(config-if)# aaa authentication port-access dot1x authenticator eapol-timeout 120
```

Resetting the EAPOL timeout to the default value:

```
switch(config-if)# no aaa authentication port-access dot1x authenticator eapol-timeout
```

aaa authentication port-access dot1x authenticator max-eapol-requests

Syntax

```
aaa authentication port-access dot1x authenticator max-eapol-requests <MAX-EAPOL-REQUESTS>  
no aaa authentication port-access dot1x authenticator max-eapol-requests
```

Description

Configures the number of EAPOL requests to send to a supplicant that must time out before authentication fails and the authentication session ends.

The `no` form of the command resets the maximum number of EAPOL requests to the default, 5.

Command context

config-if

Parameters

<MAX-EAPOL-REQUESTS>

Specifies the maximum number of EAPOL requests. Default: 5. Range: 1 to 10.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring maximum EAPOL requests on a port:

```
switch(config-if)# aaa authentication port-access dot1x authenticator max-eapol-requests 3
```

Resetting the maximum EAPOL requests on a port to default:

```
switch(config-if)# no aaa authentication port-access dot1x authenticator max-eapol-requests
```

port-access dot1x authenticator max-retries

Syntax

```
aaa authentication port-access dot1x authenticator max-retries <max-retries>  
no aaa authentication port-access dot1x authenticator max-retries
```

Description

Configures the maximum number of retries that the switch attempts to authenticate a client on a port before marking the client as unauthenticated.

The `no` form of the command resets the maximum number of retries to the default, 2.

Command context

config-if

Parameters

<max-retries>

Indicates the number of authentication attempts. Default: 2. Range: 1 to 10.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring maximum authentication attempts on a port:

```
switch(config-if)# aaa authentication port-access dot1x authenticator max-retries 5
```

Resetting the maximum authentication attempts on a port to default:

```
switch(config-if)# no aaa authentication port-access dot1x authenticator max-retries
```

aaa authentication port-access dot1x authenticator quiet-period

Syntax

```
aaa authentication port-access dot1x authenticator quiet-period <PERIOD>
no aaa authentication port-access dot1x authenticator quiet-period
```

Description

Configures the period during which the port does not try to acquire a supplicant. This period begins after the last authentication attempt, authorized by the maximum retries parameter, fails.

You can configure the number of maximum retries with the `aaa authentication port-access dot1x authenticator max-retries` command.

The `no` form of the command resets the quiet period to the default, 60 seconds.

Command context

config-if

Parameters

<PERIOD>

Specifies the quiet period (in seconds). Default: 60. Range: 0 to 65535.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring quiet period on a port:

```
switch(config-if)# aaa authentication port-access dot1x authenticator quiet-period
100
```

Resetting the quiet period on a port to default:

```
switch(config-if)# no aaa authentication port-access dot1x authenticator quiet-
period
```

aaa authentication port-access dot1x authenticator radius server-group

Syntax

```
aaa authentication port-access dot1x authenticator radius server-group <GROUP-NAME>
no aaa authentication port-access dot1x authenticator radius server-group
```

Description

Configures the switch to use an existing RADIUS server group for 802.1X authentication.

The `no` form of the command resets the server group to the default, `radius`.

Command context

config

Parameters

<GROUP-NAME>

Specifies the name of the RADIUS server group.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring the switch to use RADIUS server group `employee`:

```
switch(config)# aaa authentication port-access dot1x authenticator radius server-group employee
```

Resetting RADIUS server group configuration to default:

```
switch(config)# no aaa authentication port-access dot1x authenticator radius server-group
```

aaa authentication port-access dot1x authenticator reauth

Syntax

```
aaa authentication port-access dot1x authenticator reauth  
no aaa authentication port-access dot1x authenticator reauth
```

Description

Enables periodic reauthentication of authenticated clients on the port.

The `no` form of the command disables periodic reauthentication.

Command context

config-if

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling periodic reauthentication on a port:

```
switch(config-if)# aaa authentication port-access dot1x authenticator reauth
```

Disabling periodic reauthentication on a port:

```
switch(config-if)# no aaa authentication port-access dot1x authenticator reauth
```

aaa authentication port-access dot1x authenticator reauth-period

Syntax

```
aaa authentication port-access dot1x authenticator reauth-period <PERIOD>
no aaa authentication port-access dot1x authenticator reauth-period
```

Description

Configures the period after which the authenticated clients are reauthenticated on the port. You must enable reauthentication on the port before configuring the reauthentication period.

The `no` form of the command resets the reauthentication period to the default, 3600 seconds.

Command context

config-if

Parameters

<PERIOD>

Specifies the reauthentication period (in seconds). Default: 3600. Range: 1 to 65535.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring reauthentication period on a port:

```
switch(config-if)# aaa authentication port-access dot1x authenticator reauth-period
100
```

Resetting the reauthentication period to the default value:

```
switch(config-if)# no aaa authentication port-access dot1x authenticator reauth-
period
```

clear dot1x authenticator statistics interface

Syntax

```
clear dot1x authenticator statistics [interface <IF-NAME>]
```

Description

Clears the 802.1X authentication statistics associated with the port and all the authenticator clients attached to this port.

If no interface is specified, the statistics is cleared for all 802.1X enabled ports.

Command context

Manager (#)

Parameters

<IF-NAME>

Specifies the interface name.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Clearing authentication statistics on a port (6200 and 6300 Switch Series):

```
switch# clear dot1x authenticator statistics interface 1/1/1
```

Clearing authentication statistics on a port (6400 Switch Series):

```
switch# clear dot1x authenticator statistics interface 1/3/1
```

Clearing authentication statistics on all ports:

```
switch# clear dot1x authenticator statistics
```

show aaa authentication port-access dot1x authenticator interface client-status

Syntax

```
show aaa authentication port-access dot1x authenticator interface {all|<IF-NAME>}  
client-status [mac <MAC-ADDRESS>]
```

Description

Shows information about active 802.1X authentication sessions. The output can be filtered by interface or MAC address.

Command context

Operator (>) or Manager (#)

Parameters

all

Specifies all interfaces.

<IF-NAME>

Specifies the interface name.

<MAC-ADDRESS>

Specifies the client MAC address.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Showing client status information for all ports.

```
switch# show aaa authentication port-access dot1x authenticator interface all  
client-status
```

```
Client  FE:04:D7:50:89:37, johndoe, 1/1/1
=====

Authentication Details
-----
Status           : Authenticated
Type             : Pass-Through
EAP-Method       : MD5
Time Since Last State Change : 10s

Authentication Statistics
-----
Authentication           : 0
Authentication Timeout   : 0
EAP-Start While Authenticating : 0
EAP-Logoff While Authenticating : 0
Successful Authentication : 0
Failed Authentication     : 0
Re-Authentication       : 0
Successful Re-Authentication : 0
Failed Re-Authentication : 0
EAP-Start When Authenticated : 0
EAP-Logoff When Authenticated : 0
Re-Auths When Authenticated : 0
Cached Re-Authentication  : 0
```

```
Client  9A:B4:59:97:D0:7E, janedoe, 1/1/1
=====

Authentication Details
-----
Status           : Authenticated
Type             : Pass-Through
EAP-Method       : TLS
Time Since Last State Change : 5s

Authentication Statistics
-----
Authentication           : 0
Authentication Timeout   : 0
EAP-Start While Authenticating : 0
EAP-Logoff While Authenticating : 0
Successful Authentication : 0
Failed Authentication     : 0
Re-Authentication       : 0
Successful Re-Authentication : 0
Failed Re-Authentication : 0
EAP-Start When Authenticated : 0
EAP-Logoff When Authenticated : 0
Re-Auths When Authenticated : 0
Cached Re-Authentication  : 0
```

show aaa authentication port-access dot1x authenticator interface port-statistics

Syntax

```
show aaa authentication port-access dot1x authenticator interface {all|<IF-NAME>} port-
statistics
```

Description

Shows information about 802.1X ports. The output can be filtered by interface.

Command context

Operator (>) or Manager (#)

Parameters

all

Specifies all interfaces.

<IF-NAME>

Specifies the interface name.

Authority

Operators or Administrators or local user group members with execution rights for this command.

Operators can execute this command from the operator context (>) only.

Examples

Showing information for all ports.

```
switch# show aaa authentication port-access dot1x authenticator interface all port-  
statistics
```

```
Port 1/1/1  
=====
```

Client Details

```
Number of Clients           : 1  
Number of Authenticated Clients : 1  
Number of Unauthenticated Clients : 0
```

Statistics

```
EAPOL Frames Received           : 4  
EAPOL Frames Transmitted        : 3  
EAPOL Start Frames Received     : 1  
EAPOL Logoff Frames Received    : 0  
EAPOL Response ID Frames Received : 2  
EAPOL Response Frames Received  : 1  
EAPOL Request ID Frames Transmitted : 2  
EAPOL Request Frames Transmitted : 1  
EAPOL Invalid Frames Received   : 0  
EAPOL EAP Length Error Frames Received : 0  
EAPOL Last Received Frame Version : 0  
EAPOL Last Received Frame Client MAC : 0
```

```
Port 1/1/2  
=====
```

Client Details

```
Number of Clients           : 1  
Number of Authenticated Clients : 1  
Number of Unauthenticated Clients : 0
```

Statistics

```
EAPOL Frames Received           : 4  
EAPOL Frames Transmitted        : 3
```

EAPOL Start Frames Received	: 1
EAPOL Logoff Frames Received	: 0
EAPOL Response ID Frames Received	: 2
EAPOL Response Frames Received	: 1
EAPOL Request ID Frames Transmitted	: 2
EAPOL Request Frames Transmitted	: 1
EAPOL Invalid Frames Received	: 0
EAPOL EAP Length Error Frames Received	: 0
EAPOL Last Received Frame Version	: 0
EAPOL Last Received Frame Client MAC	: 0

aaa authentication port-access mac-auth

Syntax

```
aaa authentication port-access mac-auth {enable | disable}
```

Description

Enables or disables MAC authentication globally or at the port-level.

Command context

```
config  
config-if
```

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling MAC authentication on all interfaces:

```
switch(config)# aaa authentication port-access mac-auth  
switch(config-macauth)# enable
```

Disabling MAC authentication on all interfaces:

```
switch(config)# aaa authentication port-access mac-auth  
switch(config-macauth)# disable
```

Enabling MAC authentication on an interface:

```
switch(config-if)# aaa authentication port-access mac-auth  
switch(config-if-macauth)# enable
```

Disabling MAC authentication on an interface:

```
switch(config-if)# aaa authentication port-access mac-auth  
switch(config-if-macauth)# disable
```

aaa authentication port-access mac-auth addr-format

Syntax

```
aaa authentication port-access mac-auth addr-format {no-delimiter | single-dash |
multi-dash |multi-colon | no-delimiter-uppercase | single-dash-uppercase |
multi-dash-uppercase | multi-colon-uppercase}

no aaa authentication port-access mac-auth addr-format {no-delimiter | single-dash |
multi-dash |multi-colon | no-delimiter-uppercase | single-dash-uppercase |
multi-dash-uppercase | multi-colon-uppercase}
```

Description

Configures the MAC address format that the switch must use in the RADIUS request message.

The `no` form of the command resets the MAC address format to the default, `no-delimiter`.

Command context

config

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the MAC address format on the switch:

```
switch(config)# aaa authentication port-access mac-auth
switch(config-macauth)# addr-format single-dash
```

Resetting the MAC address format on the switch to its default:

```
switch(config)# aaa authentication port-access mac-auth
switch(config-macauth)# no addr-format
```

aaa authentication port-access mac-auth auth-method

Syntax

```
aaa authentication port-access mac-auth auth-method {chap | pap}
no aaa authentication port-access mac-auth auth-method
```

Description

Configures the RADIUS authentication method for MAC authentication.

Following are the MAC authentication methods supported:

- CHAP
- PAP



The PEAP-MSCHAPv2 method of authentication is not supported.

The `no` form of the command resets the authentication method to the default, `chap`.

Command context

config

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring the RADIUS authentication method on the switch:

```
switch# config
switch(config)# aaa authentication port-access mac-auth
switch(config-macauth)# auth-method pap
```

Resetting the RADIUS authentication method on the switch:

```
switch(config)# no aaa authentication port-access mac-auth auth-method
```

aaa authentication port-access mac-auth cached-reauth

Syntax

```
aaa authentication port-access mac-auth cached-reauth
no aaa authentication port-access mac-auth cached-reauth
```

Description

Enables cached reauthentication on a port. Cached reauthentication allows MAC reauthentications to succeed when the RADIUS server is unavailable. Users who are already authenticated, retain their currently assigned RADIUS attributes.

The `no` form of the command disables cached reauthentication.

Command context

config-if

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling cached reauthentication on a port:

```
switch(config-if)# aaa authentication port-access mac-auth
switch(config-if-macauth)# cached-reauth
```

Disabling cached reauthentication on a port:

```
switch(config-if)# aaa authentication port-access mac-auth
switch(config-if-macauth)# no cached-reauth
```

aaa authentication port-access mac-auth cached-reauth-period

Syntax

```
aaa authentication port-access mac-auth cached-reauth-period <PERIOD>
```

```
no aaa authentication port-access mac-auth cached-reauth-period
```

Description

Configures the period during which an authenticated client, which has failed to reauthenticate because the RADIUS server is unreachable, remains authenticated.

The `no` form of the command resets the cached reauthentication period to the default, 30 seconds.

Command context

```
config-if
```

Parameters

<PERIOD>

Specifies the cached reauthentication period (in seconds). Default: 30. Range: 30 to 86400.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring cached reauthentication period on a port:

```
switch(config-if)# aaa authentication port-access mac-auth  
switch(config-if-macauth)# cached-reauth-period 300
```

Resetting the cached reauthentication period to the default value:

```
switch(config-if)# aaa authentication port-access mac-auth  
switch(config-if-macauth)# no cached-reauth-period
```

aaa authentication port-access mac-auth password

Syntax

```
aaa authentication port-access mac-auth password {plaintext|ciphertext}<PASSWORD>  
no aaa authentication port-access mac-auth password
```

Description

Enables and configures the global password that the switch must use for MAC authentication. The password can be either in ciphertext or plaintext format.

The `no` form of the command disables the password for MAC authentication.

Command context

```
config
```

Parameters

<PASSWORD>

Specifies the global password to be used by all MAC authenticating devices.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the MAC authentication password:

```
switch(config)# aaa authentication port-access mac-auth  
switch(config-macauth)# password {ciphertext|plaintext}
```

Disabling the MAC authentication password:

```
switch(config)# aaa authentication port-access mac-auth  
switch(config-macauth)# no password
```

aaa authentication port-access mac-auth quiet-period

Syntax

```
aaa authentication port-access mac-auth quiet-period <PERIOD>  
no aaa authentication port-access mac-auth quiet-period
```

Description

Configures the period during which the switch does not try to authenticate a rejected client.

The `no` form of the command resets the quiet period to the default, 60 seconds.

Command context

config-if

Parameters

<PERIOD>

Specifies the quiet period (in seconds). Default: 60. Range: 0 to 65535.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring the quiet period on a port:

```
switch(config-if)# aaa authentication port-access mac-auth  
switch(config-if-macauth)# quiet-period 65
```

Resetting the quiet period on a port to default:

```
switch(config-if)# aaa authentication port-access mac-auth  
switch(config-if-macauth)# no quiet-period
```

aaa authentication port-access mac-auth radius server-group

Syntax

```
aaa authentication port-access mac-auth radius server-group <GROUP-NAME>
no aaa authentication port-access mac-auth radius server-group
```

Description

Configures the MAC authentication server group.

The `no` form of the command resets the authentication server group to the default value, `radius`.

Command context

config

Parameters

<GROUP-NAME>

Specifies the MAC authentication server group name.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring the MAC authentication server group:

```
switch# config
switch(config)# aaa authentication port-access mac-auth
switch(config-macauth)# radius server-group group1
```

aaa authentication port-access mac-auth reauth

Syntax

```
aaa authentication port-access mac-auth reauth
no aaa authentication port-access mac-auth reauth
```

Description

Enables periodic MAC reauthentication of authenticated clients on the port.

The `no` form of the command disables periodic MAC reauthentication on the port.

Command context

config-if

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling reauthentication on a port:

```
switch(config-if)# aaa authentication port-access mac-auth
switch(config-if-macauth)# reauth
```

Disabling reauthentication on a port:


```
switch(config-if)# aaa authentication port-access mac-auth  
switch(config-if-macauth)# no reauth
```

aaa authentication port-access mac-auth reauth-period

Syntax

```
aaa authentication port-access mac-auth reauth-period <PERIOD>  
no aaa authentication port-access mac-auth reauth-period
```

Description

Configures the period after which MAC authenticated clients must be reauthenticated on the port. You must first enable MAC reauthentication on the port before configuring the MAC reauthentication period.

The `no` form of the command resets the MAC reauthentication period to the default, 3600 seconds.

Command context

config-if

Parameters

<PERIOD>

Specifies the MAC reauthentication period (in seconds). Default: 3600. Range: 1 to 65535.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring the MAC reauthentication period on a port:

```
switch(config-if)# aaa authentication port-access mac-auth  
switch(config-if-macauth)# reauth-period 60
```

Resetting the MAC reauthentication period to its default:

```
switch(config-if)# aaa authentication port-access mac-auth  
switch(config-if-macauth)# no reauth-period
```

clear mac-auth statistics

Syntax

```
clear mac-auth statistics [interface <IF-NAME>]
```

Description

Clears the MAC authentication statistics associated with the port and all the authenticator state machines associated to this port.

If no interface is specified, the statistics is cleared for all MAC authentication enabled ports.

Command context

Operator (>) or Manager (#)

Parameters

<IF-NAME>

Specifies the interface name.

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Examples

Clearing MAC authentication statistics on a port (6200 and 6300 Switch Series):

```
switch# clear mac-auth statistics interface 1/1/1
```

Clearing MAC authentication statistics on a port (6400 Switch Series):

```
switch# clear mac-auth statistics interface 1/3/1
```

Clearing MAC authentication statistics on all ports:

```
switch# clear mac-auth statistics
```

show aaa authentication port-access mac-auth interface client-status

Syntax

```
show aaa authentication port-access mac-auth interface {all|<IF-NAME>}  
client-status [mac <MAC-ADDRESS>]
```

Description

Shows information about MAC authentication clients status. The output can be filtered by interface or MAC address.

Command context

Operator (>) or Manager (#)

Parameters

all

Specifies all interfaces.

<IF-NAME>

Specifies the interface name.

<MAC-ADDRESS>

Specifies the client MAC address.

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Examples

Showing client status information for all ports:

```
switch# show aaa authentication port-access mac-auth interface all client-status

Port Access Client Status Details

Client  AB:CD:DE:FF:AA:BB, 1/1/1
=====
Authentication Details
-----
      Status                               : Authenticated
      Type                                 : Pass-Through
      Auth-Method                          : CHAP
      Time Since Last State Change         : 10 secs

Authentication Statistics
-----
      Authentication                       : 1
      Authentication Timeout               : 0
      Successful Authentication            : 1
      Failed Authentication                 : 0
      Re-Authentication                    : 0
      Successful Re-Authentication         : 0
      Failed Re-Authentication             : 0
      Re-Auths When Authenticated         : 0
      Cached Re-Authentication             : 0

Client  DD:CD:AB:CS:EE:OI, 1/1/2
=====
Authentication Details
-----
      Status                               : Unauthenticated
      Type                                 : Pass-Through
      Auth-Method                          : CHAP
      Auth Failure reason                   : Server reject/ Server timeout
      Time Since Last State Change         : 15 secs

Authentication Statistics
-----
      Authentication                       : 1
      Authentication Timeout               : 0
      Successful Authentication            : 0
      Failed Authentication                 : 1
      Re-Authentication                    : 0
      Successful Re-Authentication         : 0
      Failed Re-Authentication             : 0
      Re-Auths When Authenticated         : 0
      Cached Re-Authentication             : 0
```

Showing status information for a client:

```
switch# show aaa authentication port-access mac-auth interface 1/1/1 client-status
mac ab:cd:de:ff:aa:bb

Port Access Client Status Details
```

```

Client  AB:CD:DE:FF:AA:BB, 1/1/1
=====
Authentication Details
-----
      Status                        : Authenticated
      Type                         : Pass-Through
      Auth-Method                  : CHAP
      Time Since Last State Change : 10 secs

Authentication Statistics
-----
      Authentication                : 1
      Authentication Timeout        : 0
      Successful Authentication     : 1
      Failed Authentication        : 0
      Re-Authentication            : 0
      Successful Re-Authentication : 0
      Failed Re-Authentication     : 0
      Re-Auths When Authenticated  : 0
      Cached Re-Authentication     : 0

```

show aaa authentication port-access mac-auth interface port-statistics

Syntax

```
show aaa authentication port-access mac-auth interface {all|<IF-NAME>} port-statistics
```

Description

Shows information about MAC authentication ports. The output can be filtered by interface.

Command context

Operator (>) or Manager (#)

Parameters

all

Specifies all interfaces.

<IF-NAME>

Specifies the interface name.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Showing information for all ports.

```

switch# show aaa authentication port-access mac-auth interface all port-statistics

Port 1/1/1
=====

Client Details

```

```
-----  
Number of Clients           : 3  
Number of authenticated clients : 2  
Number of unauthenticated clients : 1  
Number of authenticating clients : 0
```

Port 1/1/2

=====

Client Details

```
-----  
Number of Clients           : 4  
Number of authenticated clients : 2  
Number of unauthenticated clients : 2  
Number of authenticating clients : 0
```

port-access policy

Syntax

```
port-access policy <POLICY-NAME>
  [<SEQUENCE-NUMBER>]
  class {ip|ipv6} <CLASS-NAME>
    action {<REMARK-ACTIONS> | <POLICE-ACTIONS> | <OTHER-ACTIONS>}
  [<SEQUENCE-NUMBER>]
  comment ...
```

Description

Creates or modifies policy and policy entries. A policy is made up of one or more policy entries ordered and prioritized by sequence numbers. Each entry has an IPv4/IPv6 class and one or more policy actions associated with it.

A policy must be applied to a role using the `associate policy` command.

The `no` form of the command can be used to delete either a policy (use `no` with the policy command) or an individual policy entry (use `no` with the sequence number).

Command context

`config`

The `policy` command takes you into the `config-pa-policy` context where you enter the policy entries.

Parameters

`<POLICY-NAME>`

Specifies the name of the policy.

`<SEQUENCE-NUMBER>`

Specifies a sequence number for the policy entry. Optional. Range: 1 to 4294967295.

`comment`

Specifies the description of an entry in the policy. Optional.

`class {ip|ipv6} <CLASS-NAME>`

Specifies a type of class, `ip` for IPv4 and `ipv6` for IPv6 policy. And specifies a class name.

`<REMARK-ACTIONS>`

Remark actions can be any of the following options: `{ip-precedence <IP-PRECEDENCE-VALUE> | dscp <DSCP-VALUE> | local-priority <LOCAL-PRIORITY-VALUE>}` where:

`ip-precedence <IP-PRECEDENCE-VALUE>`

Specifies the numeric IP precedence value. Range: 0 to 7.

`dscp <DSCP-VALUE>`

Specifies a Differentiated Services Code Point (DSCP) value. Enter either a numeric value (0 to 63) or a keyword as follows:

Keyword	Status	Syntax	Description
AF11	Optional	Keyword	DSCP 10 (Assured Forwarding Class 1, low drop probability)
AF12	Optional	Keyword	DSCP 12 (Assured Forwarding Class 1, medium drop probability)
AF13	Optional	Keyword	DSCP 14 (Assured Forwarding Class 1, high drop probability)
AF21	Optional	Keyword	DSCP 18 (Assured Forwarding Class 2, low drop probability)
AF22	Optional	Keyword	DSCP 20 (Assured Forwarding Class 2, medium drop probability)
AF23	Optional	Keyword	DSCP 22 (Assured Forwarding Class 2, high drop probability)
AF31	Optional	Keyword	DSCP 26 (Assured Forwarding Class 3, low drop probability)
AF32	Optional	Keyword	DSCP 28 (Assured Forwarding Class 3, medium drop probability)
AF33	Optional	Keyword	DSCP 30 (Assured Forwarding Class 3, high drop probability)
AF41	Optional	Keyword	DSCP 34 (Assured Forwarding Class 4, low drop probability)
AF42	Optional	Keyword	DSCP 36 (Assured Forwarding Class 4, medium drop probability)
AF43	Optional	Keyword	DSCP 38 (Assured Forwarding Class 4, high drop probability)
CS0	Optional	Keyword	DSCP 0 (Class Selector 0: Default)
CS1	Optional	Keyword	DSCP 8 (Class Selector 1: Scavenger)
CS2	Optional	Keyword	DSCP 16 (Class Selector 2: OAM)
CS3	Optional	Keyword	DSCP 24 (Class Selector 3: Signaling)
CS4	Optional	Keyword	DSCP 32 (Class Selector 4: Real time)
CS5	Optional	Keyword	DSCP 40 (Class Selector 5: Broadcast video)
CS6	Optional	Keyword	DSCP 48 (Class Selector 6: Network control)
CS7	Optional	Keyword	DSCP 56 (Class Selector 7)
EF	Optional	Keyword	DSCP 46 (Expedited Forwarding)

`local-priority <LOCAL-PRIORITY-VALUE>`

Specifies a local priority value. Range: 0 to 7.

`<POLICE-ACTIONS>`

Police actions can be the following {cir kbps <RATE-BPS> exceed}

where:

`cir kbps <RATE-BPS>`

Specifies a Committed Information Rate (CIR) value in Kilobits per second. Range: 1 to 4294967295.

`exceed`

Specifies action to take on packets that exceed the rate limit.

`<OTHER-ACTIONS>`

Other actions can be the following:

`drop`

Specifies drop traffic.

Authority

Administrators or local user group members with execution rights for this command.

Usage

- An applied policy processes the packet sequentially against policy and class entries in the list, until either the last policy entry in the list has been evaluated or the packet matches an entry. If there is no match, the packet will be dropped by one of the implicit `deny all` IPv4 and IPv6 entries.
- Entering an existing `<POLICY-NAME>` value will cause the existing policy to be modified, with any new `<SEQUENCE-NUMBER>` value creating an additional policy entry, and any existing `<SEQUENCE-NUMBER>` value replacing the existing policy entry with the same sequence number.
- If no sequence number is specified, a new policy entry will be appended to the end of the entry list with a sequence number equal to the highest policy entry currently in the list plus 10. The sequence numbers may be reordered with the `port-access policy <POLICY-NAME> resequence <STARTING-SEQ-NUM> <INCREMENT>` command.
- If a policy is configured without any action, the default action, `permit`, is applied for that policy.

Examples

Creating a policy with several class entries:

```
switch(config)# port-access policy POL1
switch(config-pa-policy)# 10 class ip dns
switch(config-pa-policy)# 20 class ip dhcp
switch(config-pa-policy)# 30 class ip test action cir kbps 1024 exceed drop
switch(config-pa-policy)# exit
switch(config)# show port-access policy POL1
```

Access Policy Details:

=====

Policy Name : POL1
Policy Type : Local
Policy Status : Applied

SEQUENCE	CLASS	TYPE	ACTION
10	dns	ipv4	permit
20	dhcp	ipv4	permit
30	test	ipv4	cir kbps 1024 exceed drop

Adding a comment to an existing class entry:


```

switch(config)# port-access policy POL1
switch(config-pa-policy)# 20 comment DHCP-PERMIT
switch(config-pa-policy)# exit
switch(config)# show run port-access policy POL1

port-access policy POL1
  10 class ip dns
  20 class ip dhcp
  20 comment DHCP-PERMIT
  30 class ip test action cir kbps 1024 exceed drop

```

Removing a comment from an existing class entry:

```

switch(config)# port-access policy POL1
switch(config-pa-policy)# no 20 comment
switch(config-pa-policy)# exit
switch(config)# show run port-access policy POL1

port-access policy POL1
  10 class ip dns
  20 class ip dhcp
  30 class ip test action cir kbps 1024 exceed drop

```

Modifying a policy by replacing one class with another at the same sequence number:

```

switch(config)# port-access policy POL1
switch(config-pa-policy)# 10 class ip mds action dscp af21
switch(config-pa-policy)# exit
switch(config)# show port-access policy POL1

```

Access Policy Details:
=====

Policy Name : POL1
Policy Type : Local
Policy Status : Applied

SEQUENCE	CLASS	TYPE ACTION
10	mds	ipv4 dscp AF21
20	dhcp	ipv4 permit
30	test	ipv4 cir kbps 1024 exceed drop

Removing a class:

```

switch(config)# port-access policy POL1
switch(config-pa-policy)# no 10
switch(config-pa-policy)# exit
switch(config)# show port-access policy POL1

```

Access Policy Details:
=====

Policy Name : POL1
Policy Type : Local
Policy Status : Applied

SEQUENCE	CLASS	TYPE	ACTION
20	dhcp	ipv4	permit
30	clearpass-web	ipv4	cir kbps 1024 exceed drop

port-access policy copy

Syntax

```
port-access policy <POLICY-NAME> copy <DESTINATION-POLICY>
```

Description

Copies details of an existing policy to a new policy. Copying a policy copies all its entries as well.

Command context

```
config
```

Parameters

<POLICY-NAME>

Specifies the existing policy name.

<DESTINATION-POLICY>

Specifies the destination policy name.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Copying a policy:

```
switch(config)# port-access policy POL1 copy POL1_copy
switch(config)# show port-access policy
```

```
Access Policy Details:
=====
```

```
Policy Name   : POL1
Policy Type   : Local
Policy Status : Applied
```

SEQUENCE	CLASS	TYPE	ACTION
20	dhcp	ipv4	permit
30	test	ipv4	cir kbps 1024 exceed drop

```
Policy Name   : POL1_copy
Policy Type   : Local
Policy Status : Applied
```

SEQUENCE	CLASS	TYPE	ACTION
20	dhcp	ipv4	permit
30	test	ipv4	cir kbps 1024 exceed drop

port-access policy resequence

Syntax

```
port-access policy <POLICY-NAME> resequence <STARTING-SEQ-NUM> <INCREMENT>
```

Description

Resequences numbering in a policy.

Command context

config

Parameters

<POLICY-NAME>

Specifies the policy where you want to resequence policy entries.

<STARTING-SEQ-NUM>

Specifies the sequence number to start resequencing from. Range: 1-4294967295.

<INCREMENT>

Specifies how much to increment the sequence numbers by.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Resequencing a policy starting at 5 with an increment of 10:

```
switch(config)# port-access policy POL1 resequence 5 10
switch(config)# show port-access policy POL1
```

```
Access Policy Details:
=====
```

```
Policy Name      : POL1
Policy Type      : Local
Policy Status    : Applied
```

SEQUENCE	CLASS	TYPE	ACTION
5	dhcp	ipv4	permit
15	test	ipv4	cir kbps 1024 exceed drop

port-access policy reset

Syntax

```
port-access policy <POLICY-NAME> reset
```

Description

Resets the policy configuration to match the current hardware configuration of the policy.

Command context

config

Parameters

<POLICY-NAME>

Specifies the name of the policy to be reset.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Resetting a policy:

```
switch(config)# port-access policy POL2
switch(config-pa-policy)# 20 class ip dhcp
switch(config-pa-policy)# 40 class test2 action cir kbps 1024 exceed drop
switch(config-pa-policy)# exit
switch(config)# show port-access policy POL1-V2

Access Policy Details:
=====

Policy Name      : POL2
Policy Type      : Local
Policy Status    : Applied

SEQUENCE    CLASS                                TYPE ACTION
-----
20          dhcp                                ipv4 permit
40          test2                               ipv4 cir kbps 1024 exceed drop

switch(config)# port-access policy POLV2
switch(config-pa-policy)# 50 class ip test3 action cir kbps 1024 exceed drop
switch(config-pa-policy)# no 20
switch(config-pa-policy)# exit
switch(config)# show port-access policy POL2

Access Policy Details:
=====

Policy Name      : POL2
Policy Type      : Local
Policy Status    : Rejected

SEQUENCE    CLASS                                TYPE ACTION
-----
40          test2                               ipv4 cir kbps 1024 exceed drop
50          test3                               ipv4 cir kbps 1024 exceed drop

switch(config)# port-access policy POL2 reset
Following policy entries will be removed:
class ip test3 action cir kbps 1024 exceed drop

Following policy entries will be added:
20 class ip dhcp

Do you want to continue (y/n)? y
switch(config)# show port-access policy POL2

Access Policy Details:
=====

Policy Name      : POL1-V2
Policy Type      : Local
Policy Status    : Applied
```

SEQUENCE	CLASS	TYPE	ACTION
20	dhcp	ipv4	permit
40	test2	ipv4	cir kbps 1024 exceed drop

clear port-access policy hitcounts

Syntax

```
clear port-access policy <POLICY-NAME> hitcounts {port | client}
```

Description

Clears statistics and conform rate of policy applied on a port or client.

Command context

Operator (>) or Manager (#)

Parameters

<POLICY-NAME>

Specifies the policy name.

port

Clears the policy statistics matching policy name in the port mode.

client

Clears the policy statistics matching policy name in the client mode.

Authority

Operators or Administrators or local user group members with execution rights for this command.

Operators can execute this command from the operator context (>) only.

Examples

Clearing policy hit counts:

```
switch# show port-access policy POL6 hitcounts port

Port Access Policy Hit-Counts Details:
=====

Policy Name      : POL4
Policy Type      : Local
Policy Status    : Applied

SEQUENCE CLASS                TYPE ACTION                                CUR-RATE (kbps)
-----
3          test8              ipv4 cir kbps 1024 exceed drop            512

Class Name       : dhcp
Class Type       : ipv4

SEQUENCE CLASS-ENTRY                                HIT-COUNT
-----
```

10 match icmp any any count

0

Class Name : clearpass-web
Class Type : ipv4

SEQUENCE	CLASS-ENTRY	HIT-COUNT
----------	-------------	-----------

15	match udp any any count	15101830
----	-------------------------	----------

Class Name : web-traffic
Class Type : ipv4

SEQUENCE	CLASS-ENTRY	HIT-COUNT
----------	-------------	-----------

10	match any any any count	241
20	match any 10.1.1.1 10.1.1.2 dscp AF11 count	50

Class Name : class6
Class Type : ipv6

SEQUENCE	CLASS-ENTRY	HIT-COUNT
----------	-------------	-----------

10	match any any any count	173
20	match icmpv6 2001:db8:a::123 2001:db8:a::125 dscp AF11 count	32

switch#

switch# **clear port-access policy POL6 hitcounts port**

switch#

switch# **show port-access policy POL6 hitcounts port**

Port Access Policy Hit-Counts Details:
=====

Policy Name : POL4
Policy Type : Local
Policy Status : Applied

SEQUENCE	CLASS	TYPE	ACTION	CUR-RATE (kbps)
----------	-------	------	--------	-----------------

3	test8	ipv4	cir kbps 1024 exceed drop	512
---	-------	------	---------------------------	-----

Class Name : dhcp
Class Type : ipv4

SEQUENCE	CLASS-ENTRY	HIT-COUNT
----------	-------------	-----------

10	match icmp any any count	0
----	--------------------------	---

Class Name : clearpass-web
Class Type : ipv4

SEQUENCE	CLASS-ENTRY	HIT-COUNT
----------	-------------	-----------

15	match udp any any count	0
----	-------------------------	---

Class Name : web-traffic
Class Type : ipv4

SEQUENCE	CLASS-ENTRY	HIT-COUNT
----------	-------------	-----------

10	match any any any count	0
20	match any 10.1.1.1 10.1.1.2 dscp AF11 count	0

```
Class Name : class6
Class Type : ipv6
```

SEQUENCE	CLASS-ENTRY	HIT-COUNT
10	match any any any count	0
20	match icmpv6 2001:db8:a::123 2001:db8:a::125 dscp AF11 count	0

show port-access policy

Syntax

```
show port-access policy [<POLICY-NAME>]
```

Description

Shows various aspects of policies and their current usage. Details of a policy including the content of a specific policy is shown.

Following are the possible values for policy types:

- **Local**—User configured policy
- **Downloaded**—Downloaded user policy
- **RADIUS**—Policy obtained from the RADIUS server

Following are the possible values for policy status:

- **Applied**—Policy is successfully applied in the hardware.
- **Rejected**—Policy is not supported in the hardware.
- **In-Progress**—Policy is being processed in the hardware.
- **Failed**—Displayed when the switch fails to apply the policy configuration because the TCAM resources are unavailable or full.



If a policy is configured without any action, the `show` command will represent such an entry with the action as `permit`.

Command context

Operator (>) or Manager (#)

Parameters

<POLICY-NAME>

Specifies the policy name.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Showing information for all policies:

```
switch(config)# show port-access policy
```

```
Access Policy Details:
=====
```

```
Policy Name   : POL1
Policy Type   : Local
Policy Status : Applied
```

SEQUENCE	CLASS	TYPE	ACTION
20	dhcp	ipv4	permit
30	test	ipv4	cir kbps 1024 exceed drop

```
Policy Name   : POL1_copy
Policy Type   : Local
Policy Status : Applied
```

SEQUENCE	CLASS	TYPE	ACTION
20	dhcp	ipv4	permit
30	test	ipv4	cir kbps 1024 exceed drop

Showing information for a particular policy:

```
switch(config)# show port-access policy POL1
```

```
Access Policy Details:
=====
```

```
Policy Name   : POL1
Policy Type   : Local
Policy Status : Applied
```

SEQUENCE	CLASS	TYPE	ACTION
20	dhcp	ipv4	permit
30	test	ipv4	cir kbps 1024 exceed drop

show port-access policy hitcounts

Syntax

Syntax that shows statistical information in the form of hit counts:

```
show port-access policy <POLICY-NAME> hitcounts {port | client}
```

Description

Shows the statistics and the current rate of policy applied on the port or the client.

Command context

Operator (>) or Manager (#)

Parameters

<POLICY-NAME>

Specifies the policy name.

port

Shows policy statistics matching the policy name in the port mode.
client

Shows policy statistics matching the policy name in the client mode.

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Examples

Showing policy hit counts (statistics) with current rate:

```
switch# show port-access policy POL6 hitcounts port

Port Access Policy Hit-Counts Details:
=====

Policy Name      : POL1
Policy Type      : Local
Policy Status    : Applied

SEQUENCE CLASS          TYPE ACTION                                CUR-RATE (kbps)
-----
30      test8           ipv4 cir kbps 1024 exceed drop          512

Class Name : dhcp
Class Type : ipv4

SEQUENCE CLASS-ENTRY                                HIT-COUNT
-----
10      match icmp any any count                    982150

Class Name : clearpass-web
Class Type : ipv4

SEQUENCE CLASS-ENTRY                                HIT-COUNT
-----
70      match udp any any count                      15101830

Class Name : web-traffic
Class Type : ipv4

SEQUENCE CLASS-ENTRY                                HIT-COUNT
-----
4       match any any any count                      3194
5       match any 10.1.1.1 10.1.1.2 dscp AF11 count  1716

Class Name : class6
Class Type : ipv6

SEQUENCE CLASS-ENTRY                                HIT-COUNT
-----
10      match any any any count                      0
20      match icmpv6 2001:db8:a::123 2001:db8:a::125 dscp AF11
count                                          0
```

associate policy

Syntax

```
associate policy <POLICY-NAME>
no associate policy <POLICY-NAME>
```

Description

Associates the policy with the current role.

The `no` form of this command dissociates the policy from the role.

Command context

`config-pa-role`

The `port-access role` command takes you into the `config-pa-role` context.

Parameters

`<POLICY-NAME>`

Specifies the policy name to associate with the current role. The maximum number of characters allowed is 64.



Only those policies created by using the `port-access policy` command are allowed to be associated with a role.

Policies created using the `policy` command are not allowed to be associated with a role.

Policies that are of the downloaded type are not allowed to be associated with a role.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Associating a policy with a role:

```
switch(config)# port-access role role01
switch(config-pa-role)# associate policy policy01
```

Dissociating a policy from the role:

```
switch(config-pa-role)# no associate policy policy01
```

auth-mode

Syntax

```
auth-mode {client-mode | device-mode}
```

Description

Configures the authentication mode for the clients that are associated with the current role.

Command context

```
config
```

The `port-access role` command takes you into the `config-pa-role` context.

Parameters

```
client-mode
```

Selects client mode. In this mode, all clients connecting to the port are sent for authentication.

```
device-mode
```

Selects device mode. In this mode, only the first client connecting to the port is sent for authentication. Once this client is authenticated, the port is considered as open and all subsequent clients trying to connect on that port are not sent for authentication.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring the client authentication mode:

```
switch(config-pa-role)# auth-mode client-mode
```

cached-reauth-period

Syntax

```
cached-reauth-period [<PERIOD>]  
no cached-reauth-period
```

Description

Enables cached reauthentication, setting the period after which clients that associated with the current role must be reauthenticated.

The no form of this command disables cached authentication.

Command context

```
config-pa-role
```

The `port-access role` command takes you into the `config-pa-role` context.

Parameters

```
<PERIOD>
```

Specifies the cached reauthentication period (in seconds) for clients associated with the role. Default: None. Range: 30 to 4294967295 .

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling cached reauthentication and setting its period to 200 seconds:

```
switch(config-pa-role)# cached-reauth-period 200
```

Disabling cached reauthentication:

```
switch(config-pa-role)# no cached-reauth-period
```

client-inactivity timeout

Syntax

```
client-inactivity timeout {<CLIENT-INACTIVITY-PERIOD> | none}  
no client-inactivity timeout
```

Description

Configures the period that the switch waits for a response from a client after which it removes the client from the role.

The `no` form of the command resets the timeout period to the default.

Command context

`config-pa-role`

The `port-access role` command takes you into the `config-pa-role` context.

Parameters

<CLIENT-INACTIVITY-PERIOD>

Specifies the client inactivity time (in seconds). Default: 300. Range: 300 to 4294967295.
`none`

Specifies that the client must not be deleted because of inactivity.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring client inactivity timer for a role:

```
switch(config-pa-role)# client-inactivity timeout 3600
```

description

Syntax

```
description <ROLE-DESCRIPTION>
```

Description

Configures the role description.

Command context

config-pa-role

The `port-access role` command takes you into the `config-pa-role` context.

Parameters

<ROLE-DESCRIPTION>

Specifies the role description. Up to 255 characters.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring the role description:

```
switch(config-pa-role)# description student role
```

mtu

Syntax

mtu <MTU-SIZE>

Description

Configures the MTU size of a client for a role.

Command context

config-pa-role

The `port-access role` command takes you into the `config-pa-role` context.

Parameters

<MTU-SIZE>

Specifies the MTU size in bytes of a client for a role. Range: 68 to 9198.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring client MTU size:

```
switch(config-pa-role)# mtu 9198
```

poe-priority

Syntax

```
poe-priority {critical | high | low}  
no poe-priority
```

Description

Configures the power distribution priority for the port access roles. High power consumption can be prevented using `poe-priority` control mechanism.

The `no` form of this command restores the power distribution to its default priority.

Command context

`config-pa-role`

The `port-access role` command takes you into the `config-pa-role` context.

Parameters

`critical`

Specifies critical priority.

`high`

Specifies high priority.

`low`

Specifies low priority.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring PoE priority for a new role:

```
switch(config)# port-access role role01
switch(config-pa-role)# poe-priority critical
```

Resetting PoE priority for the role to its default:

```
switch(config)# port-access role role01
switch(config-pa-role)# no poe-priority
```

port-access role

Syntax

```
port-access role <ROLE-NAME>
no port-access role <ROLE-NAME>
```

Description

Creates a new port access role or modifies an existing role. This command takes you into the `config-pa-role` context. A maximum of 32 port access roles can be created.

The `no` form of this command deletes a role.

Command context

`config`

Parameters

`<ROLE-NAME>`

Specifies the role name. The maximum number of characters allowed is 64.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Creating a new role:

```
switch(config)# port-access role basic01
```

reauth-period

Syntax

```
reauth-period <PERIOD>  
no reauth-period
```

Description

Configures the period after which clients that associated with the current role must be reauthenticated.



The reauthentication period configured here takes precedence over the reauthentication period configured at the port level.

Command context

config-pa-role

The `port-access role` command takes you into the `config-pa-role` context.

Parameters

<PERIOD>

Specifies the reauthentication period (in seconds) for clients associated with the role. Default: None. Range: 1 to 86400.



Reauthentication period of less than 60 seconds is not recommended.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring reauthentication period:

```
switch(config-pa-role)# reauth-period 3000
```

session timeout

Syntax

```
session-timeout <SESSION-TIMEOUT>  
no session-timeout
```

Description

Configures the session timeout for the role. After the timeout period, the session will be disconnected.

Command context

config-pa-role

The `port-access role` command takes you into the `config-pa-role` context.

Parameters

<SESSION-TIMEOUT>

Specifies the session timeout (in seconds). Range: 1 to 4294967295.



A session timeout period of less than 60 seconds is not recommended.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring session timeout for a role:

```
switch(config-pa-role)# session timeout 3600
```

show aaa authentication port-access interface client-status

Syntax

```
show aaa authentication port-access interface {all | <IF-NAME>}  
client-status [mac <MAC-ADDRESS>]
```

Description

Shows information about the status of the role applied on ports.

Command context

Operator (>) or Manager (#)

Parameters

all

Specifies all interfaces.

<IF-NAME>

Specifies the instance name.

<MAC-ADDRESS>

Specifies the client MAC address.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Showing information about a client:


```

switch# show aaa authentication port-access interface all client-status mac
00:00:00:00:00:01

Port Access Client Status Details

Client 00:00:00:00:00:01
=====
Session Details
-----
Port          : 1/7/24
Session Time  : 151s

Authentication Details
-----
Status          : mac-auth Authenticated
Auth Precedence : mac-auth - Authenticated, dot1x - Not attempted

Authorization Details
-----
Role           : UserRole_1
Status         : Applied

```

show port-access role

Syntax

```
show port-access role {local | radius | name <ROLE-NAME>}
```

Description

Shows information about roles configured locally, or downloaded from RADIUS server.

Command context

Operator (>) or Manager (#)

Parameters

local

Shows information about locally configured roles.

radius

Shows information about roles downloaded from the RADIUS server.

<ROLE-NAME>

Specifies the role name.

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Examples

Showing locally configured role information:

```

switch# show port-access role local

Role Information

```

```

Name   : local_role_01
Type   : local
-----
Reauthentication Period      : 333 secs
Authentication Mode          :
Session Timeout              :
Client Inactivity Timeout    :
Tunneled Node Server Zone    :
Tunneled Node Server Secondary Role :
Access VLAN                  :
Native VLAN                  :
Allowed Trunk VLANs          :
MTU                           :
QoS Trust Mode               :
PoE Priority                  : low

Policy                        :

```

Showing information for roles downloaded from a RADIUS server:

```

switch# show port-access role radius

Role Information

Name   : RADIUS_21963402
Type   : radius
-----
Reauthentication Period      : 333 secs
Authentication Mode          :
Session Timeout              :
Client Inactivity Timeout    :
Tunneled Node Server Zone    :
Tunneled Node Server Secondary Role :
Access VLAN                  : 10
Native VLAN                  :
Allowed Trunk VLANs          :
MTU                           :
QoS Trust Mode               :
PoE Priority                  : low

Policy                        : PERMIT-ALL_87364653

```

stp-admin-edge-port

Syntax

```

stp-admin-edge-port
no stp-admin-edge-port

```

Description

Configures the port as a spanning tree administrative edge port for the role. This configuration removes the port participation from STP interactions when onboarding devices. This in turn helps in faster onboarding of devices.

The `no` form of the command disables STP edge port functionality.



If the port receives STP BPDU on the STP administrative edge configured port, the port will move to the STP state. You must configure the port as an STP administrative edge port only if you are sure that the connected device will not participate in STP interactions.

Command context

config-pa-role

The `port-access role` command takes you into the `config-pa-role` context.

Authority

Administrators or local user group members with execution rights for this command.

Example

Configuring STP edge port for a role:

```
switch(config)# port-access role role01
switch(config-pa-role)# stp-admin-edge-port
```

trust-mode

Syntax

```
trust-mode [dscp | cos | none]
no trust-mode
```

Description

Configures QoS trust mode for the role.

The `no` form of this command configures the default trust mode for the role.

Command context

config-pa-role

The `port-access role` command takes you into the `config-pa-role` context.

Parameters

dscp

Specifies to trust DSCP and preserve the 802.1p priority.

cos

Specifies to trust 802.1p priority and retain DSCP or IP-ToS.

none

Specifies to not trust any priority fields.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring DSCP trust mode for a role:

```
switch(config)# port-access role role01
switch(config-pa-role)# trust-mode dscp
```

vlan

Syntax

```
vlan {access | trunk native | trunk allowed} <VLAN-ID>
no vlan {access | trunk native | trunk allowed} <VLAN-ID>
```

Or

```
vlan {access name | trunk native name | trunk allowed name} <VLAN-NAME>
no vlan {access name | trunk native name | trunk allowed name} [<VLAN-NAME>]
```

Description

Configures VLAN IDs or VLAN names, and VLAN modes for a port access role. You can configure either VLAN IDs or VLAN names, or a combination of both for a role.

The `no` form of the command deletes the VLAN configuration from the role. For trunk allowed VLAN names, you can delete the VLAN names individually or all names at once.

Command context

`config-pa-role`

The `port-access role` command takes you into the `config-pa-role` context.

Parameters

access <VLAN-ID>

Specifies the VLAN ID for the access VLAN. Supports a single VLAN ID in the range 1 to 4094.

trunk native <VLAN-ID>

Specifies the native VLAN ID on the trunk interface. Supports a single VLAN ID in the range 1 to 4094.

trunk allowed <VLAN-ID>

Specifies the list of tagged or allowed VLANs on the trunk interface. Supports a list of VLAN IDs in the range 1 to 4094. The 6300, 6400 Switch Series support a maximum of 1024 trunk allowed VLAN IDs. The 6200 Switch Series supports a maximum of 256 trunk allowed VLAN IDs.

access name <VLAN-NAME>

Specifies the VLAN name for the access VLAN. Supports a single VLAN name with a maximum length of 32 characters.

trunk native name <VLAN-NAME>

Specifies the native VLAN name on the trunk interface. Supports a single VLAN name with a maximum length of 32 characters.

trunk allowed name <VLAN-NAME>

Specifies the tagged or allowed VLAN name on the trunk interface. Supports a single VLAN name in each CLI with a maximum length of 32 characters. The switch supports a maximum of 50 trunk allowed VLAN names.

Authority

Administrators or local user group members with execution rights for this command.

Usage

Note the following points when configuring the VLAN IDs and names for a role:

- For VLAN access and VLAN trunk native respectively, it is recommended to configure only one of either VLAN ID or name for a role. In case both VLAN ID and name are configured, then VLAN ID takes precedence and is applied with the role.
- For VLAN trunk allowed, you can collectively configure a maximum of 50 names and 1024 VLAN IDs on the 6300, 6400 Switch Series, or 256 VLAN IDs on the 6200 Switch Series. In case this limit is exceeded in the role, then that role is rejected when applying it to an onboarding device.

Examples

Configuring VLAN modes and VLAN IDs for a new role:

```
switch(config)# port-access role role01
switch(config-pa-role)# vlan trunk native 10
switch(config-pa-role)# vlan trunk allowed 11-15
switch(config-pa-role)# vlan access 50
```

Configuring VLAN modes and VLAN names for a new role:

```
switch(config)# port-access role role10
switch(config-pa-role)# vlan trunk native name hpe01
switch(config-pa-role)# vlan trunk allowed name data
switch(config-pa-role)# vlan trunk allowed name voice
switch(config-pa-role)# vlan trunk allowed name video
```

Deleting VLAN configuration from a role:

```
switch(config-pa-role)# no vlan trunk native 10
switch(config-pa-role)# no vlan trunk allowed 10-15
switch(config-pa-role)# no vlan access 50
```

Deleting trunk allowed VLAN names from a role individually:

```
switch(config-pa-role)# no vlan trunk native name hpe01
switch(config-pa-role)# no vlan trunk allowed name data
switch(config-pa-role)# no vlan trunk allowed name voice
switch(config-pa-role)# no vlan trunk allowed name video
```

Deleting trunk allowed VLAN names from a role all at once:

```
switch(config-pa-role)# no vlan trunk native name hpe01
switch(config-pa-role)# no vlan trunk allowed name
```

associate-vlan

Syntax

```
associate-vlan <VLAN-ID>  
no associate-vlan <VLAN-ID>
```

Description

Associates VLANs with an existing VLAN group.

The `no` form of this command removes the association of the VLAN with the specified VLAN group.

Command context

`config-pa-vlan-group`

Parameters

<VLAN-ID>

Specifies the VLAN or a specific set of VLANs. Range 1 to 4094.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Associating VLANs with **group1**:

```
switch(config)# port-access vlan-group group1  
switch(config-pa-vlan-group)# associate-vlan 5,10-15,20,21
```

Associating additional VLANs with **group1**:

```
switch(config)# port-access vlan-group group1  
switch(config-pa-vlan-group)# associate-vlan 30-40
```

Dissociating VLANs 10-15 from VLAN **group1**:

```
switch(config-pa-vlan-group)# no associate-vlan 10-15
```

port-access vlan-group

Syntax

```
port-access vlan-group <NAME>  
no port-access vlan-group <NAME>
```

Description

Creates the specified VLAN group (if it does not already exist) and then enters its context `config-pa-vlan-group`. For an existing VLAN group, this command enters the context of the specified VLAN group.

The `no` form of this command removes the specified VLAN group.

In order for the group to be applied to a client, VLANs associated to the group should be configured on the switch. If not, the role displays an error.

Command context

`config`

Parameters

`<NAME>`

Specifies the name of the VLAN group. Range 2 to 32 characters.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Creating VLAN **group1** and associating VLANs with it:

```
switch(config)# port-access vlan-group group1
switch(config-pa-vlan-group)# associate-vlan 5,10-15,20,21
```

Dissociating VLANs 10-15 from VLAN **group1**:

```
switch(config-pa-vlan-group)# no associate-vlan 10-15
```

show running-config port-access vlan-group

Syntax

```
show running-config port-access vlan-group
```

Description

Shows information for all configured VLAN groups.

Command context

Operator (>) or Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

Showing the port access VLAN group configuration:

```
switch# show running-config port-access vlan-group  
...  
port-access vlan-group group1  
    associate-vlan 5,20,21,30-40  
port-access vlan-group group2  
    associate-vlan 50-60,75-85  
...
```


portfilter

Syntax

```
portfilter <INTERFACE-LIST>
```

```
no portfilter [<INTERFACE-LIST>]
```

Description

Configures the specified ports so they do not egress any packets that were received on the source port specified in interface context.

The `no` form of this command removes the port filter setting from one or more ingress ports/LAGs.

Command context

```
config-if  
config-lag-if
```

Parameters

<INTERFACE-LIST>

Specifies a list of ports/LAGs to be blocked for egressing. Specify a single interface or LAG, or a range as a comma-separated list, or both. For example: 1/1/1, 1/1/3-1/1/6, lag2, lag1-lag4.

Authority

Administrators or local user group members with execution rights for this command.

Usage

When a port filter configuration is applied on the same ingress physical port/LAG, the configuration is updated with the new sets of egress ports/LAGs that are to be blocked for egressing and that are not a part of its previous configuration. Duplicate updates on an existing port filter configuration are ignored.

When egress ports/LAGs are removed from the existing port filter configuration of an ingress port/LAG, egressing is allowed again on those egress ports/LAGs for all packets originating from the ingress port/LAG.

The `no portfilter [<IF-NAME-LIST>]` command removes port filter configurations from the egress ports/LAGs listed in the <IF-NAME-LIST> parameter only. All other egress ports/LAGs in the port filter configuration of the ingress port/LAG remain intact.

If no physical ports or LAGs are provided for the `no portfilter` command, the command removes the entire port filter configuration for the ingress port/LAG.

Examples

Creating a filter that prevents packets received on port **1/1/1** from forwarding to ports **1/1/3-1/1/6** and to LAGs **1** through **4**:

```
switch(config)# interface 1/1/1
switch(config-if)# portfilter 1/1/3-1/1/6,lag1-lag4
```

Creating a filter that prevents packets received on LAG 1 from forwarding to ports 1/1/6 and LAGs 2 and 4:

```
switch(config)# interface lag 1
switch(config-lag-if)# portfilter 1/1/6,lag2,lag4
```

Removing filters from an existing configuration that allows back packets received on port 1/1/1 to forward to ports 1/1/6 and LAGs 3 and 4:

```
switch(config)# interface 1/1/1
switch(config-if)# no portfilter 1/1/6,lag3,lag4
```

Removing all filters from an existing configuration that allows back packets received on LAG 1 to forward to all the ports and LAGs:

```
switch(config)# interface lag 1
switch(config-lag-if)# no portfilter
```

show portfilter

Syntax

```
show portfilter [<IFNAME>]
```

Description

Displays filter settings for all interfaces or a specific interface.

Command context

Operator (>) or Manager (#)

Parameters

<IFNAME>

Specifies the ingress interface name.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Displaying all port filter settings on the switch:

```
switch# show portfilter
Incoming   Blocked
Interface  Outgoing Interfaces
-----
1/1/1      1/1/3-1/1/6,lag1-lag2
```

```
1/1/3      1/1/1,1/1/5,1/1/7,1/1/9,1/1/11,1/1/13,1/1/15,1/1/17,1/1/19,1/1/21,
           1/1/23,1/1/25,1/1/27,1/1/29,1/1/31,1/1/33,1/1/35
lag2       1/1/1,1/1/3-1/1/6
```

Displaying the port filter settings for port **1/1/1**:

```
switch# show portfilter 1/1/1
Incoming    Blocked
Interface   Outgoing Interfaces
-----
1/1/1       1/1/3-1/1/6,lag1-lag2
```

Displaying the port filter settings for **LAG2**:

```
switch# show portfilter lag2
Incoming    Blocked
Interface   Outgoing Interfaces
-----
lag2        1/1/1,1/1/3-1/1/6
```

port-access port-security

Syntax

```
port-access port-security {enable | disable}
```

Description

Enables or disables port security globally or at the port level.

Command context

```
config  
config-if
```

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling port security globally:

```
switch(config)# port-access port-security enable
```

Disabling port security globally:

```
switch(config)# port-access port-security disable
```

Enabling port security on a port:

```
switch(config-if)# port-access port-security enable
```

Disabling port security on a port:

```
switch(config-if)# port-access port-security disable
```

port-access port-security client-limit

Syntax

```
port-access port-security client-limit <CLIENTS>  
no port-access port-security client-limit
```

Description

Configures the maximum number of clients that are allowed on a port. After configuring the maximum clients limit, the MAC addresses of the clients can be learned by one of the following methods:

- User can manually configure all MAC addresses by using the `mac-address` command.
- User can allow the port to dynamically learn all MAC addresses.
- User can configure a fixed number of MAC addresses and allow the switch to learn the remaining addresses dynamically.

The `no` form of the command resets the number of clients to the default, 1.

Command context

`config-if`

Parameters

`<CLIENTS>`

Specifies the maximum number of clients. Default: 1. Range: 1 to 32 (6100).

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring client limit on a port:

```
switch(config-if)# port-access port-security enable
switch(config-if-port-security)# client-limit 64
```

port-access port-security mac-address

Syntax

```
port-access port-security mac-address <MAC-ADDRESS>
no port-access port-security mac-address <MAC-ADDRESS>
```

Description

Configures the MAC addresses of static clients.

The `no` form of this command removes an authorized static client from the port.

Command context

`config-if`

Parameters

`<MAC-ADDRESS>`

Specifies the MAC addresses of the static clients.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring a static client on a port:

```
switch(config-if)# port-access port-security
switch(config-if-port-security)# mac-address aa:bb:cc:dd:ee:ff
```

Deleting a static client on a port:

```
switch(config-if)# port-access port-security
switch(config-if-port-security)# no mac-address aa:bb:cc:dd:ee:ff
```

show port-access port-security interface client-status

Syntax

```
show port-access port-security interface {all|<IF-NAME>}
client-status [mac <MAC-ADDRESS>]
```

Description

Shows port security clients status information for the ports. The output can be filtered by interface or MAC address.

Command context

Operator (>) or Manager (#)

Parameters

all

Selects all interfaces.

<IF-NAME>

Specifies the interface name.

<MAC-ADDRESS>

Specifies the client MAC address.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Showing client status information for all ports:

```
switch# show port-access port-security interface all client-status
```

Port Security Client Status Details

Authorized-Clients	Type	Port
AB:CD:DE:FF:AA:BB	static	1/1/1
DD:CD:AB:CD:EE:01	dynamic	1/1/2
00:50:56:96:7e:fc	sticky-dynamic	1/3/2

Showing client status information with sticky-learning enabled for all ports:

```
switch# show port-access port-security interface all client-status
```

Port Security Client Status Details

Authorized-Clients	Type	Port
AB:CD:DE:FF:AA:BB	sticky-static	1/1/1
DD:CD:AB:CD:EE:01	sticky-dynamic	1/1/2
DE:CD:AB:BB:EE:02	sticky-dynamic	1/1/2

Showing client status information for a client:

```
switch# show port-access port-security interface 1/3/2 client-status mac  
00:50:56:96:7e:fc
```

Port Security Client Status Details

Authorized-Clients	Type	Port
00:50:56:96:7e:fc	sticky-dynamic	1/3/2

Showing client status information for a port:

```
switch# show port-access port-security interface 1/3/2 client-status
```

Port Security Client Status Details

Authorized-Clients	Type	Port
00:50:56:96:7e:fc	sticky-dynamic	1/3/2

show port-access port-security interface port-statistics

Syntax

```
show port-access port-security interface {all|<IF-NAME>} port-statistics
```

Description

Shows port security statistics for the ports in a switch. The output can be filtered by interface.

Command context

Operator (>) or Manager (#)

Parameters

all

Selects all interfaces.

<IF-NAME>

Specifies the interface name.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Showing information for all ports.

```
switch# show port-access port-security interface all port-statistics

Port 1/1/1
=====

Client Details
-----
Number of authorized clients           : 0
Number of sticky authorized clients    : 2
```

sticky-learn enable

Syntax

```
sticky-learn enable
no sticky-learn enable
```

Description

Enables sticky learning on the port. All the existing and new MACs learned on the port are made sticky. The `no` form of this command disables the sticky learning on the port.

Command context

```
config-if
```

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling sticky learning on the port:

```
switch(config)# interface 1/1/1
switch(config-if)# port-access port-security
switch(config-if-port-security)# sticky-learn enable
```

Disabling sticky learning on the port:

```
switch(config)# interface 1/1/1
switch(config-if)# port-access port-security
switch(config-if-port-security)# no sticky-learn enable
```

show port-access security violation sticky-mac-client-move interface

Syntax

```
show port-access security violation sticky-mac-client-move interface {all|<INTERFACE-NAME>}
```

Description

Shows information about the sticky-mac client move violation. The output can be filtered by interface.

Command context

Operator (>) or Manager (#)

Parameters

all

Specifies all interfaces.

<INTERFACE-NAME>

Specifies the interface name.

Authority

Operators or Administrators or local user group members with execution rights for this command.

Operators can execute this command from the operator context (>) only.

Examples

Showing information for all ports.

```
switch# show port-access port-security violation sticky-mac-client-move  
interface all
```

Sticky MAC Client Move Violation Status Details

Port	Violation	Violation-Count
1/1/1	No	0
1/1/2	Yes	10
1/1/5	No	10

Showing information for a particular port.

```
switch# show port-access port-security violation sticky-mac-client-move  
interface 1/1/1
```

Sticky MAC Client Move Violation Status Details

Port	Violation	Violation-Count
1/1/1	No	10

apply qos

Syntax

```
apply qos [queue-profile <QUEUE-NAME>] schedule-profile <SCHEDULE-NAME>
no apply qos schedule-profile <SCHEDULE-NAME>
```

Description

Applies a queue profile and schedule profile globally to all Ethernet and LAG interfaces on the switch, or applies a schedule profile to a specific interface. When applied globally, the specified schedule profile is configured only on Ethernet interfaces and LAGs that do not already have their own schedule profile.

The same profile can be applied both globally and locally to an interface. This guarantees that an interface always uses the specified profile, even if the global profile is changed.

The `no` form of this command removes the specified schedule profile from an interface and the interface uses the global schedule profile. This is the only way to remove a schedule profile override from the interface.



Interfaces may shut down briefly during reconfiguration.

Command context

```
config
config-if
config-lag-if
```

Parameters

`queue-profile <QUEUE-NAME>`

Specifies the name of the queue profile to apply. Range: 1 to 64 alphanumeric characters, including period (.), underscore (_), and hyphen (-). This parameter is not supported in the `config-if` context.

`schedule-profile <SCHEDULE-NAME>`

Specifies the name of the schedule profile to apply. Range: 1 to 64 alphanumeric characters, including period (.), underscore (_), and hyphen (-).

Authority

Administrators or local user group members with execution rights for this command.

Usage

- The switch must always have a globally-applied queue and schedule profile. To stop using a given profile, apply a different profile.
- For a queue profile to be complete and ready to be applied, all eight CoS values must be mapped to a queue.

- For a schedule profile to be complete and ready to be applied, it must define all queues specified in the queue profile. All queues must use the same algorithm, except for the highest numbered queue, which can be **strict**.
- Both the queue profile and the schedule profile must specify the same number of queues.
- Schedule profiles can be modified while applied, but only in ways where a single command will not result in the profile becoming invalid. For example, queue 7 can have the algorithm changed, and weighted queues can have their weights changed.
- Queues must be consecutively defined starting at queue number zero. For example, a four-queue profile with priority values defined for queues 0, 1, 2, 3 is valid, but a four-queue profile which defines priority values for queues 1, 3, 5, and 7 is not.
- There can be only 2, 4, or 8 queues in a queue profile configuration and the queues must be consecutively numbered starting at zero.
- All queues must use the same algorithm except for the highest numbered queue, which may be strict.

If the number of queues was changed from the previous queue profile to the new one, any Ethernet or LAG interfaces with locally applied schedule profiles will program the newly applied global schedule-profile. The *show running-config interface* command will list the existing *apply qos schedule-profile* command with a comment describing the actual profile applied:

Examples

The following commands illustrate a valid configuration where every CoS value is assigned to a queue and all assigned queues are defined:

```
switch(config)# qos trust cos
switch(config)# qos queue-profile Q1
switch(config)# map queue 0 cos 0
switch(config)# map queue 1 cos 1
switch(config)# map queue 2 cos 2
switch(config)# map queue 3 cos 3
switch(config)# map queue 4 cos 4
switch(config)# map queue 5 cos 5
switch(config)# map queue 6 cos 6
switch(config)# map queue 7 cos 7
switch(config)# qos schedule-profile S1
switch(config)# min-bandwidth queue 0 percent 5
switch(config)# min-bandwidth queue 1 percent 5
switch(config)# min-bandwidth queue 2 percent 10
switch(config)# min-bandwidth queue 3 percent 10
switch(config)# min-bandwidth queue 4 percent 20
switch(config)# min-bandwidth queue 5 percent 20
switch(config)# min-bandwidth queue 6 percent 10
switch(config)# min-bandwidth queue 7 percent 20
```

The following commands illustrate an invalid configuration because CoS 2 is not assigned to a queue:

```
switch(config)# qos trust cos
switch(config)# qos queue-profile Q1
switch(config)# map queue 0 cos 0
switch(config)# map queue 1 cos 1
switch(config)# map queue 3 cos 3
switch(config)# map queue 4 cos 4
switch(config)# map queue 5 cos 5
switch(config)# map queue 6 cos 6
switch(config)# map queue 7 cos 7
```

```
switch(config)# qos schedule-profile S1
switch(config)# min-bandwidth queue 0 percent 5
switch(config)# min-bandwidth queue 1 percent 5
switch(config)# min-bandwidth queue 3 percent 10
switch(config)# min-bandwidth queue 4 percent 20
switch(config)# min-bandwidth queue 5 percent 20
switch(config)# min-bandwidth queue 6 percent 10
switch(config)# min-bandwidth queue 7 percent 20
```

Applying the QoS profile **Q1** and the schedule profile **S1** to all interfaces that do not have an applied interface-specific schedule profile:

```
switch(config)# apply qos queue-profile Q1 schedule-profile S1
```

map queue

Syntax

```
map queue <QUEUE-NUMBER> cos <PRIORITY-NUMBER>
no map queue <QUEUE-NUMBER> [cos <PRIORITY-NUMBER>]
```

Description

Assigns a CoS value to a queue in a queue profile. By default, the larger the queue number the higher its priority. A queue without a CoS value assigned to it is not used to store packets. The same queue can be assigned multiple CoS values.

The `no` form of this command removes the specified cos value from a specific queue. If no CoS value is specified, then all CoS values are removed from the queue.

Command context

config-queue

Parameters

<QUEUE-NUMBER>

Specifies the queue number. Range: 0 to 7.

<PRIORITY-NUMBER>

Specifies the CoS value. Range: 0 to 7, where 0 is the lowest priority and 7 is the highest.

Authority

Administrators or local user group members with execution rights for this command.

Usage

The following commands illustrate a valid configuration, where every local priority value is assigned to a queue:

```
map queue 0 local-priority 0
map queue 1 local-priority 1
map queue 1 local-priority 2
map queue 3 local-priority 3
map queue 4 local-priority 4
map queue 5 local-priority 5
```

```
map queue 5 local-priority 6
map queue 5 local-priority 7
```

The following commands illustrate an invalid configuration, because local priority 2 is not assigned to a queue:

```
map queue 0 local-priority 0
map queue 1 local-priority 1
map queue 2 local-priority 3
map queue 3 local-priority 4
map queue 4 local-priority 5
map queue 5 local-priority 6
map queue 5 local-priority 7
```

Examples

Assigning priority **7** to queue **7** in profile **myprofile**:

```
switch(config)# qos queue-profile myprofile
switch(config-queue)# map queue 7 local-priority 7
```

Removing priority **7** from queue **7** in profile **myprofile**:

```
switch(config)# qos queue-profile myprofile
switch(config-queue)# no map queue 7 local-priority 7
```

min-bandwidth

Syntax

```
min-bandwidth queue <QUEUE-NUMBER> percent <VALUE>
no min-bandwidth queue <QUEUE-NUMBER>
```

Description

Assigns the Guaranteed Minimum Bandwidth (GMB) algorithm and a percentage of bandwidth to a queue. GMB allocates available bandwidth among all non-empty queues in relation to their configured minimum bandwidth. Non-empty queues are serviced first in strict order up to their minimum bandwidth. If there is any remaining bandwidth, the scheduler will strictly service any remaining non-empty queues.

The **no** form of this command only clears the algorithm for a queue if GMB has been assigned.

Command context

config-schedule

Parameters

<QUEUE-NUMBER>

Specifies the queue number. Range: 0 to 7.

<VALUE>

Specifies bandwidth percentage used for GMB scheduling. Range: 0 to 100.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Assigning queue 0 of schedule profile S1 the GMB scheduling algorithm with minimum bandwidth of 5 percent:

```
switch(config)# qos schedule-profile S1
switch(config-schedule)# min-bandwidth queue 0 percent 5
```

Removing GMB from queue 0:

```
switch(config)# qos schedule-profile s1
switch(config-schedule)# no min-bandwidth queue 0
```

name queue

Syntax

```
name queue <QUEUE-NUMBER> <DESCRIPTION>
no name queue <QUEUE-NUMBER>
```

Description

Assigns a description to a queue in a queue profile. This is for identification purposes and has no effect on configuration.

The `no` form of this command removes the description associated with a queue.

Command context

config-queue

Parameters

<QUEUE-NUMBER>

Specifies the queue number. Range: 0 to 7.

<DESCRIPTION>

Specifies a queue description for identification purposes. Range: 1 to 64 alphanumeric characters, including period (.), underscore (_), and hyphen (-).

Authority

Administrators or local user group members with execution rights for this command.

Examples

Assigning the description **priority-traffic** to queue 7:

```
switch(config)# qos queue-profile myprofile
switch(config-queue)# name queue 7 priority-traffic
```

Removing the description from queue 7:

```
switch(config)# qos queue-profile myprofile
switch(config-queue)# no name queue 7
```

qos cos

Syntax

```
qos cos <CODE-POINT>
no qos cos
```

Description

Configures a CoS PCP remark for an Ethernet or LAG interface. Packets that ingress on the interface are remarked at egress using the configured CoS PCP value.

The remark only occurs when QoS trust mode on the interface is set to `none`.

If QoS trust mode is not set to `none`, then the remark is ignored, and the following commands will show the CoS remark status as `ignored (incompatible Port Access Trust configuration)` or `not applied' (incompatible QoS global/port Trust configuration)`:

- `show running-configuration`
- `show interface <PORT-NUM>`
- `show interface <PORT-NUM> qos`

The `no` form of this command removes a CoS remark on an interface.

Command context

`config-if`

Parameters

`<CODE-POINT>`

Specifies an 802.1 VLAN priority CoS value. Range: 0 to 7.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring a CoS remark of **3** on interface **1/1/1**:

```
switch(config)# interface 1/1/1
switch(config-if)# qos trust none
switch(config-if)# qos cos 3
```

Deleting a CoS remark of **3** on interface **1/1/1**:

```
switch(config)# interface 1/1/1
switch(config-if)# no qos cos
```

qos dscp

Syntax

```
qos dscp <CODE-POINT>
no qos dscp
```

Description

Configures a differentiated services code point (DSCP) remark for an Ethernet or LAG interface. IPV4 and IPV6 packets that ingress on the interface are remarked at egress using the configured DSCP value.

The remark only occurs when QoS trust mode on the interface is set to `none`. If a DSCP remark is configured and then trust mode is subsequently set to `cos` or `dscp`, then the DSCP remark is ignored.

The following commands will show the remark status as *ignored* (incompatible Port Access Trust configuration) or *not applied* (incompatible QoS global or port trust configuration):

- `show running-configuration`
- `show interface <INTERFACE-NAME>`
- `show interface <INTERFACE-NAME> qos`

The `no` form of this command removes a CoS remark on an interface.

Command context

```
config-if
config-lag-if
```

Parameters

<CODE-POINT>

Specifies an IP differentiated services code point value. Range: 0 to 63.

Authority

Administrators or local user group members with execution rights for this command.

Usage

Order of operation for arriving IPv4 or IPv6 packets:

1. The CoS metadata is assigned from the DSCP map entry indexed by the DSCP remark value.
2. If a CoS remark is also configured along with the DSCP remark, the CoS remark value will be assigned to the packet's CoS metadata.
3. The CoS metadata and queue profile are then used to determine the queue for the packet. If the packet is transmitted with an 802.1Q VLAN tag, the PCP will be remarked to match the CoS metadata.

For arriving non-IP packets:

The CoS metadata is assigned from the DSCP map entry indexed by the DSCP remark value. This CoS value and the queue profile are used to select the queue for packet scheduling. The PCP of tagged non-IP packets will be remarked to this CoS value.

Examples

Configuring a DSCP remark of **43** on interface **1/1/1**:

```
switch(config)# interface 1/1/1
switch(config-if)# qos trust none
switch(config-if)# qos dscp 43
```


Deleting a DSCP remark of **43** on interface **1/1/1**:

```
switch(config)# interface 1/1/1
switch(config-if)# no dscp 43
```

qos dscp-map

Syntax

```
qos dscp-map <CODE-POINT> cos <COS-VALUE> [name <DESCRIPTION>]
no qos dscp-map <CODE-POINT>
```

Description

Defines the CoS value assigned to incoming packets for a specific IP differentiated services code point (DSCP) value. The DSCP map values are used to prioritize incoming packets when QoS trust mode is set to **dscp**.

The **no** form of this command restores the assignments for a code point to the default setting.

Use **show qos dscp-map** to view the current settings. To see the default DSCP map settings, use the following command:

```
switch# show qos dscp-map default
DSCP      code_point  cos   name
-----
000000    0             0     CS0
000001    1             0
000010    2             0
000011    3             0
000100    4             0
000101    5             0
...
101101    45            5
101110    46            5
101111    47            5
110000    48            6     CS6
...
111100    60            7
111101    61            7
111110    62            7
111111    63            7
```

Command context

config

Parameters

<CODE-POINT>

Specifies an IP differentiated services code point. Range: 0 to 63. Default: 0.

cos <COS-VALUE>

Specifies an 802.1p VLAN priority CoS remark value. Range: 0 to 7. Default 0.

cos <PCP-VALUE>

Specifies an optional 802.1p VLAN Priority Code Point remark value. Range: 0 to 7. Default: No remark.

name <DESCRIPTION>

Specifies a description for the DSCP setting. The name is used for identification only, and has no effect on queue configuration. Range: 1 to 64 alphanumeric characters, including period (.), underscore (_), and hyphen (-).

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting code point **41** to a CoS of **6**:

```
switch(config)# qos dscp-map 41 cos 6
```

Setting code point **41** to the default value:

```
switch(config)# no qos dscp-map 41
```

qos queue-profile

Syntax

```
qos queue-profile <NAME>  
no qos queue-profile <NAME>
```

Description

Creates a new QoS queue profile and switches to the `config-queue` context for the profile. Or, if the specified QoS queue profile exists, this command switches to the `config-queue` context for the profile. . A queue profile maps queues to CoS values. Each profile has two, four, or eight queues numbered 0 to 7. The larger the queue number, the higher its priority during transmission scheduling.

A queue profile named **factory-default** is defined by default and is automatically applied to all interfaces. It cannot be edited or deleted. Use the command `show qos queue-profile factory-default` to view this profile. Do not use `show running-configuration`, as it will only display changes from the initial values. Use `show qos queue-profile [NAME]` to view the status of all queue profiles or the settings of the named profile.

The `no` form of this command removes the specified QoS queue profile. Only profiles that are not currently applied can be removed.

Command context

`config`

Parameters

<NAME>

Specifies the name of the QoS queue profile to create or configure. Range: 1 to 64 alphanumeric characters, including period (.), underscore (_), and hyphen (-).

Authority

Administrators or local user group members with execution rights for this command.

Examples

Creating the profile **myprofile**:

```
switch(config)# qos queue-profile myprofile
switch(config-queue)#
```

Deleting the profile **myprofile**:

```
switch(config)# no qos queue-profile myprofile
```

qos schedule-profile

Syntax

```
qos schedule-profile <NAME>
no qos schedule-profile <NAME>
```

Description

Creates a QoS schedule profile and switches to the `config-schedule` context for the profile. If the specified schedule profile exists, this command switches to the `config-schedule` context for the profile. The schedule profile determines the order in which queues are selected to transmit a packet, and the amount of service defined for each queue.

Command context

config

Parameters

<NAME>

Specifies the name of the QoS schedule profile to create or configure. Range: 1 to 64 alphanumeric characters, including period (.), underscore (_), and hyphen (-).

Authority

Administrators or local user group members with execution rights for this command.

Usage

Queues in a schedule profile are numbered consecutively starting from zero. Queue zero is the lowest priority queue. The larger the queue number, the higher priority the queue has in scheduling algorithms.

A profile named **factory-default** is defined by default and applied to all interfaces. It cannot be edited or deleted. To see its settings, use the command:

```
switch# show qos schedule-profile factory-default
queue_num algorithm      percent max-bandwidth_kbps
-----
0          min-bandwidth 2
1          min-bandwidth 3
2          min-bandwidth 30
3          min-bandwidth 10
4          min-bandwidth 10
5          min-bandwidth 10
6          min-bandwidth 15
7          min-bandwidth 20
```

A profile named **strict** is predefined and cannot be edited or deleted. The strict profile services all queues of the queue profile to which it is applied, using the strict priority algorithm.

A schedule profile must be defined on all interfaces at all times.

There are two permitted configurations for a schedule profile:

1. All queues use the same scheduling algorithm (for example, GMB).
2. The highest queue number uses strict priority, and all remaining (lower) queues use the same algorithm (for example, GMB). This supports priority scheduling behavior necessary for the IETF RFC 3246 Expedited Forwarding specification (<https://tools.ietf.org/html/rfc3246>).

Only limited changes can be made to an applied schedule profile:

- The percentage of a GMB queue.
- The bandwidth of a strict queue.
- The algorithm of the highest numbered queue can be swapped between GMB and strict, and vice versa.

Applicable to REST: Any other changes will result in an unusable schedule profile, and the switch will revert to the `factory-default` profile until the profile is corrected.

The `no` form of this command removes the specified QoS schedule profile when it is not applied. Only profiles that are not currently applied to an interface can be removed.

Examples

Creating the schedule profile **myschedule**:

```
switch(config)# qos schedule-profile myschedule
switch(config-schedule)#
```

Deleting the schedule profile **myschedule**:

```
switch(config)# no qos schedule-profile myschedule
```

qos trust

Syntax

```
qos trust {none | cos | dscp}
no qos trust
```

Description

Configures one of three modes that are applied globally on all Ethernet interfaces and LAGs that have not applied their own trust mode. Trust mode determines whether VLAN or IP headers are used to assign CoS values to ingress packets.

In the `config` context:

- This command sets the trust mode that is globally applied to all interfaces that do not have a trust mode configured.
- The `no` form of this command restores all interfaces that do not currently have a trust mode configured to the default setting.

In the `config-if` context:

- This command sets the trust mode override for a specific interface.
- The `no` form of this command clears a trust mode override. The interface then uses the global setting. This is the only way to remove a trust mode override.

Command context

config
config-if
config-lag-if

Parameters

none

Ignores all packet headers. Ingress packets are assigned CoS value zero.

cos

For 802.1 VLAN-tagged packets, use the priority code point field from the outermost VLAN header to assign the CoS value. For untagged packets, the CoS value is assigned to zero. Default.

dscp

For IP packets, use the DSCP as the index into the DSCP Map table to obtain the CoS value for the packet. For non-IP packets, the CoS value is assigned to zero.

Authority

Administrators or local user group members with execution rights for this command.

Example

Setting the global trust mode to **dscp**, which is applied to all interfaces that do not already have an individual trust mode configured. An override is then applied to interface **2/2/2**, and LAG 100, setting trust mode to **cos**:

```
switch(config)# qos trust dscp
switch(config)# interface 2/2/2
switch(config-if)# qos trust cos
switch(config-if)# interface lag 100
switch(config-if)# qos trust cos
```



WARNING: QoS port remark configurations are not applied when the QoS trust mode is *mode*. This warning message is seen if a port trust command other than *trust none* is attempted when there is already a remark configuration on the port. To restore the old remark configuration, configure the port trust mode to *none*.



WARNING: QoS port remark configurations are not applied when the global QoS trust mode is *mode*. This warning message is seen if a port *no qos trust* command is attempted when there is already a remark configuration on the port and the global trust mode is not *none*. To re-apply the remark configuration, set the port trust mode to *none*.

rate-limit

Syntax

```
rate-limit {broadcast | multicast | unknown-unicast | icmp {ip-all | ipv4 | ipv6}} <RATE>
kbps
no rate-limit {broadcast | multicast | unknown-unicast | icmp}
```

Description

Sets the amount of traffic of a specific type that can ingress on an Ethernet interface, or on each port of a LAG interface. Rate limits are enforced separately on each individual member of a LAG, not on the LAG as a whole.

The `no` form of this command removes the traffic limit for the specified traffic type.

Command context

`config-if`

Parameters

```
{broadcast | multicast | unknown-unicast| icmp {ip-all | ipv4 | ipv6}}
```

Specifies the type of ingress traffic to which the rate limit applies: broadcast, multicast, unknown-unicast, or ICMP.

The multicast rate limit affects multicast and broadcast traffic. The broadcast rate limit only affects broadcast traffic. When both types are applied to the same interface, broadcast packets are limited to the lower of the two rate values. Layer 2 BPDU packets, like spanning tree, are also included in the multicast rate limit.

Unknown-unicast packets may be intended for devices whose addresses have temporarily aged out of network forwarding caches. Configuring rate limits can help provide balance between necessary and flooded traffic.

The ICMP rate limit can be configured to apply to IPv4, IPv6, or all IP traffic. Only one ICMP rate-limit can be configured at a time. Applying a new ICMP rate-limit replaces any previous ICMP rate-limit.

<RATE>

Specifies the rate limit. Range: 64 to 100000000 kbps (in steps of 64 kbps). The actual rate limit varies with steps approximately equal to the minimum value. Verify the actual rate limit using the command `show interface <INTERFACE-NAME>`.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Limiting broadcast traffic to **1024kbps** on interface **1/1/3**:

```
switch(config)# interface 1/1/3
switch(config-if)# rate-limit broadcast 1024 kbps
```

Limiting all ICMP IPv4 traffic to **10000kbps** on interface **1/1/3**:

```
switch(config)# interface 1/1/3
switch(config-if)# rate-limit icmp ipv4 1024 kbps
```

Viewing the results of the previous configuration settings:

```
switch# show interface 1/1/3 qos
Interface 1/1/3 is up
```

```

Admin state is up
qos trust cos (global)
qos queue-profile factory-default (global)
qos schedule-profile factory-default (global)
rate-limit unknown-unicast 1024 kbps (1024 actual)
rate-limit broadcast 1024 kbps (1100 actual)
rate-limit multicast 1024 kbps (1100 actual)
rate-limit icmp ip-all 1024 kbps (1024 actual)

switch# show interface 1/1/3

Interface 1/1/3 is up
Admin state is up
Link state: up for 3 minutes (since Thu Nov 26 17:56:14 UTC 2020)
Link transitions: 1
Description:
Hardware: Ethernet, MAC Address: f8:60:f0:c9:21:bc
MTU 1500
Type 1GbT
Full-duplex
qos trust cos
rate-limit unknown-unicast 1024 kbps (1024 actual)
rate-limit broadcast 1024 kbps (1100 actual)
rate-limit multicast 1024 kbps (1100 actual)
rate-limit icmp ip-all 1024 kbps (1024 actual)
Speed 1000 Mb/s
Auto-negotiation is on
Energy-Efficient Ethernet is disabled
Flow-control: off
Error-control: off
MDI mode: MDIX
VLAN Mode: access
Access VLAN: 1
Rx
    0 total packets
    0 unicast packets
    0 multicast packets
    0 broadcast packets
    0 errors
    0 CRC/FCS
    0 total bytes
    0 dropped
    0 pause
Tx
1057962 total packets
    0 unicast packets
    0 multicast packets
1058039 broadcast packets
    0 errors
    0 collision
366066962 total bytes
    0 dropped
    0 pause

```

show interface queues

Syntax

show interface <INTERFACE-NAME> queues

Description

Displays interface-level queue statistics.

Command context

Operator (>) or Manager (#)

Parameters

<INTERFACE-NAME>

Specifies the name of an Ethernet port or LAG on the switch. Format: member/slot/port or lag number.

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Usage

Statistics include:

- **Tx Bytes:** Total bytes transmitted. The byte count may include packet headers and internal metadata that are removed before the packet is transmitted. Packet headers added when the packet is transmitted may not be included. The byte count includes any packets subsequently dropped by an egress ACL .
- **Tx Packets:** Total packets transmitted. The count includes packets subsequently dropped by an egress ACL .
- **Tx Drops:** Total packets dropped by an egress queue due to insufficient capacity.

Examples

Showing queue statistics for interface **1/1/5**:

```
switch# show interface 1/1/5 queues
Interface 1/1/5 is down
Admin state is up
```

	Tx Bytes	Tx Packets	Tx Drops
Q0	0	0	3
Q1	15356	73	0
Q2	0	0	0
Q3	0	0	0
Q4	0	0	0
Q5	0	0	0
Q6	0	0	0
Q7	0	0	0

Showing queue statistics for interface **lag 1**:

```
switch# show interface lag 1 queues
Aggregate-name lag1
Aggregated-interfaces :
1/1/6 1/1/7
Speed 20000 Mb/s
```

	Tx Bytes	Tx Packets	Tx Drops
Q0	0	0	0
Q1	0	0	0
Q2	0	0	0
Q3	0	0	0
Q4	0	0	0
Q5	0	0	0
Q6	0	0	0
Q7	3450	25	0

show interface qos

Syntax

```
show interface <INTERFACE-NAME> qos
```

Description

Shows various QoS settings for a specific interface.

Command context

Operator (>) or Manager (#)

Parameters

<INTERFACE-NAME>

Specifies the name of an interface on the switch. Format: member/slot/port or lag number.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Showing QoS settings for interface **1/1/5**:

```
switch# show interface 1/1/5 qos
Interface 1/1/5 is up
Admin state is up
qos trust cos (global)
qos queue-profile factory-default (global)
qos schedule-profile factory-default (global)
qos cos 5
qos dscp 47
rate-limit broadcast 40000 kbps (40000 actual)
rate-limit icmp ip-all 10000 kbps (10000 actual)
```

show qos dscp-map

Syntax

```
show qos dscp-map [default]
```

Description

Displays the current or default global QoS dscp-map.

Command context

Operator (>) or Manager (#)

Parameters

default

Shows the factory default DSCP code point settings.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Showing the current QoS DSCP map:

```
switch# show qos dscp-map
DSCP      code_point  cos   name
-----
000000    0           0     CS0
000001    1           0
000010    2           0
000011    3           0
000100    4           0
000101    5           0
...
101101    45          5
101110    46          6     new
101111    47          5
110000    48          6     CS6
...
111101    61          7
111110    62          7
111111    63          7
```

Showing the default QoS DSCP map:

```
switch# show qos dscp-map default
DSCP      code_point  cos   name
-----
000000    0           0     CS0
000001    1           0
000010    2           0
000011    3           0
000100    4           0
000101    5           0
...
101101    45          5
101110    46          5
101111    47          5
110000    48          6     CS6
...
111100    60          7
111101    61          7
111110    62          7
111111    63          7
```

show qos queue-profile

Syntax

```
show qos queue-profile [<NAME> | factory-default]
```

Description

Shows the status of all queue profiles, or a specific queue profile.

Command context

Operator (>) or Manager (#)

Parameters

<NAME>

Specifies the name of a queue profile. Range 1 to 64 alphanumeric characters, including period (.), underscore (_), and hyphen (-).

[factory-default]

Specifies the factory default queue profile.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Usage

The status of a queue profile can be:

- Applied - The profile is actively being used by the switch.
- Complete - The profile meets the criteria to be applied.
- Incomplete - The profile does not meet the criteria to be applied.

For a queue profile to be complete and ready to be applied:

- All eight cos values must be mapped to some queue.
- There can be only 2, 4, or 8 queues.
- The queues must be consecutively numbered starting at zero.

Examples

Showing the settings of the factory default queue profile:

```
switch# show qos queue-profile factory-default
queue_num cos name
-----
0          1
1          0
2          2
3          3
4          4
5          5
6          6
7          7
```

show qos schedule-profile

Syntax

```
show qos schedule-profile [<NAME> | factory-default | strict]
```

Description

Shows the status of all schedule profiles, or a specific schedule profile.

Command context

Operator (>) or Manager (#)

Parameters

<NAME>

Specifies the name of a queue or schedule profile. Range: 1 to 64 alphanumeric characters, including period (.), underscore (_), and hyphen (-).

[factory-default]

Specifies the factory default queue profile.

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Usage

The status of a schedule profile can be:

- Applied - The profile is actively being used by one or more ports.
- Complete - The profile meets the criteria to be applied.
- Incomplete - The profile does not meet the criteria to be applied.

For a schedule profile to be complete and ready to be applied it must have:

- An algorithm for each queue defined by the applied queue profile.
- All queues must use the same algorithm except for the highest numbered queue, which may be strict.

Example

Showing the status of all schedule profiles:

```
switch# show qos schedule-profile
profile_status  profile_name
-----
applied         myschedule
complete        factory-default
complete        Test
```

Showing the configuration of factory default schedule profile:

```
switch# show qos schedule-profile factory-default
queue_num  algorithm      percent  max-bandwidth_kbps
-----
0          min-bandwidth  2
1          min-bandwidth  3
2          min-bandwidth  30
3          min-bandwidth  10
4          min-bandwidth  10
5          min-bandwidth  10
6          min-bandwidth  15
7          min-bandwidth  20
```

show qos trust

Syntax

```
show qos trust [default]
```

Description

Shows the global QoS trust settings, or the factory default settings.

Command context

Operator (>) or Manager (#)

Parameters

default

Shows the factory default QoS trust settings.

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Examples

Showing the current QoS trust settings:

```
switch# show qos trust
qos trust cos
```

Showing the default QoS trust settings:

```
switch# show qos trust default
qos trust cos
```

strict queue

Syntax

```
strict queue <QUEUE-NUMBER> [[max-bandwidth <BANDWIDTH>]]
no strict queue <QUEUE-NUMBER>
```

Description

Assigns the strict priority algorithm to a queue. Strict priority services all packets waiting in a queue, before servicing the packets in lower priority queues.

Egress queue shaping can be configured using the `max-bandwidth` option to limit the amount of traffic transmitted per output queue. The buffer associated with each egress queue stores the excess traffic to smooth the output rate. Sustained rates of traffic above the maximum bandwidth will eventually fill the output queue causing tail drops. Use the command `show interface` to determine if any tail drop errors have occurred.

The `no` form of this command removes the queue configuration from the schedule profile. To remove only egress queue shaping, re-enter the `strict queue` command without the `max-bandwidth` parameter.

Command context

config-schedule

Parameters

<QUEUE-NUMBER>

Specifies the number of the queue. Range: 0 to 7.

max-bandwidth <BANDWIDTH>

Specifies the maximum bandwidth allowed on the queue in Kbps. Range: 468 to 100000000.

Authority

Administrators or local user group members with execution rights for this command.

Usage

Either all the queues of the schedule profile can be *strict* or just the highest numbered queue. When applied to a LAG, each member Ethernet port independently schedules its egress transmissions using the strict settings. Only limited changes can be made to a *strict* queue that is part of an applied schedule profile:

- The max-bandwidth settings.
- The highest numbered queue can be swapped between *strict* and *min-bandwidth*
-
-

Any other changes or removing a queue (`no strict queue`) will result in an unusable schedule profile. If that schedule profile is applied in the interface context, the switch will revert to the schedule profile applied in the global context until the profile is corrected. If that schedule profile is applied in the global context, the switch will revert to using the factory-default profile until the profile is corrected.

Examples

Assigning strict priority to queue **7** in the schedule profile **myschedule**:

```
switch(config)# qos schedule-profile myschedule
switch(config-schedule)# strict queue 7
```

Deleting strict priority from queue **7** in the schedule profile **myschedule**:

```
switch(config)# qos schedule-profile myschedule
switch(config-schedule)# no strict queue 7
```

Assigning strict priority to queue **7** in the schedule profile **myschedule** with a maximum bandwidth of 10000 Kbps:

```
switch(config)# qos schedule-profile myschedule
switch(config-schedule)# strict queue 7 max-bandwidth 10000
```

aaa radius-attribute group

Syntax

```
aaa radius-attribute group <GROUP-NAME>
no aaa radius-attribute group <GROUP-NAME>
```

Description

Configures an existing RADIUS server group for which the configured RADIUS attributes will be included in request packets. Enters the `config-radius-attr` context.

The `no` form of this command unconfigures the RADIUS server group for the configured RADIUS attributes.

Command context

config

Parameters

<GROUP-NAME>

Specifies an existing RADIUS server group name.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring various RADIUS attributes for **rad_group1**:

```
switch(config)# aaa radius-attribute group rad_group1
switch(config-radius-attr)# nas-id value ARUBA_NAS-01
switch(config-radius-attr)# nas-id request-type authentication
switch(config-radius-attr)# tunnel-private-group-id value static
switch(config-radius-attr)# tunnel-private-group-id request-type authentication
```

Unconfiguring RADIUS attributes for **rad_group1**:

```
switch(config)# no aaa radius-attribute group rad_group1
```

nas-id request-type

Syntax

```
nas-id request-type {authentication | accounting | both}
no nas-id request-type {authentication | accounting | both}
```

Description

For the selected (by context) RADIUS server group, configures the Network Access Server (NAS) ID request type for which the attribute configured with command `nas-id value` will be included.

The `no` form of this command unconfigures the specified request type.

Command context

`config-radius-attr`

Parameters

`authentication`

Selects the authentication request type.

`accounting`

Selects the accounting request type.

`both`

Selects both the authentication and accounting request types.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring the authentication request type for **rad_group1**:

```
switch(config)# aaa radius-attribute group rad_group1
switch(config-radius-attr)# nas-id request-type authentication
```

Configuring both the authentication and accounting request types for **rad_group2**:

```
switch(config)# aaa radius-attribute group rad_group2
switch(config-radius-attr)# nas-id request-type both
```

Unconfiguring the authentication request type for **rad_group1**:

```
switch(config)# aaa radius-attribute group rad_group1
switch(config-radius-attr)# no nas-id request-type authentication
```

nas-id value

Syntax

```
nas-id value <NAS-ID>
no nas-id [value <NAS-ID>]
```

Description

For the selected (by context) RADIUS server group, configures the Network Access Server Identifier (NAS ID) (type 32, RFC 2865). The NAS ID is sent in the RADIUS access request and accounting packets to notify the source of the RADIUS access request.

The `no` form of this command unconfigures the specified NAS ID.

Command context

`config-radius-attr`

Parameters

<NAS-ID>

Specifies the FQDN or other unique identifying name of the Network Access Server (NAS). Range 1 to 253 characters.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring the Network Access Server (NAS) ID for **rad_group1**:

```
switch(config)# aaa radius-attribute group rad_group1
switch(config-radius-attr)# nas-id value ARUBA_NAS-01
```

Unconfiguring the NAS ID for **rad_group1**:

```
switch(config)# aaa radius-attribute group rad_group1
switch(config-radius-attr)# no nas-id value ARUBA_NAS-01
```

Unconfiguring both the NAS-ID value and the request type for **rad_group2**:

```
switch(config)# aaa radius-attribute group rad_group2
switch(config-radius-attr)# no nas-id
```

tunnel-private-group-id request-type

Syntax

```
tunnel-private-group-id request-type {authentication | accounting | both}
no tunnel-private-group-id request-type {authentication | accounting | both}
```

Description

For the selected (by context) RADIUS server group, configures the request type for which the attribute configured with command `tunnel-private-group-id value` will be included.

The no form of this command unconfigures the specified request type.

Command context

config-radius-attr

Parameters

authentication

Selects the authentication request type.

accounting

Selects the accounting request type.

both

Selects both the authentication and accounting request types.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring the authentication request type for **rad_group1**:

```
switch(config)# aaa radius-attribute group rad_group1
switch(config-radius-attr)# tunnel-private-group-id request-type authentication
```

Configuring both the authentication and accounting request types for **rad_group2**:

```
switch(config)# aaa radius-attribute group rad_group2
switch(config-radius-attr)# tunnel-private-group-id request-type both
```

Unconfiguring the authentication request type for **rad_group2**:

```
switch(config)# aaa radius-attribute group rad_group2
switch(config-radius-attr)# no tunnel-private-group-id request-type authentication
```

tunnel-private-group-id value

Syntax

```
tunnel-private-group-id value {static | dynamic}
no tunnel-private-group-id value {static | dynamic}
```

Description

For the selected (by context) RADIUS server group, configures the `tunnel-private-group-id value` (type 81, RFC 2868) that will be sent in RADIUS access-request packets. This is used for VLAN identification.

The `no` form of this command unconfigures specified `tunnel-private-group-id value`.

Command context

`config-radius-attr`

Parameters

`static`

Causes the switch to send (as an attribute value) the native VLAN of the client port.

`dynamic`

Causes the switch to send (as an attribute value) the client VLAN assigned by server. This is applicable during re-authentication scenarios.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring **rad_group1** for the RADIUS attribute to identify the native VLAN of the client port:

```
switch(config)# aaa radius-attribute group rad_group1
switch(config-radius-attr)# tunnel-private-group-id value static
```

Configuring **rad_group2** for the RADIUS attribute to identify the client VLAN assigned by the server:

```
switch(config)# aaa radius-attribute group rad_group2  
switch(config-radius-attr)# tunnel-private-group-id value dynamic
```

Unconfiguring (for **rad_group1**) the RADIUS attribute to identify the native VLAN of the client port:

```
switch(config)# aaa radius-attribute group rad_group1  
switch(config-radius-attr)# no tunnel-private-group-id value static
```

Unconfiguring (for **rad_group3**) both the group-ID value and request type:

```
switch(config)# aaa radius-attribute group rad_group3  
switch(config-radius-attr)# no tunnel-private-group-id
```



On the 6100 Switch Series, only the vrf named `default` is available. Replace any references to the `mgmt` or other VRFs with `default`.

radius dyn-authorization enable

Syntax

```
radius dyn-authorization enable
no radius dyn-authorization enable
```

Description

Enables RADIUS dynamic authorization. This command must be issued before the configuration set with other `radius dyn-authorization` commands takes effect.

The `no` form of this command disables RADIUS dynamic authorization.

Command context

`config`

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling RADIUS dynamic authorization:

```
switch(config)# radius dyn-authorization enable
```

radius dyn-authorization client

Syntax

```
radius dyn-authorization client {<IPV4> | <IPV6> | <HOSTNAME>}
    [secret-key [plaintext <PASSKEY> | ciphertext] <PASSKEY>]]
    [time-window <WIDTH>] [replay-protection {enable|disable}]

no radius dyn-authorization client {<IPV4> | <IPV6> | <HOSTNAME>} [vrf <VRF-NAME>]
```

Description

Configures RADIUS dynamic authorization for the specified client on the specified (or `default`) VRF.

The `no` form of this command unconfigures RADIUS dynamic authorization for the specified client on the specified (or `default`) VRF.

Command context

config

Parameters

<IPV4> | <IPV6> | <HOSTNAME>

Specifies the client IPv4 address, IPv6 address, or host name.

secret-key [plaintext <PASSKEY> | ciphertext <PASSKEY>]

Specifies the dynamic authorization server (RADIUS server) shared secret key required for client access.

Provide either a plaintext or an encrypted shared-secret passkey. As per RFC 2865, the shared-secret can be a mix of alphanumeric and special characters. Plaintext passkeys are between 1 and 32 alphanumeric and special characters.



When `secret-key` is entered without either sub-parameter, plaintext shared secret prompting occurs upon pressing Enter. Enter must be pressed immediately after the `secret-key` parameter without entering other parameters. The entered shared secret characters are masked with asterisks.

time-window <WIDTH>

Specifies the width of the synchronization window (in seconds) between the RADIUS dynamic authorization client and the RADIUS dynamic authorization server. Default 300. Range: 1 to 65535.

replay-protection {enable|disable}

Enables or disables RADIUS dynamic authorization replay protection for the specified client on the specified (or default) VRF.

vrf <VRF-NAME>

Specifies the VRF on which the identified client is connected. When omitted, VRF `default` is assumed.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring RADIUS dynamic authorization with replay protection for a client on the default VRF:

```
switch(config)# radius dyn-authorization client 1.1.2.5 replay-protection enable
```

Configuring RADIUS dynamic authorization with time window and shared secret for a client on the default VRF:

```
switch(config)# radius dyn-authorization client 1.1.2.7 time-window 8
secret-key plaintext skF82#450
```

Configuring RADIUS dynamic authorization with a prompted shared secret:

```
switch(config)# radius dyn-authorization client 1.1.2.7 secret-key
Enter the RADIUS dyn-authorization key: *****
Re-Enter the RADIUS dyn-authorization key: *****
```

radius dyn-authorization port

Syntax

radius dyn-authorization port <PORT-NUMBER>

Description

Sets the RADIUS dynamic authorization server UDP/TCP port.

Command context

config

Parameters

<PORT-NUMBER>

Specifies the UDP or TCP port. Default UDP: 3799 and TCP:2083

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the RADIUS dynamic authorization server UDP port back to its default 3799:

```
switch(config)# radius dyn-authorization port 3799
```

Setting the RADIUS dynamic authorization server TCP port back to its default 2083:

```
switch(config)# radius dyn-authorization port 2083
```

show radius dyn-authorization

Syntax

show radius dyn-authorization

Description

Shows RADIUS dynamic authorization configuration and summarized statistics for all clients configured for dynamic authorization.

Command context

Operator (>) or Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Usage

Show command output item identification:

- Radius Dynamic Authorization: Enabled or Disabled status, system wide.
- Radius Dynamic Authorization Port: The UDP or TCP port used for dynamic authorization (default 3799).
- Invalid Client Address in CoA Requests: The number of CoA (change of authorization) requests received with an incorrect DAC (dynamic authorization client) address.

- Invalid Client Address in Disconnect Requests: The number of disconnect requests received with incorrect DAC address.
- Disconnect Requests: The number of disconnect requests received from the DAC.
- Disconnect ACKs: The number of Disconnect-ACKs sent to the DAC.
- Disconnect NAKs: The number of Disconnect-NAKs sent to the DAC.
- CoA Requests: The number of CoA-requests received from the DAC.
- CoA ACKs: The number of CoA-ACKs sent to the DAC.
- CoA NAKs: The number of CoA-NAKs sent to the DAC.

Example

Showing RADIUS dynamic authorization summarized statistics for all clients configured for dynamic authorization:

```
switch# show radius dyn-authorization
Status and Counters - RADIUS Dynamic Authorization Information

RADIUS Dynamic Authorization           : Enabled
RADIUS Dynamic Authorization UDP Port  : 3799
Invalid Client Addresses in CoA Requests : 0
Invalid Client Addresses in Disconnect Requests: 0

Dynamic Authorization Client Information
=====

IP Address      : 1.1.2.1
VRF             : adm2
Replay Protection : Disabled
Time Window     : 20
Disconnect Requests : 1
Disconnect ACKs   : 1
Disconnect NAKs   : 0
CoA Requests     : 7
CoA ACKs         : 2
CoA-NAKs         : 5
Shared-Secret    :
AQBapb+HsdpqV1Q3CPCBMQTG8ekK1cA+CyD0RvfbeA8BEgikCgAAAJOWZSNzA2SWrLA=

IP Address      : 1.1.2.5
VRF             : default
Replay Protection : Enabled
Time Window     : 20
Disconnect Requests : 6
Disconnect ACKs   : 6
Disconnect NAKs   : 0
CoA Requests     : 9
CoA ACKs         : 5
CoA-NAKs         : 4
Shared-Secret    :
AQBapb+HsdpqV1Q3CPCBMQTG8ekK1cA+CyD0RvfbeA8BEgikCgAAAJOWZSNzA2SWrLA=

IP Address      : 1.1.2.7
VRF             : default
Replay Protection : Disabled
Time Window     : 8
Disconnect Requests : 6
Disconnect ACKs   : 6
```

```
Disconnect NAKs      : 0
CoA Requests         : 9
CoA ACKs             : 5
CoA-NAKs             : 4
Shared-Secret        :
AQBapb+HsdpqVlQ3CPCBMQTG8ekK1cA+CyD0RvfbeA8BEgikCgAAAJ0wZSNzA2SWrLA=
```

show radius dyn-authorization client

Syntax

```
show radius dyn-authorization client <IP-ADDR> [vrf <VRF-NAME>]
```

Description

Shows RADIUS dynamic authorization statistics for the specified client on the specified VRF.

Command context

Operator (>) or Manager (#)

Parameters

client <IP-ADDR>

Specifies the client IPv4 or IPv6 address.

vrf <VRF-NAME>

Specifies the VRF on which the identified client is connected. When omitted, VRF `default` is assumed.

Authority

Operators or Administrators or local user group members with execution rights for this command.

Operators can execute this command from the operator context (>) only.

Usage

Show command output item identification:

- **Total Requests:** The number of Disconnect and CoA (change of authorization) requests received from the DAC (dynamic authorization client).
- **Authorize Only Requests:** The number of Disconnect and CoA requests received from the DAC with an "Authorize only" Service-Type attribute.
- **Malformed Requests:** The number of malformed Disconnect and CoA requests received from the DAC.
- **Bad Authenticator Requests:** The number of Disconnect and CoA requests received from this DAC with an invalid authenticator field.
- **Dropped Requests:** The number of Disconnect and CoA requests from this DAC that have been silently discarded for reasons other than malformed, bad authenticators, or unknown type.
- **Total ACK Responses:** The number of Disconnect-ACKs sent to the DAC.
- **Total NAK Responses:** The number of Disconnect-NAKs sent to the DAC.
- **Session Not Found Responses:** The number of Disconnect-NAKs sent to the DAC because no session context could be found.
- **User Sessions Modified:** The number of user sessions for which authorization changed due to Disconnect and CoA requests received from the DAC.

Example

Showing RADIUS dynamic authorization statistics for client 1.1.2.1 on VRF default:

```
switch# show radius dyn-authorization client 1.1.2.1 vrf default
Status and Counters - RADIUS Dynamic Authorization Client Information
```

VRF Name	: default	
Authorization Client	: 1.1.2.1	
Unknown Packets	: 55	
Message-Type	Disconnect	CoA

Total Requests	2147483647	10
Authorize Only Requests	10	10
Malformed Requests	10	10
Bad Authenticator Requests	2147483647	2147483647
Dropped Requests	10	10
Total ACK Responses	10	10
Total NAK Responses	10	10
Session Not Found Responses	10	10
User Sessions Modified	20	20



On the 6100 Switch Series, only the vrf named `default` is available. Replace any references to the `mgmt` or other VRFs with `default`.

aaa accounting all-mgmt

Syntax

```
aaa accounting all-mgmt <CONNECTION-TYPE> start-stop {local | group <GROUP-LIST>}  
no aaa accounting all-mgmt <CONNECTION-TYPE>
```

Description

Defines accounting as being local (with the name `local`) (the default). Or defines a sequence of remote AAA server groups to be accessed for accounting purposes.

For remote accounting, the information is sent to the first reachable remote server that was configured with this command for remote accounting. If no remote server is reachable, local accounting remains available. Each available connection type (channel) can be configured individually as either local or using remote AAA server groups. All server groups named in your command, must exist. This command can be issued multiple times, once for each connection type. Local is always available for any connection type not configured for remote accounting.



The system accounting log is not associated with any connection type (channel) and is therefore sent to the accounting method configured on the default connection type (channel) only.

The `no` form of this command removes for the specified connection type, any defined remote AAA server group accounting sequence. Local accounting is available for connection types without a configured remote AAA server group list (whether default or for the specific connection type).

Command context

`config`

Parameters

`<CONNECTION-TYPE>`

One of these connection types (channels):

`default`

Defines a list of accounting server groups to be used for the `default` connection type. This configuration applies to all other connection types (`console`, `https-server`, `ssh`) that are not explicitly configured with this command. For example, if you do not use `aaa accounting all-mgmt console...` to define the console accounting list, then this default configuration is used for console.

`console`

Defines a list of accounting server groups to be used for the `console` connection type.

`https-server`

Defines a list of accounting server groups to be used for the `https-server` (REST, Web UI) connection type.

`ssh`

Defines a list of accounting server groups to be used for the `ssh` connection type.

`start-stop`

Selects accounting information capture at both the beginning and end of a process.

`local`

Selects local-only accounting when used without the `group` parameter.

`group <GROUP-LIST>`

Specifies the list of remote AAA server group names. Each name can be specified one time. Predefined remote AAA group names `tacacs` and `radius` are available. Although not a group name, predefined name `local` is available. User-defined TACACS+ and RADIUS server group names may also be used. The remote AAA server groups are accessed in the order that the group names are listed in this command. Within each group, the servers are accessed in the order in which the servers were added to the group. Server groups are defined using command `aaa group server` and servers are added to a server group with the command `server`.

Authority

Administrators or local user group members with execution rights for this command.

Usage

Local accounting is always active. It cannot be turned off.

Examples

Defining the default accounting sequence based on two user-defined TACACS+ server groups, then the default TACACS+ server group, and finally (if needed), local accounting.

```
switch(config)# aaa accounting all-mgmt default start-stop group tg1 tg2 tacacs local
```

Defining the console accounting sequence based on two user-defined TACACS+ server groups, then the default TACACS+ server group, and finally (if needed), local accounting.

```
switch(config)# aaa accounting all-mgmt console start-stop group tg2 tg3 tacacs local
```

Defining the `ssh` accounting sequence based on one user-defined TACACS+ server group and then the default TACACS+ server group.

```
switch(config)# aaa accounting all-mgmt ssh start-stop group tg2 tacacs
```

Defining the default accounting sequence based on two user-defined RADIUS server groups, then the default RADIUS server group, and finally (if needed), local accounting.

```
switch(config)# aaa accounting all-mgmt default start-stop group rg1 rg2 radius local
```

Defining the `https-server` accounting sequence based on one user-defined RADIUS server group and then the default RADIUS server group.

```
switch(config)# aaa accounting all-mgmt https-server start-stop group rg1 radius
```

Setting local accounting for the default connection type:

```
switch(config)# aaa accounting all-mgmt default start-stop local
```

aaa authentication allow-fail-through

Syntax

```
aaa authentication allow-fail-through  
no aaa authentication allow-fail-through
```

Description

Enables authentication fail-through. When this option is enabled, the next server/authentication method is tried after an authentication failure.

The `no` form of this command disables authentication fail-through. If the system fails to authenticate with a reachable TACACS+ or RADIUS server, the system does not attempt to authenticate with the next TACACS+/RADIUS server.

Command context

config

Authority

Administrators or local user group members with execution rights for this command.

Example

Enabling authentication fail-through:

```
switch(config)# aaa authentication allow-fail-through
```

aaa authentication login

Syntax

```
aaa authentication login <CONNECTION-TYPE> {local | group <GROUP-LIST>}  
no aaa authentication login <CONNECTION-TYPE>
```

Description

Defines authentication as being local (with the name `local`) (the default). Or defines a sequence of remote AAA server groups to be accessed for authentication purposes. Each available connection type (channel) can be configured individually as either local or using remote AAA server groups. All server groups named in your command, must exist. This command can be issued multiple times, once for each connection type. Local is always available for any connection type not configured for remote AAA authentication.



If you do not want local authentication to occur in cases where all AAA servers contacted reject the user's credentials, do not enable authentication fail-through (command `aaa authentication allow-fail-through`).

The `no` form of this command removes for the specified connection type, any defined remote AAA server group authentication sequence. Local authentication is available for connection types without a configured remote AAA server group list (whether default or for the specific connection type).

Command context

config

Parameters

<CONNECTION-TYPE>

One of these connection types (channels):

`default`

Defines a list of authentication server groups to be used for the `default` connection type. This configuration applies to all other connection types (`console`, `https-server`, `ssh`) that are not explicitly configured with this command. For example, if you do not use `aaa authentication login console...` to define the console authentication list, then this default configuration is used for `console`.

`console`

Defines a list of authentication server groups to be used for the `console` connection type.

`https-server`

Defines a list of authentication server groups to be used for the `https-server` (REST, Web UI) connection type.

`ssh`

Defines a list of authentication server groups to be used for the `ssh` connection type.

`local`

Selects local-only authentication when used without the `group` parameter.

`group` <GROUP-LIST>

Specifies the list of remote AAA server group names. Each name can be specified one time. Predefined remote AAA group names `tacacs` and `radius` are available. Although not a group name, predefined name `local` is available. User-defined TACACS+ and RADIUS server group names may also be used. The remote AAA server groups are accessed in the order that the group names are listed in this command. Within each group, the servers are accessed in the order in which the servers were added to the group. Server groups are defined using command `aaa group server` and servers are added to a server group with the command `server`.

If no AAA server is reachable, local authentication is attempted.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Defining the default authentication sequence based on two user-defined TACACS+ server groups, then the default TACACS+ server group, and finally (if needed), local authentication.

```
switch(config)# aaa authentication login default group tg1 tg2 tacacs local
```

Defining the console authentication sequence based on two user-defined TACACS+ server groups, then the default TACACS+ server group, and finally (if needed), local authentication.

```
switch(config)# aaa authentication login console group tg2 tg3 tacacs local
```

Defining the ssh authentication sequence based on one user-defined TACACS+ server group and then the default TACACS+ server group.

```
switch(config)# aaa authentication login ssh group tg2 tacacs
```

Defining the default authentication sequence based on two user-defined RADIUS server groups, then the default RADIUS server group, and finally (if needed), local authentication.

```
switch(config)# aaa authentication login default group rg1 rg2 radius local
```

Defining the https-server authentication sequence based on one user-defined RADIUS server group and then the default RADIUS server group.

```
switch(config)# aaa authentication login https-server group rg1 radius
```

Setting local authentication for the default connection type:

```
switch(config)# aaa authentication login default local
```

aaa authorization commands

Syntax

```
aaa authorization commands <CONNECTION-TYPE> {local | none}
aaa authorization commands <CONNECTION-TYPE> group <GROUP-LIST>

no aaa authorization commands <CONNECTION-TYPE>
```

Description

Defines authorization as being basic local RBAC (specified as `none`), or as full-fledged local RBAC specified as `local` (the default), or as remote TACACS+ (specified with `group <GROUP-LIST>`). Each available connection type (channel) can be configured individually. All server groups named in the command, must exist. This command can be issued multiple times, once for each connection type.

The `no` form of this command unconfigures authorization for the specified connection type, reverting to the default of `local`.



Although only TACACS+ servers are supported for remote authorization, local authorization (basic or full-fledged) can be used with remote RADIUS authentication.

Command context

config

Parameters

<CONNECTION-TYPE>

One of these connection types (channels):

`default`

Selects the `default` connection type for configuration. This configuration applies to all other connection types (`console`, `ssh`) that are not explicitly configured with this command. For example, if

you do not use `aaa authorization commands console...` to define the console authorization list, then this default configuration is used for console.

`console`

Selects the `console` connection type for configuration.

`ssh`

Selects the `ssh` connection type for configuration.

`local` (the default)

When used alone without `group <GROUP-LIST>`, selects local authorization which can be used to provide authorization for a purely local setup without any remote AAA servers and also for when RADIUS is used for remote Authentication and Accounting but Authorization is local.

When used after `group`, provides for fallback (to full-fledged local authorization) when every server in every specified TACACS+ server group cannot be reached.



If any TACACS+ server in the specified groups is reachable, but the command fails to be authorized by that server, the command is rejected and local authorization is never attempted. Local authorization is only attempted if every TACACS+ server cannot be reached.

`none`

When used alone without `group <GROUP-LIST>`, selects basic local RBAC authorization, for use with the built-in user groups (`administrators`, `operators`, `auditors`).

When used after `group`, provides for fallback (to basic local RBAC authorization) when every server in every specified TACACS+ server group cannot be reached.



With `none`, for users belonging to user-defined user groups, all commands can be executed regardless of what authorization rules are defined in such groups. For per-command local authorization, use `local` instead.

`group <GROUP-LIST>`

Specifies the list of remote AAA server group names. Predefined remote AAA group name `tacacs` is available. User-defined TACACS+ server group names may also be used. The remote AAA server groups are accessed in the order that the group names are listed in this command. Within each group, the servers are accessed in the order in which the servers were added to the group. Server groups are defined using command `aaa server group` and servers are added to a server group using command `server`.

It is recommended to always include either the special name `local` or `none` as the last name in the group list. If both `local` and `none` are omitted, and no remote AAA server is reachable (or the first reachable server cannot authorize the command), command execution for the current user will not be possible.

Authority

Administrators or local user group members with execution rights for this command.

Usage

TACACS+ server authorization considerations



Use caution when configuring authorization, as it has no fail through. If the switch is not configured properly, the switch might get into an unusable state in which all command execution is prohibited.

To prevent authorization difficulties:

- Make sure that all listed TACACS+ servers can authorize users for command execution.
- Make sure that credential database changes are promptly synchronized across all TACACS+ servers.

- Make sure either `local` or `none` is included as the last name in the group list. If both `local` and `none` are omitted, and no remote TACACS+ server is reachable (or the first reachable server cannot authorize), authorization will not be possible.
- Although not recommended, if you choose to omit both `local` and `none` from the list, and are manipulating configuration files, special caution is necessary. If the source configuration includes TACACS+ authorization and you are copying configuration from an existing switch into the running configuration of a new switch, and you have not yet configured the interface or routing information to reach the TACACS+ server, the switch will enter an unusable state, requiring hard reboot.
To avoid getting into this situation that can occur when `local` and `none` have been omitted, do either of the following:
 - In the configuration source, delete or comment-out the line configuring remote authorization. Then, after the configuration copy and paste, manually configure authorization.
 - Move the line configuring the authorization to the end of the source configuration before copying and pasting.

Examples

Defining the default authorization sequence based on a user-defined TACACS+ server group, then the default TACACS+ server group, and finally (as a precaution), `local` authorization:

```
switch(config)# aaa authorization commands default group tg1 tacacs local
All commands will fail if none of the servers in the group list are reachable.
Continue (y/n)? y
```

Defining the console authorization sequence based on two user-defined TACACS+ server groups, and finally (as a precaution), `local` authorization:

```
switch(config)# aaa authorization commands console group tg1 tg2 local
All commands will fail if none of the servers in the group list are reachable.
Continue (y/n)? y
```

Setting the authorization for default to `local`:

```
switch(config)# aaa authorization commands default local
```

Setting the authorization for the SSH interface to `none`:

```
switch(config)# aaa authorization commands ssh none
```

aaa group server

Syntax

```
aaa group server {tacacs | radius} <SERVER-GROUP-NAME>
```

```
no aaa group server {tacacs | radius} <SERVER-GROUP-NAME>
```

Description

Creates an AAA server group that is either empty or contains preconfigured RADIUS/TACACS+ servers. You can create a maximum of 28 server groups.

The `no` form of this command deletes a server group. Only a preconfigured user-defined RADIUS/TACACS+ server group can be deleted. RADIUS or TACACS+ servers that were in a deleted server group remain a part of their default server group. The default server group for TACACS+ servers is `tacacs`. The default server group for RADIUS servers is `radius`.

Command context

config

Parameters

```
server {tacacs | radius}
```

Select either `tacacs` or `radius` for the server type.

<SERVER-GROUP-NAME>

Specifies the name of the server group to be created. The name of the server group can have a maximum of 32 characters.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Creating TACACS+ server group sg1:

```
switch(config)# aaa group server tacacs sg1
```

Creating RADIUS server group sg3:

```
switch(config)# aaa group server radius sg3
```

Deleting TACACS+ server group sg1:

```
switch(config)# no aaa group server tacacs sg1
```

Deleting RADIUS server group sg3:

```
switch(config)# no aaa group server radius sg3
```

radius-server auth-type

Syntax

```
radius-server auth-type {pap | chap}  
no radius-server auth-type
```

Description

Enables the CHAP or PAP authentication protocol, which is used for communication with the RADIUS servers, at the global level. You can override this command with a fine-grained per server `auth-type` configuration.

The `no` form of this command resets the global authentication mechanism for RADIUS to PAP, which is the default authentication mechanism for RADIUS.

Command context

config

Parameters

auth-type {pap | chap}

Selects either the PAP or CHAP authentication protocol.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Authenticating CHAP:

```
switch(config)# radius-server auth-type chap
```

Authenticating PAP:

```
switch(config)# radius-server auth-type pap
```

radius-server host

Syntax

```
radius-server host {<FQDN> | <IPv4> | <IPv6>}  
    [key [plaintext <PASSKEY> | ciphertext <PASSKEY>]]  
    [timeout <TIMEOUT-SECONDS>] [port <PORT-NUMBER>]  
    [auth-type {pap | chap}] [acct-port <ACCT-PORT>] [retries <RETRY-COUNT>]  
    [tracking {enable | disable}] [tracking-mode {any | dead-only}] [vrf <VRF-NAME>]  
  
no radius-server host {<FQDN> | <IPv4> | <IPv6>} [port <PORT-NUMBER>] [vrf <VRF-NAME>]
```

Description

Adds a RADIUS server. By default, the RADIUS server is associated with the server group named `radius`. The `no` form of this command removes a previously added RADIUS server.



For enhanced security with IPsec, the alternative command `radius-server host secure ipsec` is available. The standard non-IPsec `radius-server host` command does not modify any existing IPsec configuration. If IPsec is already configured for the RADIUS server, then IPsec will remain enabled for the server.

Command context

config

Parameters

{<FQDN> | <IPv4> | <IPv6>}

Specifies the RADIUS server as:

- <FQDN>: a fully qualified domain name.
- <IPv4>: an IPv4 address.

- `<IPV6>`: an IPv6 address.

`key [plaintext <PASSKEY> | ciphertext <PASSKEY>]`

Specifies either a plaintext or an encrypted local shared-secret passkey for the server. As per RFC 2865, the shared-secret can be a mix of alphanumeric and special characters. Plaintext passkeys are between 1 and 32 alphanumeric and special characters.



When `key` is entered without either sub-parameter, plaintext passkey prompting occurs upon pressing Enter. Enter must be pressed immediately after the `key` parameter without entering other parameters. The entered passkey characters are masked with asterisks.

When `key` is omitted, the server uses the global passkey. This command requires either the global or local passkey to be set; otherwise the server will not be contacted. Command `radius-server key` is available for setting the global passkey.

`timeout <TIMEOUT-SECONDS>`

Specifies the timeout. The range is 1 to 60 seconds. If a timeout is not specified, the value from the global timeout for RADIUS is used.

`port <PORT-NUMBER>`

Specifies the authentication port number. Range: 1 to 65535. Default RADIUS: 1812.

`auth-type {pap | chap}`

Selects either PAP (default) or CHAP authentication type. If this parameter is not specified, the RADIUS global default is used.

`acct-port <ACCT-PORT>`

Specifies the UDP accounting port number. Range: 1 to 65535. Default: 1813.

`retries <RETRY-COUNT>`

Specifies the number of retry attempts for contacting the specified RADIUS server. Range is 0 to 5 attempts. If no retry value is provided, the default value of 1 is used.

`tracking {enable | disable}`

Enables or disables server tracking for the RADIUS server. Tracked servers are probed at the start of each server tracking interval to check if they are reachable.

Use command `radius-server tracking` to configure RADIUS server tracking globally.



Server tracking uses authentication request and response packets to determine server reachability status. The server tracking user name and password are used to form the request packet which is sent to the server with tracking enabled. Upon receiving a response to the request packet, the server is considered to be reachable.

`tracking-mode {any | dead-only}`

Configures tracking mode for the RADIUS server that has tracking enabled with the server. The tracking mode is used to monitor the status of RADIUS server reachability. The default tracking mode is `any`.

Sets the tracking mode to:

- `any`: track the RADIUS server irrespective of its server reachability.
- `dead-only`: track the RADIUS server only when the server is marked as unreachable.

`vrf VRF-NAME>`

Specifies the VRF name to be used for communicating with the server. If no VRF name is provided, the default VRF named `default` is used.

Authority

Administrators or local user group members with execution rights for this command.

Usage

If the fully qualified domain name is provided for the RADIUS server, a DNS server must be configured and accessible through the same VRF which is configured for the RADIUS server. This configuration is required for the resolution of the RADIUS server hostname to its IP address. If a DNS server is not available for this VRF, the RADIUS servers reachable through this VRF must be configured by means of their IP addresses only.

Examples

Adding a RADIUS server with an IPv4 address and a prompted passkey:

```
switch(config)# radius-server host 1.1.1.5 key
Enter the RADIUS server key: *****
Re-Enter the RADIUS server key: *****
```

Adding a RADIUS server with an IPv4 address and a named VRF:

```
switch(config)# radius-server host 1.1.1.1 vrf mgmt
```

Adding a RADIUS server with an IPv4 address, a port, and a named VRF:

```
switch(config)# radius-server host 1.1.1.2 port 32 vrf mgmt
```

Adding a RADIUS server with an IPv6 address:

```
switch(config)# radius-server host 2001:0db8:85a3:0000:0000:8a2e:0370:7334
```

Adding a RADIUS server with tracking enabled and tracking mode is set to dead-only:

```
switch(config)# radius-server host 1.1.1.1 tracking enable tracking-mode dead-only
```

Adding a RADIUS server with tracking disabled:

```
switch(config)# radius-server host 1.1.1.1 tracking disable
```

Deleting a RADIUS server with an IPv4 address and specified VRF:

```
switch(config)# no radius-server host 1.1.1.1 vrf mgmt
```

radius-server host secure ipsec

Syntax

Syntax for a RADIUS server that uses IPsec for authentication:

```
radius-server host {<FQDN> | <IPV4> | <IPV6>}
[key [plaintext <PASSKEY> | ciphertext <PASSKEY>]]
[timeout <TIMEOUT-SECONDS>] [port <PORT-NUMBER>]
[auth-type {pap | chap}] [acct-port <ACCT-PORT>] [retries <RETRY-COUNT>]
[tracking {enable | disable}] [tracking-mode {any | dead-only}] [vrf <VRF-NAME>]
secure ipsec authentication spi <SPI-INDEX> <AUTH-TYPE> <AUTH-KEY-TYPE> [<AUTH-KEY>]
```

```
no radius-server host {<FQDN> | <IPV4> | <IPV6>} [port <PORT-NUMBER>]
[vrf <VRF-NAME>] secure ipsec authentication
```

Syntax for a RADIUS server that uses IPsec for both authentication and encryption:

```
radius-server host {<FQDN> | <IPV4> | <IPV6>}
[key [plaintext <PASSKEY> | ciphertext <PASSKEY>]]
[timeout <TIMEOUT-SECONDS>] [port <PORT-NUMBER>]
[auth-type {pap | chap}] [acct-port <ACCT-PORT>] [retries <RETRY-COUNT>]
[tracking {enable | disable}] [tracking-mode {any | dead-only}] [vrf <VRF-NAME>]
secure ipsec encryption spi <SPI-INDEX> <AUTH-TYPE> <AUTH-KEY-TYPE>
[<AUTH-KEY>] <ENCRYPT-TYPE> <ENCRYPT-KEY-TYPE> [<ENCRYPT-KEY>]

no radius-server host {<FQDN> | <IPV4> | <IPV6>} [port <PORT-NUMBER>]
[vrf <VRF-NAME>] secure ipsec encryption
```

Description

Adds a RADIUS server that uses IPsec for enhanced security (authentication and possibly encryption). By default, the RADIUS server is associated with the server group named `radius`.

The `no` form of this command removes a previously added RADIUS (with IPsec) server.



Unless enhanced security with IPsec is required, use the `radius-server host` command instead.

Command context

config

Parameters

{<FQDN> | <IPV4> | <IPV6>}

Specifies the RADIUS server as:

- <FQDN>: a fully qualified domain name.
- <IPV4>: an IPv4 address.
- <IPV6>: an IPv6 address.

key [plaintext <PASSKEY> | ciphertext <PASSKEY>]

Selects either a plaintext or an encrypted local shared-secret passkey for the server. As per RFC 2865, shared-secret can be a mix of alphanumeric and special characters. Plaintext passkeys are between 1 and 32 alphanumeric and special characters.

When `key` is entered without either sub-parameter, plaintext passkey prompting occurs upon pressing Enter. Enter must be pressed immediately after the `key` parameter without entering other parameters. The entered passkey characters are masked with asterisks.

When `key` is omitted, the server uses the global passkey. This command requires either the global or local passkey to be set; otherwise the server will not be contacted. Command `radius-server key` is available for setting the global passkey.

timeout <TIMEOUT-SECONDS>

Specifies the timeout. The range is 1 to 60 seconds. If a timeout is not specified, the value from the global timeout for RADIUS is used.

port <PORT-NUMBER>

Specifies the authentication port number. Range: 1 to 65535. Default: 1812.

auth-type {pap | chap}

Selects either the PAP (the default) or CHAP authentication types. If this parameter is not specified, the RADIUS global default is used.

`acct-port <ACCT-PORT>`

Specifies the UDP accounting port number. Range: 1 to 65535. Default: 1813.

`retries <RETRY-COUNT>`

Specifies the number of retry attempts for contacting the specified RADIUS server. Range is 0 to 5 attempts. If no retry value is provided, the default value of 1 is used.

`tracking {enable | disable}`

Enables or disables server tracking for the RADIUS server. Tracked servers are probed at the start of each server tracking interval to check if they are reachable.

Use command `radius-server tracking` to configure RADIUS server tracking globally.



Server tracking uses authentication request and response packets to determine server reachability status. The server tracking user name and password are used to form the request packet which is sent to the server with tracking enabled. Upon receiving a response to the request packet, the server is considered to be reachable.

`tracking-mode {any | dead-only}`

Configures tracking mode for the RADIUS server that has tracking enabled with the server. The tracking mode is used to monitor the status of RADIUS server reachability. The default tracking mode is `any`.

Sets the tracking mode to:

- `any`: track the RADIUS server irrespective of its server reachability.
- `dead-only`: track the RADIUS server only when the server is marked as unreachable.

`vrf <VRF-NAME>`

Specifies the VRF name to be used for communicating with the server. If no VRF name is provided, the default VRF named `default` is used.

`spi <SPI-INDEX>`

Specifies the Security Parameters Index. The SPI is an identification tag carried in the IPsec AH header. The SPI must be unique on the switch. Range: 256 to 4294967295.

`<AUTH-TYPE>`

Specifies the authentication algorithm: `md5`, `sha1`, or `sha256`.

`<AUTH-KEY-TYPE>`

Specifies the authentication key type: `plaintext`, `hex-string`, or `ciphertext`.

`[ <AUTH-KEY> ]`

Specifies the authentication key. For `<AUTH-TYPE>` of `ciphertext`, this is the ciphertext string.

For `<AUTH-TYPE>` of `plaintext` or `hex-string`:

- `md5 (plaintext)`: 1 to 16 characters, `(hex-string)`: 2 to 32 hexadecimal digits.
- `sha1 (plaintext)`: 1 to 20 characters, `(hex-string)`: 2 to 40 hexadecimal digits.
- `sha256 (plaintext)`: 1 to 32 characters, `(hex-string)`: 2 to 64 hexadecimal digits.



When `<AUTH-KEY-TYPE>` is not followed by `<AUTH-KEY>`, plaintext authentication key prompting occurs upon pressing Enter. Enter must be pressed immediately after the `<AUTH-KEY-TYPE>` parameter without entering other parameters. The entered authentication key characters are masked with asterisks.

`<ENCRYPT-TYPE>`

Specifies the encryption algorithm: `3des`, `aes`, `des`, or `null`.

`<ENCRYPT-KEY-TYPE>`

Specifies the encryption key type: `plaintext`, `hex-string`, or `ciphertext`.

`[ <ENCRYPT-KEY> ]`

Specifies the encryption key. For `<ENCRYPT-TYPE>` of `ciphertext`, this is the ciphertext string.

For *<ENCRYPT-KEY-TYPE>* of plaintext or hex-string:

- 3des (plaintext): 24 characters, (hex-string): 48 hexadecimal digits.
- aes (plaintext): 16, 24, or 32 characters, (hex-string): 32, 48, or 64 hexadecimal digits.
- des (plaintext): 8 characters, (hex-string): 16 hexadecimal digits.



When *<ENCRYPT-KEY-TYPE>* is not followed by *<ENCRYPT-KEY>*, plaintext encryption key prompting occurs upon pressing Enter. Enter must be pressed immediately after the *<ENCRYPT-KEY-TYPE>* parameter without entering other parameters. The entered encryption key characters are masked with asterisks.

Authority

Administrators or local user group members with execution rights for this command.

Usage

If the fully qualified domain name is provided for the RADIUS server host, a DNS server must be configured and accessible through the same VRF as mentioned for the server host. This configuration is required for the resolution of the RADIUS server hostname to its IP address. If a DNS server is not available for this VRF, the RADIUS servers reachable through this VRF must be configured by means of their IP addresses only.

Examples

Adding a RADIUS server with an IPv4 address, a plaintext passkey, and IPsec authentication (md5 plaintext).

```
switch(config)# radius-server host 1.1.1.1 key plaintext 98ab vrf mgmt secure
ipsec authentication spi 261 md5 plaintext labc
```

Adding a RADIUS server with an IPv4 address and a prompted IPsec authentication (md5) plaintext authentication key.

```
switch(config)# radius-server host 1.1.1.1 secure ipsec authentication spi 261 md5
Enter the IPsec authentication key: *****
Re-Enter the IPsec authentication key: *****
```

Adding a RADIUS server with an IPv4 address, IPsec authentication (MD5 plaintext), and IPsec encryption (AES plaintext):

```
switch(config)# radius-server host 1.1.1.2 vrf mgmt secure
ipsec encryption spi 262 md5 plaintext 9xyz aes plaintext 1234567890abcdef
```

Adding a RADIUS server by providing an IPv4 address and IPsec MD5 authentication type, and then responding to prompts for the keys and encryption type:

```
switch(config)# radius-server host 1.1.1.6 secure ipsec encryption spi 262 md5
Enter the IPsec authentication key: *****
Re-Enter the IPsec authentication key: *****

Enter the IPsec encryption type (3des/aes/des/null)? aes

Enter the IPsec encryption key: *****
Re-Enter the IPsec encryption key: *****
```

Adding a RADIUS server with an IPv4 address, tracking enabled, tracking mode, IPsec authentication (MD5 plaintext), IPsec encryption (AES plaintext) is set to dead-only:

```
switch(config)# radius-server host 1.1.1.1 tracking enable tracking-mode dead-only
vrf mgmt secure ipsec encryption spi 262 md5 plaintext 9xyz
aes plaintext 1234567890abcdef
```

Removing a RADIUS server:

```
switch(config)# no radius-server host 1.1.1.1 vrf mgmt
```

Removing the ipsec configuration from a RADIUS server:

```
switch(config)# no radius-server host 1.1.1.2 vrf mgmt secure ipsec encryption
```

radius-server key

Syntax

```
radius-server key [plaintext <GLOBAL-PASSKEY> | ciphertext <GLOBAL-PASSKEY>]
```

```
no radius-server key
```

Description

Creates or modifies a RADIUS global passkey. The RADIUS global passkey is used as a shared-secret for encrypting the communication between all RADIUS servers and the switch. The RADIUS global passkey is required for authentication unless local passkeys have been set. By default, the RADIUS global passkey is empty. If the administrator has not set this key, the switch will not be able to perform RADIUS authentication. The switch will instead rely on the authentication mechanism configured with `aaa authentication login`.



When this command is entered without parameters, plaintext passkey prompting occurs upon pressing Enter. The entered passkey characters are masked with asterisks.

The `no` form of the command removes the global passkey.

Command context

config

Parameters

plaintext <GLOBAL-PASSKEY>

Specifies the RADIUS global passkey in plaintext format with a length of 1 to 31 characters. As per RFC 2865, a shared-secret can be a mix of alphanumeric and special characters.

ciphertext <GLOBAL-PASSKEY>

Specifies the RADIUS global passkey in encrypted format.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Adding the global passkey:

```
switch(config)# radius-server key plaintext mypasskey123
```

Adding the global passkey with prompting:

```
switch(config)# radius-server key
Enter the RADIUS server key: *****
Re-Enter the RADIUS server key: *****
```

Removing the global passkey:

```
switch(config)# no radius-server key
```

radius-server retries

Syntax

```
radius-server retries <0-5>
no radius-server retries
```

Description

Sets at the global level the number of retries the switch makes before concluding that the RADIUS server is unreachable.

You can override this setting with a fine-grained per RADIUS server retries configuration.

The `no` form of this command resets the RADIUS global retries to the default retries value of 1.

Command context

config

Parameters

retries <0-5>

Specifies the number of retry attempts for contacting RADIUS servers. Range is 0 to 5 retries.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

```
switch(config)# radius-server retries 3
```

radius-server timeout

Syntax

```
radius-server timeout [<1-60>]
no radius-server timeout
```

Description

Specifies the number of seconds to wait for a response from the RADIUS server before trying the next RADIUS server. If a value is not specified, a default value of 5 seconds is used. You can override this value with a fine-grained per server timeout configured for individual servers.

The `no` form of this command resets the RADIUS global authentication timeout to the default of 5 seconds.

Command context

`config`

Parameters

`timeout <1-60>`

Specifies the timeout interval of 1 to 60 seconds. The default is 5 seconds.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the RADIUS server timeout:

```
switch(config)# radius-server timeout 10
```

Resetting the timeout for the RADIUS server to the default:

```
switch(config)# no radius-server timeout
```

radius-server tracking

Syntax

`radius-server tracking interval <INTERVAL>`

`no radius-server tracking interval`

`radius-server tracking retries <RETRIES>`

`no radius-server tracking retries`

`radius-server tracking user-name <NAME>`

`[password [plaintext <PASSWORD> | ciphertext <PASSWORD>]]`

`no radius-server tracking user-name <NAME>`

Description

Configures RADIUS server tracking settings globally for all configured RADIUS servers that have tracking enabled with the `radius-server host` command on individual servers.

The `no` form of the command removes the specified configuration, reverting it to its default. The `no` form with `user-name` also clears the password (resets it to empty).

Command context

`config`

Parameters

`interval <INTERVAL>`

Specifies the time interval, in seconds, to wait before checking the server reachability status. Default: 300. Range 60 to 84600.

`retries <RETRIES>`

Specifies the number of server retries. Default: Global RADIUS retries. Range: 0 to 5.

`user-name <NAME> [password [plaintext <PASSWORD> | ciphertext <PASSWORD>]]`

Specifies the user name (and optionally a password) to be used for server checking. The default user name is `radius-tracking-user` with an empty password.

The password is optional and may be entered as `plaintext` or pasted in as `ciphertext`. The plaintext password is visible as cleartext when entered but is encrypted thereafter. Command history does show the password as cleartext.



When `password` is entered without a following sub-parameter, plaintext password prompting occurs upon pressing Enter. The entered password characters are masked with asterisks.



The user does not have to be configured on the server. Server tracking can still be performed with a user which is not configured on the server because authentication failure on the server achieves confirmation that the server is reachable.



Server tracking uses authentication request and response packets to determine server reachability status. The server tracking user name and password are used to form the request packet which is sent to the server with tracking enabled. Upon receiving a response to the request packet, the server is considered to be reachable.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring a tracking interval of 120 seconds:

```
switch(config)# radius-server tracking interval 120
```

Reverting the tracking interval to its default of 300 seconds:

```
switch(config)# no radius-server tracking interval
```

Configuring three retries:

```
switch(config)# radius-server tracking retries 3
```

Configuring user `radius-tracker` with a plaintext password.

```
switch(config)# radius-server tracking user-name radius-tracker  
password plaintext track$1
```

Configuring user `radius-tracker` with a prompted plaintext password.

```
switch(config)# radius-server tracking user-name radius-tracker password
Enter the RADIUS server tracking password: *****
Re-Enter the RADIUS server tracking password: *****
```

Reverting the tracking user name to its default of radius-tracking-user:

```
switch(config)# no radius-server tracking user-name
```

server

Syntax

```
server {<FQDN> | <IPv4> | <IPv6>} [port <PORT-NUMBER>] [vrf <VRF-NAME>]
no server {<FQDN> | <IPv4> | <IPv6>} [port <PORT-NUMBER>] [vrf <VRF-NAME>]
```

Description

Adds a TACACS+/RADIUS server to a server-group. Only the configured TACACS+/RADIUS servers are allowed to be added within the server group. If the same server name exists with multiple ports or multiple VRFs, specify the server name, port, and VRF when adding the server to the server-group.

The `no` form of this command removes a TACACS+/RADIUS server from a server-group.

Command context

config-sg

Parameters

{<FQDN> | <IPv4> | <IPv6>}

Specifies the server as:

- <FQDN>: a fully qualified domain name.
- <IPv4>: an IPv4 address.
- <IPv6>: an IPv6 address.

port <PORT-NUMBER>

Specifies the authentication port number. Range: 1 to 65535. Default TACACS+ (TCP): 49, RADIUS (UDP): 1812.

If a port number is not provided, the system searches the TACACS+/RADIUS server by host name and sets the default authentication port. Group server priority is assigned based on the sequence in which the servers are added.

vrf <VRF-NAME>

Specifies the VRF name.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Adding a server to TACACS+ server group sg1 by providing an IPv4 address, port number, and VRF name:

```
switch(config)# aaa group server tacacs sg1
switch(config-sg)# server 1.1.1.2 port 32 vrf default
```

Adding a server to TACACS+ server group sg2 by providing an IPv6 address and default VRF:

```
switch(config)# aaa group server tacacs sg2
switch(config-sg)# server 2001:0db8:85a3:0000:0000:8a2e:0370:7334 vrf default
```

Adding a server to RADIUS server group sg3 by providing an IPv4 address, port number, and VRF name:

```
switch(config)# aaa group server radius sg3
switch(config-sg)# server 1.1.1.5 port 12 vrf default
```

Adding a server to RADIUS server group sg4 by providing an IPv6 address and default VRF:

```
switch(config)# aaa group server radius sg4
switch(config-sg)# server 2001:0db8:85a3:0000:0000:8a2e:0371:7334 vrf default
```

Adding a server to RADIUS server group sg4 by providing an IPv4 address, port number, and VRF name:

```
switch(config)# aaa group server radius sg4
switch(config-sg)# server 1.1.1.6 port 32 vrf vrf_red
```

Specifying an IPv4 address when removing a TACACS+ server from server group sg1:

```
switch(config)# aaa group server tacacs sg1
switch(config-sg)# no server 1.1.1.2 port 12 vrf default
```

Specifying an IPv6 address when removing a TACACS+ server from server group sg2 with the default VRF:

```
switch(config)# aaa group server tacacs sg2
switch(config-sg)# no server 2001:0db8:85a3:0000:0000:8a2e:0370:7334 vrf default
```

show aaa accounting

Syntax

```
show aaa accounting
```

Description

Shows the accounting configuration per connection type (channel).

Command context

Operator (>) or Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Example

Configuring and then showing the accounting sequence for TACACS+ groups and local:

```
switch(config)# aaa accounting all default start-stop group tg1 tg2 tacacs local
switch(config)# aaa accounting all ssh start-stop group tg1 tg2
switch(config)# aaa accounting all console start-stop group tg4 tacacs local
switch(config)# aaa accounting all https-server start-stop local group tacacs tg3
switch(config)# exit
switch# show aaa accounting
AAA Accounting:
  Accounting Type           : all
  Accounting Mode           : start-stop

Accounting for default channel:
-----
GROUP NAME                  | GROUP PRIORITY
-----
tg1                          | 0
tg2                          | 1
tacacs                       | 2
local                        | 3
-----

Accounting for ssh channel:
-----
GROUP NAME                  | GROUP PRIORITY
-----
tg1                          | 0
tg2                          | 1
-----

Accounting for console channel:
-----
GROUP NAME                  | GROUP PRIORITY
-----
tg4                          | 0
tacacs                       | 1
local                        | 2
-----

Accounting for https-server channel:
-----
GROUP NAME                  | GROUP PRIORITY
-----
local                        | 0
tacacs                       | 1
tg3                          | 2
-----
```

Configuring and then showing the accounting sequence for RADIUS groups and local:

```
switch(config)# aaa accounting all default start-stop group rg1 rg2 radius local
switch(config)# aaa accounting all console start-stop group rg4 radius local
switch(config)# exit
switch# show aaa accounting
AAA Accounting:
  Accounting Type           : all
  Accounting Mode           : start-stop

Accounting for default channel:
```

```

-----
GROUP NAME                | GROUP PRIORITY
-----
rg1                        | 0
rg2                        | 1
radius                    | 2
local                     | 3
-----

Accounting for console channel:
-----
GROUP NAME                | GROUP PRIORITY
-----
tg4                        | 0
radius                    | 1
local                     | 2
-----

```

Configuring and then showing only local accounting for default:

```

switch(config)# aaa accounting all default start-stop local
switch(config)# exit
switch# show aaa accounting
AAA Accounting:
  Accounting Type           : all
  Accounting Mode           : start-stop

Accounting for default channel:
-----
GROUP NAME                | GROUP PRIORITY
-----
local                     | 0
-----

```

show aaa authentication

Syntax

```
show aaa authentication
```

Description

Shows the authentication configuration per connection type (channel).

Command context

Operator (>) or Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Example

Configuring TACACS+ authentication sequences and then showing the configuration per connection type (channel):

```

switch(config)# aaa authentication login default group tg1 tg2 tg3 tg4 tacacs local
switch(config)# aaa authentication login ssh group tg1 tg2
switch(config)# aaa authentication login console group tg4 tacacs local
switch(config)# aaa authentication login https-server local group tacacs tg3
switch(config)# exit

```

```
switch# show aaa authentication
```

```
AAA Authentication:
```

```

Fail-through           : Enabled
Limit Login Attempts   : Not set
Lockout Time           : 300
Minimum Password Length : Not set

```

```
Authentication for default channel:
```

```

-----
GROUP NAME                | GROUP PRIORITY
-----
tg1                        | 0
tg2                        | 1
tg3                        | 2
tg4                        | 3
tacacs                    | 4
local                     | 5
-----

```

```
Authentication for ssh channel:
```

```

-----
GROUP NAME                | GROUP PRIORITY
-----
tg1                        | 0
tg2                        | 1
-----

```

```
Authentication for console channel:
```

```

-----
GROUP NAME                | GROUP PRIORITY
-----
tg4                        | 0
tacacs                    | 1
local                     | 2
-----

```

```
Authentication for https-server channel:
```

```

-----
GROUP NAME                | GROUP PRIORITY
-----
local                     | 0
tacacs                    | 1
tg3                       | 2
-----

```

Configuring RADIUS authentication sequences and then showing the configuration per connection type (channel):

```

switch(config)# aaa authentication login default group rg1 rg2 rg3 rg4 radius local
switch(config)# aaa authentication login console group rg4 radius local
switch(config)# exit

```

```
switch# show aaa authentication
```

```
AAA Authentication:
```

```

Fail-through           : Enabled
Limit Login Attempts   : Not set
Lockout Time           : 300

```



```

Minimum Password Length      : Not set

Authentication for default channel:
-----
GROUP NAME                   | GROUP PRIORITY
-----
rg1                           | 0
rg2                           | 1
rg3                           | 2
rg4                           | 3
radius                        | 4
local                         | 5
-----

Authentication for console channel:
-----
GROUP NAME                   | GROUP PRIORITY
-----
rg4                           | 0
radius                        | 1
local                         | 2
-----

```

Configuring only default authentication and then showing the default connection type (channel):

```

switch(config)# aaa authentication login default local
switch(config)# exit
switch# show aaa authentication

AAA Authentication:
  Fail-through                : Disabled
  Limit Login Attempts        : Not set
  Lockout Time                 : 300
  Minimum Password Length     : Not set

Authentication for default channel:
-----
GROUP NAME                   | GROUP PRIORITY
-----
local                        | 0
-----

```

show aaa authorization

Syntax

```
show aaa authorization
```

Description

Shows the authorization configuration per connection type (channel).

Command context

Operator (>) or Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Example

Configuring and then showing the authorization sequence for default and console connection types (channels):

```
switch(config)# aaa authorization commands default group tg1 tacacs none
All commands will fail if none of the servers in the group list are reachable.
Continue (y/n)? y
switch(config)#
switch(config)# aaa authorization commands console group tg1 tg2 tacacs none
All commands will fail if none of the servers in the group list are reachable.
Continue (y/n)? y
switch(config)# exit
switch#
switch# show aaa authorization
Authorization for default channel:
-----
GROUP NAME                | GROUP PRIORITY
-----
tg1                        | 0
tacacs                    | 1
none                      | 2
-----

Authorization for console channel:
-----
GROUP NAME                | GROUP PRIORITY
-----
tg1                        | 0
tg2                       | 1
tacacs                    | 2
none                      | 3
-----
```

show aaa server-groups

Syntax

```
show aaa server-groups [tacacs | radius]
```

Description

Shows TACACS+ and RADIUS AAA server group information for all server types or for the specified server type.

Command context

Operator (>) or Manager (#)

Parameters

tacacs

Narrows the command output to only TACACS+ servers.

radius

Narrows the command output to only RADIUS servers.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

Showing all AAA server group information:

```
switch# show aaa server-groups

***** AAA Mechanism TACACS+ *****
-----
GROUP NAME          | SERVER NAME                               | PORT | VRF      | PRIORITY
-----
sg2                  | 2001:0db8:85a3:0000:0000:8a2e:         | 49   | default  | 1
                    | 0370:7334                               |      |          |
-----
sg1                  | 1.1.1.2                               | 12   | mgmt     | 1
-----
tacacs (default)    | FQDN.com                               | 32   | mgmt     | 1
tacacs (default)    | 1.1.1.1                               | 49   | mgmt     | 2
tacacs (default)    | 1.1.1.2                               | 12   | mgmt     | 3
tacacs (default)    | abc.com                                | 32   | vrf_red  | 4
tacacs (default)    | 2001:0db8:85a3:0000:0000:8a2e:         | 49   | default  | 5
                    | 0370:7334                               |      |          |
tacacs (default)    | 1.1.1.3                               | 32   | vrf_blue | 6
-----
***** AAA Mechanism RADIUS *****
-----
GROUP NAME          | SERVER NAME                               | PORT | VRF      | PRIORITY
-----
sg4                  | 2001:0db8:85a3:0000:0000:8a2e:         | 1812 | default  | 1
                    | 0370:7334                               |      |          |
-----
sg3                  | 1.1.1.5                               | 12   | mgmt     | 1
-----
radius (default)    | 1.1.1.4                               | 1812 | mgmt     | 1
radius (default)    | 1.1.1.5                               | 12   | mgmt     | 2
radius (default)    | abc1.com                               | 32   | mgmt     | 3
radius (default)    | 2001:0db8:85a3:0000:0000:8a2e:         | 1812 | default  | 4
                    | 0370:7334                               |      |          |
radius (default)    | 1.1.1.6                               | 32   | vrf_red  | 5
radius (default)    | 1.1.1.7                               | 32   | vrf_blue | 6
-----
```

Showing TACACS+ server group information:

```
switch# show aaa server-groups tacacs

***** AAA Mechanism TACACS+ *****
-----
GROUP NAME          | SERVER NAME                               | PORT | VRF      | PRIORITY
-----
sg2                  | 2001:0db8:85a3:0000:0000:8a2e:         | 49   | default  | 1
                    | 0370:7334                               |      |          |
-----
sg1                  | 1.1.1.2                               | 12   | mgmt     | 1
-----
tacacs (default)    | FQDN.com                               | 32   | mgmt     | 1
tacacs (default)    | 1.1.1.1                               | 49   | mgmt     | 2
tacacs (default)    | 1.1.1.2                               | 12   | mgmt     | 3
tacacs (default)    | abc.com                                | 32   | vrf_red  | 4
tacacs (default)    | 2001:0db8:85a3:0000:0000:8a2e:         | 49   | default  | 5
                    | 0370:7334                               |      |          |
```

```
tacacs (default) | 1.1.1.3 | 32 | vrf_blue | 6
-----
```

Showing RADIUS server group information:

```
switch# show aaa server-groups radius

***** AAA Mechanism RADIUS *****
-----
GROUP NAME          | SERVER NAME                               | PORT | VRF      | PRIORITY
-----
sg4                  | 2001:0db8:85a3:0000:0000:8a2e:0370:7334 | 1812 | default  | 1
-----
sg3                  | 1.1.1.5                                   | 12   | mgmt     | 1
-----
radius (default)    | 1.1.1.4                                   | 1812 | mgmt     | 1
radius (default)    | 1.1.1.5                                   | 12   | mgmt     | 2
radius (default)    | abc1.com                                  | 32   | mgmt     | 3
radius (default)    | 2001:0db8:85a3:0000:0000:8a2e:0370:7334 | 1812 | default  | 4
radius (default)    | 1.1.1.6                                   | 32   | vrf_red  | 5
radius (default)    | 1.1.1.7                                   | 32   | vrf_blue | 6
-----
```

show accounting log

Syntax

```
show accounting log [last <QTY-TO-SHOW> | all]
```

Description

Entered without optional parameters, this command shows all accounting log records for the current boot. Sensitive information is masked from the log, by being represented as asterisks.



This `show accounting log` command replaces the `show audit-log` command that is supported only in 10.00 releases.

Command context

Manager (#) or Auditor (auditor>)

Parameters

`last <QTY-TO-SHOW>`

Specifies how many most-recent accounting log records to show for the current boot. Range: 1 to 1000.
`all`

Selects for showing, all accounting records from the current boot and the previous boot.

Authority

Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only.

Usage

The log message starts with the record type, which is specific to AOS-CX. Values are the following:

USER_START

Record of a user login action.

USER_END

Record of a user logout action.

USYS_CONFIG

Record of a command executed by the user.

The three types of accounting log information are identified by the `msg=` element starting with the `rec=` item as follows:

- Exec is identified with: `msg='rec=ACCT_EXEC'`
- Command is identified with: `msg='rec=ACCT_CMD'`
- System is identified with: `msg='rec=ACCT_SYSTEM'`

The user group is indicated by `priv-lvl`, which is specific to AOS-CX. Values are the following:

Privilege level	User group
1	operators
15	administrators
19	auditors

The value of `service` indicates which user interface was used:

`service=shell`

Indicates that the log entry is a result of a CLI command.

`service=https-server`

Indicates that the log entry is a result of a REST API request or a Web UI action.

The string value of `data` identifies the CLI command or REST API request that was executed.

These elements are shown in context under *Examples*.

Examples

Showing the accounting log for the previous and current boot. Line breaks have been added for readability.

```
switch# show accounting log all

-----
Local accounting logs from previous boot
-----
----
type=DAEMON_START msg=audit(Nov 05 2018 23:00:58.607:9057) :
auditd start, ver=2.4.3 format=raw kernel=4.9.119-yocto-standard res=success
----
type=USER_START msg=audit(Nov 05 2018 23:06:42.398:42) :
msg='rec=ACCT_EXEC op=start session=CONSOLE timezone=UTC user=user1 priv-lvl=15
auth-method=LOCAL auth-type=LOCAL service=shell isconfig=no
hostname=8xxx addr=0.0.0.0 res=success'
----
type=USYS_CONFIG msg=audit(Nov 05 2018 23:06:42.399:43) :
msg='rec=ACCT_CMD op=stop session=CONSOLE timezone=UTC user=user1 priv-lvl=15
auth-method=LOCAL auth-type=LOCAL service=shell isconfig=no
```

```

data="enable" hostname=8xxx addr=0.0.0.0 res=success'
----
type=USYS_CONFIG msg=audit(Nov 05 2018 23:08:24.693:51) :
msg='rec=ACCT_CMD op=stop session=CONSOLE timezone=UTC user=user1 priv-lvl=1
auth-method=LOCAL auth-type=LOCAL service=shell isconfig=no
data="configure terminal" hostname=8xxx addr=0.0.0.0 res=success'
----
type=USYS_CONFIG msg=audit(Nov 05 2018 23:08:39.108:52) :
msg='rec=ACCT_CMD op=stop session=CONSOLE timezone=UTC user=user1 priv-lvl=15
auth-method=LOCAL auth-type=LOCAL service=shell isconfig=yes
data="https-server rest access-mode read-write"
hostname=8xxx addr=0.0.0.0 res=success'
----
type=USER_START msg=audit(Nov 05 2018 23:10:57.238:58) :
msg='rec=ACCT_EXEC op=start session=REST timezone=UTC user=admin priv-lvl=15
auth-method=LOCAL auth-type=LOCAL service=https-server
data="http-method=POST http-uri=/rest/v1/login"
hostname=8xxx addr=127.0.0.1 res=success'
----
type=USYS_CONFIG msg=audit(Nov 05 2018 23:15:11.958:75) :
msg='rec=ACCT_CMD op=stop session=CONSOLE timezone=UTC user=user1 priv-lvl=15
auth-method=LOCAL auth-type=LOCAL service=shell isconfig=yes
data="tacacs-server host 2.2.2.2" hostname=8xxx addr=0.0.0.0 res=success'
----
type=USYS_CONFIG msg=audit(Nov 05 2018 23:15:37.090:76) :
msg='rec=ACCT_CMD op=stop session=REST timezone=UTC user=admin priv-lvl=15
auth-method=LOCAL auth-type=LOCAL service=https-server
data="http-method=GET http-uri=/rest/v1/system/vrfs/mgmt/tacacs_servers"
hostname=8xxx addr=127.0.0.1 res=success'
----
type=USER_END msg=audit(Nov 05 2018 23:26:59.207:90) :
msg='rec=ACCT_EXEC op=stop session=REST timezone=UTC user=admin priv-lvl=15
auth-method=LOCAL auth-type=LOCAL service=https-server
data="http-method=POST http-uri=/rest/v1/logout"
hostname=8xxx addr=127.0.0.1 res=success'
----
type=USER_END msg=audit(Nov 05 2018 23:27:49.164:93) :
msg='rec=ACCT_EXEC op=stop session=CONSOLE timezone=UTC user=user1 priv-lvl=15
auth-method=LOCAL auth-type=LOCAL service=shell isconfig=no
hostname=8xxx addr=0.0.0.0 res=success'

-----
Local accounting logs from current boot
-----
----
type=DAEMON_START msg=audit(Nov 05 2018 23:32:05.642:626) :
auditd start, ver=2.4.3 format=raw kernel=4.9.119-yocto-standard res=success
----
type=USER_START msg=audit(Nov 05 2018 23:35:52.915:11) :
msg='rec=ACCT_EXEC op=start session=CONSOLE timezone=UTC user=admin priv-lvl=15
auth-method=LOCAL auth-type=LOCAL service=shell isconfig=no
hostname=8xxx addr=0.0.0.0 res=success'
----
type=USYS_CONFIG msg=audit(Nov 05 2018 23:35:52.917:12) :
msg='rec=ACCT_CMD op=stop session=CONSOLE timezone=UTC user=admin priv-lvl=15
auth-method=LOCAL auth-type=LOCAL service=shell isconfig=no data="enable"
hostname=8xxx addr=0.0.0.0 res=success'

```

show radius-server

Syntax

show radius-server [detail]

Description

Shows configured RADIUS servers information.

Command context

Operator (>) or Manager (#)

Parameters

detail

Selects additional RADIUS server details and global parameters for showing.

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Usage

When the `show radius-server` command shows `None` for the shared-secret, the passkey is missing.

Examples

Showing a summary of the global RADIUS configuration:

```
switch# show radius-server
***** Global RADIUS Configuration *****

Shared-Secret: None
Timeout: 60
Auth-Type: pap
Retries: 5
Tracking Time Interval (seconds): 60
Tracking Retries: 5
Tracking User-name: radius-tracking-user
Tracking Password: None
Number of Servers: 1

-----
SERVER NAME                               | PORT | VRF
-----
20.1.1.129                                | 1812 | default
1.1.1.4                                   | 1812 | default
1.1.1.5                                   | 12   | default
abc1.com                                  | 32   | default
2001:0db8:85a3:0000:0000:8a2e:0371:7334 | 1812 | default
-----
```

Showing a summary of a RADIUS server when the per-server shared key or the global RADIUS shared key is not set:

```
switch# show radius-server
***** Global RADIUS Configuration *****

Shared-Secret: None
Timeout: 60
Auth-Type: pap
Retries: 5
Tracking Time Interval (seconds): 60
```

```
Tracking Retries: 5
Tracking User-name: radius-tracking-user
Tracking Password: None
Number of Servers: 1
```

SERVER NAME	PORT	VRF
20.1.1.129	1812	default

Showing details of a global RADIUS configuration:

```
switch# show radius-server detail
***** Global RADIUS Configuration *****

Shared-Secret: AQBapb+HsdpqVlQcA+CyD0RvfbeA8BEgikCgAAAJOWZSNzA2SWrLA=
Timeout: 5
Auth-Type: pap
Retries: 5
Tracking Time Interval (seconds): 60
Tracking Retries: 5
Tracking User-name: radius-tracking-user
Tracking Password: None
Number of Servers: 1

***** RADIUS Server Information *****
Server-Name          : 20.1.1.129
Auth-Port            : 1812
Accounting-Port      : 1813
VRF                  : default
Shared-Secret        : None
Timeout              : 60
Retries              : 5
Auth-Type             : pap
Server-Group          : radius
Default-Priority      : 4
Tracking              : disabled
Tracking-Mode         : any
Reachability-Status  : N/A
ClearPass-Username    :
ClearPass-Password    : None
```

Showing details of a RADIUS server when the per-server shared key and the global RADIUS shared key are not set:

```
switch# show radius-server detail
***** Global RADIUS Configuration *****

Shared-Secret: None
Timeout: 5
Auth-Type: pap
Retries: 1
Number of Servers: 1

***** RADIUS Server Information *****
Server-Name          : 1.1.1.1
Auth-Port            : 2083
VRF                  : default
```



```
Shared-Secret (default) : None
Timeout (default)       : 5
Retries (default)       : 1
Auth-Type (default)     : pap
Server-Group (default)  : radius
Default-Priority         : 1
```

show radius-server secure ipsec

Syntax

```
show radius-server secure ipsec { server-list | host {<FQDN> | <IPv4> | <IPv6>} [port
<PORT-NUMBER>] [vrf <VRF-NAME>] }
```

Description

Shows information for one or all RADIUS servers configured with IPsec.

Command context

Operator (>) or Manager (#)

Parameters

`server-list`

Selects all servers for showing.

`host {<FQDN> | <IPv4> | <IPv6>}`

Specifies the RADIUS server as:

- `<FQDN>`: a fully qualified domain name.
- `<IPv4>`: an IPv4 address.
- `<IPv6>`: an IPv6 address.

`port <PORT-NUMBER>`

Specifies the authentication port number. Range: 1 to 65535. Default: 1812.

`vrf <VRF-NAME>`

Specifies the VRF name to be used for communicating with the server. If no VRF name is provided, the default VRF named `default` is used.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Usage

The IPsec key is shown in an exportable ciphertext format.

Examples

Showing information for RADIUS server 1.1.1.1 secured with IPsec:

```
switch# show radius-server secure ipsec host 1.1.1.1
IPsec                : enabled
Protocol              : ESP
Authentication        : MD5
```

```
Encryption      : AES
SPI             : 1234
```

Showing information for all RADIUS servers secured with IPsec:

```
switch# show radius-server secure ipsec server-list
Server          : 1.1.1.1
IPsec           : enabled
Protocol        : ESP
Authentication   : MD5
Encryption      : AES
SPI             : 1234

Server          : 1.1.1.2
IPsec           : enabled
Protocol        : ESP
Authentication   : MD5
Encryption      : AES
SPI             : 12341
```

show tacacs-server

Syntax

```
show tacacs-server [detail]
```

Description

Shows the configured TACACS+ servers.

Command context

Operator (>) or Manager (#)

Parameters

detail

Selects additional TACACS+ server details and global parameters for showing.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Showing a summary of a global TACACS+ configuration with a shared-secret:

```
switch# show tacacs-server
***** Global TACACS+ Configuration *****

Shared-Secret: AQBapb+HsdpqVlQ3CPCBMQTG8e1cA+CyD0RvfbeA8BEgikCgAAAJOWZSNzA2SWrLA=
Timeout: 5
Auth-Type: pap
Number of Servers: 5

-----
```

SERVER NAME	PORT	VRF
1.1.1.1	49	mgmt
1.1.1.2	12	mgmt
abc.com	32	vrf_blue
2001:0db8:85a3:0000:0000:8a2e:0370:7334	49	default
1.1.1.3	32	vrf_red

Showing details of a global TACACS+ configuration:

```
switch# show tacacs-server detail
***** Global TACACS+ Configuration *****

Shared-Secret: AQBapb+HsdpqV1Q3CPCBMQTG8e1cA+CyD0RvfbeA8BEgikCgAAAJOWZSNzA2SWrLA=
Timeout: 5
Auth-Type: pap
Number of Servers: 5

***** TACACS+ Server Information *****

Server-Name          : 1.1.1.2
Auth-Port            : 12
VRF                  : mgmt
Shared-Secret (default) : AQBapb+HsdpqV1Q3CPCBMQTG8eeA8BEgikCgAAAJOWZSNzA2SWrLA=
Timeout (default)    : 5
Auth-Type (default)  : pap
Server-Group         : sg1
Group-Priority       : 1

Server-Name          : 2001:0db8:85a3:0000:0000:8a2e:0370:7334
Auth-Port            : 49
VRF                  : default
Shared-Secret (default) : AQBapb+HsdpqV1Q3CPCBMQTG8eeA8BEgikCgAAAJOWZSNzA2SWrLA=
Timeout (default)    : 5
Auth-Type (default)  : pap
Server-Group         : sg2
Group-Priority       : 1

Server-Name          : 1.1.1.1
Auth-Port            : 49
VRF                  : mgmt
Shared-Secret (default) : AQBapb+HsdpqV1Q3CPCBMQTG8eeA8BEgikCgAAAJOWZSNzA2SWrLA=
Timeout (default)    : 5
Auth-Type (default)  : pap
Server-Group (default) : tacacs
Default-Priority     : 1

Server-Name          : abc.com
Auth-Port            : 32
VRF                  : vrf_red
Shared-Secret (default) : AQBapb+HsdpqV1Q3CPCBMQTG8eeA8BEgikCgAAAJOWZSNzA2SWrLA=
Timeout              : 15
Auth-Type (default)  : pap
Server-Group (default) : tacacs
Default-Priority     : 3

Server-Name          : 1.1.1.3
Auth-Port            : 32
VRF                  : vrf_blue
Shared-Secret       : AQBapfnqbSswqKC476tdUFZ+AncIRY92hDTYkQCAAAAFEaHn43vNC
Timeout              : 15
```

```
Auth-Type           : chap
Server-Group (default) : tacacs
Default-Priority     : 5
```

Showing TACACS+ server when per-server shared key and global TACACS+ shared key is not set:

```
switch# show tacacs-server
***** Global TACACS+ Configuration *****

Shared-Secret: None
Timeout: 5
Auth-Type: pap
Number of Servers: 1

-----
SERVER NAME                               | PORT | VRF
-----
1.1.1.1                                   | 49    | default
-----
```

Showing TACACS+ server details when per-server shared key and global TACACS+ shared key is not set:

```
switch# show tacacs-server detail
***** Global TACACS+ Configuration *****

Shared-Secret: None
Timeout: 5
Auth-Type: pap
Number of Servers: 1

***** TACACS+ Server Information *****
Server-Name           : 1.1.1.1
Auth-Port              : 49
VRF                    : default
Shared-Secret (default) : None
Timeout (default)      : 5
Auth-Type (default)    : pap
Server-Group (default) : tacacs
Default-Priority       : 1
```

show tacacs-server statistics

Syntax

```
show tacacs-server statistics
```

Description

Shows authentication statistics for all configured TACACS+ servers.

Command context

Operator (>) or Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Showing TACACS+ server authentication statistics:

```
switch# show tacacs-server statistics
Server Name      : tacl
Auth-Port       : 49
VRF             : mgmt
Authentication Statistics
-----
Round Trip Time      : 1
Pending Requests    : 0
Timeout            : 0
Unknown Types       : 0
Packet Dropped      : 0
Auth Start         : 8
Auth challenge      : 0
Auth Accepts       : 4
Auth Rejects       : 4
Auth reply malformed : 0
Tracking Requests   : 0
Tracking Responses   : 0
```

show tech aaa

Syntax

```
show tech aaa
```

Description

Shows the AAA configuration settings.

Command context

Manager (#)

Authority

Administrators or local user group members with execution rights for this command.

Example

Showing the AAA configuration settings:

```
switch# show tech aaa
=====
Show Tech executed on Tue Feb 14 02:19:11 2017
=====
[Begin] Feature aaa
=====

*****
Command : show aaa authentication
*****
AAA Authentication:
  Fail-through      : Enabled
  Limit Login Attempts : Not set
  Lockout Time      : 300
```

```

Minimum Password Length      : Not set

Authentication for ssh channel:
-----
GROUP NAME                    | GROUP PRIORITY
-----
local                          | 0
-----

Authentication for https-server channel:
-----
GROUP NAME                    | GROUP PRIORITY
-----
local                          | 0
-----

Authentication for console channel:
-----
GROUP NAME                    | GROUP PRIORITY
-----
local                          | 0
-----

Authentication for default channel:
-----
GROUP NAME                    | GROUP PRIORITY
-----
tacacs                         | 0
local                          | 1
-----

*****
Command : show aaa accounting
*****
AAA Accounting:
  Accounting Type               : all
  Accounting Mode               : start-stop

Accounting for default channel:
-----
GROUP NAME                    | GROUP PRIORITY
-----
local                          | 0
-----

Accounting for ssh channel:
-----
GROUP NAME                    | GROUP PRIORITY
-----
tacacs                         | 0
local                          | 1
-----

Accounting for https-server channel:
-----
GROUP NAME                    | GROUP PRIORITY
-----
tacacs                         | 0
-----

*****
Command : show aaa authorization

```

Authorization for default channel:

GROUP NAME	GROUP PRIORITY
none	0

Authorization for console channel:

GROUP NAME	GROUP PRIORITY
none	0

Authorization for ssh channel:

GROUP NAME	GROUP PRIORITY
tacacs	0
none	1

Command : show aaa server-groups

***** AAA Mechanism TACACS+ *****

GROUP NAME	SERVER NAME	PORT	PRIORITY	VRF
tacacs (default)	1.1.1.1	49	1	mgmt

***** AAA Mechanism RADIUS *****

GROUP NAME	SERVER NAME	PORT	PRIORITY	VRF
------------	-------------	------	----------	-----

Command : show tacacs-server detail

***** Global TACACS+ Configuration *****

Shared-Secret: G8ekK1cA+CyD0RvfbeA8BEgikCgAAAJ0wZSNzA2SWrLA=

Timeout: 5

Auth-Type: pap

Tracking: disabled

Tracking Time Interval (seconds): 300

Tracking User-name: tacacs-tracking-user

Tracking Password: None

Number of Servers: 1

***** TACACS+ Server Information *****

Server-Name : 1.1.1.1

Auth-Port : 49

VRF : mgmt

Shared-Secret : KCdmOMxMD26T0fQoXfJbtj9j2AUxlGn6eCAAAAF2MkfMTojqX

Timeout (default) : 5

Auth-Type (default) : pap

Server-Group (default) : tacacs

```

Default-Priority      : 1
Tracking              : disabled
Reachability-Status   : N/A

*****
Command : show radius-server detail
*****
***** Global RADIUS Configuration *****

Shared-Secret: CPCBMQTG8ekK1cA+CyD0RvfbeA8BEgikCgAAAJ0wZSNzA2SWrLA=
Timeout: 5
Auth-Type: pap
Retries: 1
Number of Servers: 0

=====
[End] Feature aaa
=====

=====
Show Tech commands executed successfully
=====

```

tacacs-server auth-type

Syntax

```

tacacs-server auth-type {pap | chap}
no tacacs-server auth-type

```

Description

Enables the CHAP or PAP authentication protocol, which is used for communication with the TACACS+ servers, at the global level. You can override this command with a fine-grained per server `auth-type` configuration.

The `no` form of this command resets the global authentication mechanism for TACACS+ to PAP, which is the default authentication mechanism for TACACS+.

Command context

config

Parameters

`auth-type {pap | chap}`

Selects either the PAP or CHAP authentication protocol.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling command for CHAP authentication:

```
switch(config)# tacacs-server auth-type chap
```

Enabling command for PAP authentication:


```
switch(config)# tacacs-server auth-type pap
```

tacacs-server host

Syntax

```
tacacs-server host {<FQDN> | <IPv4> | <IPv6>}  
    [key [plaintext <PASSKEY> | ciphertext <PASSKEY>]]  
    [timeout <TIMEOUT-SECONDS>] [port <PORT-NUMBER>]  
    [auth-type {pap | chap}] [tracking {enable | disable}] [vrf <VRF-NAME>]  
  
no tacacs-server host {<FQDN> | <IPv4> | <IPv6>}  
    [port <PORT-NUMBER>] [vrf <VRF-NAME>]
```

Description

Adds a TACACS+ server. By default, the TACACS+ server is associated with the server group named `tacacs`. The `no` form of this command removes a previously added TACACS+ server.

Command context

config

Parameters

{<FQDN> | <IPv4> | <IPv6>}

Specifies the TACACS+ server as:

- <FQDN>: a fully qualified domain name.
- <IPv4>: an IPv4 address.
- <IPv6>: an IPv6 address.

key {plaintext <PASSKEY> | ciphertext <PASSKEY>}

Specifies either a plaintext or an encrypted local shared-secret passkey for the server. As per RFC 2865, shared-secret can be a mix of alphanumeric and special characters. The length of shared-secret in plaintext format is fewer than 32 characters.

When `key` is entered without either sub-parameter, plaintext passkey prompting occurs upon pressing Enter. The entered passkey characters are masked with asterisks.

When `key` is omitted, the server uses the global passkey. This command requires either the global or local passkey to be set; otherwise the server will not be contacted. Command `radius-server key` is available for setting the global passkey.

timeout <TIMEOUT-SECONDS>

Specifies the timeout. The range is 1 to 60 seconds. The default timeout is 5 seconds.

port <PORT-NUMBER>

Specifies the TCP authentication port number. Range: 1 to 65535. Default: 49.

auth-type {pap | chap}

Selects either PAP (default) or CHAP authentication type. If this parameter is not specified, the TACACS+ global default is used.

tracking {enable | disable}

Enables or disables server tracking for the server. Tracked servers are probed at the start of each server tracking interval to check if they are reachable. Unreachable servers are skipped in favor of servers that



are proven to be reachable. Use command `tacacs-server tracking` to configure TACACS+ server tracking.

`vrf <VRF-NAME>`

Specifies the VRF name to be used for communicating with the server. If no VRF name is provided, the default VRF named `default` is used.

Authority

Administrators or local user group members with execution rights for this command.

Usage

If the fully qualified domain name is provided for the TACACS+ server, a DNS server must be configured and accessible through the same VRF which is configured for the TACACS+ server. This configuration is required for the resolution of the TACACS+ server hostname to its IP address. If a DNS server is not available for this VRF, the TACACS+ servers reachable through this VRF must be configured by means of their IP addresses only.

Examples

Adding a TACACS+ server with an IPv4 address, plaintext passkey, timeout, port, authentication type, and VRF name:

```
switch(config)# tacacs-server host 1.1.1.3 key plaintext test-123 timeout 15 port 32
auth-type chap vrf vrf_red
```

Adding a TACACS+ server with an IPv4 address and prompted plaintext passkey:

```
switch(config)# tacacs-server host 1.1.1.5 key
Enter the TACACS server key: *****
Re-Enter the TACACS server key: *****
```

Adding a TACACS+ server with an IPv4 address and a named VRF:

```
switch(config)# tacacs-server host 1.1.1.1 vrf mgmt
```

Adding a TACACS+ server with an IPv4 address, a port, and a named VRF:

```
switch(config)# tacacs-server host 1.1.1.2 port 32 vrf mgmt
```

Adding a TACACS+ server with an FQDN, a timeout, port number, and a named VRF:

```
switch(config)# tacacs-server host abc.com timeout 15 port 32 vrf vrf_blue
```

Adding a TACACS+ server with an IPv6 address:

```
switch(config)# tacacs-server host 2001:0db8:85a3:0000:0000:8a2e:0370:7334
```

Deleting a TACACS+ server with an IPv4 address and specified VRF:

```
switch(config)# no tacacs-server host 1.1.1.1 vrf mgmt
```

Deleting a TACACS+ server with an FQDN, port, and specified VRF:

```
switch(config)# no tacacs-server host abc.com port 32 vrf vrf_blue
```

tacacs-server key

Syntax

```
tacacs-server key [plaintext <GLOBAL-PASSKEY> | ciphertext <GLOBAL-PASSKEY>]
```

```
no tacacs-server key
```

Description

Creates or modifies a TACACS+ global passkey. The TACACS+ global passkey is used as a shared-secret for encrypting the communication between all TACACS+ servers and the switch. The TACACS+ global passkey is required for authentication unless local passkeys have been set. By default, the TACACS+ global passkey is empty. If the administrator has not set this key, the switch will not be able to perform TACACS+ authentication. The switch will instead rely on the authentication mechanism configured with `aaa authentication login`.



When this command is entered without parameters, plaintext passkey prompting occurs upon pressing Enter. The entered passkey characters are masked with asterisks.

The `no` form of the command removes the global passkey.

Command context

config

Parameters

plaintext <GLOBAL-PASSKEY>

Specifies the TACACS+ global passkey in plaintext format with a length of 1 to 31 characters. As per RFC 2865, shared-secret can be a mix of alphanumeric and special characters.

ciphertext <GLOBAL-PASSKEY>

Specifies the TACACS+ global passkey in encrypted format.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Adding the global passkey:

```
switch(config)# tacacs-server key plaintext mypasskey123
```

Adding the global passkey with prompting:

```
switch(config)# tacacs-server key
Enter the TACACS server key: *****
Re-Enter the TACACS server key: *****
```

Removing the global passkey:

```
switch(config)# no tacacs-server key
```

tacacs-server timeout

Syntax

```
tacacs-server timeout [<1-60>]
no tacacs-server timeout
```

Description

Specifies the number of seconds to wait for a response from the TACACS+ server before trying the next TACACS+ server. If a value is not specified, a default value of 5 seconds is used. You can override this value with a fine-grained per server timeout configured for individual servers.

The `no` form of this command resets the TACACS+ global authentication timeout to the default of 5 seconds.

Command context

config

Parameters

timeout <1-60>

Specifies the timeout interval of 1 to 60 seconds. The default is 5 seconds.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Specifying the TACACS+ server timeout:

```
switch(config)# tacacs-server timeout 10
```

Resetting the timeout for the TACACS+ server to the default:

```
switch(config)# no tacacs-server timeout
```

tacacs-server tracking

Syntax

```
tacacs-server tracking interval <INTERVAL>
no tacacs-server tracking interval
```

```
tacacs-server tracking user-name <NAME>
[password [plaintext <PASSWORD> | ciphertext <PASSWORD>]]
```

```
no tacacs-server tracking user-name <NAME>
```

Description

Configures TACACS+ server tracking settings globally for all configured TACACS+ servers that have tracking enabled with the `tacacs-server host` command on individual servers.

The `no` form of the command removes the specified configuration, reverting it to its default. The `no` form with `user-name` also clears the password (resets it to empty).

Command context

config

Parameters

interval <INTERVAL>

Specifies the time interval, in seconds, to wait before checking the server reachability status. Default: 300. Range 60 to 84600.

user-name <NAME> [password [plaintext <PASSWORD> | ciphertext <PASSWORD>]]

Specifies the user name (and optionally a password) to be used for server checking. The default user name is `tacacs-tracking-user` with an empty password.

The password is optional and may be entered as `plaintext` or pasted in as `ciphertext`. The plaintext password is visible as cleartext when entered but is encrypted thereafter. Command history does show the password as cleartext.



When `password` is entered without a following sub-parameter, plaintext password prompting occurs upon pressing Enter. The entered password characters are masked with asterisks.



The user does not have to be configured on the server. Server tracking can still be performed with a user which is not configured on the server because authentication failure on the server achieves confirmation that the server is reachable.



Server tracking uses authentication request and response packets to determine server reachability status. The server tracking user name and password are used to form the request packet which is sent to the server with tracking enabled. Upon receiving a response to the request packet, the server is considered to be reachable.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring a tracking interval of 120 seconds:

```
switch(config)# tacacs-server tracking interval 120
```

Reverting the tracking interval to its default of 300 seconds:

```
switch(config)# no tacacs-server tracking interval
```

Configuring user `tacacs-tracker` with a plaintext password.

```
switch(config)# tacacs-server tracking user-name tacacs-tracker password plaintext track$1
```

Configuring user `tacacs-tracker` with a prompted plaintext password.

```
switch(config)# tacacs-server tracking user-name tacacs-tracker password  
Enter the TACACS server tracking password: *****  
Re-Enter the TACACS server tracking password: *****
```

Reverting the tracking user name to its default of `tacacs-tracking-user`:

```
switch(config)# no tacacs-server tracking user-name
```

logging

Syntax

```
logging {<IPv4-ADDR> | <IPv6-ADDR> | <FQDN | HOSTNAME>} [ udp [<PORT-NUM>] | tcp [<PORT-NUM>] ] [severity <LEVEL>] [vrf <VRF-NAME>] [include-auditable-events] [filter <FILTER-NAME>] [ rate-limit-burst <BURST> [rate-limit-interval <INTERVAL>] ]
```

```
logging {<IPv4-ADDR> | <IPv6-ADDR> | <FQDN | HOSTNAME>} [tls [<PORT-NUM>]] [auth-mode {certificate|subject-name}] [legacy-tls-renegotiation] [severity <LEVEL>] [vrf <VRF-NAME>] [include-auditable-events] [filter <FILTER-NAME>] [ rate-limit-burst <BURST> [rate-limit-interval <INTERVAL>]]
```

```
no logging {<IPv4-ADDR> | <IPv6-ADDR> | <FQDN | HOSTNAME> }
```

Description

Enables syslog forwarding to a remote syslog server.

The `no` form of this command disables syslog forwarding to a remote syslog server.

Command context

config

Parameters

{<IPv4-ADDR> | <IPv6-ADDR> | <HOSTNAME>}

Selects the IPv4 address, IPv6 address, or host name of the remote syslog server. Required.

[udp [<PORT-NUM>] | tcp [<PORT-NUM>] | tls [<PORT-NUM>]]

Specifies the UDP port, TCP port, or TLS port of the remote syslog server to receive the forwarded syslog messages.

udp [<PORT-NUM>]

Range: 1 to 65535. Default: 514

tcp [<PORT-NUM>]

Range: 1 to 65535. Default: 1470

tls [<PORT-NUM>]

Range: 1 to 65535. Default: 6514

include-auditable-events

Specifies that auditable messages are also logged to the remote syslog server.

severity <LEVEL>

Specifies the severity of the syslog messages:

- `alert`: Forwards syslog messages with the severity of `alert` (6) and `emergency` (7).
- `crit`: Forwards syslog messages with the severity of `critical` (5) and above.
- `debug`: Forwards syslog messages with the severity of `debug` (0) and above.
- `emerg`: Forwards syslog messages with the severity of `emergency` (7) only.

- **err**: Forwards syslog messages with the severity of `err` (4) and above
- **info**: Forwards syslog messages with the severity of `info` (1) and above. Default.
- **notice**: Forwards syslog messages with the severity of `notice` (2) and above.
- **warning**: Forwards syslog messages with the severity of `warning` (3) and above.

`auth-mode`

Specifies the TLS authentication mode used to validate the certificate.

- **certificate**: Validates the peer using trust anchor certificate based authentication. Default.
- **subject-name**: Validates the peer using trust anchor certificates as well as subject-name based authentication.

`legacy-tls-renegotiation`

Enables the TLS connection with a remote syslog server supporting legacy renegotiation.

`filter <FILTER-NAME>`

Specifies the name of the filter to be applied on the syslog messages.

`rate-limit-burst <BURST>`

Specifies the rate limit for the messages sent to the remote syslog server.

`rate-limit-interval <INTERVAL>`

Specifies the rate limit interval in seconds. Default: 30 Seconds

`vrf <VRF-NAME>`

Specifies the VRF used to connect to the syslog server. Optional. Default: `default`

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling the syslog forwarding to remote syslog server 10.0.10.2:

```
switch(config)# logging 10.0.10.2
```

Enabling the syslog forwarding of messages with a severity of `err` (4) and above to TCP port 4242 on remote syslog server 10.0.10.9 with VRF `lab_vrf`:

```
switch(config)# logging 10.0.10.9 tcp 4242 severity err vrf lab_vrf
```

Disabling syslog forwarding to a remote syslog server:

```
switch(config)# no logging
```

Enabling syslog forwarding over TLS to a remote syslog server using subject-name authentication mode:

```
switch(config)# logging example.com tls auth-mode subject-name
```

Applying log filtering for syslog server forwarding:

```
switch(config)# logging 10.0.10.6 severity info filter filter_lldp_logs vrf mgmt
```


Applying log filtering and enabling the rate limit for syslog server forwarding over TCP port:

```
switch(config)# logging 10.0.10.2 tcp 3440 severity err vrf mgmt include-auditable-  
events filter filter_lldp_logs rate-limit-burst 3 rate-limit-interval 35
```

logging filter

Syntax

```
logging filter <FILTER-NAME>
```

```
[{enable | disable}]
```

```
[<SEQUENCE-ID>] {permit | deny} [event-id <EVENT-ID-RANGE>] [includes <REGEX>] [severity  
<COMPARISON-OPERATOR> <LEVEL>]
```

```
no <SEQUENCE-ID>
```

```
resequence <OLD-SEQUENCE-ID> <NEW-SEQUENCE-ID>
```

```
no logging filter <FILTER-NAME>
```

Description

Creates a filter to restrict what event or debug logs are logged. A filter can be used to either permit or deny:

- The event logs from being generated on the switch, or
- The event or debug logs generated on the switch from being forwarded to a syslog server.

A filter is identified by a filter name and can have up to 20 rules or entries, each with a different sequence number, matching criteria, and corresponding action (deny or permit). When a filter is applied on a log, the log is matched against the criteria mentioned in the rules or entries in ascending numerical order of their sequence numbers until a matching entry is found. Once a matching entry is found, its corresponding action is applied on the log. If no matching rule is found, the default action (permit) is applied.

The `no` form of this command removes the filter.

Command context

`config` and `config-logging-filter`

Parameters

<FILTER-NAME>

Specifies the unique name to identify the filter.

`enable`

Filter event logs generated on the switch.

<SEQUENCE-ID>

Specifies the filter criteria sequence number. Default: Increments by 10 from the largest sequence-id currently used in this filter.

`deny`

Prevents the matching log from being logged.

`permit`

Allows the matching log.

`<event-id>`

Matches logs by event ID. Specify an event ID or a range of event IDs. It supports a maximum of 100 event IDs.

`includes <REGEX>`

Matches the log message against a regular expression string.

`severity`

Matches the logs by severity level.

The following options are used to compare the severity:

- `eq`: Match events of severity equal to the specified.
- `ge`: Match events of severity greater than or equal to the specified.
- `gt`: Match events of severity greater than the specified.
- `le`: Match events of severity lesser than or equal to the specified.
- `lt`: Match events of severity lesser than the specified.

The following are the severity levels:

- `alert`: Logs with the severity `alert` (6).
- `crit`: Logs with the severity `critical` (5).
- `debug`: Logs with the severity `debug` (0).
- `emerg`: Logs with the severity `emergency` (7).
- `err`: Logs with the severity `err` (4).
- `info`: Logs with the severity `info` (1).
- `notice`: Logs with the severity `notice` (2).
- `warning`: Logs with the severity `warning` (3).

Authority

Administrators or local user group members with execution rights for this command.

Usage

Filtering event logs on the switch: To permit or deny event logs from being generated on the switch. In this case, the matching event logs are filtered at generation. The denied event logs are neither logged to the switch events nor forwarded to any remote syslog servers. Multiple filters can be configured, but only one filter can be applied to filter the events on the switch. Such a filter can be chosen by adding the `enable` command under its configuration. Configuring the `enable` command under a new filter automatically removes it from the filter where it was previously used.

For example:

```
logging filter low_severity_logs
enable
10 deny severity lt info
```

This configuration denies the event logs which have a severity less than info.



If a filter contains `enable` command, it is not recommended to configure this filter in the `logging` command used for remote syslog server configuration. This is because, any event logs denied by the filter are already not available for forwarding to a remote server.

A filter with `enable` command will not affect debug logs. Consider the configuration in the following example of a filter with `enable` command and two rules applied `10 permit severity ge info` and `20 deny`. This implies permit only those event logs which have severity greater than or equal to `info`.

Example:

```
logging filter low_severity_logs
enable
10 permit severity ge info
20 deny
```

Filtering event or debug logs when forwarding to a remote syslog server: The filter name must be configured in the logging command that is used to configure remote syslog server. The logs will be generated on the switch and the filter only decides whether to deny or permit the syslog forwarding for the matching log. For example: `logging 10.0.10.6 filter filter_lldp_logs`



The filter affects debug logs only when the command `debug destination syslog` is configured on the switch.



The severity mentioned in the remote syslog server configuration using logging command under configuration context has more precedence than the severity mentioned in a filter entry. If a log with `warning` severity is permitted by a filter, but the remote syslog configuration has severity `err` mentioned in it, the log will not be forwarded to the remote syslog server (since `warning(3)` is lesser than `err(4)`). On the other hand, if a log with `err` severity is permitted by a filter and the remote syslog configuration has severity `warning` mentioned in it, the log will be forwarded to the remote syslog server.

Examples

Configuring a new logging filter:

```
switch(config)# logging filter example_filter
```

To deny logs having event ID 1301 and a range of event IDs from 1305 to 1309:

```
switch(config-logging-filter)# 20 deny event-id 1301,1305-1309
```

To permit logs having event ID 1300:

```
switch(config-logging-filter)# 30 permit event-id 1300
```

To permit logs with severity greater than or equal to `err`:

```
switch(config-logging-filter)# 30 permit severity ge err
```

To deny logs with severity greater than `info`:

```
switch(config-logging-filter)# 30 deny severity gt info
```

To deny logs with event ID 1024 and a message matching the regular expression LLDP:

```
switch(config-logging-filter)# 40 deny event-id 1024 includes LLDP
```

Denying all logs:

```
switch(config-logging-filter)# 40 deny
```

Changing the sequence ID of an existing rule:

```
switch(config-logging-filter)# resequence 20 70
```

logging facility

Syntax

```
logging facility {local0 | local1 | local2 | local3  
| local4 | local5 | local6 | local7}  
no logging facility
```

Description

Sets the logging facility to be used for remote syslog messages. Default: `local7`

The `no` form of this command disables the logging facility to be used for remote syslog messages.

Command context

`config`

Parameters

```
{local0 | local1 | local2 | local3 | local4 | local5 | local6 | local7}
```

Selects the logging facility to be used for remote syslog messages. Required.

Specifies the severity of the syslog messages:

- `local0`
- `local1`
- `local2`
- `local3`
- `local4`
- `local5`
- `local6`
- `local7`

Authority

Administrators or local user group members with execution rights for this command.

Examples

Sets the `local5` logging facility to be used for remote syslog messages:

```
switch(config)# logging facility local5
```

logging persistent-storage

Syntax

```
logging persistent-storage [severity {alert|crit|debug|emerg|err|info|notice|warning}]  
no logging persistent-storage
```

Description

Enables or disables storage of logs in storage. Only logs of the specified severity and above will be preserved in the storage.

The `no` form of this command disables storage of logs in storage.

Command context

config

Parameters

severity <LEVEL>

Specifies the severity of the syslog messages:

- `alert`: Preserves syslog messages with the severity of `alert` (6) and `emergency` (7)
- `crit`: Preserves syslog messages with the severity of `critical` (5) and above. Default.
- `debug`: Preserves syslog messages with the severity of `debug` (0) and above.
- `emerg`: Preserves syslog messages with the severity of `emergency` (7) only.
- `err`: Preserves syslog messages with the severity of `err` (4) and above.
- `info`: Preserves syslog messages with the severity of `info` (1) and above.
- `notice`: Preserves syslog messages with the severity of `notice` (2) and above.
- `warning`: Preserves syslog messages with the severity of `warning` (3) and above.

Authority

Administrators or local user group members with execution rights for this command.

Usage

These logs can be copied out by using the `copy support-files all` OR `copy support-files previous-boot`.

Examples

Enabling storage of logs in storage with severity `info`:

```
switch(config)#logging persistent-storage severity info  
Logs will be written to storage and made available across reboot.  
Do you want to continue (y/n)?
```

Disabling storage of logs in storage:

```
switch(config)# no logging persistent-storage
```

show spanning-tree

Syntax

```
show spanning-tree [detail]
```

Description

Shows the spanning tree mode and information on the RPVST instances and optionally shows details on the RPVST instances.

Command context

Operator (>) or Manager (#)

Parameters

detail

Specifies showing details on the RPVST instances.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Showing spanning tree mode and RPVST instance information:

```
switch# show spanning-tree
Spanning tree status      : Enabled Protocol: RPVST
Extended System-id       : Enabled
Ignore PVID Inconsistency : Enabled
Path cost method          : Long
RPVST-MSTP Interconnect VLAN : 1
VLAN1
  Root ID    Priority    : 32768
             MAC-Address: 70:72:cf:31:c9:23
             This bridge is the root
             Hello time(in seconds):2   Max Age(in seconds):20
             Forward Delay(in seconds):15

  Bridge ID   Priority    : 32768
             MAC-Address: 70:72:cf:31:c9:23
             Hello time(in seconds):2   Max Age(in seconds):20
             Forward Delay(in seconds):15

PORT      ROLE      STATE      COST      PRIORITY  TYPE      BPDU-Tx      BPDU-Rx
TCN-Tx    TCN-Rx
-----
-----
```

1/1/1	Designated	Forwarding	20000	128	P2P Edge	100	60
20	10						
1/1/2	Designated	Forwarding	20000	128	P2P	100	60
20	10						
1/1/3	Designated	Forwarding	20000	128	Shr	100	60
20	10						
1/1/4	Designated	Forwarding	20000	128	Shr Edge	100	60
20	10						
1/1/5	Alternate	Loop-Inc	20000	128	Shr Edge	100	60
20	10						
1/1/6	Alternate	Root-Inc	20000	128	Shr Edge	100	60
20	10						

Number of topology changes : 4
 Last topology change occurred : 516 seconds ago

Showing spanning tree mode and detailed RPVST instance information:

switch# **show spanning-tree detail**

Spanning tree status : Enabled Protocol: RPVST
 Extended System-id : Enabled
 Ignore PVID Inconsistency : Enabled
 Path cost method : Long
 RPVST-MSTP Interconnect VLAN : 1
 Current Virtual Ports Count : 0
 Maximum Allowed Virtual Ports : 2048

VLAN1

Root ID Priority : 32768
 MAC-Address: 70:72:cf:31:c9:23
 This bridge is the root
 Hello time(in seconds):2 Max Age(in seconds):20
 Forward Delay(in seconds):15

Bridge ID Priority : 32768
 MAC-Address: 70:72:cf:31:c9:23
 Hello time(in seconds):2 Max Age(in seconds):20
 Forward Delay(in seconds):15

PORT	ROLE	STATE	COST	PRIORITY	TYPE	BPDU-Tx	BPDU-Rx
TCN-Tx	TCN-Rx						
1/1/1	Designated	Forwarding	20000	128	P2P Edge	100	60
20	10						
1/1/2	Designated	Forwarding	20000	128	P2P	100	60
20	10						
1/1/3	Designated	Forwarding	20000	128	Shr	100	60
20	10						
1/1/4	Designated	Forwarding	20000	128	Shr Edge	100	60
20	10						
1/1/5	Alternate	Loop-Inc	20000	128	Shr Edge	100	60
20	10						
1/1/6	Alternate	Root-Inc	20000	128	Shr Edge	100	60
20	10						

Topology change flag : False
 Number of topology changes : 1
 Last topology change occurred : 33293 seconds ago

Port 1/1/1

```

Designated Root Priority           : 32768      Address:
48:0F:CF:AF:22:1D
Designated Bridge Priority        : 32768      Address:
48:0F:CF:AF:22:1D
Designated Port                   : 1/1/1
Forwarding-State transitions      : 0
BPDUs sent 1582, received 1506
TCN_Tx: 10, TCN_Rx: 10

Port lag1
Designated Root Priority           : 32768      Address:
48:0F:CF:AF:22:1D
Designated Bridge Priority        : 32768      Address:
48:0F:CF:AF:22:1D
Designated Port                   : lag1
Forwarding-State transitions      : 0
BPDUs sent 1402, received 1316
TCN_Tx: 10, TCN_Rx: 10
Multi-chassis role                : active

```

show spanning-tree inconsistent-ports

Syntax

```
show spanning-tree inconsistent-ports [vlan <VLAN-ID>]
```

Description

Shows ports blocked by STP protection functions such as Root guard, Loop guard, BPDU guard, and RPVST guard.

Command context

Operator (>) or Manager (#)

Parameters

<VLAN-ID>

Specifies a VLAN ID number.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Showing inconsistent port information:

```

switch# show spanning-tree inconsistent-ports
VLAN ID   Blocked Port   Reason
-----
1         1/1/1         BPDU Guard
2         1/1/1         BPDU Guard
3         1/1/1         BPDU Guard
4         1/1/1         BPDU Guard
5         1/1/1         BPDU Guard
6         1/1/1         BPDU Guard
7         1/1/1         BPDU Guard

```


8	1/1/1	BPDU Guard
9	1/1/1	BPDU Guard
10	1/1/1	BPDU Guard

Showing inconsistent port information for VLANs 1 to 4:

```
switch# show spanning-tree inconsistent-ports vlan 1-4
VLAN ID    Blocked Port    Reason
-----
1          1/1/3          Root Guard
2          1/1/7          BPDU Guard
3          1/1/9          Loop Guard
4          1/1/37         RPVST Guard
```

show spanning-tree summary port

Syntax

```
show spanning-tree summary port
```

Description

Shows a summary of port-related spanning-tree configuration and status.

Command context

Operator (>) or Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Example

Showing a summary of port-related spanning tree information:

```
switch# show spanning-tree summary port

STP status           : Enabled
Protocol             : RPVST
BPDU guard Timeout value : None
BPDU guard enabled interfaces : None
BPDU filter enabled interfaces : None
Root guard enabled interfaces : None
Loop guard enabled interfaces : None
TCN guard enabled interfaces : None
RPVST filter enabled interfaces : None
RPVST guard enabled interfaces : None

Interface count by state

VLAN              Blocking Listening Learning Forwarding
-----
VLAN1              28         0         0         0
VLAN2              0         0         0         0
VLAN3              0         0         0         0
```

VLAN4	0	0	0	0
VLAN5	0	0	0	0
VLAN6	0	0	0	0
VLAN7	0	0	0	0
VLAN8	0	0	0	0
VLAN9	0	0	0	0
VLAN10	0	0	0	0

Total = 10	28	0	0	0

show spanning-tree summary root

Syntax

```
show spanning-tree summary root
```

Description

Shows the summary of spanning tree root and configurations for all VLANs.

Command context

Operator (>) or Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Example

Showing summary of spanning tree configurations:

```
switch# show spanning-tree summary root
```

STP status : Enabled
Protocol : RPVST
System ID : f8:60:f0:c9:70:40

Root bridge for VLANs : 1-10

VLAN	Priority	Root ID	Root cost	Hello Time	Max Age	Fwd Dly	Root Port
VLAN1	32768	f8:60:f0:c9:70:40	0	2	20	15	0
VLAN2	32768	f8:60:f0:c9:70:40	0	2	20	15	0
VLAN3	32768	f8:60:f0:c9:70:40	0	2	20	15	0
VLAN4	32768	f8:60:f0:c9:70:40	0	2	20	15	0
VLAN5	32768	f8:60:f0:c9:70:40	0	2	20	15	0
VLAN6	32768	f8:60:f0:c9:70:40	0	2	20	15	0
VLAN7	32768	f8:60:f0:c9:70:40	0	2	20	15	0
VLAN8	32768	f8:60:f0:c9:70:40	0	2	20	15	0
VLAN9	32768	f8:60:f0:c9:70:40	0	2	20	15	0
VLAN10	32768	f8:60:f0:c9:70:40	0	2	20	15	0

show spanning-tree vlan

Syntax

```
show spanning-tree vlan <VLAN-ID> [detail]
```

Description

Displays the spanning tree mode and information on the RPVST instance of the specified VLAN and optionally displays details on the RPVST instance for the VLAN.

Command context

Operator (>) or Manager (#)

Parameters

<VLAN-ID>

Specifies the number of a VLAN.

[detail]

Specifies displaying details on the RPVST instance for the VLAN.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Showing spanning tree mode and RPVST instance information for VLAN 2:

```
switch# show spanning-tree vlan 2
VLAN2
Spanning tree status: Enabled Protocol: RPVST
  Root ID      Priority    : 32768
                MAC-Address: 70:72:cf:76:43:2a
                This bridge is the root
                Hello time(in seconds):2   Max Age(in seconds):20
                Forward Delay(in seconds):15

  Bridge ID    Priority    : 32768
                MAC-Address: 70:72:cf:76:43:2a
                Hello time(in seconds):2   Max Age(in seconds):20
                Forward Delay(in seconds):15

PORT      ROLE      STATE      COST      PRIORITY  TYPE      BPDU-Tx  BPDU-Rx
TCN-Tx    TCN-Rx
-----
1/1/1     Designated Forwarding 20000     128       P2P Edge  100       60
20        10
1/1/2     Designated Forwarding 20000     128       P2P      100       60
20        10
1/1/3     Designated Forwarding 20000     128       Shr      100       60
20        10
1/1/4     Designated Forwarding 20000     128       Shr Edge  100       60
20        10
1/1/5     Alternate  Loop-Inc   20000     128       Shr Edge  100       60
20        10
1/1/6     Alternate  Root-Inc   20000     128       Shr Edge  100       60
20        10

Number of topology changes      : 4
Last topology change occurred   : 516 seconds ago
```

Showing spanning tree mode and detailed RPVST instance information for VLAN 2:

```
switch# show spanning-tree vlan 2 detail
VLAN2
Spanning tree status: Enabled Protocol: RPVST
  Root ID      Priority    : 32768
                MAC-Address: 70:72:cf:76:43:2a
                This bridge is the root
                Hello time(in seconds):2  Max Age(in seconds):20
                Forward Delay(in seconds):15

  Bridge ID    Priority    : 32768
                MAC-Address: 70:72:cf:76:43:2a
                Hello time(in seconds):2  Max Age(in seconds):20
                Forward Delay(in seconds):15

PORT      ROLE      STATE      COST      PRIORITY  TYPE      BPDU-Tx  BPDU-Rx
TCN-Tx    TCN-Rx
-----
1/1/1     Designated Forwarding 20000     128       P2P Edge  100       60
20        10
1/1/2     Designated Forwarding 20000     128       P2P      100       60
20        10
1/1/3     Designated Forwarding 20000     128       Shr      100       60
20        10
1/1/4     Designated Forwarding 20000     128       Shr Edge  100       60
20        10
1/1/5     Alternate  Loop-Inc   20000     128       Shr Edge  100       60
20        10
1/1/6     Alternate  Root-Inc   20000     128       Shr Edge  100       60
20        10

Topology change flag : False
Number of topology changes : 1
Last topology change occurred : 33293 seconds ago

Port 1/1/1
Designated root has priority :32768 Address: 48:0f:cf:af:22:1d
Designated bridge has priority :32768 Address: 48:0f:cf:af:22:1d
Designated port :1
Number of transitions to forwarding state : 0
BPDUs sent 1582, received 1506
TCN_Tx: 10, TCN_Rx: 10
```

spanning-tree bpdu-guard timeout

Syntax

```
spanning-tree bpdu-guard timeout <INTERVAL>
no spanning-tree bpdu-guard timeout
```

Description

Enables and configures the auto re-enable timeout in seconds for all interfaces with BPDU guard enabled. When an interface is disabled after receiving an unauthorized BPDU it will automatically be re-enabled after the timeout expires. The default is for the interface to stay disabled until manually re-enabled.

The `no` form of the command disables BPDU guard timeout on the interface. This is the default.

Command context

config

Parameters

<INTERVAL>

Specifies the re-enable timeout in seconds. Range: 1 to 65535.

Authority

Administrators or local user group members with execution rights for this command.

Example

Enabling the BPDU guard timeout on interface 1/1/1:

```
switch(config)# interface 1/1/1
switch(config-if)# spanning-tree bpdu-guard timeout 10
```

Disabling BPDU guard timeout on interface 1/1/1:

```
switch(config)# interface 1/1/1
switch(config-if)# no spanning-tree bpdu-guard
```

spanning-tree extend-system-id

Syntax

```
spanning-tree extend-system-id {enable | disable}
no spanning-tree extend-system-id
```

Description

Configures use of extended system ID. When enabled, the VLAN ID is included in spanning tree packets. When disabled, the VLAN ID is set to NULL in the spanning tree packets.

By default, extended system ID is enabled. If you disable extended system ID, the bridge identifier field in the spanning tree packet is filled with zeros.

The `no` form of this command disables extended system ID.

Command context

config

Parameters

enable

Specifies enabling use of extended system ID.

disable

Specifies disabling use of extended system ID.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling extended system ID:

```
switch# config  
switch(config)# spanning-tree extend-system-id enable
```

Disabling extended system ID:

```
switch# config  
switch(config)# spanning-tree extend-system-id disable  
switch(config)# no spanning-tree extend-system-id
```

spanning-tree ignore-pvid-inconsistency

Syntax

```
spanning-tree ignore-pvid-inconsistency {enable | disable}  
no spanning-tree ignore-pvid-inconsistency
```

Description

Configures port behavior when per-VLAN ID inconsistencies are present. For example, when the ports on both ends of a point-to-point link are untagged members of different VLANs, enabling this option allows RPVST+ to process untagged RPVST+ packets belonging to the peer's untagged VLAN as if they were received on the current device's untagged VLAN. When this option is disabled, RPVST+ blocks the link, causing traffic on the mismatched VLANs to be dropped.

If this option is enabled on multiple switches connected by hubs, there could be more than two VLANs involved in PVID mismatches that will be ignored by RPVST+.

If port VLAN memberships is misconfigured on a switch in the network, then enabling this option prevents RPVST+ from detecting the problem, which may result in packet duplication in the network since RPVST+ would not converge correctly.

This command affects all ports on the switch belonging to VLANs on which RPVST+ is enabled.

By default ignore per-VLAN ID inconsistency is disabled.

The `no` form of this command sets the ignore per-VLAN ID inconsistencies to disabled.

Command context

`config`

Parameters

`enable`

Specifies ignore per-VLAN ID inconsistencies and allow RPVST to run on mismatched links.

`disable`

Disables the ignore per-VLAN ID inconsistencies functionality.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling ignore per-VLAN ID inconsistencies:

```
switch# config  
switch(config)# spanning-tree ignore-pvid-inconsistency enable
```

Disabling ignore per-VLAN ID inconsistencies:

```
switch# config  
switch(config)# spanning-tree ignore-pvid-inconsistency disable  
switch(config)# no spanning-tree ignore-pvid-inconsistency
```

spanning-tree link-type

Syntax

```
spanning-tree link-type {point-to-point | shared}  
no spanning-tree link-type
```

Description

Configures the link type of a port.

The `no` form of this command sets the spanning tree link type to the default value of `point-to-point`.

Command context

`config-if`

Parameters

`point-to-point`

Sets the spanning tree link type as point-to-point. Use this for full-duplex ports that provide a point-to-point link to devices such as a switch, bridge, or end-node. Default.

`shared`

Sets the spanning tree link type as shared. Use this when the port is connected to a hub.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting spanning tree link type to shared:

```
switch(config)# interface 1/1/1  
switch(config-if)# spanning-tree link-type shared
```

Setting spanning tree link type to point-to-point for a port:

```
switch(config)# interface 1/1/1  
switch(config-if)# no spanning-tree link-type
```

spanning-tree mode

Syntax

```
spanning-tree mode {mstp|rpvst}
```

Description

Sets the spanning tree mode to either MSTP mode (Multiple-instance Spanning Tree Protocol) or RPVST mode (Rapid Per VLAN Spanning Tree).

The `no` form of this command sets the spanning tree mode to the default value `mstp`.

Command context

`config`

Parameters

`mstp`

Sets the mode to MSTP (Multiple-instance Spanning Tree Protocol), which applies the STP (spanning tree protocol) separately for each set of VLANs (called an MSTI - multiple spanning tree instance).

`rpvst`

Sets the mode to RPVST (Rapid Per VLAN Spanning Tree).

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling MSTP mode:

```
switch(config)# spanning-tree mode mstp
```

Enabling RPVST mode:

```
switch(config)# spanning-tree mode rpvst
```

spanning-tree pathcost-type

Syntax

```
spanning-tree pathcost-type {long | short}  
no spanning-tree pathcost-type
```

Description

Configures the spanning tree path cost type. The long mode provides support for the wider range of link speeds required by high-speed interfaces. All switches in the network must use the same path cost type or errors can occur in the spanning tree.

The `no` form of this command sets the spanning tree path cost type to the default value of `long`.

Command context

`config`

Parameters

`long`

Specifies the spanning tree path cost type as a 32-bit value, allowing port cost values to be set in the range 1-200,000,000. Default.

`short`

Specifies the spanning tree path cost type as a 16-bit value, allowing port cost values to be set in the range 1-65535.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting spanning tree path cost type to short:

```
switch# config
switch(config)# spanning-tree pathcost-type short
```

Setting spanning tree path cost type to long:

```
switch# config
switch(config)# spanning-tree pathcost-type long
```

Setting spanning tree path cost to default of long:

```
switch# config
switch(config)# no spanning-tree pathcost-type
```

spanning-tree rpvst-mstp interconnect vlan

Syntax

```
spanning-tree rpvst-mstp-interconnect-vlan <VLAN-ID>
no spanning-tree rpvst-mstp-interconnect-vlan
```

Description

Configures the VLAN that has to be used to interconnect RPVST and MSTP domains. VLAN 1 is used by default.

The `no` form of this command removes the VLAN configuration.

- It is required to create the interconnect VLAN and then configure RPVST spanning tree on it.
- The same interconnect VLAN must be kept on all the switches in the network.
- Adding or deleting the interconnect VLAN triggers a re-convergence in the network.
- Deleting a VLAN that is configured as the interconnect VLAN does not reset the value to the default.

Command context

config

Parameters

<VLAN-ID>

Specifies the number of a VLAN.

Authority

Administrators or local user group members with execution rights for this command.

Examples

This example configures VLAN 10 to be used to interconnect RPVST and MSTP domains.

```
switch#(config)# spanning-tree rpvst-mstp-interconnect-vlan 10
```

spanning-tree tcn-guard

Syntax

```
spanning-tree tcn-guard  
no spanning-tree tcn-guard
```

Description

Disables propagation of topology change notifications (TCNs) to other STP ports. Use this when you do not want topology changes to be noticed by peer devices.

The `no` form of this command, enables propagation of topology changes. By default, propagation is disabled.

Command context

```
config-if
```

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling `tcn-guard`, which disables propagation of topology changes:

```
switch(config-if)# spanning-tree tcn-guard
```

Disabling `tcn-guard`, which enables propagation of topology changes:

```
switch(config-if)# no spanning-tree tcn-guard
```

spanning-tree vlan

Syntax

```
spanning-tree vlan <VLAN-LIST> [{hello-time | forward-delay | max-age | priority} <VALUE>]  
no spanning-tree vlan <VLAN-LIST> [hello-time | forward-delay | max-age | priority]
```

Description

Creates an RPVST instance for the specified VLAN. This command also allows for configuration of RPVST instance-specific time parameters.

The `no` form of this command removes the RPVST instance associated with the specified VLAN, and configures default values for RPVST instance-specific parameters.

Command context

```
config
```

Parameters

<VLAN-LIST>

Specifies the number of a single VLAN, or a series of numbers for a range of VLANs, separated by commas (1, 2, 3, 4), dashes (1-4), or both (1-4,6).

hello-time <VALUE>

Specifies the hello-time in seconds for the RPVST instance. Range: 2-10 seconds. Default: 2 seconds.

forward-delay <VALUE>

Specifies the forward-delay time in seconds for the RPVST instance. Range: 4-30 seconds. Default: 15 seconds.

max-age <VALUE>

Specifies the maximum age time in seconds for the RPVST instance. Range: 6-40 seconds. Default: 20 seconds.

priority <VALUE>

Specifies the priority for the RPVST instance. Priority value is configured as a multiple of 4096. Range: 0-15. Default: 8 which is 32768.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Creating an RPVST instance for a list of VLANs and configuring various time parameters:

```
switch# config
switch(config)# spanning-tree vlan 2-5
switch(config)# spanning-tree vlan 2-5 hello-time 5
switch(config)# spanning-tree vlan 5 max-age 10
switch(config)# spanning-tree vlan 2-5 forward-delay 25
switch(config)# spanning-tree vlan 2-5 priority 5
```

Removing an RPVST instance for a list of VLANs and setting various time parameters to the default:

```
switch# config
switch(config)# no spanning-tree vlan 2-5
switch(config)# no spanning-tree vlan 2-5 hello-time
switch(config)# no spanning-tree vlan 2-5 forward-time
switch(config)# no spanning-tree vlan 2-5 max-age
switch(config)# no spanning-tree vlan 2-5 priority
```

spanning-tree vlan cost

Syntax

```
spanning-tree vlan <VLAN-LIST> cost <PORT-COST>
no spanning-tree vlan <VLAN-LIST> cost
```

Description

Configures the spanning tree cost for the VLAN. This is the cost to reach the root port.

The `no` form of this command sets the port cost to the default value.

Command context

config-if

Parameters

<VLAN-LIST>

Specifies the number of a single VLAN, or a series of numbers for a range of VLANs, separated by commas (1, 2, 3, 4), dashes (1-4), or both (1-4,6).

<PORT-COST>

Specifies the spanning tree cost for the VLAN. Range: 1-200,000,000. Default is calculated from the port link speed:

- 10 Mbps link speed equals a path cost of 2,000,000.
- 100 Mbps link speed equals a path cost of 200,000.
- 1 Gbps link speed equals a path cost of 20,000.
- 2 Gbps link speed equals a path cost of 10,000.
- 10 Gbps link speed equals a path cost of 2,000.
- 100 Gbps link speed equals a path cost of 200.
- 1 Tbps link speed equals a path cost of 20.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting port cost:

```
switch(config-if)# spanning-tree vlan 5 cost 100000
```

Setting port cost to the default:

```
switch(config-if)# no spanning-tree vlan 5 cost
```

spanning-tree vlan port-priority

Syntax

```
spanning-tree vlan <VLAN-LIST> port-priority <PRIORITY>  
no spanning-tree vlan <VLAN-LIST> port-priority
```

Description

Configures port priority. A port with the lowest priority number has the highest priority for use in forwarding traffic.

The `no` form of this command, sets the port priority to the default of 8.

Command context

config-if

Parameters

<VLAN-LIST>

Specifies the number of a single VLAN, or a series of numbers for a range of VLANs, separated by commas (1, 2, 3, 4), dashes (1-4), or both (1-4,6).

<PRIORITY>

Specifies the port priority. The value, configured as a multiple of 16, helps in determining the designated port. The lower a priority value, the higher the priority. Range: 1 to 15. Default: 8.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting port priority:

```
switch(config-if)# spanning-tree vlan 5 port-priority 10
```

Setting port priority to the default of 8:

```
switch(config-if)# no spanning-tree vlan 5 port-priority
```

spanning-tree trap

Syntax

```
spanning-tree trap {new-root | topology-change [vlan <VLAN-ID>] | errant-bpdu |  
root-guard-inconsistency | loop-guard-inconsistency}
```

```
no spanning-tree trap {new-root|topology-change [vlan <VLAN-ID>] | errant-bpdu |  
root-guard-inconsistency | loop-guard-inconsistency}
```

Description

This command enables SNMP traps for new root, topology change event, errant-bpdu received event, root-guard inconsistency, and loop-guard inconsistency notifications. It is disabled by default.

The `no` form of this command disables SNMP traps for new root and topology change event notifications.

Command context

config

Parameters

`new-root`

Enables SNMP notification when a new root is elected on any PVST vlan on the switch.

`topology-change`

Enables SNMP notification when a topology change event occurred in specified PVST vlan on the switch.

`errant-bpdu`

Enables SNMP notification when an errant bpdu is received by any PVST vlan on the switch.

`root-guard-inconsistency`

Enables SNMP notification when the root-guard finds the port inconsistent for any PVST vlan on the switch.

`loop-guard-inconsistency`

Enables SNMP notification when the loop-guard finds the port inconsistent for any PVST vlan on the switch.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Commands enabling spanning-tree trap features:

```
switch(config)# spanning-tree trap
    new-root          Enable notifications which are sent when a new root is
elected
    topology-change   Enable notifications which are sent when a topology
change occurs
    errant-bpdu       Enable notifications which are sent when an errant bpdu
is received
    root-guard-inconsistency Enable notifications which are sent when root guard
inconsistency occurs
    loop-guard-inconsistency Enable notifications which are sent when root guard
inconsistency occurs
switch(config)# spanning-tree trap new-root
<cr>
switch(config)# spanning-tree trap topology-change
    vlan Enable topology change notification for the specified PVST vlan id.
switch(config)# spanning-tree trap topology-change vlan
    <0-64> Enable topology change information on the specified vlan id.
switch(config)# spanning-tree trap topology-change vlan 1
<cr>
switch(config)# spanning-tree trap errant-bpdu
<cr>
switch(config)# spanning-tree trap root-guard-inconsistency
<cr>
switch(config)# spanning-tree trap loop-guard-inconsistency
<cr>
```

Commands disabling spanning-tree trap features:

```
switch(config)# no spanning-tree trap
    new-root          Disable notifications which are sent when a new root is
elected
    topology-change   Disable notifications which are sent when a topology
change occurs
    errant-bpdu       Disable notifications which are sent when an errant bpdu
is received
    root-guard-inconsistency Disable notifications which are sent when root guard
inconsistency occurs
    loop-guard-inconsistency Disable notifications which are sent when root guard
inconsistency occurs
switch(config)# no spanning-tree trap new-root
<cr>
switch(config)# no spanning-tree trap topology-change
    instance Disable topology change notification for the specified PVST vlan id.
switch(config)# no spanning-tree trap topology-change vlan
    <0-64> Disable topology change information on the specified PVST vlan id.
switch(config)# no spanning-tree trap topology-change vlan 0
<cr>
switch(config)# no spanning-tree trap errant-bpdu
<cr>
switch(config)# no spanning-tree trap root-guard-inconsistency
<cr>
switch(config)# no spanning-tree trap loop-guard-inconsistency
<cr>
```

diagnostic monitor

Syntax

```
diagnostic monitor {line-module | management-module} [<SLOT-ID>]  
no diagnostic monitor {line-module | management-module} [<SLOT-ID>]
```

Description

Enables runtime diagnostics for all modules or for a specified module. This feature is enabled by default for all modules.

The `no` form of this command disables runtime diagnostics for all modules or for a specified module.

Command context

`config`

Parameters

`line-module`

Specifies the enabling of diagnostic monitoring specific to a line module.

`management-module`

Specifies the enabling of diagnostic monitoring specific to a management module.

`<SLOT-ID>`

Specifies the slot ID of a module. Format: member/slot.

Authority

Administrators or local user group members with execution rights for this command.

Usage

When no parameters are used in the command (`diagnostic monitor` or `no diagnostic monitor`), the command applies to all modules. This command impacts the diagnostics that run periodically. It does not affect on-demand diagnostics.

Example

Enabling runtime diagnostics for a specified module:

```
switch(config)# diagnostic monitor management-module 1/1
```

diag on-demand

Syntax

```
diag on-demand {line-module | management-module} [<SLOT-ID>]
```

Description

Runs the diagnostic tests for all modules or for a specified module.

Command context

Manager (#)

Parameters

[line-module | management-module]

Selects the options for enabling or disabling runtime diagnostics for a specific module.

line-module

Specifies the enabling of diagnostic monitoring specific to a line module.

management-module

Specifies the enabling of diagnostic monitoring specific to a management module.

<SLOT-ID>

Specifies the member/slot for management modules (1/1) and line modules (1/1).

Authority

Administrators or local user group members with execution rights for this command.

Usage

When no parameters are used in the command (`diag on-demand`), the command applies to all modules.

Examples

Running diagnostic tests for all modules on a 6100 switch:

```
switch# diag on-demand
Fetching Test results. Please wait ...

Module                ID      Diagnostics Success
                   ID      Performed
-----
LineModule            1/1          13      100%
ManagementModule      1/1          13      100%
```

Running diagnostic tests for a specific module on a 6100 switch:

```
switch# diag on-demand management-module 1/1
Performing diagnostic tests. Please wait ...
Fetching Test results. Please wait ...

Module                ID      Diagnostics Success
                   ID      Performed
-----
ManagementModule      1/1          13      100%
```

show diagnostic

Syntax

show diagnostic {line-module | management-module} [<SLOT-ID>] {brief | detail}

Description

Displays the diagnostic test results for all modules or for a specified module.

Command context

Manager (#)

Parameters

[line-module | management-module]

Selects the options for enabling or disabling runtime diagnostics for a specific module.

line-module

Specifies the enabling of diagnostic monitoring specific to a line module.

management-module

Specifies the enabling of diagnostic monitoring specific to a management module.

<SLOT-ID>

Specifies the member/slot for management modules (1/1) and line modules (1/1)

Authority

Administrators or local user group members with execution rights for this command.

Usage

When no parameters are used in the command (`show diagnostic`), the command applies to all modules.

Example

Showing diagnostic test results in brief format for all modules on a 6100 switch:

```
switch# show diagnostic brief
Module           ID      Diagnostics Success
                  ID      Performed
-----
ManagementModule 1/1      13      100%
LineModule        1/1      13      100%
```

Showing diagnostic test results in brief format for a specified module on a 6100 switch:

```
switch# show diagnostic line-module brief
Module           ID      Diagnostics Success
                  ID      Performed
-----
LineModule        1/1      13      100%
```

Showing diagnostic test results in detail format for all modules on a 6100 switch:

```
switch# show diagnostic detail

Module : ManagementModule 1/1

Diagnostic      Status Error Code History Code Successive      Total Failure Total
Last Run Timestamp First Run Timestamp                               Failure Count Count          Iteration
-----
ddr_cecount     Pass   0x0      0x0                               0          0          109
```

2019-07-31 16:43:38	2019-07-31 07:44:55				
emmc	Pass	0x0	0x0	0	4
2019-07-31 16:08:04	2019-07-31 07:44:55				
fan_ctrlr	Pass	0x0	0x0	0	4
2019-07-31 16:08:04	2019-07-31 07:44:55				
fepld	Pass	0x0	0x0	0	109
2019-07-31 16:43:38	2019-07-31 07:44:54				
fru_eeeprom	Pass	0x0	0x0	0	4
2019-07-31 16:08:04	2019-07-31 07:44:54				
fru_eeeprom_ul	Pass	0x0	0x0	0	4
2019-07-31 16:08:04	2019-07-31 07:44:54				
mm_lcb	Pass	0x0	0x0	0	109
2019-07-31 16:43:37	2019-07-31 07:44:54				
pmc	Pass	0x0	0x0	0	109
2019-07-31 16:43:37	2019-07-31 07:44:54				
rdimm_spd	Pass	0x0	0x0	0	4
2019-07-31 16:08:04	2019-07-31 07:44:55				
rdimm_tmp	Pass	0x0	0x0	0	4
2019-07-31 16:08:04	2019-07-31 07:44:55				
rtc	Pass	0x0	0x0	0	4
2019-07-31 16:08:04	2019-07-31 07:44:55				
tmp1	Pass	0x0	0x0	0	4
2019-07-31 16:08:04	2019-07-31 07:44:55				
tmp2	Pass	0x0	0x0	0	4
2019-07-31 16:08:04	2019-07-31 07:44:55				

Module : LineModule 1/1

Diagnostic Last Run	Status Timestamp	Error Code First Run	History Code Timestamp	Successive Failure Count	Total Count	Total Iteration
lc_asic	Pass	0x0	0x0	0	0	108
2019-07-31 16:43:37	2019-07-31 07:46:03					
poe_ctrlr_1_q1	Pass	0x0	0x0	0	0	4
2019-07-31 16:08:16	2019-07-31 07:46:03					
poe_ctrlr_1_q2	Pass	0x0	0x0	0	0	4
2019-07-31 16:08:16	2019-07-31 07:46:04					
poe_ctrlr_1_q3	Pass	0x0	0x0	0	0	4
2019-07-31 16:08:16	2019-07-31 07:46:04					
poe_ctrlr_2_q1	Pass	0x0	0x0	0	0	4
2019-07-31 16:08:16	2019-07-31 07:46:05					
poe_ctrlr_2_q2	Pass	0x0	0x0	0	0	4
2019-07-31 16:08:16	2019-07-31 07:46:05					
poe_ctrlr_2_q3	Pass	0x0	0x0	0	0	4
2019-07-31 16:08:16	2019-07-31 07:46:05					
poe_ctrlr_3_q1	Pass	0x0	0x0	0	0	4
2019-07-31 16:08:16	2019-07-31 07:46:06					
poe_ctrlr_3_q2	Pass	0x0	0x0	0	0	4
2019-07-31 16:08:16	2019-07-31 07:46:06					
poe_ctrlr_3_q3	Pass	0x0	0x0	0	0	4
2019-07-31 16:08:17	2019-07-31 07:46:06					
poe_ctrlr_4_q1	Pass	0x0	0x0	0	0	4
2019-07-31 16:08:17	2019-07-31 07:46:07					
poe_ctrlr_4_q2	Pass	0x0	0x0	0	0	4
2019-07-31 16:08:17	2019-07-31 07:46:07					
poe_ctrlr_4_q3	Pass	0x0	0x0	0	0	4
2019-07-31 16:08:17	2019-07-31 07:46:08					

show diagnostic events

Syntax

show diagnostic events

Description

Displays the diagnostic related event logs.

Authority

Administrators or local user group members with execution rights for this command.

Example

Showing diagnostic related event logs:

```
switch# show diagnostic events
2019-08-07:17:19:21.214532|hhmd|106001|ERR|
Diagnostic mm_mcbe failed with error code 0x380 on management module 1/1
2019-08-07:17:19:21.214554|hhmd|106001|ERR|
Diagnostic pmc failed with error code 0x4 on management module 1/1
2019-08-07:17:19:21.215532|hhmd|106001|ERR|
Diagnostic ledpld failed with error code 0x4 on management module 1/1
2019-08-07:17:19:21.353221|hhmd|106001|ERR|
Diagnostic mm_mcbe failed with error code 0x380 on management module 1/1
2019-08-07:17:19:21.354421|hhmd|106001|ERR|
Diagnostic pmc failed with error code 0x4 on management module 1/1
2019-08-07:17:19:21.453221|hhmd|106001|ERR|
Diagnostic ledpld failed with error code 0x4 on management module 1/1
```

fastboot

Syntax

```
fastboot
no fastboot
```

Description

Enables fastboot for the system.

The `no` form of this command disables fastboot for the system.

Command context

```
config
```

Authority

Administrators or local user group members with execution rights for this command.

Usage

When fastboot is enabled, most tests under a Power On Self Test (POST) are skipped. By default, fastboot is enabled.

After disabling fastboot, save switch configurations and then reboot for POST to run. POST verifies the hardware functionality of various modules during boot-up. Based on the criticality of the test, the selftest module decides whether to go ahead with the boot-up sequence of a particular subsystem or interface during a POST failure.

POST runs memory built-in selftest (BISTs) and front-end port loopback tests. Memory BISTs verify the internal and external memory blocks present in the module. The memory tables are critical for proper functionality of the system so any failures in these tests results in the corresponding subsystem to be marked as "Failed" and thus that subsystem is not available for use.

Front-end port loopback tests verify the physical port front-end interface. These tests check if a particular interface can function properly. A test failure means that a particular interface has been marked as "Failed" and is now unavailable for use.

Examples

Enabling fastboot:

```
switch# configure terminal
switch(config)# fastboot
switch(config)# end
switch# show running-config
Current configuration:
!
!Version ArubaOS-CX PL.10.06.0001
```

```

module 1/1 product-number j1677a
!
!
!
!
!
!
!
vlan 1
interface 1/1/1
    no shutdown

```

Disabling fastboot:

```

switch# configure terminal
switch(config)# no fastboot
switch(config)# end
switch(config)# write mem
Configuration changes will take time to process, please be patient.
switch# show running-config
Current configuration:
!
!Version ArubaOS-CX PL.10.06.0001
module 1/1 product-number j1677a
!
!
!
no fastboot
!
!
!
!
vlan 1
interface 1/1/1
    no shutdown

```

show selftest

Syntax

```

show selftest [brief]
show selftest line-module <SLOT-ID>
show selftest line-module <SLOT-ID> interface [brief]
show selftest interface [<PORT-NUM>]

```

Description

Displays selftest results.

Command context

Manager (#)

Parameters

[brief]

Shows the selftest results as a brief description. Default.

line-module

Shows the selftest results for a line module.

<SLOT-ID>

Shows the selftest results for the slot ID of the line or fabric module.

<PORT-NUM>

Shows the selftest results for the port number.

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Examples

Displaying the output when fastboot is enabled:

```
switch# show selftest
Name      Id    Status      ErrorCode  LastRunTime
-----
LineModule 1/1  passed      0x0
LineModule 1/2  passed      0x0

switch# show selftest line-module
Name      Id    Status      ErrorCode  LastRunTime
-----
LineModule 1/1  passed      0x0
LineModule 1/2  passed      0x0

switch# show selftest line-module 1/1
Name      Id    Status      ErrorCode  LastRunTime
-----
LineModule 1/1  passed      0x0
```

Displaying the output when fastboot is enabled:

```
switch# show selftest interface 1/1/2
Name      Status      ErrorCode  LastRunTime
-----
1/1/2     skipped      0x0

switch# show selftest line-module 1/1 interface
Name      Status      ErrorCode  LastRunTime
-----
1/1/1     skipped      0x0
1/1/2     skipped      0x0
1/1/3     skipped      0x0
1/1/31    skipped      0x0
```

Displaying the output when fastboot is disabled:

Testing to register read/write:



This test is run irrespective of fastboot being enabled or disabled.

```
switch# show selftest
Name      Id    Status      ErrorCode  LastRunTime
-----
LineModule 1/1  passed      0x0        2018-02-16 18:15:53
```

clear sflow statistics

Syntax

```
clear sflow statistics {global | interface <INTERFACE-NAME>}
```

Description

This command clears the sFlow sample statistics counter to 0 either globally or for a specific interface.

Command context

```
config
```

Parameters

```
global
```

Specifies all interfaces on the switch.

```
interface <INTERFACE-NAME>
```

Specifies the name of an interface on the switch.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Clearing the global sFlow sample statistics counter to 0 globally:

```
switch(config)# clear sflow statistics global
```

Clearing the global sFlow sample statistics counter to 0 for interface 1/1/1:

```
switch(config)# clear sflow statistics interface 1/1/1
```

sflow

Syntax

```
sflow  
no sflow
```

Description

Enables the sFlow agent.

- In the `config` context, this command enables the sFlow agent globally on all interfaces.
- In an `config-if` context, this command enables the sFlow agent on a specific interface. sFlow cannot be enabled on a member of a LAG, only on the LAG.

The sFlow agent is disabled by default.

The `no` form of this command disables the sFlow agent and deletes all sFlow configuration settings, either globally, or for a specific interface.

Command context

```
config
config-if
```

Parameters

None.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling sFlow globally on all interfaces:

```
switch(config)# sflow
```

Disabling sFlow globally on all interfaces:

```
switch(config)# no sflow
```

Enabling sFlow on interface **1/1/1**:

```
switch(config)# interface 1/1/1
switch(config-if)# sflow
```

Disabling sFlow on interface **1/1/1**:

```
switch(config)# interface 1/1/1
switch(config-if)# no sflow
```

Enabling sFlow on interface **lag100**:

```
switch(config)# interface lag100
switch(config-if)# sflow
```

Disabling sFlow on interface **lag100**:

```
switch(config)# interface lag100
switch(config-if)# no sflow
```


sflow agent-ip

Syntax

```
sflow agent-ip <IP-ADDR>  
no sflow agent-ip [<IP-ADDR>]
```

Description

Defines the IP address of the sFlow agent to use in sFlow datagrams. This address must be defined for sFlow to function. HPE recommends that the address:

- can uniquely identify the switch
- is reachable by the sFlow collector
- does not change with time

The `no` form of this command deletes the IP address of the sFlow agent. This causes sFlow to stop working and no datagrams will be sent to the sFlow collector.

Command context

config

Parameters

<IP-ADDR>

Specifies an IP address in IPv4 format (`x.x.x.x`), where `x` is a decimal number from 0 to 255, or IPv6 format (`xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx`), where `x` is a hexadecimal number from 0 to F. The agent address is used to identify the switch in all sFlow datagrams sent to sFlow collectors. It is usually set to an IP address on the switch that is reachable from an sFlow collector.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the agent address to **10.10.10.100**:

```
switch(config)# sflow agent-ip 10.0.0.100
```

Setting the agent address to **2001:0db8:85a3:0000:0000:8a2e:0370:7334**:

```
switch(config)# sflow agent-ip 2001:0db8:85a3:0000:0000:8a2e:0370:7334
```

Removing the address configuration from the switch, which results in sFlow being disabled:

```
switch(config)# no sflow agent-ip
```

sflow collector

Syntax

```
sflow collector <IP-ADDR> [port <PORT>] [vrf <VRF>]  
no sflow collector <IP-ADDR> [port <PORT>] [vrf <VRF>]
```

Description

Defines a collector to which the sFlow agent sends data. Up to three collectors can be defined. At least one collector should be defined, and it must be reachable from the switch for sFlow to work.

Command context

config

Parameters

collector <IP-ADDR>

Specifies the IP address of a collector in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255, or IPv6 format (xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx), where x is a hexadecimal number from 0 to F.

port <PORT>

Specifies the UDP port on which to send information to the sFlow collector. Range: 0 to 65536. Default: 6343.

vrf <VRF>

Specifies the VRF on which to send information to the sFlow collector. The VRF must be defined on the switch. If no VRF is specified, the default VRF (default) is used.

Authority

Administrators or local user group members with execution rights for this command.

Example

Defining a collector with IP address **10.10.10.100** on UDP port **6400**:

```
switch(config)# sflow collector 10.0.0.1 port 6400
```

sflow disable

Syntax

sflow disable

Description

Disables the sFlow agent, but retains any existing sFlow configuration settings. The settings become active if the sFlow agent is re-enabled.

Command context

config

Parameters

None.

Authority

Administrators or local user group members with execution rights for this command.

Example

Disabling sFlow support:

```
switch(config)# sflow disable
```

sflow header-size

Syntax

```
sflow header-size <SIZE>  
no sflow header-size [<SIZE>]
```

Description

Sets the sFlow header size in bytes.

The `no` form of this command sets the header size to the default value of 128.

Command context

config

Parameters

header-size <SIZE>

Specifies the sFlow header size in bytes. Range: 64 to 256. Default: 128.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the header size to **64** bytes:

```
switch(config)# sflow header-size 64
```

Setting the header size to the default value of **128** bytes:

```
switch(config)# no sflow header-size
```

sflow max-datagram-size

Syntax

```
sflow max-datagram-size <SIZE>  
no sflow max-datagram-size [<SIZE>]
```

Description

Sets the maximum number of bytes that are sent in one sFlow datagram.

The `no` form of this command sets maximum number of bytes to the default value of 1400.

Command context

config

Parameters

max-datagram-size <SIZE>

Specifies the maximum datagram size in bytes. Range: 1 to 9000. Default: 1400.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the datagram size to **1000** bytes:

```
switch(config)# sflow max-datagram-size 1000
```

Setting the header size to the default value of **1400** bytes:

```
switch(config)# no sflow max-datagram-size
```

sflow polling

Syntax

```
sflow polling <INTERVAL>  
no sflow polling [<INTERVAL>]
```

Description

Defines the global polling interval for sFlow in seconds.

The `no` form of this command sets the polling interval to the default value of 30 seconds.

Command context

config

Parameters

<INTERVAL>

Specifies the polling interval in seconds. Range: 10 to 3600. Default: 30.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the polling interval to 10:

```
switch(config)# sflow polling 10
```

Setting the polling interval to the default value.

```
switch(config)# no sflow polling
```

sflow sampling

Syntax

```
sflow sampling <RATE>  
no sflow sampling [<RATE>]
```

Description

Defines the global sampling rate for sFlow in number of packets. The default sampling rate is 4096, which means that one in every 4096 packets is sampled. A warning message is displayed when the sampling rate is set to less than 4096 and proceeds only after user confirmation.

The `no` form of this command sets the sampling rate to the default value of 4096.

Command context

config

Parameters

sampling <RATE>

Specifies the sampling rate. Range: 1 to 1000000000. Default: 4096.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the sampling rate to **5000**:

```
switch(config)# sflow sampling 5000
```

Setting the sampling rate to the default:

```
switch(config)# no sflow sampling
```

Setting the sampling rate to **1000**:

```
switch(config)# sflow sampling 1000  
Setting the sFlow sampling rate lower than 4096 is not recommended and might  
affect system performance.  
Do you want to continue [y/n]? y  
switch(config)#
```

show sflow

Syntax

```
show sflow [interface <INTERFACE-NAME>]
```

Description

Shows sFlow configuration settings and statistics for all interfaces, or for a specific interface

Command context

Manager (#)

Parameters

interface <INTERFACE-NAME>

Specifies the name of an interface on the switch.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Showing sFlow information for all interfaces:

```
switch# show sflow
sFlow Global Configuration
-----
sFlow                      enabled
Collector IP/Port/Vrf      10.10.10.2/6343/default
Agent Address              10.0.0.1
Sampling Rate              1024
Polling Interval           30
Header Size                128
Max Datagram Size          1400

sFlow Status
-----
Running - Yes

sFlow enabled on Interfaces:
-----
lag100

sFlow Statistics
-----
Number of Samples          200
```

Showing sFlow information for interface **1/1/1**:

```
switch# show sflow interface 1/1/1
sFlow configuration - Interface 1/1/1
-----
sFlow                      enabled
Sampling Rate              1024
Number of Samples          30
sFlow Sampling Status      success
```

event-trap-enable

Syntax

```
event-trap-enable
```

```
no event-trap-enable
```

Description

Enables the notification of events to be sent as traps to the SNMP management stations. It is enabled by default.

The `no` form of this command disables the event traps.

Command context

```
config
```

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling the event traps:

```
switch(config)# event-trap-enable
```

Disabling the event traps:

```
switch(config)# no event-trap-enable
```

lldp trap enable

Syntax

```
lldp trap enable
```

```
no lldp trap enable
```

Description

Enables sending SNMP traps for LLDP related events from a particular interface. LLDP trap generation is enabled by default on all the interfaces and has to be disabled for interfaces on which traps are not required to be generated.

The `no` form of this command disables the LLDP trap generation.



LLDP trap generation is disabled by default at the global level and must be enabled before any LLDP traps are sent.

Command context

`config` and `config-if`

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling LLDP trap generation on global level:

```
switch(config)# lldp trap enable
```

Enabling LLDP trap generation on interface level:

```
switch(config-if)# lldp trap enable
```

Disabling LLDP trap generation on global level:

```
switch(config)# no lldp trap enable
```

Disabling LLDP trap generation on interface level:

```
switch(config-if)# no lldp trap enable
```

Displaying LLDP global configuration:

```
switch# show lldp configuration

LLDP Global Configuration
=====
LLDP Enabled                : No
LLDP Transmit Interval      : 30
LLDP Hold Time Multiplier   : 4
LLDP Transmit Delay Interval : 2
LLDP Reinit Timer Interval  : 2
LLDP Trap Enabled           : No

TLVs Advertised
=====
Management Address
Port Description
Port VLAN-ID
System Description
System Name

LLDP Port Configuration
```



```

=====
PORT          TX-ENABLED      RX-ENABLED      INTF-TRAP-ENABLED
-----
1/1/1         Yes             Yes             Yes
1/1/2         Yes             Yes             Yes
1/1/3         Yes             Yes             Yes
1/1/4         Yes             Yes             Yes
1/1/5         Yes             Yes             Yes
1/1/6         Yes             Yes             Yes
.....
.....
mgmt          Yes             Yes             Yes

```

Displaying LLDP Configuration for the interface:

```

switch# show lldp configuration 1/1/1

LLDP Global Configuration
=====
LLDP Enabled           : Yes
LLDP Transmit Interval : 30
LLDP Hold Time Multiplier : 4
LLDP Transmit Delay Interval : 2
LLDP Reinit Timer Interval : 2
LLDP Trap Enabled      : No

LLDP Port Configuration
=====
PORT          TX-ENABLED      RX-ENABLED      INTF-TRAP-ENABLED
-----
1/1/1         Yes             Yes             Yes

```

Displaying LLDP Configuration for the management interface:

```

switch# show lldp configuration mgmt

LLDP Global Configuration
=====
LLDP Enabled           : Yes
LLDP Transmit Interval : 30
LLDP Hold Time Multiplier : 4
LLDP Transmit Delay Interval : 2
LLDP Reinit Timer Interval : 2
LLDP Trap Enabled      : Yes

LLDP Port Configuration
=====
PORT          TX-ENABLED      RX-ENABLED      INTF-TRAP-ENABLED
-----
mgmt          Yes             Yes             Yes

```

rmon alarm

Syntax

```

rmon alarm index <INDEX> snmp-oid <SNMP-OID> rising-threshold <RISING-THRESHOLD>
    falling-threshold <FALLING-THRESHOLD> [sample-interval <SAMPLE-INTERVAL>] [sample-type

```

<ABSOLUTE|DELTA>]

no rmon alarm [index <INDEX>]

Description

Stores configuration entries in an alarm table that defines the sample interval, sample-type, and threshold parameters for an SNMP MIB object. Only the SNMP MIB objects that resolve to an ASN.1 primitive type of INTEGER (INTEGER, Integer32, Counter32, Counter64, Gauge32, or TimeTicks) will be monitored.

The `no` form of this command removes all RMON alarms and allows you to specify an index to remove a particular RMON alarm.

Command context

config

Parameters

index <INDEX>

Specifies the RMON alarm index. Range: 1 to 20.

snmp-oid <SNMP-OID>

Specifies the SNMP MIB object to be monitored by RMON.

rising-threshold <RISING-THRESHOLD>

Specifies the upper threshold value for the RMON alarm.

falling-threshold <FALLING-THRESHOLD>

Specifies the falling threshold value for the RMON alarm. The falling threshold must be less than the rising threshold.

sample-interval <SAMPLE-INTERVAL>

Sample interval in seconds. Default: 30.

sample-type <ABSOLUTE|DELTA>

Specifies the method of sampling of the SNMP MIB object. Default: Absolute.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring RMON for the MIB object **ifOutErrors.15** with an index **1**, rising threshold of **2147483647** and falling threshold of **-2134** using **absolute** sampling for a sample interval of **100** seconds:

```
switch(config)# rmon alarm index 1 snmp-oid ifOutErrors.15 rising-threshold
2147483647
falling-threshold -2134 sample-type absolute sample-interval 100
```

Removing RMON alarm with the index 5:

```
switch(config)# no rmon alarm index 5
```

rmon alarm {enable | disable} {index | all}

Syntax

rmon alarm {enable | disable} {index <INDEX> | all}

Description

Enables and disables the RMON alarm and its index. RMON alarm is enabled by default.

Command context

config

Parameters

enable

Enables the RMON alarm index

disable

Disables the RMON alarm index.

index <INDEX>

Specifies the RMON alarm index. Range: 1 to 20.

all

Specifies all the RMON alarms.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling or disabling all the RMON alarm:

```
switch(config)# rmon alarm enable all
switch(config)# rmon alarm disable all
```

Enabling or disabling RMON alarm by index:

```
switch(config)# rmon alarm enable index 1
switch(config)# rmon alarm disable index 1
```

show rmon alarm

Syntax

```
show rmon alarm [index <INDEX>]
```

Description

Displays the RMON alarm configurations.

Command context

config

Parameters

index <INDEX>

Specifies the RMON alarm index. Range: 1 to 20.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Showing all RMON alarm configurations:

```
switch# show rmon alarm
Index          : 1
Enabled        : true
Status         : valid
MIB object     : ifOutErrors.15
Sample type    : delta
Sampling interval : 6535 seconds
Rising threshold : 100
Falling threshold : 10
Last sampled value : 0
Last sample time : 2020-09-21 05:58:11

Index          : 3
Enabled        : true
Status         : invalid
MIB object     : IF-MIB::ifDescr.19
Sample type    : absolute
Sampling interval : 10000 seconds
Rising threshold : 4000
Falling threshold : 10
Last sampled value : 0
```

Showing RMON alarm with alarm index 1:

```
switch# show rmon alarm index 1
Index          : 1
Enabled        : true
Status         : valid
MIB object     : ifOutErrors.15
Sample type    : delta
Sampling interval : 6535 seconds
Rising threshold : 100
Falling threshold : 10
Last sampled value : 0
Last sample time : 2020-06-21 05:58:11
```

Showing disabled RMON alarm information:

```
switch# show rmon alarm
Index          : 1
Enabled        : false
Status         : valid
MIB object     : ifOutErrors.15
Sample type    : delta
Sampling interval : 6535 seconds
Rising threshold : 100
Falling threshold : 10
Last sampled value : 0
Last sample time : 2020-09-21 05:58:11

Index          : 3
Enabled        : false
Status         : invalid
MIB object     : IF-MIB::ifDescr.19
Sample type    : absolute
```

```
Sampling interval : 10000 seconds
Rising threshold  : 4000
Falling threshold : 10
Last sampled value : 0
```

show snmp agent-port

Syntax

```
show snmp agent-port
```

Description

Displays SNMP agent UDP port number.

Command context

Operator (>) or Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Example

Displaying SNMP agent UDP port number:

```
switch# show snmp agent-port
SNMP agent port : 161
```

show snmp community

Syntax

```
show snmp community
```

Description

Displays a list of all configured SNMPv1/v2c communities.

Command context

Manager (#)

Authority

Administrators or local user group members with execution rights for this command.

Usage

When a user creates a custom community before enabling an SNMP agent, AOS-CX automatically removes the default `public` community from the system.

Example

Displaying a list of all configured SNMPv1/v2c communities:

Before any community is created by user

```
switch# show snmp community
-----
SNMP communities
-----
public
```

After community is created by user

```
switch#show snmp community
-----
SNMP communities
-----
private
private2
```

show snmp system

Syntax

```
show snmp system
```

Description

Displays SNMP description, location, and contact information.

Command context

Operator (>) or Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

Displaying SNMP description, location, and contact information:

```
switch# show snmp system
SNMP system information
-----
System description : Aggregation router
System location   : Main lab
System contact    : John Smith, Lab Admin
```

show snmp trap

Syntax

```
show snmp trap
```

Description

Displays all configured SNMP traps/informs receivers.

Command context

Operator (>) or Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Example

Displaying all configured SNMP trap and informs receivers:

```
switch# show snmp trap
HOST                                PORT  TYPE  VER  COMMUNITY/USER  NAME  VRF
-----
10.10.10.10                        162   trap  v1   public          default
10.10.10.10                        162   inform v2c  public          default
10.10.10.10                        162   inform v3   name           default
```

show snmp vrf

Syntax

```
show snmp vrf
```

Description

Displays the VRF on which the SNMP agent service is running.

Command context

Operator (>) or Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Example

Displaying SNMP services enabled on VRF:

```
switch#show snmp vrf
SNMP enabled VRF
-----
mgmt
default
```

show snmpv3 context

Syntax

```
show snmpv3 context
```

Description

Displays all configured SNMP contexts.

Command context

Operator (>) or Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Examples

Displaying all configured SNMP contexts:

```
switch# show snmpv3 context
```

name	vrf	community
contextA	default	private

```
switch# show snmpv3 context
```

Name	vrf	Community	ype[Instance_id]
A	default	public	vrf

```
switch#
```

show snmpv3 engine-id

Syntax

```
show snmpv3 engine-id
```

Description

Displays the configured SNMPv3 snmp engine-id.

If the SNMPv3 engine-id is not configured, by default a unique engine-id is created by the switch using a combination of the enterprise OID value and the switch's mac address.

Command context

Operator (>) or Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Example

Displaying the configured SNMPv3 engine-id:

```
switch# show snmpv3 engine-id
SNMP engine-id : 80:00:B8:5C:08:00:09:1d:de:a5
```

show snmpv3 security-level

Syntax


```
show snmpv3 security-level
```

Description

Displays the configured SNMPv3 security level.

Command context

Operator (>) or Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Examples

Displaying the configured SNMPv3 security level:

```
switch# show snmpv3 security-level
SNMPv3 security-level : auth
```

show snmpv3 users

Syntax

```
show snmpv3 users
```

Description

Displays all configured SNMPv3 users.

For more details on the user enabled status, see [snmpv3 security-level](#).

Command context

Operator (>) or Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Example

Displaying all configured SNMPv3 users:

```
switch# show snmpv3 users
-----
User                               AuthMode  PrivMode  Context  Enabled
-----
name                               md5       none      none     False
name2                              sha       aes       none     True
```

snmp-server agent-port

Syntax

```
snmp-server agent-port <PORT>
```

```
no snmp-server agent-port [<PORT>]
```

Description

Sets the UDP port number that the SNMP master agent uses to communicate. UDP port 161 is the default port.

The `no` form of this command sets the SNMP master agent port to the default value.

Command context

```
config
```

Parameters

<PORT>

Specifies the UDP port number that the SNMP master agent will use. Range: 1 to 65535. Default: 161.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the SNMP master agent port to **2000**:

```
switch(config)# snmp-server agent-port 2000
```

Resetting the SNMP master agent port to the default value:

```
switch(config-schedule)# no snmp-server agent-port 2000
```

SNMP server community commands

snmp-server community

Syntax

```
snmp-server community <STRING>
```

```
no snmp-server community <STRING>
```

Description

Adds an SNMPv1/SNMPv2c community string. A community string is a password that controls read access to the SNMP agent. A network management program must supply this name when attempting to get SNMP information from the switch. A maximum of 10 community strings are supported. Once you create your own community string, the default community string (`public`) is deleted.

The `no` form of this command removes the specified SNMPv1/SNMPv2c community string. When no community string exists, a default community string with the value `public` is automatically defined.

Command context

```
config
```

Parameters

<STRING>

Specifies the SNMPv1/SNMPv2c community string. Range: 1 to 32 printable ASCII characters, excluding space and question mark.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the SNMPv1/SNMPv2c community string to **private**:

```
switch(config)# snmp-server community private
```

Removing SNMPv1/SNMPv2c community string **private**:

```
switch(config)# no snmp-server community private
```

access-level

Syntax

```
access-level <ro|rw>
```

```
no access-level <ro|rw>
```

Description

Changes the access level of the SNMP community. The default access level is read-only (*ro*).

The *no* form of this command changes the access level of the community to default.

Command context

config-community

Parameters

ro

Specifies Read-Only access with the SNMP community.

rw

Specifies Read-Write access with the SNMP community.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling the SNMP community:

```
switch(config)# snmp-server community my_community  
switch(config-community)#
```

Configuring the access level for the SMNP community:

```
switch(config-community) # access-level rw
```

Changing the access level of the SNMP community to default:

```
switch(config-community) # no access-level rw
```

access-list

Syntax

```
access-list <ACL-NAME>
```

```
no access-list <ACL-NAME>
```

Description

Associates an ACL with the SNMP community. If an ACL is not associated with the SNMP community, the default access is allowed for all the hosts.

The `no` form of this command removes association of the ACL with the SNMP community.

Command context

config-community

Parameters

<ACL-NAME>

Specifies the ACL name. It supports a maximum of 64 characters.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling the SNMP community:

```
switch(config) # snmp-server community my_community  
switch(config-community) #
```

Associating an ACL with the SMNP community:

```
switch(config-community) # access-list my_acl
```

Removing the associated ACL with the SNMP community:

```
switch(config-community) # no access-list my_acl
```



hitcounts for SNMP ACL will not be incremented.

Example: `show access-list hitcounts ip all` will not show the hit count of SNMP ACL.

snmp-server historical-counters-monitor

Syntax

```
snmp-server historical-counters-monitor  
  
no snmp-server historical-counters-monitor
```

Description

Enables the Remote Network Monitoring agent (`rmond`) to start collecting historical interface statistics. The `no` form of this command stops the historical interface statistics collection.

Command context

Operator (>) or Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

Enabling the `rmond` agent to start historical interface statistics collection:

```
switch(config)# snmp-server historical-counters-monitor
```

Disabling the `rmond` agent to stop historical interface statistics collection:

```
switch(config)# no snmp-server historical-counters-monitor
```

snmp-server host

Syntax

```
snmp-server host <IPv4-ADDR | IPv6-ADDR> trap version <VERSION> [community <STRING>]  
[port <UDP-PORT>] [<VRF-NAME>]  
  
no snmp-server host <IPv4-ADDR | IPv6-ADDR> trap version <VERSION> [community <STRING>]  
[port <UDP-PORT>] [<VRF-NAME>]  
  
snmp-server host <IPv4-ADDR | IPv6-ADDR> inform version v2c [community <STRING>]  
[port <UDP-PORT>] [<VRF-NAME>]  
  
no snmp-server host <IPv4-ADDR | IPv6-ADDR> inform version v2c [community <STRING>]  
[port <UDP-PORT>] [<VRF-NAME>]  
  
snmp-server host <IPv4-ADDR | IPv6-ADDR> [trap version v3 | inform version v3] user <NAME>  
[port <UDP-PORT>] [<VRF-NAME>]  
  
no snmp-server host <IPv4-ADDR | IPv6-ADDR> [trap version v3 | inform version v3] user  
<NAME>  
[port <UDP-PORT>] [<VRF-NAME>]  
Description
```

Configures a trap/informs receiver to which the SNMP agent can send SNMP v1/v2c/v3 traps or v2c informs. A maximum of 30 SNMP traps/informs receivers can be configured.

The `no` form of this command removes the specified trap/inform receiver.

Command context

config

Parameters

<IPv4-ADDR>

Specifies the IP address of a trap receiver in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255. You can remove leading zeros. For example, the address 192.169.005.100 becomes

192.168.5.100.

<IPv6-ADDR>

Specifies the IP address of a trap receiver in IPv6 format (x:x::x:x).

trap version <VERSION>

Specifies the trap notification type for SNMPv1, v2c or v3. Available options are: v1, v2c or v3.

inform version v2c

Specifies the inform notification type for SNMPv2c.

trap version v3

Specifies the trap notification type for SNMPv3.

user <NAME>

Specifies the SNMPv3 user name to be used in the SNMP trap notifications.

community <STRING>

Specifies the name of the community string to use when sending trap notifications. Range: 1 - 32 printable ASCII characters, excluding space and question mark. Default: public.

<UDP-PORT>

Specifies the UDP port on which notifications are sent. Range: 1 - 65535. Default: 162.

<VRF-NAME>

Specifies the VRF on which the SNMP agent listens for incoming requests.

Authority

Administrators or local user group members with execution rights for this command.

Examples

```
switch(config)# snmp-server host 10.10.10.10 trap version v1
switch(config)# no snmp-server host 10.10.10.10 trap version v1
switch(config)# snmp-server host a:b::c:d trap version v1
switch(config)# no snmp-server host a:b::c:d trap version v1
switch(config)# snmp-server host 10.10.10.10 trap version v2c community public
switch(config)# no snmp-server host 10.10.10.10 trap version v2c community public
switch(config)# snmp-server host a:b::c:d trap version v2c community public
switch(config)# no snmp-server host a:b::c:d trap version v2c community public
switch(config)# snmp-server host 10.10.10.10 trap version v2c community public port
5000
switch(config)# no snmp-server host 10.10.10.10 trap version v2c community public
port 5000
switch(config)# snmp-server host 10.10.10.10 trap version v2c community public port
5000 vrf default
switch(config)# no snmp-server host 10.10.10.10 trap version v2c community public
port 5000 vrf default
switch(config)# snmp-server host a:b::c:d trap version v2c community public port
5000
switch(config)# no snmp-server host a:b::c:d trap version v2c community public port
5000
switch(config)# snmp-server host 10.10.10.10 inform version v2c community public
switch(config)# no snmp-server host 10.10.10.10 inform version v2c community public
switch(config)# snmp-server host a:b::c:d inform version v2c community public
```

```

switch(config)# no snmp-server host a:b::c:d inform version v2c community public
switch(config)# snmp-server host 10.10.10.10 inform version v2c community public
port 5000
switch(config)# no snmp-server host 10.10.10.10 inform version v2c community public
port 5000
switch(config)# snmp-server host 10.10.10.10 inform version v2c community public
port 5000 vrf default
switch(config)# no snmp-server host 10.10.10.10 inform version v2c community public
port 5000 vrf default
switch(config)# snmp-server host a:b::c:d inform version v2c community public port
5000
switch(config)# no snmp-server host a:b::c:d inform version v2c community public
port 5000
switch(config)# snmp-server host 10.10.10.10 trap version v3 user Admin
switch(config)# no snmp-server host 10.10.10.10 trap version v3 user Admin
switch(config)# snmp-server host a:b::c:d trap version v3 user Admin
switch(config)# no snmp-server host a:b::c:d trap version v3 user Admin
switch(config)# snmp-server host 10.10.10.10 trap version v3 user Admin port 2000
switch(config)# no snmp-server host 10.10.10.10 trap version v3 user Admin port 2000
switch(config)# snmp-server host a:b::c:d trap version v3 user Admin port 2000
switch(config)# no snmp-server host a:b::c:d trap version v3 user Admin port 2000

```

snmp-server system-contact

Syntax

```
snmp-server system-contact <INFO>
```

```
no snmp-server system-contact [<INFO>]
```

Description

Sets SNMP contact information.

The no form of this command removes the SNMP contact information.

Command context

config

Parameters

<INFO>

Specifies SNMP contact information. Range: 1 to 128 printable ASCII characters, except for question mark (?).

Authority

Administrators or local user group members with execution rights for this command.

Examples

Defines SNMP contact information to be **John Smith, Lab Admin**:

```
switch(config)# snmp-server system-contact John Smith, Lab Admin
```

Removes SNMP contact information:

```
switch(config)# no snmp-server system-contact
```

snmp-server system-description

Syntax

```
snmp-server system-description <DESCRIPTION>
```

```
no snmp-server system-description
```

Description

Sets the SNMP system description.

The `no` form of this command removes the SNMP system description.

Command context

config

Parameters

<DESCRIPTION>

Specifies the SNMP system description. Typical content to include would be the full name and version of the following:

- Hardware type of the system
- Software operating system
- Networking software

Range: 1 to 64 printable ASCII characters, except for the question mark (?).

Authority

Administrators or local user group members with execution rights for this command.

Examples

Defines the SNMP system description to be **mainSwitch**:

```
switch(config)# snmp-server system-description mainSwitch
```

Removes the SNMP system description:

```
switch(config)# no snmp-server system-description mainSwitch
```

snmp-server system-location

Syntax

```
snmp-server system-location <INFO>
```

```
no snmp-server system-location
```

Description

Sets the SNMP location information.

The `no` form of this command removes the SNMP location information.

Command context

config

Parameters

<INFO>

Specifies the SNMP location information. Range: 1 to 128 printable ASCII characters, except for the question mark (?).

Authority

Administrators or local user group members with execution rights for this command.

Examples

Defines the SNMP location information to be **Main Lab**:

```
switch(config)# snmp-server system-location Main Lab
```

Removes the SNMP location information:

```
switch(config)# no snmp-server system-location
```

snmp-server vrf

Syntax

```
snmp-server vrf <VRF-NAME>
```

```
no snmp-server vrf <VRF-NAME>
```

Description

Configures a VRF on which the SNMP agent listens for incoming requests. By default, the SNMP agent does not listen on any VRF. 6100 only supports default VRF.

The `no` form of this command stops the SNMP agent from listening for incoming requests on the specified VRF.

Command context

config

Parameters

<VRF-NAME>

Specifies the name of a VRF.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring the SNMP agent to listen on VRF `default`.

```
switch(config)# snmp-server vrf default
```

Configuring the SNMP agent to listen on VRF `mgmt`.

Configuring the SNMP agent to listen on user-defined VRF `myvrf`.

```
switch(config)# snmp-server vrf myvrf
```

Stopping the SNMP agent from listening on VRF `default`.

```
switch(config)# no snmp-server vrf default
```

snmp-server trap-source interface vrf

Syntax

```
snmp-server trap-source {interface <IF-NAME> | <IPv4-Address> | <IPv6-Address>} [vrf <VRF-NAME>]
```

```
no snmp-server trap-source {interface <IF-NAME> | <IPv4-Address> | <IPv6-Address>} [vrf <VRF-NAME>]
```

Description

Configures SNMP trap source interface or IP address for a VRF.

The `no` form of this command removes the SNMP `trap-source` configuration for a VRF.

Command context

config

Parameters

<IF-NAME>

Specifies the source interface name. Interface name can be physical interface, loopback interface, LAG interface, or VLAN interface.

<IPv4-Address>

Specifies the IPv4 address of source interface for the SNMP trap.

<IPv6-Address>

Specifies the IPv6 address of source interface for the SNMP trap.

<VRF-NAME>

Specifies the name of a VRF associated to the source interface for the SNMP trap.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring SNMP trap source interface for a VRF.

```
switch(config)# snmp-server trap-source interface 1/1/12 vrf sample
switch(config)# snmp-server trap-source interface loopback10 vrf sample
switch(config)# snmp-server trap-source interface vlan23 vrf sample
```

Configuring SNMP trap source IP address for a VRF.

```
switch(config)# snmp-server trap-source 10.0.0.1 vrf red
switch(config)# snmp-server trap-source 1001::1 vrf red
```

snmp-server trap

Syntax

```
snmp-server trap {cpu-utilization | memory-utilization | rmon-events}
no snmp-server trap {cpu-utilization | memory-utilization | rmon-events}
```

Description

Enables the SNMP traps. The SNMP traps are enabled by default.

The `no` form of this command disables the SNMP traps.

Command context

config

Parameters

cpu-utilization

Enables the CPU utilization traps.

memory-utilization

Enables the memory utilization traps.

rmon-events

Enables the RMON event traps.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling the SNMP traps:

```
switch(config)# snmp-server trap cpu-utilization
switch(config)# snmp-server trap memory-utilization
switch(config)# snmp-server trap rmon-events
```

Disabling the SNMP traps:

```
switch(config)# no snmp-server trap cpu-utilization
switch(config)# no snmp-server trap memory-utilization
switch(config)# no snmp-server trap rmon-events
```

Displaying the SNMP trap configuration:

```
switch(config)# show running-config all | inc snmp
snmp-server trap rmon-events
snmp-server trap cpu-utilization
snmp-server trap memory-utilization
```

Displaying CPU and Memory usage:

```
switch(config)# show system
Hostname           : XXXX
System Description : XX.10.07.0001CI
System Contact     :
System Location    :
Vendor             : Aruba
Product Name       : JLXXXX XXXX Base Chassis/3xFT/18xFans/Cbl Mgr/X462 Bundle
Chassis Serial Nbr : SG6Z009068
Base MAC Address   : f40343-806400
ArubaOS-CX Version : XX.10.07.0001CI
Time Zone          : UTC
Up Time            : 8 minutes
CPU Util (%)       : 1
Memory Usage (%)   : 10
```

snmp-server trap vsx

Syntax

```
snmp-server trap vsx
```

```
no snmp-server trap vsx
```

Description

Enables sending the SNMP traps for VSX related events. VSX trap generation is disabled by default.

The `no` form of this command disables sending the SNMP traps for VSX related events.

The trap support is available for the following VSX events:

- ISL up and down
- KA up and down
- MCLAG up and down

Command context

```
config
```

Parameters

```
vsx
```

Specifies SNMP traps for VSX events.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling the VSX traps:

```
switch(config)# snmp-server trap vsx
```

```
switch(config)# show vsx configuration trap
SNMP traps : Enabled
```

Disabling the VSX traps:

```
switch(config)# no snmp-server trap vsx
```

```
switch(config)# show vsx configuration trap  
SNMP traps : Disabled
```

snmpv3 context

Syntax

```
snmpv3 context <NAME> vrf <VRF-NAME> [community <STRING>]
```

```
no snmpv3 context <NAME> [vrf <VRF-NAME>]
```

Description

Creates an SNMPv3 context on the specified VRF.

The `no` form of this command removes the specified SNMP context.

Command context

config

Parameters

<NAME>

Specifies the name of the context. Range: 1 to 32 printable ASCII characters, excluding space and question mark (?).

vrf <VRF-NAME>

Specifies the VRF associated with the context. Default: default.

community <STRING>

Specifies the SNMP community string associated with the context. Range: 1 to 32 printable ASCII characters, excluding space and question mark. Default: public.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Creating an SNMPv3 context named **newContext**:

```
switch(config)# snmpv3 context newContext
```

Creating an SNMPv3 context named **newContext** on VRF **myVrf** and with community string **private**.

```
switch(config)# snmpv3 context newContext vrf myVrf community private
```

Removing the SNMPv3 context named **newContext** on VRF **myVrf**:

```
switch(config)# no snmpv3 context newContext vrf myVrf
```

snmpv3 engine-id

Syntax

```
snmpv3 engine-id <ENGINE-ID>
no snmpv3 engine-id <ENGINE-ID>
```

Description

Configures the SNMPv3 SNMP engine-id allowing an administrator to configure a unique SNMP engine-id for the switch. This engine-id is used by the NMS management tool to identify and distinguish multiple switches on the same network.

The `no` form of this command restores the default engine-id, created by the switch using a combination of the enterprise OID value and the switch's mac address.

Command context

config

Parameters

<ENGINE-ID>

SNMPv3 SNMP engine-id in colon separated hexadecimal notation.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring the SNMPv3 engine-id:

```
switch(config)#
switch(config)# snmpv3 engine-id
WORD SNMPv3 snmp engine-id in colon seperated hexadecimal notation
switch(config)# snmpv3 engine-id 01:23:45:67:89:ab:cd:ef:01:23:45:67
```

Restoring the default SNMPv3 engine-id:

```
switch(config)# no snmpv3 engine-id
```

snmpv3 security-level

Syntax

```
snmpv3 security-level {auth | auth-privacy}
no snmpv3 security-level {auth | auth-privacy}
```

Description

Configures the SNMPv3 security level. The security level determines which SMNPv3 users defined by the command `snmpv3 user` are able to connect.

The `no` form of this command changes the security level as follows:

- `no snmpv3 security-level auth`: Sets the security level to `auth-privacy`.
- `no snmpv3 security-level auth-privacy`: Sets the security level to no authentication or privacy, allowing any SNMP user to connect.

Command context

`config`

Parameters

`auth`

SNMPv3 users that support authentication, or authentication and privacy are allowed.

`auth-privacy`

Only SNMPv3 users with both authentication and privacy are allowed. This is the highest level of SNMPv3 security. Default.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the SNMPv3 security level to authentication and privacy:

```
switch(config)# snmpv3 security-level auth-privacy
```

Setting the SNMPv3 security level to authentication only:

```
switch(config)# snmpv3 security-level auth
```

Setting the SNMPv3 security level to no authentication and no privacy:

```
switch(config)# no snmpv3 security-level auth-privacy
```

Restoring the default SNMPv3 security level to authentication and privacy:

```
switch(config)# no snmpv3 security-level auth
```

snmpv3 user

Syntax

```
snmpv3 user <NAME>
    [auth <AUTH-PROTO> auth-pass [{plaintext | ciphertext} <AUTH-PASS>]]
    [priv <PRIV-PROTO> priv-pass [{plaintext | ciphertext} <PRIV-PASS>]]
no snmpv3 user <NAME>
    [auth <AUTH-PROTO> auth-pass [{plaintext | ciphertext} <AUTH-PASS>]]
    [priv <PRIV-PROTO> priv-pass [{plaintext | ciphertext} <PRIV-PASS>]]
```

Description

Creates an SNMPv3 user and adds it to an SNMPv3 context. The SNMPv3 security level (set with command `snmpv3 security-level`) determines which users are allowed to authenticate.

The `no` form of this command removes the specified SNMPv3 user.

Command context

config

Parameters

<NAME>

Specifies the SNMPv3 username. Range 1 to 32 printable ASCII characters, excluding space and question mark (?).

auth <AUTH-PROTO>

Selects the authentication protocol used to validate user logins: `md5` or `sha1`.

auth-pass [{plaintext | ciphertext} <AUTH-PASS>]

Specifies the SNMPv3 user authentication password. Range for `plaintext` is 8 to 32 printable ASCII characters, excluding space and question mark (?). Range for `ciphertext` is 1 to 256 printable ASCII characters. Ciphertext is used when copying user configuration settings between switches.

priv <PRIV-PROTO>

Selects the SNMPv3 privacy protocol (encryption method): `aes` or `des`.

priv-pass [{plaintext | ciphertext} <PRIV-PASS>]

Specifies the SNMPv3 user privacy encryption password. Range for `plaintext` is 8 to 32 printable ASCII characters, excluding space and question mark (?). Range for `ciphertext` is 1 to 256 printable ASCII characters. Ciphertext is used when copying user configuration settings between switches.



When the authentication password is not provided on the command line, plaintext authentication password prompting occurs upon pressing Enter, followed by privacy encryption protocol prompting, and finally plaintext encryption password prompting. The entered password characters are masked with asterisks.



When the authentication type and password plus the privacy protocol (encryption method) are provided on the command line but the encryption password is not provided, plaintext encryption password prompting occurs upon pressing Enter. The entered password characters are masked with asterisks.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Defining SNMPv3 user **Admin1** using **sha** authentication and **des** privacy encryption with provided plaintext passwords:

```
switch(config)# snmpv3 user Admin1 auth sha auth-pass plaintext F82#450h
                  priv des priv-pass plaintext F82#4eva
```

Defining SNMPv3 user **Admin2** using **MD5** authentication and **AES** privacy encryption with provided authentication password and privacy encryption type but prompted encryption password:

```
switch(config)# snmpv3 user Admin2 auth md5 auth-pass plaintext F82#450h
                  priv aes priv-pass
Enter the privacy encryption key: *****
Re-Enter the privacy encryption key: *****
```


Defining SNMPv3 user **Admin2** using **MD5** authentication and **AES** privacy encryption with plaintext password prompting and privacy encryption selection:

```
switch(config)# snmpv3 user Admin2 auth md5 auth-pass
Enter the authentication password: *****
Re-Enter the authentication password: *****

Configure the privacy protocol (y/n)? y
Enter the privacy protocol (aes/des)? aes

Enter the privacy encryption key: *****
Re-Enter the privacy encryption key: *****
```

Removing SNMPv3 user **Admin1**:

```
switch(config)# no snmpv3 user Admin1
```

Creating an SNMP user on switch 1 and then creating the same user on switch 2 by copying from the switch 1 configuration:

On switch 1, configure a user named **Admin3**, and then use the `show running-config` command to display switch configuration. Save a copy of the full `snmpv3 user` command (shown by `show running-config`). This saved command is used on switch 2.

```
switch1(config)# snmpv3 user Admin3 auth sha auth-pass plaintext F82#450h
priv des priv-pass plaintext F82#4eva
switch1(config)# exit
switch1# show running-config
Current configuration:
!
!Version ArubaOS-CX xx.xx.xx.xxxxxx
!
snmpv3 user Admin3 auth sha auth-pass ciphertext AQBaf2d...FJVcZ3o=
priv des priv-pass ciphertext AQBah2p...2jfTFwQ=
ssh server vrf mgmt
!
interface mgmt
    no shutdown
    ip dhcp
vlan 1
```

On switch 2, execute the `snmpv3 user` command that you saved from switch 1 (as shown by `show running-config`). This creates the user on switch 2 with the same configuration.

```
switch2(config)# snmpv3 user Admin3 auth sha auth-pass ciphertext AQBaf2d...FJVcZ3o=
priv des priv-pass ciphertext AQBah2p...2jfTFwQ=
```

ip source-interface

Syntax

```
ip source-interface <PROTOCOL> <IP-ADDR> [vrf <VRF-NAME>]
```

```
no ip source-interface <PROTOCOL> <IP-ADDR> [vrf <VRF-NAME>]
```

Description

Configures the source-interface IPv4 address to use for the specified protocol. If a VRF is not given, the default VRF applies. If no interface option is given, the device floods through interfaces and VRFs to reach Aruba Central. Whichever reaches Aruba Central will be picked automatically.

The `no` form of this command removes all configurations.

Command context

config

Parameters

<PROTOCOL>

Specifies the protocol to configure.

all

Selects all protocols that can be configured by this command.

central

Selects Aruba Central.

dhcp_relay

Selects DHCP relay.

dns

Selects DNS.

ntp

Selects NTP.

radius

Selects radius.

sflow

Selects sFlow.

simplivity

Selects simplivity.

syslog

Selects syslog.

tacacs

Selects TACACS.

tftp

Selects TFTP.

<IP-ADDR>

Specifies the IPv4 address.

vrf <VRF-NAME>

Specifies the VRF name.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring source-interface IPv4 10.1.1.1 to use for the TFTP protocol:

```
switch(config)# ip source-interface tftp 10.1.1.1
```

Configuring source-interface IPv4 10.1.1.2 to use for the TFTP protocol on VRF green :

```
switch(config)# ip source-interface tftp 10.1.1.2 vrf green
```

Removing source-interface IPv4 10.1.1.1 configuration for the TFTP protocol:

```
switch(config)# no ip source-interface tftp 10.1.1.1
```

Removing source-interface IPv4 10.1.1.2 configuration for TFTP protocol on VRF green:

```
switch(config)# no ip source-interface tftp 10.1.1.2 vrf green
```

Configuring source-interface IPv4 10.1.1.1 to use for the DNS protocol:

```
switch(config)# ip source-interface dns 10.1.1.1
```

Configuring source-interface IPv4 10.1.1.2 to use for the DNS protocol on VRF green :

```
switch(config)# ip source-interface dns 10.1.1.2 vrf green
```

Removing source-interface IPv4 10.1.1.1 configuration for the DNS protocol:

```
switch(config)# no ip source-interface tftp 10.1.1.1
```

Removing source-interface IPv4 10.1.1.2 configuration for the DNS protocol on VRF green:

```
switch(config)# no ip source-interface dns 10.1.1.2 vrf green
```

ip source-interface interface

Syntax

```
ip source-interface <PROTOCOL> interface <IFNAME> [vrf <VRF-NAME>]
```

```
no ip source-interface <PROTOCOL> interface <IFNAME> [vrf <VRF-NAME>]
```

Description

Configures the IPv4 source-interface interface to use for the specified protocol. If a VRF is not given, the default VRF applies.

The `no` form of this command removes the specified configuration.

Command context

config

Parameters

<PROTOCOL>

Specifies the protocol to configure.

all

Selects all protocols supported by this command.

central

Selects Aruba Central.

dhcp_relay

Selects DHCP relay.

dns

Selects DNS.

ntp

Selects NTP.

radius

Selects radius.

sflow

Selects sFlow.

syslog

Selects syslog.

tacacs

Selects TACACS.

tftp

Selects TFTP.

vrf <VRF-NAME>

Specifies the VRF name.

<IFNAME>

Specifies the interface name.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring IPv4 source-interface interface 1/1/1 to use for the TFTP protocol:

```
switch(config)# ip source-interface tftp interface 1/1/1
```

Configuring IPv4 source-interface interface 1/1/2 to use for the TFTP protocol on VRF green :

```
switch(config)# ip source-interface tftp interface 1/1/2 vrf green
```

Removing IPv4 source-interface 1/1/1 configuration for the TFTP protocol:

```
switch(config)# no ip source-interface tftp interface 1/1/1
```

Removing source-interface interface 1/1/2 configuration for TFTP protocol on VRF green:

```
switch(config)# no ip source-interface tftp interface 1/1/2 vrf green
```

ipv6 source-interface

Syntax

```
ipv6 source-interface <PROTOCOL> <IPV6-ADDR> [vrf <VRF-NAME>]
```

```
no source-interface <PROTOCOL> <IPV6-ADDR> [vrf <VRF-NAME>]
```

Description

Configures the source-interface IPv6 address to use for the specified protocol. If a VRF is not given, the default VRF applies.

The `no` form of this command removes the specified protocol configuration.

Command context

config

Parameters

<PROTOCOL>

Specifies the protocol to configure.

all

Selects all protocols supported by this command.

central

Selects Aruba Central.

ntp

Selects NTP.

radius

Selects radius.

sflow

Selects sFlow.

syslog

Selects syslog.

tacacs

Selects TACACS.

tftp

Selects TFTP.

<IPV6-ADDR>

Specifies the IPv6 address.

vrf <VRF-NAME>

Specifies the VRF name.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring source-interface IPv6 1111:2222 to use for the TFTP protocol:

```
switch(config)# ipv6 source-interface tftp 1111:2222
```

Configuring source-interface IPv6 1111:3333 to use for TFTP protocol on VRF green :

```
switch(config)# ipv6 source-interface tftp 1111:3333 vrf green
```

Removing source-interface IPv6 1111:2222 configuration for TFTP protocol:

```
switch(config)# no ipv6 source-interface tftp 1111:2222
```

Removing source-interface IPv6 1111:3333 configuration for TFTP protocol on VRF green:

```
switch(config)# no ipv6 source-interface tftp 1111:3333 vrf green
```

ipv6 source-interface interface

Syntax

```
ipv6 source-interface <PROTOCOL> interface <IFNAME> [vrf <VRF-NAME>]
```

```
no ipv6 source-interface <PROTOCOL> interface <IFNAME> [vrf <VRF-NAME>]
```

Description

Configures the IPv6 source-interface interface to use for the specified protocol. If a VRF is not given, the default VRF applies.

The `no` form of this command removes all configurations.

Command context

config

Parameters

<PROTOCOL>

Specifies the protocol to configure.

all

Selects all protocols supported by this command.

central

Selects Aruba Central.

ntp

Selects NTP.

radius

Selects radius.
 sflow
 Selects sFlow.
 syslog
 Selects syslog.
 tacacs
 Selects TACACS.
 tftp
 Selects TFTP.
 <IFNAME>
 Specifies the interface name.
 vrf <VRF-NAME>
 Specifies the VRF name.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring IPv6 source-interface interface 1/1/1 to use for the TFTP protocol :

```
switch(config)# ipv6 source-interface tftp interface 1/1/1
```

Configuring IPv6 source-interface interface 1/1/2 to use for the TFTP protocol on VRF green :

```
switch(config)# ipv6 source-interface tftp interface 1/1/2 vrf green
```

Removing IPv6 source-interface interface 1/1/1 configuration for the TFTP protocol:

```
switch(config)# no ipv6 source-interface tftp interface 1/1/1
```

Removing IPv6 source-interface interface 1/1/2 configuration for the TFTP protocol on VRF green:

```
switch(config)# no ipv6 source-interface tftp interface 1/1/2 vrf green
```

show ip source-interface

Syntax

```
show ip source-interface <PROTOCOL> [vrf <VRF-NAME> | all-vrfs]
```

Description

Displays the source interface information for all VRFs or a specific VRF.

If a VRF is not specified, the default is displayed.

Command context

Manager (#)

Parameters

<PROTOCOL>

Specifies the protocol to show.

all

Shows the source interface configuration for all other protocols.

central

Shows the source interface configuration for Aruba Central.

dhcp relay

Shows the source interface configuration for DHCP relay.

dns

Shows the source interface configuration for DNS.

ntp

Shows the source interface configuration for NTP.

radius

Shows the source interface configuration for radius.

sflow

Shows the source interface configuration for sFlow.

syslog

Shows the source interface configuration for syslog.

tacacs

Shows the source interface configuration for TACACS.

tftp

Shows the source interface configuration for TFTP.

vrf <VRF-NAME>

Specifies the VRF name.

all-vrfs

Shows the source interface configuration for all VRFs.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Displaying all source-interface protocol configurations for VRF red:

```
switch# show ip source-interface all vrf red
Source-interface Configuration Information
-----
Protocol      Src-Interface      Src-IP              VRF
-----
all           1/1/1              red
switch#
```

Displaying all source-interface protocol configurations for default VRF:

```
switch# show ip source-interface all
Source-interface Configuration Information
-----
Protocol      Src-Interface      Src-IP              VRF
-----
all           1.1.1.1            default
switch#
```

Displaying all source-interface protocol configurations for all VRFs:


```
switch# show ip source-interface all all-vrfs
Source-interface Configuration Information
-----
Protocol      Src-Interface      Src-IP      VRF
-----
all
all           1/1/1/1           2.2.2.2     all-vrfs
all           1/1/1/1           1.1.1.1     default
all           1/1/1/1           1.1.1.1     red
switch#
```

show ipv6 source-interface

Syntax

```
show ipv6 source-interface <PROTOCOL> [detail] [vrf <VRF-NAME> | all-vrfs]
```

Description

Displays the IPV6 source interface information configured in the router for all VRFs or a specific VRF. If a VRF is not specified, the default is displayed.

Command context

Manager (#)

Parameters

<PROTOCOL>

Specifies the protocol to show.

all

Shows the source interface configuration for all other protocols.

central

Shows the source interface configuration for Aruba Central.

ntp

Shows the source interface configuration for NTP.

radius

Shows the source interface configuration for radius.

sflow

Shows the source interface configuration for sFlow.

syslog

Shows the source interface configuration for syslog.

tacacs

Shows the source interface configuration for TACACS.

tftp

Shows the source interface configuration for TFTP.

vrf <VRF-NAME>

Specifies the VRF name.

all-vrfs

Shows the source interface configuration for all VRF.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Displaying all IPv6 source-interface protocol configurations for default VRF:

```
switch# show ipv6 source-interface all
Source-interface Configuration Information
-----
Protocol      Src-Interface      Src-IP              VRF
-----
all           1111:2222          default
switch#
```

Displaying all IPv6 source-interface protocol configuration for VRF red:

```
switch# show ipv6 source-interface all vrf red
Source-interface Configuration Information
-----
Protocol      Src-Interface      Src-IP              VRF
-----
all           1/1/1             red
switch#
```

Displaying all IPv6 source-interface protocol configurations for all VRFs:

```
switch# show ipv6 source-interface all all-vrfs
Source-interface Configuration Information
-----
Protocol      Src-Interface      Src-IP              VRF
-----
all           2.2.2.2:3.3.3.3   all-vrfs
all           1.1.1.1:2.2.2.2   default
all           1/1/1             2::2               red
switch#
```

show running-config

Syntax

```
show running-config
```

Description

Displays the current running configuration.

Command context

Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Examples

Displaying the running configuration (only items of interest to source interface selection are shown in this example output command):



Aruba Central is the priority agent. If no command is specified for ip source-interface, Central will choose the command automatically if it is reachable on any of the known ports.

```
switch# show running-config
vrf green
ip source-interface tftp interface 1/1/2 vrf green
ip source-interface radius interface 1/1/2 vrf green
ip source-interface ntp interface 1/1/2 vrf green
ip source-interface tacacs interface 1/1/2 vrf green
ip source-interface dns interface 1/1/2 vrf green
ip source-interface central interface 1/1/2 vrf green
ip source-interface all interface 1/1/2 vrf green
ipv6 source-interface tftp 2222::3333 vrf green
ipv6 source-interface radius 2222::3333 vrf green
ipv6 source-interface ntp 2222::3333 vrf green
ipv6 source-interface tacacs 2222::3333 vrf green
ipv6 source-interface central 2222::3333 vrf green
ipv6 source-interface all 2222::3333 vrf green
ip source-interface tftp 10.20.3.1
ip source-interface radius 10.20.3.1
ip source-interface ntp 10.20.3.1
ip source-interface tacacs 10.20.3.1
ip source-interface dns 10.20.3.1
ip source-interface central 10.20.3.1
ip source-interface all 10.20.3.1
interface 1/1/1
    no shutdown
    ip address 10.20.3.1/24
interface 1/1/2
    vrf attach green
    ip address 20.1.1.1/24
    ipv6 address 2222::3333/64
interface 1/1/45
    no shutdown
    ip address 100.1.0.1/24
    ipv6 address 1111::2222/64
ip route 100.2.0.0/24 10.20.3.2
switch#
```



On the 6100 Switch Series, only the vrf named `default` is available. Replace any references to the `mgmt` or other VRFs with `default`.

ssh (client login)

Syntax

```
ssh [<USERNAME>@]{<IPv4> | <HOSTNAME>} [vrf <VRF-NAME>] [port <PORT-NUMBER>]
```

Description

Establishes a client session with an SSH server which is typically another switch.

Command context

Manager (#)

Parameters

<USERNAME>

Specifies the username that the client uses to log in to an SSH server. When omitted, the username of the current session is used.

{<IPv4> | <HOSTNAME>}

Specifies the SSH server to which the SSH client will connect.

- <IPv4>: The IPv4 address.
- <HOSTNAME>: The host name.

vrf <VRF-NAME>

Specifies the VRF to be used for the SSH client session. When omitted, the default VRF named `default` is used.

port <PORT-NUMBER>

Specifies the SSH server TCP port number. When omitted, the default TCP port 22 is used.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Establishing an SSH client session (using the management VRF) with an SSH server:

```
switch# ssh admin@10.0.11.180 vrf mgmt
```

Establishing an SSH client session (using the default VRF and a specific port) with an SSH server:

```
switch# ssh admin@10.0.11.175 port 223
```

Configuring a test user on switch 1 and then connecting to switch 1 from switch 2 using the SSH client on the mgmt VRF:

```
** Configuring a test user on switch 1 **  
switch(config)# user-group test  
switch(config-usr-grp-test)# permit cli command ".*"  
switch(config)# exit  
switch(config)# user test-user group test password plaintext tst#9J%** On switch 2,  
connecting to switch 1 using the SSH client **  
switch# ssh test-user@10.0.11.177 vrf mgmt
```



On the 6100 Switch Series, only the vrf named `default` is available. Replace any references to the `mgmt` or other VRFs with `default`.

show ssh host-key

Syntax

```
show ssh host-key [ecdsa | ed25519 | rsa]
```

Description

Shows the public host keys for the SSH server. If the key type is not provided, all available host-keys are shown.

Command context

Manager (#)

Parameters

`ecdsa`

Selects the ECDSA host-key pair.

`ed25519`

Selects the ED25519 host-key pair.

`rsa`

Selects the RSA host-key pair.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Showing the ECDSA public host-key:

```
switch# show ssh host-key ecdsa

Key Type : ECDSA      Curve : ecdsa-sha2-nistp256

ecdsa-sha2-nistp256 AAAAE2VjZHNhLXNoYTItbmlzdHAyNTYAAAAIbmlzdHAyNTYAAhtuv5rABBBGs
...
04mjVFGMVKZ87RWkyrxeQa2fAGZZEp1902K33/k3q17fA4EivRzC75YvjDu8=
```

Showing all public host keys:

```
switch# show ssh host-key

Key Type : ECDSA      Curve : ecdsa-sha2-nistp256
```

```
ecdsa-sha2-nistp256 AAAAE2VjZHNhLXNoYTItbmlzdHAyNTYAAAAIbmlzdHAyNTYAAhtuv5rABBBGs
...
O4mjVFGMVkZ87RWkyrxeQa2fAGZZEp1902K33/k3q17fA4EivRzC75YvjDu8=

Key Type : ED25519
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIGb6910Jwoe8Hkl9K5YhqijrWI3yovNbiJVq6tw4WjJr4

Key Type : RSA      Key Size : 2048
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQDdVCXlw43h4n1bwg9jI6DSBMngymCdPD0JUG42Sn9IS
...
nGSXtrNy6OmlFDJTAy+zz5Kd8d21ZLuhf07IHNgF3pff65Xc8qNJBv
```

show ssh server

Syntax

```
show ssh server [vrf default]
```

Description

Shows the SSH server configuration for the default VRF.

Command context

Operator (>) or Manager (#)

Parameters

vrf default

Specifies the default VRF.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Showing the SSH server configuration on the default VRF:

```
switch# show ssh server

SSH server configuration on VRF default :

IP Version      : IPv4 and IPv6      SSH Version      : 2.0
TCP Port        : 22                  Grace Timeout (sec) : 120
Max Auth Attempts : 6

Ciphers:
chacha20-poly1305@openssh.com, aes128-ctr, aes192-cbc,
aes128-cbc, aes192-ctr, aes256-gcm@openssh.com,
aes128-gcm@openssh.com, aes256-ctr, aes256-cbc

Host Key Algorithms:
ecdsa-sha2-nistp256, ecdsa-sha2-nistp384, ecdsa-sha2-nistp521,
ssh-ed25519, rsa-sha2-256, rsa-sha2-512, ssh-rsa

Key Exchange Algorithms:
curve25519-sha256, curve25519-sha256@libssh.org,
```

```
ecdh-sha2-nistp256,ecdh-sha2-nistp384, ecdh-sha2-nistp521,  
diffie-hellman-group-exchange-sha256,diffie-hellman-group16-sha512,  
diffie-hellman-group18-sha512,diffie-hellman-group14-sha256,  
diffie-hellman-group14-sha1
```

MACs:

```
hmac-sha1-etm@openssh.com, umac-64@openssh.com,  
umac-128@openssh.com, hmac-sha2-256,hmac-sha2-512,hmac-sha1
```

Public Key Algorithms:

```
rsa-sha2-256, rsa-sha2-512ssh-rsa, ecdsa-sha2-nistp256,  
ecdsa-sha2-nistp384, ecdsa-sha2-nistp521, ssh-ed25519,  
x509v3-rsa2048-sha256, x509v3-ssh-rsa, x509v3-sign-rsa,  
x509v3-ecdsa-sha2-nistp256, x509v3-ecdsa-sha2-nistp384,  
x509v3-ecdsa-sha2-nistp521
```

show ssh server sessions

Syntax

```
show ssh server sessions [vrf default]
```

Description

Shows the active SSH sessions on the default VRF.

Command context

Operator (>) or Manager (#)

Parameters

vrf default

Specifies the default VRF.

Authority

Administrators or local user group members with execution rights for this command.

Usage

If you provide the command with a VRF name, the command shows the active SSH session for the specified VRF. Any user can show sessions of all VRFs by using the `all-vrfs` parameter. The maximum number of sessions per VRF is five. The maximum SSH idle session timeout is 60 seconds.

Examples

Showing the active SSH sessions on the default VRF:

```
switch# show ssh server sessions  
  
SSH sessions on VRF default  
IPv4 SSH Sessions  
  Server IP       : 10.1.1.1  
  Client IP       : 10.1.1.2  
  Client Port     : 58835  
  
IPv6 SSH Sessions  
  Server IP       : FF01:0:0:0:0:0:0:FB
```



```
Client IP      : FF01:0:0:0:0:0:FC
Client Port    : 58836
```

ssh ciphers

Syntax

```
ssh ciphers <CIPHERS-LIST>
no ssh ciphers
```

Description

Configures SSH to use a set of ciphers in the specified priority order. Ciphers in SSH are used for privacy of data being transported over the connection. The first cipher type entered in the CLI is considered a first priority. Each option is an algorithm that is used to encrypt the link and each name indicates the algorithm and cryptographic parameters that are used. Only ciphers that are entered by the user are configured.

The `no` form of this command removes the configuration of ciphers and reverts SSH to use the default set of ciphers.

Command context

```
config
```

Parameters

<CIPHERS-LIST>

Valid cipher types are:

- aes128-cbc
- aes192-cbc
- aes256-cbc
- aes128-ctr
- aes192-ctr
- aes256-ctr
- aes128-gcm@openssh.com
- aes256-gcm@openssh.com
- chacha20-poly1305@openssh.com

Default set of ciphers in priority order:

1. chacha20-1305@openssh.com
2. aes128-ctr
3. aes192-ctr
4. aes256-ctr
5. aes128-gcm@openssh.com
6. aes256-gcUm@openssh.com

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring SSH to use only specified ciphers in the priority order:

```
switch(config)# ssh ciphers chacha20-poly1305@openssh.com aes256-ctr aes256-cbc
```

Reverting SSH to use the default set of ciphers:

```
switch(config)# no ssh ciphers
```

ssh host-key

Syntax

```
ssh host-key {ecdsa [ecdsa-sha2-nistp256 | ecdsa-sha2-nistp384 | ecdsa-sha2-nistp521 |  
ed25519 | rsa [bits {2048 | 4096}]] }
```

Description

Generates an SSH host-key pair.

Command context

config

Parameters

ecdsa

Selects the ECDSA host-key pair type as `ecdsa-sha2-nistp256` (the default), `ecdsa-sha2-nistp384`, or `ecdsa-sha2-nistp521`.

ed25519

Selects the ED25519 host-key pair.

rsa

Selects the RSA host-key pair. Optionally, the key bit length is selected with either `bits 2048` (the default) or `bits 4096`.

Authority

Administrators or local user group members with execution rights for this command.

Usage

When an SSH server is enabled on a VRF for the first time, host-keys are generated.

If the host-key of the given type exists, a warning message is displayed with a request to overwrite the previous host-key with the new key.

Examples

Overwriting an old ECDSA host-key with a new `ecdsa-sha2-nistp384` host-key:

```
switch(config)# ssh host-key ecdsa ecdsa-sha2-nistp384  
ecdsa host-key will be overwritten.  
Do you want to continue (y/n)?
```

Overwriting an old RSA host-key with a new RSA host-key with 2048 bits:

```
switch(config)# ssh host-key rsa bits 2048  
rsa host-key will be overwritten.  
Do you want to continue (y/n)?
```

Overwriting an ECDSA host-key with an ED25519 host-key pair:

```
switch(config)# ssh host-key ed25519  
ed25519 host-key will be overwritten.  
Do you want to continue (y/n)?
```

ssh host-key-algorithms

Syntax

```
ssh host-key-algorithms <HOST-KEY-ALGORITHMS-LIST>  
no ssh host-key-algorithms
```

Description

Configures SSH to use a set of host key algorithms in the specified priority order. Host key algorithms specify which host key types are allowed to be used for the SSH connection. The first host key entered in the CLI is considered a first priority. Each option represents a type of key that can be used. Host keys are used to verify the host that you are connecting to. This configuration allows you to control which host key types are presented to incoming clients, or which host key types to receive first from hosts. Only the host key algorithms that are specified by the user are configured.

The `no` form of this command removes the configuration of host key algorithms and reverts SSH to use the default set of algorithms.

Command context

config

Parameters

<HOST-KEY-ALGORITHMS-LIST>

Valid host key algorithms are:

- `ecdsa-sha2-nistp256`
- `ecdsa-sha2-nistp384`
- `ecdsa-sha2-nistp521`
- `rsa-sha2-256`
- `rsa-sha2-512`
- `ssh-ed25519`
- `ssh-rsa`

Default set of host key algorithms in priority order:

1. `ecdsa-sha2-nistp256`
2. `ecdsa-sha2-nistp384`
3. `ecdsa-sha2-nistp521`
4. `ssh-ed25519`
5. `rsa-sha2-256`

6. `rsa-sha2-512`
7. `ssh-rsa`

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring SSH to use only specified host key algorithms:

```
switch(config)# ssh host-key-algorithms ssh-rsa ssh-ed25519 ecdsa-sha2-nistp521
```

Reverting SSH to use the default set of host key algorithms:

```
switch(config)# no host-key-algorithms
```

ssh key-exchange-algorithms

Syntax

```
ssh key-exchange-algorithms <KEY-EXCHANGE-ALGORITHMS-LIST>  
no ssh key-exchange-algorithms
```

Description

Configures SSH to use a set of key exchange algorithm types in the specified priority order. The first key exchange type entered in the CLI is considered a first priority. Key exchange algorithms are used to exchange a shared session key with a peer securely. Each option represents an algorithm that is used to distribute a shared key in a way that prevents outside interference, manipulation, or recovery. Only the key exchange algorithms that are specified by the user are configured.

The `no` form of this command removes the configuration of key exchange algorithms and reverts SSH to use the default set of algorithms.

Command context

`config`

Parameters

<KEY-EXCHANGE-ALGORITHMS-LIST>

Valid key exchange algorithms are:

- `curve25519-sha256`
- `curve25519-sha256@libssh.org`
- `diffie-hellman-group-exchange-sha1`
- `diffie-hellman-group-exchange-sha256`
- `diffie-hellman-group14-sha1`
- `diffie-hellman-group14-sha256`
- `diffie-hellman-group16-sha512`
- `diffie-hellman-group18-sha512`
- `ecdh-sha2-nistp256`

- ecdh-sha2-nistp384
- ecdh-sha2-nistp521

Default set of key exchange algorithms in priority order:

1. curve25519-sha256
2. curve25519-sha256@libssh.org
3. ecdh-sha2-nistp256
4. ecdh-sha2-nistp384
5. ecdh-sha2-nistp521
6. diffie-hellman-group-exchange-sha256
7. diffie-hellman-group16-sha512
8. diffie-hellman-group18-sha512
9. diffie-hellman-group14-sha256
10. diffie-hellman-group-exchange-sha1

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring SSH to use a set of specified key exchange algorithms:

```
switch(config)# ssh key-exchange-algorithms ecdh-sha2-nistp256 curve25519-sha256
diffie-hellman-group-exchange-sha256
```

Reverting SSH to use the default set of key-exchange-algorithms:

```
switch(config)# no key-exchange-algorithms
```

ssh known-host remove

Syntax

```
ssh known-host remove {all | {<IPv4-ADDRESS> | <HOSTNAME> | <IPv6-ADDRESS> } }
```

Description

Clears the list of trusted SSH servers for your user account. When you download or upload a file to or from a server using SFTP, you establish a trusted SSH relationship with that server. Each user account maintains its own set of SSH server host-keys for every server to which the user previously connected.

Command context

config

Parameters

all

Clears the trusted servers list.

<IPv4-ADDRESS>

Specifies the IPv4 address of the remote device.

<HOSTNAME>

Specifies the host name of the remote device. The length of the host name can be up to 255 characters.
<IPv6-ADDRESS>

Specifies the IPv6 address of the remote device.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Clearing the trusted server list:

```
switch(config)# ssh known-host remove all
```

Removing a specified server from the trusted server list:

```
switch(config)# ssh known-host remove 1.1.1.1
```

ssh macs

Syntax

```
ssh macs <MACS-LIST>  
no ssh macs
```

Description

Configures SSH to use a set of message authentication codes (MACs) in the specified priority order. The first MAC entered in the CLI is considered a first priority. MACs maintain the integrity of each message sent across an SSH connection. Each option represents an algorithm that can be used to provide integrity between peers. Only the MAC types that are specified by the user are configured.

The `no` form of this command removes the configuration of MACs and reverts SSH to use the default set of MACs.

Command context

config

Parameters

<MACS-LIST>

Valid MAC types are:

- hmac-sha1
- hmac-sha1-96
- hmac-sha1-etm@openssh.com
- hmac-sha2-256
- hmac-sha2-512
- hmac-sha2-256-etm@openssh.com
- hmac-sha2-512-etm@openssh.com

Default set of MACs in priority order:

1. hmac-sha2-256-etm@openssh.com

2. `hmac-sha2-512-etm@openssh.com`
3. `hmac-sha1-etm@openssh.com`
4. `hmac-sha2-256`
5. `hmac-sha2-512`
6. `hmac-sha1`

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring SSH to use a set of specified MACs:

```
switch(config)# ssh macs hmac-sha2-256 hmac-sha2-512
```

Reverting SSH to use the default set of MACs:

```
switch(config)# no ssh macs
```

ssh maximum-auth-attempts

Syntax

```
ssh maximum-auth-attempts <ATTEMPTS>  
no maximum-auth-attempts
```

Description

Sets the SSH maximum number of authentication attempts.

The `no` form of the command resets the maximum to its default of 6.

Command context

config

Parameters

<ATTEMPTS>

Specifies the maximum number of SSH authentication attempts. Range: 1 to 10. Default: 6.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the maximum number of authentication attempts:

```
switch(config)# ssh maximum-auth-attempts 3
```

Resetting the maximum number of authentication attempts to its default of 6:

```
switch(config)# no maximum-auth-attempts
```

ssh public-key-algorithms

Syntax

```
ssh public-key-algorithms <PUBLIC-KEY-ALGORITHMS-LIST>
no ssh public-key-algorithms
```

Description

Configures SSH to use a set of public key algorithms in the specified priority order. The first public key type entered in the CLI is considered a first priority. Public key algorithms specify which public key types can be used for public key authentication in SSH. Each option represents a public key type that the SSH server can accept or that the SSH client can present to a server. Only the public key algorithms that are chosen by the user are configured.

The `no` form of this command removes the configuration of public key algorithms and reverts SSH to use the default set.

Command context

config

Parameters

<PUBLIC-KEY-ALGORITHMS-LIST>

Valid public key algorithm types are:

- ecdsa-sha2-nistp256
- ecdsa-sha2-nistp384
- ecdsa-sha2-nistp521
- ssh-ed25519
- ssh-rsa
- rsa-sha2-256
- rsa-sha2-512
- x509v3-ecdsa-sha2-nistp256
- x509v3-ecdsa-sha2-nistp384
- x509v3-ecdsa-sha2-nistp521
- x509v3-rsa2048-sha256
- x509v3-sign-rsa
- x509v3-ssh-rsa

Default set of public key algorithms in priority order:

1. rsa-sha2-256
2. rsa-sha2-512
3. ssh-rsa
4. ecdsa-sha2-nistp256
5. ecdsa-sha2-nistp384
6. ecdsa-sha2-nistp521
7. ssh-ed25519
8. x509v3-rsa2048-sha256
9. x509v3-ssh-rsa

10. x509v3-sign-rsa
11. x509v3-ecdsa-sha2-nistp256
12. x509v3-ecdsa-sha2-nistp384
13. x509v3-ecdsa-sha2-nistp521

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring SSH to use a set of specified public key algorithms:

```
switch(config)# ssh public-key-algorithms x509v3-ssh-rsa ssh-rsa rsa-sha2-256
```

Reverting SSH to use the default set of public key algorithms:

```
switch(config)# no ssh public-key-algorithms
```

ssh server vrf

Syntax

```
ssh server vrf <VRF-NAME>  
no ssh server vrf <VRF-NAME>
```

Description

Enables the SSH server on the specified VRF.

The `no` form of the command disables the SSH server on the specified VRF.

Command context

config

Parameters

vrf <VRF-NAME>

Specifies the VRF name.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling the SSH server on the management VRF:

```
switch(config)# ssh server vrf mgmt
```

Disabling the SSH server on the management VRF:

```
switch(config)# no ssh server vrf mgmt
```

ip route

Syntax

```
ip route <DEST-IPV4-ADDR>/<NETMASK> {<NEXTHOP-ADDR> | <NEXTHOP-PORT-LAG-VLAN> | reject | blackhole}
no ip route <DEST-IPV4-ADDR>/<NETMASK> {<NEXTHOP-ADDR> | <NEXTHOP-PORT-LAG-VLAN> | reject | blackhole}
```

Description

Adds an IPv4 static route on the default VRF.

The `no` form of this command deletes a IPv4 static route.

Command context

config

Parameters

<DEST-IPV4-ADDR>/<NETMASK>

Specifies the IPv4 route destination.

<NEXTHOP-ADDR>

Specifies the next hop address for reaching the destination in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255.

<NEXTHOP-PORT-LAG-VLAN>

Specifies the next hop as an outgoing interface.

blackhole

Specifies that packets matching the destination route are silently discarded and no ICMP error notification is sent to the sender.

reject

Specifies that packets matching the destination route are discarded and an ICMP error notification is sent to the sender.

Authority

Administrators or local user group members with execution rights for this command.

Examples

```
switch(config)# ip route 10.0.0.0/24 blackhole
switch(config)# ip route 10.0.1.0/24 reject
switch(config)# ip route 10.0.2.0/24 20.0.0.2
switch(config)# ip route 10.0.3.0/24 1/1/1
```

ip route distance

Syntax

```
ip route <DEST-IPV4-ADDR>/<NETMASK> [<NEXT-HOP-IP-ADDR>|<INTERFACE>] distance <VALUE>  
no ip route <DEST-IPV4-ADDR>/<NETMASK> [<NEXT-HOP-IP-ADDR>|<INTERFACE>] distance <VALUE>
```

Description

Configures the administrative distance for the IPv4 static route.

The `no` form of this command deletes the static route.

Command context

config

Parameters

<DEST-IPV4-ADDR>

Specifies an IP address in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255.

<MASK>

Specifies the number of bits in the address mask in CIDR format (x), where x is a decimal number from 0 to 32.

<NEXT-HOP-IP-ADDR>

Specifies the next hop IPv4 address in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255.

<INTERFACE>

Specifies the next hop as an outgoing interface.

distance <VALUE>

Specifies the administrative distance to associate with this static route. Default: 1. Range: 1-255.

Authority

Administrators or local user group members with execution rights for this command.

Examples

```
switch(config)# ip route 10.0.2.0/24 20.0.0.2 distance 4  
switch(config)# ip route 10.0.3.0/24 1/1/1 distance 6
```

```
switch(config)# no ip route 10.0.3.0/24 1/1/1 distance 6
```

ip route tag

Syntax

```
ip route <DEST-IPV4-ADDR>/<NETMASK> {<NEXTHOP-ADDR> | <NEXTHOP-PORT-LAG-VLAN> | reject |  
blackhole} [tag] <1-4294967295>  
no ip route <DEST-IPV4-ADDR>/<NETMASK> {<NEXTHOP-ADDR> | <NEXTHOP-PORT-LAG-VLAN> | reject |  
blackhole} [tag] <1-4294967295>
```

Description

Configures tag for IPv4 static route.

The `no` form of this command deletes tag for IPv4 static route.

Command context

config

Parameters

<DEST-IPV4-ADDR>/<NETMASK>

Specifies the IPv4 route destination.

<NEXTHOP-ADDR>

Specifies the next hop address for reaching the destination in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255.

<NEXTHOP-PORT-LAG-VLAN>

Specifies the next hop as an outgoing interface.

reject

Specifies that packets matching the destination route are discarded and an ICMP error notification is sent to the sender.

blackhole

Specifies that packets matching the destination route are silently discarded and no ICMP error notification is sent to the sender.

tag

Specifies and assigns tag for the route.

Authority

Administrators or local user group members with execution rights for this command.

Examples

```
switch(config)# ip route 10.1.1.1/32 20.1.1.2 tag 10
switch(config)# ip route 10.1.1.5/32 1/1/1 tag 20
```

```
switch(config)# no ip route 10.1.1.1/32 20.1.1.2 tag 10
switch(config)# no route 10.1.1.5/32 1/1/1 tag 20
```

ipv6 route

Syntax

```
ipv6 route <DEST-IPV6-ADDR>/<NETMASK> {<NEXTHOP-ADDR> | <NEXTHOP-PORT-LAG-VLAN> | reject |
blackhole}
no ipv6 route <DEST-IPV6-ADDR>/<NETMASK> {<NEXTHOP-ADDR> | <NEXTHOP-PORT-LAG-VLAN> |
reject | blackhole}
```

Description

Adds an IPv6 static route.

The no form of this command deletes an IPv6 static route on the default VRF.

Command context

config

Parameters

<DEST-IPV6-ADDR>

Specifies the route destination in IPv6 format (xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx), where x is a hexadecimal number from 0 to F.

<NETMASK>

Specifies the number of bits in the address mask in CIDR format (x), where x is a decimal number from 0 to 128.

<NEXTHOP-ADDR>

Specifies the next hop in IPv6 format (xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx), where x is a hexadecimal number from 0 to F.

<NEXTHOP-PORT-LAG-VLAN>

Specifies the next hop as an outgoing interface.

reject

Specifies that packets matching the destination route are discarded and an ICMP error notification is sent to the sender.

blackhole

Specifies that packets matching the destination route are silently discarded and no ICMP error notification is sent to the sender.

Authority

Administrators or local user group members with execution rights for this command.

Usage

- Access platforms for IPv6 routes above /64 are not supported.
- IPv6 addresses with prefixes 65-127 will not be configured in the ASIC route table.

Examples

```
switch(config)# ipv6 route 120::/124 blackhole
switch(config)# ipv6 route 121::/124 reject
switch(config)# ipv6 route 122::/124 1/1/1
switch(config)# ipv6 route 123::/124 120::1
```

```
switch(config)# no ipv6 route 122::/124 1/1/1
```

ipv6 route distance

Syntax

```
ipv6 route <DEST-IPV6-ADDR>/<MASK> [<NEXTHOP-IP-ADDR>|<INTERFACE>] distance <VALUE>
no ipv6 route <DEST-IPV6-ADDR>/<MASK> [<NEXTHOP-IP-ADDR>|<INTERFACE>] distance <VALUE>
```

Description

Configures the administrative distance for the IPv6 static route

The no form of this command deletes the static route.

Command context

config

Parameters

<DEST-IPV6-ADDR>

Specifies the route destination address in IPv6 format (xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx), where x is a hexadecimal number from 0 to F.

<MASK>

Specifies the number of bits in the address mask in CIDR format (x), where x is a decimal number from 0 to 128.

<NEXT-HOP-IP-ADDR>

Specifies the next hop in IPv6 format (xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx), where x is a hexadecimal number from 0 to F.

<INTERFACE>

Specifies the next hop as an outgoing interface.

distance <VALUE>

Specifies the administrative distance to associate with this static route. Range: 1 to 255. Default: 1.

Authority

Administrators or local user group members with execution rights for this command.

Examples

```
switch(config)# ipv6 route 122::/124 1/1/1 distance 5
switch(config)# ipv6 route 123::/124 120::1 distance 6
```

```
switch(config)# no ipv6 route 123::/124 120::1 distance 6
```

ipv6 route tag

Syntax

```
ipv6 route <DEST-IPV6-ADDR>/<NETMASK> {<NEXTHOP-ADDR> | <NEXTHOP-PORT-LAG-VLAN> | reject |
blackhole} [tag] <1-4294967295>
no ipv6 route <DEST-IPV6-ADDR>/<NETMASK> {<NEXTHOP-ADDR> | <NEXTHOP-PORT-LAG-VLAN> |
reject | blackhole} [tag] <1-4294967295>
```

Description

Configures tag for IPv6 static route.

Command context

config

Parameters

<DEST-IPV6-ADDR>

Specifies the route destination in IPv6 format (xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx), where x is a hexadecimal number from 0 to F.

<NETMASK>

Specifies the number of bits in the address mask in CIDR format (x), where x is a decimal number from 0 to 128.

<NEXTHOP-ADDR>

Specifies the next hop in IPv6 format (xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx), where x is a hexadecimal number from 0 to F.

<NEXTHOP-PORT-LAG-VLAN>

Specifies the next hop as an outgoing interface.

reject

Specifies that packets matching the destination route are discarded and an ICMP error notification is sent to the sender.

blackhole

Specifies that packets matching the destination route are silently discarded and no ICMP error notification is sent to the sender.

tag

Specifies and assigns tag for the route.

Authority

Administrators or local user group members with execution rights for this command.

Examples

```
switch(config)# ipv6 route 3001::1/128 1/1/1 tag 10
switch(config)# ipv6 route 3002::1/128 1000::2 tag 20
```

```
switch(config)# no ipv6 route 3001::1/128 1/1/1 tag 10
switch(config)# no ipv6 route 3002::1/128 1000::2 tag 20
```

show ipv4 rib

Syntax

```
show ip rib <FILTER> [vrf <VRF-NAME>]
```

Description

Shows the IPv4 Routing Information Base (RIB) of VRF with name (<VRF-NAME>). If VRF name is not specified, default VRF routes are displayed.

Command context

Operator (>) or Manager (#)

Parameters

<FILTER>

Selects filter, see Usage section.

vrf <VRF-NAME>

Specifies the VRF name.

Authority

Administrators or local user group members with execution rights for this command.

Usage

There are sub-options available within this command:

- **A.B.C.D:** Shows longest prefix match.
- **A.B.C.D/M:** Shows exact route match.
- **all-vrfs:** Shows all VRF information.
- **bgp:** Shows BGP routes only.
- **connected:** Shows connected routes only.
- **local:** Shows local routes only.
- **ospf:** Shows OSPF routes only.

- **rip:** Shows RIP routes only.
- **static:** Shows static routes only.
- **summary:** Shows aggregate count of routes per routing protocol.
- **vrf:** Specifies the VRF name.
- **selected:** Shows routes selected for forwarding only.
- **non-selected:** Shows routes not selected for forwarding only.

Examples

Showing IPv4 routes in RIB:

```
Origin Codes: R - RIP, O - OSPFv2, B - BGP
              C - connected, S - static
Type Codes:  E - External BGP, I - Internal BGP, IA - OSPF inter area
              E1 - OSPF external type 1, E2 - OSPF external type 2
* indicates selected for forwarding

VRF: default
```

Prefix	Nexthop	Interface	VRF	Origin/ Type	Distance/ Metric	Age
*10.0.0.0/30	-	1/1/1	-	S	[20/0]	0d:10h:01m:41s
*10.0.1.0/30	-	1/1/1	-	B/I	[200/0]	2d:20h:01m:42s
*10.1.64.0/18	-	loopback2	-	C	[0/0]	-
*10.2.64.0/18	10.0.0.3	lag1	-	O/E1	[110/25]	1d:05h:03m:43s
*10.2.64.0/18	20.10.0.1	vlan100	-	O/E1	[110/25]	0d:05h:03m:43s
*20.1.2.3/32	2.2.2.2	1/1/4	vrf_red	B/E	[20/0]	2d:10h:01m:45s
*30.1.3.0/24	-	reject	-	S	[1/0]	33d:10h:01m:43s
*50.10.13.0/24	-	reject	-	S	[1/0]	12d:10h:01m:44s
*61.1.1.2/32	4.4.4.4	1/1/5	-	B/I	[200/0]	1d:11h:01m:45s
*62.1.1.3/32	5.5.5.5	1/1/6	-	B/I	[200/0]	0d:12h:01m:45s
*193.0.0.2/32	50.0.0.2	1/1/2	-	S	[1/0]	0d:04h:01m:43s
193.0.0.2/32	56.0.0.3	1/1/3	-	O/E1	[110/25]	0d:04h:03m:43s

Total Route Count : 13

Showing IPv4 exact route match in RIB:

```
show ip rib 10.0.0.0/30

VRF : default
```

Prefix	: 10.0.0.0/30	VRF(egress)	: -
Nexthop	: -	Interface	: 1/1/1
Origin	: Connected	Type	: -
Distance	: 0	Metric	: 0
Age	: -	Tag	: 0
Selected	: Yes	Recursive Nexthop	: No

Showing IPv4 RIB summary:

```
show ip rib summary

IPv4 RIB Table Summary
```


VRF name :	default
Protocol	RIB Routes

connected	1010
local	1011
static	4
ospfv2	509
bgp	9014
selected	10008
non-selected	1518

show ipv6 rib

Syntax

```
show ipv6 rib <FILTER> [vrf <VRF-NAME>]
```

Description

Shows the IPv6 Routing Information Base (RIB) of VRF with name (<VRF-NAME>). If VRF name is not specified, default VRF routes are displayed.

Command context

Operator (>) or Manager (#)

Parameters

<FILTER>

Selects filter, see usage section.

vrf <VRF-NAME>

Shows routes in the VRF and specifies VRF name.

Authority

Administrators or local user group members with execution rights for this command.

Usage

There are sub-options available within this command:

- **X:X: :X:X:** Shows longest prefix match.
- **X:X: :X:X/M:** Shows exact route match.
- **all-vrfs:** Shows all VRF information.
- **bgp:** Shows BGP routes only.
- **connected:** Shows connected routes only.
- **local:** Shows local routes only.
- **ospf:** Shows OSPF routes only.
- **rip:** Shows RIP routes only.
- **static:** Shows static routes only.
- **summary:** Shows aggregate count of routes per routing protocol.
- **vrf:** Specifies the VRF name.

- **selected:** Shows routes selected for forwarding only.
- **non-selected:** Shows routes not selected for forwarding only.

Examples

Showing IPv6 routes in RIB:

```
Origin Codes: R - RIPng, O - OSPFv3, B - BGP
              C - connected, S - static
Type Codes:  E - External BGP, I - Internal BGP, IA - OSPF inter area
              E1 - OSPF external type 1, E2 - OSPF external type 2
* indicates selected for forwarding

VRF: default
```

Prefix	Nexthop	Interface	VRF	Origin/ Type	Distance/ Metric	Age
*1000::/64	-	1/1/1	-	C	[0/0]	-
*1000::8/128	-	1/1/1	-	L	[0/0]	-
*1001:db8::/32	1000::10	1/1/1	-	B/I	[200/0]	1d:20h:01m:42s
*2000::/64	fe80::3182	vlan100	-	S	[1/0]	2d:05h:03m:43s
2000::/64	fe80::1241	1/1/1	-	O/E1	[110/25]	0d:05h:03m:43s
*2000::2000:0:0:0/67	fe80::1111	lag1	Green	B/E	[20/0]	1d:10h:01m:45s
*3001::0/64	-	vlan100	-	C	[0/0]	-
*3001::1/128	-	vlan100	-	L	[0/0]	-
*6101::0/64	-	blackhole	-	S	[1/0]	12d:10h:01m:43s

Total Route Count : 9

Showing IPv6 exact route match in RIB:

```
show ipv6 rib 2000::2000:0:0:0

VRF : default
```

Prefix	: 2000::2000:0:0:0/67	VRF (egress)	: Green
Nexthop	: fe80::1111	Interface	: lag1
Origin	: BGP	Type	: External
Distance	: 20	Metric	: 0
Age	: 1d:10h:01m:45s	Tag	: 20
Selected	: Yes	Recursive Nexthop	: Yes

Showing IPv6 RIB summary:

```
show ipv6 rib summary

IPv6 RIB Table Summary

VRF name : default
Protocol  RIB Routes
-----
connected 1009
local     1010
static     3
ospfv3     508
bgp        1013
```

selected	10004
non-selected	1527

copy command-output

Syntax

```
copy command-output "<COMMAND>" {<STORAGE-URL> | <REMOTE-URL> [vrf <VRF-NAME>]}
```

Description

Copies the specified command output using TFTP, SFTP, or USB.

Command context

Manager (#)

Parameters

<COMMAND>

Specifies the command from which you want to obtain its output. Required. Users with auditor rights can specify these two commands only:

```
show accounting log
```

```
show events
```

{<STORAGE-URL> | <REMOTE-URL> [vrf <VRF-NAME>]}

Select either the storage URL or the remote URL for the destination of the copied command output. Required.

<STORAGE-URL>

Specifies the USB to copy command output.

Syntax:

```
{usb}:/<FILE>
```

<REMOTE-URL>

Specifies the URL to copy the command output.

Syntax: {tftp:// | sftp://<USER>@}{<IP> | <HOST>}[:<PORT>][;blocksize=<VAL>]/<FILE>

vrf <VRF-NAME>

Specifies the VRF name. The default VRF name is default. Optional.

Authority

Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only.

Examples

Copying the output from the `show events` command to a remote URL:

```
switch# copy command-output "show events" tftp://10.100.0.12/file
```

Copying the output from the `show events` command to a file named `events` on a USB drive:

```
switch# copy command-output "show events" usb:/events
```

copy diag-dump feature <FEATURE>

Syntax

```
copy diag-dump feature <FEATURE> {<REMOTE-URL> [vrf <VRF-NAME>] | <STORAGE-URL>}
```

Description

Copies the specified diagnostic information using TFTP, SFTP, or USB.

Command context

Manager (#)

Parameters

<FEATURE>

The name of a feature, for example `aaa`. Required.

{<REMOTE-URL> [vrf <VRF-NAME>] | <STORAGE-URL>}

Select either the remote URL or the storage URL for the destination of the copied command output. Required.

<REMOTE-URL>

Specifies the remote destination URL. Required. The syntax of the URL is the following:

Syntax: {tftp:// | sftp://<USER>@}{<IP> | <HOST>}[:<PORT>][;blocksize=<VAL>]/<FILE>
vrf <VRF-NAME>

Specifies the VRF name. If no VRF name is provided, the VRF named *default* is used. Optional.

<STORAGE-URL>

Specifies the USB to copy command output. Required.

Syntax: {usb}:/<FILE>

Authority

Administrators or local user group members with execution rights for this command.

Examples

Copying the output from the `aaa` feature to a remote URL with a specified VRF:

```
switch# copy diag-dump feature aaa tftp://10.100.0.12/diagdump.txt vrf mgmt
```

copy diag-dump local-file

Syntax

```
copy diag-dump local-file {<REMOTE-URL> [vrf <VRF-NAME>] | <STORAGE-URL>}
```

Description

Copies the diagnostic information stored in a local file using TFTP, SFTP, or USB.

Command context

Manager (#)

Parameters

{<REMOTE-URL> [vrf <VRF-NAME>] | <STORAGE-URL>}

Select either the storage URL or the remote URL for the destination of the copied command output.
Required.

<REMOTE-URL>

Specifies the URL to copy the command output.

Syntax: {tftp:// | sftp://<USER>@}{<IP> | <HOST>}[:<PORT>][;blocksize=<VAL>]/<FILE>
vrf <VRF-NAME>

Specifies the VRF name. The default VRF name is default. Optional.

<STORAGE-URL>

Specifies the USB to copy command output.

Syntax: {usb}:/<FILE>

Authority

Administrators or local user group members with execution rights for this command.

Usage

The `copy diag-dump local-file` command can be used only after the information is captured. Run the `diag-dump <FEATURE-NAME> basic local-file` command before you enter the `copy diag-dump local-file` command to capture the diagnostic information for the specified feature into the local file.

Examples

Copying the output from the local file to a remote URL:

```
switch# diag-dump aaa basic local-file
switch# copy diag-dump local-file tftp://10.100.0.12/diagdump.txt
```

Copying the output from the local file to a USB drive:

```
switch# diag-dump aaa basic local-file
switch# copy diag-dump local-file usb:/diagdump.txt
```

copy show-tech feature

Syntax

copy show-tech feature <FEATURE> {<REMOTE-URL> [vrf <VRF-NAME>] | <STORAGE-URL>}

Description

Copies show tech output using TFTP, SFTP, and USB.

Command context

Manager (#)

Parameters

{<REMOTE-URL> [vrf <VRF-NAME>] | <STORAGE-URL>}

Select either the remote URL or the storage URL for the destination of the copied command output.
Required.

<REMOTE-URL>

Specifies the URL to copy the command output. Required.

Syntax: {tftp:// | sftp://<USER>@}{<IP> | <HOST>}[:<PORT>][;blocksize=<VAL>]/<FILE>
vrf <VRF-NAME>

Specifies the VRF name. The default VRF name is default. Optional.

<STORAGE-URL>

Specifies the USB to copy command output. Required.

Syntax: {usb}:/<FILE>

Authority

Administrators or local user group members with execution rights for this command.

Example

Copying show tech output of the aaa feature using SFTP:

```
switch# copy show-tech feature aaa sftp://user@10.0.0.12/file.txt vrf mgmt
```

Copying show tech output of the config feature using SFTP on the mgmt VRF:

```
switch# copy show-tech feature config sftp://root@10.0.0.1/tech.txt vrf mgmt
```

copy show-tech local-file

Syntax

copy show-tech local-file {<REMOTE-URL> [vrf <VRF-NAME>] | <STORAGE-URL>}

Description

Copies show tech output stored in a local file.

Command context

Manager (#)

Parameters

{<REMOTE-URL> [vrf <VRF-NAME>] | <STORAGE-URL> }

Select either the remote URL or the storage URL for the destination of the copied command output. Required.

<REMOTE-URL>

Specifies the URL to copy the command output.

Syntax: {tftp:// | sftp://<USER>@}{<IP> | <HOST>}[:<PORT>][;blocksize=<VAL>]/<FILE>
vrf <VRF-NAME>

Specifies the VRF name. The default VRF name is default. Optional.

<STORAGE-URL>

Specifies the USB to copy command output.

Syntax: {usb}:/<FILE>

Authority

Administrators or local user group members with execution rights for this command.

Usage

Before entering the `copy show-tech local-file` command, run the `show tech` command with the `local-file` parameter for the specified feature.

Examples

Copying the output to a remote URL:

```
switch# copy show-tech local-file tftp://10.100.0.12/file.txt
```

Copying the output to a remote URL with a VRF:

```
switch# copy show-tech local-file tftp://10.100.0.12/file.txt vrf mgmt
```

Copying the output to a USB:

```
switch# copy show-tech local-file usb:/file
```

copy support-files

Syntax

```
copy support-files previous-boot <REMOTE-URL> [vrf <VRF-NAME>]
copy support-files all <REMOTE-URL> [vrf <VRF-NAME>]
copy support-files <REMOTE-URL> [vrf <VRF-NAME>]
copy support-files feature <FEATURE-NAME> <STORAGE-URL>
copy support-files previous-boot <STORAGE-URL>
copy support-files all <STORAGE-URL>
copy support-files <STORAGE-URL>
```

Description

Copies a set of support files to a compressed file in tar.gz format using TFTP, SFTP, or USB.

Command context

Manager (#)

Parameters

<FEATURE-NAME>

The feature name, for example, aaa.

{<REMOTE-URL> [vrf <VRF-NAME>] | <STORAGE-URL> }

Select either the remote URL or the storage URL for the destination of the copied command output.
Required.

<REMOTE-URL>

Specifies the URL to copy the command output.

Syntax: {tftp:// | sftp://<USER>@}{<IP> | <HOST>}{:<PORT>}[;blocksize=<VAL>]/<FILE>
vrf <VRF-NAME>

Specifies the VRF name. The default VRF name is default. Optional.

<STORAGE-URL>

Specifies the USB to copy command output.

Syntax: {usb}:/<FILE>

Usage

If feature name is not provided, the command collects generic system-specific support information. If a feature name is provided, the command collects feature-specific support information.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Copying the support files to a remote URL:

```
switch# copy support-files tftp://10.100.0.12/file.tar.gz
```

Copying the support files of the `lldp` feature to a remote URL with a specified VRF:

```
switch# copy support-files feature lldp tftp://10.100.0.12/file.tar.gz vrf mgmt
```

Copying the support files from the previous boot to a remote URL with a specified VRF:

```
switch# copy support-files previous-boot sftp://root@10.0.14.206/file.tar.gz vrf mgmt
```

Copying the support files to a USB:

```
switch# copy support-files usb:/file.tar.gz
```

Copying all the support files to a remote URL:

```
switch# copy support-files all sftp://root@10.0.14.216/file.tar.gz vrf mgmt
```

Copying the support files of the `config` feature to a USB:

```
switch# copy support-files feature config usb:/file.tar.gz
```

copy support-files local-file

Syntax

```
copy support-files [feature <FEATURE-NAME> | previous-boot | all ] local-file {<REMOTE-URL> [vrf <VRF-NAME>] | <STORAGE-URL>}
```

Description

Stores a set of support files as a compressed file in the switch locally and copies the preserved support files to a directory using TFTP, SFTP, or USB.



You can store only one copy of the support file locally. When you store a new support file, it overwrites the existing support file.

Command context

Manager (#)

Parameters

<FEATURE-NAME>

Specifies the feature for the support files.

<SLOT-ID>

Specifies the module slot number identifier for the support files. Range: 1/1-1/4, 1/7-1/10

<MEMBER-ID>

Specifies the VSF member identifier for the support files. Range: 1-10

<REMOTE-URL>

Specifies the URL to copy the support files.

<STORAGE-URL>

Specifies the USB to copy the support files.

<VRF-NAME>

Specifies the VRF name. The default VRF name is default.

Authority

Administrators or local user group members with execution rights for this command.

Usage

If the copy of the support files to the destination fails, an alternate option is prompted to store the collected data in the local file. This helps us to retry the copy process using `copy support-files local-file`

<REMOTE-URL/>STORAGE-URL> without the need of regenerating the file.

Examples

Copying support file to the local file:

```
switch# copy support-files local-file
switch# copy support-files feature lldp local-file
switch# copy support-files previous-boot local-file
switch# copy support-files all local-file
The operation to copy all support files could take a while to complete.
Do you want to continue (y/n)?
```

Copying local support file to a remote URL and storage URL:

```
switch# copy support-files local-file usb:/support_files_dir_path/

switch# copy support-files local-file sftp://root@10.0.14.206//support_files_dir_
path/abc.tar.gz vrf mgmt
```

copy support-log

Syntax

Description

Copies the specified support log for a daemon TFTP, SFTP, or USB.

Command context

Manager (#)

Parameters

<DAEMON-NAME>

Specifies the name of the daemon. Required.

{*<STORAGE-URL>* | *<REMOTE-URL>* [*vrf <VRF-NAME>*] }

Selects either the storage URL or the remote URL for the destination of the copied command output. Required.

<STORAGE-URL>

Specifies the USB to copy command output.

Syntax: {usb} : / *<FILE>*

<REMOTE-URL>

Specifies the URL to copy the command output.

Syntax: {tftp:// | sftp://*<USER>*@{*<IP>* | *<HOST>*}[: *<PORT>*] [; blocksizes=*<VAL>*] } / *<FILE>*

vrf <VRF-NAME>

Specifies the VRF name. If no VRF name is provided, the VRF named *default* is used. Optional.

Usage

Fast log is a high performance, per-daemon binary logging infrastructure used to debug daemon level issues by precisely capturing the per daemon/module/functionalities debug traces in real time. Fast log, also referred to as support logs, helps users to understand the feature internals and its specific happenings. The fast logs from one daemon are not overwritten by other daemon logs because fast logs are captured as part of a daemon core dump. Fast logs are enabled by default.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Copying the support log from the daemon hpe-fand to a remote URL:

```
switch# copy support-log hpe-fand tftp://10.100.0.12/file
```

Copying the support log from the daemon hpe-fand to a remote URL with a VRF named mgmt:

```
switch# copy support-log hpe-fand tftp://10.100.0.12/file vrf mgmt
```

Copying the support log from the daemon hpe-fand to a USB:

```
switch# copy support-log hpe-fand usb:/support-log
```

clear events

Syntax

```
clear events
```

Description

Clears up event logs. Using the `show events` command will only display the logs generated after the `clear events` command.

Command context

Manager (#)

Authority

Administrators or local user group members with execution rights for this command.

Examples

Clearing all generated event logs:

```
switch# show events
-----
show event logs
-----
2018-10-14:06:57:53.534384|hpe-sysmond|6301|LOG_INFO|MSTR|1|System resource
utilization poll interval is changed to 27
2018-10-14:06:58:30.805504|lldpd|103|LOG_INFO|MSTR|1|Configured LLDP tx-timer to 36
2018-10-14:07:01:01.577564|hpe-sysmond|6301|LOG_INFO|MSTR|1|System resource
utilization poll interval is changed to 49

switch# clear events

switch# show events
-----
show event logs
-----
2018-10-14:07:03:05.637544|hpe-sysmond|6301|LOG_INFO|MSTR|1|System resource
utilization poll interval is changed to 34
```

domain-name

Syntax

```
domain-name <NAME>
no domain-name [<NAME>]
```

Description

Specifies the domain name of the switch.

The `no` form of this command sets the domain name to the default, which is no domain name.

Command context

`config`

Parameters

`<NAME>`

Specifies the domain name to be assigned to the switch. The first character of the name must be a letter or a number. Length: 1 to 32 characters.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting and showing the domain name:

```
switch# show domain-name

switch# config
switch(config)# domain-name example.com
switch(config)# show domain-name
example.com
switch(config)#
```

Setting the domain name to the default value:

```
switch(config)# no domain-name
switch(config)# show domain-name

switch(config)#
```

hostname

Syntax

`hostname <HOSTNAME>`

`no hostname [<HOSTNAME>]`

Description

Sets the host name of the switch.

The `no` form of this command sets the host name to the default value, which is `switch`.

Command context

`config`

Parameters

`<HOSTNAME>`

Specifies the host name. The first character of the host name must be a letter or a number. Length: 1 to 32 characters. Default: `switch`

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting and showing the host name:

```
switch# show hostname
switch
switch# config
switch(config)# hostname myswitch
myswitch(config)# show hostname
myswitch
```

Setting the host name to the default value:

```
myswitch(config)# no hostname
switch(config)#
```

led locator

Syntax

```
led locator {on | off | flashing}
```

Description

Sets the state of the locator LED to `on`, `off` (default), or `flashing`.

Command context

Manager (#)

Parameters

`on`

Turns on the LED.

`off`

Turns off the LED, which is the default value.

`flashing`

Sets the LED to blink on and off repeatedly.

Authority

Administrators or local user group members with execution rights for this command.

Example

Setting the state of the locator LED:

```
switch# led locator flashing
```

show capacities

Syntax

```
show capacities <FEATURE>
```

Description

Shows system capacities and their values for all features or a specific feature.

Command context

Manager (#)

Parameters

<FEATURE>

Specifies a feature. For example, aaa.

Authority

Administrators or local user group members with execution rights for this command.

Usage

Capacities are expressed in user-understandable terms. Thus they may not map to a specific hardware or software resource or component. They are not intended to define a feature exhaustively.

Examples

Showing classifier-related capacities on the 6100:

```
switch# show capacities classifier

System Capacities: Filter Classifier
Capacities Name                                     Value
-----
Maximum number of Access Control Entries configurable in a system      4096
Maximum number of Access Control Lists configurable in a system         512
Maximum number of class entries configurable in a system                4096
Maximum number of classes configurable in a system                     512
Maximum number of entries in an Access Control List                    1024
Maximum number of entries in a class                                   1024
Maximum number of entries in a policy                                   64
Maximum number of classifier policies configurable in a system          512
Maximum number of policy entries configurable in a system               4096
```

Showing all available capacities on the 6100:

```
switch# show capacities

System Capacities:
Capacities Name
Value
-----
Maximum number of Access Control Entries configurable in a system
4096
Maximum number of Access Control Lists configurable in a system
512
Maximum number of class entries configurable in a system
4096
Maximum number of classes configurable in a system
```

512
 Maximum number of entries in an Access Control List
 1024
 Maximum number of entries in a class
 1024
 Maximum number of entries in a policy
 64
 Maximum number of classifier policies configurable in a system
 512
 Maximum number of policy entries configurable in a system
 4096
 Maximum number of clients supported for tracking the IP address in the system
 128
 Maximum number of dynamic VLANs that can be allowed using MVRP
 256
 Maximum number of nexthops per IP ECMP group
 1
 Maximum number of IP neighbors (IPv4+IPv6) supported in the system
 1024
 Maximum number of IPv4 neighbors(# of ARP entries) supported in the system
 1024
 Maximum number of IPv6 neighbors(# of ND entries) supported in the system
 512
 Maximum number of L2 MAC addresses supported in the system
 8192
 Maximum number of L3 Groups for IP Tunnels and ECMP Groups
 1
 Maximum number of L3 Destinations for Routes, Nexthops in Tunnels and ECMP groups
 1024
 Maximum number of configurable LAG ports
 8
 Maximum number of members supported by a LAG port
 8
 Maximum number of VLANs across ports allowed in loop-protect
 3328
 Maximum number of IGMP/MLD groups supported
 512
 Maximum number of IGMP/MLD snooping groups supported
 512
 Maximum number of Mirror Sessions configurable in a system
 4
 Maximum number of enabled Mirror Sessions in a system
 4
 Maximum number of mstp instances configurable in a system
 16
 Maximum number of Clients that can be authenticated on a port
 32
 Maximum number of Device Profiles allowed to be created on the system
 8
 Maximum number of Port Access Roles allowed to be created on the system
 32
 Maximum number of MAC Address that can be authorized on a port
 32
 Maximum number of Port Access Role VLAN IDs allowed to be created on the system
 50
 Maximum number of Port Access Role VLAN names allowed to be created on the system
 50
 Maximum number of RBAC rules per user group
 1024
 Maximum number of RPVST VLANs configurable on the system
 16
 Maximum number of RPVST VPORTs supported in a system


```
512
Maximum number of SVIs supported in the system
16
Maximum number of routes (IPv4+IPv6) on the system
512
Maximum number of IPv4 routes on the system
512
Maximum number of IPv6 routes on the system
512
Maximum number of VLANs supported in the system
512
```

Showing all available capacities for mirroring:

```
switch# show capacities mirroring

System Capacities: Filter Mirroring
Capacities Name                                     Value
-----
Maximum number of Mirror Sessions configurable in a system      4
Maximum number of enabled Mirror Sessions in a system           4
```

Showing all available capacities for MSTP:

```
switch# show capacities mstp

System Capacities: Filter MSTP
Capacities Name                                     Value
-----
Maximum number of mstp instances configurable in a system      64
```

Showing all available capacities for VLAN count:

```
switch# show capacities vlan-count

System Capacities: Filter VLAN Count
Capacities Name                                     Value
-----
Maximum number of VLANs supported in the system              4094
```

show capacities-status

Syntax

```
show capacities-status <FEATURE>
```

Description

Shows system capacities status and their values for all features or a specific feature.

Command context

Manager (#)

Parameters

<FEATURE>

Specifies the feature, for example `aaa` for which to display capacities, values, and status. Required.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Showing the system capacities status for all features:

```
switch# show capacities-status
```

System Capacities Status		
Capacities Status Name		Value
Maximum		

--		
Number of Access Control Entries currently configured	0	4096
Number of Access Control Lists currently configured	0	512
Number of class entries currently configured	0	4096
Number of classes currently configured	0	512
Number of policies currently configured	0	512
Number of policy entries currently configured	0	4096
Number of dynamic VLANs currently learnt using MVRP	0	256
Number of IP neighbor (IPv4+IPv6) entries	1	1024
Number of IPv4 neighbor(ARP) entries	1	1024
Number of IPv6 neighbor(ND) entries	0	512
Number of L3 Groups for IP Tunnels and ECMP Groups currently configured	0	1
Number of L3 Destinations for Routes, Nexthops in ECMP groups and Tunnels currently configured	0	1024
Number of Mirror Sessions currently configured	0	4
Number of Mirror Sessions currently enabled	0	4
Number of mstp instances currently configured	0	16
Number of RPVST VLANs currently configured	0	16
Number of routes (IPv4+IPv6) currently configured	1	512
Number of IPv4 routes currently configured	1	512
Number of IPv6 routes currently configured	0	512
Number of VLANs currently configured	1	512

show core-dump

Syntax

```
show core-dump all
```

Description

Shows core dump information about the specified module. When no parameters are specified, shows only the core dumps generated in the current boot of the management module. When the `all` parameter is specified, shows all available core dumps.

Command context

Manager (#)

Parameters

`all`

Shows all available core dumps.

Authority

Administrators or local user group members with execution rights for this command.

Usage

When no parameters are specified, the `show core-dump` command shows only the core dumps generated in the current boot of the management module. You can use this command to determine when any crashes are occurring in the current boot.

If no core dumps have occurred, the following message is displayed: `No core dumps are present`

To show core dump information for the standby management module, you must use the `standby` command to switch to the standby management module and then execute the `show core-dump` command.

In the output, the meaning of the information is the following:

Daemon Name

Identifies name of the daemon for which there is dump information.

Instance ID

Identifies the specific instance of the daemon shown in the **Daemon Name** column.

Present

Indicates the status of the core dump:

Yes

The core dump has completed and available for copying.

In Progress

Core dump generation is in progress. Do not attempt to copy this core dump.

Timestamp

Indicates the time the daemon crash occurred. The time is the local time using the time zone configured on the switch.

Build ID

Identifies additional information about the software image associated with the daemon.

Examples

Showing core dump information for the current boot of the active management module only:

```
switch# show core-dump
=====
Daemon Name      | Instance ID | Present   | Timestamp           | Build ID
=====
hpe-fand         | 1399        | Yes       | 2017-08-04 19:05:34 | 1246d2a
hpe-sysmond      | 957         | Yes       | 2017-08-04 19:05:29 | 1246d2a
=====
Total number of core dumps : 2
=====
```

Showing all core dumps:

```
switch# show core-dump all
=====
Management Module core-dumps
=====
Daemon Name      | Instance ID | Present   | Timestamp           | Build ID
=====
hpe-sysmond      | 513         | Yes       | 2017-07-31 13:58:05 | e70f101
=====
```

```

hpe-tempd      1048      Yes      2017-08-13 13:31:53      e70f101
hpe-tempd      1052      Yes      2017-08-13 13:41:44      e70f101

Line Module core-dumps
=====
Line Module : 1/1
=====
dune_agent_0    18958      Yes      2017-08-12 11:50:17      e70f101
dune_agent_0    18842      Yes      2017-08-12 11:50:09      e70f101
=====
Total number of core dumps : 5
=====

```

show domain-name

Syntax

show domain-name

Description

Shows the current domain name.

Command context

Manager (#)

Authority

Administrators or local user group members with execution rights for this command.

Usage

If there is no domain name configured, the CLI displays a blank line.

Example

Setting and showing the domain name:

```

switch# show domain-name

switch# config
switch(config)# domain-name example.com
switch(config)# show domain-name
example.com
switch(config)#

```

show environment fan



show environment fan is not available for JL679A.

Syntax

show environment fan

Description

Shows the status information for all fans and fan trays (if present) in the system.

Command context

Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Usage

For fan trays, `Status` is one of the following values:

`ready`

The fan tray is operating normally.

`fault`

The fan tray is in a fault event. The status of the fan tray does not indicate the status of fans.

`empty`

The fan tray is not installed in the system.

For fans:

`Speed`

Indicates the relative speed of the fan based on the nominal speed range of the fan.

`N/A`

This value is not applicable to the 6100.

`Direction`

The direction of airflow through the fan. Values are:

`left-to-right`

Air flows from the left of the system to the right of the system.

`N/A`

The fan is not installed.

`Status`

Fan status. Values are:

`uninitialized`

The fan has not completed initialization.

`ok`

The fan is operating normally.

`fault`

The fan is in a fault state.

`empty`

The fan is not installed.

Examples

Showing output for systems with fan trays for 6100 switch series:

```
switch# show environment fan
```

```
Fan information
```

Mgr/Fan	Product Name	Serial Number	Speed	Direction	Status	RPM
1/1	N/A	N/A	N/A	left-to-right	ok	N/A

show environment led

Syntax

```
show environment led
```

Description

Shows state and status information for all the configurable LEDs in the system.

Command context

Operator (>) or Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

Showing state and status for LED:

```
switch# show environment led
Mbr/Name      State      Status
-----
1/locator     off        ok
```

show environment power-supply

Syntax

```
show environment power-supply
```

Description

Shows status information about all power supplies in the switch.

Command context

Operator (>) or Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Usage

The following information is provided for each power supply:

Mbr/PSU

Shows the member and slot number of the power supply.

Product Number

Shows the product number of the power supply.

Serial Number

Shows the serial number of the power supply, which uniquely identifies the power supply.

PSU Status

The status of the power supply. Values are:

OK

Power supply is operating normally.

OK*

Power supply is operating normally, but it is the only power supply in the chassis. One power supply is not sufficient to supply full power to the switch. When this value is shown, the output of the command also shows a message at the end of the displayed data.

Absent

No power supply is installed in the specified slot.

Input fault

The power supply has a fault condition on its input.

Output fault

The power supply has a fault condition on its output.

Warning

The power supply is not operating normally.

Wattage Maximum

Shows the maximum amount of wattage that the power supply can provide.

Example

Showing the output when only one power supply is installed in the switch:

```
switch# show environment power-supply
```

	Product	Serial	PSU	Wattage
Mbr/PSU	Number	Number	Status	Maximum
1/1	N/A	N/A	OK	500

show environment temperature

Syntax

```
show environment temperature [detail]
```

Description

Shows the temperature information from sensors in the switch that affect fan control.

Command context

Operator (>) or Manager (#)

Parameters

detail

Shows detailed information from each temperature sensor.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Usage

Temperatures are shown in Celsius.

Valid values for status are the following:

normal

Sensor is within nominal temperature range.

min

Lowest temperature from this sensor.

max

Highest temperature from this sensor.

low_critical

Lowest threshold temperature for this sensor.

critical

Highest threshold temperature for this sensor.

fault

Fault event for this sensor.

emergency

Over temperature event for this sensor.

Examples

Showing current temperature information for a 6100 switch:

```
switch# show environment temperature
Temperature information
```

Mgr/Slot-Sensor	Module Type	Current temperature	Status
1/1-COMe-Daughter-Boar	line-card-module	66.45 C	normal
1/1-PCIE-Switch	line-card-module	95.82 C	normal
1/1-Processor	line-card-module	00.00 C	fault
1/1-Switch-ASIC	line-card-module	116.36 C	emergency
1/1-Switch-ASIC-Internal	line-card-module	108.25 C	critical
1/2-COMe-Daughter-Boar	line-card-module	67.29 C	normal
1/2-PCIE-Switch	line-card-module	95.82 C	normal
1/2-Processor-1	line-card-module	72.92 C	normal
1/2-Processor-2	line-card-module	73.05 C	normal
1/2-Switch-ASIC	line-card-module	97.41 C	normal
1/2-Switch-ASIC-Internal	line-card-module	97.62 C	normal
...			

show events

Syntax

```
show events [ -e <EVENT-ID> |
  -s {alert | crit | debug | emer | err | info | notice | warn} |
  -r | -a | -i <MEMBER-ID> | -n <count> |
  -c {lldp | ospf | ... | } |
  -d {lldpd | hpe-fand | ... |}]
```

Description

Shows event logs generated by the switch modules since the last reboot.

Command context

Manager (#)

Parameters

-e <EVENT-ID>

Shows the event logs for the specified event ID. Event ID range: 101 through 99999.

`-s {alert | crit | debug | emer | err | info | notice | warn}`

Shows the event logs for the specified severity. Select the severity from the following list:

- `alert`: Displays event logs with severity alert and above.
- `crit`: Displays event logs with severity critical and above.
- `debug`: Displays event logs with all severities.
- `emer`: Displays event logs with severity emergency only.
- `err`: Displays event logs with severity error and above.
- `info`: Displays event logs with severity info and above.
- `notice`: Displays event logs with severity notice and above.
- `warn`: Displays event logs with severity warning and above.

`-r`

Shows the most recent event logs first.

`-a`

Shows all event logs, including those events from previous boots.

`-n <count>`

Displays the specified number of event logs.

`-c {lldp | ospf | ... | }`

Shows the event logs for the specified event category. Enter `show event -c` for a full listing of supported categories with descriptions.

`-d {lldpd | hpe-fand | ... | }`

Shows the event logs for the specified process. Enter `show event -d` for a full listing of supported daemons with descriptions.

Authority

Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only.

Examples

Showing event logs:

```
switch# show events
-----
show event logs
-----
2016-12-01:12:37:31.733551|lacpd|15007|INFO|AMM|1|LACP system ID set to
70:72:cf:51:50:7c
2016-12-01:12:37:31.734541|intfd|4001|INFO|AMM|1|Interface port_admin set to up for
bridge_normal interface
2016-12-01:12:37:32.583256|switchd|24002|ERR|AMM|1|Failed to create VLAN 1 in
Hardware
```

Showing the most recent event logs first:

```
switch# show events -r
-----
show event logs
-----
2016-12-01:12:37:32.583256|switchd|24002|ERR|AMM|1|Failed to create VLAN 1 in
Hardware
2016-12-01:12:37:31.734541|intfd|4001|INFO|AMM|1|Interface port_admin set to up for
```

```
bridge_normal interface
2016-12-01:12:37:31.733551|lacpd|15007|INFO|AMM|1|LACP system ID set to
70:72:cf:51:50:7c
```

Showing all event logs:

```
switch# show events -a
-----
show event logs
-----
2016-12-01:12:37:31.733551|lacpd|15007|INFO|AMM|1|LACP system ID set to
70:72:cf:51:50:7c
2016-12-01:12:37:31.734541|intfd|4001|INFO|AMM|1|Interface port_admin set to up for
bridge_normal interface
2016-12-01:12:37:32.583256|switchd|24002|ERR|AMM|1|Failed to create VLAN 1 in
Hardware
```

Showing event logs related to IRDP:

```
switch# switch# show events -c irdp
2016-05-31:06:26:27.363923|hpe-rdiscd|111001|LOG_INFO|IRDP enabled on interface %s
2016-05-31:07:08:51.351755|hpe-rdiscd|111002|LOG_INFO|IRDP disabled on interface %s
```

Showing event logs related to LACP:

```
switch# show events -c lacp
-----
show event logs
-----
2016-12-01:12:37:31.733551|lacpd|15007|INFO|AMM|1|LACP system ID set to
70:72:cf:51:50:7c
```

Showing event logs as per the specified slot ID:

```
switch# show events -i 1
-----
Event logs from current boot
-----
2020-04-22T05:36:14.430166+00:00 6300 vrfmgrd[3053]: Event|5401|LOG_
INFO|MSTR|1|Created a vrf entity b1721d27-41c2-485d-9bae-2cfcbc9bd13d
2020-04-22T05:36:14.942597+00:00 6300 vrfmgrd[3053]: Event|5401|LOG_
INFO|MSTR|1|Created a vrf entity 5eb532c9-5b4d-4d83-b34a-db24ae542d4e
2020-04-22T05:36:15.886252+00:00 6300 vsfd[710]: Event|9903|LOG_INFO|MSTR|1|Master 1
boot complete
```

Showing event logs as per the specified process:

```
switch# show events -d lacpd
-----
show event logs
-----
2016-12-01:12:37:31.733551|lacpd|15007|INFO|AMM|1|LACP system ID set to
70:72:cf:51:50:7c
```

Displaying the specified number of event logs:

```
switch# show events -n 5
-----
show event logs
-----
2018-03-21:06:12:15.500603|arpmgrd|6101|LOG_INFO|AMM|-|ARPMGRD daemon has started
2018-03-21:06:12:17.734405|lldpd|109|LOG_INFO|AMM|-|Configured LLDP tx-delay to 2
2018-03-21:06:12:17.740517|lacpd|1307|LOG_INFO|AMM|-|LACP system ID set to
70:72:cf:d4:34:42
2018-03-21:06:12:17.743491|vrfmgrd|5401|LOG_INFO|AMM|-|Created a vrf entity
42cc3df7-1113-412f-b5cb-e8227b8c22f2
2018-03-21:06:12:17.904008|vrfmgrd|5401|LOG_INFO|AMM|-|Created a vrf entity
4409133e-2071-4ab8-adfe-f9662c06b889
```

show hostname

Syntax

```
show hostname
```

Description

Shows the current host name.

Command context

Manager (#)

Authority

Administrators or local user group members with execution rights for this command.

Example

Setting and showing the host name:

```
switch# show hostname
switch
switch# config
switch(config)# hostname myswitch
myswitch(config)# show hostname
myswitch
```

show images

Syntax

```
show images
```

Description

Shows information about the software in the primary and secondary images.

Command context

Operator (>) or Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

Showing the primary and secondary images on a 6100 switch:

```
switch(config)# show images
-----
ArubaOS-CX Primary Image
-----
Version : PL.10.06.0002E
Size    : 243 MB
Date    : 2020-11-25 22:00:47 PST
SHA-256 : 61fe9233b2c842e8ac1731ad46949bd63e269c5c72d69290932ef19c1ebb0730

-----
ArubaOS-CX Secondary Image
-----
Version : PL.10.07.0000E-201-gba0c336
Size    : 271 MB
Date    : 2020-11-25 21:09:08 UTC
SHA-256 : 2fdfc646a8013efcca75729584bbb0fa54604086bad3f46e1c5d4e706b8b30ee

Default Image : primary
Boot Profile Timeout : 2 seconds

-----
Management Module 1/1 (Active)
-----
Active Image      : primary
Service OS Version : PL.01.07.0003-internal
BIOS Version      : PL.01.0001
```

show module

Syntax

```
show module
```

Description

Shows information about installed line modules and management modules.



Although this switch does not have removable modules, this command will still return information about the switch, referring to management modules and line modules.

Command context

Operator (>) or Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Usage

Identifies and shows status information about the line modules and management modules that are installed in the switch.

To show the configuration information—if any—associated with that line module slot, use the `show running-configuration` command.

Status is one of the following values:

Active

This is the active management module.

Deinitializing

The switch is being deinitialized.

Diagnostic

The switch is in a state used for troubleshooting.

Down

The switch is physically present but is powered down.

Failed

The switch has experienced an error and failed.

Initializing

The switch is being initialized.

Present

The switch hardware is installed in the chassis.

Ready

The switch is available for use.

Updating

A firmware update is being applied to the switch.

Examples

Showing all installed modules:

```
switch# show module
```

```
Management Modules
```

```
=====
```

	Product		Serial	
Name	Number	Description	Number	Status
1/1	JL678A	6100F 24G 4SFP+ Swch	SG0ZKW600P	Active (local)

```
Line Modules
```

```
=====
```

	Product		Serial	
Name	Number	Description	Number	Status
1/1	JL678A	6100F 24G 4SFP+ Swch	SG0ZKW600P	Ready

show running-config

Syntax

```
show running-config [<FEATURE>] [all]
```

Description

Shows the current nondefault configuration running on the switch. No user information is displayed.

Command context

Manager (#)

Parameters

<FEATURE>

Specifies the name of a feature. For a list of feature names, enter the `show running-config` command, followed by a space, followed by a question mark (?).

all

Shows all default values for the current running configuration.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Showing the current running configuration:

```
switch> show running-config
Current configuration:
!
!Version ArubaOS-CX PL.10.06.0001-346-g56a12a8f4cf15
!export-password: default
ntp enable
!
!
!
!
ssh server vrf default
vlan 1
spanning-tree
spanning-tree instance 1 vlan 1,2,4-10
interface 1/1/1
    no shutdown
    vlan access 1
    portfilter 1/1/2-1/1/3

interface 1/1/2
    no shutdown
    vlan access 1
interface 1/1/3
    no shutdown
    vlan access 1
interface 1/1/4
    no shutdown
    vlan access 1
interface 1/1/5
    no shutdown
    vlan access 1
interface 1/1/6
    no shutdown
    vlan access 1
interface 1/1/7
    no shutdown
    vlan access 1
interface 1/1/8
    no shutdown
    vlan access 1
interface 1/1/9
    no shutdown
```

```

    vlan access 1
interface 1/1/10
    no shutdown
    vlan access 1
interface 1/1/11
    no shutdown
    vlan access 1
interface 1/1/12
    no shutdown
    vlan access 1
interface 1/1/13
    no shutdown
    vlan access 1
interface 1/1/14
    no shutdown
    vlan access 1
interface 1/1/15
    no shutdown
    vlan access 1
interface 1/1/16
    no shutdown
    vlan access 1
interface vlan 1
    ip dhcp
snmp-server vrf default
snmp-server community public
snmp-server host 1.1.1.1 inform version v2c
snmp-server host 1.1.1.1 trap version v2c
snmpv3 context A vrf default
!
!
!
!
!
https-server vrf default

```

Showing the current running configuration in json format:

```

switch> show running-config json
Running-configuration in JSON:
{
  "Monitoring_Policy_Script": {
    "system_resource_monitor_mm1.1.0": {
      "Monitoring_Policy_Instance": {
        "system_resource_monitor_mm1.1.0/system_resource_monitor_
mm1.1.0.default": {
          "name": "system_resource_monitor_mm1.1.0.default",
          "origin": "system",
          "parameters_values": {
            "long_term_high_threshold": "70",
            "long_term_normal_threshold": "60",
            "long_term_time_period": "480",
            "medium_term_high_threshold": "80",
            "medium_term_normal_threshold": "60",
            "medium_term_time_period": "120",
            "short_term_high_threshold": "90",
            "short_term_normal_threshold": "80",
            "short_term_time_period": "5"
          }
        }
      }
    }
  },

```

...
...
...
...

Show the current running configuration without default values:

```
switch(config)# show running-config
Current configuration:
!
!Version ArubaOS-CX PL.10.06.0001-346-g56a12a8f4cf15
!export-password: default
!
!
!
ssh server vrf default
vlan 1
spanning-tree
interface 1/1/1
    no shutdown
    vlan access 1
interface 1/1/2
    no shutdown
    vlan access 1
interface 1/1/3
    no shutdown
    vlan access 1
interface 1/1/4
    no shutdown
    vlan access 1
interface 1/1/5
    no shutdown
    vlan access 1
interface 1/1/6
    no shutdown
    vlan access 1
interface 1/1/7
    no shutdown
    vlan access 1
interface 1/1/8
    no shutdown
    vlan access 1
interface 1/1/9
    no shutdown
    vlan access 1
interface 1/1/10
    no shutdown
    vlan access 1
interface 1/1/11
    no shutdown
    vlan access 1
interface 1/1/12
    no shutdown
    vlan access 1
interface 1/1/13
    no shutdown
    vlan access 1
interface 1/1/14
    no shutdown
    vlan access 1
```



```

interface 1/1/15
  no shutdown
  vlan access 1
interface 1/1/16
  no shutdown
  vlan access 1
interface vlan 1
  ip dhcp
!
!
!
!
!
https-server vrf default
switch#
switch#

```

show running-config current-context

Syntax

`show running-config current-context`

Description

Shows the current non-default configuration running on the switch in the current command context.

Command context

`config` or a child of `config`. See Usage.

Authority

Administrators or local user group members with execution rights for this command.

Usage

You can enter this command from the following configuration contexts:

- Any child of the global configuration (`config`) context. If the child context has instances—such as interfaces—you can enter the command in the context of a specific instance. Support for this command is provided for one level below the `config` context. For example, entering this command for a child of a child of the `config` context not supported. If you enter the command on a child of the `config` context, the current configuration of that context and the children of that context are displayed.
- The global configuration (`config`) context. If you enter this command in the global configuration (`config`) context, it shows the running configuration of the entire switch. Use the `show running-configuration` command instead.

Examples

Showing the running configuration for the current interface:

```

switch(config-if)# show running-config current-context
interface 1/1/1
  no shutdown
  description Example interface
vlan access 1

```

```
exit
```

Showing the current running configuration for the in-band management interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)#description IN-BAND Management Interface
switch(config-if-vlan)#ip dhcp
switch(config-if-vlan)#no shutdown
switch(config-if-vlan)#end
switch#
```

Showing the current running configuration for the in-band management interface without DHCP:

```
switch(config)# interface vlan 1
switch(config-if-vlan)#description IN-BAND Management Interface
switch(config-if-vlan)#no ip dhcp
switch(config-if-vlan)#ip address 192.168.10.1/24
switch(config-if-vlan)#no shutdown
switch(config-if-vlan)#end
switch#
```

Showing the running configuration for the external storage share named `nasfiles`:

```
switch(config-external-storage-nasfiles)# show running-config current-context
external-storage nasfiles
  address 192.168.0.1
  vrf default
  username nasuser
  password ciphertext AQBapalKj+XMsZumHEwIc9OR6YcOw5Z6Bh9rV+9ZtKDKzvbaBAAAAB1CTrM=
  type scp
  directory /home/nas
  enable
switch(config-external-storage-nasfiles)#
```

show startup-config

Syntax

```
show startup-config [json]
```

Description

Shows the contents of the startup configuration.



Switches in the `factory-default` configuration do not have a startup configuration to display.

Command context

Manager (#)

Parameters

json

Display output in JSON format.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Showing the startup-configuration in JSON format:

```
switch# show startup-config json
Startup configuration:
{
  "AAA_Server_Group": {
    "local": {
      "group_name": "local"
    },
    "none": {
      "group_name": "none"
    }
  },
  ...
}
```

show system

Syntax

show system

Description

Shows general status information about the system.

Command context

Operator (>) or Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Usage

CPU utilization represents the average utilization across all the CPU cores.

System Contact, System Location, and System Description can be set with the `snmp-server` command.

Examples

Showing system information:

```
switch(config)# show system
Hostname           : switch
System Description : PL.10.xx.xxxxx
System Contact     :
System Location    :

Vendor             : Aruba
Product Name       : JL678A 6100 24G 4SFP+ Swch
Chassis Serial Nbr : CN9ZKRD058
Base MAC Address   : f860f0-c91160
```

```
ArubaOS-CX Version : PL.10.xx.xxxxxx

Time Zone           : UTC

Up Time             : 9 hours, 10 minutes
CPU Util (%)        : 3
Memory Usage (%)    : 15
```

show system resource-utilization

Syntax

```
show system resource-utilization [daemon <DAEMON-NAME>]
```

Description

Shows information about the usage of system resources such as CPU, memory, and open file descriptors.

Command context

Manager (#)

Parameters

daemon <DAEMON-NAME>

Shows the filtered resource utilization data for the process specified by <DAEMON-NAME> only.



For a list of daemons that log events, enter `show events -d ?` from a switch prompt in the manager (#) context.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Showing all system resource utilization data:

```
switch# show system resource-utilization
System Resources:
Processes: 147
CPU usage(%): 12
Memory usage(%): 13
Open FD's: 4128
mmc-type-a: Endurance utilization = 0-10%, Health = normal
mmc-type-b: Endurance utilization = 0-10%, Health = normal
switch#
```

Showing the resource utilization data for the pmd process:

```
switch# show system resource-utilization daemon pmd
Process           CPU Usage      Memory Usage    Open FD's
-----
pmd                2              1              14
```

Showing resource utilization data when system resource utilization polling is disabled:

```
switch# show system resource-utilization
System resource utilization data poll is currently disabled
```

Showing resource utilization data for a line module:

```
switch# show system resource-utilization module 1/1
System Resource utilization for line card module: 1/1
CPU usage(%): 0
Memory usage(%): 35
Open FD's: 512
```

show tech

Syntax

```
show tech [basic | <FEATURE>] [local-file]
```

Description

Shows detailed information about switch features by automatically running the `show` commands associated with the feature. If no parameters are specified, the `show tech` command shows information about all switch features. Technical support personnel use the output from this command for troubleshooting.

Command context

Manager (#)

Parameters

`basic`

Specifies showing a basic set of information.

`<FEATURE>`

Specifies the name of a feature. For a list of feature names, enter the `show tech` command, followed by a space, followed by a question mark (?).

`local-file`

Shows the output of the `show tech` command to a local text file.

Authority

Administrators or local user group members with execution rights for this command.

Usage

To terminate the output of the `show tech` command, enter **Ctrl+C**.

If the command was not terminated with **Ctrl+C**, at the end of the output, the `show tech` command shows the following:

- The time consumed to execute the command.
- The list of failed `show` commands, if any.

To get a copy of the local text file content created with the `show tech` command that is used with the `local-file` parameter, use the `copy show-tech local-file` command.

Example

Showing the basic set of system information:

```

switch# show tech basic
=====
Show Tech executed on Wed Sep  6 16:50:37 2017
=====
[Begin] Feature basic
=====

*****
Command : show core-dump all
*****
no core dumps are present

...

[End] Feature basic
=====

1 show tech command failed
=====
Failed command:
  1. show boot-history
=====
Show tech took 3.000000 seconds for execution

```

Directing the output of the **show tech basic** command to the local text file:

```

switch# show tech basic local-file
Show Tech output stored in local-file. Please use 'copy show-tech local-file'
to copy-out this file.

```

show usb

Syntax

show usb

Description

Shows the USB port configuration and mount settings.

Command context

Operator (>) or Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

If USB has not been enabled:

```
switch> show usb  
Enabled: No  
Mounted: No
```

If USB has been enabled, but no device has been mounted:

```
switch> show usb  
Enabled: Yes  
Mounted: No
```

If USB has been enabled and a device mounted:

```
switch> show usb  
Enabled: Yes  
Mounted: Yes
```

show usb file-system

Syntax

```
show usb file-system [<PATH>]
```

Description

Shows directory listings for a mounted USB device. When entered without the <PATH> parameter the top level directory tree is shown.

Command context

Operator (>) or Manager (#)

Parameters

<PATH>

Specifies the file path to show. A leading "/" in the path is optional.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Usage

Adding a leading "/" as the first character of the <PATH> parameter is optional.

Attempting to enter '.' as any part of the <PATH> will generate an invalid path argument error. Only fully-qualified path names are supported.

Examples

Showing the top level directory tree:

```
switch# show usb file-system  
/mnt/usb:  
'System Volume Information'  dir1'
```

```
/mnt/usb/System Volume Information':  
IndexerVolumeGuid  WPSettings.dat  
  
/mnt/usb/dir1:  
dir2  test1  
  
/mnt/usb/dir1/dir2:  
test2
```

Showing available path options from the top level:

```
switch# show usb file-system /  
total 64  
drwxrwxrwx 2 32768 Jan 22 16:27 'System Volume Information'  
drwxrwxrwx 3 32768 Mar  5 15:26 dir1
```

Showing the contents of a specific folder:

```
switch# show usb file-system /dir1  
total 32  
drwxrwxrwx 2 32768 Mar  5 15:26 dir2  
-rwxrwxrwx 1      0 Feb  5 18:08 test1  
  
switch# show usb file-system dir1/dir2  
total 0  
-rwxrwxrwx 1 0 Feb  6 05:35 test2
```

Attempting to enter an invalid character in the path:

```
switch# show usb file-system dir1/../../../../  
Invalid path argument
```

show version

Syntax

```
show version
```

Description

Shows version information about the network operating system software, service operating system software, and BIOS.

Command context

Operator (>) or Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

Showing version information:

```
switch(config)# show version
-----
ArubaOS-CX
(c) Copyright 2017-2020 Hewlett Packard Enterprise Development LP
-----
Version       : PL.10.xx.xxxxx
Build Date    : 2020-12-06 22:13:46 UTC
Build ID      : ArubaOS-CX:PL.10.xx.xxxxx:9e8bf51170a6:202012062115
Build SHA     : 9e8bf51170a602370f12e0bde814e8d8f49bf706
Active Image  : secondary

Service OS Version : PL.10.xx.xxxxx-internal
BIOS Version      : PL.01.0002
```

system resource-utilization poll-interval

Syntax

```
system resource-utilization poll-interval <SECONDS>
```

Description

Configures the polling interval for system resource information collection and recording such as CPU and memory usage.

Command context

config

Parameters

<SECONDS>

Specifies the poll interval in seconds. Range: 10-3600. Default: 10.

Authority

Administrators or local user group members with execution rights for this command.

Example

Configuring the system resource utilization poll interval:

```
switch(config)# system resource-utilization poll-interval 20
```

top cpu

Syntax

top cpu

Description

Shows CPU utilization information.

Command context

Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Example

Showing top CPU information:

```
switch# top cpu
top - 09:42:55 up 3 min, 3 users, load average: 3.44, 3.78, 1.70
Tasks: 76 total, 2 running, 74 sleeping, 0 stopped, 0 zombie
%Cpu(s): 31.4 us, 32.7 sy, 0.5 ni, 34.4 id, 04. wa, 0.0 hi, 0.6 si, 0.0 st
KiB Mem : 4046496 total, 2487508 free, 897040 used, 661948 buff/cache
KiB Swap: 0 total, 0 free, 0 used, 2859196 avail Mem

  PID USER      PRI  NI   VIRT   RES   SHR S  %CPU  %MEM     TIME+ COMMAND
...
```

top memory

Syntax

top memory

Description

Shows memory utilization information.

Command context

Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Example

Showing top memory:

```
switch> top memory
top - 09:42:55 up 3 min, 3 users, load average: 3.44, 3.78, 1.70
Tasks: 76 total, 2 running, 74 sleeping, 0 stopped, 0 zombie
%Cpu(s): 31.4 us, 32.7 sy, 0.5 ni, 34.4 id, 04. wa, 0.0 hi, 0.6 si, 0.0 st
KiB Mem : 4046496 total, 2487508 free, 897040 used, 661948 buff/cache
KiB Swap: 0 total, 0 free, 0 used, 2859196 avail Mem

  PID USER      PRI  NI   VIRT   RES   SHR S  %CPU  %MEM     TIME+ COMMAND
...
```

usb

Syntax

usb
no usb

Description

Enables the USB ports on the switch. This setting is persistent across switch reboots and management module failovers. Both active and standby management modules are affected by this setting.

The `no` form of this command disables the USB ports.

Command context

`config`

Authority

Administrators or local user group members with execution rights for this command.

Example

Enabling USB ports:

```
switch(config)# usb
```

Disabling USB ports when a USB drive is mounted:

```
switch(config)# no usb
```

usb mount | unmount

Syntax

`usb {mount | unmount}`

Description

Enables or disables the inserted USB drive.

Command context

Manager (#)

Parameters

`mount`

Enables the inserted USB drive.

`unmount`

Disables the inserted USB drive in preparation for removal.

Authority

Administrators or local user group members with execution rights for this command.

Usage

If USB has been enabled in the configuration, the USB port on the active management module is available for mounting a USB drive. The USB port on the standby management module is not available.

An inserted USB drive must be mounted each time the switch boots or fails over to a different management module.

A USB drive must be unmounted before removal.

The supported USB file systems are FAT16 and FAT32.

Examples

Mounting a USB drive in the USB port:

```
switch# usb mount
```

Unmounting a USB drive:

```
switch# usb unmount
```

terminal-monitor {notify | severity | filter}

Syntax

```
terminal-monitor {notify <event|debug|all> | severity <level> | filter keyword}

no terminal-monitor
```

Description

Enables the terminal monitor feature in the SSH session. It display all debug log or event log or both debug and event log messages. Terminal monitoring can be filtered with the severity options or with the help of keywords. Enabling terminal monitor without options displays both debug and event log with a severity error.

The `no` form of this command disables the terminal monitor configuration.

Parameters

`notify <event|debug|all>`

Specifies the type of log notification.

- **Event:** Displays the event log messages.
- **Debug:** Displays the debug log messages.
- **All:** Displays both event and debug log messages.

`severity <level>`

Specifies the severity level for the logs. The different severity levels are emergency, critical, error (default), warning, notice, information, alert, and debug (shows all severities).

`filter <keyword>`

Specifies the filter by applying keyword for the logs.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring terminal monitor:

```
switch# terminal-monitor
Terminal-monitor is enabled successfully

switch# terminal-monitor notify all
Terminal-monitor is enabled successfully

switch# terminal-monitor notify event severity info
Terminal-monitor is enabled successfully
```

```
switch# terminal-monitor filter lldp  
Terminal-monitor is enabled successfully
```

show terminal-monitor

Syntax

```
show terminal-monitor
```

Description

Shows whether the terminal monitoring is enabled or disabled.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Displaying terminal monitor when enabled:

```
switch# show terminal-monitor  
  
Terminal-monitor is enabled  
-----  
Notify      | Severity   | Filter  
-----  
event       | debug      | lldp  
-----
```

Displaying terminal monitor when disabled:

```
switch# show terminal-monitor  
  
Terminal-monitor is disabled
```

traceroute

Syntax

```
traceroute {IPv4-address | hostname} [ip-option loosesourceroute <IPV4-ADDR>] [dstport <NUMBER> | maxttl <NUMBER> | minttl <NUMBER> | probes <NUMBER> | timeout <TIME>] [vrf {<VRF-NAME> | mgmt}]
```

Description

Uses traceroute for the specified IPv4 address or hostname with or without optional parameters.

Command context

Operator (>) or Manager (#)

Parameters

IPv4-address

Specifies the IPv4 address of the device to use traceroute.

hostname

Specifies the hostname of the device to use traceroute.

ip-option

Specifies the IP option.

loosesourceroute <IPV4-ADDR>

Specifies the route for loose source record route. Enter one or more intermediate router IP addresses separated by ',' for loose source routing.

dstport <NUMBER>

Specifies the destination port, <1-34000>. Default: 33434

maxttl <NUMBER>

Specifies the maximum number of hops to reach the destination, <1-255>. Default: 30

minttl <NUMBER>

Specifies the Minimum number of hops to reach the destination, <1-255>. Default: 1

probes <NUMBER>

Specifies the number of probes, <1-5>. Default: 3

timeout <TIME>

Specifies the traceroute timeout in seconds, <1-60>. Default: 3 seconds

vrf <VRF-NAME>

Specifies the virtual routing and forwarding (VRF) to use, <VRF-NAME>.

mgmt

Specifies use of the management interface.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Usage

Traceroute is a computer network diagnostic tool for displaying the route (path), and measuring transit delays of packets across an Internet Protocol (IP) network. It sends a sequence of User Datagram Protocol (UDP) packets addressed to a destination host. The time-to-live (TTL) value, also known as hop limit, is used in determining the intermediate routers being traversed towards the destination.

Examples

```
switch# traceroute 10.0.10.1
traceroute to 10.0.10.1 (10.0.10.1) , 1 hops min, 30 hops max, 3 sec. timeout, 3
probes
 1  10.0.40.2  0.002ms  0.002ms  0.001ms
 2  10.0.30.1  0.002ms  0.001ms  0.001ms
 3  10.0.10.1  0.001ms  0.002ms  0.002ms

switch# traceroute localhost
traceroute to localhost (127.0.0.1), 1 hops min, 30 hops max, 3 sec. timeout, 3
probes
 1  127.0.0.1  0.018ms  0.006ms  0.003ms

switch# traceroute 10.0.10.1 maxttl 20
traceroute to 10.0.10.1 (10.0.10.1) , 1 hops min, 20 hops max, 3 sec. timeout, 3
probes
 1  10.0.40.2  0.002ms  0.002ms  0.001ms
 2  10.0.30.1  0.002ms  0.001ms  0.001ms
 3  10.0.10.1  0.001ms  0.002ms  0.002ms

switch# traceroute 10.0.10.1 minttl 1
traceroute to 10.0.10.1 (10.0.10.1) , 1 hops min, 30 hops max, 3 sec. timeout, 3
probes
 1  10.0.40.2  0.002ms  0.002ms  0.001ms
 2  10.0.30.1  0.002ms  0.001ms  0.001ms
 3  10.0.10.1  0.001ms  0.002ms  0.002ms

switch# traceroute 10.0.10.1 dstport 33434
traceroute to 10.0.10.1 (10.0.10.1) , 1 hops min, 30 hops max, 3 sec. timeout, 3
probes
 1  10.0.40.2  0.002ms  0.002ms  0.001ms
 2  10.0.30.1  0.002ms  0.001ms  0.001ms
 3  10.0.10.1  0.001ms  0.002ms  0.002ms

switch# traceroute 10.0.10.1 probes 2
traceroute to 10.0.10.1 (10.0.10.1) , 1 hops min, 30 hops max, 3 sec. timeout, 2
probes
 1  10.0.40.2  0.002ms  0.002ms
 2  10.0.30.1  0.002ms  0.001ms
 3  10.0.10.1  0.001ms  0.002ms

switch# traceroute 10.0.10.1 timeout 5
traceroute to 10.0.10.1 (10.0.10.1) , 1 hops min, 30 hops max, 5 sec. timeout, 3
probes
 1  10.0.40.2  0.002ms  0.002ms  0.001ms
 2  10.0.30.1  0.002ms  0.001ms  0.001ms
 3  10.0.10.1  0.001ms  0.002ms  0.002ms

switch# traceroute localhost vrf red
traceroute to localhost (127.0.0.1), 1 hops min, 30 hops max, 3 sec. timeout, 3
probes
 1  127.0.0.1  0.003ms  0.002ms  0.001ms

switch# traceroute localhost mgmt
traceroute to localhost (127.0.0.1), 1 hops min, 30 hops max, 3 sec. timeout, 3
probes
```



```

1    127.0.0.1  0.018ms  0.006ms  0.003ms

switch# traceroute 10.0.10.1 maxttl 20 timeout 5 minttl 1 probes 3 dstport 33434
traceroute to 10.0.10.1 (10.0.10.1) , 1 hops min, 20 hops max, 5 sec. timeout, 3
probes
 1    10.0.40.2  0.002ms  0.002ms  0.001ms
 2    10.0.30.1  0.002ms  0.001ms  0.001ms
 3    10.0.10.1  0.001ms  0.002ms  0.002ms

switch# traceroute 10.0.10.1 ip-option loosesourceroute 10.0.40.2
traceroute to 10.0.10.1 (10.0.10.1) , 1 hops min, 30 hops max, 3 sec. timeout, 3
probes
 1    10.0.40.2  0.002ms  0.002ms  0.001ms
 2    10.0.30.1  0.002ms  0.001ms  0.001ms
 3    10.0.10.1  0.001ms  0.002ms  0.002ms

switch# traceroute 10.0.10.1 ip-option loosesourceroute 10.0.40.2 maxttl 20 timeout
5 minttl 1 probes 3 dstport 33434
traceroute to 10.0.10.1 (10.0.10.1) , 1 hops min, 20 hops max, 5 sec. timeout, 3
probes
 1    10.0.40.2  0.002ms  0.002ms  0.001ms
 2    10.0.30.1  0.002ms  0.001ms  0.001ms
 3    10.0.10.1  0.001ms  0.002ms  0.002ms

```

traceroute6

Syntax

```
traceroute6 {IPv6-address | hostname} [dstport <NUMBER> | maxttl <NUMBER> |
  probes <NUMBER> | timeout <TIME>] [vrf {<VRF-NAME> | mgmt}]
```

Description

Uses traceroute for the specified IPv6 address or hostname with or without optional parameters.

Command context

Operator (>) or Manager (#)

Parameters

IPv6-address

Specifies the IPv6 address of the device to use traceroute.

hostname

Specifies the hostname of the device to use traceroute.

dstport <NUMBER>

Specifies the destination port, <1-34000>. Default: 33434

maxttl <NUMBER>

Specifies the maximum number of hops to reach the destination, <1-255>. Default: 30

probes <NUMBER>

Specifies the number of probes, <1-5>. Default: 3

timeout <TIME>

Specifies the traceroute timeout in seconds, <1-60>. Default: 3 seconds

vrf <VRF-NAME>

Specifies the virtual routing and forwarding (VRF) to use, <VRF-NAME>.

mgmt

Specifies use of the management interface.

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Usage

Traceroute is a computer network diagnostic tool for displaying the route (path), and measuring transit delays of packets across an Internet Protocol (IP) network. It sends a sequence of User Datagram Protocol (UDP) packets addressed to a destination host. The time-to-live (TTL) value, also known as hop limit, is used in determining the intermediate routers being traversed towards the destination.

Examples

```
switch# traceroute6 0:0::0:1
traceroute to 0:0::0:1 (::1) from ::1, 30 hops max, 3 sec. timeout, 3 probes, 24
byte packets
 1  localhost (::1)  0.117 ms  0.032 ms  0.021 ms

switch# traceroute6 localhost
traceroute to localhost (::1) from ::1, 30 hops max, 3 sec. timeout, 3 probes, 24
byte packets
 1  localhost (::1)  0.089 ms  0.03 ms  0.014 ms

switch# traceroute6 0:0::0:1 maxttl 30
traceroute to 0:0::0:1 (::1) from ::1, 30 hops max, 3 sec. timeout, 3 probes, 24
byte packets
 1  localhost (::1)  0.117 ms  0.032 ms  0.021 ms

switch# traceroute6 0:0::0:1 dstport 33434
traceroute to 0:0::0:1 (::1) from ::1, 30 hops max, 3 sec. timeout, 3 probes, 24
byte packets
 1  localhost (::1)  0.117 ms  0.032 ms  0.021 ms

switch# traceroute6 0:0::0:1 probes 2
traceroute to 0:0::0:1 (::1) from ::1, 30 hops max, 3 sec. timeout, 2 probes, 24
byte packets
 1  localhost (::1)  0.117 ms  0.032 ms

switch# traceroute6 0:0::0:1 timeout 3
traceroute to 0:0::0:1 (::1) from ::1, 30 hops max, 3 sec. timeout, 3 probes, 24
byte packets
 1  localhost (::1)  0.117 ms  0.032 ms  0.021 ms

switch# traceroute6 localhost vrf red
traceroute to localhost (::1) from ::1, 30 hops max, 3 sec. timeout, 3 probes, 24
byte packets
 1  localhost (::1)  0.077 ms  0.051 ms  0.054 ms

switch# traceroute6 localhost mgmt
traceroute to localhost (::1) from ::1, 30 hops max, 3 sec. timeout, 3 probes, 24
byte packets
 1  localhost (::1)  0.089 ms  0.03 ms  0.014 ms

switch# traceroute6 0:0::0:1 maxttl 30 timeout 3 probes 3 dstport 33434
traceroute to 0:0::0:1 (::1) from ::1, 30 hops max, 3 sec. timeout, 3 probes, 24
byte packets
 1  localhost (::1)  0.117 ms  0.032 ms  0.021 ms
```

clear udld statistics

Syntax

```
clear udld statistics [interface <INTERFACE-NAME>]
```

Description

Clears UDLD statistics for all interfaces or a specific interface.

Command context

Manager (#)

Authority

Administrators or local user group members with execution rights for this command.

Examples

Clearing all UDLD statistics on all interfaces:

```
switch# clear udld statistics
```

Clearing all UDLD statistics on interface 1/1/1:

```
switch# clear udld statistics interface 1/1/1
```

show udld

Syntax

```
show udld [interface <INTERFACE-NAME>]
```

Description

Displays UDLD information for all interfaces or for a specific interface.

Command context

Operator (>) or Manager (#)

Parameters

```
interface <INTERFACE-NAME>
```

Specifies the name of a logical interface on the switch, which can be:

- An Ethernet interface associated with a physical port. Use the format `member/slot/port` (for example,

1/3/1).

- UDLD runs only on physical interfaces. LAGs, tunnels, and the like are not supported. However, UDLD can be configured individually on each port of a LAG or trunk group. Configuring UDLD on a trunk group primary port enables UDLD on that port only.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Displaying all UDLD information:

```
switch# show udld

Abbreviations:
VTF - Verify-then-forward    FTV - Forward-then-verify
NOR - RFC 5171 normal        AGG - RFC 5171 aggressive

-----
Interface  UDLD      UDLD      UDLD      UDLD      Mode  Interval
           Config    State     Substate   Link
-----
1/1/1      Disabled  Inactive  Undetermined  Unblock  FTV   8000
1/1/2      Enabled   Active   Bidirectional  Unblock  FTV   7000
1/1/3      Enabled   Active   Blocked       Block    FTV   7000
1/1/4      Enabled   Inactive  Uninitialized  Unblock  NOR   7000
1/1/5      Enabled   Active   ErrDisabled   Block    AGG   7000
1/1/6      Disabled  Active   Detection     Unblock  NOR   7000
-----

Retries  Tx      Rx      Rx      Rx      Transitions
         Pkts  Pkts    Pkts  Pkts disc.  Pkts drop.
-----
4         4      54      123      123      1
7        1234567  1548421  23214    1878981    3
4         3      77871   2157      81878     1
5        50       0        0         0         0
3       150      25        0         2         1
3         6      54      123      23         1
-----
```

Displaying information for interface **1/1/1**:

```
switch# show udld interface 1/1/1

Interface 1/1/1
Config: Enabled
State: Active
Substate: Bidirectional
Link: Unblock
Version: Aruba OS
Mode: Forward then verify
Interval: 7000 milliseconds
Retries: 7
Tx: 1234567 packets
Rx: 1548421 packets, 23214 discarded packets, 1878981 dropped packets
Port transitions: 3
```

Displaying the UDLD enable interfaces information:

```
switch# show udld enabled
```

Abbreviations:

VTF - Verify-then-forward

NOR - RFC 5171 normal

FTV - Forward-then-verify

AGG - RFC 5171 aggressive

```
-----
Interface  UDLD      UDLD      UDLD      UDLD      Mode  Interval  Retries  Tx
   Rx      Rx      Rx      Transitions
   Pkts    Config  State    Substate    Link
          Pkts disc. Pkts drop.
-----
2          Enabled  Active   Bidirectional  Unblock  FTV    7000      7
1234567  1548421  23214   1878981      3
3          Enabled  Active   Blocked        Block    FTV    7000      4      3
   77871   2157     81878      1
4          Enabled  Inactive  Uninitialized  Unblock  NOR    7000      5      50
   0       0       0          0
5          Enabled  Active   ErrDisabled    Block    AGG    7000      3      150
   25      0       2          1
-----
```

udld

Syntax

```
udld [disable]
no udld
```

Description

Enables UDLD support on a physical interface. UDLD is disabled by default. UDLD is configured on a per-port basis and must be enabled at both ends of the link.

UDLD runs only on physical interfaces. LAGs, tunnels, and the like are not supported. However, UDLD can be configured individually on each port of a LAG or trunk group. Configuring UDLD on a trunk group's primary port enables UDLD on that port only.

The `no` form of this command disables UDLD support and resets all configuration values to their default settings.

Command context

```
config-if
```

Parameters

disable

Disables UDLD on the interface but retains all UDLD configuration settings.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling UDLD on interface **1/1/1**:

```
switch(config)# interface 1/1/1
switch(config-if)# udld
```

Disabling UDLD on interface 1/1/1:

```
switch(config)# interface 1/1/1
switch(config-if)# no udld
```

udld interval

Syntax

```
udld interval <TIME>
no udld interval
```

Description

Sets the packet transmission interval.

The `no` form of this command sets the packet transmission interval to the default value of 7000 ms.

The allowed values vary depending on the operation mode.

The default interval is 7000 ms (7 seconds) for both ArubaOS-Switch and RFC5171 operation modes.

Values must be specified as multiples of 10 ms (7000 ms is allowed but 7005 ms is not a valid setting).

Switch mode can be configured to support under 100ms unidirectional link detection times.



Sessions under 100ms total detection time are susceptible to increasing processing load on the system. It is advisable to experiment with values that provide adequate detection times and system/protocol stability. Aruba recommends additional testing prior to configuring these sessions on a production environment.

However, these settings are recommended for specific deployments only, such as using UDLD for Ethernet Ring Protection Switching (ERPS) link-failure detection (ERPS is not supported on the 6100). The minimum detection time appropriate for your environment depends on the specific device family and configuration on which the protocol and system load is running. Aruba recommends additional testing for these configurations. During testing, monitor for unexpected false positive detections (i.e., UDLD records a failure when there was not any) on the interfaces running UDLD. Such false positive failures are an indication that the interval configuration requires tuning and that the system load might not allow such configuration.



When configuring detection times under 100ms for LAG interfaces, consider adding the interface first to the LAG and then enabling UDLD in the interface, to avoid false positive link failure detections. Adding an interface to a LAG causes momentary control plane traffic interruption for up to 100ms, which UDLD detects as a link failure if the detection time is following the control traffic interruption interval.

Command context

config-if

Parameters

<TIME>

Specifies the packet transmission interval. Range: 200 ms to 90000 ms or 1000 ms to 90000 ms for the 6100 Switch Series (in increments of 10).

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the packet transmission interval to **1000** ms on interface **1/1/1**.

```
switch(config)# interface 1/1/1
switch(config-if)# uddl interval 1000
```

Setting the packet transmission interval on interface **1/1/1** to the default value.

```
switch(config)# interface 1/1/1
switch(config-if)# no uddl interval
```

Trying to set the packet interval to 1055 ms on interface 1 is rejected because the interval must be specified as a multiple of 10:

```
switch(config)# interface 1
switch(config-if)# uddl interval 1055
Invalid interval. The interval value must be between 20ms and 90000ms and should be
specified as a multiple of 10, for example: 20, 100, 3000 or 90000.
```

Trying to set the packet interval to less than 7000 ms on interface 1 is rejected if using the RFC5171 mode.

```
switch(config)# interface 1
switch(config-if)# uddl mode rfc5171 normal
switch(config-if)# uddl interval 1000
Invalid interval. The interval must be equal or greater than 7000ms.
```

uddl mode

Syntax

```
uddl mode aruba-os {verify-then-forward | forward-then-verify}
uddl mode rfc5171 <RFC5171-MODE>
no uddl mode
```

Description

Sets the operating mode.

The `no` form of this command sets the operating mode to the default value of `aruba-os` and `forward-then-verify`.

Command context

config-if

Parameters

```
aruba-os {verify-then-forward | forward-then-verify}
```

Selects the ArubaOS mode to use. Use this mode when interconnecting with HPE PVOS/Brocade/Foundry switches.

```
verify-then-forward
```

In this mode:

- Interfaces start as unblocked.
- Once an interface is determined to be bidirectional, it is blocked if the retry limit is reached without receiving any UDLD packets.
- Interfaces automatically unblock if a UDLD packet is received.
- On failover, the UDLD state does not change if the (interval * retries) time is around 6 seconds.

`forward-then-verify`

In this mode:

- Interfaces start as unblocked.
- Interfaces transition to the unblocked state when receiving UDLD packets.
- Once an interface is determined to be bidirectional, it is blocked if the retry limit is reached without receiving any UDLD packets.
- Interfaces automatically unblock if a UDLD packet is received.

`rfc5171 <RFC5171-MODE>`

Selects the RFC5171 mode to use. Use this mode when interconnecting with third-party switches.

`normal`

In this mode:

- Interfaces start as unblocked.
- Interfaces do not block when the retry limit is reached without receiving any UDLD packets (plus 8 extra packets sent to the peer). Instead, an event is generated.
- Interfaces automatically unblock if a UDLD packet is received.

`aggressive`

In this mode:

- Interfaces start as unblocked.
- Once an interface is determined to be bidirectional, an interface will block when the retry limit is reached without receiving any UDLD packets (plus 8 extra packets sent to the peer).
- Interfaces implement a limited/reduced errDisabled recovery mechanism. When the interface's state goes to errDisabled, a maximum of 3 attempts (5 minutes apart) are triggered to try and bring up the interface in case the remote endpoint is still sending UDLD packets. After these 3 retries, the interface will remain blocked even if UDLD packets are received. The only way to unblock the interface when this occurs is to disable (and optionally re-enable) UDLD on the interface. The retry limit is reset once the interface becomes unblocked.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the operating mode to **aruba-os** and **forward-then-verify** on interface **1/1/1**:

```
switch(config)# interface 1/1/1
switch(config-if)# udld mode aruba-os forward-then-verify
```

Setting the operating mode to **rfc5171** and **aggressive** on interface **1/1/1**:


```
switch(config)# interface 1/1/1  
switch(config-if)# udld mode rfc5171 aggressive
```

Setting the operating mode on interface **1/1/1** to the default value:

```
switch(config)# interface 1/1/1  
switch(config-if)# no udld mode
```

udld retries

Syntax

```
udld retries <COUNT>  
no retries
```

Description

Sets the UDLD retry count.

The `no` form of this command sets the retry count to the default value.

Command context

config-if

Parameters

<COUNT>

Specifies the UDLD retry count. Range: 3 to 10. Default: 4.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the UDLD retry count to **5** on interface **1/1/1**:

```
switch(config)# interface 1/1/1  
switch(config-if)# udld retries 5
```

Setting the UDLD retry count on interface **1/1/1** to the default value:

```
switch(config)# interface 1/1/1  
switch(config-if)# no udld retries
```

ip forward-protocol udp

Syntax

```
ip forward-protocol udp <IPV4-ADDR> {<PORT-NUM> | <PROTOCOL>}  
no ip forward-protocol udp
```

Description

Defines the UDP server to which the interface forwards ingress UDP broadcast packets received on a specific UDP port. A maximum of 8 UDP broadcast servers can be configured per interface.

The `no` form of this command removes traffic forwarding for the specified server and port/protocol.

Command context

config-if

Parameters

<IPV4-ADDR>

Specifies the UDP server IP address in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255.

<PORT-NUM>

Specifies the UDP port number for which traffic is forwarded.

<PROTOCOL>

Specifies the protocol name for which traffic is forwarded. Supported protocols and their port numbers are:

- dns (53): Domain Name Service
- ntp (123): Network Time Protocol
- netbios-ns (137): NetBIOS Name Service
- netbios-dgm (138): NetBIOS Datagram Service
- radius (1812): Remote Authentication Dial-In User Service
- radius-old (1645): Remote Authentication Dial-In User Service
- rip (520): Routing Information Protocol
- snmp (161): Simple Network Management Protocol
- snmp-trap (162): Simple Network Management Protocol
- tftp (69): Trivial File Transfer Protocol
- timep (37): Time Protocol

Authority

Administrators or local user group members with execution rights for this command.

Examples

On the 6400 Switch Series, interface identification differs.

Forwarding DNS traffic to server 192.168.1.10 on interface 1/1/1:

```
switch(config)# interface 1/1/1
switch(config-if)# ip udp-bcast-forward protocol udp 192.168.1.10 dns
```

Forwarding DNS traffic (port 53) to server 192.168.1.10 on interface 1/1/1:

```
switch(config)# interface 1/1/1
switch(config-if)# ip udp-bcast-forward protocol udp 192.168.1.10 53
```

ip udp-bcast-forward

Syntax

```
ip udp-bcast-forward
no ip udp-bcast-forward
```

Description

Enables UDP broadcast forwarding.

The **no** form of this command disables UDP broadcast forwarding.

Command context

config

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling UDP broadcast forwarding:

```
switch(config)# ip udp-bcast-forward
```

Disabling UDP broadcast forwarding:

```
switch(config)# no ip udp-bcast-forward
```

show ip forward-protocol udp

Syntax

```
show ip forward-protocol udp [<INTERFACE-NAME>] [vsx-peer]
```

Description

Shows the configured UDP forwarding settings for all interfaces or a specific interface.

Command context

Manager (#)

Parameters

<INTERFACE-NAME>

Specifies the name of an interface. Format: member/slot/port.

[vsx-peer]

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Showing the configured UDP forwarding settings for all interfaces:

```
switch# show ip forward-protocol udp
UDP Broadcast Forwarder : enabled
Interface: 1/1/1
IP Forward Address UDP Port
-----
2.2.2.2 1645
4.4.4.4 138
4.4.4.4 1812
1.1.1.1 53
8.1.1.1 123
8.1.1.1 137
Interface: 1/1/2
IP Forward Address UDP Port
-----
2.2.2.2 37
2.2.2.2 69
2.2.2.2 520
2.2.2.2 161
2.2.2.2 162
```

Showing the configured UDP forwarding settings for a specific interface:

```
switch# show ip forward-protocol udp interface 1/1/1
UDP Broadcast Forwarder : enabled

Interface: 1/1/1
IP Forward Address UDP Port
-----
2.2.2.2 1645
4.4.4.4 138
4.4.4.4 1812
1.1.1.1 53
8.1.1.1 123
8.1.1.1 137
```

user

Syntax

```
user <USERNAME> group {administrators | operators | auditors | <USER-GROUP>}  
    password [ciphertext <CIPHERTEXT-PASSWORD> | plaintext <PLAINTEXT-PASSWORD>]
```

```
no user <USERNAME>
```

Description

Creates a user and adds the user to one of the user groups. Users are given the privileges of their group. For the three built-in user groups (`administrators`, `operators`, `auditors`), the privileges are fixed. For user-defined local user groups, the privileges are defined by the CLI command authorization rules of the group.

When entered without either optional `ciphertext` or `plaintext` parameters, the cleartext password is prompted for twice, with the characters entered masked with "*" symbols.

The `no` form of this command removes a user account from the switch. The administrator cannot delete the user account from which they are logged in. The `admin` user cannot be deleted.

Command context

`config`

Parameters

`<USERNAME>`

Specifies the user name. Requirements:

- Must start with a lowercase letter.
- Can contain numbers and lowercase letters.
- Can include only these three special characters: hyphens (-), dots (.), and underscores (_).
- Can have a maximum of 32 characters.
- Cannot be empty.
- Cannot contain uppercase letters.
- Cannot be: `admin`, `root`, or `remote_user`.
- Cannot be Linux reserved names such as:

`daemon`, `bin`, `sys`, `sync`, `proxy`, `www-data`, `backup`, `list`, `irc`, `gnats`, `nobody`, `systemd-bus-proxy`, `sshd`, `messagebus`, `rpc`, `systemd-journal-gateway`, `systemd-journal-remote`, `systemd-journal-upload`, `systemd-timesync`, `systemd-coredump`, `systemd-resolve`, `rpcuser`, `vagrant`, `opds`, `rdanet`, `_lldpd`, `rdaadmin`, `rdaweb`, `docker_container`, `tss`.

`group`

Selects the local user group to which the new user will be assigned.

`administrators` | `operators` | `auditors`

Selects one of three built-in local user groups.

`<USER-GROUP>`

Specifies an existing user-defined local user group.

`ciphertext <CIPHERTEXT-PASSWORD>`

Specifies a ciphertext password. No password prompts are provided and the ciphertext password is validated before the configuration is applied for the user. The variable `<CIPHERTEXT-PASSWORD>` is Base64 and is typically copied from another switch using the `show running-config` command output and then pasted into this command.



The administrator cannot construct ciphertext passwords themselves. The ciphertext is only created by an AOS-CX switch. The ciphertext is created by setting a password for a user with the `user` command. The ciphertext is available for copying from the `show running-config` output and pasting into the configuration on any other AOS-CX switch. The target switch must have the same export password (default or otherwise) as the source switch.

`plaintext <PLAINTEXT-PASSWORD>`

Specifies the password without prompting. The password is visible as cleartext when entered but is encrypted thereafter. Command history does show the password as cleartext.

Authority

Administrators or local user group members with execution rights for this command.

Usage

- Up to 63 local users can be added, for a total of 64 users including the default user `admin`. A user can belong to only one group.
- The switch ships with the `admin` user account and three built-in local user groups: `administrators`, `operators`, and `auditors`. The `admin` account belongs to the `administrators` group. The Service OS also includes the administrator user `admin`. The two `admin` users are entirely distinct.
- When a local user account is removed, the user loses all active login/SSH sessions. Any calls on the existing REST session with that local user account fail with a permissions issue as soon as the user is deleted. Soon afterwards, the existing REST sessions with the deleted local user account become invalidated. If a user is viewing the GUI while their account is deleted, the user is redirected to the login page within 60 seconds. The home directory associated with the user is also removed from the switch.
- Cleartext passwords (whether entered with prompting or entered directly) must:
 - Contain only ASCII characters from hexadecimal 21 to hexadecimal 7E [`\x21-\x7E`] (decimal 33 to 126). Spaces are not allowed. When the password is entered directly without prompting, the `"` symbol (hexadecimal 3F [`\x3F`] (decimal 63)) is not permitted.
 - Contain at most 32 characters.
 - Contain at least the number of characters configured (optionally) for `minimum-password-length`.



Although empty passwords are supported, it is recommended that you use strong passwords for all production switches.



Only an administrator can change the password of a user assigned to the `operators` role.

Examples

Creating local user `jamie` in the `administrators` group with a prompted password:

```
switch(config)# user jamie group administrators password
Adding user jamie
Enter password:*****
Confirm password:*****
```

Creating user `chris` in the existing user-defined local user group `admuser2` with a cleartext password, using direct entry without prompting:

```
switch(config)# user chris group admuser2 password plaintext passWORDxJ|989
```

Creating user `alex` in the `operators` group with a ciphertext password (the ciphertext shown is a placeholder that must be replaced with actual ciphertext):

```
switch(config)# user alex group operators password ciphertext NDcDI2...8igJfA=
```

Removing user `jamie`:

```
switch(config)# no user jamie
User jamie's home directory and active sessions will be deleted.
Do you want to continue [y/n]?y
```

user-group

Syntax

```
user-group <GROUP-NAME>
no user-group <GROUP-NAME>
```

Description

If `<GROUP-NAME>` does not exist, this command creates a local user group and then enters its context. If `<GROUP-NAME>` exists, this command enters the context for the specified `<GROUP-NAME>`. Within the `<GROUP-NAME>` context, several subcommands are available for working with rules that specify what CLI commands are permitted or denied for all members of the local group.

In addition to the three built-in user groups `administrators`, `operators`, and `auditors`, up to 29 user-defined local user groups can be defined. All users can be members of only one of the up to 32 groups.

The no form of this command deletes the specified user group. All members of the deleted group lose all command authorization privilege.



Do not causally delete user-defined local user groups without understanding the implications. Although user-defined local user groups can be deleted with the respective members losing all privileges, the three built-in groups `administrators`, `operators`, and `auditors` are always available and their privileges are unchangeable.

Command context

config

Authority

Administrators or local user group members with execution rights for this command.

Subcommands

These subcommands are available within the user-defined local user group context (shown in the switch prompt as `config-usr-grp-<GROUP-NAME>`).

```
[<SEQ-NUM>] {permit | deny} cli command "<REGEX>"
no <SEQ-NUM>
```

Defines a CLI command privilege `permit` or `deny` rule. There is an implicit "`deny .*`" rule at the end of every user-defined group rule list. Members of a user-defined group without any `permit` rules have no CLI command privileges.

The `no` form of this subcommand deletes the specified (by sequence number) rule from the group.



Rule evaluation proceeds from lowest to highest sequence number until the first successful match, resulting in either CLI command permission or denial. Rule evaluation ceases upon first match. Therefore, rules for related CLI commands must be defined in most restrictive to least restrictive order.

<SEQ-NUM>

Specifies the CLI command rule sequence number. When omitted, a sequence number that is 10 greater the highest existing sequence number is auto-assigned. When no rules exist, the first auto-assigned sequence number is 10.

{permit | deny}

Sets the rule type as either `permit` or `deny`. Rule order is important. For example, these two related rules together authorize all `show` commands except for the `show aaa` commands.

```
switch(config-usr-grp-admuser2) #10 deny cli command "show aaa .*"
switch(config-usr-grp-admuser2) #20 permit cli command "show ."
```

To achieve the wanted effect in this example, the `deny` rule must precede the `permit` rule. These two rules together achieve the following:

- All `show aaa` commands match on rule 10, triggering command denial, and the immediate cessation of further rule evaluation. Matching on rule 20 is never attempted.
- All other `show` commands (excluding `show aaa` commands) match on rule 20 and are therefore permitted.

<REGEX>

Specifies the CLI command matching criteria of the rule. The criteria can be expressed as `.*` which matches all commands. Otherwise, the criteria is expressed as a POSIX-compliant regular expression (regex) string starting with an exact match command token (for example `show`) followed by a regex representing command arguments. The first word must be a string that contains only alphanumeric or hyphen characters.

For example, to allow all commands starting with the word `interface`, the regex must be `"interface ."` or just `"interface"`. Using `"interface."` (without the space) is not supported. For example, `"show ."` matches every `show` command. Consult the Extended regular expression information available at: https://pubs.opengroup.org/onlinepubs/9699919799/basedefs/V1_chap09.html#tag_09_04.

Sample matching criteria	Sample matched CLI command or specifier	Matches
<code>show .*</code>	<code>show accounting log</code>	All <code>show</code> commands
<code>bgp .*</code>	<code>bgp router-id 1.1.1.1</code>	All <code>bgp</code> commands

Sample matching criteria	Sample matched CLI command or specifier	Matches
interface .*	interface 1/1/1	All interface specifiers
vlan (3 4)	vlan 3	VLAN 3 or 4
vlan [1-9]	vlan 5	A single VLAN in the range 1 to 9
vlan ([1-9] 1[0-9])	vlan 19	A single VLAN in the range 1 to 19

```
[<SEQ-NUM>] comment <TEXT-STRING>
no <SEQ-NUM> comment
```

Adds a comment to an existing rule. The no form of this subcommand removes an existing comment.

```
switch(config-usr-grp-admuser2)# 10 comment Deny all show aaa commands.
switch(config-usr-grp-admuser2)# 20 comment Permit all other show commands.
switch(config-usr-grp-admuser2)#
switch(config-usr-grp-admuser2)# show running-config current-context
user-group admuser2
  10 comment Deny all show aaa commands.
  10 deny cli command "show aaa .*"
  20 comment Permit all other show commands.
  20 permit cli command "show .*"

```

```
include <GROUP-NAME> [no] include <GROUP-NAME>
```

Include all rules from the specified user-defined <GROUP-NAME>. Only one group can be included in the definition of another group. The content of the included group is effectively placed at the top of the rules list in the current group. If the specified <GROUP-NAME> does not exist, it is created.

The no form of this subcommand removes the specified included group from the current group. The specified included group must exist and must be included in the current group or else an error message is shown.

The name of the included group is shown at the top of the show user-group command for the group with the include.

In this example, group admuser1 is included in group admuser2. So the admuser1 rules are evaluated first and then the rules in the admuser2 group are only evaluated if no CLI command match occurs for the rules in group admuser1.

```
switch(config-usr-grp-admuser2)# include admuser1
switch(config-usr-grp-admuser2)# show user-group admuser2
User Group Summary
=====
Name           : admuser2
Type           : configuration
Included Group : admuser1
Number of Rules : 2
User Group Rules
=====
SEQUENCE NUM  ACTION  COMMAND  COMMENT
-----
10            deny    show aaa .*    Deny all show aaa commands.
20            permit  show .*      Permit all other show
commands.
```

```
resequence [<STARTING-SEQ-NUM> <INCREMENT>]
```

Resequences the CLI command authorization rules. When entered without the optional parameters the rules are resequenced with a <STARTING-SEQ-NUM> of 10 and an <INCREMENT> of 10.

<STARTING-SEQ-NUM>

Specifies the starting sequence number.

<INCREMENT>

Specifies the sequence number increment.

Resequencing the rules to start at 100 with an increment of 20:

```
switch(config-usr-grp-admuser2)# resequence 100 20
switch(config-usr-grp-admuser2)# show running-config current-context
user-group admuser2
  100 comment Deny all show aaa commands.
  100 deny cli command "show aaa .*"
  120 comment Permit all other show commands.
  120 permit cli command "show .*"

```

Resequencing the rules to the default of starting at 10 with an increment of 10:

```
switch(config-usr-grp-admuser2)# resequence
switch(config-usr-grp-admuser2)# show running-config current-context
user-group admuser2
  10 comment Deny all show aaa commands.
  10 deny cli command "show aaa .*"
  20 comment Permit all other show commands.
  20 permit cli command "show .*"

```

```
show running-config current-context
```

Shows all the commands used to configure the rules in the current group context.

```
switch(config-usr-grp-admuser2)# show running-config current-context
user-group admuser2
  10 comment Deny all show aaa commands.
  10 deny cli command "show aaa .*"
  20 comment Permit all other show commands.
  20 permit cli command "show .*"

```

```
list
```

List the subcommands available within the user-defined group context.

```
exit
```

Exits the user-defined group context.

```
end
```

Exits the user-defined group context and then the config context.

user password

Syntax

```
user <USERNAME> password [ciphertext <CIPHERTEXT-PASSWORD> | plaintext <PLAINTEXT-PASSWORD>]
```

Description

Changes a password for an account or enables the password for the admin account. When entered without either optional `ciphertext` or `plaintext` parameters, the cleartext password is prompted for twice, with the characters entered masked with "*" symbols.

Command context

`config`

Parameters

`<USERNAME>`

Specifies the corresponding user name for the password you want to change.

`ciphertext <CIPHERTEXT-PASSWORD>`

Specifies a ciphertext password. No password prompts are provided and the ciphertext password is validated before the configuration is applied for the user. The variable `<CIPHERTEXT-PASSWORD>` is Base64 and is typically copied from another switch using the `show running-config` command output and then pasted into this command.



The administrator cannot construct ciphertext passwords themselves. The ciphertext is only created by an AOS-CX switch. The ciphertext is created by setting a password for a user with the `user` command. The ciphertext is available for copying from the `show running-config` output and pasting into the configuration on any other AOS-CX switch. The target switch must have the same export password (default or otherwise) as the source switch.

`plaintext <PLAINTEXT-PASSWORD>`

Specifies the password without prompting. The password is visible as cleartext when entered but is encrypted thereafter. Command history does show the password as cleartext.

Authority

Administrators or local user group members with execution rights for this command.

Usage

The admin account is available on the switch without a password by default.

Cleartext passwords (whether entered with prompting or entered directly) must:

- Contain only ASCII characters from hexadecimal 21 to hexadecimal 7E [`\x21-\x7E`] (decimal 33 to 126). Spaces are not allowed. When the password is entered directly without prompting, the "?" symbol (hexadecimal 3F [`\x3F`] (decimal 63)) is not permitted.
- Contain at most 32 characters.
- Contain at least the number of characters configured (optionally) for `minimum-password-length`.



Although empty passwords are supported, it is recommended that you use strong passwords for all production switches.



Only an administrator can change the password of a user assigned to the `operators` role.

Examples

Enabling (or changing) a cleartext password for `admin`:

```
switch(config)# user admin password
Changing password for user admin
Enter password:*****
Confirm password:*****
```

Changing the cleartext password for user `chris`, using direct entry without prompting:

```
switch(config)# user chris password plaintext PASSwordZQ#@67
```

Changing the ciphertext password for user `alex` (the ciphertext shown is a placeholder that must be replaced with actual ciphertext):

```
switch(config)# user alex password ciphertext XqYJ36...W83D4Y=
```

service export-password

Syntax

```
service export-password
no service export-password
```

Description

Configures a nondefault export password. The export password is used to transform critical security parameters (such as password hashes) into ciphertext suitable for exporting and showing by commands such as `show running-config`. This transformation enables safe switch configuration import and export. The `no` form of this command reverts the export password to its factory default.



All factory-default switches have identical default export passwords. For security, it is recommended that you set the same nondefault export password on every switch in a group that will exchange configuration information. Only switches with identical export passwords can exchange configuration information.

Command context

`config`

Authority

Administrators or local user group members with execution rights for this command.

Usage

Prompts you twice for the new export password.

The export password must:

- Contain only ASCII characters from hexadecimal 21 to hexadecimal 7E [`\x21-\x7E`] (decimal 33 to 126). Spaces are not allowed.
- Contain at most 32 characters.
- Not be blank.

Examples

Configuring a new export password:

```
switch(config)# service export-password
Enter password:*****
Confirm password:*****
```

Reverting the export password to its factory default:

```
switch(config)# no service export-password
```

show user-group

Syntax

```
show user-group [<GROUP-NAME>]
```

Description

Shows user group information for the built-in groups plus any user-defined local user groups. When entered without *<GROUP-NAME>*, summary information is shown for all groups.

Command context

Manager (#)

Parameters

<GROUP-NAME>

Narrows the `show` command output to that of the specified group, and for local user groups, adds the User Group Rules list.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Show the list of all user groups, including built-in groups and local user groups.

```
switch# show user-group
GROUP NAME      GROUP TYPE      INCLUDED GROUP  NUMBER OF RULES
-----
administrators  built-in        n/a             n/a
admuser1        configuration    --             5
admuser2        configuration    admuser1        2
auditors        built-in        n/a             n/a
operators       built-in        n/a             n/a
```

Show detailed information for local user group `admuser2`.

```
switch(config-usr-grp-admuser2)# show user-group admuser2
User Group Summary
=====
Name           : admuser2
Type           : configuration
Included Group : admuser1
Number of Rules : 2
```

```

User Group Rules
=====
SEQUENCE NUM  ACTION      COMMAND      COMMENT
-----
10             deny        show aaa .*   Deny all show aaa commands.
20             permit     show .*       Permit all other show
commands.

```

show user information

Syntax

```
show user information
```

Description

Shows the following information for the logged-in user:

- User name.
- User authentication type: `local`, `RADIUS`, or `TACACS+`.
- User group: `administrators`, `operators`, or `<GROUP-NAME>`.
- User privilege level: For the built-in user groups and `RADIUS` or `TACACS+`, the role privilege level value is shown. For user-defined user groups, `N/A` is shown.

Command context

Operator (>) or Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Examples

Showing information for the `admin` user:

```

switch# show user information
Username           : admin
Authentication type : Local
User group         : administrators
User privilege level : 15

```

Showing information for a member of the user-defined local user group `admuser2`:

```

switch# show user information
Username           : admin2-b
Authentication type : Local
User group         : admuser2
User privilege level : N/A

```

Showing information for a member of `operators`:

```
switch# show user information
Username           : operator
Authentication type : Local
User group         : operators
User privilege level : 1
```

Showing information for remote RADIUS user `rad_user1` mapped to local user group `administrators`:

```
switch# show user information
Username           : rad_user1
Authentication type : RADIUS
User group         : administrators
User privilege level : 15
```

Showing information for remote RADIUS user `rad_user2` mapped to local user group `operators`:

```
switch# show user information
Username           : rad_user2
Authentication type : RADIUS
User group         : operators
User privilege level : 1
```

Showing information for remote TACACS+ `tac_user1` logged in with `priv-lvl 15` (mapped to user group `administrators`):

```
switch# show user information
Username           : tac_user1
Authentication type : TACACS+
User group         : administrators
User privilege level : 15
```

show user-list

Syntax

```
show user-list
```

Description

Shows all configured users and their corresponding group names.

Command context

Manager (#)

Authority

Administrators or local user group members with execution rights for this command.

Examples

Show the user list from a switch with only the admin user defined.

```
switch# show user-list
```

USER	GROUP

admin	administrators

Show the user list after adding a user to the operators built-in group.

```
switch# show user-list
```

USER	GROUP

admin	administrators
oper1	operators

Show the user list after adding a user to the auditors built-in group.

```
switch# show user-list
```

USER	GROUP

admin	administrators
oper1	operators
audit1	auditors

Show the user list after adding a total of three users to two user-defined user groups.

```
switch# show user-list
```

USER	GROUP

admin	administrators
oper1	operators
audit1	auditors
adm1a	admuser1
admin2-a	admuser2
admin2-b	admuser2

description

Syntax

```
description <DESCRIPTION>
```

Description

Specifies a descriptive for a VLAN.

Command context

```
config-vlan-<VLAN-ID>
```

Parameters

<DESCRIPTION>

Specifies a description for the VLAN.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Assigning a description to VLAN 20:

```
switch(config)# vlan 20
switch(config-vlan-20)# description primary
```

name

Syntax

```
name <VLAN-NAME>
```

Description

Associates a name with a VLAN.

Command context

```
config-vlan-<VLAN-ID>
```

Parameters

<VLAN-NAME>

Specifies a name for a VLAN. Length: 1 to 32 alphanumeric characters, including underscore (_) and hyphen (-).

Authority

Administrators or local user group members with execution rights for this command.

Examples

Assigning the name **backup** to VLAN 20:

```
switch(config)# vlan 20  
switch(config-vlan-20)# name backup
```

show capacities svi-count

Syntax

```
show capacities svi-count
```

Description

Shows the maximum number of SVIs supported by the switch.

Command context

Manager (#)

Authority

Administrators or local user group members with execution rights for this command.

Examples

Showing switch SVI capacity:

```
switch# show capacities svi-count  
System Capacities: Filter SVI count  
Capacities Name                                     Value  
-----  
Maximum number of SVIs supported in the system      16
```

show vlan

Syntax

```
show vlan [<VLAN-ID>]
```

Description

Displays configuration information for all VLANs or a specific VLAN.

Command context

Manager (#)

Parameters

<VLAN-ID>

Specifies a VLAN ID.

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Examples

Displaying configuration information for VLAN 2:

```
switch# show vlan 2
```

VLAN	Name	Status	Reason	Type	Interfaces
2	UserVLAN1	up	ok	static	1/1/1,1/1/3,1/1/5

Displaying configuration information for all defined VLANs:

```
switch# show vlan
```

VLAN	Name	Status	Reason	Type	Interfaces
1	DEFAULT_VLAN_1	up	ok	static	1/1/3-1/1/4
2	UserVLAN1	up	ok	static	1/1/1,1/1/3,1/1/5
3	UserVLAN2	up	ok	static	1/1/2-1/1/3,1/1/5-1/1/6
5	UserVLAN3	up	ok	static	1/1/3
10	TestNetwork	up	ok	static	1/1/3,1/1/5
11	VLAN11	up	ok	static	1/1/3
12	VLAN12	up	ok	static	1/1/3,1/1/6,lag1-lag2
13	VLAN13	up	ok	static	1/1/3,1/1/6
14	VLAN14	up	ok	static	1/1/3,1/1/6
20	ManagementVLAN	down	admin_down	static	1/1/3,1/1/10

show vlan port

Syntax

```
show vlan port <INTERFACE-ID>
```

Description

Displays the VLANs configured for a specific layer 2 interface.

Command context

Manager (#)

Parameters

<INTERFACE-ID>

Specifies an interface ID. Format: member/slot/port.

Authority

Operators or Administrators or local user group members with execution rights for this command.
Operators can execute this command from the operator context (>) only.

Example

Displaying the VLANs configured on interface **1/1/3**:

```
switch# show vlan port 1/1/3
```

VLAN	Name	Mode
1	DEFAULT_VLAN_1	native-untagged
2	UserVLAN1	trunk
3	UserVLAN2	trunk
5	UserVLAN3	trunk
10	TestNetwork	trunk
11	VLAN11	trunk
12	VLAN12	trunk
13	VLAN13	trunk
14	VLAN14	trunk
20	ManagementVLAN	trunk

show vlan summary

Syntax

```
show vlan summary
```

Description

Displays a summary of the VLAN configuration on the switch.

Command context

Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Displaying a summary of the VLAN configuration on the switch:

```
switch# show vlan summary
Number of existing VLANs: 11
Number of static VLANs:  11
Number of dynamic VLANs:  0
```

shutdown

Syntax

```
shutdown
```

```
no shutdown
```

Description

Disables a VLAN. (By default, a VLAN is automatically enabled when it is created with the `vlan` command.) The `no` form of this command enables a VLAN.

Command context

config-vlan-<VLAN-ID>

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling VLAN 20:

```
switch(config)# vlan 20  
switch(config-vlan-20)# no shutdown
```

Disabling VLAN 20:

```
switch(config)# vlan 20  
switch(config-vlan-20)# shutdown
```

vlan

Syntax

vlan <VLAN-LIST>

no vlan <VLAN-LIST>

Description

Creates a VLAN and changes to the `config-vlan-id` context for the VLAN. By default, the VLAN is enabled. To disable a VLAN, use the `no shutdown` command.

If the specified VLAN exists, this command changes to the `config-vlan-id` context for the VLAN. If a range of VLANs is specified, the context does not change.

The `no` form of this command removes a VLAN. VLAN 1 is the default VLAN and cannot be deleted.

Command context

config

Parameters

<VLAN-LIST>

Specifies a single ID, or a series of IDs separated by commas (2, 3, 4), dashes (2-4), or both (2-4,6). Range: 1 to . A maximum of 512 (6100 Switch Series) VLANs are supported.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Creating VLAN 20:

```
switch(config)# vlan 20  
switch(config-vlan-20)#
```

Removing VLAN 20:

```
switch(config)# no vlan 20
```

Creating VLANs **2** to **8** and **10**:

```
switch(config)# vlan 2-8,10
```

Removing VLANs **2** to **8** and **10**:

```
switch(config)# no vlan 2-8,10
```

vlan access

Syntax

```
vlan access <VLAN-ID>
```

```
no vlan access [<VLAN-ID>]
```

Description

Creates an access interface and assigns an VLAN ID to it. Only one VLAN ID can be assigned to each access interface.

VLANs can only be assigned to a non-routed (layer 2) interface or LAG interface. By default, all interfaces are routed (layer 3) when created. Use the `no routing` command to disable routing on an interface and change the interface to a layer 2 interface.

The `no` form of this command removes an access VLAN from the interface in the current context and sets it to the default VLAN ID of 1.

Command context

```
config-if
```

Parameters

<VLAN-ID>

Specifies a single ID, or a series of IDs separated by commas (2, 3, 4), dashes (2-4), or both (2-4,6). Range: 1 to . A maximum of 512 (6100 Switch Series) VLANs are supported.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring interface **1/1/2** as an access interface with VLAN ID set to **20**:

```
switch(config)# interface 1/1/2
switch(config-if)# vlan access 20
```

Removing VLAN ID **20** from interface **1/1/2**:

```
switch(config)# interface 1/1/2
```

```
switch(config-if)# no vlan access 20
```

or:

```
switch(config)# interface 1/1/2  
switch(config-if)# no vlan access
```

vlan trunk allowed

Syntax

```
vlan trunk allowed [<VLAN-LIST> | all]
```

```
no vlan trunk allowed [<VLAN-LIST>]
```

Description

Assigns a VLAN ID to a trunk interface. Multiple VLAN IDs can be assigned to a trunk interface. These VLAN IDs define which VLAN traffic is allowed across the trunk interface.

VLANs can be assigned only to a non-routed (layer 2) interface or LAG interface. By default, all interfaces are routed (layer 3) when created. Use the `no routing` command to disable routing on an interface.

The `no` form of this command removes one or more VLAN IDs from a trunk interface. When the last VLAN is removed from a trunk interface, the interface continues to operate in trunk mode, and will trunk all the VLANs currently defined on the switch, and any new VLANs defined in the future. To disable the trunk interface, use the command `shutdown`.

Command context

config-if

Parameters

<VLAN-LIST>

Specifies a single ID, or a series of IDs separated by commas (2, 3, 4), dashes (2-4), or both (2-4,6). Range: 1 to .

all

Configures the trunk interface to allow all the VLANs currently configured on the switch and any new VLANs that are configured in the future.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Assigning VLANs **2**, **3**, and **4** to trunk interface **1/1/2**:

```
switch(config)# interface 1/1/2  
switch(config-if)# vlan trunk allowed 2,3,4
```

Assigning VLAN IDs **2** to **8** to trunk interface **1/1/2**:

```
switch(config)# interface 1/1/2
switch(config-if)# vlan trunk allowed 2-8
```

Assigning VLAN IDs **2** to **8** and **10** to trunk interface **1/1/2**:

```
switch(config)# interface 1/1/2
switch(config-if)# vlan trunk allowed 2-8,10
```

Removing VLAN IDs **2**, **3**, and **4** from trunk interface **1/1/2**:

```
switch(config)# interface 1/1/2
switch(config-if)# no vlan trunk allowed 2,3,4
```

Removing all VLANs assigned to trunk interface **1/1/2**:

```
switch(config)# interface 1/1/2
switch(config-if)# no vlan trunk allowed
```

vlan trunk native

Syntax

```
vlan trunk native <VLAN-ID>
```

```
no vlan trunk native [<VLAN-ID>]
```

Description

Assigns a native VLAN ID to a trunk interface. By default, VLAN ID 1 is assigned as the native VLAN ID for all trunk interfaces. VLANs can only be assigned to a non-routed (layer 2) interface or LAG interface. Only one VLAN ID can be assigned as the native VLAN.



When a native VLAN is defined, the switch automatically executes the `vlan trunk allowed all` command to ensure that the default VLAN is allowed on the trunk. To only allow specific VLANs on the trunk, issue the `vlan trunk allowed` command specifying only specific VLANs.

The `no` form of this command removes a native VLAN from a trunk interface and assigns VLAN ID 1 as its native VLAN.

Command context

config-if

Parameters

<VLAN-ID>

Authority

Administrators or local user group members with execution rights for this command.

Examples

Assigning native VLAN ID **20** to trunk interface **1/1/2**:


```
switch(config)# interface 1/1/2
switch(config-if)# vlan trunk native 20
```

Removing native VLAN **20** from trunk interface **1/1/2** and returning to the default VLAN 1 as the native VLAN.

```
switch(config)# interface 1/1/2
switch(config-if)# no vlan trunk native 20
```

or:

```
switch(config)# interface 1/1/2
switch(config-if)# no vlan trunk native
```

Assigning native VLAN ID **20** to trunk interface **1/1/2** and then removing it from the list of allowed VLANs. (Only allow VLAN 10 on the trunk.)

```
switch(config)# interface 1/1/2
switch(config-if)# vlan trunk native 20
switch(config-if)# vlan trunk allowed 10
```

vlan trunk native tag

Syntax

```
vlan trunk native <VLAN-ID> tag
```

```
no vlan trunk native <VLAN-ID> tag
```

Description

Enables tagging on a native VLAN. Only incoming packets that are tagged with the matching VLAN ID are accepted. Incoming packets that are untagged are dropped except for BPDUs. Egress packets are tagged. The `no` form of this command removes tagging on a native VLAN.

Command context

config-if

Parameters

<VLAN-ID>

Specifies the number of a VLAN. Range: 1 to .

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling tagging on native VLAN **20** on trunk interface **1/1/2**:

```
switch(config)# interface 1/1/2
```

```
switch(config-if)# vlan trunk native 20  
switch(config-if)# vlan trunk native 20 tag
```

Removing tagging on native VLAN **20** assigned to trunk interface **1/1/2**:

```
switch(config)# interface 1/1/2  
switch(config-if)# no vlan trunk native 20 tag
```

Enabling tagging on native VLAN **20** assigned to LAG trunk interface **2**:

```
switch(config)# interface lag 2  
switch(config-lag-if)# vlan trunk native 20  
switch(config-lag-if)# vlan trunk native 20 tag
```

voice

Syntax

voice

no voice

Description

Configures a VLAN as a voice VLAN.

The **no** form of this command removes voice configuration from a VLAN.

Command context

config-vlan-<VLAN-ID>

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring VLAN 10 as a voice VLAN:

```
switch(config)# vlan 10  
switch(config-vlan-10)# voice
```

Removing voice from VLAN 10:

```
switch(config-vlan-10)# no voice
```

erase all zeroize

Syntax

```
erase all zeroize
```

Description

Restores the switch to its factory default configuration. You will be prompted before the procedure starts. Once complete, the switch will restart from the primary image with factory default settings.



Back up all data before running this command as all configuration settings will be lost.

Command context

Manager (#)

Authority

Administrators or local user group members with execution rights for this command.

Example

Restoring the switch to factory default configuration:

```
switch# erase all zeroize
This will securely erase all customer data and reset the switch
to factory defaults. This will initiate a reboot and render the
switch unavailable until the zeroization is complete.
This should take several minutes to one hour to complete.
Continue (y/n)? y
The system is going down for zeroization.

...

##### Preparing for zeroization #####

##### Storage zeroization #####
##### WARNING: DO NOT POWER OFF UNTIL #####
##### ZEROIZATION IS COMPLETE #####
##### This should take several minutes #####
##### to one hour to complete #####

##### Restoring files #####

...

We'd like to keep you up to date about:
```

- * Software feature updates
- * New product announcements
- * Special events

Please register your products now at: <https://asp.arubanetworks.com>

switch login: admin
Password:

Please configure the 'admin' user account password.
Enter new password: *****
Confirm new password: *****

show ztp information

Syntax

```
show ztp information
```

Description

Shows information about Zero Touch Provisioning (ZTP) operations performed on the switch.

Command context

Operator (>) or Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Usage

When a switch configured to use ZTP is booted from a factory default configuration, the switch contacts a DHCP server, which offers options for obtaining files used to provision the switch:

- The IP address of the TFTP server
- The name of the image file
- The name of the configuration file

The `show ztp information` command shows the options offered by the DHCP server and the status of the ZTP operation.

The status of the ZTP operation is one of the following:

Success

The ZTP operation succeeded.

One of the following is true:

- Both the running configuration and the startup configuration were updated.
- The IP address of the TFTP server was received, but the offer did not include a configuration file or a firmware image file.
- Any combination of vendor encapsulated DHCP options are received as configured, along with the firmware image and switch configuration file.
- Only vendor encapsulated DHCP options are configured and are received accordingly.

Failed - Custom startup configuration detected

The switch was booted from a configuration that is not the factory default configuration. For example, the administrator password has been set.

Failed - Timed out while waiting to receive ZTP options

Either the switch received the DHCP IPv4 address but no ZTP options were received within 1 minute or ZTP force-provision is triggered and no ZTP options are received within 3 minutes.

Failed - Detected change in running configuration

The running configuration was modified by a user while the ZTP operation was in progress.

Failed - TFTP server unreachable

The TFTP server is not reachable at the specified IP address.

Failed - TFTP server information unavailable

The image file name or config file name is provided without the TFTP server location to fetch the files from and ZTP enters failed state.

Failed - Invalid configuration file received

Either the file transfer of the configuration file failed, or the configuration file is invalid (an error occurred while attempting to apply the configuration).

Failed - Invalid image file received

Either the file transfer of the firmware image file failed, or the firmware image file is invalid (an error occurred while verifying the image).

Examples

Showing switch image download in progress after receiving ZTP options:

```
switch# show ztp information
TFTP Server       : 10.0.0.2
Image File        : TL_10_02_0001.swi
Configuration File : config_file
ZTP Status        : In-progress - Image download and verification
Aruba Central Location : secure.arubanetworks.com
Force-Provision   : Disabled
HTTP Proxy Location : http.proxy.arubanetworks.com
```

Showing switch image download failure after receiving ZTP options:

```
switch# show ztp information
TFTP Server       : 10.0.0.2
Image File        : TL_10_02_0001.swi
Configuration File : config_file
ZTP Status        : Failed - Unable to download image
Aruba Central Location : secure.arubanetworks.com
Force-Provision   : Disabled
HTTP Proxy Location : http.proxy.arubanetworks.com
```

Showing switch configuration download in progress after receiving ZTP options:

```
switch# show ztp information
TFTP Server       : 10.0.0.2
Image File        : TL_10_02_0001.swi
Configuration File : config_file
ZTP Status        : In-progress - Configuration download
Aruba Central Location : secure.arubanetworks.com
Force-Provision   : Disabled
HTTP Proxy Location : http.proxy.arubanetworks.com
```

Showing switch configuration download failure after receiving ZTP options:

```
switch# show ztp information
TFTP Server      : 10.0.0.2
Image File       : TL_10_02_0001.swi
Configuration File : config_file
ZTP Status       : Failed - Unable to download configuration
Aruba Central Location : secure.arubanetworks.com
Force-Provision  : Disabled
HTTP Proxy Location : http.proxy.arubanetworks.com
```

Showing switch failure to update start-up configuration after downloading configuration received from ZTP options:

```
switch# show ztp information
TFTP Server      : 10.0.0.2
Image File       : TL_10_02_0001.swi
Configuration File : config_file
ZTP Status       : Failed - Could not copy to start-up configuration
Aruba Central Location : secure.arubanetworks.com
Force-Provision  : Disabled
HTTP Proxy Location : http.proxy.arubanetworks.com
```

In the following example, the ZTP operation succeeded, and both an image file and a configuration file were provided.

```
switch# show ztp information
TFTP Server      : 20.1.1.4
Image File       : PL_10_06_0001BT.swi
Configuration File : bristol_maxlimit
Status           : Success
Force-Provision  : Disabled
switch#
```

In the following example, the ZTP option succeeded. A configuration file was not provided, but an image file was provided.

```
switch# show ztp information
TFTP Server      : 20.1.1.4
Image File       : NA
Configuration File : bristol_maxlimit
Status           : Success
Force-Provision  : Disabled
switch#
```

In the following example, the ZTP operation failed because the TFTP server was unreachable.

```
switch# show ztp information
TFTP Server      : 20.1.1.4
Image File       : PL_10_06_0001BT.swi
Configuration File : bristol_maxlimit
Status           : Failed - TFTP server unreachable
Force-Provision  : Disabled
switch#
```

In the following example, the ZTP operation was stopped because the switch did not receive any options from the DHCP server for ZTP within 1 minute of receiving the IP address from the server.

```
switch## show ztp information
TFTP Server      : NA
Image File       : NA
Configuration File : NA
Status           : Failed - Timed out while waiting to receive ZTP options
Force-Provision  : Disabled
switch#
```

In the following example, the ZTP operation was stopped because the switch was booted from a configuration that was not the factory default configuration.

```
switch# show ztp information
TFTP Server      : 20.1.1.4
Image File       : PL_10_06_0001BT.swi
Configuration File : bristol_maxlimit
Status           : Failed - Custom startup configuration detected
Force-Provision  : Disabled
```

ztp force provision

Syntax

```
ztp force-provision
no ztp force-provision
```

Description

Starts on-demand ZTP.

Command context

Operator (>) or Manager (#)

Authority

Administrators or local user group members with execution rights for this command.

Usage

DHCP options received are processed independent of the current state of configuration on the switch. Previous ZTP TFTP Server, Image File, Configuration File, Aruba Central Location, and HTTP Proxy location options are cleared and the switch sends a DHCP request.

Examples

In the following example, force-provision is enabled.

```
switch# configure terminal
switch(config)# ztp force-provision
```

In the following example, force-provision status is checked while enabled.


```
switch# show ztp information  
TFTP Server      : 10.0.0.2  
Image File       : TL_10_02_0001.swi  
Configuration File : ztp.cfg  
Status           : Success  
Aruba Central Location : NA  
Force-Provision   : Enabled  
HTTP Proxy Location : NA
```

In the following example, force-provision is disabled.

```
switch# configure terminal  
switch(config)# no ztp force-provision
```

In the following example, force-provision status is checked while disabled.

```
switch# show ztp information  
TFTP Server      : 10.0.0.2  
Image File       : TL_10_02_0001.swi  
Configuration File : ztp.cfg  
Status           : Success  
Aruba Central Location : NA  
Force-Provision   : Disabled  
HTTP Proxy Location : NA
```

Accessing Aruba Support

Aruba Support Services	https://www.arubanetworks.com/support-services/
Aruba Support Portal	https://asp.arubanetworks.com/
North America telephone	1-800-943-4526 (US & Canada Toll-Free Number) +1-408-754-1200 (Primary - Toll Number) +1-650-385-6582 (Backup - Toll Number - Use only when all other numbers are not working)
International telephone	https://www.arubanetworks.com/support-services/contact-support/

Be sure to collect the following information before contacting Support:

- Technical support registration number (if applicable)
- Product name, model or version, and serial number
- Operating system name and version
- Firmware version
- Error messages
- Product-specific reports and logs
- Add-on products or components
- Third-party products or components

Other useful sites

Other websites that can be used to find information:

Airheads social forums and Knowledge Base	https://community.arubanetworks.com/
Software licensing	https://lms.arubanetworks.com/
End-of-Life information	https://www.arubanetworks.com/support-services/end-of-life/
Aruba software and documentation	https://asp.arubanetworks.com/downloads

Accessing Updates

You can access updates from the Aruba Support Portal or the HPE My Networking Website.

Aruba Support Portal

<https://asp.arubanetworks.com/downloads>

If you are unable to find your product in the Aruba Support Portal, you may need to search My Networking, where older networking products can be found:

My Networking

<https://www.hpe.com/networking/support>

To view and update your entitlements, and to link your contracts and warranties with your profile, go to the Hewlett Packard Enterprise Support Center **More Information on Access to Support Materials** page:

<https://support.hpe.com/portal/site/hpsc/aae/home/>

Access to some updates might require product entitlement when accessed through the Hewlett Packard Enterprise Support Center. You must have an HP Passport set up with relevant entitlements.

Some software products provide a mechanism for accessing software updates through the product interface. Review your product documentation to identify the recommended software update method.

To subscribe to eNewsletters and alerts:

<https://asp.arubanetworks.com/notifications/subscriptions> (requires an active Aruba Support Portal (ASP) account to manage subscriptions). Security notices are viewable without an ASP account.

Warranty Information

To view warranty information for your product, go to <https://www.arubanetworks.com/support-services/product-warranties/>.

Regulatory Information

To view the regulatory information for your product, view the *Safety and Compliance Information for Server, Storage, Power, Networking, and Rack Products*, available at <https://www.hpe.com/support/Safety-Compliance-EnterpriseProducts>

Additional regulatory information

Aruba is committed to providing our customers with information about the chemical substances in our products as needed to comply with legal requirements, environmental data (company programs, product recycling, energy efficiency), and safety information and compliance data, (RoHS and WEEE). For more information, see <https://www.arubanetworks.com/company/about-us/environmental-citizenship/>.

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